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**LES C^* -ALGÈBRES
ET
LEURS REPRÉSENTATIONS**

**C^* -ALGEBRAS
AND THEIR REPRESENTATIONS**

BY

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TO MARC, ÉDITH AND FRANÇOISE

Preface to the Second Edition

This second edition was produced essentially by photographic means. This naturally ruled out any major revision. I have confined myself to making a few indispensable corrections (some of them due to J. Dieudonné), removing the problems solved since 1964, and supplementing the bibliography.

Introduction

Let H be a Hilbert space, and let $\mathcal{L}(H)$ be the set of continuous linear operators on H . Consider a subset A of $\mathcal{L}(H)$ that is stable under addition, multiplication, multiplication by scalars, and adjunction; suppose that A is closed in the operator norm. Then A is an involutive Banach algebra of a special kind. Such an algebra is called a C^* -algebra.

The theory began in 1943, when Gelfand and Naimark showed that, among involutive Banach algebras, C^* -algebras can be characterized by simple axioms. It was subsequently realized that C^* -algebras play a universal role in the study of representations of a very broad class of involutive Banach algebras: for every algebra B in this class, one can construct a C^* -algebra A such that the representations of B on a Hilbert space can be identified with the representations of A . For many questions (especially those involving ideals), A is easier to handle than B . In particular, this construction applies when B is taken to be the algebra of integrable functions on a locally compact group G . The study of the unitary representations of G is thereby reduced to that of the representations of a certain C^* -algebra, called the C^* -algebra of G .

The study of C^* -algebras occupies nearly four-fifths of this book. We present the principal results, due especially to the work of Fell, Glimm, Kadison, Kaplansky, Mackey, Segal, and others. It seemed a pity not to take advantage of the material thus accumulated, together with the material contained in my book on von Neumann algebras (*Cahiers Scientifiques*, fasc. XXV; cited as [A v N]), to say a few words about unitary representations of groups. This was all the more appropriate because group theory supplies the most interesting examples of C^* -algebras. The final pages of the book, however, by no means constitute a treatise on group representations. To gain an idea of the questions that ought to be treated in such a work, the reader may consult Mackey's lecture, *Infinite dimensional group representations* (*Bull. Amer. Math. Soc.*, vol. 69, 1963, pp. 629–686); of the sixty pages of that lecture, only the first twelve concern the questions treated in the present

book. Moreover, Gelfand and the Russian school on the one hand, and Harish-Chandra on the other, are scarcely cited here; this indicates how incomplete my account is as regards groups.

Although von Neumann algebras are examples of C^* -algebras, the theory of C^* -algebras is in fact dependent on that of von Neumann algebras, as will be seen repeatedly. To spare the reader from having to refer constantly to [A v N], the indispensable results on von Neumann algebras have been collected in Appendix A. I admit that I took the opportunity to reformulate certain statements so as to make them more readily usable. Almost always this reformulation is very easy; in the exceptional cases where it requires some work, the proof is indicated.

Results of various kinds used in the course of the text are, on the other hand, collected in Appendix B. That appendix is, of course, highly miscellaneous.

When writing [A v N], it seemed to me that most of the theorems presented there had a more or less definitive form. By contrast, the theory of C^* -algebras seems to me far from having reached a stable state.

* * *

Each section ends with supplements, given without proof or with a sketch of proof. Some of these easy supplements may serve as exercises, but it was difficult to separate them sharply from the others; I have merely marked with an asterisk those supplements whose proof is genuinely long (for example, more than three pages). The reader is asked not to be overly exacting about this classification. The supplements are not used in subsequent sections.

The bibliography contains the essentials as regards C^* -algebras. It is, however, very incomplete as regards groups.

As with [A v N], reading the present book presupposes a good knowledge of general topology, topological vector spaces, and integration. Proofs concerning C^* -algebras are given in fairly full detail; those concerning groups are more condensed and assume that the reader is familiar with the properties of convolution. It has also been assumed that the reader already knows the theory of locally compact abelian groups: although, in principle, that theory is not invoked (except for occasional points of detail), reference to the abelian case sheds much light on the problems studied.

The results are numbered lexicographically. A reference such as 4.7.5 is self-explanatory. References such as A 8 and B 15 concern Appendices A and B.

* * *

Finally, a few words on the relation between this book and those of Naimark and Rickart. The treatises of Naimark and Rickart contain, among other things, the general theory of normed algebras, which will not be found here. On the other hand, on the much more specialized subject of C^* -algebras, the present book is more detailed. There are of course developments common to all three books, but they are fairly limited. The general results on normed algebras used here are few in number and are cited in Appendix B.

C^* -Algebras and Their Representations

C^* -Algebras

§ 1. — Involutive Normed Algebras

1.1. Involutive Algebras

1.1.1. Definition. Let A be an algebra over the field $\mathbb{C}$ of complex numbers. An involution on A is a map $x \mapsto x^*$ from A into itself such that

- (i) $(x^*)^* = x$,
- (ii) $(x + y)^* = x^* + y^*$,
- (iii) $(\lambda x)^* = \overline{\lambda} x^*$,
- (iv) $(xy)^* = y^* x^*$,

for all $x, y \in A$ and $\lambda \in \mathbb{C}$. An algebra over $\mathbb{C}$ equipped with an involution is called an involutive algebra.

One often says that x^* is the adjoint of x . A subset of A stable under the involution is called self-adjoint.

Property (i) implies that an involution on A is necessarily a bijection of A onto itself.

1.1.2. Examples.

- (1) On $A = \mathbb{C}$, the map $z \mapsto \overline{z}$ (the complex conjugate of z) is an involution that makes A a commutative involutive algebra.
- (2) Let X be a locally compact space, and let A be the algebra of continuous complex-valued functions on X that vanish at infinity. Equipped with the map $f \mapsto \overline{f}$, it is a commutative involutive algebra. When X consists of a single point, this gives Example (1).
- (3) Let H be a Hilbert space, and let $A = \mathcal{L}(H)$ be the algebra of continuous endomorphisms of H . Equipped with the usual adjunction operation, it is an involutive algebra. Examples (2) and (3) will play a fundamental role. (Throughout this book, “Hilbert space” means “complex Hilbert space”.)

- (4) Let G be a unimodular locally compact group, and let A be the algebra $L^1(G)$ (with convolution). For every $f \in L^1(G)$, put

$$f^*(s) = \overline{f(s^{-1})} \quad (s \in G).$$

Equipped with the map $f \mapsto f^*$, A is an involutive algebra.

1.1.3. We now introduce terminology inspired by Example (3) above. Let A be an involutive algebra. An element $x \in A$ is called *Hermitian* if $x^* = x$, and *normal* if $xx^* = x^*x$. A Hermitian idempotent is called a *projection*. Every Hermitian element is normal. The set of Hermitian elements is a real vector subspace of A . If x and y are Hermitian, then $(xy)^* = y^*x^* = yx$, and hence xy is Hermitian if x and y commute. For every $x \in A$, the elements xx^* and x^*x are Hermitian [but a Hermitian element does not in general have either of these forms, as Example (1) above shows].

1.1.4. Every $x \in A$ can be written uniquely as $x_1 + ix_2$, with x_1, x_2 Hermitian. [In Example (1) above, this is the decomposition of a complex number into its real and imaginary parts.] Indeed, if we put

$$x_1 = \frac{1}{2}(x + x^*), \quad x_2 = \frac{1}{2i}(x - x^*),$$

then x_1 and x_2 are Hermitian and $x = x_1 + ix_2$. Conversely, if $x = x_1 + ix_2$ with x_1 and x_2 Hermitian, then $x^* = x_1 - ix_2$, so

$$x_1 = \frac{1}{2}(x + x^*), \quad x_2 = \frac{1}{2i}(x - x^*),$$

which proves the assertion. Observe that

$$\begin{aligned} xx^* &= x_1^2 + x_2^2 + i(x_2x_1 - x_1x_2), \\ x^*x &= x_1^2 + x_2^2 - i(x_2x_1 - x_1x_2); \end{aligned}$$

hence x is normal if and only if x_1 and x_2 commute.

1.1.5. If A has a left identity 1 , then, for every $x \in A$,

$$x1^* = (1x^*)^* = x^{**} = x;$$

thus 1^* is a right identity, and consequently $1 = 1^*$ is an identity element of A . If x is invertible in A , then

$$(x^{-1})^*x^* = (xx^{-1})^* = 1^* = 1, \quad x^*(x^{-1})^* = (x^{-1}x)^* = 1^* = 1;$$

hence x^* is invertible and $(x^*)^{-1} = (x^{-1})^*$. Conversely, if x^* is invertible, then $x^{**} = x$ is invertible. Since

$$(x - \lambda 1)^* = x^* - \bar{\lambda} 1$$

for every $\lambda \in \mathbb{C}$, it follows that

$$\text{Sp}_A x^* = \overline{\text{Sp}_A x}.$$

(In any unital algebra B , the *spectrum* of an element x , denoted by $\text{Sp}_B x$ or $\text{Sp } x$, is the set of scalars λ such that $x - \lambda 1$ is not invertible.)

An element $x \in A$ is called *unitary* if $xx^* = x^*x = 1$, or equivalently if x is invertible and $x = (x^*)^{-1}$. [In Example (1) above, the unitary elements are the complex numbers of modulus 1.] The unitary elements of A form a group under multiplication, the *unitary group* of A ; indeed, if x and y are unitary elements of A , then

$$(xy)^{*-1} = (y^*x^*)^{-1} = x^{*-1}y^{*-1} = xy,$$

so xy is unitary, and

$$(x^{-1})^{*-1} = (x^{*-1})^{-1} = x^{-1},$$

so x^{-1} is unitary.

1.1.6. Let A be an involutive algebra. If $\tilde{A}$ is the algebra obtained from A by adjoining an identity element, there is a unique involution on $\tilde{A}$ extending that on A , as is readily seen; it is defined by

$$(\lambda, x)^* = (\bar{\lambda}, x^*) \quad (\lambda \in \mathbb{C}, x \in A).$$

For every $x \in A$,

$$\text{Sp}'_A x = \text{Sp}_{\tilde{A}} x.$$

(In any algebra B , whether or not it has an identity, $\text{Sp}'_B x$ or $\text{Sp}' x$ denotes $\text{Sp}_{\tilde{B}} x$, where $\tilde{B}$ is the algebra obtained from B by adjoining an identity. One has $0 \in \text{Sp}'_B x$ for every $x \in B$.)

1.1.7. Let A and B be two involutive algebras. A *morphism* (respectively, an *isomorphism*) from A into B is a map (respectively, a bijection) φ from A into B such that

$$\begin{aligned} \varphi(x + y) &= \varphi(x) + \varphi(y), & \varphi(\lambda x) &= \lambda \varphi(x), \\ \varphi(xy) &= \varphi(x)\varphi(y), & \varphi(x^*) &= \varphi(x)^* \end{aligned}$$

for all $x, y \in A$ and $\lambda \in \mathbb{C}$. In particular, φ is a morphism from the algebra underlying A into the algebra underlying B . When there is a risk of confusion, we shall specify “morphism for the involutive-algebra structure” or “morphism for the algebra structure.”

1.1.8. Let A be an involutive algebra. An *involutive subalgebra* of A is a subalgebra of A stable under the involution. Every intersection of involutive subalgebras is an involutive subalgebra; hence, if M is any subset of A , there is a smallest involutive subalgebra of A containing M , namely the intersection of all involutive subalgebras of A containing M . It is called the involutive subalgebra of A generated by M ; it is the set of linear combinations of elements of the form $x_1 x_2 \cdots x_n$, where $x_1, x_2, \dots, x_n \in M \cup M^*$. When M consists of a single element x , this subalgebra is commutative if and only if x is normal.

Let A be an involutive algebra, and let B be a self-adjoint two-sided ideal of A . The involution of A induces, on passage to the quotient, an involution on the algebra A/B , and the canonical map from A onto A/B is a morphism.

Every product of involutive algebras is in the evident way an involutive algebra.

The algebra opposite to an involutive algebra, equipped with the same involution, is an involutive algebra.

1.1.9. Let A be an involutive algebra and let M be a self-adjoint subset of A . Then the commutant M' of M in A is an involutive subalgebra of A . Consequently, the bicommutant M'' of M in A is an involutive subalgebra of A containing M ; in general it differs from the involutive subalgebra of A generated by M (for example, when A is commutative). If the elements of M commute pairwise, then $M \subset M'$, hence $M' \supset M''$, and therefore M'' is commutative. Let $x \in A$ and $M = \{x, x^*\}$; then M'' is commutative if and only if x is normal.

1.1.10. Let A be an involutive algebra. If f is a linear form on A , the function

$$x \mapsto \overline{f(x^*)}$$

on A is a linear form on A , denoted by f^* and called the adjoint of f . One has

$$f^{**} = f, \quad (f + f')^* = f^* + f'^*, \quad (\lambda f)^* = \bar{\lambda} f^* \quad (\lambda \in \mathbb{C}).$$

The form f is called Hermitian if $f = f^*$. Every linear form f on A can be written uniquely as $f_1 + i f_2$, with f_1, f_2 Hermitian; namely,

$$f_1 = \frac{1}{2}(f + f^*), \quad f_2 = \frac{1}{2i}(f - f^*).$$

A linear form f on A is Hermitian if and only if it is real-valued on the set A_h of Hermitian elements of A . The map $f \mapsto f|_{A_h}$ is an isomorphism from the real vector space of Hermitian forms onto the dual of the real vector space A_h . If A is commutative and χ is a character of A , then χ^* is a character of A .

Bibliography: [108, 126].

1.2. Involutive Normed Algebras

1.2.1. Definition. An involutive normed algebra is a normed algebra A equipped with an involution $x \mapsto x^*$ such that $\|x^*\| = \|x\|$ for every $x \in A$. If, in addition, A is complete, it is called an involutive Banach algebra.

1.2.2. Examples. The four examples in 1.1.2 are examples of involutive Banach algebras when the norms are defined as follows: in Example (1), take $\|z\| = |z|$ for every $z \in \mathbb{C}$; in Example (2), take $\|f\| = \sup_{t \in X} |f(t)|$ for every $f \in A$; in Example (3), take the usual norm on $\mathcal{L}(H)$; in Example (4), take

$$\|f\| = \int_G |f(g)| dg$$

for every $f \in L^1(G)$.

1.2.3. Let A be an involutive normed algebra, and let $\tilde{A}$ be the involutive algebra obtained from A by adjoining an identity element. There are norms on $\tilde{A}$ extending that of A and making $\tilde{A}$ an involutive normed algebra (for example, one may put $\|(\lambda, x)\| = |\lambda| + \|x\|$ for $\lambda \in \mathbb{C}$, $x \in A$). Every involutive normed algebra obtained in this way is called an involutive normed algebra obtained from A by adjoining an identity element.

1.2.4. Let A and B be two involutive normed algebras. A morphism from A into B will simply mean a morphism of the underlying involutive algebras (with no condition on the norms). By contrast, an isomorphism will mean an isomorphism of the underlying involutive algebras that preserves the norms.

1.2.5. Let A be an involutive normed algebra. The closure of an involutive subalgebra is an involutive subalgebra. If $M \subset A$, the smallest closed involutive subalgebra B containing M is called the *closed involutive subalgebra generated by M* ; it is the closure of the involutive subalgebra generated by M . If M consists of a normal element, B is commutative.

The quotient of an involutive normed algebra by a closed self-adjoint two-sided ideal, the product of finitely many involutive normed algebras, the opposite of an involutive normed algebra, and the completion of an involutive normed algebra are naturally involutive normed algebras.

1.2.6. Let A be an involutive normed algebra. If f is a continuous linear form on A , then f^* is continuous and $\|f^*\| = \|f\|$, because the unit ball of A is self-adjoint. The set A_h of Hermitian elements of A is a real normed vector space. Let f be a continuous Hermitian linear form on A , and put $g = f|_{A_h}$. Then $\|f\| = \|g\|$. Indeed, it is clear that $\|f\| \geq \|g\|$. On the other hand, for every $\varepsilon > 0$, there exists $x \in A$ such that $\|x\| \leq 1$ and $|f(x)| \geq \|f\| - \varepsilon$; multiplying x by a scalar of modulus 1, we may suppose that $f(x) \geq 0$. Then

$$\left| g\left(\frac{1}{2}(x + x^*)\right) \right| = \frac{1}{2} \left| f(x) + \overline{f(x)} \right| = f(x) \geq \|f\| - \varepsilon.$$

Since $\left\| \frac{1}{2}(x + x^*) \right\| \leq 1$, it follows that $\|g\| \geq \|f\| - \varepsilon$, proving the assertion. We shall therefore identify continuous Hermitian linear forms on A with continuous real linear forms on A_h .

Bibliography: [108, 126].

1.3. C^* -Algebras

1.3.1. Definition. A C^* -algebra is an involutive Banach algebra A such that

$$\|x\|^2 = \|x^*x\|$$

for every $x \in A$.

1.3.2. Examples (1), (2), and (3) of 1.1.2 and 1.2.2 are examples of C^* -algebras. By contrast, Example (4) is not in general a C^* -algebra.

1.3.3. If A is a C^* -algebra, every closed involutive subalgebra of A is a C^* -algebra. In particular, if H is a Hilbert space, every closed involutive subalgebra of $\mathcal{L}(H)$ is a C^* -algebra. We shall see later (2.6.1) that every C^* -algebra is isomorphic to a C^* -algebra of this kind; it was in connection with this example that the theory of C^* -algebras was created.

Let $(A_i)_{i \in I}$ be a family of C^* -algebras. Let A be the set of families $(x_i)_{i \in I}$ such that $x_i \in A_i$ for every $i \in I$ and

$$\sup_{i \in I} \|x_i\| < +\infty.$$

With the operations

$$\begin{aligned}(x_i) + (y_i) &= (x_i + y_i), & \lambda(x_i) &= (\lambda x_i), \\ (x_i)(y_i) &= (x_i y_i), & (x_i)^* &= (x_i^*),\end{aligned}$$

and the norm

$$\|(x_i)\| = \sup_i \|x_i\|,$$

one verifies at once that A is a C^* -algebra, called the product C^* -algebra of the A_i . Notice that the set A is not the full Cartesian product of the A_i .

Let A be a C^* -algebra. Retain all the operations and the norm of A , except for the multiplication $(x, y) \mapsto xy$, which is replaced by $(x, y) \mapsto yx$. In other words, consider the involutive normed algebra A° opposite to A . It is immediate that A° is a C^* -algebra.

The question of quotient C^* -algebras is more delicate (see 1.8.2).

1.3.4. Let A be a Banach algebra equipped with an involution such that

$$\|x\|^2 \leq \|x^*x\|.$$

It follows that $\|x\|^2 \leq \|x^*\| \|x\|$, whence $\|x\| \leq \|x^*\|$; replacing x by x^* shows that $\|x^*\| = \|x\|$. The hypothesis then implies

$$\|x\|^2 \leq \|x^*x\| \leq \|x^*\| \|x\| = \|x\|^2,$$

so A is a C^* -algebra.

1.3.5. Let A be a C^* -algebra. For every $x \in A$,

$$\|x\| = \sup_{\|x'\| \leq 1} \|xx'\|.$$

Indeed, it is clear that $\|x'\| \leq 1$ implies $\|xx'\| \leq \|x\|$. To prove the reverse inequality, we may suppose that $\|x\| = 1$; then $\|x^*\| = 1$, and

$$\sup_{\|x'\| \leq 1} \|xx'\| \geq \|xx^*\| = \|x\|^2 = 1.$$

1.3.6. Let A be a unital C^* -algebra. One has

$$\|1\|^2 = \|1^*1\| = \|1\|,$$

and therefore $\|1\| = 0$ or 1 . Thus, if $A \neq 0$, then $\|1\| = 1$. It follows that every unitary element u of A satisfies

$$\|u\| = \|u^*u\|^{1/2} = 1$$

when $A \neq 0$.

Recall that no continuity condition was imposed in the definition of morphisms of involutive normed algebras. We shall see that, for C^* -algebras, every morphism is automatically continuous. More precisely:

1.3.7. Proposition. *Let A be an involutive Banach algebra, let B be a C^* -algebra, and let π be a morphism from A into B . Then*

$$\|\pi(x)\| \leq \|x\|$$

for every $x \in A$.

Proof. For every Hermitian element y of B , one has

$$\|y^2\| = \|y^*y\| = \|y\|^2,$$

and hence, by induction, $\|y^{2^n}\|^{2^{-n}} = \|y\|$. As $n \rightarrow +\infty$, the left-hand side tends to the spectral radius $\rho(y)$ of y (see B 1); hence

$$\rho(y) = \|y\|. \quad (1)$$

Now, for every $x \in A$,

$$\mathrm{Sp}'_B \pi(x) \subset \mathrm{Sp}'_A x, \quad \rho(\pi(x)) \leq \rho(x) \leq \|x\|.$$

Consequently, in view of (1),

$$\|\pi(x)\|^2 = \|\pi(x^*x)\| = \rho(\pi(x^*x)) \leq \|x^*x\| \leq \|x^*\| \|x\| = \|x\|^2.$$

The following proposition will make it possible, in almost every case, to reduce the study of C^* -algebras to that of C^* -algebras with an identity element.

1.3.8. Proposition. *Let A be a C^* -algebra, and let $\tilde{A}$ be the involutive algebra obtained from A by adjoining an identity element. There is a unique norm on $\tilde{A}$ extending that of A and making $\tilde{A}$ a C^* -algebra.*

Proof. Uniqueness follows from 1.3.7. We prove existence. Suppose first that A has an identity element e , and denote the identity element of $\tilde{A}$ by 1. In $\tilde{A}$, the spaces A and $\mathbb{C}(1 - e)$ are supplementary self-adjoint two-sided ideals; consequently there is an isomorphism of the involutive algebra $\tilde{A}$ onto the involutive algebra $\mathbb{C} \times A$ that carries A onto $\{0\} \times A$. We may therefore equip $\mathbb{C} \times A$ with its product C^* -algebra structure. Suppose from now on that A has no identity

element. Every element x of $\widetilde{A}$ defines an operator L_x of left multiplication on the two-sided ideal A of $\widetilde{A}$; put

$$\|x\| = \|L_x\|.$$

For $x \in A$, this gives back the given norm on A , by 1.3.5. It is also clear that $x \mapsto \|x\|$ is a seminorm on $\widetilde{A}$ and that $\|xy\| \leq \|x\| \|y\|$. This seminorm is a norm. Indeed, let $x = \lambda 1 - x'$ ($\lambda \in \mathbb{C}$, $x' \in A$) be an element of $\widetilde{A}$ such that $\|L_x\| = 0$, that is, $xy = 0$ for every $y \in A$, and let us prove that $x = 0$. If $\lambda \neq 0$, then for every $y \in A$,

$$0 = \lambda^{-1}xy = y - \lambda^{-1}x'y;$$

thus $\lambda^{-1}x'$ is a left identity of A , and A has an identity element by 1.1.5, contrary to the hypothesis. Therefore $\lambda = 0$, so $x \in A$, and $\|x\| = 0$ implies $x = 0$. We have thus proved that $x \mapsto \|x\|$ is a norm on $\widetilde{A}$. Since A is complete and has codimension one in $\widetilde{A}$, the latter is complete. It remains to prove, by 1.3.4, that

$$\|x\|^2 \leq \|x^*x\|$$

for every $x \in \widetilde{A}$, and it suffices to do so when $\|x\| = 1$. For every $r < 1$, there is a $y \in A$ such that $\|y\| \leq 1$ and $\|xy\|^2 \geq r$; then, since $xy \in A$,

$$\|x^*x\| \geq \|y^*(x^*x)y\| = \|(xy)^*(xy)\| = \|xy\|^2 \geq r,$$

and hence $\|x^*x\| \geq 1$.

1.3.9. Proposition. *Let A be a C^* -algebra.*

- (i) *If h is a Hermitian element of A , every point of $\text{Sp}' h$ is real.*
- (ii) *If A has an identity element and u is a unitary element of A , every point of $\text{Sp } u$ has modulus 1.*

Proof. By 1.3.8, in proving both parts we may suppose that A has an identity element (and that $A \neq 0$). By 1.3.6,

$$\|u\| = \|u^{-1}\| = 1;$$

thus $\rho(u) \leq 1$ and $\rho(u^{-1}) \leq 1$, so both $\text{Sp } u$ and $\text{Sp}(u^{-1}) = (\text{Sp } u)^{-1}$ lie in the closed unit disk of $\mathbb{C}$, which proves (ii). On the other hand,

$$(\exp(ih))^* = \left(1 + ih + \frac{i^2 h^2}{2!} + \cdots\right)^* = 1 + (-i)h + \frac{(-i)^2 h^2}{2!} + \cdots = \exp(-ih),$$

so $\exp(ih)$ is unitary. Hence, if $z \in \operatorname{Sp} h$, then $|\exp(iz)| = 1$ (B 4), and therefore $z \in \mathbb{R}$. (Throughout the book, $\mathbb{R}$ denotes the set of real numbers.)

The following proposition shows that, in the case of C^* -algebras, there is no need to be concerned with the ambient algebra when defining the spectrum of an element.

1.3.10. Proposition. *Let A be a C^* -algebra, let B be a C^* -subalgebra, and let $x \in B$.*

- (i) $\operatorname{Sp}'_A x = \operatorname{Sp}'_B x$.
- (ii) *If A has an identity element belonging to B , then $\operatorname{Sp}_A x = \operatorname{Sp}_B x$.*

Proof. By adjoining an identity element, one sees that (i) follows from (ii). We prove (ii). If x is Hermitian, then $\operatorname{Sp}_B x \subset \mathbb{R}$ by 1.3.9, and hence $\operatorname{Sp}_B x = \operatorname{Sp}_A x$ by B 2. In the general case, if $x \in B$ is invertible in A , then xx^* is invertible in A , hence in B by what precedes, and therefore x is right-invertible in B . Similarly, x is left-invertible in B , and is thus invertible in B . Applying this result to $x - \lambda 1$, where $\lambda \in \mathbb{C}$, proves (ii).

C^* -algebras are called completely regular algebras in [108], and B^* -algebras by many authors.

Bibliography: [36, 108, 123, 125, 126].

1.4. Commutative C^* -Algebras

Their theory is completed by the following theorem:

1.4.1. Theorem. *Let A be a commutative C^* -algebra, let S be its spectrum (which is a locally compact space), and let B be the C^* -algebra of continuous complex-valued functions on S that vanish at infinity.*

- (i) *Every character of A is Hermitian.*
- (ii) *The Gelfand transform is an isomorphism of the C^* -algebra A onto the C^* -algebra B .*

Proof. Let χ be a character of A . If y is a Hermitian element of A , then

$$\chi(y) \in \operatorname{Sp}' y \subset \mathbb{R}$$

by 1.3.9. If x is any element of A , write $x = x_1 + ix_2$ with x_1, x_2 Hermitian; then

$$\chi(x^*) = \chi(x_1 - ix_2) = \chi(x_1) - i\chi(x_2) = \overline{\chi(x)},$$

which proves (i).

In other words, if $\mathcal{G}$ denotes the Gelfand transform, then

$$\mathcal{G}(x^*) = \overline{\mathcal{G}(x)}$$

for every $x \in A$. Moreover, $\mathcal{G}(A)$ separates the points of S , and at every point of S there is a function in $\mathcal{G}(A)$ that does not vanish (B 3). The Stone–Weierstrass theorem therefore shows that $\mathcal{G}(A)$ is dense in B . To complete the proof, it remains only to show that $\mathcal{G}$ is isometric. But $\|\mathcal{G}(y)\| = \|y\|$ for Hermitian y by 1.3.7, formula (1); hence, for every $x \in A$,

$$\|x\|^2 = \|x^*x\| = \|\mathcal{G}(x^*x)\| = \|\overline{\mathcal{G}(x)}\mathcal{G}(x)\| = \|\mathcal{G}(x)\|^2.$$

1.4.2. Retain the preceding notation. If $f \in B$, then $g(f)$ has a meaning for every continuous complex-valued function g on

$$f(S) \cup \{0\} = \text{Sp}'_B f$$

that vanishes at 0, and it represents an element of B . Thus, if $x \in A$, one may define $h(x) \in A$ for every continuous complex-valued function h on $\text{Sp}'_A x$ that vanishes at 0. In 1.5.1 we shall see that such a “functional calculus” can be defined for normal elements of an arbitrary C^* -algebra (not necessarily commutative). It will be a very useful tool in what follows.

1.4.3. Proposition. *Let A be a unital commutative C^* -algebra, let S be its spectrum, and let $x \in A$. Suppose that the C^* -subalgebra of A generated by 1 and x is equal to A . Then*

$$\chi \longmapsto \chi(x)$$

is a homeomorphism from S onto $\text{Sp}_A x$.

Proof. This map is continuous and has image $\text{Sp}_A x$ (B 3). Let $\chi, \chi' \in S$ satisfy $\chi(x) = \chi'(x)$. Since χ and χ' are continuous and Hermitian, the set A' of all $y \in A$ such that $\chi(y) = \chi'(y)$ is a C^* -subalgebra of A containing x . Thus $A' = A$ and $\chi' = \chi$. The map in question is injective and hence is a homeomorphism, since S is compact.

Bibliography: [1, 41, 108, 126].

1.5. Functional Calculus in C^* -Algebras

1.5.1. Theorem. *Let A be a unital C^* -algebra, let x be a normal element of A , let $S = \text{Sp}_A x$, and let A' be the C^* -algebra of continuous complex-valued functions on S . There is a unique morphism φ from A' into A such that*

$$\varphi(1) = 1, \quad \varphi(\iota) = x,$$

where ι denotes the function $z \mapsto z$ on S . This morphism is isometric. Its image is the C^* -subalgebra of A generated by 1 and x , and hence consists of normal elements.

Proof. The polynomials in z and $\bar{z}$ are dense in A' , and every morphism from A' into A is continuous by 1.3.7; this proves uniqueness of φ . Let B be the commutative C^* -subalgebra of A generated by 1 and x , let T be its spectrum, let C be the C^* -algebra of continuous complex-valued functions on T , and let $\mathcal{G}_B$ be the Gelfand transform for B , which is an isomorphism of B onto C by 1.4.1. Proposition 1.4.3 gives a homeomorphism from T onto $\text{Sp}_B x = S$ by 1.3.10, and hence an isomorphism $\psi : A' \rightarrow C$ carrying ι to $\mathcal{G}_B x$ [for every $\chi \in T$,

$$(\mathcal{G}_B x)(\chi) = \chi(x) = \iota(\chi(x)).$$

] Consider the composite isomorphism

$$A' \xrightarrow{\psi} C \xrightarrow{\mathcal{G}_B^{-1}} B.$$

Composing it with the canonical injection of B into A gives a morphism from A' into A with the asserted properties.

1.5.2. Definition. *Let A be a unital C^* -algebra. If x is a normal element of A and f is a continuous complex-valued function on $\text{Sp}_A x$, the element $\varphi(f)$ of Theorem 1.5.1 is denoted by $f(x)$.*

The fact that φ is an isometric isomorphism is expressed by the following formulas, where f and g are continuous complex-valued functions on $\text{Sp}_A x$:

$$(f + g)(x) = f(x) + g(x), \tag{1}$$

$$(fg)(x) = f(x)g(x), \tag{2}$$

$$\overline{f}(x) = f(x)^*, \tag{3}$$

$$\|f(x)\| = \|f\|. \tag{4}$$

If f is the restriction to $\text{Sp}_A x$ of a polynomial $z \mapsto P(z, \bar{z})$ in z and $\bar{z}$, then

$$f(x) = P(x, x^*),$$

where the symbol $P(x, x^*)$ has its usual algebraic meaning (recall that $xx^* = x^*x$).

With the notation above,

$$\text{Sp}_A f(x) = \text{Sp}_B f(x) = \text{Sp}_{A'} f = f(S),$$

or, equivalently,

$$\text{Sp}_A f(x) = f(\text{Sp}_A x). \quad (5)$$

1.5.3. Proposition. *Let A and B be two unital C^* -algebras, let φ be a morphism from A into B carrying 1 to 1, and let x be a normal element of A , so that $\varphi(x)$ is normal in B . Let f be a continuous complex-valued function on $\text{Sp}_A x$. Denoting its restriction to $\text{Sp}_B \varphi(x)$ again by f , one has*

$$\varphi(f(x)) = f(\varphi(x)).$$

Proof. Let C be the C^* -algebra of continuous complex-valued functions on $\text{Sp}_A x$. The maps

$$f \mapsto \varphi(f(x)), \quad f \mapsto f(\varphi(x))$$

are morphisms from C into B and take the same values when f is one of the functions $z \mapsto 1$, $z \mapsto z$, or $z \mapsto \bar{z}$. Since the C^* -subalgebra of C generated by these functions is all of C , the two morphisms are equal.

1.5.4. Corollary. *Let A be a unital commutative C^* -algebra, let $x \in A$, let $\mathcal{G}$ be the Gelfand transform for A , and let f be a continuous complex-valued function on $\text{Sp}_A x$. Then*

$$\mathcal{G}(f(x)) = f \circ \mathcal{G}(x).$$

This follows, for example, from 1.5.3 applied to $\varphi = \mathcal{G}$.

1.5.5. Corollary. *Let A be a unital C^* -algebra, let x be a normal element of A , let C be the C^* -algebra of continuous complex-valued functions on $\text{Sp } x$, let $f \in C$, let C' be the C^* -algebra of continuous complex-valued functions on*

$$\text{Sp } f(x) = f(\text{Sp } x),$$

and let $g \in C'$. Then $g \circ f \in C$ and

$$(g \circ f)(x) = g(f(x)).$$

Proof. The map $g \mapsto (g \circ f)(x)$ is a morphism from C' into A that carries 1 to 1 and the function $z \mapsto z$ to $f(x)$. By the uniqueness assertion in Theorem 1.5.1,

$$(g \circ f)(x) = g(f(x)).$$

1.5.6. Proposition. *Let A be a C^* -algebra, let $x \in A$ be normal, let $S = \text{Sp}'_A x$, and let A' be the C^* -algebra of continuous complex-valued functions on S that vanish at 0. There is a unique morphism φ from A' into A such that*

$$\varphi(\iota) = x,$$

where ι denotes the function $z \mapsto z$ on S . This morphism is isometric. Its image is the C^* -subalgebra of A generated by x , and hence consists of normal elements.

Proof. Since the polynomials in z and $\bar{z}$ without constant terms are dense in A' , uniqueness of φ is immediate. Existence follows from Theorem 1.5.1 upon adjoining an identity element to A .

1.5.7. By adjoining an identity element, all the results of this subsection extend immediately, with the evident modifications, to C^* -algebras without an identity element. We shall therefore refer to these results even when the existence of an identity element has not been assumed.

In particular, let x be a Hermitian element of the C^* -algebra A . Its spectrum is real. Consider the continuous functions of a real variable

$$t \mapsto f_1(t) = \sup(t, 0), \quad t \mapsto f_2(t) = \sup(-t, 0), \quad t \mapsto f_3(t) = |t|.$$

Put

$$x^+ = f_1(x), \quad x^- = f_2(x), \quad |x| = f_3(x).$$

These are Hermitian elements of A (indeed, of the C^* -subalgebra of A generated by x), because $\overline{f_1} = f_1$, $\overline{f_2} = f_2$, and $\overline{f_3} = f_3$. Since f_1, f_2, f_3 take nonnegative values, one has

$$\text{Sp}'(x^+) \geq 0, \quad \text{Sp}'(x^-) \geq 0, \quad \text{Sp}'(|x|) \geq 0. \quad (1)$$

Since

$$f_1(t) - f_2(t) = t, \quad f_1(t) + f_2(t) = f_3(t), \quad f_1(t)f_2(t) = 0,$$

one has

$$x = x^+ - x^-, \quad |x| = x^+ + x^-, \quad x^+x^- = x^-x^+ = 0. \quad (2)$$

The norm of $|x|$ is equal to that of x , and the norms of x^+ and x^- are at most that of x . Observe, for 1.6.4, that x^+ and x^- are the squares of two Hermitian elements whose product is zero: it suffices to consider the functions $\sqrt{f_1}$ and $\sqrt{f_2}$.

1.5.8. Let A be a C^* -algebra. For every integer $n > 0$, one has $A = A^n$ (the set of linear combinations of products of n elements of A). Indeed, it suffices to prove that every Hermitian element x of A is a product of n elements of A . If $f_1, \dots, f_n$ are continuous real-valued functions of a real variable such that

$$f_1(t)f_2(t)\cdots f_n(t) = t, \quad f_1(0) = \cdots = f_n(0) = 0,$$

then

$$x = f_1(x)f_2(x)\cdots f_n(x).$$

Bibliography: [41, 108, 126].

1.6. Positive Elements in C^* -Algebras

1.6.1. Proposition. *Let A be a C^* -algebra, and let x be a Hermitian element of A . The following conditions are equivalent:*

- (i) $\text{Sp}'_A x \geq 0$.
- (ii) x has the form yy^* for some $y \in A$.
- (iii) x has the form h^2 for some Hermitian $h \in A$.

Moreover, the set P of elements satisfying these conditions is a closed convex cone such that

$$P \cap (-P) = \{0\}.$$

Proof. For the proof, let P denote the set of Hermitian elements of A satisfying condition (i).

(i) $\Rightarrow$ (iii): if $\text{Sp}'_A x \geq 0$, one may form $h = x^{1/2}$, which is a Hermitian element of A , and $x = h^2$.

(iii) $\Rightarrow$ (i): if $x = h^2$ with h Hermitian, then

$$\text{Sp}'_A x = (\text{Sp}'_A h)^2 \geq 0,$$

since $\text{Sp}'_A h$ is real.

(iii) $\Rightarrow$ (ii) is evident.

To prove (ii) $\Rightarrow$ (iii) and the final assertion of the proposition, we shall need the following lemmas.

1.6.2. Lemma. Suppose that A has an identity element. If $x \in A$ is Hermitian and $\|1 - x\| \leq 1$, then $x \in P$. If $x \in P$ and $\|x\| \leq 1$, then $\|1 - x\| \leq 1$.

Proof. Considering the C^* -subalgebra of A generated by x , we reduce first to the case where A is commutative and then, by 1.4.1, to the case where A is the C^* -algebra of continuous complex-valued functions on a compact space. The lemma is then evident.

1.6.3. Lemma. Suppose that A has an identity element. Let x be a Hermitian element of A . Then $x \in P$ if and only if

$$\| \|x\|1 - x \| \leq \|x\|.$$

Proof. We may suppose that $x \neq 0$ and hence, by scaling, that $\|x\| = 1$. The assertion then follows at once from Lemma 1.6.2.

1.6.4. Return to the setting of 1.6.1. To prove that P is a closed convex cone with $P \cap (-P) = \{0\}$, we may suppose, by 1.3.8, that A has an identity element. Lemma 1.6.3 implies that P is closed. It is clear that $x \in P$ and $\lambda \geq 0$ imply $\lambda x \in P$. Let $x, y \in P$, and let us show that $x + y \in P$. We may suppose that $\|x\| \leq 1$ and $\|y\| \leq 1$. Then $\|1 - x\| \leq 1$ and $\|1 - y\| \leq 1$ by 1.6.2; hence

$$\left\| 1 - \frac{1}{2}(x + y) \right\| = \frac{1}{2} \|1 - x + 1 - y\| \leq \frac{1}{2} (\|1 - x\| + \|1 - y\|) \leq 1.$$

Thus $\frac{1}{2}(x + y) \in P$ by 1.6.2, and $x + y \in P$. If $x \in P \cap (-P)$, then $\text{Sp } x = \{0\}$, so $\rho(x) = 0$ and hence $x = 0$ by 1.3.7, formula (1).

Finally, without assuming that A has an identity element, let us prove (ii) $\Rightarrow$ (iii) in 1.6.1. Let $y \in A$. Write

$$(yy^*)^+ = u^2, \quad (yy^*)^- = v^2,$$

where u, v are Hermitian elements of A with $uv = 0$, as in 1.5.7. Then

$$(vy)(vy)^* = v(yy^*)v = vu^2v - v^4 = -v^4 \in -P,$$

because (iii) $\Rightarrow$ (i). Write $vy = s + it$ with s, t Hermitian. Then

$$\begin{aligned} (vy)^*(vy) &= -(vy)(vy)^* + (s + it)(s - it) + (s - it)(s + it) \\ &= -(vy)(vy)^* + 2s^2 + 2t^2 \in P, \end{aligned}$$

since $-(vy)(vy)^* \in P, s^2 \in P, t^2 \in P$, and P is a convex cone. Therefore $(vy)(vy)^* \in P$, because in every algebra the spectrum of a product does not depend on the order of its factors (B 26). Thus

$$(vy)(vy)^* \in P \cap (-P) = \{0\}.$$

It follows that $v^4 = 0$, hence $v = 0$ and $yy^* = u^2$. Proposition 1.6.1 is proved.

1.6.5. Definition. Let A be a C^* -algebra. An element x of A is called positive, and one writes $x \geq 0$, if it is Hermitian and satisfies the three equivalent conditions of Proposition 1.6.1. The set of positive elements of A is denoted by A^+ .

If B is a C^* -subalgebra of A and $x \in B$, then, by 1.3.10, the condition $x \geq 0$ has the same meaning relative to A as it does relative to B .

Since A^+ is a convex cone and $A^+ \cap (-A^+) = \{0\}$, the relation $x - y \geq 0$ is an order relation on A_h , compatible with its real vector-space structure; this relation is denoted by $x \geq y$ or $y \leq x$. If z is a Hermitian element of A , then $z^+ \geq 0$, $z^- \geq 0$, and $|z| \geq 0$ by 1.5.7, formulas (1); z^+ (respectively, z^-) is called the positive (respectively, negative) part of z . By 1.5.7, formulas (2), every Hermitian element of A is the difference of two elements of A^+ .

1.6.6. Let A be the C^* -algebra of continuous complex-valued functions vanishing at infinity on a locally compact space T , and let $f \in A$. The relation $f \geq 0$ in A has its usual meaning, by condition (i) of Proposition 1.6.1 and the fact that

$$\text{Sp}' f = f(T) \cup \{0\}.$$

1.6.7. Let H be a Hilbert space, let A be the C^* -algebra $\mathcal{L}(H)$, and let $x \in A$. We show that $x \geq 0$ is equivalent to

$$\langle x\xi, \xi \rangle \geq 0 \quad \text{for every } \xi \in H,$$

that is, to the usual definition of operators ≥ 0 . If $x = y^*y$ with $y \in A$, then

$$\langle x\xi, \xi \rangle = \langle y^*y\xi, \xi \rangle = \|y\xi\|^2 \geq 0 \quad (\xi \in H).$$

Conversely, suppose that $\langle x\xi, \xi \rangle \geq 0$ for every $\xi \in H$. For every $\eta \in H$,

$$\begin{aligned} 0 &\leq \langle x(x^-\eta), x^-\eta \rangle \\ &= \langle (x^+ - x^-)(x^-\eta), x^-\eta \rangle \\ &= -\langle (x^-)^2\eta, x^-\eta \rangle = -\langle (x^-)^3\eta, \eta \rangle. \end{aligned}$$

Since $(x^-)^3 \geq 0$, one also has $\langle (x^-)^3\eta, \eta \rangle \geq 0$; hence this inner product is zero for every η , so $(x^-)^3 = 0$, $x^- = 0$, and finally $x = x^+ \geq 0$.

1.6.8. Let A be a C^* -algebra, and let $a, b, x \in A$. If $a \leq b$, then

$$x^*ax \leq x^*bx.$$

Indeed, there is a $y \in A$ such that $b - a = y^*y$, whence

$$x^*bx - x^*ax = x^*(y^*y)x = (yx)^*(yx) \geq 0.$$

Suppose that A has an identity element, and let $b \in A$ satisfy $b \geq 1$ (so that b is invertible). Applying what precedes with $x = b^{-1/2}$, one obtains $1 \geq b^{-1}$. More generally, let a and b be invertible elements of A^+ such that $0 \leq a \leq b$. Then $a^{-1} \geq b^{-1}$; indeed, by what precedes, successively,

$$1 \leq a^{-1/2}ba^{-1/2}, \quad 1 \geq a^{1/2}b^{-1}a^{1/2}, \quad a^{-1} \geq b^{-1}.$$

1.6.9. Let A be a C^* -algebra, and let $x, y \in A^+$ satisfy $y \leq x$. Then

$$\|y\| \leq \|x\|.$$

Indeed, we may suppose that A has an identity element. By 1.4.1, for example, $x \leq \|x\|1$. Hence

$$0 \leq y \leq \|x\|1,$$

and therefore $\|y\| \leq \|x\|$, again by 1.4.1.

1.6.10. Let A and B be two C^* -algebras, and let φ be a morphism from A into B . Clearly,

$$\varphi(A^+) \subset \varphi(A) \cap B^+.$$

Conversely, let $y \in \varphi(A) \cap B^+$; there is an $x \in A$ such that $y = \varphi(x)$, and

$$y = (y^*y)^{1/2} = \varphi((x^*x)^{1/2}),$$

so $y \in \varphi(A^+)$. Therefore

$$\varphi(A^+) = \varphi(A) \cap B^+.$$

Bibliography: [36, 80, 108, 126, 132, 142].

1.7. Approximate Identities in C^* -Algebras

1.7.1. Let A be a C^* -algebra. We shall say that an approximate identity (u_λ) of A (B 29) is *increasing and directed* if $u_\lambda \geq 0$ for every λ , and if $\lambda \leq \mu$ implies $u_\lambda \leq u_\mu$.

1.7.2. Proposition. *Let A be a C^* -algebra, and let m be a dense two-sided ideal of A . There exists an increasing directed approximate identity of A consisting of elements of m . If A is separable, this approximate identity may moreover be indexed by $\{1, 2, \dots\}$.*

Proof. Let $\tilde{A}$ be the C^* -algebra obtained from A by adjoining an identity element. Let Λ be the set of finite subsets of m , ordered by inclusion. For

$$\lambda = \{x_1, \dots, x_n\} \in \Lambda,$$

put

$$v_\lambda = x_1 x_1^* + \dots + x_n x_n^* \in m, \quad u_\lambda = v_\lambda \left(\frac{1}{n} + v_\lambda \right)^{-1}$$

(the element u_λ is calculated in $\tilde{A}$, but in fact $u_\lambda \in m$). Since the real-valued function

$$t \mapsto t \left(\frac{1}{n} + t \right)^{-1}$$

lies between 0 and 1 for $t \geq 0$, one has

$$0 \leq u_\lambda \leq 1.$$

Moreover,

$$\sum_{i=1}^n [(u_\lambda - 1)x_i][(u_\lambda - 1)x_i]^* = (u_\lambda - 1)v_\lambda(u_\lambda - 1) = \frac{1}{n^2}v_\lambda \left(\frac{1}{n} + v_\lambda \right)^{-2}.$$

The real-valued function

$$t \mapsto t \left(\frac{1}{n} + t \right)^{-2}$$

is at most $n/4$. Therefore

$$\sum_{i=1}^n [(u_\lambda - 1)x_i][(u_\lambda - 1)x_i]^* \leq \frac{1}{4n}.$$

For $i = 1, 2, \dots, n$, it follows that

$$[(u_\lambda - 1)x_i][(u_\lambda - 1)x_i]^* \leq \frac{1}{4n},$$

and hence, by 1.6.9,

$$\|(u_\lambda - 1)x_i\|^2 \leq \frac{1}{4n}.$$

Thus $\|(u_\lambda - 1)x\| \rightarrow 0$ for every $x \in m$, and consequently for every $x \in A$, since $\overline{m} = A$ and $\|u_\lambda\| \leq 1$; moreover,

$$\|xu_\lambda - x\| = \|u_\lambda x^* - x^*\| \rightarrow 0.$$

Thus (u_λ) is an approximate identity.

Let $\lambda, \mu \in \Lambda$ with $\lambda \leq \mu$. Write

$$\lambda = \{x_1, \dots, x_n\}, \quad \mu = \{x_1, \dots, x_p\}, \quad p \geq n;$$

then $v_\lambda \leq v_\mu$. Hence, by 1.6.8,

$$\left(\frac{1}{n} + v_\lambda\right)^{-1} \geq \left(\frac{1}{n} + v_\mu\right)^{-1}.$$

For every real number $t \geq 0$,

$$\frac{1}{n} \left(\frac{1}{n} + t\right)^{-1} \geq \frac{1}{p} \left(\frac{1}{p} + t\right)^{-1};$$

therefore

$$\frac{1}{n} \left(\frac{1}{n} + v_\mu\right)^{-1} \geq \frac{1}{p} \left(\frac{1}{p} + v_\mu\right)^{-1}.$$

Altogether this gives

$$1 - \frac{1}{n} \left(\frac{1}{n} + v_\lambda\right)^{-1} \leq 1 - \frac{1}{n} \left(\frac{1}{n} + v_\mu\right)^{-1} \leq 1 - \frac{1}{p} \left(\frac{1}{p} + v_\mu\right)^{-1},$$

that is, $u_\lambda \leq u_\mu$. The approximate identity (u_λ) is increasing and directed.

Suppose that A is separable. There is a sequence $(y_1, y_2, \dots)$ dense in m . Put

$$u_n = u_{\{y_1, y_2, \dots, y_n\}}.$$

The preceding argument shows that, for every i , $\|u_n y_i - y_i\| \rightarrow 0$ as $n \rightarrow +\infty$. Since $\|u_n\| \leq 1$, it follows that $u_n x \rightarrow x$ for every $x \in A$, and the proof is completed as above.

1.7.3. Let A be a C^* -algebra, and let I be a right ideal of A . There is a family (u_λ) in $I \cap A^+$, indexed by a directed set, such that:

- 1° $\|u_\lambda\| \leq 1$;
- 2° $\lambda \leq \mu$ implies $u_\lambda \leq u_\mu$;
- 3° for every $x \in \bar{I}$, one has $\|u_\lambda x - x\| \rightarrow 0$.

The proof of 1.7.2 applies word for word.

Bibliography: [16, 134].

1.8. Quotient of a C^* -Algebra

1.8.1. Proposition. *Let A be a C^* -algebra, let B be an involutive normed algebra, and let φ be an injective morphism from A into B . Then*

$$\|\varphi(x)\| \geq \|x\|$$

for every $x \in A$.

Proof. Let $x \in A$. If we show that

$$\|\varphi(x^*x)\| \geq \|x^*x\|,$$

then

$$\|x\|^2 = \|x^*x\| \leq \|\varphi(x^*x)\| = \|\varphi(x)^*\varphi(x)\| \leq \|\varphi(x)\|^2,$$

which proves the proposition. We may therefore suppose from now on that x is Hermitian. Replacing φ by its restriction to the C^* -subalgebra of A generated by x , we may suppose that A is commutative. Replacing B by $\varphi(A)$, we may suppose that B is commutative, and then replace B by its completion. We may also adjoin identity elements to A and B . In short, we are reduced to the case where A and B are commutative, complete, and unital.

Let S and T be the spectra of A and B , which are compact spaces. For every $\chi \in T$, the composite $\chi \circ \varphi$ is a character of A , that is, an element of S , which we denote by $\varphi'(\chi)$. If $x \in A$, then

$$\varphi'(\chi)(x) = \chi(\varphi(x))$$

is a continuous function of χ , so φ' is a continuous map from T into S , and $\varphi'(T)$ is a compact subset of S . If $\varphi'(T) \neq S$, there are continuous complex-valued functions f, g on S such that

$$fg = 0, \quad f \neq 0, \quad g = 1 \text{ on } \varphi'(T).$$

By 1.4.1, f and g are the Gelfand transforms of two elements $x, y \in A$ such that

$$xy = 0, \quad x \neq 0, \quad \chi(\varphi(y)) = 1 \quad \text{for every } \chi \in T.$$

Then $\varphi(y)$ is invertible in B by B 3, while $\varphi(x)\varphi(y) = 0$ and $\varphi(x) \neq 0$, a contradiction. Hence $\varphi'(T) = S$. It follows that, for every $x \in A$,

$$\|x\| = \sup_{\chi \in S} |\chi(x)| = \sup_{\chi \in T} |\varphi'(\chi)(x)|$$

$$= \sup_{\chi \in T} |\chi(\varphi(x))| \leq \|\varphi(x)\|.$$

1.8.2. Proposition. *Let A be a C^* -algebra, and let I be a closed two-sided ideal of A . Then I is self-adjoint, and A/I , equipped with its natural involutive-algebra structure and the quotient norm, is a C^* -algebra.*

Proof. Let (u_λ) be a family with the properties of 1.7.3. If $x \in I$, then

$$\|x^*u_\lambda - x^*\| = \|u_\lambda x - x\| \longrightarrow 0,$$

and $x^*u_\lambda \in I$, so $x^* \in \bar{I} = I$; hence $I = I^*$.

It is known that A/I is an involutive algebra satisfying the axioms of a Banach algebra. Denote the canonical map from A onto A/I by $x \mapsto \dot{x}$. To prove that A/I is a C^* -algebra, it is enough, by 1.3.4, to prove that

$$\|\dot{x}\|^2 \leq \|\dot{x}\dot{x}^*\|.$$

One has

$$\|\dot{x}\| = \lim_{\lambda} \|x - u_\lambda x\|. \quad (1)$$

Indeed, if $y \in I$, then $u_\lambda y \rightarrow y$, and therefore

$$\begin{aligned} \lim_{\lambda} \|x - u_\lambda x\| &= \lim_{\lambda} \|x - u_\lambda x + y - u_\lambda y\| \\ &= \lim_{\lambda} \|(1 - u_\lambda)(x + y)\| \leq \|x + y\|, \end{aligned}$$

where the calculation is made in $\tilde{A}$. Thus

$$\|\dot{x}\| \leq \lim_{\lambda} \|x - u_\lambda x\| \leq \inf_{y \in I} \|x + y\| = \|\dot{x}\|,$$

which proves (1). Now, for every $z \in I$,

$$\begin{aligned} \|\dot{x}\|^2 &= \lim_{\lambda} \|x - u_\lambda x\|^2 \\ &= \lim_{\lambda} \|(x - u_\lambda x)(x - u_\lambda x)^*\| \\ &= \lim_{\lambda} \|xx^* - u_\lambda xx^* - xx^*u_\lambda + u_\lambda xx^*u_\lambda\| \\ &= \lim_{\lambda} \|xx^* + z - u_\lambda z - u_\lambda xx^* - xx^*u_\lambda - zu_\lambda + u_\lambda zu_\lambda + u_\lambda xx^*u_\lambda\| \\ &= \lim_{\lambda} \|(1 - u_\lambda)(xx^* + z)(1 - u_\lambda)\| \\ &\leq \|xx^* + z\|. \end{aligned}$$

Therefore

$$\|\dot{x}\|^2 \leq \|\dot{x}\dot{x}^*\|.$$

1.8.3. Corollary. *Let A and B be C^* -algebras, let φ be a morphism from A into B , and let I be the kernel of φ . Consider the canonical factorization*

$$A \longrightarrow A/I \xrightarrow{\psi} \varphi(A) \longrightarrow B.$$

Then I is closed in A , $\varphi(A)$ is closed in B , and ψ is an isometric isomorphism of the C^ -algebra A/I onto the C^* -algebra $\varphi(A)$.*

Proof. Since φ is continuous by 1.3.7, I is closed; and A/I is a C^* -algebra by 1.8.2. The morphism $A/I \rightarrow B$ induced by φ on passing to the quotient is injective, and hence isometric by 1.3.7 and 1.8.1. Thus $\varphi(A)$ is complete and consequently closed in B .

1.8.4. Corollary. *Let A be a C^* -algebra, let B be a C^* -subalgebra of A , and let I be a closed two-sided ideal of A . Then $B + I$ is a C^* -subalgebra of A , and the C^* -algebras*

$$(B + I)/I \quad \text{and} \quad B/(B \cap I)$$

are canonically isomorphic.

Proof. Let φ be the canonical morphism $A \rightarrow A/I$, and let ψ be the restriction of φ to B . Then

$$\psi(B) = (B + I)/I$$

is closed in A/I by 1.8.3, so $B + I$ is closed in A . Consider the canonical factorization of ψ :

$$B \longrightarrow B/(B \cap I) \longrightarrow \psi(B) \longrightarrow A/I.$$

By 1.8.3, the morphism

$$B/(B \cap I) \longrightarrow \psi(B) = (B + I)/I$$

is an isomorphism of C^* -algebras.

1.8.5. Proposition. *Let A be a C^* -algebra, let I be a closed two-sided ideal of A , and let J be a closed two-sided ideal of I . Then J is a closed two-sided ideal of A .*

Proof. By 1.5.8, $J = J^3$, and hence

$$AJA = AJ^3A \subset IJI \subset J.$$

Bibliography: [41, 74, 108, 126, 135]. The proofs of 1.8.2 and 1.7.3 were communicated to me orally by F. Combes.

1.9. Supplements

1.9.1. Let A be a C^* -algebra. Retain the norm and all the operations of A except scalar multiplication $(\lambda, x) \mapsto \lambda x$, which is replaced by $(\lambda, x) \mapsto \bar{\lambda}x$. The resulting C^* -algebra $\bar{A}$ is called the conjugate of A .

1.9.2. Let A be a C^* -algebra. If $x \in A$ is normal, then

$$\|x\| = \rho(x).$$

(Use 1.4.1.) [126].

1.9.3. Let A be a C^* -algebra.

- a. If every maximal commutative C^* -subalgebra of A has an identity element, then A has an identity element.
- b. If every maximal commutative C^* -subalgebra of A is finite-dimensional, then A is finite-dimensional. [116]

1.9.4. Let A be a C^* -algebra. If the conditions $x \in A^+$, $y \in A^+$, and $x \geq y$ imply $x^2 \geq y^2$, then A is commutative. [117]

*** 1.9.5.** Let A be a unital Banach algebra equipped with an involution such that

$$\|x^*x\| = \|x^*\|\|x\|$$

for every $x \in A$. Then A is a C^* -algebra. [49, 120]

1.9.6. Problem. Does a unital C^* -algebra A of dimension > 1 , whose only closed two-sided ideals are 0 and A , have projections distinct from 0 and 1 ? [79]

1.9.7. Let A be a unital C^* -algebra, and let $x \in A$ be not left-invertible. Then x^*x is not invertible in the C^* -subalgebra B generated by 1 and x^*x ; consequently there exist $y_1, y_2, \dots \in B$ such that

$$\|y_n\| = 1, \quad \|y_n x^* x\| \longrightarrow 0$$

(use 1.4.1). Thus, if $x \in A$ is not invertible, then x is a topological zero divisor. [126]

1.9.8. Let A be a unital C^* -algebra, let N be the set of normal elements of A , let $x \in N$, and let V be a neighborhood of 0 in $\mathbb{C}$. There is a neighborhood U of x in N such that, for every $y \in U$,

$$\operatorname{Sp} y \subset \operatorname{Sp} x + V, \quad \operatorname{Sp} x \subset \operatorname{Sp} y + V.$$

[126]

* **1.9.9.** Let A be a unital C^* -algebra, let H be a Hilbert space, and let φ be a linear map from A into $\mathcal{L}(H)$ such that

$$\varphi(A^+) \subset \mathcal{L}(H)^+.$$

Then, for every Hermitian $x \in A$,

$$\varphi(x^2) \geq \varphi(x)^2.$$

[66]

* **1.9.10.** Let A and B be two unital C^* -algebras.

a. Let $\rho : A \rightarrow B$ be a bijective linear map such that

$$\rho(x^*) = \rho(x)^*, \quad \rho(x^n) = \rho(x)^n$$

for Hermitian x and every integer $n > 0$. Then $\rho(A^+) = B^+$, ρ is isometric, ρ carries commuting pairs to commuting pairs, and $\rho(1) = 1$.

b. Let $\tau : A \rightarrow B$ be a bijective linear isometry. Then τ is the product of a map ρ having the properties in (a) and left multiplication by the element $\tau(1)$ of B .

c. Let $\tau : A \rightarrow B$ be a linear isometry such that $\tau(1) = 1$. Then

$$\tau(x^*) = \tau(x)^*$$

for every $x \in A$. [3, 65]

* **1.9.11.** Let A be a C^* -algebra.

a. Every derivation of A is continuous.

b. Let D be a derivation of A , and let x be a normal element of A . If

$$x(Dx) = (Dx)x,$$

then $Dx = 0$. In particular, every derivation of a commutative C^* -algebra is zero. It follows that a closed commutative two-sided ideal in a C^* -algebra is central.

c. Let D be a derivation of A , and let x be a normal element of A such that $Dx = 0$. Then $Dx^* = 0$. [79, 129, 130]

d. Let D be a derivation of A , and let I be a closed two-sided ideal of A . Then

$$D(I) \subset I.$$

(Use 1.5.8.)

- e. Let H be an infinite-dimensional Hilbert space, let A be the C^* -algebra of compact operators on H , and let $x \in \mathcal{L}(H)$ with $x \notin A + \mathbb{C}1$. The derivation

$$y \mapsto xy - yx$$

of A is not inner.

1.9.12. Let A be a C^* -algebra.

- a. Let I, J be two closed two-sided ideals of A . The product ideal IJ is equal to $I \cap J$. (If $x \in (I \cap J)^+$, then $x^{1/2} \in (I \cap J)^+$.) [21]

One has

$$(I + J)^+ = I^+ + J^+.$$

[293, 365]

- b. If $I \cap J = 0$, the canonical map

$$I + J \longrightarrow I \times J$$

is an isomorphism of C^* -algebras.

- c. Let (I_λ) be a family of closed two-sided ideals of A with intersection I . Let ω_λ (respectively, ω) be the canonical morphism of A onto A/I_λ (respectively, A/I). For every $x \in A$,

$$\|\omega(x)\| = \sup_{\lambda} \|\omega_\lambda(x)\|.$$

[29]

1.9.13. Let A be a C^* -algebra. A C^* -seminorm on A is a seminorm N such that

$$N(x) \leq \|x\|, \quad N(xy) \leq N(x)N(y), \quad N(x^*x) = N(x)^2$$

for all $x, y \in A$. The set $\mathcal{N}(A)$ of C^* -seminorms on A is compact for the topology of pointwise convergence. If I is a closed two-sided ideal of A and $x \in A$, let $N_I(x)$ be the norm of the canonical image of x in A/I . Then

$$I \mapsto N_I$$

is a bijection from the set $\mathcal{J}(A)$ of closed two-sided ideals of A onto $\mathcal{N}(A)$. If $I, J \in \mathcal{J}(A)$, then

$$N_{I \cap J} = \sup(N_I, N_J)$$

(use 1.9.12(b)). A C^* -seminorm N is called *extremal* if it cannot be the supremum of two C^* -seminorms without being equal to one of them. A nonzero C^* -seminorm N_I is extremal if and only if I is prime. [In an arbitrary algebra R , a

two-sided ideal J is called prime if $J \neq R$ and the relation $J'J'' \subset J$, where J', J'' are two-sided ideals of R , implies $J' \subset J$ or $J'' \subset J$.] If A is separable, the set E of extremal C^* -seminorms is a G_δ in $\mathcal{N}(A)$. [The set of pairs

$$(N, N') \in \mathcal{N}(A) \times \mathcal{N}(A)$$

such that $N \leq N'$ or $N' \leq N$ is compact. Its complement is a countable union of compact sets; its image in $\mathcal{N}(A)$ under the continuous map

$$(N, N') \longmapsto \sup(N, N')$$

is the complement of E .] [20, 29]

1.9.14. Let $(A_i)_{i \in I}$ be a family of C^* -algebras. Let A be the set of

$$x = (x_i) \in \prod_{i \in I} A_i$$

such that, for every $\varepsilon > 0$, one has $\|x_i\| \leq \varepsilon$ except for finitely many indices i . With

$$\|x\| = \sup_i \|x_i\|,$$

A becomes naturally a C^* -algebra, called the restricted product of the A_i .

1.9.15. The radical R of a C^* -algebra is zero. [If $x \in R$, then $1 + \lambda x^*x$ is invertible in $\tilde{A}$ for every $\lambda \in \mathbb{C}$, so $\text{Sp}'(x^*x) = \{0\}$ and $x = 0$.] [133]

§ 2. — Positive Functionals and Representations

One of the most important notions in this book is that of a representation. Given an involutive Banach algebra A , it would be difficult to establish directly the existence of representations of A . We shall instead define a correspondence between representations of A and positive functionals on A , and in particular between irreducible representations and pure positive functionals. The classical tools of functional analysis (the Hahn–Banach and Krein–Milman theorems) will then make it possible to prove the existence of positive functionals, and even of pure positive functionals. This is the essential idea of this section.

In the various statements, varying hypotheses are imposed on the involutive algebras under consideration. On a first reading, one may suppose throughout that they are C^* -algebras.

2.1. Positive Functionals

2.1.1. Definition. *Let A be an involutive algebra. A linear functional f on A is said to be positive if $f(x^*x) \geq 0$ for every $x \in A$. If A is an involutive normed algebra, a continuous positive linear functional f on A such that $\|f\| = 1$ is called a state of A .*

Let A be an involutive algebra, and let f and g be two linear functionals on A . We say that f majorizes g , and write $f \geq g$ or $g \leq f$, if $f - g$ is positive. This defines on the dual of A a preorder relation compatible with its real vector-space structure. If A is a C^* -algebra, this relation is in fact an order relation: for if $f \geq 0$ and $f \leq 0$, then f vanishes on A^+ , hence on the set of hermitian elements of A (1.6.5), and consequently $f = 0$.

Let Ω be a locally compact space, and let A be the C^* -algebra of continuous complex-valued functions on Ω that vanish at infinity. A continuous linear functional on A is nothing other than a bounded measure μ on Ω . To say that this functional is positive means exactly that the measure μ is positive.

2.1.2. Let A be an involutive algebra and f a positive functional on A . For $x, y \in A$, put

$$(x \mid y) = f(y^*x).$$

This inner product is linear in x , conjugate-linear in y , and $(x \mid x) \geq 0$ for every x . Thus A acquires the structure of a pre-Hilbert space.

In particular, this proves that

$$\overline{f(y^*x)} = f(x^*y) \quad (x \in A, y \in A), \quad (1)$$

$$|f(y^*x)|^2 \leq f(x^*x)f(y^*y) \quad (x \in A, y \in A). \quad (2)$$

If A has an identity element, then, on setting $y = 1$ in (1) and (2), we obtain

$$f(x^*) = \overline{f(x)}, \quad (3)$$

$$|f(x)|^2 \leq f(1)f(x^*x). \quad (4)$$

Let H be the separated pre-Hilbert space canonically obtained from the pre-Hilbert space A : thus $H = A/N$, where N is the set of all $x \in A$ such that $f(x^*x) = 0$. By (2), N is also the set of all $x \in A$ such that $f(yx) = 0$ for every $y \in A$. Hence N is a left ideal of A .

[If one put $(x \mid y) = f(xy^*)$, one would obtain another pre-Hilbert space structure on A , with analogous properties.]

2.1.3. Lemma. *Let A be a Banach algebra with identity, let $x' \in A$ satisfy $\|x'\| \leq 1$, and set $x = 1 + x'$. The series*

$$1 + \frac{1}{2}x' + \frac{1}{2!} \frac{1}{2} \left(\frac{1}{2} - 1 \right) x'^2 + \dots \\ + \frac{1}{n!} \frac{1}{2} \left(\frac{1}{2} - 1 \right) \dots \left(\frac{1}{2} - n + 1 \right) x'^n + \dots$$

converges to an element $y \in A$ such that $y^2 = x$. If A is equipped with an isometric involution and x is hermitian, then y is hermitian.

The series

$$1 + \frac{1}{2}\|x'\| + \frac{1}{2!} \left| \frac{1}{2} \left(\frac{1}{2} - 1 \right) \right| \|x'\|^2 + \dots \\ + \frac{1}{n!} \left| \frac{1}{2} \left(\frac{1}{2} - 1 \right) \dots \left(\frac{1}{2} - n + 1 \right) \right| \|x'\|^n + \dots$$

converges, which proves the existence of y . If y^2 is calculated, one obtains a power series in x' whose coefficients are known from the classical case $A = \mathbb{C}$; hence $y^2 = 1 + x' = x$. If A is equipped with an isometric involution and x is hermitian, y appears as a limit of hermitian elements and is therefore hermitian, since the involution is continuous.

2.1.4. Proposition. *Let A be an involutive Banach algebra having an identity 1 such that $\|1\| = 1$. If f is a positive linear functional on A , then f is continuous and $\|f\| = f(1)$.*

If $x \in A$ is hermitian and $\|x\| \leq 1$, then $1 - x$ can be written in the form y^*y (Lemma 2.1.3); hence $f(1 - x) \geq 0$ and $f(x) \leq f(1)$. If $x' \in A$ and $\|x'\| \leq 1$, then $\|x'^*x'\| \leq 1$, and therefore, by 2.1.2, formula (4),

$$|f(x')|^2 \leq f(1)f(x'^*x') \leq f(1)^2.$$

This proves that f is continuous and that $\|f\| \leq f(1)$. It is clear that

$$f(1) \leq \|f\|, \quad \text{and hence} \quad \|f\| = f(1).$$

2.1.5. Proposition. *Let A be an involutive Banach algebra admitting an approximate identity (B 29), let $\tilde{A}$ be the involutive algebra obtained from A by adjoining an identity element, and let f be a continuous positive linear functional on A .*

- (i) $f(x^*) = \overline{f(x)}$ and $|f(x)|^2 \leq \|f\|f(x^*x)$ for every $x \in A$.
- (ii) $|f(y^*xy)| \leq \|x\|f(y^*y)$ for all $x, y \in A$.
- (iii) $\|f\| = \sup_{x \in A, \|x\| \leq 1} f(x^*x)$.
- (iv) If $(x_i)_{i \in I}$ is a family of elements of A , indexed by a directed set, such that $\|x_i\| \leq 1$ and $f(x_i) \rightarrow \|f\|$, then $f(x_i^*x_i) \rightarrow \|f\|$.
- (v) If $(u_i)_{i \in I}$ is an approximate identity of A , then $f(u_i) \rightarrow \|f\|$ and $f(u_i^*u_i) \rightarrow \|f\|$.
- (vi) f extends uniquely to a positive functional $\tilde{f}$ on $\tilde{A}$ such that $\tilde{f}(1) = \|f\|$. Every positive functional on $\tilde{A}$ extending f majorizes $\tilde{f}$.
- (vii) With the notation of (iv), $x_i \rightarrow 1$ for the pre-Hilbert space structure defined by $\tilde{f}$ on $\tilde{A}$; consequently, A is dense in $\tilde{A}$ for this structure.

For every $x \in A$, we have

$$\begin{aligned} \overline{f(x^*)} &= \lim_i \overline{f(x^*u_i)} = \lim_i \overline{f((u_i^*x)^*)} = \lim_i f(u_i^*x) = f(x), \\ |f(x)|^2 &= \lim_i |f(xu_i)|^2 \leq f(x^*x) \lim_i f(u_i^*u_i) \leq \|f\|f(x^*x). \end{aligned}$$

This proves (i), and the implications (i) $\Rightarrow$ (iv) $\Rightarrow$ (iii) are clear. Let us prove (vi). The uniqueness of $\tilde{f}$ is immediate. We prove its existence. For every element $(\lambda, x) = \lambda + x$ of $\tilde{A}$ ($\lambda \in \mathbb{C}$, $x \in A$), put

$$\tilde{f}(\lambda + x) = \lambda\|f\| + f(x).$$

Then $\tilde{f}$ is a linear functional on $\tilde{A}$ extending f , and, by (i),

$$\begin{aligned} \tilde{f}((\lambda + x)^*(\lambda + x)) &= f(x^*x + \bar{\lambda}x + \lambda x^*) + |\lambda|^2\|f\| \\ &= f(x^*x) + 2\operatorname{Re}(\bar{\lambda}f(x)) + |\lambda|^2\|f\| \end{aligned}$$

$$\begin{aligned}
&\geq f(x^*x) - 2|\lambda|\|f\|^{1/2}f(x^*x)^{1/2} + |\lambda|^2\|f\| \\
&= \left(f(x^*x)^{1/2} - |\lambda|\|f\|^{1/2}\right)^2 \geq 0.
\end{aligned}$$

Thus $\tilde{f}$ is positive. Clearly, $\tilde{f}(1) = \|f\|$. Let g be a positive functional on $\tilde{A}$ extending f . One may give $\tilde{A}$ the structure of an involutive Banach algebra extending that of A and satisfying $\|1\| = 1$. By 2.1.4, $g(1) = \|g\| \geq \|f\|$, whence

$$\begin{aligned}
g((\lambda + x)^*(\lambda + x)) &= f(x^*x + \bar{\lambda}x + \lambda x^*) + |\lambda|^2g(1) \\
&\geq f(x^*x + \bar{\lambda}x + \lambda x^*) + |\lambda|^2\|f\| \\
&= \tilde{f}((\lambda + x)^*(\lambda + x)).
\end{aligned}$$

Hence $g \geq \tilde{f}$, and (vi) is proved. If $y \in A$, the functional $x \mapsto \tilde{f}(y^*xy)$ on $\tilde{A}$ is positive, because $\tilde{f}(y^*(x^*x)y) = \tilde{f}((xy)^*(xy)) \geq 0$. By 2.1.4, its norm is $f(y^*y)$, which proves (ii).

With the notation of (iv),

$$\begin{aligned}
\tilde{f}((x_i - 1)^*(x_i - 1)) &= f(x_i^*x_i) - \overline{f(x_i)} - f(x_i) + \|f\| \\
&\longrightarrow \|f\| - \|f\| - \|f\| + \|f\| = 0,
\end{aligned}$$

which proves (vii).

Denote by $\|\cdot\|'$ the seminorm of the pre-Hilbert space $\tilde{A}$. Let $\varepsilon > 0$. By (vii), there exists $x \in A$ such that $\|x\| \leq 1$ and $\|x - 1\|' < \varepsilon$. Next there exists $j_0 \in I$ such that $j \geq j_0$ implies $\|u_jx - x\| < \varepsilon$, whence

$$\|u_jx - x\|' = f((u_jx - x)^*(u_jx - x))^{1/2} \leq \|f\|^{1/2}\|u_jx - x\| \leq \varepsilon\|f\|^{1/2}.$$

Finally, using (ii) for $\tilde{f}$, we have

$$\|u_jx - u_j\|' = \tilde{f}((x - 1)^*u_j^*u_j(x - 1))^{1/2} \leq \|u_j^*u_j\|^{1/2}\|x - 1\|' \leq \varepsilon.$$

Thus $j \geq j_0$ implies $\|u_j - 1\|' \leq \varepsilon(2 + \|f\|^{1/2})$. It follows that $f(u_j) = (u_j | 1)$ tends to $(1 | 1) = \tilde{f}(1) = \|f\|$, and hence, by (iv), that $f(u_j^*u_j)$ tends to $\|f\|$. This completes the proof.

2.1.6. With the notation of 2.1.5, $\tilde{f}$ is called the *canonical extension* of f to $\tilde{A}$.

Corollary. Let f and g be two continuous positive functionals on A , and let $\tilde{f}$ and $\tilde{g}$ be their canonical extensions to $\tilde{A}$. Then

$$\|f + g\| = \|f\| + \|g\|, \quad \widetilde{f + g} = \tilde{f} + \tilde{g}.$$

The first formula follows from 2.1.5(v), and the second follows from the first.

Consequently, the set of states of A is a convex subset of the dual of A .

2.1.7. Retain the notation of 2.1.5. Let Q be the set of continuous positive functionals on A , and let $\tilde{Q}$ be the set of positive functionals g on $\tilde{A}$ such that $g(1) = \|g|_A\|$. Then $f \mapsto \tilde{f}$ is a bijection from Q onto $\tilde{Q}$ that preserves order and addition. If a positive functional on $\tilde{A}$ is majorized by an element of $\tilde{Q}$, it belongs to $\tilde{Q}$. Indeed, suppose $g = g_1 + g_2$, where $g \in \tilde{Q}$ and g_1, g_2 are positive functionals on $\tilde{A}$, and let f, f_1, f_2 be the restrictions of g, g_1, g_2 to A . We have

$$g_1 + g_2 = g = \tilde{f} = \tilde{f}_1 + \tilde{f}_2$$

and

$$g_1(1) \geq \tilde{f}_1(1), \quad g_2(1) \geq \tilde{f}_2(1),$$

whence

$$g_1(1) = \tilde{f}_1(1), \quad g_2(1) = \tilde{f}_2(1).$$

Thus $g_1, g_2 \in \tilde{Q}$.

2.1.8. Let A be a C^* -algebra and f a positive functional on A . Then f is continuous. Indeed, let $(x_1, x_2, \dots)$ be a sequence of elements of A^+ such that $\|x_i\| \leq 1$, and let us show that the numbers $f(x_i)$ are bounded. For every sequence $(\lambda_1, \lambda_2, \dots)$ of nonnegative numbers such that $\sum_{i=1}^{\infty} \lambda_i < +\infty$, the series $\sum_{i=1}^{\infty} \lambda_i x_i$ converges to an element $x \in A$. For every integer $n \geq 1$,

$$\sum_{i=n+1}^{\infty} \lambda_i x_i \geq 0$$

by 1.6.1, and hence

$$\sum_{i=1}^n \lambda_i f(x_i) = f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq f(x).$$

Thus $\sum_{i=1}^{\infty} \lambda_i f(x_i) < +\infty$. Since the sequence (λ_i) is arbitrary, this proves that the $f(x_i)$ are bounded. It follows that

$$M = \sup_{x \in A^+, \|x\| \leq 1} f(x) < +\infty.$$

If x is hermitian and has norm at most 1, then

$$|f(x)| \leq |f(x^+)| + |f(x^-)| \leq 2M.$$

If x is arbitrary and has norm at most 1, then

$$|f(x)| \leq \left| f\left(\frac{1}{2}(x + x^*)\right) \right| + \left| f\left(\frac{1}{2i}(x - x^*)\right) \right| \leq 4M.$$

Therefore f is continuous.

In what follows, we shall constantly speak of positive functionals on C^* -algebras. The reader should bear in mind that the continuity of these functionals is automatic.

2.1.9. Proposition. *Let A be a unital C^* -algebra and f a continuous linear functional on A . Then f is positive if and only if $\|f\| = f(1)$.*

If $f \geq 0$, then $\|f\| = f(1)$ by 2.1.4. Conversely, suppose that $\|f\| = f(1)$. Let $x \in A^+$, and suppose that $f(x)$ is not nonnegative. We may at once reduce to the case $f(1) = 1$. There exists a closed disk $|z - z_0| \leq \rho$ in $\mathbb{C}$ that contains $\text{Sp } x$ but does not contain $f(x)$. The spectrum of the normal element $x - z_0$ is then contained in the disk $|z| \leq \rho$, so that $\|x - z_0\| \leq \rho$. Hence

$$|f(x) - z_0| = |f(x) - z_0 f(1)| = |f(x - z_0)| \leq \|f\| \|x - z_0\| \leq \rho,$$

which is a contradiction. (Another proof can be based on B 28.)

The word *state* is borrowed from physics.

Bibliography: [3, 41, 42, 52, 105, 108, 126, 134]. The result of 2.1.8 has been generalized to involutive Banach algebras admitting an approximate identity [210].

2.2. Representations

2.2.1. Definition. *Let A be an involutive algebra and H a Hilbert space. A representation of A in H is a homomorphism of the involutive algebra A into the involutive algebra $\mathcal{L}(H)$.*

In other words, a representation of A in H is a map π from A into $\mathcal{L}(H)$ such that

$$\begin{aligned} \pi(x + y) &= \pi(x) + \pi(y), & \pi(\lambda x) &= \lambda \pi(x), \\ \pi(xy) &= \pi(x)\pi(y), & \pi(x^*) &= \pi(x)^* \end{aligned}$$

for $x, y \in A$ and $\lambda \in \mathbb{C}$.

The Hilbert-space dimension of H is called the dimension of π and is denoted by $\dim \pi$. The space H is called the space of π and is denoted by H_π .

Two representations π and π' of A in H and H' are said to be *equivalent*, and we write $\pi \sim \pi'$, if there exists a Hilbert-space isomorphism $U : H \rightarrow H'$ that carries $\pi(x)$ into $\pi'(x)$ for every $x \in A$. This gives the notion of a class of representations. (For convenience of language, we shall often identify representations with classes of representations.)

2.2.2. An *intertwining operator* for π and π' is a continuous linear operator $T : H \rightarrow H'$ such that $T\pi(x) = \pi'(x)T$ for every $x \in A$. The operator U in the preceding definition is an intertwining operator. The set of intertwining operators for π and π' is a vector space whose dimension is called the *intertwining number* of π and π' . It is equal to the intertwining number of π' and π . Indeed, let $T : H \rightarrow H'$ be an intertwining operator for π and π' . Then $T^* : H' \rightarrow H$ is an intertwining operator for π' and π , since

$$T^*\pi'(x) = (\pi'(x^*)T)^* = (T\pi(x^*))^* = \pi(x)T^*.$$

It follows that

$$T^*T\pi(x) = T^*\pi'(x)T = \pi(x)T^*T.$$

Thus $|T| = (T^*T)^{1/2}$ commutes with $\pi(A)$. Let $T = U|T|$ be the polar decomposition of T . For every $x \in A$,

$$U\pi(x)|T| = U|T|\pi(x) = T\pi(x) = \pi'(x)T = \pi'(x)U|T|. \quad (1)$$

If $\ker T = 0$, then $|T|(H)$ is dense in H , and (1) implies

$$U\pi(x) = \pi'(x)U. \quad (2)$$

If, in addition, $T(H) = H'$, or equivalently $\ker T^* = 0$, then U is an isomorphism of H onto H' , and (2) proves that π and π' are equivalent.

2.2.3. Let $(\pi_i)_{i \in I}$ be a family of representations of A in Hilbert spaces H_i , and let H be the Hilbert direct sum of the H_i . If, for every $x \in A$, the numbers $\|\pi_i(x)\|$ are bounded (this is the case, by 1.3.7, when A is an involutive Banach algebra), one can form the continuous linear operator $\pi(x)$ on H whose restriction to each H_i is $\pi_i(x)$. Then $x \mapsto \pi(x)$ is a representation of A in H , called the *Hilbert direct sum* of the π_i , and denoted by $\bigoplus_{i \in I} \pi_i$, or by $\pi_1 \oplus \pi_2 \oplus \cdots \oplus \pi_n$ for a family $(\pi_1, \pi_2, \dots, \pi_n)$ of representations.

If $(\pi_i)_{i \in I}$ is a family of representations of A , all equal to a representation π , and if $\text{Card } I = c$, then $\bigoplus \pi_i$ is denoted by $c\pi$. Any representation equivalent to a representation of this form is called a *multiple* of π .

2.2.4. Let π be a representation of A in H . If a closed vector subspace K of H is invariant under $\pi(A)$, the restrictions of the $\pi(x)$ to K define a *subrepresentation* π' of A in K , denoted by π_K , or by π_E if $E = P_K$. Then $H \ominus K$ is invariant under $\pi(A)$, for if $\xi \in K$ and $\eta \in H \ominus K$, then $\pi(x^*)\xi \in K$ for every $x \in A$, and hence

$$(\pi(x)\eta \mid \xi) = (\eta \mid \pi(x)^*\xi) = (\eta \mid \pi(x^*)\xi) = 0.$$

Thus $\pi(x)\eta \in H \ominus K$, and P_K commutes with $\pi(A)$. If π'' is the subrepresentation of π defined by $H \ominus K$, then $\pi \sim \pi' \oplus \pi''$.

Let ρ, ρ' be two representations of A . If ρ' is equivalent to a subrepresentation of ρ , we say that ρ' is *contained in* ρ , or that ρ *contains* ρ' , and write $\rho' \leq \rho$ or $\rho \geq \rho'$.

2.2.5. Let π be a representation of A in H , and let $\xi \in H$. The closure of $\pi(A)\xi$ is a closed vector subspace of H invariant under $\pi(A)$. If this subspace is H , then ξ is said to be *cyclic* for π .

2.2.6. Proposition. *Let π be a representation of A in H . Let K be the closed vector subspace of H spanned by the vectors $\pi(x)\xi$ ($x \in A$, $\xi \in H$). Let K' be the closed vector subspace of H formed by the vectors $\xi \in H$ annihilated by all $\pi(x)$. Then K and K' are invariant under $\pi(A)$ and $\pi(A)'$, are orthogonal, and have sum H .*

It is clear that K and K' are invariant under $\pi(A)$. An operator commuting with $\pi(x)$ leaves invariant the kernel and range of $\pi(x)$, and hence K and K' are invariant under $\pi(A)'$. Since $\pi(A)(K) \subset K$,

$$\pi(A)(H \ominus K) \subset H \ominus K;$$

but $\pi(A)(H) \subset K$, so $\pi(A)(H \ominus K) = 0$, and consequently $H \ominus K \subset K'$. If $\xi \in K'$, $x \in A$, and $\eta \in H$, then

$$(\xi \mid \pi(x)\eta) = (\pi(x^*)\xi \mid \eta) = 0,$$

and therefore $\xi \in H \ominus K$. This proves that $K' = H \ominus K$.

The space K is called the *essential subspace* of π . The representation π is said to be *nondegenerate* if $K = H$. What precedes proves that every representation of A is, in a unique way, the Hilbert direct sum of a zero representation and a nondegenerate representation.

2.2.7. It is clear that a Hilbert direct sum of nondegenerate representations is nondegenerate, and that a representation admitting a cyclic vector is nondegenerate. Conversely:

Proposition. *Every nondegenerate representation of A is a Hilbert direct sum of representations admitting cyclic vectors.*

Let π be a nondegenerate representation of A in H , with $H \neq 0$. It is enough to prove that π has a subrepresentation on a nonzero space that admits a cyclic vector (for one then applies Zorn's theorem). Let ξ be a nonzero element of H . The closure K of $\pi(A)\xi$ is nonzero and invariant under π . Put $L = H \ominus K$, and write $\xi = \xi_1 + \xi_2$, where $\xi_1 \in K$ and $\xi_2 \in L$. If $x \in A$, then $\pi(x)\xi_1 \in K$, $\pi(x)\xi_2 \in L$, and $\pi(x)\xi_1 + \pi(x)\xi_2 = \pi(x)\xi \in K$; hence $\pi(x)\xi_2 = 0$, and consequently $\pi(x)\xi = \pi(x)\xi_1$. Thus K is the closure of $\pi(A)\xi_1$.

2.2.8. Let π be a representation of A in H , let $\overline{H}$ be the conjugate Hilbert space of H ($\overline{H} = H$, but on passing from H to $\overline{H}$, multiplication by $\lambda \in \mathbb{C}$ is replaced by multiplication by $\overline{\lambda}$, and inner products are replaced by their complex conjugates), and let A° be the opposite involutive algebra of A . It is readily verified that the map $x \mapsto \overline{\pi(x^*)}$ is a representation of A° in $\overline{H}$, denoted by $\overline{\pi}^\circ$.

2.2.9. Let $\widetilde{A}$ be the involutive algebra obtained from A by adjoining an identity element, and let π be a representation of A in H . Then π extends uniquely to a representation $\widetilde{\pi}$ of $\widetilde{A}$ in H such that $\widetilde{\pi}(1) = 1$. This extension is called canonical.

2.2.10. Let A be an involutive Banach algebra and π a representation of A in H . Recall from 1.3.7 that $\|\pi(x)\| \leq \|x\|$ for every $x \in A$. If A has an approximate identity (u_i) and π is nondegenerate, then $\pi(u_i)$ converges strongly to 1. Indeed, for every $y \in A$ and $\xi \in H$, $\pi(u_i)(\pi(y)\xi) = \pi(u_i y)\xi$ converges strongly to $\pi(y)\xi$, since

$$\|\pi(u_i y) - \pi(y)\| \leq \|u_i y - y\| \longrightarrow 0;$$

the vectors $\pi(y)\xi$ form a total set in H , while $\|\pi(u_i)\| \leq \|u_i\| \leq 1$. Hence $\pi(u_i)\eta$ converges strongly to η for every $\eta \in H$.

Bibliography: [41, 108, 126].

2.3. Topologically Irreducible Representations

2.3.1. Proposition. *Let A be an involutive algebra, H a Hilbert space, and π a representation of A in H . The following conditions are equivalent:*

- (i) *The only closed vector subspaces of H invariant under $\pi(A)$ are 0 and H .*
- (ii) *The commutant of $\pi(A)$ in $\mathcal{L}(H)$ consists only of scalar operators.*

(iii) Every nonzero vector of H is cyclic for π , or else π is the zero representation of dimension 1.

(i) $\Rightarrow$ (iii): suppose (i) holds, and let $\xi \in H$, $\xi \neq 0$. If $\pi(A)\xi$ is not dense in H , its closure is 0 by (i), and hence $\pi(A)\xi = 0$. Then $\mathbb{C}\xi$ is invariant under $\pi(A)$, so $H = \mathbb{C}\xi$ and π is the zero representation of dimension 1.

(iii) $\Rightarrow$ (i): suppose (iii) holds. Let $K \neq 0$ be a closed vector subspace of H invariant under $\pi(A)$. We must prove that $K = H$. This is clear if $\dim \pi = 1$. Otherwise every nonzero vector of H is cyclic for π . Let $\xi \in K$, $\xi \neq 0$. Then

$$\pi(A)\xi \subset K \quad \text{and} \quad \overline{\pi(A)\xi} = H,$$

so $K = H$.

(ii) $\Rightarrow$ (i): suppose (ii) holds. Let K be a closed vector subspace of H invariant under $\pi(A)$. Then P_K commutes with $\pi(A)$ by 2.2.4; hence P_K is scalar and therefore $P_K = 0$ or 1, that is, $K = 0$ or H .

(i) $\Rightarrow$ (ii): suppose (i) holds. Let $T \in \mathcal{L}(H)$ commute with $\pi(A)$, and let us prove that T is scalar. Since $T + T^*$ and $i(T - T^*)$ commute with $\pi(A)$, it is enough to consider the case in which T is hermitian. The spectral projections of T then commute with $\pi(A)$, and hence, by (i), are all 0 or 1. Thus T is scalar.

2.3.2. Definition. Let A be an involutive algebra, H a Hilbert space, and π a representation of A in H . The representation π is said to be topologically irreducible if $H \neq 0$ and if π satisfies the equivalent conditions of 2.3.1.

Such a representation is either the zero representation of dimension 1, or it is nonzero and nondegenerate. We denote by $\widehat{A}$ the set of classes of nonzero topologically irreducible representations of A .

Recall that irreducibility of π in the algebraic sense means that the only vector subspaces of H invariant under $\pi(A)$ are $\{0\}$ and H . If $\dim \pi = +\infty$, this is a much stronger condition than topological irreducibility. We shall see, however, in 2.8.4 that, if A is a C^* -algebra, the two notions are equivalent.

A finite-dimensional representation π of A is a Hilbert direct sum of topologically irreducible representations (2.3.5). We shall see in 8.5.2 that this result extends, to a certain extent, to infinite-dimensional representations, but in a much subtler form. This is one reason why we shall be particularly interested below in topologically irreducible representations. Given an involutive algebra A , an essential problem is to list all its topologically irreducible representations (up to equivalence). It may very well happen that there is no nonzero topologically irreducible representation of A (or even no nonzero representation at all). We

shall nevertheless see in 2.7.3 that every C^* -algebra has sufficiently many topologically irreducible representations.

2.3.3. Let A be a separable involutive normed algebra, and let π be a topologically irreducible representation of A in a Hilbert space H . Then H is separable. This is clear if π is the zero representation of dimension 1. Otherwise, let ξ be a nonzero vector of H ; if (x_n) is a dense sequence in A , then the vectors $\pi(x_n)\xi$ are dense in H . The same reasoning shows that, for every involutive normed algebra B , the dimensions of the topologically irreducible representations of B are bounded above by a fixed cardinal.

2.3.4. Let A be an involutive algebra, let π, π' be topologically irreducible representations of A , and let n be their intertwining number. If $n = 0$, then π and π' are inequivalent. If $n > 0$, let $T : H_\pi \rightarrow H_{\pi'}$ be a nonzero intertwining operator. By 2.2.2, T^*T and TT^* are nonzero scalar operators, and π, π' are equivalent.

2.3.5. Let A be an involutive algebra. We analyze its finite-dimensional representations. What follows may be regarded both as a special case of later theorems and as a special case of theorems of pure algebra; but, to make the reader's task easier, we prefer to give a direct analysis now.

Let π be a finite-dimensional representation of A . Then

$$\pi = \pi_1 \oplus \cdots \oplus \pi_n,$$

where the π_i are irreducible. This is clear if $\dim \pi = 0$ (with $n = 0$). Suppose $\dim \pi = q$ and the assertion proved when $\dim \pi < q$. If π is irreducible, the assertion is again clear. Otherwise $\pi = \pi' \oplus \pi''$, with $\dim \pi' < q$ and $\dim \pi'' < q$, and it is enough to apply the induction hypothesis. The decomposition $\pi = \pi_1 \oplus \cdots \oplus \pi_n$ is not unique (for example, the representation $\lambda \mapsto \lambda 1$ of $\mathbb{C}$ in $\mathbb{C}^n$ admits, for $n > 1$, infinitely many decompositions into one-dimensional representations). We shall nevertheless establish a uniqueness result. Let ρ_1, ρ_2 be two irreducible subrepresentations of π , and let P_1, P_2 be the projections of H_π onto H_{ρ_1} and H_{ρ_2} . They commute with $\pi(A)$. Hence the restriction of P_2 to H_{ρ_1} is an intertwining operator for ρ_1 and ρ_2 . Thus, if H_{ρ_1} and H_{ρ_2} are not orthogonal, then $\rho_1 \sim \rho_2$ by 2.3.4. This proves that every irreducible subrepresentation of π is equivalent to one of the π_i . Regrouping the π_i , we see that

$$\pi = \nu_1 \oplus \cdots \oplus \nu_m,$$

where each ν_i is a multiple $p_i \nu'_i$ of an irreducible representation ν'_i , and the ν'_i are pairwise inequivalent. If ρ is an irreducible subrepresentation of π , the preced-

ing argument shows that H_ρ is orthogonal to every H_{v_j} except one. Hence H_ρ is contained in that one H_{v_i} . This proves that each subspace H_{v_i} is uniquely determined: it is the subspace of H_π spanned by the spaces of the subrepresentations of π equivalent to v'_i .

Thus, in a decomposition $\pi = p_1 v'_1 \oplus \cdots \oplus p_m v'_m$ of π , with the v'_i irreducible and pairwise inequivalent, the integers p_i , the classes of the v'_i , and the spaces of the $p_i v'_i$ are uniquely determined.

Bibliography: [108, 126].

2.4. Positive Functionals and Representations

2.4.1. Proposition. *Let A be an involutive algebra.*

- (i) *If π is a representation of A in H , and if $\xi \in H$, then $x \mapsto (\pi(x)\xi \mid \xi)$ is a positive functional on A .*
- (ii) *Let π, π' be representations of A in H, H' , and let ξ (respectively ξ') be cyclic for π (respectively π'). If*

$$(\pi(x)\xi \mid \xi) = (\pi'(x)\xi' \mid \xi') \quad \text{for every } x \in A,$$

there exists a unique isomorphism from H onto H' that carries π into π' and ξ into ξ' .

We have

$$(\pi(x^*x)\xi \mid \xi) = (\pi(x)^*\pi(x)\xi \mid \xi) = \|\pi(x)\xi\|^2 \geq 0,$$

which proves (i). Now assume the hypotheses of (ii). For arbitrary $x, y \in A$,

$$(\pi(x)\xi \mid \pi(y)\xi) = (\pi(y^*x)\xi \mid \xi) = (\pi'(y^*x)\xi' \mid \xi') = (\pi'(x)\xi' \mid \pi'(y)\xi').$$

Since the $\pi(x)\xi$ (respectively the $\pi'(x)\xi'$) are dense in H (respectively in H'), there exists an isomorphism $U : H \rightarrow H'$ such that $U(\pi(x)\xi) = \pi'(x)\xi'$ for every $x \in A$. Let us show that U carries π into π' , that is, $U\pi(x) = \pi'(x)U$ for every $x \in A$. For every $y \in A$,

$$(U\pi(x))(\pi(y)\xi) = U\pi(xy)\xi = \pi'(xy)\xi' = \pi'(x)(\pi'(y)\xi') = (\pi'(x)U)(\pi(y)\xi).$$

Since the vectors $\pi(y)\xi$ are dense in H , this indeed gives $U\pi(x) = \pi'(x)U$. Moreover, for every $x \in A$,

$$(\xi' \mid \pi'(x)\xi') = (\xi \mid \pi(x)\xi) = (U\xi \mid U\pi(x)\xi) = (U\xi \mid \pi'(x)\xi'),$$

and therefore $\xi' = U\xi$. Finally, the asserted uniqueness of U is immediate, since its values are prescribed on $\pi(A)\xi$.

2.4.2. Retain the preceding notation. The functional $x \mapsto (\pi(x)\xi \mid \xi)$ on A is called the *functional defined by π and ξ* . With π fixed and ξ varying in H , one obtains the *functionals associated with π* . If S is a set of representations of A , the functionals associated with the various elements of S are called the *functionals associated with S* .

Let H be a Hilbert space, B an involutive subalgebra of $\mathcal{L}(H)$, and $\xi \in H$. Write ω_ξ for the positive functional on B defined by the identity representation of B and ξ , that is, $x \mapsto (x\xi \mid \xi)$. A positive functional on B is called a *vector functional* if it is equal to ω_ξ for a suitable $\xi \in H$.

2.4.3. Let A be an involutive Banach algebra admitting an approximate identity (u_i) , let π be a nondegenerate representation of A in H , let $\xi \in H$, and let f be the positive functional defined by π and ξ . Then $\|f\| = (\xi \mid \xi)$. Indeed, by 2.1.5(v),

$$\|f\| = \lim_i f(u_i) = \lim_i (\pi(u_i)\xi \mid \xi),$$

and $\pi(u_i)$ converges strongly to 1 by 2.2.10. It follows that, if $\tilde{A}$ denotes the involutive algebra obtained from A by adjoining an identity, and if $\tilde{f}$ and $\tilde{\pi}$ are the canonical extensions of f and π to $\tilde{A}$, then

$$\tilde{f}(x) = (\tilde{\pi}(x)\xi \mid \xi) \quad \text{for every } x \in \tilde{A}.$$

In particular, if π is the nondegenerate identity representation of a C^* -subalgebra A of $\mathcal{L}(H)$ that does not contain the identity operator, then the canonical extension to $\tilde{A} = A + \mathbb{C}1$ of $\omega_\xi|_A$ is $\omega_\xi|_{\tilde{A}}$.

2.4.4. Proposition. *Let A be an involutive Banach algebra having an approximate identity, let $\tilde{A}$ be the involutive algebra obtained from A by adjoining an identity element, let f be a continuous positive functional on A , let $\tilde{f}$ be its canonical extension to $\tilde{A}$, let N be the left ideal of $\tilde{A}$ formed by the $x \in \tilde{A}$ such that $\tilde{f}(x^*x) = 0$, let A'_f be the separated pre-Hilbert space $\tilde{A}/N$, and let A_f be the Hilbert-space completion of A'_f . For every $x \in \tilde{A}$, let $\pi'(x)$ be the operator on $\tilde{A}/N$ induced on passing to the quotient by left multiplication by x in $\tilde{A}$. Let ξ be the canonical image of 1 in A'_f .*

- (i) *Every $\pi'(x)$ extends uniquely to a continuous linear operator $\pi(x)$ on A_f .*
- (ii) *The map $x \mapsto \pi(x)$, $x \in A$, is a representation of A in A_f .*
- (iii) *ξ is cyclic for $\pi(A)$.*

(iv) $f(x) = (\pi(x)\xi \mid \xi)$ for every $x \in A$.

By 2.1.5(ii), for $x, y \in \widetilde{A}$,

$$\begin{aligned} (\pi'(x)\pi'(y)\xi \mid \pi'(x)\pi'(y)\xi) &= \widetilde{f}(y^*x^*xy) \\ &\leq \|x^*x\| \widetilde{f}(y^*y) \\ &= \|x^*x\|(\pi'(y)\xi \mid \pi'(y)\xi), \end{aligned}$$

which proves (i). It is clear that the $\pi'(x)$, and hence the $\pi(x)$, are representations as far as the algebraic structure is concerned. For $x, y, z \in \widetilde{A}$,

$$\begin{aligned} (\pi(x)\pi(y)\xi \mid \pi(z)\xi) &= \widetilde{f}(z^*xy) = \widetilde{f}((x^*z)^*y) \\ &= (\pi(y)\xi \mid \pi(x^*)\pi(z)\xi), \end{aligned}$$

and therefore $\pi(x)^* = \pi(x^*)$, which proves (ii). The set $\pi(A)\xi$ is the canonical image of A in A'_f , and hence is dense in A_f by 2.1.5(vii); this proves (iii). Finally, for every $x \in \widetilde{A}$,

$$(\pi(x)\xi \mid \xi) = \widetilde{f}(1^*x1) = \widetilde{f}(x),$$

which proves (iv).

The representation π and vector ξ are said to be *defined by f* . We denote them by π_f and ξ_f .

2.4.5. Retain the preceding notation. Let M be the left ideal of A formed by the $x \in A$ such that $f(x^*x) = 0$. The canonical image of A in A'_f is naturally identified with A/M , and is dense in A_f . Thus A_f may be defined as the completion of the separated pre-Hilbert space A/M , and, for $x \in A$, $\pi(x)$ may be defined as the continuous extension of left multiplication by x on A/M . This avoids introducing $\widetilde{A}$ and $\widetilde{f}$. From this point of view, however, the definition of ξ would be less easy.

2.4.6. Let A be an involutive Banach algebra having an approximate identity, let f be a continuous positive functional on A , and let π, ξ be the representation and vector defined by f . Then the positive functional defined by π, ξ is f , by 2.4.4(iv).

Conversely, start with a representation π of A in a Hilbert space and a cyclic vector ξ for π . Let f be the positive functional defined by π, ξ ; it is continuous because π is continuous. Let π', ξ' be the representation and vector defined by f . We have

$$(\pi(x)\xi \mid \xi) = f(x) = (\pi'(x)\xi' \mid \xi') \quad \text{for every } x \in A,$$

and ξ' is cyclic for π' . By 2.4.1(ii), there is a unique isomorphism from H_π onto $H_{\pi'}$ carrying π into π' and ξ into ξ' .

In particular, let π be a nonzero topologically irreducible representation of A . Every nonzero vector of H_π is cyclic for π ; hence π is defined by any nonzero functional associated with π .

2.4.7. Proposition. *Let A be an involutive Banach algebra having an approximate identity, and let π be a representation of A . For every $x \in A$,*

$$\|\pi(x)\|^2 = \sup f(x^*x),$$

where f ranges over the positive functionals associated with π such that $\|f\| \leq 1$.

By 2.2.6, we may reduce to the case in which π is nondegenerate. By 2.4.3, the positive functionals associated with π and of norm at most 1 are the functionals $\omega_\xi \circ \pi$, where $\xi \in H_\pi$ and $\|\xi\| \leq 1$. But

$$\begin{aligned} \|\pi(x)\|^2 &= \sup_{\|\xi\| \leq 1} (\pi(x)\xi \mid \pi(x)\xi) \\ &= \sup_{\|\xi\| \leq 1} (\omega_\xi \circ \pi)(x^*x). \end{aligned}$$

2.4.8. Proposition. *Let A be an involutive Banach algebra having an approximate identity, let f be a continuous positive functional on A , and let π, ξ be the representation and vector defined by f .*

- (i) *If $x_0 \in A$, then the functional $x \mapsto f(x_0^*xx_0)$ is associated with π .*
- (ii) *If f' is a positive functional associated with π , then f' is a norm limit of functionals $x \mapsto f(x_0^*xx_0)$, where $x_0 \in A$.*

If $x_0 \in A$, then

$$f(x_0^*xx_0) = (\pi(x_0^*xx_0)\xi \mid \xi) = (\pi(x)\pi(x_0)\xi \mid \pi(x_0)\xi),$$

which proves (i).

Let $\xi' \in H_\pi$ and $f' = \omega_{\xi'} \circ \pi$. For every $\varepsilon > 0$, there exists $x_0 \in A$ such that $\|\pi(x_0)\xi - \xi'\| \leq \varepsilon$. Then, for every $x \in A$,

$$\begin{aligned} |f'(x) - f(x_0^*xx_0)| &= |(\pi(x)\xi' \mid \xi') - (\pi(x)\pi(x_0)\xi \mid \pi(x_0)\xi)| \\ &\leq \|\pi(x)\xi'\| \|\xi' - \pi(x_0)\xi\| \\ &\quad + \|\pi(x)\xi' - \pi(x)\pi(x_0)\xi\| \|\pi(x_0)\xi\| \\ &\leq \|x\| \|\xi'\| \varepsilon + \|x\| \varepsilon (\|\xi'\| + \varepsilon) \\ &= \|x\| (2\varepsilon \|\xi'\| + \varepsilon^2), \end{aligned}$$

and $2\varepsilon\|\xi'\| + \varepsilon^2$ is arbitrarily small.

2.4.9. Let A be a C^* -algebra, let I be a closed two-sided ideal of A , let $B = A/I$, and let $\omega : A \rightarrow B$ be the canonical homomorphism. If f is a positive functional on A such that $f(I) = 0$, then f induces by passage to the quotient a positive functional g on B . For every $x \in A$,

$$(\pi_g(\omega(x))\xi_g \mid \xi_g) = g(\omega(x)) = f(x),$$

and hence $\pi_g \circ \omega$ and ξ_g identify, respectively, with π_f and ξ_f by 2.4.1.

Proposition. Let A be a C^* -algebra, f a positive functional on A , and I a closed two-sided ideal of A . Then $f(I) = 0$ if and only if $\pi_f(I) = 0$.

We have $\ker \pi_f \subset \ker f$, so $\pi_f(I) = 0$ implies $f(I) = 0$. If $f(I) = 0$, then, with the preceding notation, $\pi_f \sim \pi_g \circ \omega$, and consequently $\pi_f(I) = 0$.

2.4.10. Corollary. $\ker \pi_f$ is the largest closed two-sided ideal of A contained in $\ker f$.

2.4.11. Corollary. Let A be a C^* -algebra and let f, g be two positive functionals on A . The following conditions are equivalent:

- (i) $\ker \pi_f \subset \ker \pi_g$.
- (ii) g vanishes on $\ker \pi_f$.

Bibliography: [41, 42, 52, 105, 108, 126, 134].

2.5. Pure Functionals and Irreducible Representations

We know how to associate representations with positive functionals. When does this procedure yield topologically irreducible representations? We shall now solve this problem.

2.5.1. Proposition. Let A be an involutive algebra, let π be a representation of A in H , let $\xi \in H$, and let f be the positive functional on A defined by π and ξ .

- (i) If T is a hermitian operator on H , commuting with the $\pi(x)$, and satisfying $0 \leq T \leq 1$, then the functional

$$x \mapsto (\pi(x)T\xi \mid T\xi) = (\pi(x)\xi \mid T^2\xi)$$

on A is a positive functional f_T majorized by f .

- (ii) If ξ is cyclic for π , the map $T \mapsto f_T$ is injective.
- (iii) If A is an involutive Banach algebra having an approximate identity, then every continuous positive functional on A majorized by f is of the form f_T .

Since $f_T = \omega_{T\xi} \circ \pi$, we have $f_T \geq 0$. If $x \in A$,

$$\begin{aligned} f_T(x^*x) &= (\pi(x^*x)T\xi \mid T\xi) = \|\pi(x)T\xi\|^2 \\ &= \|T\pi(x)\xi\|^2 \leq \|\pi(x)\xi\|^2 = f(x^*x), \end{aligned}$$

so $f_T \leq f$. This proves (i).

If $f_T = f_{T'}$, then

$$(\pi(x)\xi \mid T^2\xi) = (\pi(x)\xi \mid T'^2\xi)$$

for every $x \in A$, and hence $T^2\xi = T'^2\xi$ if ξ is cyclic for $\pi(A)$. Since ξ is then separating for the commutant of $\pi(A)$ (A 14), it follows that $T^2 = T'^2$, and hence $T = T'$, because $T, T' \geq 0$. This proves (ii).

Let g be a positive functional on A majorized by f . For $x, y \in A$, we have

$$|g(y^*x)|^2 \leq g(x^*x)g(y^*y) \leq f(x^*x)f(y^*y) = \|\pi(x)\xi\|^2\|\pi(y)\xi\|^2.$$

Thus, on the vector subspace $\pi(A)\xi$ of H , the formula

$$(\pi(x)\xi \mid \pi(y)\xi)_0 = g(y^*x)$$

defines a unique continuous sesquilinear form, plainly hermitian and positive. Hence there exists on $X = \pi(A)\xi$ a hermitian operator T_0 such that $0 \leq T_0 \leq 1$ and

$$g(y^*x) = (\pi(x)\xi \mid T_0\pi(y)\xi).$$

For $x, y, z \in A$,

$$\begin{aligned} (\pi(y)\xi \mid T_0\pi(z)\pi(x)\xi) &= g((zx)^*y) = g(x^*(z^*y)) \\ &= (\pi(z^*y)\xi \mid T_0\pi(x)\xi) \\ &= (\pi(y)\xi \mid \pi(z)T_0\pi(x)\xi), \end{aligned}$$

and therefore $T_0\pi(z) = \pi(z)T_0$ on X . Moreover, X is invariant under $\pi(A)$, so P_X commutes with $\pi(A)$. Thus the operator T_0P_X , extended by zero on $H \ominus X$, is a hermitian operator on H , commuting with every $\pi(z)$. Let T be its positive square root. We have $0 \leq T \leq 1$, and

$$\begin{aligned} g(y^*x) &= (\pi(x)\xi \mid T^2\pi(y)\xi) = (\pi(x)T\xi \mid \pi(y)T\xi) \\ &= (\pi(y^*x)T\xi \mid T\xi) = f_T(y^*x). \end{aligned}$$

Finally, suppose that A is an involutive Banach algebra admitting an approximate identity (u_i) , and that g is continuous. Then

$$g(y^*) = \lim_i g(y^*u_i), \quad f_T(y^*) = \lim_i f_T(y^*u_i), \quad \text{so} \quad g = f_T.$$

2.5.2. Definition. Let A be an involutive normed algebra and f a continuous positive functional on A . The functional f is said to be **pure** if $f \neq 0$ and if every continuous positive functional on A majorized by f is of the form λf , where $0 \leq \lambda \leq 1$. We denote by $P(A)$ the set of pure states of A .

Let Ω be a locally compact space, and let A be the C^* -algebra of continuous complex-valued functions on Ω vanishing at infinity. The pure positive functionals on A identify with the positive measures on Ω whose support consists of one point, that is, with the measures

$$f \longmapsto \lambda f(\omega) \quad (\lambda > 0, \omega \text{ a fixed point of } \Omega).$$

It follows that the pure states of a commutative C^* -algebra are the characters of that algebra.

2.5.3. Let A be an involutive Banach algebra admitting an approximate identity, let $\tilde{A}$ be an involutive Banach algebra obtained from A by adjoining an identity element, let f be a continuous positive functional on A , and let $\tilde{f}$ be its canonical extension to $\tilde{A}$. Then f is pure if and only if $\tilde{f}$ is pure. Indeed, first, the conditions $f = 0$ and $\tilde{f} = 0$ are equivalent. Further, as g ranges over the continuous positive functionals on A majorized by f , $\tilde{g}$ ranges over the positive functionals on $\tilde{A}$ majorized by $\tilde{f}$, by 2.1.7. Finally, g is of the form λf , with $0 \leq \lambda \leq 1$, if and only if $\tilde{g} = \lambda \tilde{f}$.

2.5.4. Proposition. Let A be an involutive Banach algebra having an approximate identity, let f be a continuous positive functional on A , and let π be the representation of A defined by f . Then π is nonzero and topologically irreducible if and only if f is pure.

Let ξ be the vector of H_π defined by f . Suppose f is pure. Let E be a projection on H_π commuting with $\pi(A)$. The functional

$$x \longmapsto (\pi(x)E\xi \mid E\xi)$$

on A is continuous, positive, and majorized by f , by 2.5.1(i); it is therefore equal to λf , with $0 \leq \lambda \leq 1$. Hence, for every $x \in A$,

$$(\pi(x)E\xi \mid E\xi) = (\pi(x)\lambda^{1/2}\xi \mid \lambda^{1/2}\xi).$$

By 2.5.1(ii), $E = \lambda^{1/2}1$, and therefore $E = 0$ or 1 . On the other hand, there exist $x \in A$ with $f(x) \neq 0$, and hence $(\pi(x)\xi \mid \xi) \neq 0$. This proves that π is nonzero and topologically irreducible.

Conversely, suppose that π is nonzero and topologically irreducible. Then there exist $x \in A$ such that $(\pi(x)\xi \mid \xi) \neq 0$, so $f \neq 0$. Let g be a continuous positive functional on A majorized by f . By 2.5.1(iii), there exists a hermitian operator $T \in \pi(A)'$ such that $0 \leq T \leq 1$ and

$$g(x) = (\pi(x)T\xi \mid T\xi)$$

for every $x \in A$. Since π is topologically irreducible, we have $T = \mu 1$, with $0 \leq \mu \leq 1$, whence $g = \mu^2 f$. Thus f is pure, and the proposition is proved.

Proposition 2.5.4 defines a canonical map

$$P(A) \longrightarrow \widehat{A}.$$

By 2.4.6 and 2.5.4, this map is surjective. If $\pi \in \widehat{A}$, its inverse image in $P(A)$ is the set of states associated with π (on this point, see 2.5.7).

2.5.5. Proposition. *Let A be an involutive Banach algebra having an approximate identity, and let B be the set of continuous positive functionals on A of norm at most 1.*

- (i) *B is convex and compact for the weak topology $\sigma(A', A)$ of the dual A' of A .*
- (ii) *The extreme points of B are 0 and the pure states.*
- (iii) *B is the weakly closed convex hull of 0 and the set of pure states.*

The set B is a weakly closed convex subset of the unit ball of A' , and that ball is weakly compact. This proves (i).

Let us show that 0 is an extreme point of B . If $f \in B$ and $-f \in B$, then $f(x^*x) = 0$ for every $x \in A$, and hence

$$|f(x)|^2 \leq \|f\|f(x^*x) = 0$$

by 2.1.5; thus $f = 0$.

Let f be a pure state. Suppose

$$f = \lambda f_1 + (1 - \lambda)f_2, \quad 0 < \lambda < 1, \quad f_1, f_2 \in B.$$

Then λf_1 is majorized by f , so $\lambda f_1 = \mu f$, with $0 \leq \mu \leq 1$. Since

$$1 = \|f\| = \lambda\|f_1\| + (1 - \lambda)\|f_2\|$$

and $\|f_1\| \leq 1$, $\|f_2\| \leq 1$, necessarily $\|f_1\| = \|f_2\| = 1$, and consequently $\lambda = \mu$ and $f_1 = f$, $f_2 = f$. Thus f is an extreme point of B .

Conversely, let f be an extreme point of B , distinct from 0. Clearly $\|f\| = 1$. Suppose $f = f_1 + f_2$, where f_1, f_2 are nonzero continuous positive functionals. Put $\|f_1\| = \lambda$, so that $\|f_2\| = 1 - \lambda$, and set

$$g_1 = \lambda^{-1}f_1, \quad g_2 = (1 - \lambda)^{-1}f_2.$$

Then $f = \lambda g_1 + (1 - \lambda)g_2$, where $g_1, g_2 \in B$. Since f is extreme, $f = g_1 = g_2$. Hence $f_1 = \lambda f$ and $f_2 = (1 - \lambda)f$, so f is a pure state. This proves (ii).

Finally, (iii) follows from (i), (ii), and the Krein–Milman theorem.

2.5.6. Retain the preceding notation, and suppose in addition that A has an identity element. Then B is the set of positive functionals f on A such that $f(1) \leq 1$, and the state space $E(A)$ of A is the set of positive functionals f on A such that $f(1) = 1$. It follows that $E(A)$ is convex and weakly compact, and that the set of extreme points of $E(A)$ is the set of extreme points of B belonging to $E(A)$, namely $P(A)$. Thus $E(A)$ is the closed convex hull of $P(A)$.

2.5.7. Proposition. *Let A be an involutive algebra, let π be a nonzero topologically irreducible representation of A , let ξ_1, ξ_2 be two vectors of H_π , and let f_1, f_2 be the positive functionals defined by (π, ξ_1) and (π, ξ_2) . Then $f_1 = f_2$ if and only if there exists a complex number λ , with $|\lambda| = 1$, such that $\xi_2 = \lambda \xi_1$.*

If $\xi_2 = \lambda \xi_1$ with $|\lambda| = 1$, it is clear that $f_1 = f_2$. Conversely, suppose $f_1 = f_2$. Since ξ_1, ξ_2 are cyclic by 2.3.1, there exists by 2.4.1(ii) an automorphism U of H_π , commuting with the $\pi(x)$, such that $U\xi_1 = \xi_2$. But U is a scalar operator λ by 2.3.1, whence $\xi_2 = \lambda \xi_1$ and $|\lambda| = 1$.

In particular, the canonical map $P(A) \rightarrow \widehat{A}$ is bijective if and only if every topologically irreducible representation of A has dimension 1. When A is a C^* -algebra, Theorem 2.7.3 will imply that this condition holds if and only if A is commutative.

Bibliography: [41, 42, 52, 105, 108, 126, 134].

2.6. Existence of Representations of C^* -Algebras

2.6.1. Theorem. *Let A be a C^* -algebra. There exists an isometric representation of A in a Hilbert space.*

Let A_h be the real Banach space formed by the hermitian elements of A , and let x be a nonzero element of A . Then $x^*x \in A^+ \setminus \{0\}$ by 1.6.1, and A^+ is a closed convex cone by 1.6.1. Hence there exists a continuous real linear functional f_x on A_h such that $f_x(y) \geq 0$ for $y \in A^+$ and $f_x(-x^*x) < 0$, by B 5. Extend f_x to a hermitian functional on A . Then f_x is a positive functional on A , and $f_x(x^*x) > 0$; consequently, the representation π_x defined by f_x , in accordance with 2.4.4, satisfies $\pi_x(x) \neq 0$. Let π be the Hilbert direct sum of the π_x , for $x \in A$, $x \neq$

0. Then π is injective, and hence isometric by 1.3.7 and 1.8.1. This proves the theorem.

Thus, as announced, the closed involutive subalgebras of an algebra $\mathcal{L}(H)$, with H a Hilbert space, give the most general examples of C^* -algebras.

2.6.2. Proposition. *Let A be a C^* -algebra and $x \in A$. The following conditions are equivalent:*

- (i) $x \geq 0$.
- (ii) *For every representation π of A , the operator $\pi(x)$ is nonnegative.*
- (iii) *For every positive functional f on A , one has $f(x) \geq 0$.*
- (i) $\Rightarrow$ (iii) is clear.

(iii) $\Rightarrow$ (ii): let π be a representation of A and $\xi \in H_\pi$. The functional $y \mapsto (\pi(y)\xi \mid \xi)$ on A is positive. Hence, if (iii) holds, then $(\pi(x)\xi \mid \xi) \geq 0$, so $\pi(x) \geq 0$.

(ii) $\Rightarrow$ (i): let π be an isometric representation of A , furnished by 2.6.1. If $\pi(x) \geq 0$, then $\pi(x)$ is positive relative to $\pi(A)$ by 1.6.5; hence x is positive relative to A .

2.6.3. Corollary. *Let A be a C^* -algebra, let A_h be the set of hermitian elements of A , let B be the set of positive functionals on A of norm at most 1, and let $P \subset B$ be the set of pure states of A . Give B and P the weak topology. Let $C(B)$ and $C(P)$ be the sets of continuous real-valued functions on B and P . For each $x \in A_h$, let F_x be the continuous real-valued function*

$$f \mapsto \langle f, x \rangle$$

on B , and let G_x be its restriction to P . The map $x \mapsto F_x$ (respectively $x \mapsto G_x$) is an isometric isomorphism of the ordered Banach space A_h onto a vector subspace of the ordered Banach space $C(B)$ (respectively $C(P)$).

It is clear that these maps are linear, and that

$$x \geq 0 \implies F_x \geq 0 \implies G_x \geq 0.$$

Conversely, suppose that $G_x \geq 0$, that is, $f(x) \geq 0$ for every $f \in P$. By 2.5.5, we then have $f(x) \geq 0$ for every $f \in B$, and therefore $x \geq 0$ by 2.6.2. On the other hand, realize A on a Hilbert space H by 2.6.1, and let $x \in A_h$ satisfy $\|x\| = 1$. There exist unit vectors $\xi \in H$ such that $|(x\xi \mid \xi)|$ is arbitrarily close to 1, hence functionals $f \in B$ such that $|f(x)|$ is arbitrarily close to 1, and hence, by 2.5.5, functionals $f \in P$ such that $|f(x)|$ is arbitrarily close to 1. Consequently,

$$1 \leq \|G_x\| \leq \|F_x\| \leq 1.$$

The corollary follows.

When A is commutative, the map $x \mapsto G_x$ is just the Gelfand isomorphism, restricted to A_h . Another generalization of the Gelfand isomorphism to the non-commutative case, more satisfactory in certain respects, will be studied later in 10.5.4.

2.6.4. Corollary. *Let A be a C^* -algebra and g a continuous hermitian linear functional on A . There exist positive functionals f, f' on A such that*

$$g = f - f', \quad \|g\| = \|f\| + \|f'\|.$$

Retain the notation of 2.6.3, and identify A_h with a vector subspace of $C(B)$. By the Hahn–Banach theorem, g extends to a linear functional μ on $C(B)$ such that $\|\mu\| = \|g\|$ (cf. 1.2.6). The measure μ on B can be written $\mu = \mu^+ - \mu^-$, where μ^+, μ^- are positive measures such that

$$\|\mu\| = \|\mu^+\| + \|\mu^-\|.$$

Let $f = \mu^+|_{A_h}$ and $f' = \mu^-|_{A_h}$. Then $g = f - f'$, and

$$\|g\| \leq \|f\| + \|f'\| \leq \|\mu^+\| + \|\mu^-\| = \|\mu\| = \|g\|,$$

whence

$$\|g\| = \|f\| + \|f'\|.$$

We shall see later in 12.3.4 that the decomposition in 2.6.4 is unique.

Bibliography: [41, 42, 55, 70, 105, 108, 126, 134, 150].

2.7. Enveloping C^* -Algebra of an Involutive Banach Algebra

2.7.1. Proposition. *Let A be an involutive Banach algebra having an approximate identity. Let R be the set of representations of A , R' the set of topologically irreducible representations of A , B the set of continuous positive functionals on A of norm at most 1, and P the set of pure states of A . For every $x \in A$,*

$$\sup_{\pi \in R} \|\pi(x)\| = \sup_{\pi \in R'} \|\pi(x)\| = \sup_{f \in B} f(x^*x)^{1/2} = \sup_{f \in P} f(x^*x)^{1/2}. \quad (1)$$

Denote by $\|x\|'$ the common value of these four expressions. Then $\|x\|' \leq \|x\|$, and $x \mapsto \|x\|'$ is a seminorm on A such that

$$\|xy\|' \leq \|x\|'\|y\|', \quad \|x^*\|' = \|x\|', \quad \|x^*x\|' = \|x\|'^2$$

for all $x, y \in A$.

Denote by a, b, c, d , in succession, the four numbers appearing in (1).

$d \leq b$: let $f \in P$. Then f is associated with a representation $\pi \in R'$ by 2.5.4, and

$$f(x^*x) \leq \|\pi(x)\|^2$$

by 2.4.7.

$b \leq a$ is clear.

$a \leq c$ follows from 2.4.7.

$c \leq d$: let $f \in B$. By 2.5.5, $f(x^*x)$ belongs to the closed interval generated by 0 and the numbers $g(x^*x)$, where $g \in P$. Hence

$$f(x^*x) \leq \sup_{g \in P} g(x^*x).$$

By 1.3.7, $\|\pi(x)\| \leq \|x\|$ for every $\pi \in R$, so $\|x\|' \leq \|x\|$. Moreover, each function $x \mapsto \|\pi(x)\|$ is a seminorm on A ; hence $x \mapsto \|x\|'$ is a seminorm on A . We have

$$\|\pi(x^*)\| = \|\pi(x)\|, \quad \|\pi(x^*x)\| = \|\pi(x)\|^2$$

for every $\pi \in R$, and therefore

$$\|x^*\|' = \|x\|', \quad \|x^*x\|' = \|x\|'^2.$$

Finally, for $x, y \in A$ and every $\pi \in R$,

$$\|\pi(xy)\| \leq \|\pi(x)\| \|\pi(y)\| \leq \|x\|' \|y\|',$$

and hence $\|xy\|' \leq \|x\|' \|y\|'$.

2.7.2. Let I be the set of $x \in A$ such that $\|x\|' = 0$. It is a closed self-adjoint two-sided ideal of A . The map $x \mapsto \|x\|'$ induces, on passing to the quotient, a norm on A/I . With this norm, A/I satisfies all the axioms of a C^* -algebra except that in general it is not complete. The completion B of A/I is a C^* -algebra called the *enveloping C^* -algebra* of A . The canonical map from A into B is a norm-decreasing homomorphism of involutive algebras and has dense image in B .

2.7.3. When A is a C^* -algebra, $\|x\|' = \|x\|$ by 2.6.1, and A identifies with its enveloping C^* -algebra. In this case, Proposition 2.7.1 gives, as announced, the existence of sufficiently many topologically irreducible representations, and considerably strengthens Theorem 2.6.1:

Theorem. *Let A be a C^* -algebra. There exists a family (π_i) of topologically irreducible representations of A such that*

$$\|x\| = \sup_i \|\pi_i(x)\|$$

for every $x \in A$.

2.7.4. Proposition. *Let A be an involutive Banach algebra having an approximate identity, let B be the enveloping C^* -algebra of A , and let $\tau : A \rightarrow B$ be the canonical map.*

- (i) *If π is a representation of A , there exists one and only one representation ρ of B such that $\pi = \rho \circ \tau$; moreover, $\rho(B)$ is the C^* -algebra generated by $\pi(A)$.*
- (ii) *The map $\pi \mapsto \rho$ is a bijection from the set of representations of A onto the set of representations of B .*
- (iii) *The representation ρ is nondegenerate (respectively topologically irreducible) if and only if π is nondegenerate (respectively topologically irreducible).*

Let π be a representation of A . With the notation of 2.7.2, π vanishes on I and induces, on passing to the quotient, a representation π' of A/I such that $\|\pi'(z)\| \leq \|z\|$ for every $z \in A/I$, where $\|\cdot\|$ denotes the norm in B . Hence π' extends to a representation ρ of B , and $\pi = \rho \circ \tau$. The uniqueness assertion in (i) follows from the density of $\tau(A)$ in B . This also implies that $\pi(A)$ is norm-dense in $\rho(B)$. Since $\rho(B)$ is a C^* -algebra by 1.8.3, it is the C^* -algebra generated by $\pi(A)$. Assertions (ii) and (iii) are immediate.

Proposition 2.7.4 shows that B solves a universal problem. It also shows that, in most questions concerning representations of involutive Banach algebras having an approximate identity, one may restrict oneself to the case of C^* -algebras.

2.7.5. Proposition. *Retain the notation A, B, τ of 2.7.4.*

- (i) *If f is a continuous positive functional on A , there exists one and only one positive functional g on B such that $f = g \circ \tau$. Moreover, $\|g\| = \|f\|$.*
- (ii) *The map $f \mapsto g$ is a bijection from the set of continuous positive functionals on A onto the set of positive functionals on B .*
- (iii) *Let M be a bounded set of continuous positive functionals on A . The map $f \mapsto g$, restricted to M , is bicontinuous for the weak topologies $\sigma(A', A)$ and $\sigma(B', B)$, where A' and B' denote the duals of A and B .*

Let f be a continuous positive functional on A . For every $x \in A$, by 2.1.5(i),

$$|f(x)| \leq \|f\|^{1/2} f(x^*x)^{1/2} \leq \|f\| \|x\|'.$$

It follows that there exists a continuous linear functional g on B such that $f = g \circ \tau$ and $\|g\| \leq \|f\|$. If $y \in B$, there exists a sequence (x_n) in A such that $\tau(x_n) \rightarrow y$, and hence

$$g(y^*y) = \lim_n f(x_n^*x_n) \geq 0;$$

thus g is positive. If $x \in A$ and $\|x\| \leq 1$, then

$$|f(x)| = |g(\tau(x))| \leq \|g\| \|\tau(x)\| \leq \|g\| \|x\| \leq \|g\|,$$

so $\|f\| \leq \|g\|$. The uniqueness of g follows from the density of $\tau(A)$ in B . This proves (i), and (ii) is immediate. Let $M \subset A'$ be a set of continuous positive functionals on A , and let $N \subset B'$ be its image under $f \mapsto g$. It is clear that $f \mapsto g$ is bicontinuous from M onto N for the weak topologies $\sigma(A', A)$ and $\sigma(B', \tau(A))$. If M is bounded, then N is bounded by (i), and consequently $\sigma(B', \tau(A))$ and $\sigma(B', B)$ coincide on N , since $\tau(A)$ is dense in B .

2.7.6. Let A be an involutive Banach algebra admitting an approximate identity, and let Q be the set of continuous positive functionals on A . For every $f \in Q$, let π_f be the representation defined by f . Then

$$\pi = \bigoplus_{f \in Q} \pi_f$$

is called the *universal representation* of A . It is nondegenerate. By 2.2.7 and 2.4.6, every nondegenerate representation of A is a Hilbert direct sum of representations π_f , and hence $\|x\|' = \|\pi(x)\|$ for every $x \in A$. Thus, if B is the enveloping C^* -algebra of A , the representation of B corresponding to π is an isomorphism of B onto $\overline{\pi(A)}$.

Bibliography: [27, 42, 52, 105, 108, 126, 134].

2.8. A Transitivity Theorem

2.8.1. Lemma. Let H be a Hilbert space, let $(\xi_1, \dots, \xi_n)$ be an orthonormal system in H , and let $\eta_1, \dots, \eta_n$ be vectors of H of norm at most r . There exists $b \in \mathcal{L}(H)$, of norm at most $(2n)^{1/2}r$, such that

$$b\xi_1 = \eta_1, \dots, b\xi_n = \eta_n.$$

If there exists a hermitian element $h \in \mathcal{L}(H)$ such that $h\xi_1 = \eta_1, \dots, h\xi_n = \eta_n$, then b may also be chosen hermitian.

Let K be the vector subspace of H spanned by $\xi_1, \dots, \xi_n, \eta_1, \dots, \eta_n$. We define b to leave K invariant and to vanish on $H \ominus K$. Thus we may reduce to the case $K = H$. Let $(\xi_1, \dots, \xi_n, \xi_{n+1}, \dots, \xi_m)$ be an orthonormal basis of H . Let b be the operator on H whose matrix relative to $(\xi_1, \dots, \xi_m)$ is chosen as follows:

- a. its first n columns are the coordinates of $\eta_1, \dots, \eta_n$ relative to the ξ_i ;
- b. if h exists, its first n rows are the conjugates of its first n columns (this is possible because the existence of h implies that the square matrix formed by the first n rows and first n columns is hermitian);
- c. the entries not yet chosen are zero.

Then

$$b\xi_1 = \eta_1, \dots, b\xi_n = \eta_n$$

and

$$\|b\|^2 \leq \text{Tr}(b^*b) \leq 2(\|\eta_1\|^2 + \dots + \|\eta_n\|^2) \leq 2nr^2.$$

If h exists, b is hermitian by construction.

2.8.2. Lemma. *Let $H_1, \dots, H_p$ be Hilbert spaces, let $H = H_1 \oplus \dots \oplus H_p$, and let A be a C^* -algebra of operators on H , strongly dense in the von Neumann algebra*

$$\mathcal{L}(H_1) \times \dots \times \mathcal{L}(H_p).$$

Let $(\xi_1, \dots, \xi_n)$ be an orthonormal system in H , and let $\eta_1, \dots, \eta_n$ be vectors of H of norm at most r . Suppose that, for $i = 1, \dots, n$, the vectors ξ_i and η_i belong to the same summand H_{j_i} . There exists $b \in A$, of norm at most $3n^{1/2}r$, such that

$$b\xi_1 = \eta_1, \dots, b\xi_n = \eta_n.$$

If there exists a hermitian element $h \in \mathcal{L}(H)$ such that $h\xi_1 = \eta_1, \dots, h\xi_n = \eta_n$, then b may also be chosen hermitian in A .

By the Kaplansky density theorem, the unit ball of A (respectively the unit ball of the hermitian part of A) is strongly dense in the unit ball of $\mathcal{L}(H_1) \times \dots \times \mathcal{L}(H_p)$ (respectively its hermitian part). Thus there first exists $y_0 \in \mathcal{L}(H_1) \times \dots \times \mathcal{L}(H_p)$ such that

$$y_0\xi_1 = \eta_1, \dots, y_0\xi_n = \eta_n, \quad \|y_0\| \leq (2n)^{1/2}r,$$

by applying 2.8.1 successively in $H_1, \dots, H_p$; and then there exists $x_0 \in A$ such that

$$\|x_0\xi_i - y_0\xi_i\| \leq \frac{1}{2}r \quad (i = 1, \dots, n), \quad \|x_0\| \leq (2n)^{1/2}r.$$

Next there exists $y_1 \in \mathcal{L}(H_1) \times \cdots \times \mathcal{L}(H_p)$ such that

$$y_1 \xi_i = \eta_i - x_0 \xi_i \quad (i = 1, \dots, n), \quad \|y_1\| \leq \frac{1}{2}(2n)^{1/2}r,$$

by 2.8.1; and then there exists $x_1 \in A$ such that

$$\|x_1 \xi_i - y_1 \xi_i\| \leq \frac{1}{4}r \quad (i = 1, \dots, n), \quad \|x_1\| \leq \frac{1}{2}(2n)^{1/2}r.$$

Inductively, we construct elements

$$y_k \in \mathcal{L}(H_1) \times \cdots \times \mathcal{L}(H_p), \quad x_k \in A,$$

such that

$$\begin{aligned} y_k \xi_i &= \eta_i - x_0 \xi_i - \cdots - x_{k-1} \xi_i, & \|y_k\| &\leq \frac{1}{2^k}(2n)^{1/2}r, \\ \|x_k \xi_i - y_k \xi_i\| &\leq \frac{1}{2^{k+1}}r, & \|x_k\| &\leq \frac{1}{2^k}(2n)^{1/2}r \end{aligned}$$

for $i = 1, \dots, n$. (If h exists, the y_k and x_k may be chosen hermitian.) Then $x_0 + x_1 + \cdots$ converges in norm to an element $b \in A$ such that

$$b \xi_1 = \eta_1, \dots, b \xi_n = \eta_n,$$

and

$$\|b\| \leq \left(1 + \frac{1}{2} + \frac{1}{4} + \cdots\right) (2n)^{1/2}r \leq 3n^{1/2}r.$$

2.8.3. Theorem. *Let A be a C^* -algebra, let $\tilde{A}$ be the C^* -algebra obtained from A by adjoining an identity element, let $\pi_1, \dots, \pi_p$ be representations of A in Hilbert spaces $H_1, \dots, H_p$, and let $\tilde{\pi}_1, \dots, \tilde{\pi}_p$ be their canonical extensions to $\tilde{A}$. Suppose that the π_j are nonzero, topologically irreducible, and pairwise inequivalent.*

(i) *Let $T_1 \in \mathcal{L}(H_1), \dots, T_p \in \mathcal{L}(H_p)$, and let $K_1, \dots, K_p$ be finite-dimensional vector subspaces of $H_1, \dots, H_p$. There exists $x \in A$ such that*

$$\pi_j(x)|_{K_j} = T_j|_{K_j} \quad (j = 1, \dots, p).$$

(ii) *Let $T_1 \in \mathcal{L}(H_1), \dots, T_p \in \mathcal{L}(H_p)$ be hermitian operators, and let $K_1, \dots, K_p$ be finite-dimensional vector subspaces of $H_1, \dots, H_p$. There exists a hermitian element $x \in A$ such that*

$$\pi_j(x)|_{K_j} = T_j|_{K_j} \quad (j = 1, \dots, p).$$

(iii) Let $T_1 \in \mathcal{L}(H_1), \dots, T_p \in \mathcal{L}(H_p)$ be unitary operators, and let $K_1, \dots, K_p$ be finite-dimensional vector subspaces of $H_1, \dots, H_p$. There exists a unitary element $x \in \tilde{A}$ such that

$$\tilde{\pi}_j(x)|_{K_j} = T_j|_{K_j} \quad (j = 1, \dots, p).$$

Let

$$H = H_1 \oplus \dots \oplus H_p, \quad \pi = \pi_1 \oplus \dots \oplus \pi_p.$$

By 1.8.3, $\pi(A)$ is a C^* -subalgebra of $\mathcal{L}(H)$ commuting with the projections $E_j = P_{H_j}$. Let B be the von Neumann algebra generated by $\pi(A)$. We have

$$\pi(A) \subset \mathcal{L}(H_1) \times \dots \times \mathcal{L}(H_p), \quad \text{and hence} \quad B \subset \mathcal{L}(H_1) \times \dots \times \mathcal{L}(H_p).$$

Since each π_j is nondegenerate, π is nondegenerate; hence $\pi(A)$ is strongly dense in B . Since π_j is topologically irreducible, E_j is a minimal projection of $\pi(A)' = B'$, and the von Neumann algebra induced by B on H_j is $\mathcal{L}(H_j)$. Let $F_j \in B \cap B'$ be the central support of E_j (A 10). Let j, k be distinct indices. If F_j and F_k are not orthogonal, there exist nonzero projections $E'_j, E'_k \in B'$, respectively dominated by E_j, E_k , and equivalent relative to B' , by A 44. Since E_j, E_k are minimal projections of B' , we have $E'_j = E_j$ and $E'_k = E_k$. Hence there exists a partial isometry in B' having E_j, E_k as its initial and final projections; but then π_j and π_k are equivalent, a contradiction. Thus the F_j are pairwise orthogonal. Now $F_j \geq E_j$ for every j , and consequently $E_j = F_j \in B$. Since $B_{E_j} = \mathcal{L}(H_j)$, it follows that

$$B \supset \mathcal{L}(H_1) \times \dots \times \mathcal{L}(H_p),$$

and therefore, finally,

$$B = \mathcal{L}(H_1) \times \dots \times \mathcal{L}(H_p).$$

Let $T_j \in \mathcal{L}(H_j)$, and let K_j be a finite-dimensional vector subspace of H_j , for $j = 1, \dots, p$. By 2.8.2, there exists $x \in A$ such that $\pi_j(x)|_{K_j} = T_j|_{K_j}$ for all j . If the T_j are hermitian, x may be chosen so that $\pi(x)$ is hermitian, by 2.8.2. Since

$$\pi(x) = \pi\left(\frac{1}{2}(x + x^*)\right),$$

x itself may be chosen hermitian.

Suppose that the T_j are unitary. For $j = 1, \dots, p$, there is a finite-dimensional vector subspace $K'_j \supset K_j$ of H_j , and a unitary operator $T'_j \in \mathcal{L}(H_j)$, leaving K'_j invariant, such that $T'_j|_{K_j} = T_j|_{K_j}$. There is then a hermitian operator $T''_j \in \mathcal{L}(H_j)$, leaving K'_j invariant, such that

$$\exp(iT''_j)|_{K'_j} = T'_j|_{K'_j}.$$

By what precedes, there exists a hermitian element $y \in \tilde{A}$ such that $\tilde{\pi}_j(y)|_{K'_j} = T'_j|_{K'_j}$ for every j . Then $x = \exp(iy)$ is a unitary element of $\tilde{A}$, and $\tilde{\pi}_j(x)|_{K_j} = T_j|_{K_j}$ for every j .

2.8.4. Corollary. *Every topologically irreducible representation of a C^* -algebra is algebraically irreducible.*

It is enough to apply Theorem 2.8.3 with $p = 1$ and $\dim K_1 = 1$.

Henceforth we shall simply speak of irreducible representations of a C^* -algebra, without further qualification.

2.8.5. Corollary. *Let A be a C^* -algebra, let f be a pure positive functional on A , and let N be the left ideal formed by the $x \in A$ such that $f(x^*x) = 0$. Then A/N , equipped with the inner product induced by $f(y^*x)$, is complete, and hence is equal to the space of the representation defined by f .*

Indeed, let π be this representation; it is topologically irreducible by 2.5.4. By construction of π , A/N is a vector subspace of H_π invariant under π . But $A/N \neq 0$, and therefore $A/N = H_\pi$ by 2.8.4.

2.8.6. Corollary. *Let A be a C^* -algebra, let $\tilde{A}$ be the C^* -algebra obtained from A by adjoining an identity element, and let f_1, f_2 be two pure states of A . The representations π_1, π_2 defined by f_1, f_2 are equivalent if and only if there exists a unitary element $u \in \tilde{A}$ such that*

$$f_2(x) = f_1(u^*xu) \quad \text{for every } x \in A.$$

Let ξ_1 be the vector of H_{π_1} defined by f_1 . The states of A that define representations equivalent to π_1 are the states associated with π_1 , by 2.4.6; that is, the functionals

$$x \longmapsto (\pi_1(x)\xi \mid \xi),$$

where ξ is a unit vector of H_{π_1} , by 2.4.3. But the unit vectors of H_{π_1} are precisely the vectors $\tilde{\pi}_1(u)\xi_1$, where u is unitary in $\tilde{A}$, by 2.8.3; here $\tilde{\pi}_1$ denotes the canonical extension of π_1 to $\tilde{A}$. Finally,

$$\begin{aligned} (\pi_1(x)\tilde{\pi}_1(u)\xi_1 \mid \tilde{\pi}_1(u)\xi_1) &= (\tilde{\pi}_1(u^*xu)\xi_1 \mid \xi_1) \\ &= f_1(u^*xu). \end{aligned}$$

Bibliography: [49, 69, 126].

2.9. Ideals in C^* -Algebras

2.9.1. Proposition. *Let A be a C^* -algebra, let f be a positive functional on A , let M be its kernel, and let N be the set of $x \in A$ such that $f(x^*x) = 0$. Then*

$$M \supset N + N^*,$$

and $M = N + N^$ if f is pure.*

Since

$$|f(x)|^2 \leq \|f\| f(x^*x),$$

we have $N \subset M$, and hence $N^* \subset M^* = M$. Thus $N + N^* \subset M$.

Now suppose f is pure. Let $\tilde{A}$ be the C^* -algebra obtained from A by adjoining an identity, and let $\tilde{f}$ be the canonical extension of f to $\tilde{A}$. Let π, ξ be the irreducible representation and vector defined by f . Let $b \in M$, and let η be its canonical image in H_π . By construction of π, ξ ,

$$(\eta \mid \xi) = \tilde{f}(1^*b) = 0.$$

Thus there exists a hermitian operator on H_π that annihilates ξ and fixes η . By 2.8.3, there exists a hermitian element $a \in A$ such that $\pi(a)\xi = 0$ and $\pi(a)\eta = \eta$; in other words, $a \in N$ and $b - ab \in N$. Then

$$b^* = (b - ab)^* + b^*a \in N^* + N.$$

Consequently $M = M^* \subset N^* + N$.

2.9.2. Lemma. *Let A be a unital C^* -algebra, let L be a closed left ideal of A , and let $x \in A^+$. If, for every $\varepsilon > 0$, there exists a positive element $u_\varepsilon \in L$ such that*

$$x \leq u_\varepsilon + \varepsilon 1,$$

then $x \in L$.

Put $t_\varepsilon = u_\varepsilon^{1/2}$. This is a limit of polynomials in u_ε without constant terms, and hence belongs to L . We have

$$\begin{aligned} & \|x^{1/2}(t_\varepsilon + \sqrt{\varepsilon})^{-1}t_\varepsilon - x^{1/2}\|^2 \\ &= \|x^{1/2}\sqrt{\varepsilon}(t_\varepsilon + \sqrt{\varepsilon})^{-1}\|^2 \\ &= \varepsilon \|(t_\varepsilon + \sqrt{\varepsilon})^{-1}x(t_\varepsilon + \sqrt{\varepsilon})^{-1}\|. \end{aligned}$$

Since $0 \leq x \leq t_\varepsilon^2 + \varepsilon$, we have, by 1.6.8,

$$\begin{aligned} 0 &\leq (t_\varepsilon + \sqrt{\varepsilon})^{-1} x (t_\varepsilon + \sqrt{\varepsilon})^{-1} \\ &\leq (t_\varepsilon + \sqrt{\varepsilon})^{-1} (t_\varepsilon^2 + \varepsilon) (t_\varepsilon + \sqrt{\varepsilon})^{-1} \\ &= (t_\varepsilon^2 + \varepsilon) (t_\varepsilon^2 + 2\sqrt{\varepsilon} t_\varepsilon + \varepsilon)^{-1} \leq 1. \end{aligned}$$

Therefore

$$\|x^{1/2} (t_\varepsilon + \sqrt{\varepsilon})^{-1} t_\varepsilon - x^{1/2}\|^2 \leq \varepsilon.$$

But

$$x^{1/2} (t_\varepsilon + \sqrt{\varepsilon})^{-1} t_\varepsilon \in L.$$

Since L is closed, $x^{1/2} \in L$, and hence $x \in L$.

2.9.3. Lemma. *Let A be a C^* -algebra and L a closed left ideal of A . Then L is the closed left ideal generated by $L \cap A^+$.*

This follows at once from 1.7.3.

2.9.4. Lemma. *Let A be a C^* -algebra, and let L, L' be two closed left ideals of A such that $L \subset L'$. Suppose that every positive functional on A that vanishes on L also vanishes on L' . Then $L = L'$.*

We may suppose that A has an identity element. Let $a \in L' \cap A^+$, and let $\varepsilon > 0$. The set S_ε of positive functionals f on A such that

$$f(1) = \|f\| = 1, \quad f(a) \geq \varepsilon$$

is weakly compact. If $f \in S_\varepsilon$, then f is not identically zero on L' , and hence not on L ; thus there exists $x_f \in L$ such that $f(x_f) \neq 0$. Consequently there exists a weak neighborhood U_f of f in S_ε such that $g(x_f) \neq 0$ for $g \in U_f$. By compactness of S_ε , there is a finite open cover $(U_i)_{1 \leq i \leq m}$ of S_ε and elements $a_1, \dots, a_m \in L$ such that

$$0 < |f(a_i)|^2 \leq f(a_i^* a_i) \quad (f \in U_i).$$

Thus

$$f(a_1^* a_1 + \dots + a_m^* a_m) > 0 \quad (f \in S_\varepsilon).$$

Multiplying the a_i by sufficiently large scalars, we obtain

$$f(a_1^* a_1 + \dots + a_m^* a_m) \geq f(a) \quad (f \in S_\varepsilon).$$

Hence

$$f(a_1^* a_1 + \dots + a_m^* a_m + \varepsilon 1 - a) \geq 0$$

for every positive functional f . It follows from 2.6.2 that

$$a \leq a_1^* a_1 + \cdots + a_m^* a_m + \varepsilon 1.$$

Therefore $a \in L$ by 2.9.2. Thus L and L' have the same positive elements, and hence are equal by 2.9.3.

2.9.5. Theorem. *Let A be a C^* -algebra.*

- (i) *Let f be a positive functional on A , and let N_f be the left ideal formed by the $x \in A$ such that $f(x^*x) = 0$. Then N_f is regular maximal if and only if f is pure.*
- (ii) *The map $f \mapsto N_f$ defines a bijection from the set of pure states onto the set of regular maximal left ideals.*
- (iii) *Every closed left ideal of A is the intersection of the regular maximal left ideals that contain it.*

a. Let f be a pure positive functional on A . By 2.8.4 and 2.8.5, the canonical representation π of A in A/N_f is nonzero and algebraically irreducible. There exists $\xi \in H_\pi = A/N_f$ such that N_f is the set of all $x \in A$ for which $\pi(x)\xi = 0$. Hence N_f is a regular maximal left ideal of A .

b. Let L be a closed left ideal of A . Let S be the set of positive functionals on A , of norm at most 1, that vanish on L . If $f \in S$, then $N_f \supset L$, because $x^*x \in L$ for every $x \in L$, and hence $f(x^*x) = 0$. By 2.9.4,

$$L = \bigcap_{f \in S} N_f.$$

Moreover, S is convex and weakly compact, and therefore is the weakly closed convex hull of the set S' of its extreme points. Consequently, if $x \in A$ is such that $f(x^*x) = 0$ for every $f \in S'$, then the same holds for every $f \in S$. This proves that

$$\bigcap_{f \in S} N_f = \bigcap_{f \in S'} N_f.$$

Finally, if $f \in S$ can be written $f = f_1 + f_2$, with f_1, f_2 positive, then, for every $x \in L$,

$$0 \leq f_1(x^*x) + f_2(x^*x) = f(x^*x) = 0.$$

Thus $f_1(x^*x) = f_2(x^*x) = 0$, and consequently $f_1(x) = f_2(x) = 0$; hence $f_1, f_2 \in S$. This proves that the nonzero elements of S' are pure functionals. In view of part *a*, the equality

$$L = \bigcap_{f \in S'} N_f$$

proves (iii).

c. At the same time, we see that if L is regular maximal, and hence closed by B 1, then $L = N_f$ for some pure f . Thus the map considered in (ii) is surjective. If f, f' are pure states such that $N_f = N_{f'}$, Proposition 2.9.1 shows that f and f' have the same kernel. Hence $f = \lambda f'$, with $\lambda \geq 0$. Since $\|f\| = \|f'\| = 1$, we have $f = f'$, which completes the proof of (ii).

d. Finally, let f be a positive functional such that N_f is regular maximal. Then $N_f = N_{f'}$, where f' is pure. The kernel of f' , which is

$$N_{f'} + N_{f'}^* = N_f + N_f^*,$$

is contained in the kernel of f , by 2.9.1. Therefore f and f' are proportional, so f is pure. Together with part a, this proves (i).

2.9.6. Recall that two representations π_1, π_2 of an algebra on vector spaces E_1, E_2 are said to be *algebraically equivalent* if there exists an isomorphism from E_1 onto E_2 that carries π_1 into π_2 .

Corollary. *Let A be a C^* -algebra.*

- (i) *Every algebraically irreducible representation of the noninvolutive algebra A on a complex vector space is algebraically equivalent to a representation of the C^* -algebra A on a Hilbert space.*
- (ii) *Let π, π' be two irreducible representations of the C^* -algebra A on Hilbert spaces H, H' . If π, π' are algebraically equivalent, then they are equivalent in the sense of 2.2.1.*

Let π be a nonzero algebraically irreducible representation of A on a complex vector space. There exists a regular maximal left ideal L of A such that π identifies with the regular representation of A on A/L . By 2.9.5, there is then a pure positive functional f on A such that $L = N_f$. The space A/L can therefore be given a Hilbert-space structure for which π is a representation of the C^* -algebra A . This proves (i).

Let π, π' be two nonzero irreducible representations of the C^* -algebra A on Hilbert spaces H, H' . Let ξ (respectively ξ') be a unit vector of H (respectively H'), and let L (respectively L') be the set of $x \in A$ such that $\pi(x)\xi = 0$ (respectively $\pi'(x)\xi' = 0$). Put

$$f(x) = (\pi(x)\xi \mid \xi), \quad f'(x) = (\pi'(x)\xi' \mid \xi').$$

Then L (respectively L') is the set of $x \in A$ such that $f(x^*x) = 0$ (respectively $f'(x^*x) = 0$). If π, π' are algebraically equivalent, ξ, ξ' may be chosen so that $L = L'$. Then $f = f'$ by 2.9.5, and hence $\pi \sim \pi'$ by 2.4.1. This proves (ii).

We may therefore henceforth speak of classes of irreducible representations of a C^* -algebra without specifying whether the equivalence is purely algebraic or unitary.

2.9.7. Recall that a *primitive two-sided ideal* of an algebra A is the kernel of a nonzero algebraically irreducible representation of A on a vector space. We denote by $\text{Prim}(A)$ the set of primitive two-sided ideals of A .

Theorem. *Let A be a C^* -algebra.*

- (i) *The primitive two-sided ideals of A are the kernels of the nonzero topologically irreducible representations of A on a Hilbert space.*
- (ii) *Every closed two-sided ideal of A is the intersection of the primitive two-sided ideals that contain it.*

Assertion (i) follows from 2.8.4 and 2.9.6(i).

Let I be a closed two-sided ideal of A . Then A/I is a C^* -algebra by 1.8.2. Hence there exists a family (π_i) of irreducible representations of A , vanishing on I , whose kernels have intersection I , by 2.7.3. But $\ker \pi_i$ is primitive for every i .

Theorem 2.9.7 defines a canonical map

$$\widehat{A} \longrightarrow \text{Prim}(A)$$

which is surjective. We shall see in 4.3.7 very broad classes of cases in which this map is bijective.

Bibliography: [36, 69, 74, 126, 135].

2.10. Extension of Representations of C^* -Algebras

2.10.1. Lemma. *Let A be a unital C^* -algebra, and let B be a self-adjoint vector subspace of A such that $1 \in B$. Let F be the set of linear functionals g on B such that*

$$g(x^*) = \overline{g(x)} \quad (x \in B), \quad g(x) \geq 0 \quad (x \in B \cap A^+), \quad g(1) = 1.$$

- (i) *Every element of F extends to a state of A .*
- (ii) *Every extreme element of F extends to a pure state of A .*

- (iii) Let g be an extreme element of F . If there exists only one pure state f of A extending g , then f is the only state of A extending g , and, for every hermitian element $x \in A$,

$$f(x) = \sup_{\substack{y=y^* \in B \\ y \leq x}} g(y).$$

Let A_h (respectively B_h) be the set of hermitian elements of A (respectively B). Let $g \in F$, and let g' be its restriction to B_h ; it is real and nonnegative on $B \cap A^+$. The element 1 is an interior point, relative to B_h , of the positive cone. Hence, by B 6, g' extends to a real linear functional f' on A_h that is nonnegative on A^+ . Next f' extends to a hermitian linear functional f on A , and f is a state of A . The functionals g and $f|_B$ coincide on B_h , and hence on $B_h + iB_h = B$. This proves (i).

Let g be an extreme element of F , and let K be the set of states of A that extend g . By (i), K is nonempty, and it is clearly convex and weakly compact. Choose an extreme point f of K . We shall show that f is a pure state. Suppose f_1, f_2 are states of A such that

$$f = \frac{1}{2}(f_1 + f_2).$$

Then

$$g = \frac{1}{2}((f_1|_B) + (f_2|_B)).$$

Since g is extreme in F , $f_1|_B = f_2|_B = g$, so $f_1, f_2 \in K$. Since f is extreme in K , $f = f_1 = f_2$. This proves (ii).

Let g be an extreme element of F , and suppose there exists only one pure state f of A extending g . By what precedes, K has only the one extreme point f , and hence $K = \{f\}$. Let $x \in A_h$, $x \notin B$, and put

$$\alpha = \sup_{\substack{y \in B_h \\ y \leq x}} g(y).$$

There exists one and only one linear functional g_1 on $B + \mathbb{C}x$ extending g and satisfying $g_1(x) = \alpha$. It is clear that

$$g_1(y^*) = \overline{g_1(y)} \quad (y \in B + \mathbb{C}x).$$

If $y + \lambda x \geq 0$, where $y \in B_h$ and $\lambda \in \mathbb{R}$, then $g_1(y + \lambda x) \geq 0$ (this is evident if $\lambda = 0$; if $\lambda > 0$, then $-\lambda^{-1}y \leq x$, hence $g(-\lambda^{-1}y) \leq \alpha$, that is, $\lambda\alpha + g(y) \geq 0$; if $\lambda < 0$, then $-\lambda^{-1}y \geq x$, hence $g(-\lambda^{-1}y) \geq \alpha$, that is, $\lambda\alpha + g(y) \geq 0$). By what

precedes, g_1 extends to a state of A , which necessarily coincides with f . Thus $f(x) = g_1(x) = \alpha$.

2.10.2. Proposition. *Let A be a C^* -algebra, B a C^* -subalgebra of A , and ρ a representation of B on a Hilbert space K . There exist a Hilbert space H containing K as a subspace (with the norm of H inducing that of K), and a representation π of A on H , such that $\rho(x) = \pi(x)|_K$ for every $x \in B$. If ρ is irreducible, π may be chosen irreducible.*

Let $\tilde{A}$ be the C^* -algebra obtained from A by adjoining an identity element. Let $\tilde{B} = B + \mathbb{C}1$, which is a C^* -subalgebra of $\tilde{A}$. The representation ρ extends uniquely to a representation $\tilde{\rho}$ of $\tilde{B}$ such that $\tilde{\rho}(1) = 1$, and $\tilde{\rho}$ is irreducible if and only if ρ is irreducible. We are therefore reduced to the case in which A has an identity element and $1 \in B$. Moreover, by 2.2.6 and 2.2.7, we may suppose that ρ has a cyclic unit vector ξ . Set

$$g(x) = (\rho(x)\xi \mid \xi) \quad (x \in B).$$

Then g is a state of B , pure if ρ is irreducible. Let f be a state of A extending g , pure if ρ is irreducible (2.10.1). Let $\pi = \pi_f$ and $\eta = \xi_f$, and put $H_0 = \overline{\pi(B)\eta} \subset H_\pi$. For each $x \in B$, let $\pi'(x)$ be the restriction of $\pi(x)$ to H_0 . Then π' is a representation of B on H_0 , η is cyclic for π' , and, for every $x \in B$,

$$(\pi'(x)\eta \mid \eta) = f(x) = g(x) = (\rho(x)\xi \mid \xi).$$

Consequently there is an isomorphism from K onto H_0 which carries ξ to η and ρ to π' (2.4.1). Thus we may identify K with H_0 , and ρ with π' . Finally, if ρ is irreducible, f is a pure state by construction, and hence π is irreducible.

2.10.3. Lemma. *Let A be a C^* -algebra, I a closed two-sided ideal of A , π a representation of A on H , and H' a closed vector subspace of H invariant under $\pi(I)$; for each $x \in I$, let $\rho(x) = \pi(x)|_{H'}$. Suppose that the representation ρ of I on H' is nondegenerate.*

- (i) H' is invariant under $\pi(A)$; let π' be the subrepresentation of π defined by H' .
- (ii) $\pi'(I)$ is strongly dense in $\pi'(A)$.

Let (u_λ) be an approximate identity of the C^* -algebra I . Since ρ is nondegenerate, $\rho(u_\lambda)$ converges strongly to 1 (2.2.10). If $x \in A$, then $xu_\lambda \in I$; hence, for every $\xi \in H'$,

$$\pi(x)\rho(u_\lambda)\xi = \pi(x)\pi(u_\lambda)\xi = \pi(xu_\lambda)\xi = \rho(xu_\lambda)\xi \in H'.$$

Passing to the limit gives $\pi(x)\xi \in H'$. This proves (i). Moreover, $\pi'(x)$ is the strong limit of $\pi'(xu_\lambda)$, which proves (ii).

2.10.4. Proposition. *Let A be a C^* -algebra, I a closed two-sided ideal of A , and ρ a nondegenerate representation of I on H .*

- (i) *There exists a unique representation π of A on H extending ρ .*
- (ii) *$\rho(I)$ is strongly dense in $\pi(A)$.*

By 2.10.2, there exist a Hilbert space $H' \supset H$ and a representation π_1 of A on H' such that $\rho(x) = \pi_1(x)|_H$ for every $x \in I$. By 2.10.3, H is invariant under $\pi_1(A)$. Taking for π the subrepresentation of π_1 defined by H gives the existence assertion in (i), and (ii) follows from 2.10.3(ii). Finally, let ν be another representation of A on H extending ρ . If (u_λ) is an approximate identity of I , then for every $x \in A$, $\rho(xu_\lambda)$ converges strongly both to $\pi(x)$ and to $\nu(x)$; hence $\pi(x) = \nu(x)$.

Bibliography: [35, 45, 108, 134].

2.11. Passage to an Ideal and to a Quotient Algebra

Let A be a C^* -algebra. As we shall see, the choice of a closed two-sided ideal of A determines a decomposition of the sets $P(A)$, $\widehat{A}$, and $\text{Prim}(A)$.

2.11.1. Lemma. *Let A be a C^* -algebra, I a closed two-sided ideal of A , π a representation of A on a Hilbert space H , K the essential subspace of $\pi|_I$, and B the strong closure of $\pi(A)$.*

- (i) $P_K \in B \cap B'$.
- (ii) *The subrepresentation of π defined by $H \ominus K$ vanishes on I .*
- (iii) *If π' denotes the subrepresentation of π defined by K , then $\pi'(I)$ is strongly dense in $\pi'(A)$.*

The operator P_K is in the strong closure of $\pi(I)$, and hence $P_K \in B$. By 2.10.3, K is invariant under $\pi(A)$, and hence $P_K \in B'$. Assertion (ii) is evident, and (iii) follows from 2.10.3.

2.11.2. Proposition. *Let A be a C^* -algebra, I a closed two-sided ideal of A , and let $\widehat{A}_I$ (respectively $\widehat{A}^I$) be the set of $\pi \in \widehat{A}$ such that $\pi(I) = 0$ (respectively $\pi(I) \neq 0$).*

- (i) *For every $\pi \in \widehat{A}_I$, let π' be the representation of A/I induced from π by passage to the quotient. Then $\pi \mapsto \pi'$ is a bijection from $\widehat{A}_I$ onto $\widehat{A/I}$.*
- (ii) *The map $\pi \mapsto \pi|_I$ is a bijection from $\widehat{A}^I$ onto $\widehat{I}$.*

Assertion (i) is evident. We prove (ii). Let $\pi \in \widehat{A}^I$ and adopt the notation of 2.11.1. Since $K \neq \{0\}$, irreducibility of π gives $K = H$. Then 2.11.1(iii) proves that $\pi|_I$ is irreducible (see also 2.11.3). The map $\pi \mapsto \pi|_I$ from $\widehat{A}^I$ to $\widehat{I}$ is bijective by 2.10.4.

2.11.3. Lemma. *Let A be an algebra over a commutative field, and let π be an irreducible representation of A on a vector space E .*

- (i) *Let I be a two-sided ideal of A . If $\pi(I) \neq 0$, then $\pi|_I$ is irreducible.*
- (ii) *Let I_1, I_2 be two two-sided ideals of A such that $\pi(I_1) \neq 0$ and $\pi(I_2) \neq 0$. Then $\pi(I_1 I_2) \neq 0$.*

The set of elements of E annihilated by $\pi(I)$ is invariant under $\pi(A)$ and is distinct from E , and is therefore zero. Thus, if ξ is a nonzero element of E , then $\pi(I)\xi \neq 0$. Since $\pi(I)\xi$ is invariant under $\pi(A)$, we have $\pi(I)\xi = E$, which proves (i). Moreover, what precedes shows that $\pi(I_2)E = E$, and hence $\pi(I_1)\pi(I_2)E = E$; therefore $\pi(I_1 I_2) \neq 0$.

2.11.4. Lemma. *Let I_1, I_2 be two two-sided ideals of the algebra A , and let I be a primitive ideal of A . If $I \supset I_1 I_2$ (in particular, if $I \supset I_1 \cap I_2$), then $I \supset I_1$ or $I \supset I_2$.*

Suppose that $I \not\supset I_1$ and $I \not\supset I_2$. Applying 2.11.3(ii) to an irreducible representation π of A with kernel I , we obtain $\pi(I_1 I_2) \neq 0$, and hence $I \not\supset I_1 I_2$.

2.11.5. Proposition. *Let A be a C^* -algebra, I a closed two-sided ideal of A , and let $\text{Prim}_I(A)$ (respectively $\text{Prim}^I(A)$) be the set of primitive two-sided ideals of A that contain I (respectively do not contain I).*

- (i) *The map $J \mapsto J/I$ is a bijection from $\text{Prim}_I(A)$ onto $\text{Prim}(A/I)$.*
- (ii) *The map $J \mapsto J \cap I$ is a bijection from $\text{Prim}^I(A)$ onto $\text{Prim}(I)$.*

Assertion (i) is evident. We prove (ii). If $J \in \text{Prim}^I(A)$, then J is the kernel of an irreducible representation π of A such that $\pi(I) \neq 0$; hence $\pi|_I \in \widehat{I}$ (2.11.2) and $J \cap I = \ker(\pi|_I) \in \text{Prim}(I)$. If $J' \in \text{Prim}(I)$, there exists $\pi' \in \widehat{I}$ such that $J' = \ker \pi'$, and then there exists $\pi \in \widehat{A}$ such that $\pi' = \pi|_I$ (2.10.4); thus $J' = (\ker \pi) \cap I$, and the map in (ii) is surjective. Let $J_1, J_2 \in \text{Prim}^I(A)$ and suppose that $J_1 \cap I = J_2 \cap I$. We have $J_2 \supset J_1 \cap I$ and $J_2 \not\supset I$, and hence $J_2 \supset J_1$ by 2.11.4. Similarly $J_1 \supset J_2$, so $J_1 = J_2$: the map in (ii) is injective.

2.11.6. Before establishing for pure states a result analogous to Propositions 2.11.2 and 2.11.5, we first establish some properties of arbitrary positive forms.

Let A be a C^* -algebra, I a closed two-sided ideal of A , and $\omega: A \rightarrow A/I$ the canonical homomorphism. Since $\omega(A^+) = (A/I)^+$, the map $k \mapsto k \circ \omega$ is a bijection from the set of positive forms on A/I onto the set of positive forms on A that vanish on I . By the general properties of Banach spaces, this bijection preserves norms and is bicontinuous for the weak topologies.

2.11.7. Proposition. *Let A be a C^* -algebra, I a closed two-sided ideal of A , and f a positive form on A .*

- (i) *There is a unique decomposition $f = f_1 + f_2$, where f_1, f_2 are positive forms on A such that $\|f_1\| = \|f_1|_I\|$ and $f_2(I) = 0$.*
- (ii) *The pair (π_f, ξ_f) is identified with $(\pi_{f_1} \oplus \pi_{f_2}, \xi_{f_1} + \xi_{f_2})$; π_{f_2} vanishes on I , and $\pi_{f_1}|_I$ is nondegenerate.*

Put $\pi_f = \pi$. Let K_1 be the essential subspace of $\pi|_I$, let $K_2 = H_\pi \ominus K_1$, and set

$$\xi_1 = P_{K_1}\xi_f, \quad \xi_2 = P_{K_2}\xi_f,$$

with π_1 and π_2 the subrepresentations of π defined by K_1 and K_2 (2.11.1(ii)). Then ξ_i is cyclic for π_i ; hence, if $f_i = \omega_{\xi_i} \circ \pi_i$, the pair (π_i, ξ_i) is identified with (π_{f_i}, ξ_{f_i}) (2.4.6). We have $\pi_2(I) = 0$, and consequently $f_2(I) = 0$. The representations π_1 and $\pi_1|_I$ are nondegenerate, and therefore

$$\|f_1|_I\| = \|\xi_1\|^2 = \|f_1\|$$

by 2.4.3. The equality $\xi_f = \xi_1 + \xi_2$ gives $f = f_1 + f_2$. In particular, $f_2 = 0$ if $\|f\| = \|f|_I\|$.

It remains to prove the uniqueness assertion in (i). Let f'_1, f'_2 be positive forms on A such that

$$f = f'_1 + f'_2, \quad \|f'_1\| = \|f'_1|_I\|, \quad f'_2(I) = 0.$$

The preceding argument, applied to f'_1 in place of f , proves that $\pi_{f'_1}|_I$ is nondegenerate. Thus, if (u_λ) is an approximate identity of I , then for every $x \in A$, $f'_1(x) = \lim_\lambda f'_1(xu_\lambda)$; likewise $f_1(x) = \lim_\lambda f_1(xu_\lambda)$. But f_1 and f'_1 coincide on I , since $f_2(I) = f'_2(I) = 0$. Hence $f_1 = f'_1$ and $f_2 = f'_2$.

2.11.8. Proposition. *Let A be a C^* -algebra, I a closed two-sided ideal of A , and let $P_I(A)$ (respectively $P^I(A)$) be the set of pure states of A that vanish on I (respectively do not vanish on I).*

- (i) *For every $f \in P_I(A)$, let f' be the positive form on A/I induced from f by passage to the quotient. Then $f \mapsto f'$ is a bijection from $P_I(A)$ onto $P(A/I)$.*
- (ii) *The map $f \mapsto f|_I$ is a bijection from $P^I(A)$ onto $P(I)$.*

Let $f \in P_I(A)$, and denote by ω the canonical homomorphism from A onto A/I . If f' majorizes a positive form g' on A/I , then f majorizes $g = g' \circ \omega$; hence $g = \lambda f$, with $0 \leq \lambda \leq 1$, so $g' = \lambda f'$ and $f' \in P(A/I)$. It is clear that the map $f \mapsto f'$ is injective. We show that it is surjective. Let $h \in P(A/I)$ and $f = h \circ \omega$. If f majorizes a positive form g , then g vanishes on I^+ and hence on I , so $g = k \circ \omega$,

where k is a positive form majorized by h . Thus $k = \lambda h$, with $0 \leq \lambda \leq 1$, whence $g = \lambda f$ and $f \in P_I(A)$. Since $h = f'$, surjectivity has been proved.

Let $f \in P^I(A)$. In the decomposition $f = f_1 + f_2$ of 2.11.7, f_1 and f_2 are proportional to f . But f_2 vanishes on I whereas f does not, so $f_2 = 0$ and $f = f_1$. Hence $\|f|_I\| = \|f\| = 1$, and $f|_I$ is a state of I . By 2.10.3 and 2.11.7(ii), $\pi_f|_I$ is irreducible, and therefore $f|_I \in P(I)$. If $f, f' \in P^I(A)$ and $f|_I = f'|_I$, there is an isomorphism from H_{π_f} onto $H_{\pi_{f'}}$ which carries $\xi_{f|_I}$ (that is, ξ_f) to $\xi_{f'|_I}$ (that is, $\xi_{f'}$), and carries $\pi_{f|_I}$ to $\pi_{f'|_I}$, and hence π_f to $\pi_{f'}$ (2.10.4); consequently $f = f'$, and the map in (ii) is injective. Finally, if $g \in P(I)$, the irreducible representation π_g of I extends to an irreducible representation π of A (2.10.4); since $\|\xi_g\| = 1$, $f = \omega_{\xi_g} \circ \pi$ is a pure state of A extending g . Therefore $f \in P^I(A)$ and $f|_I = g$, which proves that the map in (ii) is surjective.

2.11.9. The six bijections in Propositions 2.11.2, 2.11.5, and 2.11.8 are called canonical.

In summary, we have the following diagram of canonical maps:

$$\begin{array}{ccccc}
 P(A/I) & \longrightarrow & P(A) & \longleftarrow & P(I) \\
 \downarrow & & \downarrow & & \downarrow \\
 \widehat{A/I} & \longrightarrow & \widehat{A} & \longleftarrow & \widehat{I} \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Prim}(A/I) & \longrightarrow & \text{Prim}(A) & \longleftarrow & \text{Prim}(I).
 \end{array}$$

The vertical arrows are surjective maps. On each line, the two horizontal arrows are injections onto complementary subsets of the middle set. It is easy to verify that this diagram is commutative.

Bibliography: [21, 64, 77, 135].

2.12. Complements

2.12.1. Let A be a unital C^* -algebra, and let f, g be two pure states of A such that $\|f - g\| < 2$. Then $\pi_f \simeq \pi_g$. [49]

2.12.2. Let A be a unital C^* -algebra, let $E(A)$ be the set of states of A , and let $f \in E(A)$. The following conditions are equivalent:

- (i) for every neighborhood U of f in $E(A)$, there exist $\delta > 0$ and $a \in A$ such that $0 \leq a \leq 1$, $f(a) = 1$, and $g(a) < 1 - \delta$ for $g \in E(A) \setminus U$;

- (ii) for every closed G_δ subset S of $E(A)$ containing f , there exists $a \in A$ such that $\|a\| = |f(a)|$ and S contains the set of $g \in E(A)$ for which $\|a\| = |g(a)|$;
- (iii) f is a pure state.

[46]

2.12.3. Let A be a C^* -algebra, and let K_n be the set of states f of A such that $\dim \pi_f = n < +\infty$. Then $P(A) \cap K_n$ is an open subset of K_n . [52]

2.12.4. Let A be a von Neumann algebra, $\mathfrak{Z}$ its center, and Ω the spectrum of $\mathfrak{Z}$. For every $\omega \in \Omega$, let I_ω be the norm-closed two-sided ideal of A generated by $\ker \omega$. For every $T \in A$, let T_ω be its canonical image in A/I_ω .

- a. If f is a pure state of A , then $f|_{\mathfrak{Z}}$ is an element ω of Ω and f vanishes on I_ω .
- b. For every $T \in A$, the function $\omega \mapsto \|T_\omega\|$ is continuous on Ω .
- c. In A/I_ω , the product of two nonzero two-sided ideals is nonzero.

[45]

2.12.5. Let A be a unital involutive Banach algebra. Suppose that, for every continuous Hermitian linear form f on A , there exist two positive forms g, h such that

$$f = g - h, \quad \|f\| = \|g\| + \|h\|.$$

Then A is a C^* -algebra. [55]

***2.12.6.** Let A_1 and A_2 be two unital C^* -algebras, and let $\overline{P(A_i)}$ be the weak closure of $P(A_i)$ in the dual of A_i . Let $\mathcal{E}_i$ be the set of functions $f \mapsto f(x)$ on $\overline{P(A_i)}$, with $x \in A_i$. If there exists a homeomorphism from $\overline{P(A_1)}$ onto $\overline{P(A_2)}$ which carries $\mathcal{E}_1$ onto $\mathcal{E}_2$, then there exists a linear isometric bijection $\varphi: A_1 \rightarrow A_2$ such that

$$\varphi(x^*) = \varphi(x)^* \quad (x \in A_1), \quad \varphi(x^2) = \varphi(x)^2 \quad (x = x^* \in A_1).$$

(See 1.9.10.) [66]

2.12.7. Let A be a unital C^* -algebra, let $\overline{P(A)}$ be the weak closure of $P(A)$ in the dual of A , and let $\mathcal{E}$ be the set of functions $f \mapsto f(x)$ on $\overline{P(A)}$, with $x \in A$. If the product of any two elements of $\mathcal{E}$ belongs to $\mathcal{E}$, then A is commutative. [66]

2.12.8. Let A be a unital C^* -algebra, let f be a state of A , and let N be the set of $x \in A$ such that $f(x^*x) = 0$. If $\ker f = N + N^*$, then f is pure. [69]

***2.12.9.** A unital C^* -algebra A is isomorphic to a von Neumann algebra if and only if it satisfies the following conditions:

- (i) every increasing, bounded, directed family of Hermitian elements of A has a supremum;
- (ii) for every nonzero $x \in A^+$, there exists a state f of A with $f(x) \neq 0$ such that, for every increasing directed family (y_i) in A^+ having supremum $y \in A^+$,

$$f(y) = \sup_i f(y_i).$$

[68, 151, 152]

***2.12.10.** Let A be a unital C^* -algebra, and let X be the compact space $\overline{P(A)}$. For each $x \in A$, let φ_x be the function $f \mapsto f(x)$ on $\overline{P(A)}$. Let $\mathcal{E}$ be the set of all φ_x , $x \in A$. Under the map $x \mapsto \varphi_x$, every state g of A becomes a positive linear form on $\mathcal{E}$; in general this form extends in infinitely many ways to a positive measure on X . Denote by N_g the collection of subsets of X that are negligible for all these measures.

If π is a representation of A , denote by N_π the intersection of the $N_{h \circ \pi}$, where h ranges over the normal states of the weak closure of $\pi(A)$.

Let H be a separable Hilbert space, A a C^* -subalgebra of $\mathcal{L}(H)$ containing 1, B the weak closure of A , and π a representation of A on a separable Hilbert space. The representation π extends to an ultraweakly continuous representation of B on H_π if and only if $N_\pi \supset N_\iota$, where ι denotes the identity representation of A . [70]

2.12.11. Let A be a C^* -algebra and G a topological group. For every $g \in G$, let ζ_g be an automorphism of A . Suppose that $g \mapsto \zeta_g$ is a representation of G , and that $g \mapsto \zeta_g(x)$ is continuous for every $x \in A$. A state f of A is said to be *stationary* for ζ if $f(\zeta_g(x)) = f(x)$ for every $x \in A$ and every $g \in G$. Suppose this is the case. There exists a unique continuous unitary representation ρ of G on H_{π_f} such that:

- 1° $\pi_f(\zeta_g(x)) = \rho(g)\pi_f(x)\rho(g)^{-1}$ for every $x \in A$ and every $g \in G$;
- 2° if η_f denotes the canonical map from A into H_{π_f} , then $\eta_f(\zeta_g(x)) = \rho(g)\eta_f(x)$ for every $x \in A$ and every $g \in G$.

[138]

2.12.12. Let A be a unital C^* -algebra, let x be a normal element of A , and let $\lambda \in \text{Sp}_A x$. There exist an irreducible representation π of A and a nonzero $\xi \in H_\pi$

such that $\pi(x)\xi = \lambda\xi$. (Apply 2.10.2 to the C^* -subalgebra B of A generated by 1 and x , and to the character of B whose value at x is λ .) [134]

2.12.13. Let A be a C^* -algebra without an identity element. Then 0 belongs to the weak closure of $P(A)$. [Indeed, let $x_1, \dots, x_n \in A^+$ and $\varepsilon > 0$; we must construct $f \in P(A)$ such that $f(x_i) \leq \varepsilon$ for $i = 1, \dots, n$. Replacing $x_1, \dots, x_n$ by $x_1 + \dots + x_n$, we may suppose $n = 1$; using 2.5.5, it is enough to construct a state g of A such that $g(x_1) \leq \varepsilon$. Realize A on a Hilbert space H by a nondegenerate representation. Then x_1 is not invertible in $A + \mathbb{C}1$, and hence is not invertible in $\mathcal{L}(H)$; there is a unit vector $\xi \in H$ such that $(x_1\xi \mid \xi) \leq \varepsilon$, and we may take $g = \omega_\xi$.] This was pointed out to me by J. Glimm.

2.12.14. Let A be a C^* -algebra, let I, J be two closed two-sided ideals of A , and let h be a state of A that vanishes on $I \cap J$. Then $h = \lambda f + \mu g$, where $\lambda \geq 0, \mu \geq 0, \lambda + \mu = 1$, and f (respectively g) is a state of A that vanishes on I (respectively J). (Reduce to the case $I \cap J = 0$. The essential subspaces of $\pi_h|_I$ and $\pi_h|_J$ are then orthogonal. Apply 2.11.7.) [135]

2.12.15. Let A and B be C^* -algebras, let π, π' be faithful representations of A , and let ρ, ρ' be faithful representations of B . Let D (respectively D') be the C^* -algebra of operators on $H_\pi \otimes H_\rho$ (respectively $H_{\pi'} \otimes H_{\rho'}$) generated by the $\pi(x) \otimes \rho(y)$ (respectively $\pi'(x) \otimes \rho'(y)$), where $x \in A, y \in B$. There is a unique isomorphism from D onto D' which carries $\pi(x) \otimes \rho(y)$ to $\pi'(x) \otimes \rho'(y)$ for all $x \in A$ and $y \in B$. Thus D , considered as an abstract C^* -algebra, depends only on A and B and is called the C^* -tensor product of A and B . [172, 173, 174, 184]

***2.12.16.**

- a. Let A and B be C^* -algebras, and let μ be a linear map from A into B . We say that μ is positive if $\mu(A^+) \subset B^+$. Let $A^{(n)}$ be the C^* -algebra of n -by- n matrices with entries in A . Applying μ to each entry of such a matrix gives a map $\mu^{(n)}: A^{(n)} \rightarrow B^{(n)}$. We say that μ is completely positive if $\mu^{(n)}$ is positive for every n . In general complete positivity is strictly stronger than positivity. The two notions are equivalent when A is commutative.
- b. Let A be a unital C^* -algebra, H a Hilbert space, and $\mu: A \rightarrow \mathcal{L}(H)$ a linear map such that $\mu(1) = 1$. There exist a Hilbert space K containing H as a subspace, and a homomorphism ρ from A into $\mathcal{L}(K)$, such that

$$\mu(x) = P_H \rho(x)|_H \quad (x \in A),$$

if and only if μ is completely positive.

[114]

2.12.17. Let A be a C^* -algebra, let A' be the Banach dual of A , and let A_h and A'_h be the spaces of Hermitian elements of A and A' , which are ordered vector spaces. The following conditions are equivalent: 1° A_h is a lattice; 2° A'_h is a lattice; 3° A is commutative. [38]

2.12.18. Let A be a C^* -algebra, let f be a pure state of A , and let N_f be the set of $x \in A$ such that $f(x^*x) = 0$, so that A/N_f , equipped with the inner product induced by $f(y^*x)$, is a Hilbert space. The norm of this space is also the norm of the quotient Banach space A/N_f . [156]

***2.12.19.** Let A be a unital C^* -algebra, and let f be a positive form on A .

a. Let E_f be the set of linear forms on A of the form

$$x \longmapsto f(x_0 x x_0^*), \quad x_0 \in A.$$

Let F_f be the norm closure of E_f in the dual A' of A . If g is a positive form on A , the following conditions are equivalent:

(i) $g \in F_f$;

(ii) $\pi_g \leq \pi_f$;

(iii) there exists $\xi \in H_{\pi_f}$ such that $g(x) = (\pi_f(x)\xi \mid \xi)$ for every $x \in A$.

b. If f is pure, then $F_f = E_f$.

c. Let E'_f be the set of positive forms on A majorized by a multiple λf of f , where $\lambda \geq 0$. Let F'_f be the norm closure of E'_f in A' . Then F'_f is the set of forms

$$x \longmapsto (\pi_f(x)\eta \mid \eta) \quad \text{on } A, \quad \eta \in \overline{\pi_f(A)'\xi_f}.$$

[71]

2.12.20. Let A be an involutive Banach algebra possessing an approximate identity, and let f be a continuous positive form on A . Denote by $L^2(f)$ the set of $g \in A'$ such that

$$\sup_{\substack{x \in A \\ f(x^*x) \neq 0}} |g(x)| f(x^*x)^{-1/2} < +\infty.$$

Let H_f be the space of π_f , and let η_f be the canonical map from A into H_f . By transposition, η_f defines a map $\zeta_f: H_f \rightarrow A'$ such that

$$\langle \zeta_f(\xi), x \rangle = (\eta_f(x) \mid \xi) \quad (\xi \in H_f, x \in A).$$

The map ζ_f is a semilinear bijection from H_f onto $L^2(f)$ which carries ξ_f to f . For every $x \in A$, let $\pi'(x)$ be the transpose of left multiplication by x on A , which is a continuous linear operator on A' . Then ζ_f carries $\pi_f(x)$ to $\pi'(x^*)|_{L^2(f)}$. Finally, ζ_f is isometric if $L^2(f)$ is equipped with the norm

$$\|g\|_{L^2(f)} = \sup_{\substack{x \in A \\ f(x^*x) \neq 0}} |g(x)| f(x^*x)^{-1/2}.$$

[163]

2.12.21. If A is a C^* -algebra whose only nilpotent element is 0, then A is commutative. [Let $x, y \in A$, with $x = x^*$, and let f, g be continuous functions on $\text{Sp}' x$ such that $fg = 0$. Then $(f(x)yg(x))^2 = 0$, and hence $f(x)yg(x) = 0$. Therefore $f(\pi(x))Tg(\pi(x)) = 0$ for every irreducible representation π of A and every $T \in \mathcal{L}(H_\pi)$; consequently $\pi(A)$ consists of scalars.] (I. Kaplansky, unpublished; communicated by R. V. Kadison.)

§ 3. — Spectrum of a C^* -Algebra

Our two principal aims will now be the following: 1° the decomposition of a representation into irreducible representations; 2° the structure of C^* -algebras. For both problems, it is important to generalize to arbitrary C^* -algebras the spectrum of a commutative C^* -algebra.

We already know three candidates: the sets $P(A)$, $\widehat{A}$, and $\text{Prim}(A)$. When A is commutative, these sets reduce to the set of characters of A : this is clear for $\widehat{A}$, it was noted for $P(A)$ (2.5.2), and for $\text{Prim}(A)$ it follows from the fact that the characters of a commutative C^* -algebra are in bijective correspondence with their kernels. So far only $P(A)$ has been equipped with a topology, namely the weak topology (the topology defined by the norm of the dual of A is almost never useful). Unfortunately, this space is too large (it has no tendency to be finite-dimensional when A is the C^* -algebra of a Lie group), and it is not locally compact. In this section we shall define topologies on $\widehat{A}$ and on $\text{Prim}(A)$. In favorable cases these two spaces are homeomorphic. [In unfavorable cases, $\text{Prim}(A)$ may reduce to a single point while $\widehat{A}$ is very complicated, thereby reflecting the complexity of A ; it is for this admittedly not very convincing reason that preference is given to $\widehat{A}$ over $\text{Prim}(A)$.] These spaces are locally quasi-compact. In general they are not Hausdorff, but we shall see that this defect is rather limited. Moreover, in the examples this failure of separation corresponds well to certain structural peculiarities of the algebras under consideration.

3.1. Jacobson Topology

3.1.1. Let A be an algebra over a commutative field. For every subset T of $\text{Prim}(A)$, let $I(T)$ be the intersection of the elements of T . This set is a two-sided ideal of A . Let $\overline{T}$ be the set of primitive ideals of A containing $I(T)$.

Lemma.

- (i) $\overline{\emptyset} = \emptyset$;
- (ii) $T \subset \overline{T}$ for $T \subset \text{Prim}(A)$;
- (iii) $\overline{\overline{T}} = \overline{T}$ for $T \subset \text{Prim}(A)$;
- (iv) $\overline{T_1 \cup T_2} = \overline{T_1} \cup \overline{T_2}$ for $T_1, T_2 \subset \text{Prim}(A)$.

Assertions (i) and (ii) are evident. Clearly $I(\overline{T}) = I(T)$, whence $\overline{\overline{T}} = \overline{T}$. Let $I_1 = I(T_1)$ and $I_2 = I(T_2)$. We have $I(T_1 \cup T_2) = I_1 \cap I_2$. Hence, by 2.11.4, $\overline{T_1 \cup T_2}$ is the set of primitive ideals of A containing I_1 or I_2 , which proves (iv).

There exists a unique topology on $\text{Prim}(A)$ such that, for every $T \subset \text{Prim}(A)$, $\overline{T}$ is the closure of T for this topology.

Definition. *This topology is called the Jacobson topology on $\text{Prim}(A)$.*

3.1.2. Proposition. *Let A be an algebra, and let T be a subset of $\text{Prim}(A)$. The set T is closed if and only if it is the set of primitive ideals containing some given subset of A .*

If T is closed, then $T = \overline{T}$, so T is the set of primitive ideals of A containing $I(T)$. Let $M \subset A$. If T is the set of primitive ideals of A containing M , then $I(T) \supset M$, and hence $\overline{T} \subset T$ and $\overline{T} = T$.

3.1.3. Recall that a topological space is called a T_0 -space if, for any two distinct points of the space, one of the two points has a neighborhood that does not contain the other.

Proposition. *The space $\text{Prim}(A)$ is a T_0 -space.*

Let I_1, I_2 be two distinct points of $\text{Prim}(A)$. For example, suppose $I_1 \not\subset I_2$. Then the set of $I \in \text{Prim}(A)$ which contain I_1 is a closed set T (3.1.2) such that $I_1 \in T$ and $I_2 \notin T$.

3.1.4. Proposition. *Let $I \in \text{Prim}(A)$. The singleton $\{I\}$ is closed in $\text{Prim}(A)$ if and only if I is a maximal primitive ideal.*

Indeed, the closure of $\{I\}$ consists of the primitive ideals of A containing I .

3.1.5. Let A be an algebra, and let $\widehat{A}$ be the set of equivalence classes of nonzero irreducible representations of A . The map $\pi \mapsto \ker \pi$ is a canonical surjection from $\widehat{A}$ onto $\text{Prim}(A)$.

Definition. *The spectrum of A is the set $\widehat{A}$ equipped with the inverse image topology of the Jacobson topology under the canonical map $\widehat{A} \rightarrow \text{Prim}(A)$.*

3.1.6. Proposition. *The following three conditions are equivalent:*

- (i) $\widehat{A}$ is a T_0 -space;
- (ii) two irreducible representations of A having the same kernel are equivalent;
- (iii) the canonical map $\widehat{A} \rightarrow \text{Prim}(A)$ is a homeomorphism.

(ii) $\Rightarrow$ (iii) is evident; (iii) $\Rightarrow$ (i) follows from 3.1.3; for (i) $\Rightarrow$ (ii), let π_1, π_2 be two representations with the same kernel and belonging to $\widehat{A}$. Every open subset of $\widehat{A}$ containing π_1 also contains π_2 ; hence $\pi_1 = \pi_2$ if A is a T_0 -space.

3.1.7. Lemma. *Suppose that A has an identity element. Let I be a maximal two-sided ideal of A . Then I is a primitive ideal.*

Let M be a maximal left ideal containing I . Let π be the canonical representation of A on A/M , which is irreducible. Since $IA \subset I \subset M$, the kernel I' of π contains I and does not contain 1. Therefore $I' = I$, and I is primitive.

3.1.8. Proposition. *If A has an identity element, then $\text{Prim}(A)$ and $\widehat{A}$ are quasi-compact (that is, they satisfy the Borel–Lebesgue axiom without necessarily being Hausdorff).*

It is enough to prove this for $\text{Prim}(A)$. Let (T_i) be a family of closed subsets of $\text{Prim}(A)$ with empty intersection. If $\sum I(T_i) \neq A$, the ideal $\sum I(T_i)$ is contained in a maximal two-sided ideal I ; by 3.1.7, I is primitive. But $I \in T_i$ for every i , because every T_i is closed, which is a contradiction. Therefore $\sum I(T_i) = A$. Hence there exist $x_1 \in I(T_{i_1}), \dots, x_n \in I(T_{i_n})$ such that $1 = x_1 + \dots + x_n$. This implies $I(T_{i_1}) + \dots + I(T_{i_n}) = A$, and consequently $T_{i_1} \cap \dots \cap T_{i_n} = \emptyset$.

3.1.9. Let A be an algebra and φ an automorphism of A . For every irreducible representation π of A , $\pi \circ \varphi^{-1}$ is an irreducible representation of A whose kernel is the image under φ of the kernel of π . We thus obtain bijections

$$\text{Prim}(\varphi): \text{Prim}(A) \longrightarrow \text{Prim}(A), \quad \widehat{\varphi}: \widehat{A} \longrightarrow \widehat{A},$$

which are plainly homeomorphisms. If φ_1, φ_2 are two automorphisms of A , then

$$\text{Prim}(\varphi_1 \varphi_2) = \text{Prim}(\varphi_1) \text{Prim}(\varphi_2), \quad \widehat{\varphi_1 \varphi_2} = \widehat{\varphi_1} \widehat{\varphi_2}.$$

Many authors say “dual” instead of “spectrum,” but this causes confusion with the dual of the vector space A or of the Banach space A .

Bibliography: [64].

3.2. Spectrum of an Ideal and of a Quotient Algebra

We shall restrict ourselves to the case of C^* -algebras, although the following proposition can be extended to the general case.

3.2.1. Proposition. *Let A be a C^* -algebra and I a closed two-sided ideal of A . The canonical bijections (see 2.11.2 and 2.11.5)*

$$\begin{aligned} \text{Prim}_I(A) &\longrightarrow \text{Prim}(A/I), & \widehat{A}_I &\longrightarrow \widehat{A/I}, \\ \text{Prim}^I(A) &\longrightarrow \text{Prim}(I), & \widehat{A}^I &\longrightarrow \widehat{I} \end{aligned}$$

are homeomorphisms. The sets $\text{Prim}_I(A)$ and $\widehat{A}_I$ are closed in $\text{Prim}(A)$ and $\widehat{A}$, respectively. The sets $\text{Prim}^I(A)$ and $\widehat{A}^I$ are open in $\text{Prim}(A)$ and $\widehat{A}$, respectively.

By 3.1.2, $\text{Prim}_I(A)$ is closed in $\text{Prim}(A)$, and therefore $\text{Prim}^I(A)$ is open; consequently, in $\widehat{A}$, $\widehat{A}_I$ is closed and $\widehat{A}^I$ is open. It follows immediately from the definition of the Jacobson topology that the bijection $\text{Prim}_I(A) \rightarrow \text{Prim}(A/I)$ is a homeomorphism, and hence so is the bijection $\widehat{A}_I \rightarrow \widehat{A/I}$. To prove that the bijection $\text{Prim}^I(A) \rightarrow \text{Prim}(I)$ is a homeomorphism, it is enough to prove the following: if (J_λ) is a family of elements of $\text{Prim}^I(A)$ and J is an element of $\text{Prim}^I(A)$, then

$$J \supset \bigcap_{\lambda} J_{\lambda} \iff J \cap I \supset \bigcap_{\lambda} (J_{\lambda} \cap I).$$

The forward implication is evident. Suppose that

$$J \cap I \supset \bigcap_{\lambda} (J_{\lambda} \cap I).$$

Then $J \supset (\bigcap_{\lambda} J_{\lambda}) \cap I$, while $J \not\supset I$; hence, by 2.11.4, $J \supset \bigcap_{\lambda} J_{\lambda}$, which proves the asserted equivalence. The fact that the bijection $\widehat{A}^I \rightarrow \widehat{I}$ is a homeomorphism follows at once.

3.2.2. Let A be a C^* -algebra. If I is a closed two-sided ideal of A , we shall identify $\widehat{A/I}$ with a closed subset of $\widehat{A}$, and $\widehat{I}$ with the complementary open subset, by means of Proposition 3.2.1.

Proposition. *Let A be a C^* -algebra. Then $I \mapsto \widehat{I}$ is a bijection from the set of closed two-sided ideals of A onto the set of open subsets of $\widehat{A}$. Moreover, $I_1 \subset I_2 \iff \widehat{I}_1 \subset \widehat{I}_2$.*

Let U be an open subset of $\widehat{A}$, and put $F = \widehat{A} \setminus U$. Since every primitive ideal of A is closed (2.9.7), the intersection of the kernels of the $\pi \in F$ is a closed two-sided ideal I of A . Since F is closed in $\widehat{A}$, every irreducible representation of A whose kernel contains I belongs to F . Therefore $\widehat{I} = \widehat{A} \setminus F = U$, and the map in the proposition is surjective. Let I_1, I_2 be two closed two-sided ideals of A . If $\widehat{I}_1 = \widehat{I}_2$, the primitive ideals of A containing I_1 are precisely those containing I_2 . But every closed two-sided ideal of a C^* -algebra is the intersection of the primitive ideals

containing it (2.9.7). Hence $I_1 = I_2$, and the map $I \mapsto \widehat{I}$ is injective. The assertion $I_1 \subset I_2 \iff \widehat{I}_1 \subset \widehat{I}_2$ is evident.

3.2.3. Recall that, in a C^* -algebra, the sum of two closed two-sided ideals is a closed two-sided ideal (1.8.4).

Proposition. *Let A be a C^* -algebra, and let I, J be two closed two-sided ideals of A . Then*

$$\widehat{I+J} = \widehat{I} \cup \widehat{J}, \quad \widehat{I \cap J} = \widehat{I} \cap \widehat{J}.$$

In particular, $I \cap J = 0$ if and only if $\widehat{I} \cap \widehat{J} = \emptyset$.

This follows from the preceding proposition, since $I + J$ and $I \cap J$ are the supremum and the infimum of $\{I, J\}$ in the ordered set of closed two-sided ideals of A .

3.2.4. Let A be a C^* -algebra and B the C^* -algebra obtained from A by adjoining an identity element. Then A is a closed two-sided ideal of codimension 1 in B , and is the kernel of the one-dimensional representation ω of B defined by

$$\omega(\lambda 1 + x) = \lambda \quad (\lambda \in \mathbb{C}, x \in A).$$

The only primitive ideal of B containing A is A itself. The only irreducible representation of B with kernel A is ω . Thus $\{A\}$ is closed in $\text{Prim}(B)$, and $\{\omega\}$ is closed in $\widehat{B}$. By 3.2.1, the spaces $\text{Prim}(A)$ and $\widehat{A}$ are identified with $\text{Prim}(B) \setminus \{A\}$ and $\widehat{B} \setminus \{\omega\}$, respectively.

Suppose that A has an identity element e . For every $\pi \in \widehat{B} \setminus \{\omega\}$, $\pi(e)$ is a nonzero idempotent commuting with $\pi(A)$, and hence $\pi(e) = 1$; thus $\pi(e-1) = 0$. It follows that, for every $I \in \text{Prim}(B) \setminus \{A\}$, one has $e-1 \in I$, whereas $e-1 \notin A$. Thus A is not in the closure of $\text{Prim}(B) \setminus \{A\}$ in $\text{Prim}(B)$. In other words, $\{A\}$ is not only closed but also open in $\text{Prim}(B)$; hence $\{\omega\}$ is not only closed but also open in $\widehat{B}$.

Conversely, suppose that $\{\omega\}$ is open in $\widehat{B}$. By 3.2.2 and 3.2.3, B is the direct sum of two closed two-sided ideals, one of which is A . Since B has an identity element, so does A .

Bibliography: [77].

3.3. Norm and Topology

3.3.1. Lemma. *Let A be a C^* -algebra, let a be a Hermitian element of A , and let L be a closed set of real numbers.*

- (i) *The set Z' of $\pi \in \widehat{A}$ such that $\text{Sp}' \pi(a) \subset \{0\} \cup L$ is closed in $\widehat{A}$.*
- (ii) *If A has an identity element, the set Z of $\pi \in \widehat{A}$ such that $\text{Sp} \pi(a) \subset L$ is closed in $\widehat{A}$.*

Let $\rho \in \overline{Z'}$. Suppose that $\text{Sp}' \rho(a)$ contains a point $\alpha \notin \{0\} \cup L$. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function which is nonzero at α and vanishes on $\{0\} \cup L$. Then

$$\pi(f(a)) = f(\pi(a))$$

vanishes for $\pi \in Z'$ and is nonzero for $\pi = \rho$, contradicting the assumption $\rho \in \overline{Z'}$. This proves (i). Assertion (ii) is proved in the same way.

3.3.2. Proposition. *Let A be a C^* -algebra and $x \in A$. The function $\pi \mapsto \|\pi(x)\|$ is lower semicontinuous on $\widehat{A}$.*

We are immediately reduced to the case where A has an identity element. Moreover, since $\|\pi(x)\|^2 = \|\pi(x^*x)\|$, we may suppose $x \in A^+$. Let $k \geq 0$, and let Z be the set of $\pi \in \widehat{A}$ such that $\|\pi(x)\| \leq k$. The set Z is also the set of $\pi \in \widehat{A}$ such that $\text{Sp} \pi(x) \subset [0, k]$. By 3.3.1, Z is closed, which proves the proposition.

3.3.3. Lemma. *Let A be a C^* -algebra, let (x_i) be a dense family of elements of A , and let Z_i be the set of $\pi \in \widehat{A}$ such that $\|\pi(x_i)\| > 1$. Then the Z_i form a base for the topology of $\widehat{A}$.*

Let U be an open subset of $\widehat{A}$ and let $\pi \in U$. Let $I \subset A$ be the intersection of the kernels of the representations belonging to $\widehat{A} \setminus U$. There exists $x \in I$ such that $\|\pi(x)\| = 2$. Choose an index i such that $\|x - x_i\| < 1$. Then $\|\pi(x_i)\| > 1$, whereas $\|\rho(x_i)\| < 1$ for $\rho \in \widehat{A} \setminus U$. Thus $\pi \in Z_i \subset U$. Since Z_i is open by 3.3.2, this proves the lemma.

3.3.4. Proposition. *If A is separable, the topology of $\widehat{A}$ has a countable base.*

This follows at once from 3.3.3.

3.3.5. Lemma. *Let A be a unital involutive algebra, let x be a Hermitian element of A , and let $\lambda \in \mathbb{R}$. For $\lambda \in \text{Sp}_A x$, it is necessary and sufficient that there exist a primitive ideal I of A such that $\lambda \in \text{Sp}_{A/I} \dot{x}$, where $\dot{x}$ denotes the canonical image of x in A/I .*

If $x - \lambda$ is invertible in A , its image is invertible in A/I for every primitive ideal I . Suppose that $x - \lambda$ is not invertible. Then $x - \lambda$ is not left invertible (for

a Hermitian element, left invertibility is equivalent to right invertibility), and is therefore contained in a maximal left ideal L . The kernel I of the canonical representation of A on A/L is a primitive ideal of A , and the image of $x - \lambda$ in A/I is not invertible.

3.3.6. Lemma. *Let A be a C^* -algebra. For every $x \in A$, the function $\pi \mapsto \|\pi(x)\|$ on $\widehat{A}$ attains its upper bound $\|x\|$.*

Replacing x by x^*x , we are reduced to the case $x \in A^+$. We may also suppose that A has an identity element. Then $\|x\| \in \text{Sp}_A x$. By 3.3.5, there exists $\pi \in \widehat{A}$ such that $\|x\|$ belongs to the spectrum of the operator $\pi(x)$; hence

$$\|x\| \leq \|\pi(x)\| \leq \|x\|.$$

3.3.7. Proposition. *Let A be a C^* -algebra, $x \in A$, and $\alpha > 0$. The set Z of $\pi \in \widehat{A}$ such that $\|\pi(x)\| \geq \alpha$ is quasi-compact.*

Let (Z_i) be a decreasing directed family of nonempty subsets of Z that are relatively closed in Z . For every i , let J_i be the intersection of the kernels of the elements of Z_i . The canonical image of x modulo J_i has norm at least α . The J_i form an increasing directed family; let J be the closure of their union, which is a closed two-sided ideal of A . The definition of the norm in a quotient normed space shows that the canonical image of x modulo J has norm at least α . By 3.3.6, there exists $\pi \in \widehat{A}$ whose kernel contains J and such that $\|\pi(x)\| \geq \alpha$. We have $\pi \in Z$, and π belongs to the closure of Z_i for every i ; hence $\pi \in \bigcap_i Z_i$, which proves the proposition.

3.3.8. Recall that a topological space is said to be locally quasi-compact if every point has a neighborhood base consisting of quasi-compact neighborhoods.

Corollary. *Let A be a C^* -algebra. Its spectrum $\widehat{A}$ is locally quasi-compact.*

Let $\pi \in \widehat{A}$ and let U be an open neighborhood of π in $\widehat{A}$. Since $\widehat{A} \setminus U$ is closed, there exists $x \in A$ such that $\pi(x) \neq 0$ and $\rho(x) = 0$ for $\rho \in \widehat{A} \setminus U$. Let V (respectively W) be the set of $\rho \in \widehat{A}$ such that

$$\|\rho(x)\| > \frac{1}{2}\|\pi(x)\| \quad (\text{respectively } \|\rho(x)\| \geq \frac{1}{2}\|\pi(x)\|).$$

By 3.3.2, V is an open neighborhood of π . Thus W is a neighborhood of π contained in U , and W is quasi-compact by 3.3.7.

3.3.9. Corollary. *Let A be a C^* -algebra such that $\widehat{A}$ is Hausdorff. For every $x \in A$, the function $\pi \mapsto \|\pi(x)\|$ is continuous on $\widehat{A}$.*

The set Z of 3.3.7 is then closed, and hence the function $\pi \mapsto \|\pi(x)\|$ is upper semicontinuous. Now apply 3.3.2.

Bibliography: [27, 28, 29, 77, 126].

3.4. Second Definition of the Topology on the Spectrum

3.4.1. Lemma. *Let A be a unital C^* -algebra, let A_h be the set of Hermitian elements of A , let $E(A)$ be the set of states of A , and let Q be a subset of $E(A)$. Suppose that, if $x \in A_h$ satisfies $f(x) \geq 0$ for every $f \in Q$, then $x \in A^+$. Under these conditions:*

- (i) *the weak closure of Q contains $P(A)$;*
- (ii) *the weakly closed convex hull of Q is $E(A)$.*

Let Q° be the polar of Q in A_h . If $x \in A_h$, then

$$\begin{aligned} x \in Q^\circ &\iff f(x) \leq 1 && \text{for every } f \in Q, \\ &\iff f(1 - x) \geq 0 && \text{for every } f \in Q, \\ &\iff 1 - x \in A^+, \\ &\iff f(x) \leq 1 && \text{for every } f \in E(A). \end{aligned}$$

The weakly closed convex hull of Q is plainly contained in $E(A)$; it is equal to the bipolar of Q , and hence, by what precedes, contains $E(A)$. This proves (ii), and (ii), together with B 14, implies that $\overline{Q}$ contains the set of extreme points of $E(A)$, namely $P(A)$ (2.5.6).

3.4.2. Proposition. *Let A be a C^* -algebra, and let $(\pi_i)_{i \in I}$ be a family of representations of A on spaces H_i .*

- (i) *Every state of A that vanishes on $\bigcap_i \ker \pi_i$ is a weak limit of states of the form*

$$\omega_{\xi_1} \circ \pi_{i_1} + \cdots + \omega_{\xi_n} \circ \pi_{i_n},$$

where $i_1, \dots, i_n \in I$ and $\xi_1 \in H_{i_1}, \dots, \xi_n \in H_{i_n}$.

- (ii) *Every pure state of A that vanishes on $\bigcap_i \ker \pi_i$ is a weak limit of states of the form $\omega_\xi \circ \pi_i$, where $i \in I$ and $\xi \in H_i$.*

By 2.2.6, we may suppose that every π_i is nondegenerate. Let $H = \bigoplus_i H_i$. Passing to the quotient by $\bigcap_i \ker \pi_i$, we may suppose that $\bigcap_i \ker \pi_i = 0$. The representation $\rho = \bigoplus_i \pi_i$ is faithful, and we identify A with the C^* -subalgebra $\rho(A)$ of $\mathcal{L}(H)$; thus the identity representation of A is nondegenerate. By 2.4.3, we may replace A by $A + \mathbb{C}1$. Let Q be the set of states of A of the form $\omega_\xi \circ \pi_i$, where $i \in I$, $\xi \in H_i$, and $\|\xi\| = 1$. If x is a Hermitian element of A such that

$(\omega_\xi \circ \pi_i)(x) \geq 0$ for every $i \in I$ and every unit vector $\xi \in H_i$, then $\pi_i(x) \geq 0$ for every i , and hence $x \geq 0$. It is therefore enough to apply 3.4.1.

3.4.3. Corollary. *Let A be a C^* -algebra and f a state of A . The states of A that vanish on $\ker \pi_f$ are weak limits of states of the form*

$$x \mapsto \sum_i f(y_i^* x y_i), \quad y_1, \dots, y_n \in A.$$

The pure states of A that vanish on $\ker \pi_f$ are weak limits of states of the form $x \mapsto f(y^ x y)$, where $y \in A$.*

This follows from 3.4.2 and 2.4.8(ii).

3.4.4. Theorem. *Let A be a C^* -algebra, π a representation of A , and S a set of representations of A . The following conditions are equivalent:*

- (i) *the kernel of π contains the intersection of the kernels of the elements of S ;*
- (ii) *every positive form on A associated with π is a weak limit of linear combinations of positive forms associated with S ;*
- (iii) *every state of A associated with π is a weak limit of states that are sums of positive forms associated with S .*

(iii) $\Rightarrow$ (ii) is evident. For (ii) $\Rightarrow$ (i), suppose that (ii) holds. If $a \in \bigcap_{\rho \in S} \ker \rho$, every positive form associated with S vanishes at a^*a , and hence every positive form associated with π vanishes at a^*a ; thus $\pi(a) = 0$. For (i) $\Rightarrow$ (iii), if $\ker \pi \supset \bigcap_{\rho \in S} \ker \rho$, every state of A associated with π vanishes on $\bigcap_{\rho \in S} \ker \rho$, and we apply 3.4.2(i).

3.4.5. It is clear that, if π is contained in one of the elements of S , the conditions of Theorem 3.4.4 are satisfied. The converse is false, even if S consists of a single element. We therefore make the following definition.

Definition. *If π and S satisfy the equivalent conditions (i), (ii), and (iii) of Theorem 3.4.4, we say that π is weakly contained in S .*

If S and T are two sets of representations of A , we say that T is weakly contained in S if every element of T is weakly contained in S .

3.4.6. Definition. *Let A be a C^* -algebra and π a representation of A . The support of π is the set of $\rho \in \widehat{A}$ that are weakly contained in π .*

Using condition (i) of 3.4.4, we see that this support S is a closed subset of $\widehat{A}$. Since $\ker \pi$ is the intersection of the primitive ideals containing it, π is weakly contained in S .

3.4.7. Let A be a C^* -algebra, π a representation of A , S a set of representations of A , $\tilde{A}$ the C^* -algebra obtained from A by adjoining an identity element, $\tilde{\pi}$ the canonical extension of π to $\tilde{A}$, and $\tilde{S}$ the set of canonical extensions to $\tilde{A}$ of the elements of S . If $\tilde{\pi}$ is weakly contained in $\tilde{S}$, it is clear that π is weakly contained in S [condition (i) of Theorem 3.4.4]. The converse is true if π is nondegenerate. Indeed, let $x \in A$ and $\lambda \in \mathbb{C}$, and suppose that $\rho(x) = \lambda 1$ for every $\rho \in S$. Then $\rho(xy - \lambda y) = 0$ for every $y \in A$ and every $\rho \in S$, and hence $\pi(xy - \lambda y) = 0$ for every $y \in A$. Thus $(\pi(x) - \lambda 1)\pi(y) = 0$ for every $y \in A$, and consequently $\pi(x) = \lambda 1$ because π is nondegenerate.

3.4.8. Let A be a C^* -algebra. To every set S of representations of A we associate the set $Q(S)$ of positive forms on A that are weak limits of sums of positive forms associated with S . It is clear that $Q(S)$ is a weakly closed convex cone in the dual of A . For T to be weakly contained in S , it is necessary and sufficient that $Q(T) \subset Q(S)$.

3.4.9. Proposition. *Retain the notation of Theorem 3.4.4. If π has a cyclic vector ξ , the conditions of that theorem are also equivalent to the following condition:*

(ii') *The positive form $x \mapsto f(x) = (\pi(x)\xi \mid \xi)$ on A is a weak limit of linear combinations of positive forms associated with S .*

It is clear that (ii) implies (ii'). Suppose that (ii') holds, and let us show that (ii) holds. By 2.4.8(ii), it is enough to show that, for every $y \in A$, the form

$$x \longmapsto g(x) = f(y^*xy)$$

is a weak limit of linear combinations of positive forms associated with S . But f is a weak limit of forms f_i that are linear combinations of positive forms associated with S . Thus g is a weak limit of the forms

$$x \longmapsto g_i(x) = f_i(y^*xy).$$

By 2.4.8(i), each g_i is a linear combination of positive forms associated with S .

3.4.10. Theorem. *Let A be a C^* -algebra, $\pi \in \widehat{A}$, and $S \subset \widehat{A}$. The following conditions are equivalent:*

- (i) $\pi \in \overline{S}$;
- (ii) π is weakly contained in S ;
- (iii) *one of the nonzero positive forms associated with π is a weak limit of positive forms associated with S ;*
- (iv) *every state associated with π is a weak limit of states associated with S .*

(i) $\iff$ (ii) follows immediately from condition (i) of Theorem 3.4.4; (iv) $\implies$ (iii) is evident; (iii) $\implies$ (ii) follows from 3.4.9; and (ii) $\implies$ (iv) follows from 3.4.2(ii).

3.4.11. Theorem. *Let A be a C^* -algebra. The canonical map $P(A) \rightarrow \widehat{A}$ is continuous and open.*

Let S be a subset of $\widehat{A}$, let Q be its inverse image in $P(A)$, let f be a point of $P(A)$, and let π be its image in $\widehat{A}$. For π to belong to $\overline{S}$, it is necessary and sufficient that f belong to $\overline{Q}$ (3.4.10). Hence, for S to be closed in $\widehat{A}$, it is necessary and sufficient that Q be closed in $P(A)$. This first proves that the canonical map $P(A) \rightarrow \widehat{A}$ is continuous. Let U be an open subset of $P(A)$ and let V be its image in $\widehat{A}$. We show that V is open.

Let $\rho \in V$, and let g be a point of U whose image in $\widehat{A}$ is ρ . Then g does not belong to the closure of $P(A) \setminus U$; a fortiori, g does not belong to the closure of the inverse image in $P(A)$ of $\widehat{A} \setminus V$. Thus ρ does not belong to the closure of $\widehat{A} \setminus V$, and consequently V is open. This proves that the canonical map $P(A) \rightarrow \widehat{A}$ is open.

3.4.12. The second announced definition of the topology of $\widehat{A}$ is therefore the following: it is the quotient topology of the topology of $P(A)$ under the canonical map $P(A) \rightarrow \widehat{A}$. We see that, when A is commutative, the topological space $\widehat{A}$ is indeed the usual spectrum of A .

3.4.13. Corollary. *Let A be a C^* -algebra. The space $\widehat{A}$ is a Baire space.*

Let $(V_1, V_2, \dots)$ be a decreasing sequence of open dense subsets of $\widehat{A}$. Let U_n be the inverse image of V_n in $P(A)$. By 3.4.11, U_n is an open dense subset of $P(A)$. Since $P(A)$ is a Baire space (B 14), $\bigcap_n U_n$ is dense in $P(A)$. By 3.4.11, $\bigcap_n V_n$, which is the image of $\bigcap_n U_n$ in $\widehat{A}$, is dense in $\widehat{A}$.

Bibliography: [3, 11, 27, 48, 77, 154].

3.5. Third Definition of the Topology on the Spectrum

3.5.1. For every cardinal n , we shall call the Hilbert space of families (ξ_i) of complex numbers such that

$$\sum_i |\xi_i|^2 < +\infty,$$

indexed by a set of cardinality n (for example, the least ordinal segment of cardinality n), the *model Hilbert space of dimension n* . This choice will be immaterial in what follows; it is intended only to fix ideas.

Let n be a cardinal, let H_n be the model Hilbert space of dimension n , let A be a C^* -algebra, let $\text{Rep}_n(A)$ be the set of representations of A on H_n , and let $\text{Irr}_n(A)$ be the set of nonzero irreducible representations of A on H_n . There is an evident canonical map from $\text{Irr}_n(A)$ into $\widehat{A}$, which assigns to each element of $\text{Irr}_n(A)$ its class. The canonical image of $\text{Irr}_n(A)$ in $\widehat{A}$ is denoted $\widehat{A}_n$: it is the set of classes of nonzero irreducible representations of dimension n .

3.5.2. We equip $\text{Rep}_n(A)$ with the topology of pointwise weak convergence on A . Thus $\pi_\lambda \rightarrow \pi$, where $\pi_\lambda, \pi \in \text{Rep}_n(A)$, means that

$$(\pi_\lambda(a)\xi \mid \eta) \longrightarrow (\pi(a)\xi \mid \eta)$$

for all $a \in A$ and $\xi, \eta \in H_n$. This topology is identical with the topology of pointwise strong convergence. Indeed, if $\pi_\lambda \rightarrow \pi$ in the preceding sense, then, for every $a \in A$ and every $\xi \in H_n$,

$$\begin{aligned} \|\pi_\lambda(a)\xi - \pi(a)\xi\|^2 &= (\pi_\lambda(a)\xi \mid \pi_\lambda(a)\xi) - 2\text{Re}(\pi_\lambda(a)\xi \mid \pi(a)\xi) + (\pi(a)\xi \mid \pi(a)\xi) \\ &= (\pi_\lambda(a^*a)\xi \mid \xi) - 2\text{Re}(\pi_\lambda(a)\xi \mid \pi(a)\xi) + (\pi(a^*a)\xi \mid \xi) \\ &\longrightarrow (\pi(a^*a)\xi \mid \xi) - 2\text{Re}(\pi(a)\xi \mid \pi(a)\xi) + (\pi(a^*a)\xi \mid \xi) = 0. \end{aligned}$$

3.5.3. Let $\pi_\lambda, \pi \in \text{Rep}_n(A)$, and suppose that π has a cyclic vector ξ . In order that $\pi_\lambda \rightarrow \pi$, it is enough that $\pi_\lambda(a)\xi \rightarrow \pi(a)\xi$ for every $a \in A$. Indeed, in that case, for all $a, b \in A$,

$$\begin{aligned} \|\pi_\lambda(a)\pi(b)\xi - \pi(a)\pi(b)\xi\| &\leq \|\pi_\lambda(a)\pi(b)\xi - \pi_\lambda(a)\pi_\lambda(b)\xi\| \\ &\quad + \|\pi_\lambda(a)\pi_\lambda(b)\xi - \pi(a)\pi(b)\xi\| \\ &\leq \|a\| \|\pi(b)\xi - \pi_\lambda(b)\xi\| + \|\pi_\lambda(ab)\xi - \pi(ab)\xi\| \longrightarrow 0. \end{aligned}$$

Since the $\pi(b)\xi$ are dense in H_n and $\|\pi_\lambda(a)\| \leq \|a\|$, it follows that $\pi_\lambda(a)\eta \rightarrow \pi(a)\eta$ for every $\eta \in H_n$.

3.5.4. Let $\pi_\lambda, \pi \in \text{Rep}_n(A)$, and let A' be a dense subset of A . In order that $\pi_\lambda \rightarrow \pi$, it is enough that $(\pi_\lambda(a)\xi \mid \eta) \rightarrow (\pi(a)\xi \mid \eta)$ for all $a \in A'$ and $\xi, \eta \in H_n$; this follows immediately from

$$\|\pi_\lambda(a) - \pi_\lambda(a')\| \leq \|a - a'\|.$$

3.5.5. In $\text{Rep}_n(A)$, equivalence of representations is an open equivalence relation. Indeed, let G be the group of unitary operators on H_n . For $U \in G$ and $\pi \in \text{Rep}_n(A)$, let $U\pi \in \text{Rep}_n(A)$ be the representation defined by

$$(U\pi)(x) = U\pi(x)U^{-1} \quad (x \in A).$$

It is immediately seen that, for each U , the map $\pi \mapsto U\pi$ is a homeomorphism of $\text{Rep}_n(A)$ onto itself. Hence the saturation of an open subset Ω of $\text{Rep}_n(A)$, being the union of the $U\Omega$ for $U \in G$, is open in $\text{Rep}_n(A)$.

3.5.6. Lemma. *Let H be a Hilbert space, let $\xi_1, \dots, \xi_n$ be vectors of H , and let $\varepsilon > 0$. There exists $\varepsilon' > 0$ with the following property: if $\eta_1, \dots, \eta_n \in H$ satisfy*

$$|(\eta_i | \eta_j) - (\xi_i | \xi_j)| \leq \varepsilon' \quad \text{for all } i, j,$$

then there exists a unitary operator U on H such that $\|U\eta_i - \xi_i\| \leq \varepsilon$ for every i .

It is enough to prove the following. Let $(\eta_1^p, \eta_2^p, \dots, \eta_n^p)$, $p = 1, 2, \dots$, be a sequence of systems of vectors such that

$$(\eta_i^p | \eta_j^p) \longrightarrow (\xi_i | \xi_j) \quad \text{for all } i, j.$$

Then, denoting by G the unitary group of H , we have

$$\inf_{U \in G} (\|U\eta_1^p - \xi_1\| + \dots + \|U\eta_n^p - \xi_n\|) \longrightarrow 0 \quad (p \rightarrow +\infty).$$

This is evident for $n = 0$. Suppose it has been proved when one starts with $n - 1$ vectors.

Let H' be the vector subspace of H spanned by $\xi_1, \dots, \xi_{n-1}$, and let $H'' = H \ominus H'$. By the induction hypothesis, there exist $U_1, U_2, \dots \in G$ such that

$$\|U_p \eta_1^p - \xi_1\| + \dots + \|U_p \eta_{n-1}^p - \xi_{n-1}\| \longrightarrow 0.$$

Put

$$\xi'_n = P_{H'} \xi_n, \quad \xi''_n = P_{H''} \xi_n, \quad \eta_n'^p = P_{H'} U_p \eta_n^p, \quad \eta_n''^p = P_{H''} U_p \eta_n^p.$$

For $1 \leq i \leq n - 1$, we have

$$\begin{aligned} |((\xi'_n - \eta_n'^p) | \xi_i)| &= |((\xi_n - U_p \eta_n^p) | \xi_i)| \\ &\leq |(\xi_n | \xi_i) - (\eta_n^p | \eta_i^p)| + |(U_p \eta_n^p | U_p \eta_i^p) - (U_p \eta_n^p | \xi_i)| \\ &\leq |(\xi_n | \xi_i) - (\eta_n^p | \eta_i^p)| + \|\eta_n^p\| \|U_p \eta_i^p - \xi_i\| \longrightarrow 0. \end{aligned}$$

Therefore $\eta_n'^p \rightarrow \xi_n'$ in the finite-dimensional space H' . At the same time,

$$\|U_p \eta_n^p\|^2 = \|\eta_n^p\|^2 \longrightarrow \|\xi_n\|^2, \quad \|\eta_n''^p\|^2 \longrightarrow \|\xi_n''\|^2.$$

There is therefore a unitary operator V_p on H , acting as the identity on H' , such that $V_p \eta_n''^p \rightarrow \xi_n''$. For $1 \leq i \leq n-1$,

$$\|P_{H''} U_p \eta_i^p\| \longrightarrow \|P_{H''} \xi_i\| = 0, \quad \|V_p U_p \eta_i^p - U_p \eta_i^p\| \longrightarrow 0.$$

Consequently

$$\|V_p U_p \eta_1^p - \xi_1\| + \cdots + \|V_p U_p \eta_{n-1}^p - \xi_{n-1}\| \longrightarrow 0.$$

Moreover,

$$\|V_p U_p \eta_n^p - \xi_n\|^2 = \|\eta_n'^p - \xi_n'\|^2 + \|V_p \eta_n''^p - \xi_n''\|^2 \longrightarrow 0.$$

Thus

$$\inf_{U \in G} \sum_{i=1}^n \|U \eta_i^p - \xi_i\| \longrightarrow 0 \quad (p \rightarrow +\infty).$$

3.5.7. Lemma. *Let H be a Hilbert space, A a C^* -algebra, π_0 a nonzero irreducible representation of A on a closed vector subspace H_0 of H , and S the set of elements of $\widehat{A}$ whose dimension is at most $\dim H$. Let $a_1, \dots, a_p \in A$, let $\xi_1, \dots, \xi_n \in H_0$, let $\varepsilon > 0$, and let $\tau_0 \in S$ be the class of π_0 . There exists a neighborhood V of τ_0 in S with the following property: if $\tau \in V$, there exists a representation π of A on a closed vector subspace H_π of H such that:*

- 1° *the class of π is τ ;*
- 2° *$\|\pi(a_i)P_{H_\pi} \xi_j - \pi_0(a_i)\xi_j\| \leq \varepsilon$ for all i, j ;*
- 3° *if $\dim \tau = \dim H$, then $H_\pi = H$.*

We are immediately reduced to the case in which A has an identity element, $a_1 = 1$, $\|a_i\| \leq 1$, and $\xi_1 \neq 0$.

1° Suppose first that $n = 1$. Let $\varepsilon_1 > 0$. There exists a neighborhood W of τ_0 in $\widehat{A}$ with the following property: if $\tau \in W$, there exist a representation ρ of A of class τ and a vector $\xi \in H_\rho$ such that

$$|(\rho(a_j^* a_i) \xi \mid \xi) - (\pi_0(a_j^* a_i) \xi_1 \mid \xi_1)| \leq \varepsilon_1 \quad \text{for all } i, j. \quad (1)$$

This follows from Theorem 3.4.11. If $\tau \in V = W \cap S$, we may suppose that H_ρ is a closed vector subspace of H , equal to H if $\dim \tau = \dim H$. For $x \in A$, extend

$\rho(x)$ to an operator $\rho'(x) \in \mathcal{L}(H)$ which is zero on $H \ominus H_\rho$. Inequality (1) may be written

$$|(\rho(a_i)\xi \mid \rho(a_j)\xi) - (\pi_0(a_i)\xi_1 \mid \pi_0(a_j)\xi_1)| \leq \varepsilon_1.$$

If ε_1 has been suitably chosen, 3.5.6 gives a unitary operator U on H such that

$$\|U\rho(a_i)\xi - \pi_0(a_i)\xi_1\| \leq \frac{\varepsilon}{2} \quad \text{for every } i.$$

In particular, taking $i = 1$, we obtain $\|U\xi - \xi_1\| \leq \varepsilon/2$. Therefore

$$\begin{aligned} \|U\rho'(a_i)U^{-1}\xi_1 - \pi_0(a_i)\xi_1\| &\leq \|U\rho'(a_i)U^{-1}\xi_1 - U\rho(a_i)\xi\| \\ &\quad + \|U\rho(a_i)\xi - \pi_0(a_i)\xi_1\| \\ &\leq \|a_i\| \|\xi_1 - U\xi\| + \frac{\varepsilon}{2} \leq \varepsilon. \end{aligned}$$

We may therefore take for π the representation $x \mapsto U\rho'(x)U^{-1}$, acting on $U(H_\rho)$.

2° Now let n be arbitrary. There exist $b_1, \dots, b_n \in A$ such that

$$\|\xi_j - \pi_0(b_j)\xi_1\| \leq \frac{\varepsilon}{4}.$$

By 1°, there exists a neighborhood V of τ_0 in S with the following property: if $\tau \in V$, there exists a representation π of A , of class τ , on a closed vector subspace of H (equal to H if $\dim \tau = \dim H$), such that

$$\begin{aligned} \|\pi(a_i b_j)P_{H_\pi}\xi_1 - \pi_0(a_i b_j)\xi_1\| &\leq \frac{\varepsilon}{4}, \\ \|\pi(b_j)P_{H_\pi}\xi_1 - \pi_0(b_j)\xi_1\| &\leq \frac{\varepsilon}{4} \end{aligned} \quad \text{for all } i, j.$$

Then

$$\begin{aligned} &\|\pi(a_i)P_{H_\pi}\xi_j - \pi_0(a_i)\xi_j\| \\ &\leq \|\pi(a_i)P_{H_\pi}(\xi_j - \pi_0(b_j)\xi_1)\| \\ &\quad + \|\pi(a_i)P_{H_\pi}\pi_0(b_j)\xi_1 - \pi(a_i)\pi(b_j)P_{H_\pi}\xi_1\| \\ &\quad + \|\pi(a_i b_j)P_{H_\pi}\xi_1 - \pi_0(a_i b_j)\xi_1\| \\ &\quad + \|\pi_0(a_i)\pi_0(b_j)\xi_1 - \pi_0(a_i)\xi_j\| \\ &\leq \frac{\varepsilon}{4}\|a_i\| + \frac{\varepsilon}{4}\|a_i\| + \frac{\varepsilon}{4} + \frac{\varepsilon}{4}\|a_i\| \leq \varepsilon. \end{aligned}$$

3.5.8. Theorem. *Let A be a C^* -algebra and n a cardinal. The canonical map from $\text{Irr}_n(A)$ onto $\widehat{A}_n$ is continuous and open.*

Let Q be a subset of $\text{Irr}_n(A)$, let $\pi_0 \in \text{Irr}_n(A)$, and let S and τ_0 be their images in $\widehat{A}_n$. If π_0 belongs to the closure of Q , every positive form associated with π_0 is a weak limit of positive forms associated with Q , and hence τ_0 belongs to the closure of S . This proves that the map $\text{Irr}_n(A) \rightarrow \widehat{A}_n$ is continuous. Moreover, every neighborhood of π_0 in $\text{Irr}_n(A)$ contains a neighborhood constructed as follows: choose $a_1, \dots, a_p \in A$, $\xi_1, \dots, \xi_m \in H_n$, and $\varepsilon > 0$, and take the set of $\pi \in \text{Irr}_n(A)$ such that

$$\|\pi(a_i)\xi_j - \pi_0(a_i)\xi_j\| \leq \varepsilon \quad \text{for all } i, j.$$

By 3.5.7, the image of such a neighborhood in $\widehat{A}_n$ is a neighborhood of τ_0 in $\widehat{A}_n$. This proves that the map $\text{Irr}_n(A) \rightarrow \widehat{A}_n$ is open.

Thus the topology of $\widehat{A}_n$ is the quotient topology of the topology of $\text{Irr}_n(A)$ under the canonical map $\text{Irr}_n(A) \rightarrow \widehat{A}_n$.

3.5.9. Proposition. *Let A be a C^* -algebra and $x \in A^+$. For every $\pi \in \widehat{A}$, $\text{Tr } \pi(x)$ is a nonnegative number, finite or not. The function $\pi \mapsto \text{Tr } \pi(x)$ is lower semicontinuous on $\widehat{A}$.*

Let H be a Hilbert space such that every irreducible representation of A can be realized on a closed vector subspace of H (2.3.3). Let $\pi_0 \in \widehat{A}$ be realized on a closed vector subspace H_0 of H . Let $\alpha < \text{Tr } \pi_0(x)$, and choose an orthonormal system $(\xi_1, \dots, \xi_n)$ in H_0 such that

$$\sum_{j=1}^n (\pi_0(x)\xi_j \mid \xi_j) - \alpha = \beta > 0.$$

By 3.5.7, there exists a neighborhood V of π_0 in $\widehat{A}$ such that every $\pi \in V$ can be realized on a closed vector subspace of H in such a way that

$$\|\pi(x)P_{H_\pi}\xi_j - \pi_0(x)\xi_j\| \leq \frac{\beta}{n} \quad \text{for every } j.$$

Then

$$\text{Tr } \pi(x) \geq \sum_{j=1}^n (\pi(x)P_{H_\pi}\xi_j \mid \xi_j) \geq \sum_{j=1}^n (\pi_0(x)\xi_j \mid \xi_j) - n \frac{\beta}{n} = \alpha.$$

Bibliography: [14, 27, 28].

3.6. Finite-Dimensional Representations

3.6.1. Let k be a commutative field, let n be an integer > 0 , and let $M_n(k)$ be the algebra of n -by- n matrices over k . This algebra has dimension n^2 . Hence, if $r > n^2$, then

$$\sum_{\sigma \in \mathfrak{S}_r} \varepsilon_\sigma X_{\sigma(1)} X_{\sigma(2)} \cdots X_{\sigma(r)} = 0 \quad (1)$$

for all $X_1, \dots, X_r \in M_n(k)$, where $\mathfrak{S}_r$ denotes the symmetric group of $\{1, 2, \dots, r\}$ and ε_σ is the sign of the permutation σ . Let $r(n)$ be the least integer r for which identity (1) holds in $M_n(k)$.

3.6.2. Lemma. *One has $r(n) \geq r(n-1) + 2$.*

Let $t = r(n-1) - 1$. Choose $X_1, \dots, X_t \in M_{n-1}(k)$ such that

$$Y = \sum_{\sigma \in \mathfrak{S}_t} \varepsilon_\sigma X_{\sigma(1)} \cdots X_{\sigma(t)} \neq 0.$$

Write $Y = (y_{ij})$; there exist h, l such that $y_{hl} \neq 0$. Let X'_i (respectively Y') be the matrix obtained from X_i (respectively Y) by adjoining an n th row and an n th column of zeros. Let X'_{t+1} (respectively X'_{t+2}) be the matrix in $M_n(k)$ all of whose entries are zero except the entry in the l th row and n th column (respectively the entry in the n th row and n th column), which is equal to 1. In the sum

$$\sum_{\tau \in \mathfrak{S}_{t+2}} \varepsilon_\tau X'_{\tau(1)} X'_{\tau(2)} \cdots X'_{\tau(t+2)}, \quad (1)$$

all terms in which X'_{t+2} is not in the last position vanish (because $X'_{t+2} X'_j = 0$ for $j < t+2$); among the terms ending in X'_{t+2} , those in which X'_{t+2} is not preceded by X'_{t+1} vanish (because $X'_j X'_{t+2} = 0$ for $j < t+1$). The sum (1) is therefore equal to

$$Y' X'_{t+1} X'_{t+2} = Y' X'_{t+1},$$

and this matrix is nonzero because $y_{hl} \neq 0$. Hence $r(n) > t + 2 = r(n-1) + 1$.

3.6.3. Proposition. *Let A be a C^* -algebra and n an integer > 0 . Let ${}_n\widehat{A}$ be the set of $\pi \in \widehat{A}$ such that $\dim \pi \leq n$, and let $\widehat{A}_n$ be the set of $\pi \in \widehat{A}$ such that $\dim \pi = n$.*

- (i) ${}_n\widehat{A}$ is closed in $\widehat{A}$, and $\widehat{A}_n$ is open in ${}_n\widehat{A}$.
- (ii) Let I_n be the intersection of the kernels of the $\pi \in {}_n\widehat{A}$. Then ${}_n\widehat{A}$ is canonically homeomorphic to the spectrum of A/I_n , and $\widehat{A}_n$ is canonically homeomorphic to the spectrum of I_{n-1}/I_n .

Let $l = r(n)$ and $x_1, \dots, x_l \in A$. The element

$$\sum_{\sigma \in \mathfrak{S}_l} \varepsilon_\sigma x_{\sigma(1)} \cdots x_{\sigma(l)}$$

belongs to the kernel of every representation in ${}_n\widehat{A}$, and hence to I_n . Let $\pi \in \widehat{A}$ satisfy $\ker \pi \supset I_n$. Then

$$\sum_{\sigma \in \mathfrak{S}_l} \varepsilon_\sigma \pi(x_{\sigma(1)}) \cdots \pi(x_{\sigma(l)}) = \pi \left(\sum_{\sigma \in \mathfrak{S}_l} \varepsilon_\sigma x_{\sigma(1)} \cdots x_{\sigma(l)} \right) = 0$$

for all $x_1, \dots, x_l \in A$. Since $\pi(A)$ is strongly dense in $\mathcal{L}(H_\pi)$, it follows that

$$\sum_{\sigma \in \mathfrak{S}_l} \varepsilon_\sigma u_{\sigma(1)} \cdots u_{\sigma(l)} = 0$$

for all $u_1, \dots, u_l \in \mathcal{L}(H_\pi)$. By 3.6.2, H_π contains no finite-dimensional vector subspace of dimension $> n$; hence $\dim H_\pi \leq n$, and $\pi \in {}_n\widehat{A}$. Thus ${}_n\widehat{A}$ is closed in $\widehat{A}$. Consequently $\widehat{A}_n = {}_n\widehat{A} \setminus {}_{n-1}\widehat{A}$ is open in ${}_n\widehat{A}$. Assertion (ii) follows from (i) and 3.2.1.

3.6.4. Proposition.

- (i) *The topology of $\widehat{A}_n$ is locally compact.*
- (ii) *This topology is the coarsest topology making all the functions $\pi \mapsto \operatorname{Tr} \pi(x)$ on $\widehat{A}_n$, $x \in A$, continuous.*

Let H_n be the model Hilbert space of dimension n . For every $x \in A$, the map $\pi \mapsto \pi(x)$ is continuous on $\operatorname{Irr}_n(A)$; since $n < +\infty$, it follows that $\pi \mapsto \operatorname{Tr} \pi(x)$ is continuous on $\operatorname{Irr}_n(A)$. Since it takes the same value on equivalent representations, it induces a continuous function φ_x on $\widehat{A}_n$ (3.5.8). Moreover, if π and π' are two inequivalent irreducible representations of A on H_n , there exists $x \in A$ such that $\operatorname{Tr} \pi(x) \neq \operatorname{Tr} \pi'(x)$ (2.8.3 or B 27). Thus the functions φ_x separate the points of $\widehat{A}_n$. This first proves that the topology of $\widehat{A}_n$ is Hausdorff. On the other hand, $\widehat{A}_n$ is identified with the spectrum of a C^* -algebra [3.6.3(ii)], and is therefore locally compact (3.3.8). Let $Z = \widehat{A}_n \cup \{0\}$ be the Alexandroff compactification of $\widehat{A}_n$. By 3.3.7, the functions φ_x extend to continuous functions ψ_x on Z that take the value 0 at the point 0. The functions ψ_x separate the points of Z ; indeed, by what precedes, it is enough to prove that, for every $\pi \in \widehat{A}_n$, there exists $x \in A$ such that $\psi_x(\pi) \neq 0$. But if $y \in A$ satisfies $\pi(y) \neq 0$, then

$$\psi_{y^*y}(\pi) = \operatorname{Tr} \pi(y^*y) = \operatorname{Tr} (\pi(y)^* \pi(y)) > 0.$$

Thus the topology on Z defined by the functions ψ_x is Hausdorff and coarser than the topology of Z , and hence is identical with it. This proves (ii).

Bibliography: [27, 75, 77].

3.7. Complements on the Spaces $\text{Rep}_n(A)$

3.7.1. Proposition. *Let A be a separable C^* -algebra and n a cardinal $\leq \aleph_0$. The topological space $\text{Rep}_n(A)$ is Polish.*

The model Hilbert space H_n of dimension n is separable. Let $\mathcal{L}_s(H_n)$ be $\mathcal{L}(H_n)$ equipped with the strong topology. It is quasi-complete (B 11). Let $\mathcal{L}_s(A, \mathcal{L}_s(H_n))$ be the space of continuous linear maps from A into $\mathcal{L}_s(H_n)$, equipped with the topology of pointwise convergence. Every closed equicontinuous subset of $\mathcal{L}_s(A, \mathcal{L}_s(H_n))$ is a complete uniform subspace of $\mathcal{L}_s(A, \mathcal{L}_s(H_n))$ (B 12). In particular, the set of linear maps π from A into $\mathcal{L}_s(H_n)$ such that $\|\pi(x)\| \leq \|x\|$ for every $x \in A$ is a complete uniform subspace B of $\mathcal{L}_s(A, \mathcal{L}_s(H_n))$.

Let $(x_i)_{i \in I}$ be a dense sequence in A such that $\|x_i\| \leq 1$ for every i . Let $(\xi_j)_{j \in J}$ be a dense sequence in H_n such that $\|\xi_j\| \leq 1$ for every j . The uniform structure of B has as a fundamental system of entourages the entourages defined by the inequalities

$$\|\pi(x_i)\xi_j - \pi'(x_i)\xi_j\| \leq \frac{1}{k} \quad (i \in I, j \in J, k = 1, 2, \dots).$$

This uniform structure is therefore metrizable. The map

$$\pi \longmapsto (\pi(x_i)\xi_j)_{i \in I, j \in J}$$

from B onto a subspace of $H_n^{I \times J}$ is bicontinuous, and hence B is separable. In short, B , equipped with the topology of pointwise strong convergence, is a Polish space. Finally, $\text{Rep}_n(A)$ is the set of $\pi \in B$ such that

$$\pi(xy) = \pi(x)\pi(y), \quad \pi(x^*) = \pi(x)^* \quad (x, y \in A).$$

The second condition is equivalent to

$$(\pi(x^*)\xi \mid \eta) = (\xi \mid \pi(x)\eta) \quad (x \in A, \xi, \eta \in H_n).$$

Thus $\text{Rep}_n(A)$ is a closed subspace of B , and hence is a Polish space.

3.7.2. Lemma. *Let E, F be two complex Banach spaces with F separable, let M be a topological space, and let $m \mapsto U_m$ be a continuous map from M into $\mathcal{L}_s(E, F)$ (the*

space of continuous linear maps from E into F , equipped with the topology of pointwise convergence). Let n be an integer. The set of $m \in M$ such that $\text{codim } \overline{U_m(E)} \leq n$ is a G_δ subset of M .

Let $f_1, f_2, \dots, f_k$ be elements of F . Let R be the set of sequences $r = (r_1, r_2, \dots, r_k)$ of complex numbers such that

$$\frac{1}{2} \leq |r_1|^2 + \dots + |r_k|^2 \leq 1.$$

For every $r \in R$, every $e \in E$, and every integer $p > 0$, the set $M(r, e, p)$ of $m \in M$ such that

$$\|r_1 f_1 + \dots + r_k f_k - U_m(e)\| < \frac{1}{p}$$

is open in M . Now $f_1, \dots, f_k$ are linearly dependent modulo $\overline{U_m(E)}$ if and only if, for every p , there exist $r \in R$ and $e \in E$ such that

$$\|r_1 f_1 + \dots + r_k f_k - U_m(e)\| < \frac{1}{p};$$

in other words, if and only if

$$m \in \bigcap_{p=1}^{\infty} \bigcup_{\substack{r \in R \\ e \in E}} M(r, e, p).$$

[Indeed, if this is so, we may suppose that, as $p \rightarrow +\infty$, $(r_1, \dots, r_k)$ tends to a limit. Then $U_m(e)$ has a limit which belongs to $\overline{U_m(E)}$ and is a linear combination, with coefficients not all zero, of $f_1, \dots, f_k$. Conversely, if $f_1, \dots, f_k$ are linearly dependent modulo $\overline{U_m(E)}$, then for every p there exist $r = (r_1, \dots, r_k) \in R$ and $e \in E$ such that

$$\|r_1 f_1 + \dots + r_k f_k - U_m(e)\| < \frac{1}{p},$$

which proves the assertion.]

Thus the set of points $m \in M$ for which $f_1, \dots, f_k$ are linearly dependent modulo $\overline{U_m(E)}$ is a G_δ subset of M . Let $(g_1, g_2, \dots)$ be a dense sequence in F . We have $\text{codim } \overline{U_m(E)} \leq n$ if and only if $g_{i_1}, g_{i_2}, \dots, g_{i_{n+1}}$ are linearly dependent modulo $\overline{U_m(E)}$ for all indices $i_1, \dots, i_{n+1}$. Since the family of systems $(i_1, \dots, i_{n+1})$ is countable, the set in Lemma 3.7.2 appears as a countable intersection of G_δ sets, and hence is a G_δ set.

3.7.3. Lemma. *Let A be a C^* -algebra, let n be a cardinal $\leq \aleph_0$, let M be a topological space, let $m \mapsto \pi_m$ and $m \mapsto \pi'_m$ be two continuous maps from M into $\text{Rep}_n(A)$, and let p be an integer ≥ 0 . The set of $m \in M$ for which the intertwining number of π_m and π'_m is at most p is a G_δ subset of M .*

Let H_n be the model Hilbert space of dimension n , let K be the unit ball of H_n , and let B be the unit ball of A . Denote by Y the Banach space of families $(\mu_{ijk})_{i \in B, j \in K, k \in K}$ of complex numbers such that

$$\|(\mu_{ijk})\| = \sum_{\substack{i \in B \\ j \in K, k \in K}} |\mu_{ijk}| < +\infty.$$

Its dual is the Banach space X of families $(\lambda_{ijk})_{i \in B, j \in K, k \in K}$ of complex numbers such that

$$\|(\lambda_{ijk})\| = \sup_{i, j, k} |\lambda_{ijk}| < +\infty.$$

We denote by P the predual of the von Neumann algebra $\mathcal{L}(H_n)$ (A 23). It is separable.

For every $m \in M$, let V_m be the continuous linear map from $\mathcal{L}(H_n)$ into X defined by

$$V_m(T) = (((\pi_m(i)T - T\pi'_m(i))j \mid k))_{i \in B, j \in K, k \in K}.$$

The intertwining number of π_m and π'_m is then the dimension of $\ker V_m$. For $T \in \mathcal{L}(H_n)$ and $(\mu_{ijk}) \in Y$, writing x_i, ξ_j in place of i, j , we have

$$\begin{aligned} \langle (\mu_{ijk}), V_m(T) \rangle &= \sum_{ijk} \mu_{ijk} ((\pi_m(x_i)T - T\pi'_m(x_i))\xi_j \mid \xi_k) \\ &= \sum_{ijk} \mu_{ijk} ((T\xi_j \mid \pi_m(x_i^*)\xi_k) - (T\pi'_m(x_i)\xi_j \mid \xi_k)). \end{aligned}$$

Given two vectors $\eta, \zeta \in H_n$, denote by $\varphi(\eta, \zeta)$ the element of P defined by

$$\langle \varphi(\eta, \zeta), T \rangle = (T\eta \mid \zeta); \quad \|\varphi(\eta, \zeta)\| = \|\eta\| \|\zeta\|.$$

Then

$$\langle (\mu_{ijk}), V_m(T) \rangle = \left\langle \sum_{ijk} \mu_{ijk} (\varphi(\xi_j, \pi_m(x_i^*)\xi_k) - \varphi(\pi'_m(x_i)\xi_j, \xi_k)), T \right\rangle.$$

Thus V_m is the transpose of a continuous linear map U_m from Y into P , defined by

$$U_m((\mu_{ijk})) = \sum_{ijk} \mu_{ijk} (\varphi(\xi_j, \pi_m(x_i^*)\xi_k) - \varphi(\pi'_m(x_i)\xi_j, \xi_k)), \quad (1)$$

and the intertwining number of π_m and π'_m is $\text{codim } \overline{U_m(Y)}$. By 3.7.2, to prove 3.7.3 it is enough to prove that the map $m \mapsto U_m$ from M into $\mathcal{L}_s(Y, P)$ is continuous. For fixed $(\mu_{ijk}) \in Y$, we therefore show that the element (1) of P depends continuously on m in the norm of P . Since

$$\|\mu_{ijk}(\varphi(\xi_j, \pi_m(x_i^*)\xi_k) - \varphi(\pi'_m(x_i)\xi_j, \xi_k))\| \leq 2|\mu_{ijk}|,$$

it is enough to prove that each term $\varphi(\xi_j, \pi_m(x_i^*)\xi_k)$ and $\varphi(\pi'_m(x_i)\xi_j, \xi_k)$ depends continuously on m . But the maps $m \mapsto \pi_m(x_i^*)\xi_k$ and $m \mapsto \pi'_m(x_i)\xi_j$ from M into H_n are continuous by the definition of the topology of $\text{Rep}_n(A)$, and the map $(\eta, \zeta) \mapsto \varphi(\eta, \zeta)$ from $H_n \times H_n$ into P is continuous by the equality $\|\varphi(\eta, \zeta)\| = \|\eta\| \|\zeta\|$.

3.7.4. Proposition. *Let A be a separable C^* -algebra and n a cardinal $\leq \aleph_0$. The topological space $\text{Irr}_n(A)$ is Polish.*

Apply 3.7.3 with $M = \text{Rep}_n(A)$, $\pi_\rho = \pi'_\rho = \rho$ for every $\rho \in \text{Rep}_n(A)$, and $p = 1$. If $n > 0$, the set thus obtained in $M = \text{Rep}_n(A)$ is $\text{Irr}_n(A)$, possibly with the zero representation added when $n = 1$. Thus $\text{Irr}_n(A)$ is a G_δ subset of $\text{Rep}_n(A)$, which is Polish (3.7.1). Hence $\text{Irr}_n(A)$ is Polish (B 15).

3.7.5. Apply 3.7.3 with $M = \text{Rep}_n(A)$, $\pi_\rho = \rho$, π'_ρ the zero representation of A on H_n , and $p = 0$. We obtain the set of nondegenerate representations of A on H_n . If A is a separable C^* -algebra, this set is a G_δ subset of $\text{Rep}_n(A)$, and hence is a Polish space.

Bibliography: [12, 89].

3.8. Mackey Borel Structure

3.8.1. Let A be a separable C^* -algebra. For every cardinal $n \leq \aleph_0$, we equip $\text{Rep}_n(A)$ and $\text{Irr}_n(A)$ with the Borel structures (B 19) underlying their topologies. These are the coarsest Borel structures for which the functions

$$\pi \longmapsto (\pi(x)\xi \mid \eta) \quad (x \in A, \xi, \eta \in H_n)$$

are Borel. Let $\text{Rep}(A)$ (respectively $\text{Irr}(A)$) be the union of the $\text{Rep}_n(A)$ (respectively $\text{Irr}_n(A)$), for $n = 1, 2, \dots, \aleph_0$, equipped with the Borel sum structure of the structures on the $\text{Rep}_n(A)$ (respectively $\text{Irr}_n(A)$). The Borel space $\text{Irr}(A)$ is a Borel subset of $\text{Rep}(A)$ by the proof of 3.7.4; and the structure induced on $\text{Irr}(A)$ by the

Borel structure of $\text{Rep}(A)$ is the Borel structure of $\text{Irr}(A)$. By 3.7.1 and 3.7.4, the Borel spaces $\text{Rep}(A)$ and $\text{Irr}(A)$ are standard (B 20).

3.8.2. Since A is separable, the canonical map $\text{Irr}(A) \rightarrow \widehat{A}$ that assigns to every element of $\text{Irr}(A)$ its class is surjective (2.3.3).

Definition. Let A be a separable C^* -algebra. The Mackey Borel structure on $\widehat{A}$ is the quotient structure of the Borel structure of $\text{Irr}(A)$ under the canonical map $\text{Irr}(A) \rightarrow \widehat{A}$.

This Borel structure is not standard in general (see, however, 4.6.1).

3.8.3. Proposition. Let A be a separable C^* -algebra. The Mackey Borel structure on $\widehat{A}$ is finer than the Borel structure underlying the topology of $\widehat{A}$.

Let Ω be an open subset of $\widehat{A}$. For $n = 1, 2, \dots, \aleph_0$, the inverse image of Ω in $\text{Irr}_n(A)$ is open (3.5.8), and hence Borel. Thus the inverse image of Ω in $\text{Irr}(A)$ is Borel, so Ω is Borel for the Mackey structure. It follows that every subset of $\widehat{A}$ which is Borel for the topology of $\widehat{A}$ is Borel for the Mackey structure.

3.8.4. Proposition. Let A be a separable C^* -algebra. Every point of $\widehat{A}$ is Borel for the Mackey structure.

Let $\pi_0 \in \text{Irr}_n(A)$ and let τ_0 be its image in $\widehat{A}$. The set of $\pi \in \text{Irr}_n(A)$ for which the intertwining number of π_0 and π is at most 0 is a G_δ subset of $\text{Irr}_n(A)$ (3.7.3). Hence, by 2.3.4, the set of $\pi \in \text{Irr}_n(A)$ inequivalent to π_0 is a G_δ subset of $\text{Irr}_n(A)$. Therefore the equivalence class of π_0 in $\text{Irr}_n(A)$ is Borel, and so $\{\tau_0\}$ is Borel in $\widehat{A}$ for the Mackey structure.

Bibliography: [28, 91].

3.9. Complements

3.9.1. Let A be a C^* -algebra.

- a. If the zero ideal is primitive, there is a dense point in $\widehat{A}$. In particular, any two nonempty open subsets of $\widehat{A}$ intersect.
- b. If the zero ideal is not primitive and A is separable, there exist two disjoint nonempty open subsets of $\widehat{A}$. (Use 3.3.2 and 3.4.13.)
- c. If A is separable and I is a closed two-sided ideal of A , it is equivalent to say that I is primitive or that I is prime (1.9.13). Problem: can the separability hypothesis be removed in b and c?
- *d. There exist separable C^* -algebras A such that the zero ideal is primitive and such that, in $\widehat{A}$, every nonempty open subset is non-Hausdorff. [11]

3.9.2. Let A be a C^* -algebra.

- a. Let $\mathfrak{N}(A)$ be the compact space of C^* -seminorms (1.9.13). Let Φ be the canonical map $I \mapsto N_I$ (1.9.13). Equip $\text{Prim}(A)$ with the Borel structure underlying its topology. Then Φ is a Borel isomorphism from $\text{Prim}(A)$ onto a subset of $\mathfrak{N}(A)$.
- b. If A is separable, $\text{Prim}(A)$ is a standard Borel space. (Use 1.9.13 and 3.9.1.)
- c. If A is separable, the Borel structure of $\text{Prim}(A)$ is the quotient structure of that of $\text{Rep}(A)$ under the canonical map $\text{Rep}(A) \rightarrow \text{Prim}(A)$. (Use b and B 22.) We may therefore speak of the Borel structure of $\text{Prim}(A)$ without further qualification. [20]

3.9.3. Let X be a locally quasi-compact space, and let $\mathcal{F}(X)$ be the set of closed subsets of X . For every quasi-compact subset C of X and every finite set $\mathcal{O}$ of nonempty open subsets of X , let $U(C, \mathcal{O})$ be the set of $Y \in \mathcal{F}(X)$ such that

$$Y \cap C = \emptyset, \quad Y \cap B \neq \emptyset \quad \text{for every } B \in \mathcal{O}.$$

Then the sets $U(C, \mathcal{O})$ form a base for a topology on $\mathcal{F}(X)$, and $\mathcal{F}(X)$ thereby becomes a compact space.

Now let A be a C^* -algebra. For every $F \in \mathcal{F}(\widehat{A})$, let N_F be the function on A defined by

$$N_F(x) = \sup_{\pi \in F} \|\pi(x)\|.$$

This is a C^* -seminorm on A (1.9.13). The map $F \mapsto N_F$ is a homeomorphism from $\mathcal{F}(\widehat{A})$ onto the compact space $\mathfrak{N}(A)$ of 1.9.13. [29]

3.9.4. Let X be a topological space. A point b of X is said to be *separated in* X if, for every point b' of X that is not adherent to b , the points b and b' have disjoint neighborhoods. Let A be a C^* -algebra.

- a. For $\pi_0 \in \widehat{A}$ to be separated in $\widehat{A}$, it is necessary and sufficient that, for every $x \in A$, the function $\pi \mapsto \|\pi(x)\|$ on $\widehat{A}$ be continuous at π_0 .
- b. If A is separable, the set of separated points of $\widehat{A}$ is a dense G_δ subset of $\widehat{A}$. (Use a and B 18.) [14]

***3.9.5.** Let A be a C^* -algebra, and let $\widetilde{A}$ be the C^* -algebra obtained from A by adjoining an identity element 1. For every representation π of A , let $\widetilde{\pi}$ be the canonical extension of π to $\widetilde{A}$. If I is a closed two-sided ideal of A , let I_1 be the set of $x + \lambda 1$, with $x \in A$ and $\lambda \in \mathbb{C}$, such that

$$xy + \lambda y \in I \quad \text{for every } y \in A.$$

Then I_1 is a closed two-sided ideal of $\tilde{A}$ such that $A \cap I_1 = I$. If $I = \ker \pi$ and π is nondegenerate, then $I_1 = \ker \tilde{\pi}$. Moreover, let (π_α) be a family of nondegenerate representations of A . Then π is weakly contained in the set of the π_α if and only if $\tilde{\pi}$ is weakly contained in the set of the $\tilde{\pi}_\alpha$. [27]

***3.9.6.** Let A be a C^* -algebra, let n be an integer > 0 , and let $(\pi_i)_{i \in I}$ be a family of elements of $\hat{A}$ of dimension n , indexed by a directed set. Let $\beta_1, \dots, \beta_r$ be points of $\hat{A}$. Suppose that $\{\beta_1, \dots, \beta_r\}$ is the set of limits of all subnets of (π_i) . Then there exist a directed subset J of I and integers $m_1, \dots, m_r > 0$ such that

$$\sum_{k=1}^r m_k \dim \beta_k \leq n, \quad \lim_{j \in J} \text{Tr } \pi_j(x) = \sum_{k=1}^r m_k \text{Tr } \beta_k(x) \quad \text{for every } x \in A.$$

[27]

3.9.7. Let B be a C^* -algebra and A a C^* -subalgebra of B such that, for every $\pi \in \hat{B}$, $\pi|_A$ is irreducible. The relation $\pi_1|_A \simeq \pi_2|_A$ on $\hat{B}$ is an equivalence relation R . Then $\hat{A}$ is canonically identified with the space $\hat{B}/R$. [29]

3.9.8. Let A be a C^* -algebra, let H be a Hilbert space, and let $\text{Rep}(A, H)$ be the set of representations of A on H . For $\pi \in \text{Rep}(A, H)$, $\varepsilon > 0$, $a_1, \dots, a_p \in A$, and $\xi_1, \dots, \xi_n$ in the essential subspace of π , let $V(\varepsilon, a_1, \dots, a_p, \xi_1, \dots, \xi_n, \pi)$ be the set of $\pi' \in \text{Rep}(A, H)$ such that

$$\|\pi'(a_i)\xi_j - \pi(a_i)\xi_j\| \leq \varepsilon \quad \text{for every } i \text{ and every } j.$$

Let $\mathcal{T}$ be the topology on $\text{Rep}(A, H)$ for which the sets just defined form a fundamental system of neighborhoods of π . This topology is not Hausdorff, but it induces a Hausdorff topology on the set of nondegenerate representations of A on H . We say that $\pi \in \text{Rep}(A, H)$ and $\pi' \in \text{Rep}(A, H)$ are almost equivalent if there exists a partially isometric operator U , whose initial and final subspaces are the essential subspaces of π and π' , respectively, and such that

$$U\pi(x) = \pi'(x)U \quad \text{for every } x \in A.$$

Let Q be the quotient space of $\text{Rep}(A, H)$ by this equivalence relation. The set Q is canonically identified with the set of classes of nondegenerate representations of A whose dimension is at most $\dim H$. Let $\pi \in Q$ and $S \subset Q$. For π to be weakly contained in S , it is necessary and sufficient that π belong to the closure of the set of Hilbert sums of finitely many elements of S . If $\dim H$ has been chosen

sufficiently large, then $\widehat{A} \subset Q$; the topology induced on $\widehat{A}$ by that of Q is the topology studied in this section. [31]

3.9.9. Let A be a C^* -algebra and G a topological group. Let ω be a homomorphism of G into the group of automorphisms of A , continuous for the topology of pointwise convergence. For every $\pi \in \widehat{A}$ and every $s \in G$, let $s \cdot \pi \in \widehat{A}$ be the representation

$$x \mapsto \pi(\omega(s)^{-1}x)$$

of A . Then $\pi \mapsto s \cdot \pi$ is a homeomorphism of $\widehat{A}$ onto $\widehat{A}$, and the map $(s, \pi) \mapsto s \cdot \pi$ from $G \times \widehat{A}$ into $\widehat{A}$ is continuous. [47]

3.9.10. Let $H = L^2([0, 1])$, and let A be the C^* -algebra of continuous complex functions on $[0, 1]$. For every $f \in A$, let $\pi(f)$ be the operator of multiplication by f on H . Then π is a representation of A on H ; every character of A is weakly contained in π , but no character of A is contained in π .

3.9.11. Let A be a C^* -algebra such that $\text{Prim}(A)$ is Hausdorff. For every $x \in A$, the function $\pi \mapsto \|\pi(x)\|$ is continuous on $\widehat{A}$. [77]

3.9.12. Let A be a C^* -algebra and Z its center. Suppose that, whenever two primitive ideals I and J satisfy $I \cap Z = J \cap Z$, one has $I = J$. Under these conditions, $I \mapsto I \cap Z$ is a homeomorphism of $\text{Prim}(A)$ onto $\text{Prim}(Z)$. [74]

3.9.13. Let A be a separable C^* -algebra and B a C^* -subalgebra of A . Then

$$\aleph_0 \text{ Card } \widehat{A} \geq \text{Card } \widehat{B}.$$

(For every $\pi \in \widehat{A}$, let E_π be the set of $\rho \in \widehat{B}$ which are contained in $\pi|_B$. Then E_π is countable, and $\bigcup_{\pi \in \widehat{A}} E_\pi = \widehat{B}$.)

3.9.14. Let A be an algebra over a commutative field, let $\text{Prim}(A)$ be the set of primitive ideals of A , equipped with the Jacobson topology, and let $\widehat{A}$ be the set of classes of irreducible representations of A , equipped with the inverse-image topology of that of $\text{Prim}(A)$ under the canonical map $\widehat{A} \rightarrow \text{Prim}(A)$. Let I be a two-sided ideal of A . Then, using the same notation as in 3.2.1, there are canonical homeomorphisms

$$\text{Prim}_I(A) \longrightarrow \text{Prim}(A/I), \quad \widehat{A}_I \longrightarrow \widehat{A/I},$$

$$\text{Prim}^I(A) \longrightarrow \text{Prim}(I), \quad \widehat{A}^I \longrightarrow \widehat{I}.$$

[77]

3.9.15. Problem. Given two C^* -algebras A, A' , classify the C^* -algebras B having the following properties:

1° B has a closed two-sided ideal I isomorphic to A ;

2° B/I is isomorphic to A' ;

3° $\widehat{I}$ is dense in $\widehat{B}$.

[262, 307]

§ 4. — Liminal C^* -Algebras

The two problems mentioned at the beginning of Section 3 are far from being solved in general.

For certain special classes of C^* -algebras, on the other hand, one comes close to a satisfactory answer. These are the classes of C^* -algebras that we shall now introduce.

4.1. The Algebra of Compact Operators

4.1.1. Let H be a Hilbert space. The compact operators on H form a self-adjoint two-sided ideal of $\mathcal{L}(H)$, and a norm limit of compact operators is compact. The set of compact operators on H is therefore a C^* -algebra, which we shall denote by $\mathcal{LC}(H)$.

In the study of finite-dimensional algebras over the complex field, it is known that the algebras $\mathcal{L}(H)$, with H a finite-dimensional vector space, play a fundamental role. In the study of C^* -algebras, this role is played not by the algebras $\mathcal{L}(H)$, with H a Hilbert space, but by the algebras $\mathcal{LC}(H)$. [Recall that $\mathcal{L}(H) = \mathcal{LC}(H)$ if H is finite-dimensional.]

A C^* -algebra A is called *elementary* if there exists a Hilbert space H such that the C^* -algebras A and $\mathcal{LC}(H)$ are isomorphic.

4.1.2. Proposition. *Let H be a Hilbert space, $A = \mathcal{LC}(H)$, and T the vector space of trace-class operators on H . For every $t \in T$, let f_t be the linear form $x \mapsto \text{Tr}(tx)$ on A . Then $t \mapsto f_t$ is a linear bijection of T onto the dual of the Banach space A . The form f_t is hermitian (respectively positive) if and only if t is hermitian (respectively positive).*

We have

$$|\text{Tr}(tx)| \leq (\text{Tr}|t|)\|x\| \quad (A32),$$

so f_t is a continuous linear form on A . The map $t \mapsto f_t$ is clearly linear. If $f_t = 0$, then $0 = f_t(t^*) = \text{Tr}(tt^*)$, hence $t = 0$, and the map $t \mapsto f_t$ is injective. Let f be a continuous linear form on A ; we show that there exists $t \in T$ such that $f = f_t$. It suffices to do so under the assumption that f is hermitian. Let $A^0 \subset A$ be the set of Hilbert–Schmidt operators; it carries a natural Hilbert-space structure, and the canonical injection of A^0 into A is continuous. Thus the restriction f' of f

to A^0 is a continuous linear form on the Hilbert space A^0 . Consequently, there exists $t_0 \in A^0$ such that

$$f'(x) = (x | t_0) = \text{Tr}(xt_0^*) \quad \text{for every } x \in A^0.$$

For every $x \in A^0$, we have

$$\begin{aligned} \text{Tr}(xt_0) &= \overline{\text{Tr}(t_0^*x^*)} = \overline{\text{Tr}(x^*t_0^*)} = \overline{f'(x^*)} \\ &= f'(x) = \text{Tr}(xt_0^*), \end{aligned}$$

hence $t_0 = t_0^*$. Take the spectral decomposition of t_0 : there are pairwise orthogonal rank-one projections e_i on H and real numbers λ_i such that

$$\sum_i \lambda_i^2 < +\infty, \quad t_0 = \lambda_1 e_1 + \lambda_2 e_2 + \cdots.$$

Put $\varepsilon_i = 1$ if $\lambda_i \geq 0$, and $\varepsilon_i = -1$ if $\lambda_i < 0$. Then

$$\begin{aligned} |\lambda_1| + \cdots + |\lambda_n| &= \text{Tr}((\varepsilon_1 e_1 + \cdots + \varepsilon_n e_n)t_0) \\ &= f(\varepsilon_1 e_1 + \cdots + \varepsilon_n e_n) \\ &\leq \|f\| \|\varepsilon_1 e_1 + \cdots + \varepsilon_n e_n\| = \|f\|. \end{aligned}$$

Thus $\sum_i |\lambda_i| < +\infty$, and hence $t_0 \in T$. The forms f and f_{t_0} agree on A^0 by construction of t_0 ; therefore $f = f_{t_0}$, since A^0 is dense in A . This proves that $t \mapsto f_t$ is a bijection of T onto the dual of the Banach space A .

For $t \in T$ and $x \in A$, we have

$$f_{t^*}(x) = \text{Tr}(t^*x) = \overline{\text{Tr}(tx^*)} = \overline{f_t(x^*)},$$

and hence

$$f_{t^*} = (f_t)^*. \quad (1)$$

This proves that f_t is hermitian if and only if t is hermitian. If $t \geq 0$, then $t^{1/2}$ is a Hilbert–Schmidt operator, and for every $x \in A$ we have

$$f_t(x^*x) = \text{Tr}(t^{1/2}x^*xt^{1/2}) = \text{Tr}((xt^{1/2})^*(xt^{1/2})) \geq 0;$$

thus $f_t \geq 0$. Finally, let $t \in T$ be such that $f_t \geq 0$. If ξ is a unit vector and e is the orthogonal projection onto $\mathbb{C}\xi$, then

$$0 \leq f_t(e) = \text{Tr}(te) = \text{Tr}(ete) = (t\xi | \xi).$$

This proves that $t \geq 0$.

4.1.3. Corollary. *The positive forms on A are the forms*

$$x \mapsto \lambda_1(x\xi_1 \mid \xi_1) + \lambda_2(x\xi_2 \mid \xi_2) + \cdots,$$

where $(\xi_1, \xi_2, \dots)$ is an orthonormal system in H , $\lambda_i \geq 0$, and $\sum_i \lambda_i < +\infty$.

The positive trace-class operators on H are the operators of the form

$$\lambda_1 P_{\mathbb{C}\xi_1} + \lambda_2 P_{\mathbb{C}\xi_2} + \cdots,$$

where $(\xi_1, \xi_2, \dots)$ is an orthonormal system in H , $\lambda_i \geq 0$, and $\sum_i \lambda_i < +\infty$. Since

$$\text{Tr}(P_{\mathbb{C}\xi_i} x) = \text{Tr}(P_{\mathbb{C}\xi_i} x P_{\mathbb{C}\xi_i}) = (x\xi_i \mid \xi_i),$$

the corollary follows at once from 4.1.2.

4.1.4. Corollary. *The pure positive forms on A are the forms $x \mapsto (x\xi \mid \xi)$, where ξ is a nonzero vector of H .*

Since the identity representation of A is irreducible when $H \neq 0$, the forms $x \mapsto (x\xi \mid \xi)$, with $\xi \neq 0$, are pure (2.5.4). Conversely, if f is a pure positive form on A , all positive forms dominated by f are proportional to f ; hence, by 4.1.3, f is of the form $x \mapsto (x\xi \mid \xi)$.

4.1.5. Corollary. *Every nonzero irreducible representation of A is equivalent to the identity representation.*

This follows from 2.5.4, 4.1.4, and 2.4.1 (ii).

4.1.6. Corollary. *Let $B \neq 0$ be a C^* -subalgebra of A , and suppose that the identity representation of B is irreducible. Then $B = A$.*

The commutant of B in $\mathcal{L}(H)$ consists only of scalars. Since $B \neq 0$, it follows from A 12 that B is ultra-strongly dense in $\mathcal{L}(H)$, and therefore in A . With the notation of 4.1.3, the form

$$x \mapsto \sum_i \lambda_i(x\xi_i \mid \xi_i) = \sum_i \omega_{\lambda_i^{1/2}\xi_i}(x)$$

is ultra-strongly continuous on A ; thus every continuous linear form on A is ultra-strongly continuous on A . What precedes proves that every continuous linear form on A which vanishes on B vanishes identically. The Hahn–Banach theorem then shows that $B = A$.

4.1.7. Corollary. *The only closed two-sided ideals of A are 0 and A .*

By 4.1.5, the zero ideal is the only primitive ideal of A . But every closed two-sided ideal of A is an intersection of primitive ideals [2.9.7 (ii)].

4.1.8. Corollary. *Let H and H' be two Hilbert spaces, and let φ be an isomorphism of the C^* -algebra $\mathcal{LC}(H)$ onto the C^* -algebra $\mathcal{LC}(H')$.*

- (i) *There exists an isomorphism U of H onto H' which implements φ .*
- (ii) *Let U' be an isomorphism of H onto H' . Then U' implements φ if and only if $U' = \lambda U$, where λ is a complex number of modulus 1.*

Since φ is a nonzero irreducible representation of $\mathcal{LC}(H)$ when $H, H' \neq 0$, assertion (i) follows from 4.1.5. For U' to implement φ , it is necessary and sufficient that $U'^{-1}U$ implement the identity automorphism of $\mathcal{LC}(H)$; in other words, that it commute with $\mathcal{LC}(H)$; in other words, that it be scalar. This proves (ii).

4.1.9. Retain the notation of 4.1.8. We see that φ maps the set of Hilbert–Schmidt elements (respectively the set of elements of finite rank, of rank one, and so on) of $\mathcal{LC}(H)$ onto the set of Hilbert–Schmidt elements (respectively the set of elements of finite rank, of rank one, and so on) of $\mathcal{LC}(H')$. Given an elementary C^* -algebra A , we may therefore speak of the Hilbert–Schmidt elements, and so on, of A , independently of the realization of A as an algebra of compact operators.

Similarly, the Hilbert dimension a of H is intrinsically attached to the C^* -algebra $\mathcal{LC}(H)$; the latter is said to have rank a^2 . For finite a , this recovers the usual notion of the rank of an algebra. Notice that, for infinite a , one has $a^2 = a$.

4.1.10. Corollary. *Let A be a C^* -algebra and π an irreducible representation of A . If $\pi(A) \cap \mathcal{LC}(H_\pi) \neq 0$, then $\pi(A) \supset \mathcal{LC}(H_\pi)$, and every irreducible representation of A having the same kernel as π is equivalent to π .*

Let $A' = \pi^{-1}(\mathcal{LC}(H_\pi))$. Then A' is a closed two-sided ideal of A . We have $\pi(A') \neq 0$, and hence $\pi(A')$ is irreducible on H_π [2.11.2 (ii)]. Therefore $\pi(A') = \mathcal{LC}(H_\pi)$ by 4.1.6. Every representation of A having the same kernel as π is the composite of π and an injective representation of $\pi(A)$. To prove the second assertion of 4.1.10, it therefore suffices to establish the following point. Let H be a Hilbert space and B a C^* -subalgebra of $\mathcal{L}(H)$ containing $\mathcal{LC}(H)$; then every injective irreducible representation ρ of B is equivalent to the identity representation. Since $\rho|_{\mathcal{LC}(H)}$ is equivalent to the identity representation of $\mathcal{LC}(H)$ by 4.1.5, we may suppose that $H_\rho = H$ and that $\rho|_{\mathcal{LC}(H)}$ is the identity map of $\mathcal{LC}(H)$. Then 2.10.4 (i) proves that ρ is the identity map of B .

4.1.11. Corollary. *Let A be a C^* -algebra and π a nonzero irreducible representation of A . Suppose that $\pi(A) \subset \mathcal{LC}(H_\pi)$.*

- (i) $\pi(A) = \mathcal{LC}(H_\pi)$.
- (ii) *The kernel of π is a maximal closed two-sided ideal of A .*

Assertion (i) follows from 4.1.10; assertion (ii) follows from (i) and 4.1.7.

Bibliography: [74, 77, 105, 106, 108, 126].

4.2. Liminal C^* -Algebras

4.2.1. Definition. *A C^* -algebra A is called liminal if, for every irreducible representation π of A and every $x \in A$, the operator $\pi(x)$ is compact.*

4.2.2. Let H be a Hilbert space. The C^* -algebra $\mathcal{LC}(H)$ is liminal (4.1.5), and the zero ideal of this algebra is primitive if $H \neq 0$. Conversely, if A is a liminal C^* -algebra in which the zero ideal is primitive, then A is elementary by 4.1.11 (i).

Every commutative C^* -algebra is liminal.

Later we shall see (10.4.5, 13.11.12, 15.5.6) other examples of liminal C^* -algebras.

4.2.3. Let A be a liminal C^* -algebra. Every primitive ideal of A is a maximal closed two-sided ideal [4.1.11 (ii)]. If π is a nonzero irreducible representation of A , then $\pi(A) = \mathcal{LC}(H_\pi)$ [4.1.11 (i)].

4.2.4. Proposition. *Let A be a liminal C^* -algebra. Every C^* -subalgebra of A and every quotient C^* -algebra of A are liminal.*

This is clear for a quotient C^* -algebra A/I of A , since every irreducible representation of A/I is identified with an irreducible representation of A . Now let B be a C^* -subalgebra of A , and let π be an irreducible representation of B . There exist a Hilbert space $K \supset H_\pi$ and an irreducible representation ρ of A on K such that

$$\pi(x) = \rho(x)|_{H_\pi} \quad \text{for every } x \in B$$

(2.10.2). Since A is liminal, the operators $\rho(x)$ are compact, and therefore the operators $\pi(x)$ are compact. Thus B is liminal.

4.2.5. Proposition. *Let A be a liminal C^* -algebra, let $\pi_1, \dots, \pi_n$ be pairwise inequivalent elements of $\widehat{A}$, and let $T_1, \dots, T_n$ be compact operators on $H_{\pi_1}, \dots, H_{\pi_n}$, respectively. There exists $z \in A$ such that*

$$\pi_1(z) = T_1, \dots, \pi_n(z) = T_n.$$

This is clear for $n = 1$; suppose it has been proved for $n - 1$. There exist $x, y \in A$ such that

$$\pi_1(x) = T_1, \dots, \pi_{n-1}(x) = T_{n-1}, \quad \pi_n(y) = T_n.$$

Put $I_k = \ker \pi_k$. By 4.1.10 and 4.2.3, $I_n \not\supset I_k$ for $k = 1, \dots, n - 1$; hence

$$I_n \not\supset I = I_1 \cap \dots \cap I_{n-1} \quad (2.11.4).$$

There exist $x' \in I$ and $y' \in I_n$ such that $x + x' = y + y' = z$. Then $\pi_k(z) = T_k$ for $k = 1, \dots, n$.

4.2.6. Proposition. *Let A be a C^* -algebra. Let I be the set of $x \in A$ such that $\pi(x)$ is compact for every irreducible representation π of A . Then I is the largest closed two-sided liminal ideal of A .*

It is clear that I is a closed two-sided ideal of A . Let ρ be an irreducible representation of I . There exists an irreducible representation π of A on H_ρ which extends ρ (2.10.4). Then $\rho(x) = \pi(x)$ is compact for every $x \in I$, and hence I is liminal. Finally, let J be a closed two-sided liminal ideal of A . We show that $J \subset I$, that is, that $\sigma(x)$ is compact for every irreducible representation σ of A and every $x \in J$. This is clear if $\sigma(J) = 0$. Otherwise, $\sigma|_J$ is irreducible [2.11.2 (ii)], and $\sigma(x)$ is compact because J is liminal.

Kaplansky introduced liminal C^* -algebras in 1951 under the name CCR-algebras (here CCR stands for “completely continuous representations”; “completely continuous” is synonymous with “compact”). Kaplansky and Glimm also introduced the notations GCR and NGCR for the C^* -algebras called below postliminal and antiliminal. These notions are fundamental, but the accepted terminology seemed to me somewhat unaesthetic.

Bibliography: [77]. The proof of 4.2.5 was pointed out to me by J. Tomiyama.

4.3. Postliminal C^* -Algebras

4.3.1. Definition. *A C^* -algebra A is called postliminal if every nonzero quotient C^* -algebra of A has a nonzero closed two-sided liminal ideal.*

These are the most important C^* -algebras for what follows. Every liminal C^* -algebra is postliminal (4.2.4), but the converse is not true.

Definition. A C^* -algebra is called *antiliminal* if it has no nonzero closed two-sided liminal ideal.

By 1.8.5, such a C^* -algebra has no nonzero closed two-sided postliminal ideal.

For examples of postliminal or antiliminal C^* -algebras, see the Complements (Sections 4, 5, 9, 10, 13, and 17).

4.3.2. Definition. Let A be a C^* -algebra. A composition series of A is an increasing family $(I_\rho)_{0 \leq \rho \leq \alpha}$ of closed two-sided ideals of A , indexed by the ordinals between 0 and some ordinal α , and having the following properties:

- (i) $I_0 = 0$ and $I_\alpha = A$;
- (ii) if $\rho \leq \alpha$ is a limit ordinal, then I_ρ is the closure of $\bigcup_{\rho' < \rho} I_{\rho'}$.

4.3.3. Proposition. Let A be a C^* -algebra. There exist an ordinal α and an increasing family $(I_\rho)_{0 \leq \rho \leq \alpha}$ of closed two-sided ideals of A having the following properties:

- (i) $I_0 = 0$, and A/I_α is antiliminal;
 - (ii) if $\rho \leq \alpha$ is a limit ordinal, then I_ρ is the closure of $\bigcup_{\rho' < \rho} I_{\rho'}$;
 - (iii) if $\rho < \alpha$, then $I_{\rho+1}/I_\rho$ is the largest liminal two-sided ideal of A/I_ρ , and is nonzero.
- Moreover, α and the family (I_ρ) are uniquely determined by the preceding conditions.

The I_ρ are constructed by transfinite induction. If ρ is a limit ordinal, condition (ii) prescribes the construction of I_ρ . Otherwise $\rho = \rho' + 1$, and condition (iii) prescribes the construction of I_ρ , unless $A/I_{\rho'}$ is antiliminal, in which case the construction stops with $\alpha = \rho'$.

4.3.4. Proposition. Let A be a C^* -algebra. The following conditions are equivalent:

- (i) A is postliminal;
- (ii) A has a composition series $(J_\rho)_{0 \leq \rho \leq \beta}$ such that all the $J_{\rho+1}/J_\rho$ are postliminal;
- (iii) A has a composition series $(I_\rho)_{0 \leq \rho \leq \alpha}$ such that all the $I_{\rho+1}/I_\rho$ are liminal.

(i) $\Rightarrow$ (iii): suppose that A is postliminal. With the notation of 4.3.3, we cannot have $A/I_\alpha \neq 0$; hence $I_\alpha = A$, and $(I_\rho)_{0 \leq \rho \leq \alpha}$ is a composition series of A with the required properties.

(iii) $\Rightarrow$ (ii) is clear.

(ii) $\Rightarrow$ (i): suppose there exists a composition series $(J_\rho)_{0 \leq \rho \leq \beta}$ such that the $J_{\rho+1}/J_\rho$ are postliminal. Let $J \neq A$ be a closed two-sided ideal of A . We must show that A/J has a nonzero liminal two-sided ideal. There is a least ordinal $\rho \leq \beta$ such that $J_\rho \not\subset J$. If $\rho' < \rho$, then $J_{\rho'} \subset J$; hence the closure of $\bigcup_{\rho' < \rho} J_{\rho'}$ is

contained in J , so ρ is not a limit ordinal. Write $\rho = \rho' + 1$ and put $I = J_\rho \cap J$. Then J_ρ/I is identified with a closed two-sided ideal of A/J (1.8.4). Moreover, J_ρ/I is a nonzero quotient of the postliminal C^* -algebra $J_{\rho'+1}/J_{\rho'}$, and hence has a nonzero liminal two-sided ideal. By 1.8.5, this ideal is also a two-sided ideal of A/J .

4.3.5. Proposition. *Let A be a postliminal C^* -algebra. Every C^* -subalgebra of A and every quotient C^* -algebra of A are postliminal.*

This is clear for quotient C^* -algebras. Now let B be a C^* -subalgebra of A . Let $(I_\rho)_{0 \leq \rho \leq \alpha}$ be a composition series of A such that the $I_{\rho+1}/I_\rho$ are liminal (4.3.4). Then

$$(I_{\rho+1} \cap B)/(I_\rho \cap B)$$

is identified with a C^* -subalgebra of $I_{\rho+1}/I_\rho$ (1.8.4), and hence is liminal (4.2.4). To complete the proof by 4.3.4, we show that $(I_\rho \cap B)_{0 \leq \rho \leq \alpha}$ is a composition series of B . It suffices to show that, if ρ is a limit ordinal and $x \in I_\rho \cap B$, then x is a limit of points of $\bigcup_{\rho' < \rho} (I_{\rho'} \cap B)$. Let $\varepsilon > 0$. There is a $\rho' < \rho$ such that the distance from x to $I_{\rho'}$ is less than ε . Since $B/(I_{\rho'} \cap B)$ is identified with $(B + I_{\rho'})/I_{\rho'}$, there is a point of $I_{\rho'} \cap B$ whose distance from x is less than ε , proving our assertion.

4.3.6. Proposition. *Let A be a C^* -algebra. The ideal I_α of 4.3.3 is the largest postliminal two-sided ideal of A , and the smallest closed two-sided ideal for which the corresponding quotient of A is antiliminal.*

The ideal I_α is postliminal by 4.3.4. Let I be a postliminal two-sided ideal of A . Then $I/(I \cap I_\alpha)$ is identified with a postliminal two-sided ideal of A/I_α . Since A/I_α is antiliminal, $I/(I \cap I_\alpha) = 0$, and hence $I \subset I_\alpha$. Finally, let J be a closed two-sided ideal of A such that A/J is antiliminal. Then $I_\alpha/(J \cap I_\alpha)$ is identified with a postliminal two-sided ideal of A/J , so $I_\alpha/(J \cap I_\alpha) = 0$ and $J \supset I_\alpha$.

Proposition 4.3.6 reduces, to some extent, the study of arbitrary C^* -algebras to the study of postliminal C^* -algebras on the one hand and antiliminal C^* -algebras on the other. Very little is known about antiliminal C^* -algebras. By contrast, fairly detailed information is available about postliminal C^* -algebras.

4.3.7. Theorem. *Let A be a postliminal C^* -algebra.*

- (i) *For every nonzero irreducible representation π of A , $\pi(A)$ contains all the compact operators on H_π .*
- (ii) *If π_1 and π_2 are two irreducible representations of A with the same kernel, then π_1 and π_2 are equivalent.*

Let π be a nonzero irreducible representation of A . Then $A/\ker \pi$ contains a nonzero liminal two-sided ideal $I/\ker \pi$. The restriction of π to I is nonzero and irreducible [2.11.2 (ii)], and therefore $\pi(I) = \mathcal{LC}(H_\pi)$ because $I/\ker \pi$ is liminal (4.2.3). This proves (i); assertion (ii) follows from (i) and 4.1.10.

Theorem 4.3.7 has a converse (9.1).

4.3.8. Let A be a separable C^* -algebra and $(I_\rho)_{0 \leq \rho \leq \alpha}$ a composition series of A . If the $I_{\rho+1}/I_\rho$ are nonzero, the family (I_ρ) is countable. Indeed, for every ordinal $\rho < \alpha$, choose $x_\rho \in I_{\rho+1}$ whose distance from I_ρ is at least 1. Then the mutual distances of the x_ρ are at least 1, and hence the family of the x_ρ is countable.

4.3.9. Let A be a C^* -algebra and $\tilde{A}$ the C^* -algebra obtained from A by adjoining an identity element. If A is postliminal, then $\tilde{A}$ is postliminal (4.3.4). If $\tilde{A}$ is postliminal, then A is postliminal (4.3.5). If $\tilde{A}$ is antiliminal, then A is obviously antiliminal. Suppose that A is antiliminal. If A has an identity element, then $\tilde{A}$ has a one-dimensional ideal, and so is not antiliminal. But if A has no identity element, then $\tilde{A}$ is antiliminal; for if $\tilde{A}$ had a nonzero liminal two-sided ideal I , then $I \cap A = 0$, so I would be a complement of A in $\tilde{A}$; consequently A would be a quotient of $\tilde{A}$ and would have an identity element.

Bibliography: [11, 28, 46, 77].

4.4. Spectrum of a Postliminal Algebra

4.4.1. If A is a postliminal C^* -algebra, Theorem 4.3.7 shows that the canonical map $\hat{A} \rightarrow \text{Prim}(A)$ is a homeomorphism. Thus $\hat{A}$ is a T_0 -space (3.1.3). If, moreover, A is liminal, all the primitive ideals of A are maximal (4.2.3), and hence the points of $\hat{A}$ are closed (3.1.4).

4.4.2. Lemma. *Let A be a C^* -algebra, let $\pi_0 \in \hat{A}$, and let $x \in A^+$ be such that the function $\pi \mapsto \text{Tr } \pi(x)$ on $\hat{A}$ is finite and continuous at π_0 .*

- (i) *If $y \in A^+$ is dominated by x , the function $\pi \mapsto \text{Tr } \pi(y)$ on $\hat{A}$ is continuous at π_0 .*
- (ii) *If $\pi_0(x) \neq 0$, there exists $z \in A^+$ such that $\pi(z)$ is a rank-one projection for every π in a neighborhood of π_0 .*

Put $y' = x - y \in A^+$. The functions $\pi \mapsto \text{Tr } \pi(y)$ and $\pi \mapsto \text{Tr } \pi(y')$ are lower semicontinuous (3.5.9); their finite sum is continuous at π_0 , and hence both functions are continuous at π_0 .

To prove (ii), we may suppose that $\|\pi_0(x)\| = 1$. Let K be a one-dimensional eigenspace of $\pi_0(x)$ corresponding to the eigenvalue 1. The irreducible algebra $\pi_0(A)$ contains the nonzero compact operator $\pi_0(x)$, and hence contains $\mathcal{LC}(H_{\pi_0})$ (4.1.10); consequently, there exists $z_1 \in A^+$ such that $\pi_0(z_1) = P_K$. Put $z_2 = f(z_1)$, where $f(t) = t$ for $0 \leq t \leq 1$ and $f(t) = 1$ for $t \geq 1$. Then $\|z_2\| \leq 1$ and $\pi_0(z_2) = f(P_K) = P_K$. Put

$$z_3 = x^{1/2} z_2 x^{1/2} \leq x;$$

then

$$\pi_0(z_3) = \pi_0(x)^{1/2} P_K \pi_0(x)^{1/2} = P_K.$$

By (i), $\text{Tr } \pi(z_3) \leq 5/4$ in a neighborhood of π_0 ; on the other hand, $\|\pi(z_3)\| \geq 3/4$ in a neighborhood of π_0 (3.3.2). Thus, in a neighborhood V of π_0 , the largest eigenvalue of $\pi(z_3)$ is at least $3/4$ and has multiplicity one, while all the following eigenvalues are at most $5/4 - 3/4 = 1/2$. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous nonnegative function equal to 0 for $t \leq 1/2$ and to 1 for $t \geq 3/4$, and put $z = g(z_3) \in A^+$. Then $\pi(z) = g(\pi(z_3))$ is a rank-one projection for every $\pi \in V$.

4.4.3. Lemma. *Let A be a liminal C^* -algebra. Let I be the set of $x \in A$ such that $\pi(x)$ has finite rank for every $\pi \in \widehat{A}$. Then I is a dense self-adjoint two-sided ideal of A .*

It is clear that I is a self-adjoint two-sided ideal of A . For $n = 1, 2, \dots$, let $f_n: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function which vanishes in a neighborhood of 0 (depending on n), and such that $f_n(t)$ tends uniformly to t as $n \rightarrow +\infty$. For every hermitian element x of A , one has $f_n(x) \rightarrow x$ as $n \rightarrow +\infty$. On the other hand, for every $\pi \in \widehat{A}$, the operator $\pi(x)$ is compact, and therefore its eigenvalues, each counted with its multiplicity, tend to zero. Thus

$$\pi(f_n(x)) = f_n(\pi(x))$$

has only finitely many nonzero eigenvalues, each of finite multiplicity; hence $\pi(f_n(x))$ has finite rank. Therefore I is dense in A .

4.4.4. Lemma. *Let A be a nonzero postliminal C^* -algebra. Then A has a nonzero closed two-sided ideal I with the following property: for any two distinct points π_1, π_2 of the open subset $\widehat{I}$ of $\widehat{A}$, there exists $x \in I^+$ such that*

- a. $\pi(x)$ has rank at most one for every $\pi \in \widehat{I}$;
- b. the function $\pi \mapsto \text{Tr } \pi(x)$ is continuous on $\widehat{I}$;
- c. $\text{Tr } \pi_1(x) = 1$ and $\text{Tr } \pi_2(x) = 0$.

We may suppose that A is liminal. By 4.4.3, there exists a nonzero $x_1 \in A^+$ such that $\pi(x_1)$ has finite rank for every $\pi \in \widehat{A}$; thus the function

$$\pi \longmapsto f(\pi) = \text{Tr } \pi(x_1)$$

on $\widehat{A}$ is finite everywhere. This function is lower semicontinuous and not identically zero, and hence remains at least some $\varepsilon > 0$ on a nonempty open subset U of $\widehat{A}$. Since U is a Baire space (3.2.2 and 3.4.13), f has a point of continuity π_0 in U (B 18). By 4.4.2 (ii), there exist $x_2 \in A^+$ and an open neighborhood V of π_0 such that $\pi(x_2)$ is a rank-one projection for every $\pi \in V$. Let I be the closed two-sided ideal of A such that $\widehat{I} = V$. Let π_1, π_2 be two distinct points of V . There exists $x_3 \in I^+$ such that

$$\pi_1(x_3) = \pi_1(x_2), \quad \pi_2(x_3) = 0$$

(4.2.4 and 4.2.5). Put

$$x = x_2^{1/2} x_3 x_2^{1/2} \in I^+.$$

Since $x \leq \|x_3\|x_2$, the function $\pi \mapsto \text{Tr } \pi(x)$ is continuous on V by 4.4.2 (i). Moreover,

$$\text{Tr } \pi_1(x) = \text{Tr } \pi_1(x_2)^2 = 1, \quad \text{Tr } \pi_2(x) = 0.$$

Finally, the rank of $\pi(x)$ is at most the rank of $\pi(x_2)$, and hence at most one, for every $\pi \in V$.

4.4.5. Theorem. *Let A be a postliminal C^* -algebra. There exists in $\widehat{A}$ a dense open subset which is locally compact.*

With the notation of 4.4.4, the points of $\widehat{I}$ are separated by continuous real-valued functions, and hence $\widehat{I}$ is Hausdorff. The set of Hausdorff subsets of $\widehat{A}$ is clearly inductive under inclusion. Let $S \subset \widehat{A}$ be a maximal Hausdorff open subset. Suppose that there is a nonempty open subset U of $\widehat{A}$ disjoint from S . Then U is identified with the spectrum of a nonzero closed two-sided ideal of A , which is a postliminal C^* -algebra. By the beginning of the proof, U contains a nonempty Hausdorff open subset S' . It is immediate that $S \cup S'$ is open and Hausdorff in $\widehat{A}$, which contradicts the maximality of S . Thus $\overline{S} = \widehat{A}$. By 3.3.8, S is locally compact.

Bibliography: [16, 77].

4.5. Continuous-Trace C^* -Algebras

4.5.1. Lemma. *Let A be a C^* -algebra and let $\mathfrak{p}$ be a subset of A^+ such that:*

- a. $\mathfrak{p} + \mathfrak{p} \subset \mathfrak{p}$;
- b. if $x \in A$ and $xx^* \in \mathfrak{p}$, then $x^*x \in \mathfrak{p}$;
- c. every element of A^+ dominated by an element of $\mathfrak{p}$ belongs to $\mathfrak{p}$.

Then:

- (i) the set $\mathfrak{n} = \{x \in A : xx^* \in \mathfrak{p}\}$ is a self-adjoint two-sided ideal of A ;
- (ii) the two-sided ideal $\mathfrak{m} = \mathfrak{n}^2$ is the set of linear combinations of elements of $\mathfrak{m}^+ = \mathfrak{m} \cap A^+$, and it has the same closure as $\mathfrak{n}$ in A ;
- (iii) $\mathfrak{m}^+ = \mathfrak{p}$.

It is clear that $x \in \mathfrak{n}$ implies $x^* \in \mathfrak{n}$. Suppose $x, y \in \mathfrak{n}$. We have

$$\begin{aligned} 2xx^* + 2yy^* - (x+y)(x+y)^* &= xx^* + yy^* - xy^* - yx^* \\ &= (x-y)(x-y)^* \geq 0, \end{aligned}$$

and hence

$$(x+y)(x+y)^* \leq 2xx^* + 2yy^* \in \mathfrak{p}.$$

Thus $(x+y)(x+y)^* \in \mathfrak{p}$, so $x+y \in \mathfrak{n}$. If $x \in \mathfrak{n}$ and $z \in A$, then

$$xzz^*x^* \leq \|zz^*\|xx^* \in \mathfrak{p};$$

therefore $(xz)(xz)^* \in \mathfrak{p}$, so $xz \in \mathfrak{n}$. Thus $\mathfrak{n}$ is a right ideal of A , and consequently a two-sided ideal because $\mathfrak{n} = \mathfrak{n}^*$. This proves (i).

Every element of $\bar{\mathfrak{n}}$ is a linear combination of products of two elements of $\bar{\mathfrak{n}}$ (1.5.8), and hence is a limit of elements of $\mathfrak{m}$. Thus $\bar{\mathfrak{n}} = \bar{\mathfrak{m}}$. Let $x \in \mathfrak{m}$. Write $x = \sum_j a_j b_j^*$, with $a_j, b_j \in \mathfrak{n}$. Then

$$\begin{aligned} 4x &= \sum_j (a_j + b_j)(a_j + b_j)^* - \sum_j (a_j - b_j)(a_j - b_j)^* \\ &\quad + i \sum_j (a_j + ib_j)(a_j + ib_j)^* - i \sum_j (a_j - ib_j)(a_j - ib_j)^*, \end{aligned}$$

and

$$(a_j + b_j)(a_j + b_j)^*, \dots, (a_j - ib_j)(a_j - ib_j)^* \in \mathfrak{m} \cap A^+.$$

This proves (ii).

If the preceding element x is hermitian, then

$$4x = \sum_j (a_j + b_j)(a_j + b_j)^* - \sum_j (a_j - b_j)(a_j - b_j)^* \leq \sum_j (a_j + b_j)(a_j + b_j)^* \in \mathfrak{p}.$$

If $x \geq 0$, it follows that $x \in \mathfrak{p}$. Hence $\mathfrak{m}^+ \subset \mathfrak{p}$. Conversely, let $x \in \mathfrak{p}$. Then $x^{1/2} \in \mathfrak{n}$, and hence $x = x^{1/2}x^{1/2} \in \mathfrak{m}^+$. This proves (iii).

4.5.2. Let A be a C^* -algebra. Let $\mathfrak{p}$ be the set of $x \in A^+$ such that the function $\pi \mapsto \text{Tr } \pi(x)$ is finite and continuous on $\widehat{A}$. It is clear that $\mathfrak{p} + \mathfrak{p} \subset \mathfrak{p}$. If $x \in A$ is such that $xx^* \in \mathfrak{p}$, then $x^*x \in \mathfrak{p}$, since

$$\text{Tr } \pi(x^*x) = \text{Tr } \pi(xx^*) \quad \text{for every } \pi \in \widehat{A}.$$

If $x \in \mathfrak{p}$ and $y \in A^+$ is dominated by x , then $y \in \mathfrak{p}$ by 4.4.2 (i). Lemma 4.5.1 therefore proves that $\mathfrak{p}$ is the positive part of a self-adjoint two-sided ideal $\mathfrak{m}$ of A , namely the set of linear combinations of elements of $\mathfrak{p}$. If $x \in \mathfrak{m}$, then $\pi(x)$ is a trace-class operator for every $\pi \in \widehat{A}$, and the function $\pi \mapsto \text{Tr } \pi(x)$ is continuous on $\widehat{A}$. We write $\mathfrak{m} = \mathfrak{m}(A)$.

Definition. A C^* -algebra A is called a *continuous-trace algebra* if the ideal $\mathfrak{m}(A)$ is dense in A .

Since $\overline{\mathfrak{m}(A)}$ is the intersection of the primitive ideals of A which contain it, this is equivalent to saying that, for every $\pi \in \widehat{A}$, there exists $x \in \mathfrak{m}(A)$ such that $\pi(x) \neq 0$.

By 4.5.1 (ii), it is also equivalent to say that the set of $x \in A$ such that the function

$$\pi \mapsto \text{Tr}(\pi(x)\pi(x)^*)$$

is finite and continuous on $\widehat{A}$ is dense in A . This set is a self-adjoint two-sided ideal $\mathfrak{n}$ such that $\mathfrak{n}^2 = \mathfrak{m}(A)$.

Examples of continuous-trace C^* -algebras will be given later (10.5.8 and 10.9.4).

4.5.3. Proposition. Let A be a continuous-trace C^* -algebra.

- (i) A is liminal.
- (ii) $\widehat{A}$ is Hausdorff.
- (iii) For every $\pi_0 \in \widehat{A}$, there exists $y \in A^+$ such that $\pi(y)$ is a rank-one projection for every π in a neighborhood of π_0 .

Let $\pi \in \widehat{A}$. For every $x \in \mathfrak{m}(A)$, $\pi(x)$ is a trace-class operator. Hence, for every $x \in \overline{\mathfrak{m}(A)} = A$, $\pi(x)$ is a norm limit of trace-class operators and is therefore compact. Thus A is liminal. Assertion (iii) follows from 4.4.2 (ii). Let π_1 and π_2 be two distinct points of $\widehat{A}$, let $I = \ker \pi_1$, and choose $x \in \mathfrak{m}(A)$ such that $\pi_2(x) \neq 0$. By 4.2.5, there exists $x_1 \in I$ such that $\pi_2(x_1) = \pi_2(x)^*$. Then $x_1x \in \mathfrak{m}(A)$, so the function $\pi \mapsto \text{Tr } \pi(x_1x)$ is finite and continuous on $\widehat{A}$; moreover,

$$\text{Tr } \pi_1(x_1x) = \text{Tr}(0) = 0, \quad \text{Tr } \pi_2(x_1x) = \text{Tr}(\pi_2(x)^*\pi_2(x)) \neq 0.$$

Thus $\widehat{A}$ is Hausdorff.

4.5.4. Proposition. *Let A be a C^* -algebra satisfying conditions (ii) and (iii) of 4.5.3. Then A is a continuous-trace C^* -algebra.*

Let $\pi_0 \in \widehat{A}$. There exist an open neighborhood V of π_0 and $y \in A^+$ such that $\pi(y)$ is a rank-one projection for every $\pi \in V$. Since $\widehat{A}$ is Hausdorff, it is locally compact (3.3.8), and there is an open neighborhood W of π_0 such that $\overline{W} \subset V$. Let I be the closed two-sided ideal of A such that $\widehat{I} = W$. Since $\pi_0 \in W$, the restriction $\pi_0|_I$ is irreducible. Hence there is a hermitian element x of I such that $\pi_0(x)$ acts identically on the one-dimensional subspace $\pi_0(y)(H_{\pi_0})$ (2.8.3); replacing x by x^+ , we may suppose that $x \in I^+$. Put

$$z = y^{1/2}xy^{1/2} \in I^+.$$

The function

$$\pi \longmapsto f(\pi) = \text{Tr } \pi(z)$$

on $\widehat{A}$ vanishes on $\widehat{A} - W$, and hence is continuous at every point of $\widehat{A} - V$. On the other hand, $z \leq \|x\|y$, so f is continuous at every point of V by 4.4.2 (i). Thus $z \in \mathfrak{m}(A)^+$. Since $\pi_0(z) = \pi_0(y) \neq 0$, this proves that A is a continuous-trace C^* -algebra.

4.5.5. Theorem. *Let A be a postliminal C^* -algebra. Then A has a composition series $(I_\rho)_{0 \leq \rho \leq \alpha}$ such that the quotients $I_{\rho+1}/I_\rho$ are continuous-trace C^* -algebras.*

If $A \neq 0$, then A has a nonzero closed two-sided ideal I_1 which is a continuous-trace C^* -algebra (4.4.4). If $I_1 \neq A$, then A/I_1 has a nonzero closed two-sided ideal I_2/I_1 which is a continuous-trace C^* -algebra. Continuing transfinitely, one obtains the composition series (I_ρ) .

This theorem reduces, to some extent, the study of the structure of postliminal C^* -algebras to the study of the structure of continuous-trace C^* -algebras. The latter will be investigated more deeply in Section 10.

4.5.6. If a C^* -algebra A has a composition series $(I_\rho)_{0 \leq \rho \leq \alpha}$, the $\widehat{I}_\rho$ are increasing open subsets of $\widehat{A}$, and, for a limit ordinal ρ ,

$$\widehat{I}_\rho = \bigcup_{\rho' < \rho} \widehat{I}_{\rho'};$$

this follows at once from 3.2.2. Thus $\widehat{A}$ is the union of the $\widehat{I}_{\rho+1} - \widehat{I}_\rho$. If A is postliminal and the I_ρ are chosen so that the $I_{\rho+1}/I_\rho$ have continuous trace, then

the spaces $\widehat{I}_{\rho+1} - \widehat{I}_\rho$ are locally compact [4.5.3 (ii)]. The existence of such a family of open subsets of $\widehat{A}$ also follows easily from Theorem 4.4.5.

4.5.7. Let A be a separable postliminal C^* -algebra. Let $(I_\rho)_{0 \leq \rho \leq \alpha}$ be a composition series of A such that the $I_{\rho+1}/I_\rho$ are nonzero continuous-trace C^* -algebras. Then the family (I_ρ) is countable (4.3.8). Thus $\widehat{A}$ is the union of the countable family of subsets $\widehat{I}_{\rho+1} - \widehat{I}_\rho$. Each of these subsets is the intersection of an open subset and a closed subset of $\widehat{A}$. Moreover, each space $\widehat{I}_{\rho+1} - \widehat{I}_\rho$ is locally compact and has a countable base, since

$$\widehat{I}_{\rho+1} - \widehat{I}_\rho = \widehat{I_{\rho+1}/I_\rho}$$

and $I_{\rho+1}/I_\rho$ is a separable C^* -algebra (3.3.4).

Bibliography: [16, 29].

4.6. Borel Structure on the Spectrum of a Postliminal C^* -Algebra

4.6.1. Proposition. *Let A be a separable postliminal C^* -algebra. The Mackey Borel structure on $\widehat{A}$ is the Borel structure underlying its topology, and makes $\widehat{A}$ a standard Borel space.*

Let S_1 be the Mackey Borel structure on $\widehat{A}$, and let S_2 be the Borel structure underlying its topology. Let $\widehat{A}_1$ and $\widehat{A}_2$ be the corresponding Borel spaces. By 4.5.7, $\widehat{A}_2$ is the union of a sequence (X_n) of disjoint Borel subsets, each of which, equipped with the structure induced by S_2 , is a standard Borel space. Therefore $\widehat{A}_2$, which is the sum of the Borel spaces X_n , is standard. On the other hand, S_1 is finer than S_2 (3.8.3). Since $\widehat{A}_1$ is a quotient of the standard Borel space $\text{Irr}(A)$ (3.8.1), it follows that $S_1 = S_2$ (B 22).

Thus, when A is a separable postliminal C^* -algebra, we may speak of the Borel space $\widehat{A}$ without risk of ambiguity.

4.6.2. Proposition. *Let A be a separable postliminal C^* -algebra. There exists a Borel map $\xi \mapsto \pi(\xi)$ from $\widehat{A}$ into $\text{Irr}(A)$ such that, for every $\xi \in \widehat{A}$, the representation $\pi(\xi)$ belongs to the class ξ .*

Let $\widehat{A}_p$ be the canonical image of $\text{Irr}_p(A)$ in $\widehat{A}$. The topological space $\text{Irr}_p(A)$ is Polish (3.7.4), and the canonical map $\text{Irr}_p(A) \rightarrow \widehat{A}_p$ is continuous and open (3.5.8). On the other hand, $\widehat{A}$ is the union of a sequence (X_n) of disjoint sub-

sets, each of which is the intersection of an open and a closed subset of $\widehat{A}$, and is Hausdorff (4.5.7). Let $\text{Irr}_{p,n}(A)$ be the inverse image of X_n in $\text{Irr}_p(A)$. Then $\text{Irr}_{p,n}(A)$ is the intersection of an open and a closed subset of $\text{Irr}_p(A)$, and hence is a G_δ ; consequently $\text{Irr}_{p,n}(A)$ is a Polish space (B 15). The saturation in $\text{Irr}_{p,n}(A)$ of an open subset is open, since equivalence in $\text{Rep}_p(A)$ is defined by a group of homeomorphisms (3.5.5). Finally, since X_n is Hausdorff, the equivalence classes are closed in $\text{Irr}_{p,n}(A)$. Hence, by B 17, there is a Borel subset $B_{p,n}$ of $\text{Irr}_{p,n}(A)$ which meets each equivalence class of $\text{Irr}_{p,n}(A)$ in exactly one point. The union B of the $B_{p,n}$ is a Borel subset of $\text{Irr}(A)$ which meets each equivalence class of $\text{Irr}(A)$ in exactly one point. Let π be the map from $\widehat{A}$ into $\text{Irr}(A)$ such that $\pi(\xi)$ is the unique representative of ξ belonging to B . Let ψ be the canonical map $\text{Irr}(A) \rightarrow \widehat{A}$. Since $\psi|_B$ is a Borel bijection of the standard Borel space B onto the standard Borel space $\widehat{A}$ (4.6.1), $\psi|_B$ is a Borel isomorphism of B onto $\widehat{A}$ (B 21). Therefore π is a Borel map from $\widehat{A}$ into $\text{Irr}(A)$.

Bibliography: [12, 15, 28].

4.7. Complements

4.7.1. Let H be a Hilbert space, t a trace-class operator on H , and f_t the linear form $x \mapsto \text{Tr}(tx)$ on $\mathcal{LC}(H)$. Then

$$\|f_t\| = \text{Tr } |t|.$$

***4.7.2.** Let A be a separable C^* -algebra and π an irreducible representation of A . If $\pi(A) \cap \mathcal{LC}(H_\pi) = 0$, there exists a family, of the cardinality of the continuum, of pairwise inequivalent irreducible representations of A having the same kernel as π . [199]

4.7.3. If a separable C^* -algebra A is such that $\widehat{A}$ consists of one point, then A is isomorphic to $\mathcal{LC}(H)$ for some Hilbert space H . (Use 4.7.2.) [106, 127] Problem: can the separability hypothesis be omitted?

4.7.4.

- a. Let A be a C^* -algebra and I a closed two-sided ideal of A . If I and A/I are antiliminal, then A is antiliminal.
- b. A C^* -subalgebra of an antiliminal C^* -algebra A need not be antiliminal (it may be commutative!). But every closed two-sided ideal of A is antiliminal.

- c. A quotient C^* -algebra of an antiliminal C^* -algebra need not be antiliminal. [By 4.3.9, it suffices to construct a nonunital antiliminal C^* -algebra. For example, if B is an antiliminal C^* -algebra, one may form the restricted product of the sequence $(B, B, B, \dots)$.] [46]

***4.7.5.** Some antiliminal C^* -algebras A have a family (π_i) of finite-dimensional irreducible representations for which

$$\bigcap_i \ker \pi_i = 0. \quad [57]$$

4.7.6. For $i = 1, 2, \dots$, let H_i be a Hilbert space of dimension i . Then the product C^* -algebra of the $\mathcal{L}(H_i)$ is neither postliminal nor antiliminal. [77]

4.7.7.

- a. Let $p_1 < p_2 < \dots$ be a sequence of integers such that p_i divides p_{i+1} , and let H be an infinite-dimensional Hilbert space. There exists a sequence of C^* -subalgebras $A_1, A_2, \dots$ of $\mathcal{L}(H)$ such that:

- (i) A_i is a factor of type I_{p_i} ;
- (ii) $A_i \subset A_{i+1}$.

Let A be the closure of $\bigcup_i A_i$. Then A is a unital, separable, antiliminal C^* -algebra whose only closed two-sided ideals are 0 and A . The data of the p_i determine A up to isomorphism.

- *b.* There exist sequences (p_i) and (p'_i) which give rise to nonisomorphic C^* -algebras. [44]

4.7.8. Let A be a postliminal C^* -algebra and I its largest closed two-sided liminal ideal. Then $\widehat{I}$ is dense in $\widehat{A}$. (Use 1.9.12 b.) Every $\pi \in \widehat{I}$ has as its kernel a minimal primitive ideal of A . Problem: is every minimal primitive ideal of A the kernel of some $\pi \in \widehat{I}$? [11]

***4.7.9.** There exist separable liminal C^* -algebras A such that the set of nonseparated points of $\widehat{A}$ (3.9.4) is dense in $\widehat{A}$. There exist liminal C^* -algebras A such that $\widehat{A}$ has no separated point. [14] Problem: if A is postliminal, is the set of separated points of $\widehat{A}$ contained in a Hausdorff open subset of $\widehat{A}$?

4.7.10. Let A be a separable liminal C^* -algebra and X a quasi-compact subset of $\widehat{A}$. Then X is a G_δ subset of $\widehat{A}$. [Let $(U_1, U_2, \dots)$ be a base for the topology of $\widehat{A}$. For every $\pi \in \widehat{A} - X$, there is a finite covering of X by finitely many of the U_i which do not contain π .] By contrast, if A is separable and postliminal, a quasi-compact subset of $\widehat{A}$ need not be Borel in $\widehat{A}$.

***4.7.11.** Let A be a C^* -algebra. The set I of $x \in A$ such that the rank of $\pi(x)$ is finite and bounded as π ranges over $\widehat{A}$ is a self-adjoint two-sided ideal of A . Suppose that I is dense in A . Let (π_i) be a family of elements of $\widehat{A}$ indexed by a directed set. Let $\rho_1, \dots, \rho_r \in \widehat{A}$ be such that, for every $x \in I$,

$$\lim_i \operatorname{Tr} \pi_i(x) = \sum_{k=1}^r \operatorname{Tr} \rho_k(x).$$

Then $\{\rho_1, \dots, \rho_r\}$ is the set of limits of (π_i) in $\widehat{A}$. [27] The C^* -algebra A is liminal, and $\widehat{A}$ has an open covering (U_i) such that, for every i , every family of elements of $\widehat{A}$ indexed by a directed set has at most a finite number of distinct limits in U_i . (Fell, unpublished.)

4.7.12.

a. For every C^* -algebra A , put $J(A) = \overline{\mathfrak{m}(A)}$ (4.5.2). There exist an ordinal α and an increasing family $(J_\rho)_{0 \leq \rho \leq \alpha}$ of closed two-sided ideals of A with the following properties:

- (a) $J_0 = 0$ and $J(A/J_\alpha) = 0$;
- (b) if $\rho \leq \alpha$ is a limit ordinal, then J_ρ is the closure of $\bigcup_{\rho' < \rho} J_{\rho'}$;
- (c) if $\rho < \alpha$, then $J_{\rho+1}/J_\rho = J(A/J_\rho) \neq 0$.

Moreover, α and the family (J_ρ) are uniquely determined by the preceding conditions. Put $J_\alpha = K(A)$.

b. For $K(A) = A$, it is necessary and sufficient that $J(A/I) \neq 0$ for every closed two-sided ideal $I \neq A$ of A . In this case A is called a *generalized continuous-trace C^* -algebra*.

c. Suppose that A has generalized continuous trace. For every $\rho < \alpha$, the set $\widehat{J}_{\rho+1} - \widehat{J}_\rho$ is open and dense in $\widehat{A} - \widehat{J}_\rho$, and each point of $\widehat{J}_{\rho+1} - \widehat{J}_\rho$ has in $\widehat{A} - \widehat{J}_\rho$ a fundamental system of closed neighborhoods. The algebra A is liminal (use 4.7.15), but a liminal algebra need not have generalized continuous trace (4.7.9).

d. Let A be a postliminal C^* -algebra, let α be an ordinal, and let $(V_\rho)_{0 \leq \rho \leq \alpha}$ be an increasing family of open subsets of $\widehat{A}$ with the following properties:

- 1° $V_0 = \emptyset$ and $V_\alpha = \widehat{A}$;
- 2° if $\rho \leq \alpha$ is a limit ordinal, then $V_\rho = \bigcup_{\rho' < \rho} V_{\rho'}$;
- 3° every point of $V_{\rho+1} - V_\rho$ has in $\widehat{A} - V_\rho$ a fundamental system of closed neighborhoods.

Then A has generalized continuous trace. In particular, if $\widehat{A}$ is Hausdorff, then A has generalized continuous trace. [16]

4.7.13. Let A be a continuous-trace C^* -algebra, and let $x \in A$ be such that the rank of $\pi(x)$ is finite and bounded as π ranges over $\widehat{A}$. Then the function $\pi \mapsto \text{Tr } \pi(x)$ is continuous on $\widehat{A}$. [29]

4.7.14.

- a. If A is a liminal C^* -algebra, the C^* -algebra $\widetilde{A}$ obtained from A by adjoining an identity element need not be liminal. [Example: $A = \mathcal{LC}(H)$, with H an infinite-dimensional Hilbert space. In this example the spectrum of $\widetilde{A}$ consists of two points π_1, π_2 , and its closed subsets are $\emptyset$, $\{\pi_1\}$, and $\{\pi_1, \pi_2\}$.]
- b. If a liminal C^* -algebra A has an identity element, all its irreducible representations are finite-dimensional. [77]

4.7.15. Let A be a postliminal C^* -algebra. For A to be liminal, it is necessary and sufficient that every point of $\widehat{A}$ be closed. (Use 4.3.7 (i).) [46]

***4.7.16.** Let A be a C^* -algebra.

- a. For every weak limit of pure states to be proportional to a pure state, it is necessary and sufficient that the following conditions hold:
 - (i) A is liminal;
 - (ii) $\widehat{A}$ is Hausdorff;
 - (iii) $A/J(A)$ (4.7.12) is commutative.
 [46]
- b. Suppose that A is unital and has no nonzero one-dimensional irreducible representation. For $P(A)$ to be weakly closed in the dual of A , it is necessary and sufficient that A be a product of finitely many C^* -algebras $A_1, \dots, A_n$ having the following property: for every $\pi \in \widehat{A}_i$, $\dim \pi$ is finite and independent of π . [169]

4.7.17. Let α be an ordinal. Let E be the set of maps $f: [0, \alpha] \rightarrow \mathbb{N}$ such that $f(\rho) = 0$ except for finitely many ρ . Let $H = \ell_E^2$, with its canonical orthonormal basis $(\xi_f)_{f \in E}$. If $S \subset [0, \alpha]$, let $E(S)$ be the set of $f \in E$ which vanish outside S . Let $\rho \leq \alpha$. We identify E with $E([0, \rho]) \times E([\rho, \alpha])$. For $g \in E([0, \rho])$, let H_g be the Hilbert subspace of H with orthonormal basis $(\xi_{(g,h)})_{h \in E([\rho, \alpha])}$; for $g, g' \in E([0, \rho])$, let $U_{g,g'}$ be the isomorphism from H_g onto $H_{g'}$ which maps $\xi_{(g,h)}$ to $\xi_{(g',h)}$ for every $h \in E([\rho, \alpha])$. Let B_ρ be the set of $x \in \mathcal{L}(H)$ such that:

- 1° $x(H_g) \subset H_g$ for every $g \in E([0, \rho])$;
- 2° the operators $x|_{H_g}$ are compact and are carried into one another by the $U_{g,g'}$.

Let C_ρ be the norm closure of $\sum_{\sigma < \rho} B_\sigma$. The C_ρ are postliminal. For C_α , the series of ideals in 4.3.3 is $(C_\rho)_{0 \leq \rho \leq \alpha}$.

4.7.18. Let H be a Hilbert space and let A be a commutative von Neumann algebra on H not contained in $\mathcal{LC}(H)$. Then $B = \mathcal{LC}(H) + A$ is postliminal but not liminal.

4.7.19. Let A be the C^* -algebra of complex two-by-two matrices. Let B be the C^* -algebra of sequences $x = (x_1, x_2, \dots)$ of elements of A such that x_n tends, as $n \rightarrow +\infty$, to a diagonal matrix

$$\begin{pmatrix} \lambda(x) & 0 \\ 0 & \mu(x) \end{pmatrix}.$$

Then $\widehat{B}$ consists of the maps $x \mapsto x_n$, $x \mapsto \lambda(x)$, and $x \mapsto \mu(x)$. The topological space $\widehat{B}$ is not Hausdorff. The algebra B has generalized continuous trace (4.7.12). [74]

4.7.20. Let A be a C^* -algebra. For every subset M of A , let $R(M)$ [respectively $L(M)$] denote the set of $x \in A$ such that $Mx = 0$ [respectively $xM = 0$]. The following conditions are equivalent:

- (i) [respectively (i')]: For every closed left (respectively right) ideal I of A , one has $L(R(I)) = I$ [respectively $R(L(I)) = I$].
 - (ii) [respectively (ii')]: The sum of the minimal left (respectively right) ideals of A is dense in A .
 - (iii) A is isomorphic to a C^* -subalgebra of an algebra $\mathcal{LC}(H)$.
 - (iv) There exists a family (A_i) of elementary C^* -algebras such that A is isomorphic to the restricted product of the A_i .
 - (v) [respectively (v')]: For every $x \in A$, left (respectively right) multiplication by x on A is a weakly compact operator.
 - (vi) The spectrum of every maximal commutative C^* -subalgebra of A is discrete.
 - (vii) For every hermitian $x \in A$, every nonzero point of $\text{Sp } x$ is isolated in $\text{Sp } x$.
- Such a C^* -algebra is called *dual*. A dual C^* -algebra is liminal. [77, 118, 119]

4.7.21. Let A be a C^* -algebra. For all the operators of left and right multiplication on A by elements of A to be compact, it is necessary and sufficient that there exist a family (A_i) of finite-dimensional C^* -algebras such that A is isomorphic to the restricted product of the A_i . [74]

4.7.22. Let H be a separable infinite-dimensional Hilbert space.

- a. The only closed two-sided ideals of $\mathcal{L}(H)$ are 0, $\mathcal{LC}(H)$, and $\mathcal{L}(H)$.
- b. The C^* -algebra $\mathcal{L}(H)/\mathcal{LC}(H)$ is antiliminal, nonseparable, not isomorphic to a von Neumann algebra, and has a family of $\mathfrak{c}$ orthogonal projections, where $\mathfrak{c}$ denotes the cardinality of the continuum.
- c. Every representation of $\mathcal{L}(H)$ can be written

$$\pi = \pi_0 \oplus \left(\bigoplus_{\sigma \in I} \pi_\sigma \right),$$

where π_0 vanishes on $\mathcal{LC}(H)$ and each π_σ is equivalent to the identity representation of $\mathcal{L}(H)$. Moreover, either $H_{\pi_0} = 0$ or H_{π_0} is nonseparable. Consequently, there is no C^* -subalgebra of $\mathcal{L}(H)$ complementary to $\mathcal{LC}(H)$. (In fact, there is no closed vector subspace of $\mathcal{L}(H)$ complementary to $\mathcal{LC}(H)$.)

- d. $\text{Card } \widehat{\mathcal{L}(H)} = 2^{\mathfrak{c}}$. (Use 2.10.1, 2.8.6, and a commutative C^* -subalgebra A of $\mathcal{L}(H)$ with $\text{Card } \widehat{A} = 2^{\mathfrak{c}}$.) [26, 34, 73, 78, 105, 159]

4.7.23. Let A be a unital C^* -algebra having minimal right ideals and a least nonzero closed two-sided ideal. For every subset P of A , let P' be the set of $x \in A$ such that $xP = 0$. Let $\mathcal{F}$ be the set of all P' as P ranges over the subsets of A . Suppose that, if $J_1, J_2 \in \mathcal{F}$ and $J_1 J_2^* = 0$, then $J_1 + J_2 \in \mathcal{F}$. Under these conditions, A is isomorphic to a C^* -algebra $\mathcal{L}(H)$. [183]

4.7.24. Problem. Let A be a continuous-trace C^* -algebra. Is the set of $x \in A$ such that $\pi(x)$ has bounded rank on $\widehat{A}$ and vanishes outside a compact subset of $\widehat{A}$ the smallest self-adjoint two-sided ideal $\mathfrak{m}$ of A such that $\overline{\mathfrak{m}} = A$?

4.7.25. Problem. Construct a liminal C^* -algebra A having no finite composition series (I_i) such that the $\widehat{I_{i+1}/I_i}$ are Hausdorff.

§ 5. — Type of a Representation

Throughout this section, A denotes an involutive algebra.

5.1. Comparison of Representations and Comparison of Projections

5.1.1. Given a family (π_i) of representations of A , we may always regard the π_i as subrepresentations of a single representation ρ of A (for example, $\rho = \bigoplus_i \pi_i$). Let E_i be the projection in $\rho(A)'$ corresponding to π_i . As we shall see, relations among the π_i can often be interpreted as relations among the projections E_i in the von Neumann algebra $\rho(A)'$.

5.1.2. Proposition. *Let ρ be a representation of A , let $\mathcal{B}$ be the von Neumann algebra generated by $\rho(A)$, and let π, π' be two subrepresentations of ρ corresponding to two projections E, E' of $\mathcal{B}'$. Let T be a continuous linear map from $E(H_\rho)$ into $E'(H_\rho)$. Then T is an intertwining operator for π and π' if and only if $TE \in \mathcal{B}'$.*

Suppose that T is an intertwining operator for π and π' . Let $x \in A$ and $\xi \in H_\rho$. We show that $TE\rho(x)\xi = \rho(x)TE\xi$; it will follow that $TE \in \mathcal{B}'$. If $\xi \in E(H_\rho)$, then $\rho(x)\xi \in E(H_\rho)$, and hence

$$TE\rho(x)\xi = T\rho(x)\xi = T\pi(x)\xi = \pi'(x)T\xi = \rho(x)T\xi = \rho(x)TE\xi.$$

If $\xi \in (1 - E)(H_\rho)$, then $\rho(x)\xi \in (1 - E)(H_\rho)$, so $E\rho(x)\xi = E\xi = 0$; hence

$$TE\rho(x)\xi = 0 = \rho(x)TE\xi.$$

Conversely, if $TE \in \mathcal{B}'$, then, for every $x \in A$ and every $\xi \in E(H_\rho)$,

$$T\pi(x)\xi = TE\rho(x)\xi = \rho(x)TE\xi = \pi'(x)T\xi,$$

so T is an intertwining operator for π and π' .

5.1.3. Corollary. *Retain the notation $\rho, \mathcal{B}, \pi, \pi', E, E'$ of 5.1.2. Then $\pi \simeq \pi'$ if and only if $E \sim E'$ relative to the von Neumann algebra $\mathcal{B}'$ (A 41).*

Indeed, $\pi \simeq \pi'$ means that there exists an intertwining operator for π and π' which is an isomorphism from $E(H_\rho)$ onto $E'(H_\rho)$. By 5.1.2, this is equivalent to the existence of a partially isometric operator in $\mathcal{B}'$ whose initial and final projections are E and E' .

5.1.4. Corollary. *Retain the notation $\rho, \mathcal{B}, \pi, \pi', E, E'$ of 5.1.2. Then $\pi \leq \pi'$ if and only if $E < E'$ relative to the von Neumann algebra $\mathcal{B}'$ (A 41).*

Indeed, $\pi \leq \pi'$ means that there exists a projection $E'_1 \in \mathcal{B}'$ dominated by E' such that π is equivalent to the subrepresentation of ρ defined by E'_1 ; by 5.1.3, this means that $E \sim E'_1$ relative to $\mathcal{B}'$.

5.1.5. Corollary. *Let π and π_1 be two representations of A . If $\pi \leq \pi_1$ and $\pi_1 \leq \pi$, then $\pi \simeq \pi_1$.*

This follows from 5.1.1, 5.1.3, 5.1.4, and A 42.

Bibliography: [89, 90].

5.2. Disjointness

5.2.1. Proposition. *Let π and π_1 be two representations of A . The following conditions are equivalent:*

- (i) *the only intertwining operator for π and π_1 is 0;*
- (ii) *if τ (respectively τ_1) is a subrepresentation of π (respectively π_1) on a nonzero space, then τ and τ_1 are inequivalent.*

Suppose, in addition, that π and π_1 are subrepresentations of a representation ρ of A . Let $\mathcal{B}$ be the von Neumann algebra generated by $\rho(A)$, and let E, E_1 be the projections in $\mathcal{B}'$ corresponding to π, π_1 . Then conditions (i) and (ii) are also equivalent to the following condition:

- (iii) *the central supports of E and E_1 in $\mathcal{B}'$ are orthogonal.*

We may suppose from the outset that π and π_1 are subrepresentations of ρ (5.1.1). We prove (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii): if π and π_1 have equivalent subrepresentations τ and τ_1 on nonzero spaces, then there is a nonzero intertwining operator for π and π_1 , namely a partially isometric operator whose initial and final subspaces are H_τ and H_{τ_1} .

(ii) $\Rightarrow$ (iii): if the central supports of E and E_1 are not orthogonal, there exist nonzero projections $F, F_1 \in \mathcal{B}'$ which are equivalent relative to $\mathcal{B}'$ and satisfy $F \leq E$ and $F_1 \leq E_1$ (A 44). The corresponding subrepresentations of π and π_1 are equivalent (5.1.3).

(iii) $\Rightarrow$ (i): suppose that the central supports G, G_1 of E, E_1 are orthogonal. Let T be an intertwining operator for π and π_1 , and put $S = TE \in \mathcal{B}'$ (5.1.2). The projections onto $H_\pi \ominus \ker S$ and onto $\overline{S(H_\pi)}$ belong to $\mathcal{B}'$ and are equivalent relative to $\mathcal{B}'$ (A 45), and therefore have the same central support (A 43). But the

projection onto $H_\pi \ominus \ker S$ is dominated by E , hence by G , while the projection onto $\overline{S(H_\pi)}$ is dominated by E_1 , hence by G_1 . Since $GG_1 = 0$, it follows that $S(H_\pi) = 0$, and hence $S = 0$ and $T = 0$.

5.2.2. Definition. *If two representations π, π_1 of A satisfy the equivalent conditions of 5.2.1, they are said to be disjoint.*

We then write $\pi \perp \pi_1$. This relation is symmetric. If π and π_1 are topologically irreducible, to say that they are disjoint means simply that they are inequivalent (2.3.4). If π and π_1 are finite-dimensional representations, to say that they are disjoint means that no irreducible representation occurs in both of their decompositions into irreducible representations; this follows from condition (ii) of 5.2.1.

5.2.3. Proposition. *Let π be a representation of A and $(\pi_i)_{i \in I}$ a family of representations of A . If π and π_i are disjoint for every $i \in I$, then π and $\bigoplus_i \pi_i$ are disjoint.*

Indeed, let H_i be the space of π_i , let H be the Hilbert sum of the H_i , and let P_i be the projection of H onto H_i . Let T be an intertwining operator for π and $\bigoplus_i \pi_i$. For every i , $P_i T$ is an intertwining operator for π and π_i , and therefore vanishes. Thus $T = 0$, so π and $\bigoplus_i \pi_i$ are disjoint.

5.2.4. Proposition. *Let ρ be a representation of A , let $\mathcal{B}$ be the von Neumann algebra generated by $\rho(A)$, and let E be a projection in $\mathcal{B}'$. Let π, π_1 be the subrepresentations of ρ corresponding to $E, 1 - E$. Then π and π_1 are disjoint if and only if E belongs to the center of $\mathcal{B}$.*

Let G, G_1 be the central supports of $E, 1 - E$. We have $G \geq E$ and $G_1 \geq 1 - E$. Therefore

$$\begin{aligned} \pi \perp \pi_1 &\iff GG_1 = 0 \\ &\iff G = E \text{ and } G_1 = 1 - E \\ &\iff E \text{ belongs to the center of } \mathcal{B}. \end{aligned}$$

5.2.5. Corollary. *Let ρ be a representation of A , and let $\mathcal{B}$ be the von Neumann algebra generated by $\rho(A)$. The following conditions are equivalent:*

- (i) *it is impossible to write ρ as a Hilbert sum of two disjoint subrepresentations on nonzero spaces;*
- (ii) *$\mathcal{B}$ is a factor;*
- (iii) *$\mathcal{B}' = \rho(A)'$ is a factor.*

By 5.2.4, condition (i) says that the only projections in the common center of $\mathcal{B}$ and $\mathcal{B}'$ are 0 and 1, that is, that this center consists only of scalars.

5.2.6. Definition. *A representation of A satisfying the conditions of 5.2.5 is called a factor representation.*

By 2.2.6, such a representation is either zero or nondegenerate.

If ρ is topologically irreducible, it is clear that ρ is a factor representation. The converse is false. For example, a Hilbert sum of equivalent irreducible representations is a factor representation (5.3.4); for finite-dimensional representations, this is in fact the most general example of a factor representation, by condition (i) of 5.2.5.

5.2.7. Proposition. *Let ρ be a representation of A , let $\mathcal{B}$ be the von Neumann algebra generated by $\rho(A)$, let $\mathfrak{Z}$ be the center of $\mathcal{B}$, let $E \neq 0$ be a projection in $\mathcal{B}'$, let π be the corresponding subrepresentation of ρ , and let F be the central support of E . Then π is a factor representation if and only if F is a minimal projection of $\mathfrak{Z}$.*

The von Neumann algebra generated by $\pi(A)$ is the induced von Neumann algebra $\mathcal{B}_E$ (A 15), which is isomorphic to $\mathcal{B}_F$ (A 20); the center of $\mathcal{B}_F$ is $\mathfrak{Z}_F$ (A 15). Thus π is a factor representation if and only if $\mathfrak{Z}_F$ consists only of scalars, that is, if and only if F is a minimal projection of $\mathfrak{Z}$ (A 34).

5.2.8. Let ρ be a representation of A , let $\mathcal{B}$ be the von Neumann algebra generated by $\rho(A)$, and let $\mathfrak{Z}$ be the center of $\mathcal{B}$. To decompose ρ as a Hilbert sum of topologically irreducible representations is to write the identity operator on H_ρ as a sum of pairwise orthogonal minimal projections of $\mathcal{B}'$. To decompose ρ as a Hilbert sum of pairwise disjoint factor representations is to write 1 as a sum of pairwise orthogonal minimal projections of $\mathfrak{Z}$, by 5.2.4 and 5.2.7. The second problem, concerning $\mathfrak{Z}$, is in principle easier to solve than the first, which concerns $\mathcal{B}'$.

Let (F_j) be the family of minimal projections of $\mathfrak{Z}$; these are pairwise distinct and hence pairwise orthogonal. If $1 = \sum_j F_j$, we obtain a decomposition of ρ as a Hilbert sum of pairwise disjoint factor representations. The problem of writing ρ as a Hilbert sum of topologically irreducible representations is thus reduced to the case in which ρ is a factor representation. We shall see in 5.4.11 that, in the important special case of type I representations, this problem is then quite easy to solve. For representations not of type I, the problem, even after this reduction, is on the contrary very difficult.

In general, however, one has $F = \sum_j F_j \neq 1$. The preceding decomposition is effective only for a subrepresentation of ρ , and it may unfortunately happen that $F = 0$. Later we shall obtain a satisfactory theory by replacing the decom-

position of ρ into a Hilbert sum of pairwise disjoint factor representations by a decomposition into a direct integral.

5.2.9. Proposition. *Let π and π_1 be factor representations of A . One of the relations*

$$\pi \perp \pi_1, \quad \pi \leq \pi_1, \quad \pi_1 \leq \pi$$

holds.

We may regard π and π_1 as subrepresentations of a representation ρ of A . Let $\mathcal{B}$ be the von Neumann algebra generated by $\rho(A)$, let E, E_1 be the projections in $\mathcal{B}'$ corresponding to π, π_1 , and let F, F_1 be their central supports. These are zero or minimal projections of $\mathcal{B} \cap \mathcal{B}'$ (5.2.7); hence F, F_1 are either orthogonal, in which case $\pi \perp \pi_1$, or equal. In the latter case, relative to the factor $\mathcal{B}'_F$, either $E < E_1$ or $E_1 < E$ (A 46); hence either $\pi \leq \pi_1$ or $\pi_1 \leq \pi$ (5.1.4).

Bibliography: [89, 90].

5.3. Quasi-Equivalence

5.3.1. Proposition. *Let π and π_1 be two representations of A , and let $\mathcal{B}, \mathcal{B}_1$ be the von Neumann algebras generated by $\pi(A), \pi_1(A)$, respectively. The following conditions are equivalent:*

- (i) *no subrepresentation of π on a nonzero space is disjoint from π_1 , and no subrepresentation of π_1 on a nonzero space is disjoint from π ;*
- (ii) *there exists an isomorphism Φ of $\mathcal{B}$ onto $\mathcal{B}_1$ such that*

$$\pi_1(x) = \Phi(\pi(x)) \quad \text{for every } x \in A;$$

- (iii) *there exists a multiple π_2 of π and, in $\pi_2(A)'$, a projection of central support 1 such that the corresponding subrepresentation of π_2 is equivalent to π_1 ;*
- (iv) *there exist a multiple of π and a multiple of π_1 which are equivalent.*

Suppose, in addition, that π and π_1 are subrepresentations of a representation ρ of A . Let $\mathcal{C}$ be the von Neumann algebra generated by $\rho(A)$, and let E, E_1 be the projections in $\mathcal{C}'$ corresponding to π, π_1 . Then conditions (i)–(iv) are also equivalent to the following condition:

- (v) *the central supports F, F_1 of E, E_1 are equal.*

(ii) $\Rightarrow$ (iii): by A 22, Φ is the composite of an amplification Ψ of $\mathcal{B}$ onto a von Neumann algebra $\mathcal{B}_2$, an induction Ψ' of $\mathcal{B}_2$ onto $\mathcal{B}_1$ defined by a projection $E \in \mathcal{B}'_2$ of central support 1, and a spatial isomorphism Ψ'' . Put

$$\pi_2(x) = \Psi(\pi(x)) \quad \text{for every } x \in A.$$

Then π_2 is a multiple of π . Moreover,

$$\Psi''^{-1}(\pi_1(x)) = \Psi''^{-1}(\Phi(\pi(x))) = \Psi'(\pi_2(x))$$

is the operator induced by $\pi_2(x)$ on $E(H_{\pi_2})$. Thus π_1 is equivalent to the subrepresentation of π_2 defined by E .

(iii) $\Rightarrow$ (ii): suppose condition (iii) holds, and let $\mathcal{B}_2$ be the von Neumann algebra generated by $\pi_2(A)$. There is clearly an isomorphism of $\mathcal{B}$ onto $\mathcal{B}_2$ which carries $\pi(x)$ to $\pi_2(x)$ for every $x \in A$. Moreover, if $E \in \mathcal{B}'_2$ is a projection of central support 1, induction from $\mathcal{B}_2$ to $(\mathcal{B}_2)_E$ is an isomorphism (A 20). Hence there is an isomorphism of $\mathcal{B}_2$ onto $\mathcal{B}_1$ which carries $\pi_2(x)$ to $\pi_1(x)$ for every $x \in A$.

(iii) $\Rightarrow$ (iv): suppose condition (iii) holds. There exists a cardinal n such that $\pi_1 \leq n\pi$, and we may suppose that n is infinite. Since (iii) $\Leftrightarrow$ (ii), condition (iii) is symmetric in π, π_1 ; hence there also exists an infinite cardinal n_1 such that $\pi \leq n_1\pi_1$. Put $p = \sup(n, n_1)$. Then

$$p\pi \leq pn_1\pi_1 = p\pi_1 \leq pn\pi = p\pi,$$

and hence $p\pi_1 \simeq p\pi$ by 5.1.5.

(iv) $\Rightarrow$ (i): suppose that $p\pi \simeq p\pi_1$. Let τ be a subrepresentation of π on a nonzero space. Then τ is not disjoint from $p\pi \simeq p\pi_1$, and therefore is not disjoint from π_1 (5.2.3). Similarly, every subrepresentation of π_1 on a nonzero space is not disjoint from π .

Now suppose that π and π_1 are subrepresentations of ρ .

(i) $\Rightarrow$ (v): if $F \neq F_1$, there is a nonzero projection G in the center of C dominated by F and orthogonal to F_1 , after interchanging F and F_1 if necessary. We have $GE \neq 0$, for otherwise the central support of E would be orthogonal to G ; and GE is orthogonal to F_1 , so the central support of GE is orthogonal to F_1 . Thus there exists a subrepresentation of π on a nonzero space which is disjoint from π_1 .

(v) $\Rightarrow$ (ii): the map $S \mapsto S_E$ (respectively $T \mapsto T_{E_1}$) from C_F onto C_E (respectively from C_{F_1} onto C_{E_1}) is an isomorphism (A 20). Hence, if $F = F_1$, there is an isomorphism from $C_E = \mathcal{B}$ onto $C_{E_1} = \mathcal{B}_1$ which carries $\rho(x)_E = \pi(x)$ to $\rho(x)_{E_1} = \pi_1(x)$ for every $x \in A$.

5.3.2. Definition. *If the equivalent conditions of 5.3.1 hold, π and π_1 are said to be quasi-equivalent. We then write $\pi \approx \pi_1$.*

It is clear from condition (ii) of 5.3.1 that this is an equivalence relation. If π and π_1 are finite-dimensional, then $\pi \approx \pi_1$ means that the same irreducible

representations occur in the decompositions of π and π_1 into irreducible components, though possibly with different multiplicities; this follows from condition (i) of 5.3.1.

5.3.3. Proposition.

- (i) If $\pi \simeq \pi_1$, then $\pi \approx \pi_1$.
- (ii) If π and π_1 are topologically irreducible and $\pi \approx \pi_1$, then $\pi \simeq \pi_1$.

Assertion (i) is clear. If π and π_1 are topologically irreducible and quasi-equivalent, they are not disjoint by condition (i) of 5.3.1, and hence are equivalent by condition (ii) of 5.2.1.

5.3.4. Proposition. *Let π and π_1 be two quasi-equivalent representations of A . If π is a factor representation, then π_1 is a factor representation.*

This follows at once from condition (ii) of 5.3.1 and condition (ii) of 5.2.5.

On the set of irreducible representations, quasi-equivalence coincides with equivalence (5.3.3). On the set of factor representations, quasi-equivalence appears to be the “right” equivalence relation.

5.3.5. Proposition. *Let π be a factor representation of A . Every subrepresentation of π on a nonzero space is quasi-equivalent to π .*

Since the only nonzero projection in the center of $\pi(A)'$ is 1, this follows from condition (iii) of 5.3.1.

5.3.6. Corollary. *Two factor representations of A are either quasi-equivalent or disjoint.*

This follows from 5.2.9 and 5.3.5.

5.3.7. Corollary. *Let A be a separable involutive Banach algebra. Every factor representation of A is quasi-equivalent to a factor representation on a separable space.*

Let π be a factor representation of A and ξ a nonzero vector of H_π . Then the closure K of $\pi(A)\xi$ is separable and invariant under $\pi(A)$. The subrepresentation of π defined by K is quasi-equivalent to π by 5.3.5.

5.3.8. Let π, π_1 be two quasi-equivalent representations of A on separable spaces. Then

$$\aleph_0\pi \simeq \aleph_0\pi_1.$$

It suffices to repeat the proof of 5.3.1, (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv), and use A 22.

The customary notation for quasi-equivalence is $\pi \sim \pi_1$. Since this conflicts with the notation $E \sim E_1$ for projections, we have preferred to modify it slightly.

Bibliography: [89, 90].

5.4. Representations of Type I

5.4.1. Proposition. *Let π be a representation of A . The following conditions are equivalent:*

- (i) *the von Neumann algebra generated by $\pi(A)$ is of type I;*
 - (ii) *the von Neumann algebra $\pi(A)'$ is of type I;*
 - (iii) *π is quasi-equivalent to a representation π_1 such that $\pi_1(A)'$ is commutative.*
- (iii) $\Rightarrow$ (i): suppose condition (iii) holds. Since $\pi_1(A)'$ is commutative, $\pi_1(A)$ generates a von Neumann algebra of type I (A 35). Hence $\pi(A)$ generates a von Neumann algebra of type I, by condition (ii) of 5.3.1.

(i) $\Rightarrow$ (ii) follows from A 52.

(ii) $\Rightarrow$ (iii): suppose condition (ii) holds. There exists a projection $E \in \pi(A)'$ of central support 1 such that $\pi(A)'_E$ is commutative (A 35). The subrepresentation π_1 of π defined by E is quasi-equivalent to π , by condition (iii) of 5.3.1, and $\pi_1(A)' = \pi(A)'_E$ is commutative.

5.4.2. Definition. *If a representation π satisfies the equivalent conditions of 5.4.1, it is said to be of type I.*

Every finite-dimensional representation is of type I. Indeed, on a finite-dimensional Hilbert space every von Neumann algebra is of type I.

5.4.3. Every representation quasi-equivalent to a representation of type I is of type I. Every subrepresentation of a representation of type I is of type I, since every von Neumann algebra induced by a von Neumann algebra of type I is of type I (A 53). Every Hilbert sum $\rho = \bigoplus_i \pi_i$ of representations π_i of type I is of type I. Indeed, let $\mathcal{B}$ be the von Neumann algebra generated by $\rho(A)$, let $E_i \in \mathcal{B}'$ be the projection corresponding to π_i , and let F_i be its central support. Then $\mathcal{B}_{E_i}$ is of type I, and so $\mathcal{B}_{F_i}$, which is isomorphic to $\mathcal{B}_{E_i}$ (A 20), is of type I. Therefore the largest projection F of $\mathcal{B} \cap \mathcal{B}'$ for which $\mathcal{B}_F$ is of type I (A 39) equals 1, since $\sup_i F_i = 1$.

5.4.4. *Let π be a representation of A . The following conditions are equivalent:*

- (i) *$\pi(A)'$ is commutative;*
- (ii) *in every decomposition $\pi = \pi_1 \oplus \pi_2$, the representations π_1 and π_2 are disjoint.*

By 5.2.4, condition (ii) says that every projection in $\pi(A)'$ belongs to the center of $\pi(A)'$, and hence that $\pi(A)'$ is commutative.

5.4.5. Definition. If π satisfies the equivalent conditions of 5.4.4, it is said to be *multiplicity-free*.

Let π be a finite-dimensional representation, and let

$$\pi = p_1\pi_1 \oplus \cdots \oplus p_n\pi_n$$

be the decomposition of π into multiples of inequivalent irreducible representations $\pi_1, \dots, \pi_n$. Then π is multiplicity-free if and only if $p_1 = \cdots = p_n = 1$, by condition (ii) of 5.4.4.

5.4.6. A multiplicity-free representation is of type I. Every representation π of type I is quasi-equivalent to a multiplicity-free representation π' [condition (iii) of 5.4.1]; moreover, π' is determined by π up to equivalence. Indeed, we have the following result, which generalizes 5.3.3 (ii).

Proposition. Let π, ρ be two multiplicity-free representations of A . If $\pi \approx \rho$, then $\pi \simeq \rho$.

Let $(E_i, F_i)_{i \in I}$ be a maximal family among the families having the following properties:

- 1° $E_i \in \pi(A)'$ and $F_i \in \rho(A)'$;
- 2° the E_i are nonzero pairwise orthogonal projections, and the F_i are nonzero pairwise orthogonal projections;
- 3° the subrepresentation of π defined by E_i is equivalent to the subrepresentation of ρ defined by F_i .

Put $E = \sum_i E_i$ and $F = \sum_i F_i$. Then the subrepresentation π_1 of π defined by E is equivalent to the subrepresentation ρ_1 of ρ defined by F . Suppose that $E \neq 1$. Since $\pi \approx \rho$, there exist a nonzero projection $E' \in \pi(A)'$ dominated by $1 - E$ and a nonzero projection $F' \in \rho(A)'$ such that the subrepresentations π', ρ' of π, ρ defined by E', F' are equivalent, by condition (i) of 5.3.1. Since $\rho(A)'$ is commutative, we have

$$F' = F'_1 + F'_2, \quad F'_1 \leq F, \quad F'_2 \leq 1 - F,$$

where F'_1, F'_2 are projections in $\rho(A)'$. Hence there are projections E'_1, E'_2 in $\pi(A)'$ such that $E' = E'_1 + E'_2$ and such that the subrepresentations of π and ρ defined by E'_j, F'_j are equivalent for $j = 1, 2$. The maximality of the family (E_i, F_i) implies that $F'_2 = E'_2 = 0$. Thus $F' \leq F$, and consequently π' is equivalent to a

subrepresentation of ρ_1 , hence to a subrepresentation of π_1 , and therefore to a subrepresentation of π . But this contradicts condition (ii) of 5.4.4. Hence $E = 1$, and similarly $F = 1$. Thus $\pi \simeq \rho$.

5.4.7. Proposition. *Let π be a representation of A and n a cardinal. The following conditions are equivalent:*

- (i) $\pi(A)'$ is a von Neumann algebra of type I_n (A 47);
- (ii) π is a Hilbert sum of n equivalent multiplicity-free representations.

Suppose condition (ii) holds. There exists a multiplicity-free representation ρ of A such that $\pi = n\rho$. Then H_π is identified with the Hilbert space $H_\rho \otimes K$, where K is a Hilbert space of dimension n , in such a way that $\pi(x)$ is identified with $\rho(x) \otimes 1$ for every $x \in A$ (A 17). Then

$$\pi(A)' = \rho(A)' \bar{\otimes} \mathcal{L}(K) \quad (\text{A18}).$$

Since $\rho(A)'$ is commutative, $\pi(A)'$ is of type I_n (A 47). The converse is proved similarly.

5.4.8. Definition. *If π satisfies the equivalent conditions of 5.4.7, it is said to have multiplicity n .*

A representation of multiplicity n is of type I. A multiplicity-free representation is simply a representation of multiplicity 1. The number n in 5.4.7 is determined by π (A 47). Moreover, if $\pi \simeq n\pi'$ with π' multiplicity-free, the class of π' is determined by π . Indeed, if $\pi \simeq n\pi'_1$ with π'_1 multiplicity-free, then $\pi' \approx \pi'_1$ by condition (iv) of 5.3.1, and hence $\pi' \simeq \pi'_1$ by 5.4.6.

Let π be a finite-dimensional representation, and let

$$\pi = p_1\pi_1 \oplus \cdots \oplus p_r\pi_r$$

be the decomposition of π into multiples of inequivalent irreducible representations $\pi_1, \dots, \pi_r$. Then π has multiplicity n if and only if $p_1 = \cdots = p_r = n$, by condition (ii) of 5.4.7.

5.4.9. Proposition. *Let π be a representation of type I of A . There exists a family $(E_j)_{j \in J}$ of nonzero pairwise orthogonal projections in $\pi(A)'$ whose sum is 1, such that:*

- a. the subrepresentations π_j of π defined by the E_j are pairwise disjoint;
- b. π_j has multiplicity n_j ;
- c. the n_j are pairwise distinct.

The family $(E_j)_{j \in J}$ is unique up to a bijection of the index set. The E_j belong to the center of $\pi(A)'$.

Let $\mathfrak{Z}$ be the center of $\pi(A)'$. In view of 5.2.4, the proposition says that there is a family $(E_j)_{j \in J}$ of nonzero pairwise orthogonal projections in $\mathfrak{Z}$, with sum 1, such that the von Neumann algebras $\pi(A)'_{E_j}$ are of type I_{n_j} , where the n_j are pairwise distinct; and that this family (E_j) is unique up to a bijection of the index set. Since $\pi(A)'$ is of type I by condition (ii) of 5.4.1, this follows from A 50.

What precedes completely reduces the study of representations of type I to that of multiplicity-free representations.

5.4.10. Retain the notation of 5.4.9. Let U be a unitary operator on H_π such that

$$U\pi(A)U^{-1} = \pi(A).$$

Then $U\mathfrak{Z}U^{-1} = \mathfrak{Z}$, so the $E'_j = UE_jU^{-1}$ are nonzero pairwise orthogonal projections in $\mathfrak{Z}$, with sum 1, such that $\pi(A)'_{E'_j}$ is of type I_{n_j} . Hence $E'_j = E_j$. In other words, U commutes with the E_j .

5.4.11. Proposition. *Let π be a representation of A . The following conditions are equivalent:*

- (i) π is a factor representation of type I;
- (ii) π is quasi-equivalent to a topologically irreducible representation;
- (iii) π is of the form $n\pi'$, with π' topologically irreducible.

Then π has multiplicity n .

(iii) $\Rightarrow$ (ii) follows from condition (iv) of 5.3.1.

(ii) $\Rightarrow$ (i) follows from 5.3.4 and condition (iii) of 5.4.1.

(i) $\Rightarrow$ (iii): if π is a factor representation of type I, 5.4.9 shows that π has multiplicity n for some cardinal n . Hence $\pi = n\rho$, with ρ multiplicity-free by condition (ii) of 5.4.7, and ρ is a factor representation because it is quasi-equivalent to π . Thus $\rho(A)'$ is a commutative factor, and hence consists only of scalars; therefore ρ is topologically irreducible.

5.4.12. Let π be a representation of A , and let $\mathfrak{Z}$ be the center of $\pi(A)'$. We have seen that π is a Hilbert sum of pairwise disjoint factor representations if and only if the identity operator is a sum of minimal projections of $\mathfrak{Z}$. Although this situation is far from being the general one, it is useful to interpret in this case the definitions in 5.4.2, 5.4.5, and 5.4.8.

Proposition. *Let (π_i) be a family of pairwise disjoint factor representations of A , and let $\pi = \bigoplus_i \pi_i$.*

- (i) π is of type I if and only if every π_i is of type I.

- (ii) π has multiplicity n if and only if every π_i has multiplicity n .
- (iii) π is multiplicity-free if and only if every π_i is topologically irreducible.

Assertion (i) follows from 5.4.3; assertion (ii) follows from the fact that a product of von Neumann algebras $\mathcal{B}_i$ is of type I_n if and only if each $\mathcal{B}_i$ is of type I_n (A 50); assertion (iii) is a special case of (ii).

5.4.13. We saw in 2.3.5 that a finite-dimensional representation is a Hilbert sum of irreducible representations. More generally:

Proposition. *Let π be a representation of A such that $\pi(x)$ is compact for every $x \in A$.*

- (i) *The representation π is a Hilbert sum of a family $(\pi_i)_{i \in I}$ of irreducible representations.*
- (ii) *If $i \in I$ and $\pi_i(A) \neq 0$, there are only finitely many indices $j \in I$ such that $\pi_j \simeq \pi_i$.*

We show that, if $H_\pi \neq 0$, then π has an irreducible subrepresentation. We may suppose that $\pi \neq 0$. Let $\mathfrak{B}$ be the von Neumann algebra generated by $\pi(A)$, and let $\mathfrak{J}$ be its center. There exists a hermitian $x \in A$ such that $\pi(x) \neq 0$. Since $\pi(x)$ is compact, it has a nonzero eigenvalue, and the corresponding spectral projection E is nonzero and of finite rank. We have $E \in \mathfrak{B}$. Let F be a projection of minimum rank among the nonzero projections of $\mathfrak{B}$ which are dominated by E . Then F is a minimal projection of $\mathfrak{B}$. Its central support F' is a minimal projection of $\mathfrak{J}$. The induced von Neumann algebra $\mathfrak{B}_{F'}$ is therefore a factor possessing a minimal projection, hence a factor of type I (A 36). Thus π has a factorial subrepresentation of type I on a nonzero space, and consequently, by 5.4.11, an irreducible subrepresentation on a nonzero space.

This being established, let $(E_i)_{i \in I}$ be a maximal family of nonzero, pairwise orthogonal projections of $\pi(A)'$ such that the corresponding subrepresentations of π are irreducible. If $\sum_i E_i \neq 1$, let π' be the subrepresentation of π corresponding to $1 - \sum_i E_i$. For every $x \in A$, $\pi'(x)$ is compact. The preceding result, applied to π' , then contradicts the maximality of (E_i) . Hence $\sum_i E_i = 1$, which proves (i). If $x \in A$ and $i \in I$ are such that $\pi_i(x) \neq 0$, and if there are infinitely many indices $j \in I$ such that $\pi_j \simeq \pi_i$, it is clear that $\pi(x)$ is not compact. This proves (ii).

Bibliography: [77, 89, 90].

5.5. Involutive Algebras of Type I

5.5.1. We shall see that the decomposition of representations is simpler in the case of representations of type I. Hence the interest of the following notion.

Definition. An involutive algebra is said to be of type I if all its representations are of type I.

For algebras of type I, the notion of a factor representation may therefore be set aside in favor of the more natural notion of a topologically irreducible representation (5.4.11). Unfortunately, there are C^* -algebras that are not of type I, and for these the consideration of factor representations sometimes leads to simpler theorems than does the consideration of topologically irreducible representations.

5.5.2. Theorem. Every postliminal C^* -algebra is of type I.

Let A be a nonzero postliminal C^* -algebra. By 4.4.4, there is a nonzero $x \in A^+$ such that $\pi(x)$ is zero or of rank 1 for every $\pi \in \widehat{A}$. Then $\pi(xAx) = \pi(x)\pi(A)\pi(x)$ is commutative for every $\pi \in \widehat{A}$; hence xAx is commutative.

Now let π be a representation of A . Let (E_i) be a maximal family of nonzero, pairwise orthogonal projections in the center of $\pi(A)'$ such that the corresponding subrepresentations of π are of type I. By 5.4.3, it is enough to prove that $\sum_i E_i = 1$. Suppose $E = 1 - \sum_i E_i \neq 0$, and let ρ be the subrepresentation of π defined by E . If $\rho(A) = 0$, we could add E to the family (E_i) , contrary to its maximality. Thus $\rho(A) \neq 0$. On the other hand, $\rho(A)$ is postliminal (4.3.5). By the beginning of the proof, there exists $x \in A^+$ such that $\rho(x)$ is nonzero and $\rho(x)\rho(A)\rho(x)$ is commutative. If $\mathfrak{C}$ denotes the von Neumann algebra generated by $\rho(A)$, then $\rho(x)\mathfrak{C}\rho(x)$ is commutative. Let

$$\rho(x) = \int_0^{+\infty} \lambda \, de_\lambda$$

be the spectral decomposition of $\rho(x)$. We have $e_\lambda \in \mathfrak{C}$ for every λ ; hence

$$y_\varepsilon = \int_\varepsilon^{+\infty} \lambda^{-1} \, de_\lambda \in \mathfrak{C} \quad (\varepsilon > 0).$$

If ε is sufficiently small, $\rho(x)y_\varepsilon = y_\varepsilon\rho(x)$ is a nonzero projection F of $\mathfrak{C}$. We have $F\mathfrak{C}F \subset \rho(x)\mathfrak{C}\rho(x)$, so that $F\mathfrak{C}F$ is commutative. Let $G \in \mathfrak{C} \cap \mathfrak{C}'$ be the central support of F . The von Neumann algebra $\mathfrak{C}_G$ is of type I (A 35). The projection on H_π which is zero on $(1 - E)(H_\pi)$ and agrees with G on $E(H_\pi)$ is therefore a nonzero element of the center of $\pi(A)'$, orthogonal to the E_i , and the corresponding subrepresentation of π is of type I. This contradicts the maximality of (E_i) . Hence $\sum_i E_i = 1$, and π is of type I.

Later, in 9.1, we shall prove a converse to Theorem 5.5.2.

5.5.3. Corollary. *Let A be a postliminal C^* -algebra. Every factor representation of A is quasi-equivalent to an irreducible representation.*

This follows from 5.5.2 and 5.4.11.

Bibliography: [76, 77].

5.6. Representations of Types II and III

5.6.1. Besides von Neumann algebras of type I, there are the more complicated von Neumann algebras of type II or type III (A 38). A representation π of A is said to be of type II (respectively, of type III) if the von Neumann algebra generated by $\pi(A)$ is of type II (respectively, III). This is equivalent to saying that $\pi(A)'$ is a von Neumann algebra of type II (respectively, III) (A 52).

5.6.2. Every representation quasi-equivalent to a representation of type II (respectively, III) is of type II (respectively, III). Every subrepresentation of a representation of type II (respectively, III) is of type II (respectively, III). Every Hilbert direct sum of representations of type II (respectively, III) is of type II (respectively, III). This follows by the same arguments as for representations of type I.

5.6.3. A representation on a nonzero space cannot be of two different types at once (A 38). It follows that two representations of distinct types are disjoint.

5.6.4. Proposition. *Let π be a representation of A . There exist projections $E_I, E_{II}, E_{III} \in \pi(A)'$, uniquely determined by the following conditions:*

- 1° E_I, E_{II}, E_{III} are pairwise orthogonal and have sum 1;
- 2° the subrepresentations of π defined by E_I, E_{II}, E_{III} are respectively of types I, II, and III.

These projections belong to the center of $\pi(A)'$.

This follows from the corresponding property of von Neumann algebras (A 39).

Thus the study of arbitrary representations of A is reduced to that of representations of type I, representations of type II, and representations of type III. This, however, does not simplify matters very much.

5.6.5. Proposition. *Let π be a representation of A . In order that π be a Hilbert direct sum of a representation of type II and a representation of type III, it is necessary and*

sufficient that every subrepresentation of π be a Hilbert direct sum of two equivalent representations.

For π to be a Hilbert direct sum of a representation of type II and a representation of type III, it is necessary and sufficient that the projection E_I of 5.6.4 be zero, in other words, that the von Neumann algebra $\pi(A)'$ be continuous (A 39). For this it is necessary and sufficient that every projection of $\pi(A)'$ be a sum of two orthogonal equivalent projections of $\pi(A)'$ (A 48). It then suffices to apply 5.1.3.

5.6.6. Let π be a representation of type I of A , and let C be the quasi-equivalence class of π . The different equivalence classes contained in C are distinguished by multiplicities (5.4.9). For representations not of type I there is an analogous, but more complicated, theory, which we shall not present. We do, however, have the following very simple result.

Proposition. *Let π_1, π_2 be two representations of type III of A on separable spaces. Then $\pi_1 \approx \pi_2$ implies $\pi_1 \simeq \pi_2$.*

Indeed, let $\mathfrak{B}_1, \mathfrak{B}_2$ be the von Neumann algebras generated by $\pi_1(A)$ and $\pi_2(A)$. If $\pi_1 \approx \pi_2$, there is an isomorphism Φ of $\mathfrak{B}_1$ onto $\mathfrak{B}_2$ which transforms $\pi_1(x)$ into $\pi_2(x)$ for every $x \in A$. But $\mathfrak{B}_1, \mathfrak{B}_2$ are of type III and act on separable spaces. Thus Φ is implemented by an isomorphism of H_{π_1} onto H_{π_2} (A 51).

5.6.7. The classification of von Neumann algebras also includes the definition of von Neumann algebras of type II_1 , of type II_∞ , semifinite, and so forth. A representation π of A is said to be of type II_1 , of type II_∞ , finite, and so forth, if $\pi(A)$ generates a von Neumann algebra of type II_1 , of type II_∞ , finite, and so forth.

Bibliography: [89, 90].

5.7. Complements

5.7.1. A finite-dimensional C^* -algebra A is a finite product of C^* -algebras $\mathcal{L}(H_i)$, where each H_i is a finite-dimensional Hilbert space. (The irreducible representations of A are finite-dimensional, and A admits an injective finite-dimensional representation which is a Hilbert direct sum of pairwise disjoint irreducible representations.)

5.7.2. Let A be an involutive algebra, and let π and ρ be representations of A . There exist a projection E in the center of $\pi(A)'$ and a projection F in the center of $\rho(A)'$ such that

$$\pi_E \approx \rho_F, \quad \pi_{1-E} \perp \rho, \quad \pi \perp \rho_{1-F}.$$

[90]

5.7.3. Let A be an involutive algebra and let π be a factor representation of A on a separable space. There exists a function m which assigns to every representation ρ on a separable space quasi-equivalent to π a number $m(\rho) \in [0, +\infty]$, with the following properties:

- (i) $m(\rho_1) = m(\rho_2)$ is equivalent to $\rho_1 \simeq \rho_2$;
- (ii) for every sequence $\rho_1, \rho_2, \dots$ of representations on separable spaces quasi-equivalent to π ,

$$m\left(\bigoplus_i \rho_i\right) = \sum_i m(\rho_i).$$

Moreover, the function m is unique up to multiplication by a constant > 0 . [90]

5.7.4. Let A be a C^* -algebra and J a closed two-sided ideal of A . In order that A be of type I, it is necessary and sufficient that J and A/J be of type I. (Use 2.11.2.) [184]

5.7.5. Let $\mathfrak{C}$ be a continuous von Neumann algebra, and let B be a C^* -algebra weakly dense in $\mathfrak{C}$. Then B is antiliminal. (Use 2.11.1 and 5.5.2.) A von Neumann algebra of type I can have weakly dense C^* -subalgebras which are antiliminal (consider an irreducible representation of a factor of type II_1). [45]

5.7.6. Let A be a C^* -algebra and π a representation of A .

- a. A representation π is called *homogeneous* if, for every projection $E \in \pi(A)'$, π_E has the same kernel as π . One then has $\|\pi(x)E\| = \|\pi(x)\|$ for every $x \in A$.
- b. A factor representation is homogeneous. The converse is false.
- c. For every closed two-sided ideal I of A , let $Q(I)$ be the projection of H_π onto the essential subspace of $\pi|_I$. As I varies, the $Q(I)$ generate a von Neumann subalgebra $\mathfrak{B}$ of the center of $\pi(A)'$, called the *ideal center associated with π* . In order that π be homogeneous, it is necessary and sufficient that $\mathfrak{B}$ reduce to the scalars.
- d. The kernel of a homogeneous representation is prime. [20]

*e. For every open subset U of $\widehat{A}$, let $I(U)$ be the closed two-sided ideal of A with spectrum U , and put $Q'(U) = Q(I(U))$. Then $U \mapsto Q'(U)$ extends to a map Q'' from the set of Borel subsets of $\widehat{A}$ into $\mathfrak{B}$ such that

$$Q''(X_1 \cup X_2 \cup \cdots) = \sum_i Q''(X_i)$$

whenever $X_1, X_2, \dots$ are pairwise disjoint Borel subsets of $\widehat{A}$. [47]

***5.7.7.** Let A be a unital C^* -algebra. We use the notation of 2.12.19. Let f, g be positive functionals on A ; we write $g < f$ if $F_g \subset F_f$, or, equivalently, if $g \in F_f$. The relation $F_g = F_f$ is an equivalence relation, and hence gives a notion of a class of positive functionals.

- a. π_f is factorial if and only if the relation $<$ totally preorders F_f . In that case f is called *primary*.
- b. Suppose that f is primary. In order that π_f be of type I, it is necessary and sufficient that there exist a pure state g such that $g < f$. In order that π_f be of type III, it is necessary and sufficient that the class of f be minimal and maximal in the set of classes of primary positive functionals and that f not be a homomorphism of A into $\mathbb{C}$. [71]

§ 6. — Traces and Representations

Let A be a C^* -algebra. In Section 2, we associated with every positive functional f on A a representation π_f . All representations are obtained by this procedure, or at least all those that admit a cyclic vector (recall that every nondegenerate representation is a sum of representations admitting cyclic vectors). But the map $f \mapsto \pi_f$ is by no means injective. We now wish to parametrize representations bijectively. Let π and π' be two finite-dimensional representations of A . If

$$\mathrm{Tr} \pi(x) = \mathrm{Tr} \pi'(x) \quad (x \in A),$$

then π and π' are equivalent. It is this fact that we shall partially generalize. In fact, only traceable representations, or rather their quasi-equivalence classes, can be parametrized.

6.1. Traces

6.1.1. Definition. Let A be a C^* -algebra. A trace on A^+ is a function $f: A^+ \rightarrow [0, +\infty]$ satisfying the following axioms:

- (i) if $x, y \in A^+$, then $f(x + y) = f(x) + f(y)$;
- (ii) if $x \in A^+$ and $\lambda \geq 0$, then $f(\lambda x) = \lambda f(x)$ (with the convention $0 \cdot +\infty = 0$);
- (iii) if $z \in A$, then $f(z z^*) = f(z^* z)$.

A trace f is said to be finite if $f(x) < +\infty$ for every $x \in A^+$.

A trace f is said to be semifinite if, for every $x \in A^+$, $f(x)$ is the supremum of the numbers $f(y)$ for $y \in A^+$ such that $y \leq x$ and $f(y) < +\infty$.

[In the definition of semifinite traces, it is plainly enough to consider those x for which $f(x) = +\infty$.]

We shall be concerned almost exclusively with traces on A^+ which are lower semi-continuous.

If A is a von Neumann algebra, these definitions agree with the familiar ones (see, however, A 28).

6.1.2. Proposition. Let A be a C^* -algebra and let f be a trace on A^+ .

- (i) Let $\mathfrak{n}$ be the set of $x \in A$ such that $f(x x^*) < +\infty$. Then $\mathfrak{n}$ is a self-adjoint two-sided ideal of A ; the two-sided ideal $\mathfrak{m} = \mathfrak{n}^2$ is the set of linear combinations of elements of $\mathfrak{m}^+ = \mathfrak{m} \cap A^+$, and $\mathfrak{m}^+$ is the set of $x \in A^+$ such that $f(x) < +\infty$. The ideals $\mathfrak{m}$ and $\mathfrak{n}$ have the same closure in A .
- (ii) There is a unique linear functional f' on $\mathfrak{m}$ which agrees with f on $\mathfrak{m}^+$.
- (iii) One has

$$\begin{aligned} f'(x^*) &= \overline{f'(x)} && \text{for every } x \in \mathfrak{m}, \\ f'(zx) &= f'(xz) && \text{for all } x \in \mathfrak{m}, z \in A, \\ f'(uv) &= f'(vu) && \text{for all } u, v \in \mathfrak{n}. \end{aligned}$$

Let $\mathfrak{p}$ be the set of $x \in A^+$ such that $f(x) < +\infty$. Applying Lemma 4.5.1 to $\mathfrak{p}$ gives (i). Assertion (ii) then follows immediately from (i). The functional f' is real on the self-adjoint elements of $\mathfrak{m}$, so $f'(x^*) = \overline{f'(x)}$ for every $x \in \mathfrak{m}$. If $u \in \mathfrak{n}$, then $uu^* \in \mathfrak{m}$ and $u^*u \in \mathfrak{m}$, and hence

$$f'(uu^*) = f(uu^*) = f(u^*u) = f'(u^*u).$$

Applying this to $u + v, u - v, u + iv, u - iv$, we deduce that $f'(uv^*) = f'(v^*u)$ for $u, v \in \mathfrak{n}$. Finally, let $z \in A$ and $x \in \mathfrak{m}$. We have $x = \sum_j u_j v_j$, with $u_j, v_j \in \mathfrak{n}$, and therefore

$$\begin{aligned} f'\left(z \sum_j u_j v_j\right) &= \sum_j f'((zu_j)v_j) = \sum_j f'(v_j(zu_j)) \\ &= \sum_j f'((v_j z)u_j) = \sum_j f'(u_j(v_j z)) = f'\left(\left(\sum_j u_j v_j\right)z\right). \end{aligned}$$

This proves the proposition.

By an abuse of language, the ideal $\mathfrak{m}$ will be called the *ideal of definition* of f , and will be denoted by $\mathfrak{m}_f$.

6.1.3. If $\overline{\mathfrak{m}_f} = A$ and f is lower semicontinuous, then f is semifinite. Indeed, let (u_i) be an increasing directed approximate identity of A , consisting of elements of $\mathfrak{m}_f$ (1.7.2). Let $x \in A^+$. The elements $x^{1/2}u_i x^{1/2}$ form an increasing directed family of elements of A^+ , bounded above by x (1.6.8), and tending to x ; hence

$$\lim_i f(x^{1/2}u_i x^{1/2}) \geq f(x),$$

and consequently $f(x^{1/2}u_i x^{1/2}) \rightarrow f(x)$, since $f(x^{1/2}u_i x^{1/2}) \leq f(x)$. But

$$x^{1/2}u_i x^{1/2} \in \mathfrak{m}_f, \quad f(x^{1/2}u_i x^{1/2}) < +\infty,$$

which proves the assertion.

6.1.4. Let A and B be two C^* -algebras, let ρ be a homomorphism from A into B , and let f be a trace on B^+ . Then $(f \circ \rho)|_{A^+}$ is a trace on A^+ . If f is lower semicontinuous, $(f \circ \rho)|_{A^+}$ is lower semicontinuous. If f is finite, $(f \circ \rho)|_{A^+}$ is finite. If f is semifinite, $(f \circ \rho)|_{A^+}$ need not be semifinite in general. Nevertheless, if A is a closed two-sided ideal of B and ρ is the canonical injection of A into B , then $(f \circ \rho)|_{A^+} = f|_{A^+}$ is semifinite; for if $x \in A^+$ and if $y \in B^+$ is bounded above by x , then $y \in A^+$, as is seen by considering the canonical map of B onto B/A .

6.1.5. Lemma. Let A be a C^* -algebra, π a representation of A , $\mathfrak{U}$ the von Neumann algebra generated by $\pi(A)$, t a normal trace on $\mathfrak{U}^+$, and $\mathfrak{m}$ a self-adjoint two-sided ideal of A such that $\pi(\mathfrak{m})$ is contained in $\mathfrak{m}_t$ and is strongly dense in $\mathfrak{U}$. Then $f = (t \circ \pi)|_{A^+}$ is semifinite and lower semicontinuous; moreover, if (u_i) is an increasing directed approximate identity of $\overline{\mathfrak{m}}$, then

$$f(x) = \lim_i f(x^{1/2}u_i x^{1/2}) \quad (x \in A^+).$$

Since $\pi(\mathfrak{m})$ is strongly dense in $\mathfrak{U}$, $\pi|_{\overline{\mathfrak{m}}}$ is nondegenerate, and hence $\pi(u_i)$ converges strongly to 1 (2.2.10). Let $x \in A^+$. The operators $\pi(x)^{1/2}\pi(u_i)\pi(x)^{1/2}$ form an increasing directed family in $\mathfrak{U}^+$ which converges strongly to $\pi(x)$. Since t is normal,

$$t(\pi(x)^{1/2}\pi(u_i)\pi(x)^{1/2}) \longrightarrow t(\pi(x)),$$

that is, $f(x^{1/2}u_ix^{1/2}) \rightarrow f(x)$. We may take the u_i in $\mathfrak{m}$ (1.7.2); then

$$\pi(x^{1/2}u_ix^{1/2}) \in \pi(\mathfrak{m}) \subset \mathfrak{m}_t, \quad f(x^{1/2}u_ix^{1/2}) < +\infty,$$

so f is semifinite. Since t is lower semicontinuous for the weak topology, and *a fortiori* for the topology induced by the norm, f is lower semicontinuous.

Bibliography: [16].

6.2. Bitraces

6.2.1. Definition. Let A be a C^* -algebra. A *bitrace* of A is a function $s : \mathfrak{n} \times \mathfrak{n} \rightarrow \mathbb{C}$, where $\mathfrak{n}$ is a self-adjoint two-sided ideal of A , satisfying the following axioms:

- (i) $s(x, y)$ is linear in x and conjugate-linear in y , $s(y, x) = \overline{s(x, y)}$, and $s(x, x) \geq 0$ (in other words, s is a positive Hermitian sesquilinear form);
- (ii) $s(y, x) = s(x^*, y^*)$ for $x, y \in \mathfrak{n}$;
- (iii) $s(zx, y) = s(x, z^*y)$ for $x, y \in \mathfrak{n}$, $z \in A$;
- (iv) for every $z \in A$, the map $x \mapsto zx$ from $\mathfrak{n}$ into $\mathfrak{n}$ is continuous for the pre-Hilbert structure defined by s ;
- (v) the elements xy , where $x, y \in \mathfrak{n}$, are dense in $\mathfrak{n}$ for the pre-Hilbert structure defined by s .

The ideal $\mathfrak{n}$ is called the *ideal of definition* of s and is denoted by $\mathfrak{n}_s$.

It follows readily from the axioms that $s(xz, y) = s(x, yz^*)$ for $x, y \in \mathfrak{n}$, $z \in A$, and that, for every $z \in A$, the map $x \mapsto xz$ from $\mathfrak{n}$ into $\mathfrak{n}$ is continuous for the pre-Hilbert structure defined by s .

6.2.2. Let N_s be the set of $x \in \mathfrak{n}_s$ such that $s(x, x) = 0$. By (i), it is a vector subspace of $\mathfrak{n}_s$; by (ii), it is self-adjoint; and by (iii), it is a left ideal of A , hence a two-sided ideal of A . Let Λ_s be the canonical map from $\mathfrak{n}_s$ onto $\mathfrak{n}_s/N_s$. The form s passes to the quotient and defines on $\mathfrak{n}_s/N_s$ a separated pre-Hilbert-space structure. We denote its Hilbert-space completion by H_s and its inner product by $(\cdot | \cdot)$. By (ii), the map $x \mapsto x^*$ passes to the quotient and defines an isometric map of

$\mathfrak{n}_s/N_s$ onto itself. Extending it by continuity, we obtain a conjugate-linear map $J_s: H_s \rightarrow H_s$ such that $J_s^2 = 1$ and

$$(J_s a \mid J_s b) = (b \mid a) \quad (a, b \in H_s).$$

Let $z \in A$. The maps $x \mapsto zx$ and $x \mapsto xz$ on $\mathfrak{n}_s$ pass to the quotient $\mathfrak{n}_s/N_s$ and extend by continuity to bounded linear operators $\lambda_s(z)$ and $\rho_s(z)$ on H_s . It is easy to see that λ_s is a representation of A on H_s ; for example, by (iii), for $x, y \in \mathfrak{n}_s$,

$$\begin{aligned} (\lambda_s(z^*)\Lambda_s x \mid \Lambda_s y) &= (\Lambda_s(z^*x) \mid \Lambda_s y) = s(z^*x, y) = s(x, zy) \\ &= (\Lambda_s x \mid \Lambda_s(zy)) = (\Lambda_s x \mid \lambda_s(z)\Lambda_s y) \\ &= (\lambda_s(z)^*\Lambda_s x \mid \Lambda_s y), \end{aligned}$$

whence $\lambda_s(z^*) = \lambda_s(z)^*$. Similarly, ρ_s is a representation on H_s of the C^* -algebra A° opposite to A . The representations λ_s and ρ_s are nondegenerate by (v). For $x \in \mathfrak{n}_s$ and $z \in A$, we have

$$\rho_s(z^*)\Lambda_s x^* = \Lambda_s(x^*z^*) = \Lambda_s((zx)^*) = J_s \lambda_s(z) J_s \Lambda_s x^*,$$

and hence

$$\rho_s(z^*) = J_s \lambda_s(z) J_s.$$

Since J_s is an isomorphism of H_s onto the conjugate Hilbert space, the representations $\bar{\lambda}_s^\circ$ (2.2.8) and ρ_s are equivalent.

6.2.3. The axioms (i)–(v) imply that the involutive algebra $\mathfrak{n}_s/N_s$, equipped with the inner product, is a Hilbert algebra (A 54). The left and right von Neumann algebras $\mathfrak{U}_s, \mathfrak{B}_s$ associated with $\mathfrak{n}_s/N_s$ are the von Neumann algebras generated by $\lambda_s(\mathfrak{n}_s)$ and $\rho_s(\mathfrak{n}_s)$; each is the commutant of the other (A 54). The von Neumann algebras generated by $\lambda_s(A)$ and $\rho_s(A)$ are *a priori* larger than the preceding ones; but since they plainly commute, they are in fact equal to $\mathfrak{U}_s$ and $\mathfrak{B}_s$, respectively.

6.2.4. The Hilbert algebra $\mathfrak{n}_s/N_s$ defines on $\mathfrak{U}_s^+$ a natural trace, denoted by t_s (A 60). This trace is normal. For every $x \in \mathfrak{n}_s$, we have

$$t_s(\lambda_s(x)\lambda_s(x)^*) = (\Lambda_s x \mid \Lambda_s x) \quad (\text{A 60}),$$

and hence

$$t_s(\lambda_s(xx^*)) = s(x, x) < +\infty. \quad (1)$$

If t'_s denotes the linear extension of $t_s|_{\mathfrak{m}_{t_s}^+}$ to $\mathfrak{m}_{t_s}$, then, for every $x, y \in \mathfrak{n}_s$,

$$t'_s(\lambda_s(x)\lambda_s(y)^*) = (\Lambda_s x \mid \Lambda_s y) \quad (\text{A 60}),$$

and hence

$$t'_s(\lambda_s(xy^*)) = s(x, y). \quad (2)$$

In the same way one can define a trace on $\mathfrak{B}_s^+$, which can also be obtained from t_s with the aid of J_s , but we shall not need it.

Bibliography: [54, 57].

6.3. Maximal Bitraces

6.3.1. Lemma. *Retain the preceding notation. Let $q = m_{t_s}$ and $q' = q^{1/2}$ (A 9). If, for $x, y \in \lambda_s^{-1}(q')$, one puts*

$$\tilde{s}(x, y) = t'_s(\lambda_s(xy^*)),$$

then $\tilde{s}$ is a bitrace extending s , and the pre-Hilbert space n_s is dense in the pre-Hilbert space $n_{\tilde{s}}$.

If $x \in n_s$, then $\lambda_s(xx^*) \in m_{t_s}$, and therefore $\lambda_s(x) \in q'$, whence $x \in \lambda_s^{-1}(q')$. This and formula (2) prove that $\tilde{s}$ extends s . Since q' is the completed Hilbert algebra corresponding to the Hilbert algebra $\lambda_s(n_s)$ (A 60 and A 61), the image under λ_s of $\lambda_s^{-1}(q')$ is a Hilbert algebra in which $\lambda_s(n_s)$ is dense. It follows that $\tilde{s}$ is a bitrace and that the pre-Hilbert space n_s is dense in the pre-Hilbert space $n_{\tilde{s}}$.

6.3.2. Lemma. *Let s' be a bitrace extending s . Then*

$$n_s \subset n_{s'} \subset n_{\tilde{s}'} \subset n_{\tilde{s}}, \quad \tilde{s}'(x, x) \geq \tilde{s}(x, x) \quad (x \in n_{\tilde{s}'}).$$

Plainly $N_s = N_{s'} \cap n_s$, and hence there is an isometric linear map $n_s/N_s \rightarrow n_{s'}/N_{s'}$. This map extends to a linear isometry $T: H_s \rightarrow H_{s'}$. Put $K = T(H_s)$. Since n_s is a two-sided ideal of A , K is invariant under the operators $\lambda_{s'}(z)$ and $\rho_{s'}(z)$ for every $z \in A$; hence the orthogonal projection of $H_{s'}$ onto K belongs to the center of $\mathfrak{U}_{s'}$. On identifying H_s with K by means of T , $\lambda_{s'}(z)$ induces $\lambda_s(z)$ on H_s . Let $B' \subset H_{s'}$ be the Hilbert algebra of elements bounded relative to the Hilbert algebra $n_{s'}/N_{s'}$, and let B be the orthogonal projection of B' onto H_s . Then the left von Neumann algebra associated with B is the von Neumann algebra induced by $\mathfrak{U}_{s'}$ on H_s , and the natural trace defined by B is the trace induced by $t_{s'}$ (A 65). Now n_s/N_s is contained in B' and is dense in H_s , hence is contained in B and is dense in B . Thus n_s/N_s defines the same von Neumann algebras as B and the same natural trace. In other words, $\mathfrak{U}_s$ and t_s are respectively the von

Neumann algebra induced by $\mathfrak{U}_{s'}$ on H_s and the trace induced by $t_{s'}$ on $\mathfrak{U}_s^+$. Now let $x \in \mathfrak{n}_{\widetilde{s}'}$. By what precedes,

$$t_s(\lambda_s(xx^*)) \leq t_{s'}(\lambda_{s'}(xx^*)) < +\infty;$$

hence $x \in \mathfrak{n}_{\widetilde{s}}$ and $\widetilde{s}(x, x) \leq \widetilde{s}'(x, x)$.

6.3.3. Definition. A bitrace is said to be maximal if there is no bitrace properly extending it.

6.3.4. Proposition. Let s be a bitrace.

- (i) The bitrace s is maximal if and only if $s = \widetilde{s}$.
- (ii) The bitrace $\widetilde{s}$ is maximal.

If s' is a bitrace extending $\widetilde{s}$, hence extending s , then $\mathfrak{n}_{s'} \subset \mathfrak{n}_{\widetilde{s}}$ by 6.3.2; hence $s' = \widetilde{s}$, so $\widetilde{s}$ is maximal. Consequently, if $s = \widetilde{s}$, then s is maximal. The converse is clear.

6.3.5. Definition. Let s be a bitrace. The bitrace $\widetilde{s}$ is called the canonical maximal extension of s .

In general, s has maximal extensions other than $\widetilde{s}$: it is enough to consider the case in which A is the product of $\mathfrak{n}_s$ and another C^* -algebra.

6.3.6. Equip $\mathfrak{n}_s$ with the norm induced by that of A . In general the map $x \mapsto \Lambda_s x$ from $\mathfrak{n}_s$ into H_s is not continuous. But the map

$$(x, y) \mapsto \Lambda_s(xy) = \lambda_s(x)\Lambda_s y = \rho_s(y)\Lambda_s x$$

from $\mathfrak{n}_s \times \mathfrak{n}_s$ into H_s is separately continuous, since the representations λ_s and ρ_s of A are continuous. It follows from this and from axiom (v) for bitraces that, if $E \subset \mathfrak{n}_s$ is norm-dense in $\mathfrak{n}_s$, the vectors $\Lambda_s(xy)$, $x, y \in E$, are dense in H_s . In particular, if A is separable, then H_s is separable.

Bibliography: [54, 57].

6.4. Relations Between Traces and Bitraces

6.4.1. Lemma. Let A be a C^* -algebra and let f be a lower semicontinuous trace on A^+ . Use the notation of 6.1.2. For $x, y \in \mathfrak{n}$, put

$$s(x, y) = f'(xy^*) = f'(y^*x).$$

Then s is a bitrace of A .

It is clear that s is sesquilinear. For $x, y \in \mathfrak{n}$ and $z \in A$, we have, by 6.1.2(iii),

$$\begin{aligned} s(y, x) &= f'(yx^*) = \overline{f'(xy^*)} = \overline{s(x, y)}, \\ s(x, x) &= f'(xx^*) \geq 0, \\ s(x^*, y^*) &= f'(x^*y) = s(y, x), \\ s(zx, y) &= f'(yx^*z^*) = f'((z^*y)^*x) = s(x, z^*y), \\ s(zx, zx) &= f'(x^*z^*zx) \leq \|z^*z\|f'(x^*x) = \|z^*z\|s(x, x). \end{aligned}$$

Thus axioms (i)–(iv) for bitraces are satisfied. Now let (u_i) be an increasing directed approximate identity of the C^* -algebra $\bar{\mathfrak{n}}$. For every $x \in \mathfrak{n}$,

$$\begin{aligned} s(u_ix - x, u_ix - x) &= s(u_ix, u_ix) - s(u_ix, x) - s(x, u_ix) + s(x, x) \\ &\leq \|u_i^*u_i\|s(x, x) - f'(x^*u_ix) - f'(x^*u_ix) + s(x, x) \\ &\leq 2[f'(x^*x) - f'(x^*u_ix)]. \end{aligned}$$

But the x^*u_ix form an increasing directed family of elements of A^+ tending to x^*x , and hence $f'(x^*u_ix) \rightarrow f'(x^*x)$ because f is lower semicontinuous. Thus x is the limit of u_ix for the pre-Hilbert structure defined by s . By 1.7.2, the u_i may be chosen in $\mathfrak{n}$.

The bitrace s is called the *bitrace associated with f* .

6.4.2. Conversely, let A be a C^* -algebra and let s be a bitrace of A . For every $x \in A^+$, put

$$f(x) = t_s(\lambda_s(x)).$$

We have $\lambda_s(\mathfrak{n}_s^2) \subset \mathfrak{m}_{t_s}$, and $\lambda_s(\mathfrak{n}_s^2)$ is strongly dense in $\mathfrak{U}_s$. Hence f is a semifinite lower semicontinuous trace on A^+ (6.1.5), which will be called the trace associated with s .

6.4.3. Lemma. *Let A be a C^* -algebra, s a bitrace of A , and f its associated trace. The bitrace s' associated with f is the canonical maximal extension of s .*

Let $q = \mathfrak{m}_{t_s}$ and $q' = q^{1/2}$. For every $x \in A$, we have

$$\begin{aligned} x \in \mathfrak{n}_{s'} &\iff f(xx^*) < +\infty \iff t_s(\lambda_s(xx^*)) < +\infty \\ &\iff \lambda_s(x) \in q' \iff x \in \lambda_s^{-1}(q'). \end{aligned}$$

Therefore $\mathfrak{n}_{s'} = \mathfrak{n}_{\tilde{s}}$, and for $x \in \mathfrak{n}_{s'}$,

$$s'(x, x) = f(xx^*) = t_s(\lambda_s(xx^*)) = \tilde{s}(x, x).$$

6.4.4. Lemma. *Let A be a C^* -algebra, let f be a semifinite lower semicontinuous trace on A^+ , and let s be the bitrace associated with f . Then s is maximal, and the trace associated with s is f .*

Let f' be the trace associated with s , and let s' be the bitrace associated with f' . By 6.4.3, s' is the canonical maximal extension of s . For $y \in \mathfrak{n}_s$, we have

$$f'(yy^*) = s'(y, y) = s(y, y) = f(yy^*).$$

Consequently, if $x \in A^+$ and $f(x) < +\infty$, then $f'(x) = f(x)$. Let $x \in A^+$ satisfy $f(x) = +\infty$. For every finite real number α , there exists $x_1 \in A^+$ such that $x_1 \leq x$ and $\alpha \leq f(x_1) < +\infty$, since f is semifinite. Hence

$$f'(x) \geq f'(x_1) = f(x_1) \geq \alpha, \quad \text{so that} \quad f'(x) = +\infty.$$

This proves that $f' = f$. Therefore $s' = s$, and s is maximal.

6.4.5. Proposition. *Let A be a C^* -algebra. Let E be the set of semifinite lower semicontinuous traces on A^+ , and let E' be the set of maximal bitraces of A . The maps*

$$\text{trace} \longmapsto \text{associated bitrace}, \quad \text{bitrace} \longmapsto \text{associated trace}$$

define mutually inverse bijections $E \rightarrow E'$ and $E' \rightarrow E$.

This follows from 6.4.1, 6.4.3, and 6.4.4.

Let f be a semifinite lower semicontinuous trace on A^+ , and let s be the associated maximal bitrace. We shall write

$$N_f = N_s, \quad \mathfrak{n}_f = \mathfrak{n}_s, \quad \dots, \quad \mathfrak{U}_f = \mathfrak{U}_s, \quad t_f = t_s;$$

and, conversely, $\mathfrak{m}_s = \mathfrak{m}_f$. Recall that $\overline{\mathfrak{m}_f} = \overline{\mathfrak{n}_f}$ [6.1.2(i)].

Bibliography: [16].

6.5. Sum of Two Traces

6.5.1. Let A be a C^* -algebra, and let f, f' be two traces on A^+ . It is clear that $f + f'$ is a trace. If f and f' are lower semicontinuous, then $f + f'$ is lower semicontinuous. Suppose that f and f' are semifinite, and let us prove that $f + f'$ is semifinite. Let $x \in A^+$ satisfy $(f + f')(x) = +\infty$, and let α be a finite real number. For example, suppose $f(x) = +\infty$. There is $y \in A^+$ such that $y \leq x$ and $\alpha \leq f(y) < +\infty$. If $f'(y) < +\infty$, then $\alpha \leq (f + f')(y) < +\infty$. If $f'(y) = +\infty$,

there exists $z \in A^+$ such that $z \leq y$ and $\alpha \leq f'(z) < +\infty$; hence $z \leq x$ and $\alpha \leq (f + f')(z) < +\infty$. In both cases we see that $f + f'$ is semifinite.

6.5.2. The set of traces on A^+ is plainly ordered by the relation $f \leq f'$, that is, by the relation

$$f(x) \leq f'(x) \quad \text{for every } x \in A^+.$$

Suppose f, f' are semifinite and lower semicontinuous, and let s, s' be the associated bitraces. Then $f \leq f'$ is equivalent to

$$\mathfrak{n}_{s'} \subset \mathfrak{n}_s, \quad s(x, x) \leq s'(x, x) \quad \text{for every } x \in \mathfrak{n}_{s'}.$$

6.5.3. Lemma. *Let A be a C^* -algebra and let s, s' be two maximal bitraces of A . Suppose there exists an involutive subalgebra B of A , dense in A and contained in $\mathfrak{n}_s$ and $\mathfrak{n}_{s'}$, such that s and s' agree on $B \times B$. Then $s = s'$.*

By 6.3.6, B is dense in $\mathfrak{n}_s$ (respectively, $\mathfrak{n}_{s'}$) for the pre-Hilbert structure of $\mathfrak{n}_s$ (respectively, $\mathfrak{n}_{s'}$). Since s and s' agree on $B \times B$, there is an isomorphism Θ from H_s onto $H_{s'}$ which, for every $x \in B$, transforms $\Lambda_s x$ into $\Lambda_{s'} x$. For $x, y \in B$, we have

$$\Theta \lambda_s(x) \Lambda_s y = \Theta \Lambda_s(xy) = \Lambda_{s'}(xy) = \lambda_{s'}(x) \Lambda_{s'} y = \lambda_{s'}(x) \Theta \Lambda_s y;$$

hence Θ transforms $\lambda_s(x)$ into $\lambda_{s'}(x)$ for every $x \in B$, and consequently for every $x \in A$. Moreover, since Θ defines an isomorphism from the Hilbert algebra $\lambda_s(B)$, dense in $\mathfrak{n}_s/N_s$, onto the Hilbert algebra $\lambda_{s'}(B)$, dense in $\mathfrak{n}_{s'}/N_{s'}$, Θ transforms $\mathfrak{U}_s$ into $\mathfrak{U}_{s'}$ and t_s into $t_{s'}$. Let f, f' be the traces associated with s, s' . For every $x \in A^+$,

$$f(x) = t_s(\lambda_s(x)) = t_{s'}(\lambda_{s'}(x)) = f'(x).$$

Since s and s' are maximal, $s = s'$ by 6.4.3.

6.5.4. Lemma. *Let A be a C^* -algebra, let f, f' be two semifinite lower semicontinuous traces on A^+ , and let s, s' be the associated bitraces. Suppose that $f' \leq f$ on $(\overline{\mathfrak{n}_s})^+$.*

(i) *There exists a unique $T \in \mathcal{L}(H_s)$ such that*

$$s'(x, y) = (T \Lambda_s x \mid \Lambda_s y) \quad (x, y \in \mathfrak{n}_s).$$

(ii) *One has $0 \leq T \leq 1$ and $T \in \mathfrak{U}_s \cap \mathfrak{B}_s$.*

(iii) *If*

$$f_1(x) = t_s(T \lambda_s(x)) \quad (x \in A^+),$$

then f_1 is a semifinite lower semicontinuous trace on A^+ , is majorized by f' , and agrees with f' on $(\overline{\mathfrak{n}_s})^+$.

For $x \in \mathfrak{n}_s$, we have $s'(x, x) \leq s(x, x)$. Hence, for $x, y \in \mathfrak{n}_s$, $s'(x, y)$ depends only on $\Lambda_s x$ and $\Lambda_s y$, and is continuous in these two variables for the Hilbert-space structure of H_s . This proves the existence of a $T \in \mathcal{L}(H_s)$ satisfying (i). Its uniqueness follows because $\Lambda_s(\mathfrak{n}_s)$ is dense in H_s . We have $0 \leq s'(x, x) \leq s(x, x)$ for every $x \in \mathfrak{n}_s$, and therefore

$$0 \leq (T\Lambda_s x \mid \Lambda_s x) \leq (\Lambda_s x \mid \Lambda_s x), \quad \text{so that} \quad 0 \leq T \leq 1.$$

If $z \in A$ and $x, y \in \mathfrak{n}_s$, then

$$\begin{aligned} (T\lambda_s(z)\Lambda_s x \mid \Lambda_s y) &= (T\Lambda_s(zx) \mid \Lambda_s y) = s'(zx, y) = s'(x, z^*y) \\ &= (T\Lambda_s x \mid \Lambda_s(z^*y)) = (T\Lambda_s x \mid \lambda_s(z^*)\Lambda_s y) \\ &= (\lambda_s(z)T\Lambda_s x \mid \Lambda_s y), \end{aligned}$$

and hence $T\lambda_s(z) = \lambda_s(z)T$. In the same way one sees that $T\rho_s(z) = \rho_s(z)T$, whence $T \in \mathfrak{U}_s \cap \mathfrak{B}_s$.

The function $S \mapsto t_s(TS)$ on $\mathfrak{U}_s^+$ is a normal trace t_1 . We have $\lambda_s(\mathfrak{m}_s) \subset \mathfrak{m}_{t_1}$; indeed, if $x \in \mathfrak{m}_s^+$, then $t_s(\lambda_s(x)) < +\infty$, and therefore

$$t_1(\lambda_s(x)) \leq \|T\|t_s(\lambda_s(x)) < +\infty.$$

Moreover, $\lambda_s(\mathfrak{m}_s)$ is strongly dense in $\mathfrak{U}_s$; consequently f_1 is a semifinite lower semicontinuous trace on A^+ (6.1.5). For $x \in \mathfrak{n}_s$,

$$f_1(xx^*) = t_s(T\lambda_s(x)\lambda_s(x^*)).$$

Since $T \in \mathfrak{U}_s$, the vector $T\Lambda_s x$ is a bounded element relative to the Hilbert algebra $\lambda_s(\mathfrak{n}_s)$ (A 59), and

$$t_s(T\lambda_s(x)\lambda_s(x^*)) = (T\Lambda_s x \mid \Lambda_s x) = s'(x, x) = f'(xx^*).$$

Thus $f_1|_{(\overline{\mathfrak{n}_s})^+}$ and $f'|_{(\overline{\mathfrak{n}_s})^+}$ are semifinite lower semicontinuous traces on $(\overline{\mathfrak{n}_s})^+$ (6.1.4), finite and equal on $(\mathfrak{n}_s^2)^+$ by what precedes, and hence equal by 6.5.3. On the other hand, let (u_i) be an increasing directed approximate identity of $\overline{\mathfrak{n}_s}$. By 6.1.5, for every $x \in A^+$,

$$f_1(x) = \lim_i f_1(x^{1/2}u_i x^{1/2}) = \lim_i f'(x^{1/2}u_i x^{1/2}) \leq f'(x),$$

because $x^{1/2}u_i x^{1/2} \in (\overline{\mathfrak{n}_s})^+$ and $x^{1/2}u_i x^{1/2} \leq x$.

6.5.5. Proposition. *Let A be a C^* -algebra, and let f, f' be two semifinite lower semicontinuous traces on A^+ . In order that $f' \leq f$, it is necessary and sufficient that there exist a semifinite lower semicontinuous trace f'' on A^+ such that $f = f' + f''$.*

The condition is plainly sufficient. Suppose $f' \leq f$, and use Lemma 6.5.4 and its notation. Put

$$f''(x) = t_s((1 - T)\lambda_s(x)) \quad (x \in A^+).$$

By 6.1.5, f'' is a semifinite lower semicontinuous trace on A^+ . For every $x \in A^+$,

$$f_1(x) + f''(x) = t_s(T\lambda_s(x)) + t_s((1 - T)\lambda_s(x)) = t_s(\lambda_s(x)) = f(x).$$

By 6.5.4, $f' + f''$ agrees with f on $(\overline{n_s})^+$. If $x \in A^+$ and $x \notin (\overline{n_s})^+$, then

$$f'(x) + f''(x) \geq f_1(x) + f''(x) = f(x) = +\infty,$$

so again $f'(x) + f''(x) = f(x)$.

6.5.6. Proposition. *Let A be a C^* -algebra, and let f, f' be two lower semicontinuous traces on A^+ such that $f' \leq f$. Suppose $\mathfrak{m}_f$ is dense in A , so that f and f' are semifinite (6.1.3). There exists a unique lower semicontinuous trace f'' on A^+ such that $f = f' + f''$.*

The existence of f'' follows from 6.5.5. Suppose that we have two decompositions

$$f = f' + f'', \quad f = f' + f_1''.$$

For $x \in \mathfrak{m}_f^+$, the numbers $f(x), f'(x), f''(x), f_1''(x)$ are finite, and hence $f''(x) = f_1''(x)$. Therefore $f'' = f_1''$ by 6.5.3.

6.5.7. The uniqueness of f'' fails when $\overline{\mathfrak{m}_f} \neq A$, even if f is semifinite (6.9.3).

Bibliography: [16, 57].

6.6. Traces and Representations

6.6.1. Definition. *Let A be a C^* -algebra. A traced representation of A is a pair (π, t) having the following properties:*

- (i) π is a nondegenerate representation of A on a Hilbert space;
- (ii) t is a faithful normal trace on $\mathfrak{U}^+$, where $\mathfrak{U}$ denotes the von Neumann algebra generated by $\pi(A)$;
- (iii) $\pi(A) \cap \mathfrak{n}_t$ is weakly dense in the von Neumann algebra $\mathfrak{U}$.

These conditions imply that t is semifinite, and hence that $\mathfrak{U}$ is semifinite. Thus π is a Hilbert direct sum of a representation of type I and a representation of type II.

6.6.2. Let (π, t) and (π', t') be two traced representations of A , and let $\mathfrak{U}, \mathfrak{U}'$ be the von Neumann algebras generated by $\pi(A)$ and $\pi'(A)$. We say that (π, t) and (π', t') are quasi-equivalent (respectively, equivalent) if there exists an isomorphism (respectively, a spatial isomorphism) from $\mathfrak{U}$ onto $\mathfrak{U}'$ which transforms π into π' and t into t' . Then π and π' are quasi-equivalent (respectively, equivalent).

6.6.3. Let A be a C^* -algebra and (π, t) a traced representation of A . Then

$$f = (t \circ \pi)|_{A^+}$$

is a semifinite lower semicontinuous trace on A^+ (6.1.5). We shall say that f , and the corresponding maximal bitrace, are associated with (π, t) . If (π, t) is replaced by a quasi-equivalent traced representation, f does not change.

6.6.4. Conversely, let f be a semifinite lower semicontinuous trace on A^+ , and let s be the corresponding bitrace. Then (λ_s, t_s) is a traced representation of A , called the traced representation associated with f , or with s . If λ_s is of type I, of type II, and so forth, then f and s are said to be of type I, of type II, and so forth.

6.6.5. Proposition.

- (i) Let f be a semifinite lower semicontinuous trace on A^+ , and let (π, t) be the traced representation associated with f . Then the trace associated with (π, t) is f .
- (ii) Let (π, t) be a traced representation of A , and let f be the trace associated with (π, t) . Then the traced representation associated with f is quasi-equivalent to (π, t) .

Let f, π, t be as in (i), let s be the bitrace associated with f , and let f' be the trace associated with (π, t) . Then $f' = (t_s \circ \lambda_s)|_{A^+}$, and therefore $f' = f$ by 6.4.4.

Let (π, t) be a traced representation of A , and let $\mathfrak{C}$ be the von Neumann algebra generated by $\pi(A)$. Recall the following facts, which have nothing to do with π . Equipped with the inner product

$$(S, T) \longmapsto t'(ST^*),$$

$\mathfrak{n}_t$ is a completed Hilbert algebra, where t' denotes the linear extension to $\mathfrak{m}_t$ of $t|_{\mathfrak{m}_t^+}$; λ_t is an isomorphism from the von Neumann algebra $\mathfrak{C}$ onto the von Neumann algebra $\mathfrak{U}_t$ on H_t ; and λ_t transforms t into the natural trace θ defined on $\mathfrak{U}_t^+$ by the Hilbert algebra $\mathfrak{n}_t$ (A 61). Now let s be the bitrace of A associated with (π, t) . We have $\mathfrak{n}_s = \pi^{-1}(\mathfrak{n}_t)$ and, for $x, y \in \mathfrak{n}_s$,

$$s(x, y) = t'(\pi(xy^*)).$$

Thus $\pi|_{\mathfrak{n}_s}$ passes to the quotient and defines an isomorphism Φ from the Hilbert algebra $\mathfrak{n}_s/N_s$ onto the Hilbert algebra $\pi(\mathfrak{n}_s) \subset \mathfrak{n}_t$. Since $\pi(\mathfrak{n}_s)$ generates the von Neumann algebra $\mathfrak{C}$ by the definition of traced representations, $\lambda_t(\pi(\mathfrak{n}_s))$ generates the von Neumann algebra $\mathfrak{U}_t$; hence, by A 55, $\pi(\mathfrak{n}_s)$ is dense in the completed Hilbert algebra $\mathfrak{n}_t$. Therefore Φ extends to an isomorphism, still denoted by Φ , from H_s onto H_t . It is easy to see that, for every $z \in A$, Φ transforms $\lambda_s(z)$ into $\lambda_t(\pi(z))$. Moreover, Φ transforms t_s into θ . Now (λ_s, t_s) is the traced representation associated with s , or with f . We see that it is equivalent to $(\lambda_t \circ \pi, \theta)$. Since λ_t is an isomorphism from $\mathfrak{C}$ onto $\mathfrak{U}_t$ which transforms t into θ , (λ_s, t_s) is quasi-equivalent to (π, t) .

6.6.6. Corollary. *There is a canonical bijection from the set of semifinite lower semicontinuous traces on A^+ onto the set of quasi-equivalence classes of traced representations of A .*

6.6.7. In fact, it is not the traced representations of A that are of interest, but the representations themselves. We therefore make the following definition.

Definition. *Let A be a C^* -algebra, let π be a representation of A , and let $\mathfrak{U}$ be the von Neumann algebra generated by $\pi(A)$. The representation π is said to be traceable if there exists a trace t on $\mathfrak{U}^+$ such that (π, t) is a traced representation.*

A representation quasi-equivalent to a traceable representation is traceable.

We shall study a case in which there is little difference between traced and traceable representations.

Bibliography: [16, 54, 57].

6.7. Characters and Traceable Factor Representations

6.7.1. Definition. *Let A be a C^* -algebra. A character of A is a nonzero semifinite lower semicontinuous trace f on A^+ satisfying the following condition: every semifinite lower semicontinuous trace on A^+ that is majorized by f is proportional to f on $(\mathfrak{m}_f)^+$.*

6.7.2. Recall that, if $\mathfrak{F}$ is a semifinite factor, there exist faithful normal semifinite traces t on $\mathfrak{F}^+$, and all these traces are proportional. Thus the ideal $\mathfrak{n}_t$ of $\mathfrak{F}$ is uniquely determined; its elements are called the Hilbert–Schmidt operators relative to $\mathfrak{F}$ (A 32).

Proposition. *Let A be a C^* -algebra, let π be a nonzero factor representation of A , and let $\mathfrak{F}$ be the factor generated by $\pi(A)$. The following conditions are equivalent:*

- (i) π is traceable;
- (ii) $\mathfrak{F}$ is semifinite and $\pi(A)$ contains a nonzero Hilbert–Schmidt operator relative to $\mathfrak{F}$.

(i) $\Rightarrow$ (ii) is clear. Suppose (ii) holds, and let t be a faithful normal semifinite trace on $\mathfrak{F}^+$. By hypothesis, $\mathfrak{n} = \pi(A) \cap \mathfrak{n}_t$ is a nonzero two-sided ideal of $\pi(A)$. Hence the weak closure B of $\mathfrak{n}$ is a nonzero weakly closed two-sided ideal of the weak closure of $\pi(A)$, that is, of $\mathfrak{F}$. Since $\mathfrak{F}$ is a factor, $B = \mathfrak{F}$ (A 9).

6.7.3. Theorem. *Let A be a C^* -algebra, let f be a semifinite lower semicontinuous trace on A^+ , and let (π, t) be the traced representation associated with f . In order that π be nonzero and factorial, it is necessary and sufficient that f be a character.*

Suppose that π is nonzero and factorial. Then f is nonzero. Let f' be a semifinite lower semicontinuous trace on A^+ , majorized by f on $(\mathfrak{m}_f)^+$. Apply Lemma 6.5.4. With the notation of that lemma, T is scalar because π is factorial. By 6.5.4(iii), f' is proportional to f on $(\mathfrak{m}_f)^+$.

Conversely, suppose that f is a character. Then π is nonzero. Let T be an element of $\mathfrak{U}_f \cap \mathfrak{B}_f$ such that $0 \leq T \leq 1$. If

$$f'(x) = t_f(T\lambda_f(x)) \quad (x \in A^+),$$

then, by 6.1.5, f' is a semifinite lower semicontinuous trace on A^+ , and it is clear that f' is majorized by f . Thus there exists $\lambda \in [0, 1]$ such that $f' = \lambda f$ on $(\mathfrak{m}_f)^+$. For every $x \in \mathfrak{n}_f$,

$$\begin{aligned} (T\Lambda_f x \mid \Lambda_f x) &= t_f(T\lambda_f(x)\lambda_f(x^*)) = f'(xx^*) \\ &= \lambda f(xx^*) = \lambda(\Lambda_f x \mid \Lambda_f x), \end{aligned}$$

whence $T = \lambda$. This proves that $\mathfrak{U}_f$ is a factor.

6.7.4. Corollary. *There is a canonical bijective correspondence between:*

- a. the set of quasi-equivalence classes of nonzero traceable factor representations of A ;
- b. the set of characters of A , defined up to a factor > 0 .

For every character f of A , λ_f is a nonzero traceable factor representation (6.7.3). Every nonzero traceable factor representation of A is obtained in this way up to quasi-equivalence (6.6.5 and 6.7.3). Finally, let f, f' be two characters of A such that λ_f and $\lambda_{f'}$ are quasi-equivalent. There exists an isomorphism Φ

from $\mathfrak{U}_f$ onto $\mathfrak{U}_{f'}$ which transforms λ_f into $\lambda_{f'}$. Since $\mathfrak{U}_f$ and $\mathfrak{U}_{f'}$ are factors, Φ transforms t_f into $\alpha t_{f'}$ for some $\alpha \in (0, +\infty)$. Hence, for every $x \in A^+$,

$$f(x) = t_f(\lambda_f(x)) = \alpha t_{f'}(\lambda_{f'}(x)) = \alpha f'(x).$$

In view of this result, one may speak, with a slight abuse of language, of the character of a nonzero traceable factor representation. This character is defined only up to a factor > 0 .

6.7.5. Let A be a C^* -algebra and let π be a nonzero irreducible (hence factorial) representation of A . In order that π be traceable, it is necessary and sufficient that

$$\pi(A) \supset \mathcal{LC}(H_\pi).$$

Indeed, this condition is plainly sufficient, and it is necessary by 4.1.10. [The character of π may be normalized by using the usual trace on $\mathcal{L}(H_\pi)^+$.] In particular, every irreducible representation of a postliminal C^* -algebra is traceable (4.3.7), and hence every factor representation of a postliminal C^* -algebra is traceable (5.5.3).

6.7.6. Let A be a postliminal C^* -algebra, and let π be a nonzero irreducible representation of A . For every $x \in A^+$, put

$$f_\pi(x) = \text{Tr } \pi(x).$$

Then f_π is a character of A (6.7.3 and 6.7.5). If π' is another irreducible representation of A and $f_\pi = f_{\pi'}$, then π and π' are quasi-equivalent (6.6.6), and hence equivalent (5.3.3). Finally, if g is a character of A , then λ_g is nonzero and factorial (6.7.3); it is therefore quasi-equivalent to a nonzero irreducible representation π (5.5.3), and consequently g is proportional to f_π (6.7.4).

6.7.7. Retain the notation of 6.7.6, and let $f = f_\pi$. Since $\pi(A) \supset \mathcal{LC}(H_\pi)$, λ_f passes to the quotient and defines an isomorphism from $\mathfrak{n}_f/N_f$ onto the Hilbert algebra B of Hilbert–Schmidt operators on H_π . Identify $\mathfrak{n}_f/N_f$ with B by this isomorphism. Then, for every $x \in A$, $\lambda_f(x)$ is the operator of left multiplication by $\pi(x)$ on B , and $\bar{\rho}_f^\circ(x)$ is the operator of right multiplication by $\pi(x)^*$ on the conjugate space of B . If B is now canonically identified with $H_\pi \otimes \overline{H_\pi}$, then $\lambda_f(x)$ becomes $\pi(x) \otimes 1$ and $\bar{\rho}_f^\circ(x)$ becomes $1 \otimes \pi(x)$. Thus $\mathfrak{U}_f$ is identified with $\mathcal{L}(H_\pi) \otimes \mathbb{C}$, $\mathfrak{V}_f$ with $\mathbb{C} \otimes \mathcal{L}(\overline{H_\pi})$, and J_f with the canonical involution of $H_\pi \otimes \overline{H_\pi}$. Finally,

$$t_f(T \otimes 1) = \text{Tr } T \quad \text{for every } T \in \mathcal{L}(H_\pi)^+.$$

6.7.8. Let A be a commutative C^* -algebra. By 6.7.6, the characters of A in the sense of 6.7.1 are the functions $x \mapsto \alpha g(x)$ on A^+ , where $\alpha > 0$ and g is a character of A in the classical sense. There is therefore a slight inconsistency in the terminology.

6.7.9. The reader will have noticed an analogy between the developments of this section and those of Section 2; the correspondence is as follows:

Section 2	Section 6
Positive functionals	Semifinite lower semicontinuous traces
Pure positive functionals	Characters
Representations	Traceable representations
Irreducible representations	Traceable factor representations

Unfortunately, the analogy breaks down at an essential point. Whereas in Section 2 one can prove the existence of pure positive functionals by the Krein–Milman theorem, characters do not always exist (see 6.9.2, 6.9.10, and 9.5.7). The question has been settled only for postliminal C^* -algebras, since in that case irreducible representations, whose existence has been proved, are automatically traceable.

On the other hand, the theories of Sections 2 and 6 have a common special case, which we now study.

Bibliography: [54, 57].

6.8. Finite Traces

6.8.1. Let A be a C^* -algebra and let g be a linear functional on A . We say that g is *central* if $g(xy) = g(yx)$ for all $x, y \in A$. Suppose g is positive and central. Then the restriction f of g to A^+ is plainly a finite trace. Conversely, let f be a finite trace on A^+ . Apply Proposition 6.1.2 and its notation. We have $m = A$ (1.5.8), and hence f' is a linear functional on A extending f , and is therefore positive and central [6.1.2(iii)]. Thus we have established a bijective correspondence between central positive functionals g on A and finite traces f on A^+ ; we shall sometimes identify f and g . By 2.1.8, finite traces are automatically continuous.

6.8.2. Let f be a lower semicontinuous trace on A^+ , and let s be the associated bitrace. We have $m_f = \mathfrak{n}_s^2 \subset \mathfrak{n}_s$. By 1.5.8, the conditions $m_f = A$ and $\mathfrak{n}_s = A$ are equivalent. The condition $m_f = A$ plainly means that f is finite.

6.8.3. Proposition. *Let A be a C^* -algebra, let f be a central positive functional on A , and put $g = f|_{A^+}$.*

- (i) *One has $\pi_f = \lambda_g$.*
- (ii) *The trace t_g on $\mathfrak{U}_g^+$ is finite. If t'_g denotes its linear extension to $\mathfrak{U}_g$, then*

$$t'_g(\pi_f(z)) = (\pi_f(z)\xi_f \mid \xi_f) \quad (z \in A).$$

- (iii) *$\mathfrak{U}_g$ and $\mathfrak{B}_g$ are finite von Neumann algebras.*
- (iv) *For every $z \in A$,*

$$\lambda_g(z)\xi_f = \rho_g(z)\xi_f = \Lambda_g z, \quad J_g \xi_f = \xi_f.$$

- (v) *The vector ξ_f is separating and cyclic for $\mathfrak{U}_g$ and $\mathfrak{B}_g$.*

Let s be the bitrace associated with g . The space $H(\pi_f)$ is the Hilbert space obtained by separation and completion of A equipped with the inner product

$$f(y^*x) = f(xy^*) = s(x, y).$$

Thus $H(\pi_f) = H_s = H_f$. The representations π_f and λ_g are both obtained from the left regular representation of A , and hence $\pi_f = \lambda_g$. For $x, y \in A$, writing t'_g for the linear extension of t_g to $\mathfrak{m}_{t_g}$, we have

$$\begin{aligned} t'_g(\lambda_g(x)\lambda_g(y)^*) &= s(x, y) = f(xy^*) \\ &= (\pi_f(xy^*)\xi_f \mid \xi_f) = (\pi_f(x)\pi_f(y)^*\xi_f \mid \xi_f). \end{aligned} \quad (1)$$

Passing first to weak limits, we deduce that

$$(ST\xi_f \mid \xi_f) = (TS\xi_f \mid \xi_f) \quad (S, T \in \mathfrak{U}_g).$$

Thus $S \mapsto (S\xi_f \mid \xi_f)$ is a finite normal trace on $\mathfrak{U}_g^+$. Moreover, (1) proves that $t'_g(\pi_f(z)) = (\pi_f(z)\xi_f \mid \xi_f)$ for every $z \in A$.

Let $\tilde{A}$ be the C^* -algebra obtained from A by adjoining an identity element 1; let $\tilde{f}$ be the canonical extension of f to $\tilde{A}$, and let $\tilde{g} = \tilde{f}|_{\tilde{A}^+}$. The space $H(\pi_f)$ is also the Hilbert space obtained by separation and completion of $\tilde{A}$ equipped with the inner product $\tilde{f}(y^*x) = \tilde{f}(xy^*)$, and ξ_f is the canonical image of 1. Let $z \in A$. Since the operators of left and right multiplication by z are continuous on the pre-Hilbert space $\tilde{A}$, as is the involution, we have

$$\lambda_g(z)\xi_f = \lambda_{\tilde{g}}(z \cdot 1) = \Lambda_g z, \quad \rho_g(z)\xi_f = \rho_{\tilde{g}}(1 \cdot z) = \Lambda_g z,$$

and

$$J_g \xi_f = \Lambda_{\tilde{g}} 1^* = \Lambda_{\tilde{g}} 1 = \xi_f.$$

It follows that ξ_f is cyclic for $\mathfrak{U}_g$ and $\mathfrak{B}_g$, and therefore separating for $\mathfrak{U}_g = \mathfrak{B}'_g$ and $\mathfrak{B}_g = \mathfrak{U}'_g$. The normal trace $S \mapsto (S\xi_f \mid \xi_f)$ on $\mathfrak{U}_g^+$ is consequently faithful, and it agrees on $\pi_f(A^+)$ with the faithful normal trace t_g ; hence the two traces are equal (A 29). Thus t_g is a finite trace, and the von Neumann algebras $\mathfrak{U}_g, \mathfrak{B}_g$ are finite.

6.8.4. Corollary. *Let A be a C^* -algebra, and let f be a semifinite lower semicontinuous trace on A^+ . In order that t_f be finite on $\mathfrak{U}_f^+$, it is necessary and sufficient that f be finite.*

The necessity is clear, since $f(x) = t_f(\lambda_f(x))$ for every $x \in A^+$. The sufficiency follows from 6.8.3.

6.8.5. Corollary. *Let A be a C^* -algebra and let f be a character of A . In order that f be of finite type, it is necessary and sufficient that f be finite.*

This is a special case of 6.8.4.

6.8.6. Corollary. *There is a canonical bijective correspondence between*

- a. the set of quasi-equivalence classes of nonzero finite-type factor representations of A ;*
- b. the set of finite characters of norm 1 of A .*

This follows from 6.7.4 and 6.8.5.

Thus the character of a nonzero factor representation of finite type can be normalized by requiring it to have norm 1. This normalization is sometimes incompatible with that of 6.7.5.

6.8.7. Proposition. *Let A be a C^* -algebra, and let C be the set of central positive functionals on A of norm at most 1.*

- (i) C is convex and compact for the weak topology of the dual of A .*
- (ii) The extreme points of C are 0 and the finite characters of norm 1.*
- (iii) C is the weakly closed convex hull of 0 and of the set of finite characters of norm 1.*

Assertion (i) is immediate; (iii) follows from (i), (ii), and the Krein–Milman theorem. On the other hand, by 6.7.1, a finite character of A is a nonzero finite trace f on A^+ such that every finite trace majorized by f is proportional to f . This being so, the proof of (ii) is exactly the same as the corresponding proof of 2.5.5.

Bibliography: [54, 57, 111].

6.9. Complements

6.9.1. Let A be a C^* -algebra, let f be a semifinite lower semicontinuous trace on A^+ , and let s be the bitrace associated with f . We shall see that the choice of a closed two-sided ideal J of A defines a decomposition of f .

a. Let $f' = f|_{J^+}$, and let s' be the bitrace of J associated with f' . Then s' is a bitrace of A . Let s_1 be its canonical maximal extension and let f_1 be the associated trace on A^+ .

b. One has

$$H_{f_1} = H_{f'} \subset H_f, \quad \mathfrak{U}_{f_1} = \mathfrak{U}_{f'}, \quad t_{f_1} = t_{f'}.$$

For every $x \in A$, $\lambda_{f_1}(x)$ is the restriction of $\lambda_f(x)$ to $H_{f'}$. If $x \in A^+$, then

$$f_1(x) = t_f(P\lambda_f(x)),$$

where P denotes the orthogonal projection of H_f onto $H_{f'}$.

c. Let (u_i) be an increasing directed approximate identity of $\overline{\mathfrak{m}_{f'}}$. If $x \in A^+$, then

$$f_1(x) = \lim_i f'(x^{1/2}u_i x^{1/2}).$$

d. For every $x \in A^+$, put

$$g(x) = t_f((1 - P)\lambda_f(x)).$$

Then g is a semifinite lower semicontinuous trace on A^+ which vanishes on J^+ , and $f = f_1 + g$. [16]

6.9.2. Let H be a separable infinite-dimensional Hilbert space, and put

$$A = \mathcal{L}(H)/\mathcal{LC}(H).$$

Let t be a trace on A^+ such that $\mathfrak{m}_t \neq 0$. Then $t = 0$. [Every two-sided ideal of A is equal to A , so $\mathfrak{m}_t = A$. Let e_1, e_2 be two infinite-rank orthogonal projections with sum 1 on H , and let f_1, f_2 be their canonical images in A . Then $t(f_1) = t(1)$ and $t(f_2) = t(1)$, whence $t(1) = 2t(1)$ and $t(1) = 0$.] [46]

6.9.3. Let A be the C^* -algebra of continuous complex-valued functions on $[0, 1]$. For $x \in A^+$, put

$$f(x) = \int_0^1 x(t)t^{-1} dt.$$

Then f is a semifinite lower semicontinuous trace, but $\overline{m_f} \neq A$. If, for $x \in A^+$, one puts $f'(x) = x(0)$, then f' is a finite continuous trace and $f + f' = f$. [16]

***6.9.4.** Let A be a C^* -algebra. Two traceable irreducible representations of A having the same kernel are equivalent (6.7.5), but two traceable factor representations of A having the same kernel may be disjoint. [57]

6.9.5. Return to the C^* -algebra B of 4.7.18. The identity representation of B is irreducible; let f be its character. Let f' be a character obtained by composing the canonical map $B \rightarrow B/\mathcal{LC}(H)$ with a character of the commutative C^* -algebra $B/\mathcal{LC}(H)$. Then f majorizes f' without being proportional to it.

6.9.6. Let A be a C^* -algebra and let f be a character of A . In order that there exist a family (f_i) of pure positive functionals on A such that $f = \sum_i f_i$, it is necessary and sufficient that f be of type I. [309]

6.9.7. Let A be a separable C^* -algebra.

- a. The set E of $\pi \in \text{Rep}(A)$ for which π is irreducible and traceable is a Borel subset of $\text{Rep}(A)$.
- b. Let E', E'' be the canonical images of E in $\widehat{A}$ and $\text{Prim}(A)$. Then E' is Borel for the Mackey Borel structure, and E'' is Borel in $\text{Prim}(A)$ (see 3.9.2). The set E' is the largest subset of $\widehat{A}$ on which the restriction of $\widehat{A} \rightarrow \text{Prim}(A)$ is injective (4.7.2). In view of 3.9.2, E' is a standard Borel subset of $\widehat{A}$ on which the Mackey Borel structure is the Borel structure underlying the topology. [57]

6.9.8. Let A be a separable C^* -algebra. The set of $\pi \in \text{Rep}(A)$ such that $\pi(A) = \mathcal{LC}(H_\pi)$ is a Borel subset of $\text{Rep}(A)$ (Guichardet, unpublished).

6.9.9. Every representation of a postliminal C^* -algebra is traceable. (Use the proof of 5.5.2.)

6.9.10. Let E_1 be the class of separable C^* -algebras all of whose representations are of type I. Let E'_1 be the class of separable C^* -algebras admitting sufficiently many representations of type I [that is, such that, for every nonzero x in the algebra, there exists a representation π of type I of the algebra such that $\pi(x) \neq 0$]. By successively replacing “of type I” by “semifinite,” “finite,” and “traceable,” one obtains the classes $E_2, E'_2, E_3, E'_3, E_4, E'_4$ of C^* -algebras. What can be said about these classes?

The class E_1 is well known (see 9.1). We have $E_1 = E_2 = E_4$ by 6.9.9, 9.1, and 9.5.4. The class E'_1 (and *a fortiori* E'_2) is the class of all separable C^* -algebras, since an irreducible representation is of type I. Next, $E'_3 \subset E'_2 = E'_1$ is the class of separable C^* -algebras all of whose irreducible representations are finite-dimensional. The classes E_4, E'_4 are distinct from the class of all separable C^* -algebras, but are poorly understood. [200]

§ 7. — Quasi-Spectrum

Let A be a separable C^* -algebra. In Section 3, we defined the Mackey Borel structure on the set of irreducible representations of A , or rather on the set of their equivalence classes. We shall now define a Borel structure on the set of factor representations of A , or rather on the set of their quasi-equivalence classes. This will be useful in Section 8.

7.1. The Space of Factor Representations

7.1.1. Let n be a cardinal $\leq \aleph_0$, and let H_n be the model Hilbert space of dimension n (3.5.1). For every $r \geq 0$, denote by $\mathcal{L}_r(H_n)$ the closed ball of center 0 and radius r in $\mathcal{L}(H_n)$.

Lemma. *Let d be a metric on $\mathcal{L}_1(H_n)$ compatible with the weak topology (B 9), let $S \in \mathcal{L}_1(H_n)$, let $a, b \geq 0$, and let $x_1, \dots, x_m \in A$. Let Y be the set of $\pi \in \text{Rep}_n(A)$ having the following property: there exist $T_1, \dots, T_m \in \mathcal{L}_b(H_n) \cap \pi(A)'$ such that*

$$\pi(x_1)T_1 + \dots + \pi(x_m)T_m \in \mathcal{L}_1(H_n)$$

and

$$d(\pi(x_1)T_1 + \dots + \pi(x_m)T_m, S) \leq a.$$

Then Y is closed in $\text{Rep}_n(A)$.

Let (π^i) be a net of elements of Y , indexed by a directed set, and let $\pi \in \text{Rep}_n(A)$, with $\pi^i \rightarrow \pi$ in $\text{Rep}_n(A)$. We prove that $\pi \in Y$. For each i , there exist

$$T_1^i, \dots, T_m^i \in \mathcal{L}_b(H_n) \cap \pi^i(A)'$$

such that

$$\pi^i(x_1)T_1^i + \dots + \pi^i(x_m)T_m^i \in \mathcal{L}_1(H_n)$$

and

$$d(\pi^i(x_1)T_1^i + \cdots + \pi^i(x_m)T_m^i, S) \leq a.$$

Since $\mathcal{L}_b(H_n)$ is weakly compact, we may, on passing to a subnet, suppose that $T_1^i, \dots, T_m^i$ converge weakly to $T_1, \dots, T_m \in \mathcal{L}_b(H_n)$. Let $x \in A$ and $\varphi, \psi \in H_n$. We have

$$\begin{aligned} (T_1^i \pi^i(x) \varphi \mid \psi) &= (\pi^i(x) \varphi \mid T_1^{i*} \psi) \longrightarrow (\pi(x) \varphi \mid T_1^* \psi) = (T_1 \pi(x) \varphi \mid \psi), \\ (\pi^i(x) T_1^i \varphi \mid \psi) &= (T_1^i \varphi \mid \pi^i(x^*) \psi) \longrightarrow (T_1 \varphi \mid \pi(x^*) \psi) = (\pi(x) T_1 \varphi \mid \psi) \quad (3.5.2). \end{aligned}$$

Therefore

$$(T_1 \pi(x) \varphi \mid \psi) = (\pi(x) T_1 \varphi \mid \psi), \quad \text{and hence } T_1 \in \pi(A)'.$$

Similarly, $T_2, \dots, T_m \in \pi(A)'$. Moreover,

$$((\pi^i(x_1)T_1^i + \cdots + \pi^i(x_m)T_m^i) \varphi \mid \psi) \longrightarrow ((\pi(x_1)T_1 + \cdots + \pi(x_m)T_m) \varphi \mid \psi).$$

Thus $\pi^i(x_1)T_1^i + \cdots + \pi^i(x_m)T_m^i$ converges weakly to $\pi(x_1)T_1 + \cdots + \pi(x_m)T_m$, whence

$$\pi(x_1)T_1 + \cdots + \pi(x_m)T_m \in \mathcal{L}_1(H_n)$$

and

$$d(\pi(x_1)T_1 + \cdots + \pi(x_m)T_m, S) \leq a.$$

Therefore $\pi \in Y$.

7.1.2. Lemma. *Let d be a metric on $\mathcal{L}_1(H_n)$ compatible with the weak topology, let $(x_1, x_2, \dots)$ be a dense sequence in A , and let $(S_1, S_2, \dots)$ be a weakly dense sequence in $\mathcal{L}_1(H_n)$ (B 9). Let k, m, p be integers. Let $Y_{k,m,p}$ be the set of $\pi \in \text{Rep}_n(A)$ having the following property: there exist $T_1, T_2, \dots, T_m \in \mathcal{L}_m(H_n) \cap \pi(A)'$ such that*

$$\pi(x_1)T_1 + \cdots + \pi(x_m)T_m \in \mathcal{L}_1(H_n)$$

and

$$d(\pi(x_1)T_1 + \cdots + \pi(x_m)T_m, S_k) \leq \frac{1}{p}.$$

- (i) $Y_{k,m,p}$ is closed in $\text{Rep}_n(A)$.
- (ii) The set

$$Y = \bigcap_{k,p} \bigcup_m Y_{k,m,p}$$

is the set of nonzero factor representations in $\text{Rep}_n(A)$.

Assertion (i) follows from 7.1.1.

For every $\pi \in \text{Rep}_n(A)$, let $N(\pi)$ be the set of linear combinations of operators of the form RR' , where $R \in \pi(A)$ and $R' \in \pi(A)'$. Let $\mathfrak{M}(\pi)$ be the weak closure of $N(\pi)$. Then π is nonzero and factorial if and only if $\mathfrak{M}(\pi) = \mathcal{L}(H_n)$.

First suppose $\pi \in \mathcal{Y}$. We shall prove that $S_k \in \mathfrak{M}(\pi)$ for every k , and this will imply that $\mathfrak{M}(\pi) = \mathcal{L}(H_n)$. Let p be a positive integer. There exists m such that $\pi \in Y_{k,m,p}$. Hence there exist $T_1, \dots, T_m \in \mathcal{L}_m(H_n) \cap \pi(A)'$ such that

$$\pi(x_1)T_1 + \dots + \pi(x_m)T_m \in \mathcal{L}_1(H_n)$$

and

$$d(\pi(x_1)T_1 + \dots + \pi(x_m)T_m, S_k) \leq \frac{1}{p}.$$

Since $\pi(x_1)T_1 + \dots + \pi(x_m)T_m \in N(\pi)$, we have

$$d(N(\pi) \cap \mathcal{L}_1(H_n), S_k) \leq \frac{1}{p}.$$

Since p is arbitrary, this proves that $S_k \in \mathfrak{M}(\pi)$. Thus π is nonzero and factorial.

Conversely, suppose $\pi \in \text{Rep}_n(A)$ and $\mathfrak{M}(\pi) = \mathcal{L}(H_n)$. Let k, p be positive integers. We shall prove that there exists an m such that $\pi \in Y_{k,m,p}$; this will complete the proof. By A 13, there exists $P \in N(\pi)$ such that $\|P\| < 1$ and $d(P, S_k) < 1/p$. We have

$$P = \pi(y_1)T_1 + \dots + \pi(y_q)T_q,$$

where

$$y_1, \dots, y_q \in A, \quad T_1, \dots, T_q \in \pi(A)'.$$

There then exist $x_{i_1}, \dots, x_{i_q}$ sufficiently close to $y_1, \dots, y_q$ that

$$\|\pi(x_{i_1})T_1 + \dots + \pi(x_{i_q})T_q\| \leq 1$$

and

$$d(\pi(x_{i_1})T_1 + \dots + \pi(x_{i_q})T_q, S_k) \leq \frac{1}{p}.$$

If $m \geq i_1, \dots, i_q, \|T_1\|, \dots, \|T_q\|$, then $\pi \in Y_{k,m,p}$, as required.

7.1.3. Let A be a separable C^* -algebra. We denote by $\text{Fac}_n(A)$ the set of nonzero factor representations of A on H_n , and by $\text{Fac}(A)$ the union of the $\text{Fac}_n(A)$ for $n = 1, 2, \dots, \aleph_0$.

7.1.4. Theorem. *Let A be a separable C^* -algebra. The set $\text{Fac}(A)$ is Borel in $\text{Rep}(A)$.*

This follows immediately from 7.1.2.

Thus $\text{Fac}(A)$, equipped with the Borel structure induced by that of $\text{Rep}(A)$, is a standard Borel space, since $\text{Rep}(A)$ is standard (3.8.1).

Bibliography: [25].

7.2. Definition of the Quasi-Spectrum

7.2.1. We denote by $\check{A}$ the set of quasi-equivalence classes of nonzero factor representations of A . If A is separable, every factor representation is quasi-equivalent to a factor representation on a separable space (5.3.7); hence the evident canonical map

$$\text{Fac}(A) \longrightarrow \check{A}$$

is surjective.

7.2.2. Definition. *Let A be a separable C^* -algebra. The Mackey Borel structure on $\check{A}$ is the quotient structure of that on $\text{Fac}(A)$ under the canonical map $\text{Fac}(A) \rightarrow \check{A}$. The Borel space $\check{A}$ is called the quasi-spectrum of A .*

7.2.3. Proposition. *Let A be a separable C^* -algebra. Let B be a Borel subset of $\text{Fac}(A)$ which meets each quasi-equivalence class in at most one point, and let $\psi: \text{Fac}(A) \rightarrow \check{A}$ be the canonical map. Then $\psi(B)$ is Borel in $\check{A}$, and $\psi|_B$ is an isomorphism of the Borel space B onto the Borel space $\psi(B)$.*

Let $\mathcal{E}(\pi, \pi')$ be the intertwining number of two representations π, π' of A . By 3.7.3, the set of pairs $(\pi, \pi') \in \text{Fac}(A) \times \text{Fac}(A)$ such that $\mathcal{E}(\pi, \pi') \leq 0$ is Borel in $\text{Fac}(A) \times \text{Fac}(A)$. Its complement R is the set of pairs (π, π') in that product such that $\pi \approx \pi'$ (5.3.6). Thus $S = R \cap (\text{Fac}(A) \times B)$ is Borel in $\text{Fac}(A) \times \text{Fac}(A)$. Let p be the first projection from this product onto $\text{Fac}(A)$. If $(\pi, \pi') \in S$ and $(\pi_1, \pi'_1) \in S$ satisfy $p(\pi, \pi') = p(\pi_1, \pi'_1)$, then $\pi' \approx \pi = \pi_1 \approx \pi'_1$, and hence $\pi' = \pi'_1$ by the hypothesis on B . Thus $p|_S$ is injective. Consequently $p(S)$ is a Borel subset of $\text{Fac}(A)$ (B 21). But $p(S)$ is the saturation of B in $\text{Fac}(A)$ for quasi-equivalence. Hence $\psi(B)$ is Borel in $\check{A}$. The map $\psi|_B$ from B onto $\psi(B)$ is a Borel bijection. To prove that it is a Borel isomorphism, it is enough to prove that the image $\psi(B')$ of every Borel subset B' of B is Borel. This follows from the preceding argument applied to B' in place of B .

7.2.4. Corollary. *Every point of $\check{A}$ is Borel in $\check{A}$.*

The quasi-spectrum has hitherto been called the quasi-dual.
Bibliography: [25].

7.3. Relations Between Spectrum and Quasi-Spectrum

7.3.1. Let A be a separable C^* -algebra. For $p = 1, 2, \dots, \infty$, let $\text{FacI}_p(A)$ be the set of $\pi \in \text{Fac}(A)$ for which $\pi(A)'$ is a factor of type I_p . We have $\text{FacI}_1(A) = \text{Irr}(A)$. Let $\text{FacI}(A)$ be the union of the $\text{FacI}_p(A)$. Then $\text{FacI}(A)$ is the set of $\pi \in \text{Fac}(A)$ for which $\pi(A)'$ is of type I, that is, the set of representations in $\text{Fac}(A)$ which are of type I. The sets $\text{FacI}_p(A)$ are saturated under equivalence of representations, but not under quasi-equivalence. In contrast, $\text{FacI}(A)$ is saturated under quasi-equivalence. A representation in $\text{Rep}(A)$ belongs to $\text{FacI}_p(A)$ if and only if it is equivalent to a representation of the form $p\pi$, where $\pi \in \text{Irr}(A)$ (5.4.11). Thus all the sets $\text{FacI}_p(A)$ have the same canonical image in $\check{A}$ as $\text{Irr}(A)$. Denote this image by $\check{A}_I$; it is the set of quasi-equivalence classes of factor representations of type I.

7.3.2. Recall that, on the set of unitary operators on a Hilbert space, the weak and strong topologies coincide and are compatible with the group structure. This being so, we have the following lemma.

Lemma. *Let H be a separable Hilbert space and let $\mathcal{G}$ be the group of unitary operators on H . With the strong (or weak) topology, $\mathcal{G}$ is a Polish group.*

Indeed, the unit ball $\mathcal{L}_1(H)$ of $\mathcal{L}(H)$ is compact and metrizable for the weak topology, and is therefore a Polish space. It is enough to prove that $\mathcal{G}$ is a G_δ in $\mathcal{L}_1(H)$. If (ξ_i) is a dense sequence in the unit sphere of H , then $\mathcal{G}$ is the set of $T \in \mathcal{L}_1(H)$ such that

$$\|T\xi_i\| > 1 - \frac{1}{j}, \quad \|T^*\xi_i\| > 1 - \frac{1}{j}$$

for all integers i, j . Each of these inequalities defines a weakly open subset of $\mathcal{L}_1(H)$, since for every $\xi \in H$ the map $T \mapsto \|T\xi\|$ is lower semicontinuous for the weak topology, and $T \mapsto T^*$ is a homeomorphism for the weak topology.

7.3.3. Lemma. *Let A be a separable C^* -algebra, let p be a cardinal $\leq \aleph_0$, and let B be a Borel subset of $\text{Irr}(A)$ saturated under equivalence. Let C be the set of elements of $\text{FacI}_p(A)$ quasi-equivalent to an element of B . Then C is Borel in $\text{Rep}(A)$.*

Let H_n be the model Hilbert space of dimension n . The sets $B \cap \text{Rep}_n(A)$ are saturated under equivalence and are Borel. It is enough to argue on each of these sets. In other words, we shall henceforth suppose that the elements of B act on one and the same H_n . Then the elements of C act on H_{np} . We give the proof for $n = p = \aleph_0$; the other cases are handled similarly and somewhat more simply. Put $H = H_n = H_{np}$.

Let $(K_1, K_2, \dots)$ be an infinite sequence of pairwise orthogonal closed infinite-dimensional vector subspaces of H whose Hilbert direct sum is H . Let J_n be an isomorphism from H onto K_n . For every $T \in \mathcal{L}(H)$, the operators $J_n T J_n^{-1}$ define, by linearity and continuity, an operator $T^\sim$ on H , and $T \mapsto T^\sim$ is an isomorphism from $\mathcal{L}(H)$ onto a factor $\mathfrak{F}$ of type I_∞ : namely, the set of bounded linear operators on H which leave the K_n invariant and whose restrictions to the K_n are carried into one another by the isomorphisms $J_n J_m^{-1}$. The commutant $\mathfrak{F}'$ of $\mathfrak{F}$ is also a factor of type I_∞ . Let π be a representation of A on H . Denote by $\pi^\sim$ the representation $x \mapsto \pi(x)^\sim$ of A on H . If $\pi \in \text{Irr}(A)$, then $\pi^\sim(A)$ generates the factor $\mathfrak{F}$. The map $\pi \mapsto \pi^\sim$ is plainly injective. For every ξ belonging to one of the K_i and every $x \in A$, the map $\pi \mapsto \pi^\sim(x)\xi$ is continuous. Thus $\pi \mapsto \pi^\sim$ is Borel. By B 21, the image $B^\sim$ of B under this map is a Borel subset of $\text{Rep}(A)$ contained in $\text{Fac}I_\infty(A)$.

The set C in the statement is the union of the $UB^\sim U^{-1}$ as U ranges over the group $\mathcal{G}$ of unitary operators on H (5.3.3 and 5.4.11). In other words, C is the image of $\mathcal{G} \times B^\sim$ under the map

$$\Theta: (U, \sigma) \mapsto U\sigma U^{-1} \quad \text{from } \mathcal{G} \times B^\sim \text{ into } \text{Rep}(A).$$

For every $x \in A$ and every $\xi \in H$, the map

$$(U, \sigma) \mapsto U\sigma(x)U^{-1}\xi \quad \text{from } \mathcal{G} \times \text{Rep}_\infty(A) \text{ into } H$$

is continuous, where $\mathcal{G}$ is equipped with the strong topology; hence Θ is Borel. We shall prove that, for $U \in \mathcal{G}$, the following conditions are equivalent, provided $B \neq \emptyset$, to which case we may plainly restrict ourselves:

- (1) $U\mathfrak{F}U^{-1} = \mathfrak{F}$;
- (2) $U = VW$, where V, W are unitary, $V \in \mathfrak{F}$ and $W \in \mathfrak{F}'$;
- (3) $UB^\sim U^{-1} = B^\sim$;
- (4) there exist $\pi_1, \pi_2 \in B^\sim$ such that $U\pi_1 U^{-1} = \pi_2$.

(4) $\Rightarrow$ (1): let $\pi_1, \pi_2 \in B^\sim$ satisfy $U\pi_1 U^{-1} = \pi_2$. Then $U\pi_1(A)U^{-1} = \pi_2(A)$, and, on taking weak closures, $U\mathfrak{F}U^{-1} = \mathfrak{F}$.

(1) $\Rightarrow$ (2): if $U\mathfrak{F}U^{-1} = \mathfrak{F}$, then U defines an automorphism of $\mathfrak{F}$. But every automorphism of a factor of type I is implemented by a unitary operator belonging to that factor (A 37). There is therefore a unitary $V \in \mathfrak{F}$ such that $W = V^{-1}U$ implements the identity automorphism of $\mathfrak{F}$, that is, $W \in \mathfrak{F}'$, and $U = VW$.

(2) $\Rightarrow$ (3): let $U = VW$, where V, W are unitary, $V \in \mathfrak{F}$, and $W \in \mathfrak{F}'$. Let $\pi \in B$. We have $W\pi(x)^\sim W^{-1} = \pi(x)^\sim$ for every $x \in A$. Moreover, V is of the form $T^\sim$ for a unitary operator T on H . Hence

$$V\pi(x)^\sim V^{-1} = (T\pi(x)T^{-1})^\sim,$$

and consequently

$$U\pi^\sim U^{-1} = (T\pi T^{-1})^\sim.$$

But $T\pi T^{-1} \in B$, since B is saturated under equivalence. Hence $U\pi^\sim U^{-1} \in B^\sim$, and finally $UB^\sim U^{-1} = B^\sim$.

(3) $\Rightarrow$ (4) is evident.

Conditions (1)–(4) define a subgroup $\mathcal{G}'$ of $\mathcal{G}$. From condition (1), it is clear that $\mathcal{G}'$ is closed in $\mathcal{G}$. By 7.3.2 and B 17, there is a Borel subset Σ of $\mathcal{G}$ which meets each left coset of $\mathcal{G}'$ in exactly one point. We prove that

$$\Theta(\mathcal{G} \times B^\sim) = \Theta(\Sigma \times B^\sim).$$

It is clear that the left-hand side contains the right-hand side. Conversely, let $U \in \mathcal{G}$ and $\sigma \in B^\sim$. We have $U = U_1 U_2$, with $U_1 \in \Sigma$ and $U_2 \in \mathcal{G}'$. Hence

$$\Theta(U, \sigma) = U_1(U_2\sigma U_2^{-1})U_1^{-1} \in \Theta(\Sigma \times B^\sim)$$

by condition (3), proving the asserted equality.

We next prove that the restriction of Θ to $\Sigma \times B^\sim$ is injective. Let $\pi_1, \pi_2 \in B^\sim$ and $V_1, V_2 \in \Sigma$ satisfy

$$V_1\pi_1 V_1^{-1} = V_2\pi_2 V_2^{-1}.$$

Then $V_2^{-1}V_1$ satisfies condition (4), so V_1 and V_2 belong to the same left coset modulo $\mathcal{G}'$. Hence $V_1 = V_2$, and then $\pi_1 = \pi_2$. It follows that $C = \Theta(\Sigma \times B^\sim)$ is Borel in $\text{Rep}(A)$ (B 21).

7.3.4. Proposition. *Let A be a separable C^* -algebra. Each set $\text{FacI}_p(A)$ is Borel in $\text{Rep}(A)$.*

This follows from 7.3.3, applied with $B = \text{Irr}(A)$.

7.3.5. Equip the sets $\text{FacI}_p(A)$ and

$$\text{FacI}(A) = \bigcup_p \text{FacI}_p(A)$$

with the Borel structures induced by that of $\text{Rep}(A)$, or of $\text{Fac}(A)$. The resulting Borel spaces are standard.

7.3.6. The equivalence relation induced on $\text{Irr}(A)$ by quasi-equivalence is just equivalence of representations (5.3.3). Thus the canonical injection i of $\text{Irr}(A)$ into $\text{FacI}(A)$ passes to the quotient and defines a bijection Φ from $\widehat{A}$ onto $\check{A}_I$ (7.3.1).

Theorem. *Let A be a separable C^* -algebra.*

- (i) $\check{A}_I$ is a Borel subset of $\check{A}$.
- (ii) *The bijection Φ is an isomorphism of $\widehat{A}$ onto $\check{A}_I$ for the Mackey Borel structures.*

Assertion (i) follows from 7.3.4. We prove (ii). Let

$$q: \text{Irr}(A) \longrightarrow \widehat{A}, \quad p: \text{FacI}(A) \longrightarrow \check{A}_I$$

be the canonical maps. The diagram is

$$\begin{array}{ccc} \text{Irr}(A) & \xrightarrow{i} & \text{FacI}(A) \\ q \downarrow & & \downarrow p \\ \widehat{A} & \xrightarrow{\Phi} & \check{A}_I. \end{array}$$

Let S be a Borel subset of $\check{A}_I$. Then $p^{-1}(S)$ is a Borel subset of $\text{FacI}(A)$ saturated under quasi-equivalence. Thus $i^{-1}(p^{-1}(S)) = q^{-1}(\Phi^{-1}(S))$ is a Borel subset of $\text{Irr}(A)$ saturated under equivalence. Hence $\Phi^{-1}(S)$ is Borel in $\widehat{A}$. Conversely, let T be a Borel subset of $\widehat{A}$. Then $q^{-1}(T)$ is a Borel subset of $\text{Irr}(A)$ saturated under equivalence. The set $p^{-1}(\Phi(T))$ is the saturation of $i(q^{-1}(T))$ under quasi-equivalence. Applying 7.3.3 with $B = q^{-1}(T)$, we see that $p^{-1}(\Phi(T))$ is Borel in $\text{FacI}(A)$. Therefore $\Phi(T)$ is Borel in $\check{A}_I$. This proves the theorem.

Henceforth we shall identify the Borel space $\widehat{A}$ with the Borel subset $\check{A}_I$ of $\check{A}$.

7.3.7. In particular, if A is a separable postliminal C^* -algebra, the Borel spaces $\widehat{A}$ and $\check{A}$ are identified (5.5.3).

Bibliography: [15].

7.4. Finite Part of the Quasi-Spectrum

7.4.1. It is not very clear what topology should be placed on $\check{A}$. We shall, however, define a subset $\check{A}_f$ of $\check{A}$ on which there is a fairly satisfactory topology.

Let A be a separable C^* -algebra, let H_n be the model Hilbert space of dimension n , let $\text{Fac}_n(A)$ be the set of nonzero finite-type factor representations of A

on H_n , and let $\text{Facf}(A)$ be the union of the $\text{Facf}_n(A)$ for $n = 1, 2, \dots, \aleph_0$. [In fact, $\text{Facf}_n(A) = \text{Fac}_n(A)$ for $n < \aleph_0$.] Denote by $\check{A}_f$ the canonical image of $\text{Facf}(A)$ in $\check{A}$, that is, the set of quasi-equivalence classes of nonzero finite-type factor representations of A .

7.4.2. On the other hand, let C be the convex set of central positive functionals on A of norm at most 1. Then C , equipped with the weak topology, is a compact metrizable space. The set D of nonzero extreme points of C is a G_δ in C (B 13), and is therefore a Polish space for the weak topology. Moreover, D is the set of finite characters of norm 1 of A (6.8.7). By 6.8.6, the map

$$g \longmapsto \text{the quasi-equivalence class of } \lambda_g$$

is a bijection from D onto $\check{A}_f$. We equip $\check{A}_f$ with the topology transported from the weak topology on D by this bijection. Thus $\check{A}_f$ becomes a Polish space.

7.4.3. Proposition. *Let A be a separable C^* -algebra.*

- (i) $\check{A}_f$ is a Borel subset of $\check{A}$.
- (ii) The Borel structure induced on $\check{A}_f$ by that of $\check{A}$ is the Borel structure underlying the topology of $\check{A}_f$.
- (iii) The Borel space $\check{A}_f$ is standard.

We shall construct a Borel map Ω from D into $\text{Rep}(A)$ such that, for every $g \in D$, $\Omega(g)$ is a finite-type factor representation with character g . If $g, g' \in D$ and $g \neq g'$, the representations $\Omega(g)$ and $\Omega(g')$ are then not quasi-equivalent. Since D and $\text{Rep}(A)$ are standard Borel spaces, $\Omega(D)$ is a Borel subset of $\text{Rep}(A)$ meeting each quasi-equivalence class in at most one point, and Ω is a Borel isomorphism from D onto $\Omega(D)$ (B 21). By 7.2.3, if ψ denotes the canonical map from $\text{Fac}(A)$ onto $\check{A}$, then $\psi(\Omega(D))$ is a Borel subset of $\check{A}$, and $\psi|_{\Omega(D)}$ is a Borel isomorphism from $\Omega(D)$ onto $\psi(\Omega(D))$. But $\psi(\Omega(D)) = \check{A}_f$ by 6.8.6. Proposition 7.4.3 will therefore be proved.

Let $(x_1, x_2, \dots)$ be a dense sequence in A . Retain the notation of 6.2.2. For every $g \in D$, the vectors $\Lambda_g x_n$ are dense in H_g , since g is continuous. The functions

$$g \longmapsto (\Lambda_g x_n \mid \Lambda_g x_p) = g(x_n x_p^*)$$

are continuous on D . Hence, by A 98, there is on the field $g \mapsto H_g$ one and only one Borel-field structure for which the vector fields $g \mapsto \Lambda_g x_n$ are Borel. Let D_p be the set of $g \in D$ such that

$$\dim H_g = p \quad (p = 1, 2, \dots, \aleph_0);$$

this is a Borel subset of D (A 96). It is enough to construct Ω on each D_p . There exists an isomorphism $g \mapsto V(g)$ from the field $g \mapsto H_g$ over D_p onto the constant field defined by the model Hilbert space H_p (A 96). It is then enough to set

$$\Omega(g)(x) = V(g)\lambda_g(x)V(g)^{-1}$$

for every $x \in A$. Since $g \mapsto \lambda_g(x)$ is a Borel operator field, the map $g \mapsto \Omega(g)(x)$ is Borel; thus Ω is a Borel map from D into $\text{Rep}(A)$.

7.4.4. It follows from 7.4.3 that $\text{Facf}(A)$ and $\text{Facf}_n(A)$ are Borel subsets of $\text{Fac}(A)$.

Bibliography: [57].

7.5. Complements

7.5.1. Let A be a C^* -algebra, I a closed two-sided ideal of A , and ω the canonical homomorphism from A onto A/I . Let E be the set of $\rho \in \check{A}$ that vanish on I . Then

$$\pi \longmapsto \pi \circ \omega$$

defines a bijection from $(\check{A}/I)$ onto E , and $\pi \mapsto \pi|_I$ defines a bijection from $\check{A} \setminus E$ onto $\check{I}$. (Use 2.11.1.) If A is separable, E is a Borel subset of $\check{A}$, and the preceding bijections are Borel isomorphisms. (Use the existence of an approximate identity in I .) In particular, $\check{A}$ is identified with the Borel sum of a standard space and a space $\check{B}$, where B is a separable antiliminal C^* -algebra. [17]

7.5.2. Let A be a separable C^* -algebra. Let M be the set of $\pi \in \text{Rep}(A)$ that are homogeneous (5.7.6). Then M is a Borel subset of $\text{Rep}(A)$. [20]

7.5.3. Let A be a separable C^* -algebra. On $\widehat{A} \cap \check{A}_f$, the topologies induced by the topology of $\widehat{A}$ and by that of $\check{A}_f$ do not necessarily coincide (take for A the example of 4.7.19).

7.5.4. Problems: let A be a separable C^* -algebra. Is the set of $\pi \in \text{Fac}(A)$ for which π is traceable (respectively, of type II) a Borel subset of $\text{Rep}(A)$? Is the set E of traceable elements of $\check{A}$ a standard Borel space (cf. 6.9.7)? In the absence of a reasonable topology on $\check{A}$, is there a reasonable topology on E ? [57]

§ 8. — Integration and Disintegration of Representations

We now have enough tools to address directly one of the two problems mentioned at the beginning of §3: the decomposition of a representation into irreducible representations. As has already been said, this will not be a decomposition into a Hilbert-space direct sum of irreducible representations, but a subtler decomposition (which we shall call disintegration). This can be anticipated from the following very simple example. Let A be the C^* -algebra of continuous complex-valued functions on $[0, 1]$; for every $f \in A$, let $\pi(f)$ be the operator of multiplication by f in the Hilbert space $H = L^2([0, 1])$. Then π is a representation of A on H . The irreducible representations of A are its characters $f \mapsto f(t_0)$, where $t_0 \in [0, 1]$. To decompose π as a Hilbert-space direct sum of irreducible representations would therefore amount to finding an orthonormal basis of H each of whose elements is a simultaneous eigenvector of all the $\pi(f)$. But no vector (except 0) is a simultaneous eigenvector of all the $\pi(f)$. If we cease to speak rigorously, however, the Dirac functions at the different points of $[0, 1]$ may be regarded as pairwise orthogonal simultaneous eigenvectors of the $\pi(f)$, and every element of H may be written as an integral linear combination of these Dirac functions.

Throughout this section, A denotes a separable C^* -algebra.

8.1. Integration of Representations

8.1.1. Let Z be a Borel space, μ a positive measure (B 30) on Z , and $\zeta \mapsto H(\zeta)$ a μ -measurable field of Hilbert spaces over Z (A 69). For every $\zeta \in Z$, let $\pi(\zeta)$ be a representation of A on $H(\zeta)$; we say that $\zeta \mapsto \pi(\zeta)$ is a field of representations of A .

Definition. The field of representations $\zeta \mapsto \pi(\zeta)$ is called measurable if, for every $x \in A$, the operator field $\zeta \mapsto \pi(\zeta)(x)$ is measurable (A 77).

8.1.2. We have $\|\pi(\zeta)(x)\| \leq \|x\|$ for every $x \in A$ and every $\zeta \in Z$. If the field $\zeta \mapsto \pi(\zeta)$ is measurable, then, for every $x \in A$, we may form the continuous operator

$$\pi(x) = \int_Z^{\oplus} \pi(\zeta)(x) d\mu(\zeta)$$

on the Hilbert space

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta).$$

Proposition. *The map $x \mapsto \pi(x)$ is a representation of A on H .*

For $x, y \in A$, we have

$$\begin{aligned} \pi(x + y) &= \int_Z^{\oplus} \pi(\zeta)(x + y) d\mu(\zeta) \\ &= \int_Z^{\oplus} (\pi(\zeta)(x) + \pi(\zeta)(y)) d\mu(\zeta) \\ &= \int_Z^{\oplus} \pi(\zeta)(x) d\mu(\zeta) + \int_Z^{\oplus} \pi(\zeta)(y) d\mu(\zeta) \\ &= \pi(x) + \pi(y). \end{aligned}$$

Likewise,

$$\pi(\lambda x) = \lambda \pi(x), \quad \pi(xy) = \pi(x)\pi(y), \quad \pi(x^*) = \pi(x)^*.$$

8.1.3. Definition. *With the preceding notation, π is called the direct integral of the $\pi(\zeta)$, and we write*

$$\pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta).$$

8.1.4. Every operator $\pi(x)$ is decomposable, and hence commutes with the algebra $\mathfrak{J}$ of diagonalizable operators (A 80). In other words, $\mathfrak{J} \subset \pi(A)'$.

8.1.5. Let K be the essential subspace of π (2.2.6). Then P_K commutes with $\pi(A)'$, hence commutes with $\mathfrak{J}$; consequently there is a measurable field $\zeta \mapsto E(\zeta) \in \mathcal{L}(H(\zeta))$ of projections such that

$$P_K = \int_Z^{\oplus} E(\zeta) d\mu(\zeta).$$

Let $(u_1, u_2, \dots)$ be an approximate identity of A (1.7.2). We have $\pi(u_n) \rightarrow P_K$ strongly; hence, after passing if necessary to a subsequence of (u_n) , $\pi(\zeta)(u_n) \rightarrow E(\zeta)$ almost everywhere (A 79). Thus, almost everywhere, $E(\zeta)(H(\zeta))$ is the essential subspace of $\pi(\zeta)$. In particular, π is nondegenerate if and only if almost all the $\pi(\zeta)$ are nondegenerate.

8.1.6. Proposition. Let $\zeta \mapsto \pi(\zeta)$ and $\zeta \mapsto \pi'(\zeta)$ be two measurable fields of representations of A relative to the same measurable field $\zeta \mapsto H(\zeta)$. Let

$$\pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta), \quad \pi' = \int_Z^{\oplus} \pi'(\zeta) d\mu(\zeta).$$

Then $\pi = \pi'$ if and only if $\pi(\zeta) = \pi'(\zeta)$ almost everywhere.

The condition is plainly sufficient. Suppose that $\pi = \pi'$. Let (x_i) be a dense sequence in A . We have $\pi(\zeta)(x_i) = \pi'(\zeta)(x_i)$ outside a null set N_i . Then $N = \bigcup_i N_i$ is null, and, for $\zeta \notin N$, we have $\pi(\zeta)(x_i) = \pi'(\zeta)(x_i)$ for every i , hence $\pi(\zeta) = \pi'(\zeta)$.

It follows from 8.1.6 that the $\pi(\zeta)$ are determined by π up to null sets.

8.1.7. Proposition. Let $\zeta \mapsto \pi(\zeta)$ be a measurable field of representations of A on the $H(\zeta)$. Let

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta), \quad \pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta).$$

Suppose that there exists a representation π_0 on a Hilbert space H_0 such that $\pi(\zeta) \simeq \pi_0$ for every ζ , and suppose that μ is standard. Then there exists an isomorphism from H onto $L^2(Z, \mu) \otimes H_0$ which carries $\pi(x)$ to $1 \otimes \pi_0(x)$ for every $x \in A$. Thus π is a multiple of π_0 .

This is a special case of A 83.

8.1.8. Let $H_1, H_2, \dots, H_{\aleph_0}$ be the model Hilbert spaces of dimensions $1, 2, \dots, \aleph_0$.

Proposition. Suppose that Z is the union of pairwise disjoint Borel sets $Z_1, Z_2, \dots, Z_{\aleph_0}$, and that, over Z_n , the field $\zeta \mapsto H(\zeta)$ reduces to the constant field defined by H_n . Let $\zeta \mapsto \pi(\zeta)$ be a field of representations of A on the $H(\zeta)$. Then $\zeta \mapsto \pi(\zeta)$ is measurable if and only if, almost everywhere, it agrees with a Borel map from Z into $\text{Rep}(A)$.

The condition is plainly sufficient. Conversely, let (ξ_j) be an orthonormal basis of H_n , and let (x_i) be a dense sequence in A . If the field $\zeta \mapsto \pi(\zeta)$ is measurable, the functions

$$\zeta \mapsto (\pi(\zeta)(x_i)\xi_j \mid \xi_k)$$

are measurable on Z_n , and hence agree almost everywhere with Borel functions. By changing $\zeta \mapsto \pi(\zeta)$ on a null subset of Z_n , we may arrange that all the maps

$$\zeta \mapsto (\pi(\zeta)(x_i)\xi_j \mid \xi_k)$$

are Borel. Then, for every $x \in A$ and all $\xi, \eta \in H_n$, the map $\zeta \mapsto (\pi(\zeta)(x)\xi \mid \eta)$ is Borel; hence $\zeta \mapsto \pi(\zeta)$ is a Borel map from Z_n into $\text{Rep}_n(A)$.

Bibliography: [20, 52, 88, 91, 95, 96, 101, 108, 110, 137, 161].

8.2. Equivalence of Two Integrals of Representations

8.2.1. Proposition. *Let Z be a Borel space, μ a positive measure on Z , $\zeta \mapsto H(\zeta)$ a measurable field of Hilbert spaces over Z , and $\zeta \mapsto \pi(\zeta)$ a measurable field of representations of A on the $H(\zeta)$. Put*

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta), \quad \pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta),$$

and let $\mathfrak{Z}$ be the algebra of diagonalizable operators on H . Let $\tilde{\mu}$ be a positive measure on Z equivalent to μ , and put

$$\tilde{H} = \int_Z^{\oplus} H(\zeta) d\tilde{\mu}(\zeta), \quad \tilde{\pi} = \int_Z^{\oplus} \pi(\zeta) d\tilde{\mu}(\zeta),$$

with $\tilde{\mathfrak{Z}}$ the algebra of diagonalizable operators on $\tilde{H}$. The canonical isomorphism from H onto $\tilde{H}$ (A 75) carries π to $\tilde{\pi}$ and $\mathfrak{Z}$ to $\tilde{\mathfrak{Z}}$.

Let

$$r(\zeta) = \frac{d\tilde{\mu}(\zeta)}{d\mu(\zeta)}.$$

The canonical isomorphism U from H onto $\tilde{H}$ is the isomorphism which carries

$$x = \int_Z^{\oplus} x(\zeta) d\mu(\zeta) \in H$$

to

$$Ux = \int_Z^{\oplus} r(\zeta)^{-1/2} x(\zeta) d\tilde{\mu}(\zeta).$$

It is clear that U carries $\mathfrak{Z}$ to $\tilde{\mathfrak{Z}}$. Let $a \in A$. Then

$$\begin{aligned} \tilde{\pi}(a)Ux &= \int_Z^{\oplus} \pi(\zeta)(a)r(\zeta)^{-1/2}x(\zeta) d\tilde{\mu}(\zeta) \\ &= U \int_Z^{\oplus} \pi(\zeta)(a)x(\zeta) d\mu(\zeta) = U\pi(a)x, \end{aligned}$$

so U carries π to $\tilde{\pi}$.

8.2.2. Proposition. Let Z be a Borel space, μ a positive measure on Z , $\zeta \mapsto H(\zeta)$ a μ -measurable field of Hilbert spaces over Z , and $\zeta \mapsto \pi(\zeta)$ a μ -measurable field of representations of A on the $H(\zeta)$. Put

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta), \quad \pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta),$$

and let $\mathfrak{Z}$ be the algebra of diagonalizable operators on H . Define analogously

$$Z_1, \mu_1, \zeta_1 \mapsto H_1(\zeta_1), \zeta_1 \mapsto \pi_1(\zeta_1), H_1, \pi_1, \mathfrak{Z}_1.$$

Suppose that there exist:

- 1° a Borel μ -null subset N of Z and a Borel μ_1 -null subset N_1 of Z_1 ;
- 2° a Borel isomorphism η from $Z \setminus N$ onto $Z_1 \setminus N_1$ carrying μ to μ_1 ;
- 3° an η -isomorphism $\zeta \mapsto V(\zeta)$ from the field $\zeta \mapsto H(\zeta)$ over $Z \setminus N$ onto the field $\zeta_1 \mapsto H_1(\zeta_1)$ over $Z_1 \setminus N_1$ (A 70, A 74), such that $V(\zeta)$ carries $\pi(\zeta)$ to $\pi_1(\eta(\zeta))$ for every ζ .

Then

$$V = \int_Z^{\oplus} V(\zeta) d\mu(\zeta)$$

carries $\mathfrak{Z}$ to $\mathfrak{Z}_1$ and π to π_1 .

Denote by $1_{H(\zeta)}$ and $1_{H_1(\zeta_1)}$ the identity operators on $H(\zeta)$ and $H_1(\zeta_1)$. If $f \in L^\infty(Z, \mu)$ and f_1 is the function on $Z_1 \setminus N_1$ transported by η from $f|_{Z \setminus N}$, then V carries

$$\int_Z^{\oplus} f(\zeta) 1_{H(\zeta)} d\mu(\zeta) \quad \text{to} \quad \int_{Z_1}^{\oplus} f_1(\zeta_1) 1_{H_1(\zeta_1)} d\mu_1(\zeta_1);$$

hence V carries $\mathfrak{Z}$ to $\mathfrak{Z}_1$. On the other hand, let $a \in A$ and

$$x = \int_Z^{\oplus} x(\zeta) d\mu(\zeta) \in H.$$

Then

$$\begin{aligned} V\pi(a)x &= V \left(\int_Z^{\oplus} \pi(\zeta)(a)x(\zeta) d\mu(\zeta) \right) \\ &= \int_{Z_1}^{\oplus} V(\eta^{-1}(\zeta_1))\pi(\eta^{-1}(\zeta_1))(a)x(\eta^{-1}(\zeta_1)) d\mu_1(\zeta_1) \\ &= \int_{Z_1}^{\oplus} \pi_1(\zeta_1)(a)V(\eta^{-1}(\zeta_1))x(\eta^{-1}(\zeta_1)) d\mu_1(\zeta_1) \\ &= \pi_1(a)Vx, \end{aligned}$$

so V carries π to π_1 .

8.2.3. Proposition. *Retain the notation $Z, \mu, H(\zeta), \pi(\zeta), H, \pi, \mathfrak{Z}$ and $Z_1, \dots, \mathfrak{Z}_1$ of 8.2.2. Suppose that N, N_1 and η exist with properties 1° and 2° of 8.2.2. Suppose further that, for every $\zeta \in Z \setminus N$, the representations $\pi(\zeta)$ and $\pi_1(\eta(\zeta))$ are equivalent. If, in addition, μ is standard, there exists an isomorphism V from H onto H_1 carrying $\mathfrak{Z}$ to $\mathfrak{Z}_1$ and π to π_1 .*

There exists (A 82) an η -isomorphism $(V(\zeta))_{\zeta \in Z \setminus N}$ from the field $(H(\zeta))_{\zeta \in Z \setminus N}$ onto the field $(H_1(\zeta_1))_{\zeta_1 \in Z_1 \setminus N_1}$ such that $V(\zeta)$ carries $\pi(\zeta)(x)$ to $\pi_1(\eta(\zeta))(x)$ for every $\zeta \in Z$ and every $x \in D$, where D is a countable dense subset of A ; hence it carries $\pi(\zeta)$ to $\pi_1(\eta(\zeta))$. The proposition now follows from 8.2.2.

8.2.4. Propositions 8.2.1 and 8.2.2 have a converse:

Proposition. *Retain the notation $Z, \mu, H(\zeta), \pi(\zeta), H, \pi, \mathfrak{Z}$ and $Z_1, \dots, \mathfrak{Z}_1$ of 8.2.2. Suppose that there is an isomorphism U from H onto H_1 carrying $\mathfrak{Z}$ to $\mathfrak{Z}_1$ and π to π_1 . If μ and μ_1 are standard, there exist:*

- 1° a Borel μ -null set N in Z and a Borel μ_1 -null set N_1 in Z_1 ;
- 2° a Borel isomorphism η from $Z \setminus N$ onto $Z_1 \setminus N_1$ carrying μ to a measure $\tilde{\mu}_1$ equivalent to μ_1 ;
- 3° an η -isomorphism $\zeta \mapsto V(\zeta)$ from the field $\zeta \mapsto H(\zeta)$ over $Z \setminus N$ onto the field $\zeta_1 \mapsto H_1(\zeta_1)$ over $Z_1 \setminus N_1$, such that $V(\zeta)$ carries $\pi(\zeta)$ to $\pi_1(\eta(\zeta))$ for every $\zeta \in Z \setminus N$.

Moreover, U is the composition of

$$\int_Z^{\oplus} V(\zeta) d\mu(\zeta)$$

and the canonical isomorphism from $\int_{Z_1}^{\oplus} H_1(\zeta_1) d\tilde{\mu}_1(\zeta_1)$ onto H_1 .

Since U carries $\mathfrak{Z}$ to $\mathfrak{Z}_1$, there exist $N, N_1, \eta, \tilde{\mu}_1$ and $\zeta \mapsto V(\zeta)$ with all the properties in 8.2.4, except for the assertion that $V(\zeta)$ carries $\pi(\zeta)$ to $\pi_1(\eta(\zeta))$ for $\zeta \in Z \setminus N$ (A 85). Since U is the composition of $\int_Z^{\oplus} V(\zeta) d\mu(\zeta)$ and the canonical isomorphism from $\int_{Z_1}^{\oplus} H_1(\zeta_1) d\tilde{\mu}_1(\zeta_1)$ onto H_1 , U carries π to

$$\int_{Z_1}^{\oplus} \pi'_1(\zeta_1) d\mu_1(\zeta_1),$$

where $\pi'_1(\zeta_1)$ is the transform of $\pi(\eta^{-1}(\zeta_1))$ by $V(\eta^{-1}(\zeta_1))$ (8.2.1 and 8.2.2). Since U carries π to π_1 , we have $\pi_1(\zeta_1) = \pi'_1(\zeta_1)$ almost everywhere (8.1.6). By adjoin-

ing a suitable null set to N , we therefore see that $V(\zeta)$ carries $\pi(\zeta)$ to $\pi_1(\eta(\zeta))$ for $\zeta \in Z \setminus N$.

Bibliography: [52, 88, 91, 95, 96, 101, 108, 110, 137, 161].

8.3. Disintegration of Representations

8.3.1. Lemma. *Let Z be a Borel space, μ a positive measure on Z , $\zeta \mapsto H(\zeta)$ a μ -measurable field of Hilbert spaces over Z ,*

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta),$$

and π a representation of A on H . Suppose that every operator $\pi(x)$, $x \in A$, is decomposable. Then, for every $\zeta \in Z$, there exists a representation $\pi(\zeta)$ of A on $H(\zeta)$ such that

$$\pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta).$$

Let $(x_1, x_2, \dots)$ be a sequence of distinct points of A , dense in A . By enlarging this sequence if necessary, we may suppose that the set of the x_i is an involutive subalgebra B of A over the (countable) field of rational complex numbers. For each i , choose arbitrarily a measurable field $\zeta \mapsto T_i(\zeta)$ such that

$$\pi(x_i) = \int_Z^{\oplus} T_i(\zeta) d\mu(\zeta).$$

Let N_i be the null set of $\zeta \in Z$ such that $\|T_i(\zeta)\| > \|\pi(x_i)\|$. The union N of the N_i is null. On the other hand, there are countably many relations among the x_i of the forms

$$x_k = \lambda_1 x_i + \lambda_2 x_j, \quad x_k = x_i x_j, \quad x_j = x_i^*,$$

where λ_1, λ_2 are rational complex numbers. Almost everywhere these relations imply, respectively,

$$T_k(\zeta) = \lambda_1 T_i(\zeta) + \lambda_2 T_j(\zeta), \quad T_k(\zeta) = T_i(\zeta) T_j(\zeta), \quad T_j(\zeta) = T_i(\zeta)^*.$$

There is therefore a null subset N' of Z such that, for $\zeta \notin N'$, the map $x_i \mapsto T_i(\zeta)$ is a homomorphism from B into $\mathcal{L}(H(\zeta))$, regarded as an involutive algebra over the field of rational complex numbers. Replace the $T_i(\zeta)$ by 0 for $\zeta \in N \cup N'$. Then, for every $\zeta \in Z$,

$$\|T_i(\zeta)\| \leq \|\pi(x_i)\| \leq \|x_i\|$$

for every i , and the map $x_i \mapsto T_i(\zeta)$ is a homomorphism from B into $\mathcal{L}(H(\zeta))$. This homomorphism extends uniquely to a representation $\pi(\zeta)$ of A on $H(\zeta)$. Let $x \in A$, and write

$$\pi(x) = \int_Z^{\oplus} T(\zeta) d\mu(\zeta).$$

There exists a sequence (x_{n_i}) of elements of B tending to x . Almost everywhere,

$$\|T(\zeta) - \pi(\zeta)(x_{n_i})\| \leq \|x - x_{n_i}\|;$$

hence $\pi(\zeta)(x_{n_i})$ tends to $T(\zeta)$ almost everywhere. Thus $T(\zeta) = \pi(\zeta)(x)$ almost everywhere, and consequently

$$\pi(x) = \int_Z^{\oplus} \pi(\zeta)(x) d\mu(\zeta).$$

8.3.2. Theorem. *Let H be a separable Hilbert space, π a representation of A on H , and $\mathfrak{Z}$ a commutative von Neumann subalgebra of $\pi(A)'$. There exist a standard Borel space Z , a bounded positive measure μ on Z , a measurable field $\zeta \mapsto H(\zeta)$ of Hilbert spaces over Z , a measurable field $\zeta \mapsto \pi(\zeta)$ of representations of A on the $H(\zeta)$, and an isomorphism from H onto*

$$\int_Z^{\oplus} H(\zeta) d\mu(\zeta)$$

which carries $\mathfrak{Z}$ to the algebra of diagonalizable operators and π to

$$\int_Z^{\oplus} \pi(\zeta) d\mu(\zeta).$$

There exist (A 84) a standard Borel space Z , a bounded positive measure μ on Z , a measurable field $\zeta \mapsto H(\zeta)$ of Hilbert spaces over Z , and an isomorphism from H onto $\int_Z^{\oplus} H(\zeta) d\mu(\zeta)$ which carries $\mathfrak{Z}$ to the algebra of diagonalizable operators. Since $\mathfrak{Z} \subset \pi(A)'$, every operator $\pi(x)$, $x \in A$, is decomposable. It remains only to apply 8.3.1.

Bibliography: [52, 88, 91, 95, 96, 101, 108, 110, 137, 161].

8.4. Central Disintegration

8.4.1. Lemma. *Let Z be a Borel space, μ a positive measure on Z , $\zeta \mapsto H(\zeta)$ a measurable field of Hilbert spaces over Z , $\zeta \mapsto \pi(\zeta)$ a measurable field of representations of A on the $H(\zeta)$, $\mathcal{A}(\zeta)$ the strong closure of $\pi(\zeta)(A)$,*

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta), \quad \pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta),$$

$\mathfrak{Z}$ the algebra of diagonalizable operators, and $\mathcal{A}$ the strong closure of $\pi(A)$. Suppose that $\mathfrak{Z} \subset \mathcal{A}$.

(i) We have

$$\mathcal{A} = \int_Z^{\oplus} \mathcal{A}(\zeta) d\mu(\zeta).$$

(ii) If $\mathfrak{Z}$ is the center of $\mathcal{A}$, then $\pi(\zeta)$ is a factor representation almost everywhere.

(iii) If μ is standard, there exists a null subset N of Z such that, whenever $\zeta, \zeta' \in Z \setminus N$ and $\zeta \neq \zeta'$, one has $\pi(\zeta) \perp \pi(\zeta')$.

Since $\mathfrak{Z} \subset \mathcal{A}$, π is nondegenerate, and hence $\pi(\zeta)$ is nondegenerate almost everywhere (8.1.5). Thus $\mathcal{A}$ (respectively, $\mathcal{A}(\zeta)$) is also the von Neumann algebra generated by $\pi(A)$ (respectively, $\pi(\zeta)(A)$). Observe that $\mathfrak{Z} \subset \pi(A)' = \mathcal{A}'$, so $\mathfrak{Z}$ is contained in the center of $\mathcal{A}$.

Let (x_i) be a dense sequence in A . Then $\mathcal{A}(\zeta)$ is the von Neumann algebra generated by the $\pi(\zeta)(x_i)$. On the other hand, $\mathcal{A}$ may be regarded as the von Neumann algebra generated by the $\pi(x_i)$ and by $\mathfrak{Z}$. But

$$\pi(x_i) = \int_Z^{\oplus} \pi(\zeta)(x_i) d\mu(\zeta).$$

Therefore, by A 88,

$$\mathcal{A} = \int_Z^{\oplus} \mathcal{A}(\zeta) d\mu(\zeta).$$

If $\mathfrak{Z}$ is the center of $\mathcal{A}$, then $\mathcal{A}(\zeta)$ is a factor almost everywhere (A 89); thus $\pi(\zeta)$ is a factor representation almost everywhere.

Let

$$T = \int_Z^{\oplus} T(\zeta) d\mu(\zeta) \in \mathcal{A}.$$

Since T is a strong limit of elements of $\pi(A)$, there are a null subset $N(T)$ of Z and a sequence $(x_1, x_2, \dots)$ in A such that, for $\zeta \in Z \setminus N(T)$, $\pi(\zeta)(x_i)$ tends strongly to $T(\zeta)$ (A 13, A 79). Let $\zeta, \zeta' \in Z \setminus N(T)$, and let R be an intertwining operator for $\pi(\zeta)$ and $\pi(\zeta')$. The equality

$$R\pi(\zeta)(x_i) = \pi(\zeta')(x_i)R,$$

valid for every i , implies

$$RT(\zeta) = T(\zeta')R.$$

This established, suppose that Z is standard, and let $(Y_1, Y_2, \dots)$ be a sequence of Borel subsets of Z separating the points of Z . Let φ_i be the characteristic function of Y_i ; it defines an element E_i of $\mathfrak{Z}$. The union N of the $N(E_i)$ is null. Let

$\zeta, \zeta' \in Z \setminus N$ with $\zeta \neq \zeta'$, and let R be an intertwining operator for $\pi(\zeta)$ and $\pi(\zeta')$. By what was proved above, for every i ,

$$RE_i(\zeta) = E_i(\zeta')R, \quad \text{that is,} \quad \varphi_i(\zeta)R = \varphi_i(\zeta')R.$$

There is an i such that $\varphi_i(\zeta) = 1$ and $\varphi_i(\zeta') = 0$, whence $R = 0$. Thus $\pi(\zeta) \perp \pi(\zeta')$. This proves (iii).

8.4.2. Theorem. *Let H be a separable Hilbert space, π a nondegenerate representation of A on H , and $\mathfrak{Z}$ the center of the von Neumann algebra generated by $\pi(A)$.*

- (i) *There exist a standard measure μ on $\check{A}$, a μ -measurable field $\zeta \mapsto H(\zeta)$ of Hilbert spaces over $\check{A}$, a μ -measurable field $\zeta \mapsto \pi(\zeta)$ of nonzero factor representations of A on the $H(\zeta)$ such that $\pi(\zeta) \in \zeta$, and an isomorphism W from H onto*

$$\int_{\check{A}}^{\oplus} H(\zeta) d\mu(\zeta)$$

which carries $\mathfrak{Z}$ to the algebra of diagonalizable operators and π to

$$\int_{\check{A}}^{\oplus} \pi(\zeta) d\mu(\zeta).$$

- (ii) *Let $\mu_1, \zeta \mapsto H_1(\zeta), \zeta \mapsto \pi_1(\zeta)$, and W_1 have the same properties. Then μ and μ_1 are equivalent. After changing the $H_1(\zeta)$ and the $\pi_1(\zeta)$ on a null set, there exists an isomorphism $\zeta \mapsto V(\zeta)$ from the field $\zeta \mapsto H(\zeta)$ onto the field $\zeta \mapsto H_1(\zeta)$ such that $V(\zeta)$ carries $\pi(\zeta)$ to $\pi_1(\zeta)$ for every ζ .*

Apply Theorem 8.3.2 to π and $\mathfrak{Z}$, retaining its notation $Z, \mu, \zeta \mapsto H(\zeta), \zeta \mapsto \pi(\zeta)$. There is an isomorphism from H onto $\int_Z^{\oplus} H(\zeta) d\mu(\zeta)$ which carries $\mathfrak{Z}$ to the algebra of diagonalizable operators and π to $\int_Z^{\oplus} \pi(\zeta) d\mu(\zeta)$. Since π is nondegenerate, $\mathfrak{Z}$ is contained in the strong closure of $\pi(A)$. By 8.4.1, almost all the $\pi(\zeta)$ are factor representations and are pairwise disjoint. By 8.1.5, almost all the $\pi(\zeta)$ are nonzero. After changing $H(\zeta)$ and $\pi(\zeta)$ on a null set, we may therefore suppose that the $\pi(\zeta)$ are nonzero, pairwise disjoint factor representations. Next, by isomorphisms on the $H(\zeta)$, we may suppose that $\zeta \mapsto \pi(\zeta)$ is a Borel map from Z into $\text{Fac}(A)$ (8.1.8). Since the $\pi(\zeta)$ are pairwise disjoint, this map is injective; it is therefore (B 21) a Borel isomorphism θ from Z onto a Borel subset B of $\text{Fac}(A)$ which meets each quasi-equivalence class in at most one point. Let ψ be the restriction to B of the canonical map $\text{Fac}(A) \rightarrow \check{A}$; it is a Borel isomorphism from B onto a Borel subset C of $\check{A}$ (7.2.3). Thus $\psi \circ \theta$ is a Borel isomorphism

from Z onto C . We may therefore replace Z by C and suppose that the quasi-equivalence class of $\pi(\zeta)$ is ζ for every $\zeta \in C$. This proves (i), on extending μ to $\check{A}$ by $\mu(\check{A} \setminus C) = 0$.

Introduce the notation $\mu_1, \zeta \mapsto H_1(\zeta), \zeta \mapsto \pi_1(\zeta)$, and W_1 of (ii). Apply 8.2.4, with $Z = Z_1 = \check{A}$, obtaining $N, N_1, \tilde{\mu}_1, \eta$, and the $V(\zeta)$. For $\zeta \in Z \setminus N$, $V(\zeta)$ carries $\pi(\zeta)$ to $\pi_1(\eta(\zeta))$. Hence the quasi-equivalence classes ζ and $\eta(\zeta)$ of $\pi(\zeta)$ and $\pi_1(\eta(\zeta))$ are equal. Thus η is the identity map on $Z \setminus N$. Consequently μ and μ_1 are equivalent, and (ii) is proved.

8.4.3. The class of the measure μ on $\check{A}$ is therefore uniquely determined by π .

Definition. It is called the *measure class associated with π* .

The disintegration

$$\pi \simeq \int_{\check{A}}^{\oplus} \pi(\zeta) d\mu(\zeta)$$

of Theorem 8.4.2 is called, with a slight abuse of language, the *central disintegration of π* .

8.4.4. Proposition. Let π, ρ be two nondegenerate representations of A on separable Hilbert spaces. Then π and ρ are quasi-equivalent if and only if their associated measure classes on $\check{A}$ are equal.

Suppose that the measure classes on $\check{A}$ associated with π and ρ are equal. Let $\mathfrak{V}$ (respectively, $\mathfrak{Z}$) be the center of the von Neumann algebra $\mathcal{A}$ (respectively, $\mathcal{B}$) generated by $\pi(A)$ (respectively, $\rho(A)$). There are a standard measure μ on $\check{A}$, a measurable field $\zeta \mapsto H(\zeta)$ (respectively, $\zeta \mapsto K(\zeta)$) of Hilbert spaces over $\check{A}$, and a measurable field $\zeta \mapsto \pi(\zeta)$ (respectively, $\zeta \mapsto \rho(\zeta)$) of factor representations of A on the $H(\zeta)$ (respectively, the $K(\zeta)$), such that

$$\pi(\zeta) \in \zeta \quad (\text{respectively, } \rho(\zeta) \in \zeta),$$

$$\pi = \int_{\check{A}}^{\oplus} \pi(\zeta) d\mu(\zeta) \quad \left[\text{respectively, } \rho = \int_{\check{A}}^{\oplus} \rho(\zeta) d\mu(\zeta) \right],$$

and $\mathfrak{V}$ (respectively, $\mathfrak{Z}$) is the algebra of diagonalizable operators (all this up to isomorphisms). Since $\pi(\zeta) \approx \rho(\zeta)$ for every ζ , there is an isomorphism $\Phi(\zeta)$ from the von Neumann algebra $\mathcal{A}(\zeta)$ generated by $\pi(\zeta)(A)$ onto the von Neumann algebra $\mathcal{B}(\zeta)$ generated by $\rho(\zeta)(A)$ such that

$$\Phi(\zeta)(\pi(\zeta)(x)) = \rho(\zeta)(x) \quad \text{for every } x \in A.$$

We have

$$\mathcal{A} = \int_{\check{A}}^{\oplus} \mathcal{A}(\zeta) d\mu(\zeta), \quad \mathcal{B} = \int_{\check{A}}^{\oplus} \mathcal{B}(\zeta) d\mu(\zeta)$$

by 8.4.1. Let us show that $\zeta \mapsto \Phi(\zeta)$ is a measurable field of isomorphisms (A 91). Let

$$T = \int_{\check{A}}^{\oplus} T(\zeta) d\mu(\zeta) \in \mathcal{A}.$$

There is a sequence (x_i) in A such that $\pi(x_i)$ tends strongly to T . On passing to a subsequence, we may suppose that $\pi(\zeta)(x_i)$ tends strongly to $T(\zeta)$ almost everywhere. Then

$$\Phi(\zeta)(\pi(\zeta)(x_i)) \longrightarrow \Phi(\zeta)(T(\zeta))$$

strongly (A 1, A 27). Since the fields

$$\zeta \longmapsto \Phi(\zeta)(\pi(\zeta)(x_i)) = \rho(\zeta)(x_i)$$

are measurable, the field $\zeta \mapsto \Phi(\zeta)(T(\zeta))$ is measurable, which proves the assertion. Hence

$$\Phi = \int_{\check{A}}^{\oplus} \Phi(\zeta) d\mu(\zeta)$$

is an isomorphism from $\mathcal{A}$ onto $\mathcal{B}$ (A 91) which, for every $x \in A$, carries

$$\pi(x) = \int_{\check{A}}^{\oplus} \pi(\zeta)(x) d\mu(\zeta) \quad \text{to} \quad \int_{\check{A}}^{\oplus} \rho(\zeta)(x) d\mu(\zeta) = \rho(x).$$

Thus $\pi \approx \rho$.

Conversely, suppose that $\pi \approx \rho$. It is clear that if

$$\pi \simeq \int_{\check{A}}^{\oplus} \pi(\zeta) d\mu(\zeta)$$

and n is a cardinal $\leq \aleph_0$, then

$$n\pi \simeq \int_{\check{A}}^{\oplus} n\pi(\zeta) d\mu(\zeta).$$

Thus the measure classes associated with π and $n\pi$ are equal. But $\aleph_0\pi \simeq \aleph_0\rho$ (5.3.8), so the measure classes associated with π and ρ are equal.

8.4.5. Proposition. *Let π, ρ be two nondegenerate representations of A on separable Hilbert spaces, and let C, D be the measure classes on $\check{A}$ associated with π, ρ . Then ρ is quasi-equivalent to a subrepresentation of π if and only if D is based on C .*

Suppose that ρ is quasi-equivalent to a subrepresentation ρ' of π . Let $E \in \pi(A)'$ be the projection corresponding to ρ' , let $\mathfrak{Z}$ be the center of $\pi(A)'$, let $E_1 \in \mathfrak{Z}$ be the central support of E , and let ρ_1 be the subrepresentation of π corresponding to E_1 . We have $\rho \approx \rho' \approx \rho_1$ (condition (v) of 5.3.1), so D is also the measure class associated with ρ_1 (8.4.4). Now let

$$\pi = \int_{\check{A}}^{\oplus} \pi(\zeta) d\mu(\zeta)$$

be the central disintegration of π ; then $\mu \in C$. Let Y be the μ -measurable subset of $\check{A}$ whose characteristic function φ_Y corresponds to E_1 . We have

$$\rho_1 = \int_{\check{A}}^{\oplus} \pi(\zeta) d(\varphi_Y \mu)(\zeta).$$

Thus D is the class of $\varphi_Y \mu$, and hence D is based on C .

Conversely, suppose that D is based on C . Choose $\mu \in C$. There is a μ -measurable subset Y of $\check{A}$ such that $\varphi_Y \mu \in D$, where φ_Y denotes the characteristic function of Y . Let

$$\pi = \int_{\check{A}}^{\oplus} \pi(\zeta) d\mu(\zeta)$$

be the central disintegration of π . Let E_1 be the diagonalizable projection corresponding to φ_Y , and let ρ_1 be the subrepresentation of π corresponding to E_1 . The measure class associated with ρ_1 is the class of $\varphi_Y \mu$, hence is D . Therefore $\rho \approx \rho_1$ (8.4.4).

8.4.6. The second part of the preceding proof shows that, if a measure class C on $\check{A}$ is associated with a representation of A on a separable Hilbert space, then every measure class based on C is also associated with a representation of A .

8.4.7. Proposition. *Let π, ρ, C, D be as in 8.4.5. Then π and ρ are disjoint if and only if C and D are mutually singular.*

Indeed, to say that π and ρ are not disjoint means that there is a nonzero representation quasi-equivalent both to a subrepresentation of π and to a subrepresentation of ρ . By 8.4.5 and 8.4.6, this is equivalent to the existence of a nonzero measure class based both on C and on D , that is, to C and D not being mutually singular.

8.4.8. Proposition. *Let π be a nondegenerate representation of A on a separable Hilbert space, and let*

$$\pi = \int_{\check{A}}^{\oplus} \pi(\zeta) d\mu(\zeta)$$

be its central disintegration. Then π is of type I if and only if almost all the $\pi(\zeta)$ are of type I; equivalently, if and only if μ is concentrated on $\widehat{A}$.

Let $\mathcal{A}$ and $\mathcal{A}(\zeta)$ be the von Neumann algebras generated by $\pi(A)$ and $\pi(\zeta)(A)$. We have

$$\mathcal{A} = \int_{\widehat{A}}^{\oplus} \mathcal{A}(\zeta) d\mu(\zeta)$$

by 8.4.1(i); consequently $\mathcal{A}$ is of type I if and only if almost all the $\mathcal{A}(\zeta)$ are of type I (A 90).

Bibliography: [25, 56, 109].

8.5. Disintegration into Irreducible Representations

8.5.1. Lemma. *Let Z be a Borel space, μ a positive measure on Z , $\zeta \mapsto H(\zeta)$ a measurable field of Hilbert spaces over Z , $\zeta \mapsto \pi(\zeta)$ a measurable field of representations of A on the $H(\zeta)$,*

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta), \quad \pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta),$$

and $\mathfrak{J}$ the algebra of diagonalizable operators. The following two conditions are equivalent:

- (i) *$\mathfrak{J}$ is a maximal commutative von Neumann subalgebra of $\pi(A)'$;*
- (ii) *almost all the $\pi(\zeta)$ are irreducible.*

Let $\mathcal{A}$ and $\mathcal{A}(\zeta)$ be the von Neumann algebras generated by $\pi(A)$ and $\pi(\zeta)(A)$. Let (x_i) be a dense sequence in A . Then the $\pi(x_i)$ generate the von Neumann algebra $\mathcal{A}$, the $\pi(\zeta)(x_i)$ generate the von Neumann algebra $\mathcal{A}(\zeta)$, and

$$\pi(x_i) = \int_Z^{\oplus} \pi(\zeta)(x_i) d\mu(\zeta).$$

Hence, by A 86, $\mathfrak{J}$ is a maximal commutative subalgebra of $\pi(A)' = \mathcal{A}'$ if and only if $\mathcal{A}(\zeta) = \mathcal{L}(H(\zeta))$ almost everywhere, that is, if and only if $\pi(\zeta)$ is irreducible almost everywhere.

8.5.2. Theorem. *Let H be a separable Hilbert space, π a representation of A on H , and $\mathfrak{J}$ a maximal commutative von Neumann subalgebra of $\pi(A)'$. There exist a standard Borel space Z , a bounded positive measure μ on Z , a measurable field $\zeta \mapsto H(\zeta)$ of*

Hilbert spaces over Z , a measurable field $\zeta \mapsto \pi(\zeta)$ of irreducible representations of A on the $H(\zeta)$, and an isomorphism from H onto

$$\int_Z^{\oplus} H(\zeta) d\mu(\zeta)$$

which carries $\mathfrak{J}$ to the algebra of diagonalizable operators and π to

$$\int_Z^{\oplus} \pi(\zeta) d\mu(\zeta).$$

This follows from 8.3.2 and 8.5.1.

8.5.3. Recall that, when π and $\mathfrak{J}$ are given, the disintegration of π provided by Theorem 8.5.2 has a certain uniqueness property (8.2.4). But when π is given and $\mathfrak{J}$ is changed, this disintegration may change completely, to the point that no irreducible representation occurs in both disintegrations (cf. 18.9.8). (In this respect, central disintegration behaves much more satisfactorily.) This pathological circumstance will, however, be excluded for postliminal C^* -algebras.

Bibliography: [52, 88, 89, 91, 95, 96, 101, 108, 110, 137, 161].

8.6. Case of Postliminal C^* -Algebras

8.6.1. Let A be a separable postliminal C^* -algebra. Then $\check{A} = \widehat{A}$ is a standard Borel space (4.6.1 and 7.3.7). Let $\widehat{A}_n$, for $n = 1, 2, \dots, \aleph_0$, be the set of elements of $\widehat{A}$ of dimension n ; this is a Borel subset of $\widehat{A}$ (3.6.3(i)). Let H_n be the model Hilbert space of dimension n . There is one and only one Borel field $\zeta \mapsto H(\zeta)$ of Hilbert spaces over $\widehat{A}$ (A 96) which reduces over $\widehat{A}_n$ to the constant field defined by H_n . We shall call it the canonical field. For every positive measure μ on $\widehat{A}$, the canonical field $\zeta \mapsto H(\zeta)$ automatically has the structure of a μ -measurable field (A 97).

8.6.2. Lemma. *Retain the preceding notation.*

- (i) *There is a field $\zeta \mapsto \pi(\zeta)$, $\zeta \in \widehat{A}$, of representations of A on the $H(\zeta)$ such that $\pi(\zeta)$ belongs to the equivalence class ζ for every $\zeta \in \widehat{A}$, and which is measurable for every positive measure on $\widehat{A}$.*
- (ii) *Let $\zeta \mapsto \pi_1(\zeta)$ be another field of representations with the properties in (i). For every positive measure μ on $\widehat{A}$,*

$$\int_{\widehat{A}}^{\oplus} \pi(\zeta) d\mu(\zeta) \simeq \int_{\widehat{A}}^{\oplus} \pi_1(\zeta) d\mu(\zeta).$$

There is a Borel map $\zeta \mapsto \pi(\zeta)$ from $\widehat{A}$ into $\text{Irr}(A)$ such that, for every $\zeta \in \widehat{A}$, $\pi(\zeta)$ belongs to the class ζ (4.6.2). For every positive measure on $\widehat{A}$, the field $\zeta \mapsto \pi(\zeta)$ is measurable (8.1.8). This proves (i). Assertion (ii) follows from Proposition 8.2.3, applied with

$$Z = Z_1 = \widehat{A}, \quad \mu = \mu_1, \quad H(\zeta) = H_1(\zeta), \quad N = N_1 = \emptyset,$$

and η the identity map of $\widehat{A}$.

8.6.3. Lemma 8.6.2(ii) shows that every positive measure μ on $\widehat{A}$ determines a well-defined equivalence class of representations of A on a separable space (cf. A 73).

Definition. This equivalence class of representations is denoted by

$$\int_{\widehat{A}}^{\oplus} \zeta \, d\mu(\zeta).$$

8.6.4. Proposition. Let A be a separable postliminal C^* -algebra, and let μ be a positive measure on $\widehat{A}$. Let $\zeta \mapsto \pi(\zeta)$, $\zeta \in \widehat{A}$, be a μ -measurable field of representations of A on the spaces $H(\zeta)$ introduced in 8.6.1, such that $\pi(\zeta)$ belongs to the class ζ for every $\zeta \in \widehat{A}$. Put

$$\pi = \int_{\widehat{A}}^{\oplus} \pi(\zeta) \, d\mu(\zeta),$$

and let $\mathfrak{Z}$ be the algebra of diagonalizable operators. Then $\pi(A)' = \mathfrak{Z}$.

For every measurable subset Y of $\widehat{A}$, denote by E_Y the diagonalizable projection corresponding to the characteristic function of Y . Every open subset of $\widehat{A}$ is identified with the spectrum $\widehat{I}$ of a closed two-sided ideal I of A (3.2.2). Let $(x_1, x_2, \dots)$ be an approximate identity of the C^* -algebra I (1.7.2). If $\zeta \in \widehat{A} \setminus \widehat{I}$, then $\pi(\zeta)(x_n) = 0$ for every n . If $\zeta \in \widehat{I}$, then $\pi(\zeta)|_I$ is a nondegenerate representation of I , so $\pi(\zeta)(x_n)$ tends strongly to 1. Hence $\pi(x_n)$ tends strongly to $E_{\widehat{I}}$ (A 79). Thus $E_{\widehat{I}}$ belongs to the von Neumann algebra $\mathcal{A}$ generated by $\pi(A)$. By passage to the limit, it follows that $E_Y \in \mathcal{A}$ for every Borel subset Y of $\widehat{A}$. Thus $\mathfrak{Z} \subset \mathcal{A}$, and therefore $\mathfrak{Z}' \supset \mathcal{A}'$. On the other hand, by 8.5.1, $\mathfrak{Z}$ is a maximal commutative subalgebra of $\mathcal{A}'$, that is, $\mathfrak{Z}' \cap \mathcal{A}' \subset \mathfrak{Z}$. Since $\mathfrak{Z}' \supset \mathcal{A}'$, it follows that $\mathcal{A}' \subset \mathfrak{Z}$. The inclusion $\mathfrak{Z} \subset \mathcal{A}'$ is clear, so $\mathcal{A}' = \mathfrak{Z}$.

8.6.5. Theorem. Let A be a separable postliminal C^* -algebra.

(i) For every positive measure μ on $\widehat{A}$,

$$\int_{\widehat{A}}^{\oplus} \zeta \, d\mu(\zeta)$$

is a multiplicity-free nondegenerate representation, and the measure class on $\check{A} = \widehat{\widehat{A}}$ associated with this representation is the class of μ .

(ii) Let μ and μ_1 be two positive measures on $\widehat{A}$. Then

$$\int_{\widehat{A}}^{\oplus} \zeta d\mu(\zeta) \simeq \int_{\widehat{A}}^{\oplus} \zeta d\mu_1(\zeta)$$

if and only if μ and μ_1 are equivalent.

(iii) Every multiplicity-free nondegenerate representation of A on a separable Hilbert space is equivalent to

$$\int_{\widehat{A}}^{\oplus} \zeta d\mu(\zeta)$$

for a bounded positive measure μ on $\widehat{A}$.

Let

$$\pi = \int_{\widehat{A}}^{\oplus} \zeta d\mu(\zeta).$$

By 8.6.4, $\pi(A)'$ is commutative, so π is multiplicity-free. By 8.1.5, π is nondegenerate. Since $\mathfrak{Z}$ is the center of the von Neumann algebra generated by $\pi(A)$ (8.6.4), the class of μ is the measure class associated with π .

If μ and μ_1 are equivalent, then

$$\int_{\widehat{A}}^{\oplus} \zeta d\mu(\zeta) \simeq \int_{\widehat{A}}^{\oplus} \zeta d\mu_1(\zeta)$$

by 8.2.1. Conversely, if these representations are equivalent, then μ and μ_1 are equivalent by 8.4.2(ii).

Finally, let π be a multiplicity-free nondegenerate representation of A on a separable Hilbert space. Let μ be a bounded member of the measure class associated with π , and put

$$\rho = \int_{\widehat{A}}^{\oplus} \zeta d\mu(\zeta).$$

By (i), ρ is multiplicity-free, and the measure class associated with ρ is the class of μ . Hence $\rho \approx \pi$ (8.4.4), and therefore $\rho \simeq \pi$ (5.4.6).

8.6.6. Theorem. *Let A be a separable postliminal C^* -algebra.*

(i) *Let π be a nondegenerate representation of A on a separable Hilbert space. There exist pairwise mutually singular positive measures $\mu_1, \mu_2, \dots, \mu_{\aleph_0}$ on $\widehat{A}$ such that*

$$\pi \simeq \int_{\widehat{A}}^{\oplus} \zeta d\mu_1(\zeta) \oplus 2 \int_{\widehat{A}}^{\oplus} \zeta d\mu_2(\zeta) \oplus \cdots \oplus \aleph_0 \int_{\widehat{A}}^{\oplus} \zeta d\mu_{\aleph_0}(\zeta).$$

- (ii) If $\mu'_1, \mu'_2, \dots, \mu'_{\aleph_0}$ have the same properties, then μ_n and μ'_n are equivalent for every n .
- (iii) The measure class on $\check{A} = \widehat{A}$ associated with π is the class of

$$\mu_1 + \mu_2 + \dots + \mu_{\aleph_0}.$$

Since H_π is separable,

$$\pi = \pi_1 \oplus \pi_2 \oplus \dots \oplus \pi_{\aleph_0},$$

where π_n has multiplicity n (with the convention $\infty = \aleph_0$), and the π_n are pairwise disjoint (5.4.9, 5.5.2). Thus

$$\pi \simeq \rho_1 \oplus 2\rho_2 \oplus \dots \oplus \aleph_0 \rho_{\aleph_0},$$

where the ρ_n are pairwise disjoint and multiplicity-free. By 8.6.5,

$$\rho_n = \int_{\widehat{A}}^{\oplus} \zeta d\mu_n(\zeta),$$

where μ_n is a positive measure on $\widehat{A}$. Since the ρ_n are pairwise disjoint, the μ_n are pairwise mutually singular (8.4.7).

Let $\mu'_1, \mu'_2, \dots, \mu'_{\aleph_0}$ have the properties of (i), and put

$$\rho'_n = \int_{\widehat{A}}^{\oplus} \zeta d\mu'_n(\zeta).$$

We have

$$\pi \simeq \rho'_1 \oplus 2\rho'_2 \oplus \dots \oplus \aleph_0 \rho'_{\aleph_0},$$

and the ρ'_n are pairwise disjoint (8.4.7). By 5.4.9, $n\rho_n \simeq n\rho'_n$. Thus μ_n and μ'_n are equivalent for every n .

Let

$$\mu = \mu_1 + \mu_2 + \dots + \mu_{\aleph_0}.$$

Then

$$\pi \approx \int_{\widehat{A}}^{\oplus} \zeta d\mu(\zeta),$$

so the measure class on $\widehat{A}$ associated with π is the class of μ .

8.6.7. We see that, to know a representation of A up to equivalence, one must know a measure class and certain *multiplicities*. In central disintegration, only

the measure class mattered. This is because the representations were studied up to quasi-equivalence and multiplicity questions therefore no longer arose (recall that a representation is quasi-equivalent to all its multiples).

8.6.8. Proposition. *Let A be a separable postliminal C^* -algebra. Let π be a nondegenerate representation of A on a separable Hilbert space, let μ be a member of the measure class associated with π , and let S be the support of μ (the smallest closed subset of $\widehat{A}$ whose complement is null). Let $\rho \in \widehat{A}$.*

- (i) ρ is weakly contained in π if and only if $\rho \in S$.
- (ii) ρ is contained in π if and only if $\mu(\{\rho\}) > 0$.

Write

$$\pi = \int_{\widehat{A}}^{\oplus} \pi(\zeta) d\mu(\zeta),$$

where $\pi(\zeta)$ belongs to the class ζ .

Suppose that $\rho \notin S$. Since S is closed, there exists $x \in A$ such that $\rho(x) \neq 0$ and $\sigma(x) = 0$ for $\sigma \in S$. Then $\pi(\zeta)(x) = 0$ for almost every ζ , and hence $\pi(x) = 0$. Therefore ρ is not weakly contained in π .

Suppose that $\rho \in S$. Let $x \in A$ satisfy $\pi(x) = 0$. Let U be the open subset of $\widehat{A}$ consisting of the ζ such that $\pi(\zeta)(x) \neq 0$. Since $\pi(\zeta)(x) = 0$ almost everywhere, U is null, and is therefore disjoint from S . Thus $\rho(x) = 0$, and ρ is weakly contained in π . This proves (i).

Let ε_ρ be the Dirac measure at ρ . To say that $\mu(\{\rho\}) > 0$ means that every μ -null set is ε_ρ -null; in other words, ε_ρ is based on μ , or equivalently ρ is quasi-equivalent to a subrepresentation ρ' of π (8.4.5). But every representation quasi-equivalent to an irreducible representation contains it (5.2.1, 5.3.1). The preceding condition therefore also means that ρ is equivalent to a subrepresentation of π .

8.6.9. Retain the notation of 8.6.8. We see that the support of π (3.4.6) is precisely the support of μ . We shall say that a subset P of $\widehat{A}$ carries π if P carries μ , that is, if $\mu(\widehat{A} \setminus P) = 0$. Naturally S carries π , but in general there are strictly smaller sets than S which carry π . Nevertheless, if $\rho \in \widehat{A}$ is such that $\mu(\{\rho\}) > 0$, then every set carrying π necessarily contains ρ .

Bibliography: [31, 56, 91].

8.7. An Interlude

Since the decomposition of representations into irreducible representations was one of our principal objectives, we shall pause for a moment to summarize the main results obtained.

The situation is ideal for finite-dimensional representations: each one decomposes as a finite sum of irreducible representations, and any two such decompositions are isomorphic.

Let us pass to infinite-dimensional representations, restricting ourselves (because little better is known) to separable C^* -algebras and representations on separable spaces. The most elementary examples (even commutative ones) show that the notion of Hilbert-space direct sum must be replaced by that of direct integral. This corresponds to the well-known fact that the decomposition of Hermitian operators in infinite dimension involves *infinitesimal* spectral projections.

If A is a separable postliminal C^* -algebra (an important case for applications to group theory), there is a fairly satisfactory theory. The set $\widehat{A}$ of irreducible representations is a locally quasi-compact space with a countable base, which is almost locally compact (and is actually locally compact if A is a continuous-trace C^* -algebra). The Borel structure defined by the topology, which equals the Mackey structure, is standard. Every representation is uniquely a sum of representations of multiplicities $1, 2, \dots, \infty$. Every representation of multiplicity n is the sum of n equivalent multiplicity-free representations, whose class is well determined. And every multiplicity-free nondegenerate representation π can be written as an integral

$$\int_{\widehat{A}}^{\oplus} \zeta \, d\mu_{\pi}(\zeta),$$

where the class of μ_{π} is completely determined by π . The map

$$\pi \longmapsto \text{the class of } \mu_{\pi}$$

defines a bijection from the set of classes of multiplicity-free nondegenerate representations onto the set of measure classes on $\widehat{A}$.

If A is a separable non-postliminal C^* -algebra, matters are less satisfactory. The topological space $\widehat{A}$, the Borel structure underlying its topology, and the Mackey Borel structure (now distinct from the preceding one) have pathological properties (cf. 9.5.6). Every representation can indeed be written as an integral of irreducible representations, but uniqueness is completely lost. Moreover, since representations of types II and III exist (9.5.4), there can be no question of

reducing to multiplicity-free representations. Introduce the quasi-spectrum $\check{A}$, which is not a topological space, but a Borel space (and which reduces to the Borel space $\widehat{A}$ when A is postliminal). This Borel space is at least as pathological as $\widehat{A}$, since it contains $\widehat{A}$ as a Borel subset. But every representation π can be written canonically as an integral over $\check{A}$ with respect to a standard measure whose properties reflect those of π fairly well. This defines a bijective correspondence between the quasi-equivalence classes of representations of A and certain classes (unfortunately not all) of standard measures on $\check{A}$. We repeat that π has a canonical decomposition over $\check{A}$, but not a unique decomposition, since uniqueness already fails over $\widehat{A}$. For example, if π is a factor representation, its canonical decomposition is written $\pi = \pi$; but π can sometimes be written in many ways as an integral of irreducible representations.

It remains to ask whether a more satisfactory theory might be obtained by restricting to traceable representations. Only very partial results are known in this direction.

8.8. Disintegration of a Positive Functional and of a Trace

8.8.1. Theorem. *Let f be a positive functional on A , let H be the space of π_f , let Z be a Borel space, let μ be a positive measure on Z , let $\zeta \mapsto H(\zeta)$ be a measurable field of Hilbert spaces over Z , and let $\zeta \mapsto \pi(\zeta)$ be a measurable field of irreducible representations of A on the $H(\zeta)$, such that*

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta), \quad \pi_f = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta).$$

Let $\zeta \mapsto \xi(\zeta)$ be a vector field such that

$$\xi_f = \int_Z^{\oplus} \xi(\zeta) d\mu(\zeta).$$

- (i) *We have $\xi(\zeta) \neq 0$ for almost every ζ ; let $f(\zeta)$ be the pure positive functional, defined for almost every ζ by $\pi(\zeta)$ and $\xi(\zeta)$.*
- (ii) *For every $x \in A$, the function $\zeta \mapsto f(\zeta)(x)$ is integrable, and*

$$f(x) = \int_Z f(\zeta)(x) d\mu(\zeta).$$

Let $Y \subset Z$ be the measurable set of $\zeta \in Z$ such that $\xi(\zeta) \neq 0$, and let E_Y be the corresponding diagonalizable projection. For every $x \in A$,

$$E_Y \pi_f(x) \xi_f = \int_Z^{\oplus} \varphi_Y(\zeta) \pi(\zeta)(x) \xi(\zeta) d\mu(\zeta),$$

where φ_Y denotes the characteristic function of Y . Since $\varphi_Y(\zeta) = 0$ implies $\xi(\zeta) = 0$, we see that $E_Y \pi_f(x) \xi_f = \pi_f(x) \xi_f$. But ξ_f is cyclic for π_f ; hence $E_Y = 1$, and consequently $\xi(\zeta) \neq 0$ almost everywhere.

We have

$$f(\zeta)(x) = (\pi(\zeta)(x) \xi(\zeta) \mid \xi(\zeta)),$$

so the function $\zeta \mapsto f(\zeta)(x)$ is integrable, and its integral is

$$\int_Z (\pi(\zeta)(x) \xi(\zeta) \mid \xi(\zeta)) d\mu(\zeta) = (\pi_f(x) \xi_f \mid \xi_f) = f(x).$$

8.8.2. Theorem. *Let f be a semifinite lower semicontinuous trace on A^+ , defining the objects*

$$H_f, J_f, \lambda_f, \rho_f, \mathfrak{U}_f, \mathfrak{B}_f, t_f.$$

(i) *There exist:*

1° *a standard positive measure μ on $\check{A}$;*

2° *for every $\zeta \in \check{A}$, a character $f(\zeta)$;*

3° *a measurable-field structure of Hilbert spaces on the field $\zeta \mapsto H_{f(\zeta)}$,*

with the following properties:

a. *H_f is identified with*

$$\int_{\check{A}}^{\oplus} H_{f(\zeta)} d\mu(\zeta);$$

the fields

$$\begin{aligned} \zeta &\mapsto J_{f(\zeta)}, & \zeta &\mapsto \lambda_{f(\zeta)}, & \zeta &\mapsto \rho_{f(\zeta)}, \\ \zeta &\mapsto \mathfrak{U}_{f(\zeta)}, & \zeta &\mapsto \mathfrak{B}_{f(\zeta)}, & \zeta &\mapsto t_{f(\zeta)} \end{aligned}$$

are measurable, and

$$\begin{aligned} J_f &= \int_{\check{A}}^{\oplus} J_{f(\zeta)} d\mu(\zeta), & \lambda_f &= \int_{\check{A}}^{\oplus} \lambda_{f(\zeta)} d\mu(\zeta), & \rho_f &= \int_{\check{A}}^{\oplus} \rho_{f(\zeta)} d\mu(\zeta), \\ \mathfrak{U}_f &= \int_{\check{A}}^{\oplus} \mathfrak{U}_{f(\zeta)} d\mu(\zeta), & \mathfrak{B}_f &= \int_{\check{A}}^{\oplus} \mathfrak{B}_{f(\zeta)} d\mu(\zeta), & t_f &= \int_{\check{A}}^{\oplus} t_{f(\zeta)} d\mu(\zeta). \end{aligned}$$

The algebra $\mathfrak{U}_f \cap \mathfrak{B}_f$ is the algebra of diagonalizable operators.

b. Almost everywhere,

$$\lambda_{f(\zeta)} \in \zeta \quad \text{and} \quad \bar{\rho}_{f(\zeta)}^0 \in \zeta.$$

c. For every $x \in A^+$, the function $\zeta \mapsto f(\zeta)(x)$ is measurable and

$$f(x) = \int_{\check{A}} f(\zeta)(x) d\mu(\zeta).$$

(ii) Suppose that a measure μ' , characters $f'(\zeta)$, and a measurable-field structure of Hilbert spaces on the field $\zeta \mapsto H_{f'(\zeta)}$ are given, with the same properties as in (i). Then μ and μ' are equivalent. Write $\mu' = d\mu$, where $d(\zeta) > 0$ almost everywhere. We have

$$f(\zeta) = d(\zeta)f'(\zeta) \quad \text{almost everywhere.}$$

After changing the $f'(\zeta)$ on a null set, there is an isomorphism from the measurable field $\zeta \mapsto H_{f(\zeta)}$ onto the measurable field $\zeta \mapsto H_{f'(\zeta)}$ which carries

$$\begin{array}{lll} J_{f(\zeta)} \text{ to } J_{f'(\zeta)}, & \lambda_{f(\zeta)} \text{ to } \lambda_{f'(\zeta)}, & \rho_{f(\zeta)} \text{ to } \rho_{f'(\zeta)}, \\ \mathfrak{U}_{f(\zeta)} \text{ to } \mathfrak{U}_{f'(\zeta)}, & \mathfrak{B}_{f(\zeta)} \text{ to } \mathfrak{B}_{f'(\zeta)}, & t_{f(\zeta)} \text{ to } d(\zeta)t_{f'(\zeta)}. \end{array}$$

Proof of (i). Let B be the Hilbert algebra $\mathfrak{n}_f/N_f$, dense in H_f , and let B' be the corresponding completed Hilbert algebra. There exist (A 84, A 94, and A 95):

- 1° a standard Borel space S ;
 - 2° a positive measure μ on S ;
 - 3° a measurable field $\zeta \mapsto H(\zeta)$ of nonzero Hilbert spaces over S ;
 - 4° a measurable field $\zeta \mapsto B(\zeta)$ of completed Hilbert algebras in the $H(\zeta)$,
- such that H_f is identified with

$$\int_S^{\oplus} H(\zeta) d\mu(\zeta),$$

$\mathfrak{U}_f \cap \mathfrak{B}_f$ with the algebra of diagonalizable operators, B' with

$$\int_S^{\oplus} B(\zeta) d\mu(\zeta),$$

$\mathfrak{U}_f$ and $\mathfrak{B}_f$ with

$$\int_S^{\oplus} \mathfrak{U}(\zeta) d\mu(\zeta) \quad \text{and} \quad \int_S^{\oplus} \mathfrak{B}(\zeta) d\mu(\zeta)$$

respectively, where $\mathfrak{U}(\zeta)$ and $\mathfrak{B}(\zeta)$ denote the left and right von Neumann algebras associated with $B(\zeta)$; J_f with

$$\int_S^\oplus J(\zeta) d\mu(\zeta),$$

where $J(\zeta)$ denotes the involution defined by $B(\zeta)$; and t_f with

$$\int_S^\oplus t(\zeta) d\mu(\zeta),$$

where $t(\zeta)$ denotes the natural trace on $\mathfrak{U}(\zeta)^+$ defined by $B(\zeta)$. Since the algebra of diagonalizable operators is the common center of $\mathfrak{U}_f$ and $\mathfrak{B}_f$, the algebras $\mathfrak{U}(\zeta)$ and $\mathfrak{B}(\zeta)$ are factors for almost every ζ (A 89).

By 8.3.1, there is a measurable field $\zeta \mapsto \lambda(\zeta)$ (respectively, $\zeta \mapsto \rho(\zeta)$) of representations of A (respectively, A°) on the $H(\zeta)$ such that

$$\lambda_f = \int_S^\oplus \lambda(\zeta) d\mu(\zeta), \quad \rho_f = \int_S^\oplus \rho(\zeta) d\mu(\zeta).$$

Let (x_i) be a dense sequence in A . Since the $\lambda_f(x_i)$ generate the von Neumann algebra $\mathfrak{U}_f$, the $\lambda(\zeta)(x_i)$ generate $\mathfrak{U}(\zeta)$ for almost every ζ (A 88). Thus, almost everywhere, $\mathfrak{U}(\zeta)$ is the von Neumann algebra generated by $\lambda(\zeta)(A)$. Similarly, almost everywhere, $\mathfrak{B}(\zeta)$ is the von Neumann algebra generated by $\rho(\zeta)(A)$.

If $x \in A^+$, the function $\zeta \mapsto t(\zeta)(\lambda(\zeta)(x))$ is measurable, and (A 92)

$$f(x) = t_f(\lambda_f(x)) = \int_S t(\zeta)(\lambda(\zeta)(x)) d\mu(\zeta) = \int_S f(\zeta)(x) d\mu(\zeta),$$

where

$$f(\zeta)(x) = t(\zeta)(\lambda(\zeta)(x)) \quad \text{for every } x \in A^+.$$

Let (y_i) be a sequence dense in $\mathfrak{n}_f$ for its pre-Hilbert structure (6.3.6). Since the $\lambda_f(y_i)$ generate the von Neumann algebra $\mathfrak{U}_f$, the $\lambda(\zeta)(y_i)$ generate, for almost every ζ , the von Neumann algebra $\mathfrak{U}(\zeta)$. On the other hand,

$$\int_S f(\zeta)(y_i y_i^*) d\mu(\zeta) = f(y_i y_i^*) < +\infty,$$

so $f(\zeta)(y_i y_i^*) < +\infty$ almost everywhere. Hence, for almost every ζ , $(\lambda(\zeta), t(\zeta))$ is a nonzero traced factor representation of A . Consequently almost all the $f(\zeta)$ are characters (6.7.3).

Write

$$\Lambda_f y_i = \int_S^\oplus y_i(\zeta) d\mu(\zeta).$$

For arbitrary i and j ,

$$\begin{aligned} \int_S^\oplus \lambda(\zeta)(y_i)y_j(\zeta) d\mu(\zeta) &= \lambda_f(y_i)\Lambda_f y_j \\ &= (\Lambda_f y_i)(\Lambda_f y_j) = \int_S^\oplus y_i(\zeta)y_j(\zeta) d\mu(\zeta). \end{aligned}$$

Therefore

$$\lambda(\zeta)(y_i)y_j(\zeta) = y_i(\zeta)y_j(\zeta)$$

almost everywhere. Thus $\lambda(\zeta)(y_i)$ is almost everywhere the continuous operator extending left multiplication by $y_i(\zeta)$; an analogous result holds for $\rho(\zeta)(y_i)$. It follows that, for almost every ζ and every i ,

$$\begin{aligned} (y_i(\zeta) \mid y_i(\zeta)) &= t(\zeta)(\lambda(\zeta)(y_i)\lambda(\zeta)(y_i^*)) \\ &= t(\zeta)(\lambda(\zeta)(y_i y_i^*)) = f(\zeta)(y_i y_i^*). \end{aligned}$$

If the y_i have been chosen to form an involutive algebra over the field of rational complex numbers, the equality

$$(y_i(\zeta) \mid y_i(\zeta)) = f(\zeta)(y_i y_i^*)$$

shows that $H(\zeta)$ is identified with a closed vector subspace of $H_{f(\zeta)}$. We may also require the y_i to be chosen so that every product $x_i y_j$ is some y_k . Then the subspace $H(\zeta)$ of $H_{f(\zeta)}$ is almost everywhere invariant under $\lambda_{f(\zeta)}$ and $\rho_{f(\zeta)}$; since $f(\zeta)$ is a character, $\lambda_{f(\zeta)}(A)$ and $\rho_{f(\zeta)}(A)$ generate the von Neumann algebra $\mathcal{L}(H_{f(\zeta)})$, and hence $H(\zeta) = H_{f(\zeta)}$ almost everywhere. We have

$$J(\zeta)y_i(\zeta) = (J_f \Lambda_f y_i)(\zeta) = y_i^*(\zeta) = J_{f(\zeta)} y_i(\zeta)$$

almost everywhere, and therefore $J(\zeta) = J_{f(\zeta)}$ almost everywhere. Likewise,

$$\lambda(\zeta)(y_i)y_j(\zeta) = y_i(\zeta)y_j(\zeta) = \lambda_{f(\zeta)}(y_i)y_j(\zeta).$$

Thus $\lambda(\zeta)$ and $\lambda_{f(\zeta)}$ agree on $\mathfrak{n}_f$ almost everywhere, and consequently agree on A , since $\lambda(\zeta)$ and $\lambda_{f(\zeta)}$ are factor representations that are nonzero on $\mathfrak{n}_f$ (2.10.4, 2.11.1). Similarly, $\rho(\zeta) = \rho_{f(\zeta)}$. It follows that

$$\mathfrak{U}(\zeta) = \mathfrak{U}_{f(\zeta)}, \quad \mathfrak{B}(\zeta) = \mathfrak{B}_{f(\zeta)}.$$

Finally, for every $x \in A^+$,

$$t(\zeta)(\lambda(\zeta)(x)) = f(\zeta)(x) = t_{f(\zeta)}(\lambda_{f(\zeta)}(x)) = t_{f(\zeta)}(\lambda(\zeta)(x))$$

almost everywhere. Writing this for $x = x_1, x = x_2, \dots$ and applying A 29, we conclude that $t(\zeta) = t_{f(\zeta)}$ almost everywhere.

After changing the $\lambda(\zeta)$ on a null set, the $\lambda(\zeta)$ are pairwise disjoint (8.4.1). Exactly as in the proof of 8.4.2, we may replace S by a standard Borel subset of $\check{A}$ and arrange that

$$\lambda_{f(\zeta)} = \lambda(\zeta) \in \zeta \quad \text{almost everywhere.}$$

Then

$$\bar{\rho}_{f(\zeta)}^0 \approx \lambda_{f(\zeta)} \implies \bar{\rho}_{f(\zeta)}^0 \in \zeta,$$

and all the assertions in (i) are proved.

Proof of (ii). The equivalence of μ and μ' follows from 8.4.2(ii). Write $\mu' = d\mu$, where $d(\zeta) > 0$ almost everywhere. For almost every ζ , $f(\zeta)$ and $f'(\zeta)$ are the characters of two quasi-equivalent representations, and hence there is a number $k(\zeta) > 0$ such that

$$f'(\zeta) = k(\zeta)f(\zeta).$$

For every $x \in A^+$,

$$\int_{\check{A}} f(\zeta)(x) d\mu(\zeta) = f(x) = \int_{\check{A}} f'(\zeta)(x) d\mu'(\zeta) = \int_{\check{A}} k(\zeta)f(\zeta)(x) d(\zeta) d\mu(\zeta).$$

Hence, for $x = y_i y_i^*$,

$$\int_{\check{A}} (y_i(\zeta) | y_i(\zeta)) d\mu(\zeta) = \int_{\check{A}} (y_i(\zeta) | y_i(\zeta)) k(\zeta) d(\zeta) d\mu(\zeta). \quad (1)$$

The equalities

$$y_1(\zeta) = y_2(\zeta) = \dots = 0$$

define a μ -null subset of $\check{A}$. Since the functions

$$\zeta \mapsto (y_i(\zeta) | y_i(\zeta)) k(\zeta) d(\zeta)$$

are μ -measurable, we first see that kd is measurable. Since the linear combinations of the functions $\zeta \mapsto (y_i(\zeta) | y_i(\zeta))$ are dense in $L^1(\mu)$, equality (1) then proves that $kd = 1$ almost everywhere. Thus

$$f(\zeta) = d(\zeta)f'(\zeta) \quad \text{almost everywhere.}$$

The map $x \mapsto d(\zeta)^{1/2}x$ on $\mathfrak{n}_{f(\zeta)}$ then defines, almost everywhere, an isomorphism $V(\zeta)$ from $H_{f(\zeta)}$ onto $H_{f'(\zeta)}$. This gives an isomorphism from the field $\zeta \mapsto H_{f(\zeta)}$ onto the field $\zeta \mapsto H_{f'(\zeta)}$ which plainly has the properties in (ii).

8.8.3. Retain the notation of 8.8.2. Let (z_i) be a sequence of elements of $\mathfrak{m}_f^+$. The functions $\zeta \mapsto f(\zeta)(z_i)$ are μ -integrable, and hence finite almost everywhere. Thus there exists a null subset N of $\check{A}$ such that

$$z_i \in \bigcap_{\zeta \in \check{A} \setminus N} \mathfrak{m}_{f(\zeta)} \quad \text{for every } i.$$

In particular, after changing the $f(\zeta)$ on a null set,

$$\mathfrak{m}_f \cap \left(\bigcap_{\zeta \in \check{A}} \mathfrak{m}_{f(\zeta)} \right)$$

is dense in $\mathfrak{m}_f$ for the norm of A .

8.8.4. The measure μ and the characters $f(\zeta)$ constructed in 8.8.2 have the following additional property. Let $x \in \mathfrak{n}_f$, let h be a bounded μ -measurable function on $\check{A}$, and let Δ_h be the corresponding diagonalizable operator. Then

$$(\Delta_h \Lambda_f x \mid \Lambda_f x) = \int_{\check{A}} h(\zeta) f(\zeta) (x^* x) d\mu(\zeta).$$

Indeed, write

$$\Lambda_f x = \int_{\check{A}}^{\oplus} x(\zeta) d\mu(\zeta).$$

Then

$$(\Delta_h \Lambda_f x \mid \Lambda_f x) = \int_{\check{A}} h(\zeta) (x(\zeta) \mid x(\zeta)) d\mu(\zeta).$$

In the proof of 8.8.2(i), however, we saw that

$$(x(\zeta) \mid x(\zeta)) = f(\zeta)(x^* x) \quad \text{almost everywhere.}$$

8.8.5. Let A be a separable postliminal C^* -algebra. Let $\zeta \mapsto K(\zeta)$ be the canonical field of Hilbert spaces over $\widehat{A}$ (8.6.1). Choose a field $\zeta \mapsto \pi(\zeta)$ of representations of A on the $K(\zeta)$ such that $\pi(\zeta)$ belongs to the class ζ for every ζ and which is measurable for every positive measure on $\widehat{A}$ (8.6.2). For simplicity, write ζ instead of $\pi(\zeta)$. Let $K(\zeta)$ be the Hilbert space conjugate to $K(\zeta)$. Denote by $\zeta \otimes 1$ the representation

$$x \longmapsto \zeta(x) \otimes 1$$

of A on $K(\zeta) \otimes \overline{K(\zeta)}$, and by $1 \otimes \overline{\zeta}$ the representation

$$x \mapsto 1 \otimes \overline{\zeta(x)}$$

of A on the same space. Let J_ζ be the canonical involution of $K(\zeta) \otimes \overline{K(\zeta)}$, and let t_ζ be the trace

$$T \otimes 1 \mapsto \text{Tr } T$$

on $(\mathcal{L}(K(\zeta)) \otimes \mathbb{C})^+$.

Theorem. *Let f be a lower semicontinuous trace on A^+ such that $\mathfrak{m}_f$ is dense in A . There exist a positive measure μ on $\widehat{A}$ and an isomorphism W from H_f onto*

$$\int_{\widehat{A}}^{\oplus} (K(\zeta) \otimes \overline{K(\zeta)}) d\mu(\zeta)$$

with the following properties:

(i) W carries

$$\begin{aligned} J_f &\text{ to } \int_{\widehat{A}}^{\oplus} J_\zeta d\mu(\zeta), & \lambda_f &\text{ to } \int_{\widehat{A}}^{\oplus} (\zeta \otimes 1) d\mu(\zeta), \\ \bar{\rho}_f^\circ &\text{ to } \int_{\widehat{A}}^{\oplus} (1 \otimes \overline{\zeta}) d\mu(\zeta), & \mathfrak{U}_f &\text{ to } \int_{\widehat{A}}^{\oplus} (\mathcal{L}(K(\zeta)) \otimes \mathbb{C}) d\mu(\zeta), \\ \mathfrak{B}_f &\text{ to } \int_{\widehat{A}}^{\oplus} (\mathbb{C} \otimes \mathcal{L}(\overline{K(\zeta)})) d\mu(\zeta), & t_f &\text{ to } \int_{\widehat{A}}^{\oplus} t_\zeta d\mu(\zeta), \end{aligned}$$

and carries $\mathfrak{U}_f \cap \mathfrak{B}_f$ to the algebra of diagonalizable operators.

(ii) If $x \in A^+$, the function $\zeta \mapsto \text{Tr } \zeta(x)$ on $\widehat{A}$ is lower semicontinuous, and

$$f(x) = \int_{\widehat{A}} \text{Tr } \zeta(x) d\mu(\zeta).$$

Introduce all the notation of 8.8.2. Since A is postliminal, the Borel space $\check{A}$ is identified with the Borel space $\widehat{A}$ (7.3.7). For $x \in A^+$ and $\zeta \in \widehat{A}$, put

$$f'(\zeta)(x) = \text{Tr } \zeta(x).$$

By 8.8.2(i)b, for almost every ζ there is a number $k(\zeta) > 0$ such that $f'(\zeta) = k(\zeta)f(\zeta)$. For every $x \in A^+$, the function $\zeta \mapsto f(\zeta)(x)$ is measurable (8.8.2(i)c), while $\zeta \mapsto f'(\zeta)(x)$ is lower semicontinuous (3.5.9). For every $\zeta_0 \in \widehat{A}$ there exists $x \in \mathfrak{m}_f$ such that $f'(\zeta_0)(xx^*) > 0$, and therefore throughout an open neighborhood V_{ζ_0} of ζ_0 . Moreover, the function $\zeta \mapsto f(\zeta)(xx^*)$ is integrable (8.8.2(i)c),

and hence finite almost everywhere. Thus $\zeta \mapsto k(\zeta)$ is measurable on V_{ζ_0} . Covering $\widehat{A}$ by a sequence of sets V_{ζ_0} (3.3.4), we see that k is measurable. Replacing μ by $k^{-1}\mu$, we may henceforth suppose that $f'(\zeta) = f(\zeta)$ for every ζ .

There is then an isomorphism from $H_{f(\zeta)}$ onto $K(\zeta) \otimes \overline{K(\zeta)}$ which carries

$$\lambda_{f(\zeta)} \text{ to } \zeta \otimes 1, \quad \bar{\rho}_{f(\zeta)}^0 \text{ to } 1 \otimes \bar{\zeta},$$

$$\mathfrak{U}_{f(\zeta)} \text{ to } \mathcal{L}(K(\zeta)) \otimes \mathbb{C}, \quad \mathfrak{B}_{f(\zeta)} \text{ to } \mathbb{C} \otimes \mathcal{L}(\overline{K(\zeta)}),$$

and $J_{f(\zeta)}$ to J_ζ , $t_{f(\zeta)}$ to t_ζ (6.7.7). In particular,

$$\dim H_{f(\zeta)} = \dim(K(\zeta) \otimes \overline{K(\zeta)}) \quad \text{for every } \zeta.$$

It follows from A 72 that the measurable fields $\zeta \mapsto H_{f(\zeta)}$ and $\zeta \mapsto K(\zeta) \otimes \overline{K(\zeta)}$ are isomorphic. We may therefore identify them. For every $\zeta \in \widehat{A}$, let $\mathcal{E}(\zeta)$ be the set of automorphisms of the Hilbert space $K(\zeta) \otimes \overline{K(\zeta)}$ which carry $\lambda_{f(\zeta)}$ to $\zeta \otimes 1$, $\bar{\rho}_{f(\zeta)}^0$ to $1 \otimes \bar{\zeta}$, and $J_{f(\zeta)}$ to J_ζ (and therefore carry $\mathfrak{U}_{f(\zeta)}$ to $\mathcal{L}(K(\zeta)) \otimes \mathbb{C}$, $\mathfrak{B}_{f(\zeta)}$ to $\mathbb{C} \otimes \mathcal{L}(\overline{K(\zeta)})$, and $t_{f(\zeta)}$ to t_ζ). We have $\mathcal{E}(\zeta) \neq \emptyset$. Hence there exists (A 82) a measurable operator field $\zeta \mapsto W(\zeta)$ such that $W(\zeta) \in \mathcal{E}(\zeta)$ for every ζ . The identification of the fields can therefore be made so that

$$\lambda_{f(\zeta)} = \zeta \otimes 1, \quad \bar{\rho}_{f(\zeta)}^0 = 1 \otimes \bar{\zeta}, \quad J_{f(\zeta)} = J_\zeta.$$

The theorem now follows from 8.8.2(i).

8.8.6. Theorem. *Retain the hypotheses and notation of 8.8.5. Let μ' be a positive measure on $\widehat{A}$ such that, for every $x \in A^+$,*

$$f(x) = \int_{\widehat{A}} \text{Tr } \zeta(x) d\mu'(\zeta).$$

Then $\mu = \mu'$.

Let G be an open subset of $\widehat{A}$, let φ_G be its characteristic function, let I be the closed two-sided ideal of A such that $\widehat{I} = G$, and let $(u_1, u_2, \dots)$ be an increasing approximate identity of the C^* -algebra I (1.7.2). Let $x \in A^+$. If $\zeta \in \widehat{A} \setminus G$, then

$$\zeta(x^{1/2}u_n x^{1/2}) = 0 \quad \text{for every } n;$$

if $\zeta \in G$, then

$$\zeta(x^{1/2}u_n x^{1/2}) = \zeta(x)^{1/2} \zeta(u_n) \zeta(x)^{1/2}$$

increases strongly to $\zeta(x)$, and therefore $\text{Tr } \zeta(x^{1/2}u_n x^{1/2})$ increases to $\text{Tr } \zeta(x)$. But

$$\int_{\widehat{A}} \text{Tr } \zeta(x^{1/2}u_n x^{1/2}) d\mu(\zeta) = f(x^{1/2}u_n x^{1/2}) = \int_{\widehat{A}} \text{Tr } \zeta(x^{1/2}u_n x^{1/2}) d\mu'(\zeta).$$

Passing to the limit gives

$$\int_{\widehat{A}} \varphi_G(\zeta) \text{Tr } \zeta(x) d\mu(\zeta) = \int_{\widehat{A}} \varphi_G(\zeta) \text{Tr } \zeta(x) d\mu'(\zeta).$$

If $x \in \mathfrak{m}_f^+$, this formula extends first to the case where G is a closed subset of $\widehat{A}$, and then, by passage to the limit, to the case where G is an arbitrary Borel subset of $\widehat{A}$. Thus, for $x \in \mathfrak{m}_f^+$, the measures

$$\text{Tr } \zeta(x) d\mu(\zeta) \quad \text{and} \quad \text{Tr } \zeta(x) d\mu'(\zeta)$$

are equal. Let $(x_1, x_2, \dots)$ be a dense sequence in $\mathfrak{m}_f^+$. The set of $\zeta \in \widehat{A}$ such that $\text{Tr } \zeta(x_i) = +\infty$ for at least one i is a Borel subset of $\widehat{A}$ which is null for both μ and μ' . On the other hand, for every $\zeta_0 \in \widehat{A}$ there is an i such that $\text{Tr } \zeta_0(x_i) > 0$, and hence throughout a neighborhood of ζ_0 . Thus $\mu = \mu'$ in a neighborhood of every ζ_0 , and consequently $\mu = \mu'$.

8.8.7. Retain the hypotheses and notation of 8.8.5. Let $x \in \mathfrak{n}_f$, let h be a bounded μ -measurable function on $\widehat{A}$, and let Δ_h be the corresponding diagonalizable operator. By 8.8.4,

$$(\Delta_h \Lambda_f x \mid \Lambda_f x) = \int_{\widehat{A}} h(\zeta) \text{Tr}(\zeta(x)^* \zeta(x)) d\mu(\zeta).$$

Bibliography: [52, 53, 57, 94, 95, 96, 101, 108, 110, 137, 148, 161, 162, 163, 164, 171, 179, 186].

8.9. Complements

***8.9.1.** Let A be a separable C^* -algebra, Z a standard Borel space, μ a positive measure on Z , $\zeta \mapsto H(\zeta)$ a μ -measurable field of Hilbert spaces over Z , and $\zeta \mapsto \pi(\zeta)$ a μ -measurable field of representations of A on the $H(\zeta)$. Put

$$\pi = \int_Z^\oplus \pi(\zeta) d\mu(\zeta),$$

and let $\mathfrak{Z}$ be the algebra of diagonalizable operators. Then the $\pi(\zeta)$ are homogeneous almost everywhere if and only if $\mathfrak{Z}$ contains the ideal center $\mathfrak{Y}$ associated with π (5.7.6). The $\pi(\zeta)$ are homogeneous and have pairwise distinct kernels, after deletion of a null set, if and only if $\mathfrak{Z} = \mathfrak{Y}$. [20]

8.9.2. Let A be a separable C^* -algebra. If A is not postliminal, there exist standard measures on $\hat{A}$ whose class is associated with no representation of A on a separable space. [25, 199]

***8.9.3.** Theorems 8.8.5 and 8.8.6 can be generalized to nonseparable postliminal C^* -algebras. [16]

8.9.4.

- a.* Let A be a separable C^* -algebra. The canonical Borel field of Hilbert spaces over $\hat{A}$ is defined as in 8.6.1. For every standard positive measure μ on $\hat{A}$, there is a Borel map $\zeta \mapsto \pi(\zeta)$ from $\hat{A}$ into $\text{Irr}(A)$ such that $\pi(\zeta) \in \zeta$ almost everywhere. The class of

$$\int_{\hat{A}}^{\oplus} \pi(\zeta) d\mu(\zeta)$$

depends only on the class C of μ ; denote it by $\mathcal{R}(C)$.

- b.* For every class R of type I representations of A , denote by $C(R)$ the associated class of standard measures on $\hat{A}$. If R is a class of multiplicity-free representations of A , then

$$R = \mathcal{R}(C(R)).$$

- c.* If C is a class of standard measures on $\hat{A}$ and $\mathcal{R}(C)$ is of type I, then $\mathcal{R}(C)$ is multiplicity-free and

$$C(\mathcal{R}(C)) = C.$$

- *d.* If A is not postliminal, there exist two distinct classes C, C' of standard measures on $\hat{A}$ such that $\mathcal{R}(C) = \mathcal{R}(C')$ is a type II factor representation. [87, 91, 185, 199]

Problems: Which classes C of standard measures on $\hat{A}$ are such that $\mathcal{R}(C)$ is of type I? Which equivalence classes of representations have the form $\mathcal{R}(C)$, where C is a class of standard measures? Is every representation quasi-equivalent to an element of a class $\mathcal{R}(C)$, where C is a class of standard measures?

***8.9.5.** Let A be a separable C^* -algebra. Define the canonical Borel field of Hilbert spaces over $\check{A}$ to be equal to the canonical field over $\hat{A}$, and over $\check{A} \setminus \hat{A}$ to be the

constant field defined by a Hilbert space of dimension $\aleph_0$. If μ is a standard positive measure on $\check{A}$, there is a Borel map $\zeta \mapsto \pi(\zeta)$ from $\check{A}$ into $\text{Fac}(A)$ such that $\pi(\zeta) \in \zeta$ almost everywhere. The quasi-equivalence class of

$$\int_{\check{A}}^{\oplus} \pi(\zeta) d\mu(\zeta)$$

depends only on the class C of μ ; denote it by $\mathcal{R}'(C)$. For every quasi-equivalence class R of representations of A , denote by $C'(R)$ the associated class of standard measures on $\check{A}$. We have

$$\mathcal{R}'(C'(R)) = R.$$

If A is not postliminal, there exist classes C of standard measures on $\check{A}$ such that

$$C'(\mathcal{R}'(C)) \neq C.$$

[20, 25, 91, 199]

8.9.6. Let A be a separable C^* -algebra, let π be a representation of A on a separable Hilbert space, and let

$$\pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta)$$

be a disintegration of π into irreducible representations. If π is a type I factor representation, then almost all the $\pi(\zeta)$ are equivalent. [90]

§ 9. — Type I C^* -Algebras

9.1. Statement of the Theorem. Beginning of the Proof

Theorem. *Let A be a separable C^* -algebra. The following conditions are equivalent:*

- (i) A is of type I;
- (ii) every factor representation of A is of type I;
- (iii) A is postliminal;
- (iv) if π is an irreducible representation of A , then $\pi(A)$ contains the set of compact operators on H_π ;
- (v) if π_1 and π_2 are two irreducible representations of A with the same kernel, then π_1 and π_2 are equivalent.

(For the case in which A is not separable, cf. 9.5.9.)

The scheme of the proof is as follows:

$$(iii) \Rightarrow (iv) \Rightarrow (v) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (i) \Rightarrow (ii).$$

$$(iii) \Rightarrow (iv) : \text{cf. 4.3.7,}$$

$$(iv) \Rightarrow (v) : \text{cf. 4.1.10,}$$

$$(iii) \Rightarrow (i) : \text{cf. 5.5.2,}$$

$$(i) \Rightarrow (ii) : \text{clear.}$$

$(v) \Rightarrow (ii)$: suppose (v) holds. Let π be a factor representation of A , and let us prove that it is of type I. We may suppose that $H_\pi = H$ is separable (5.3.7). Apply Theorem 8.5.2, whose notation we adopt. For every μ -measurable subset Y of Z , let E_Y be the corresponding diagonalizable projection; it commutes with $\pi(A)$, and hence defines a subrepresentation π_Y of π . If $\mu(Y) \neq 0$, then $E_Y \neq 0$, so $\pi_Y \approx \pi$ (5.3.5), and therefore

$$\|\pi(x)\| = \|\pi_Y(x)\| = \|E_Y \pi(x)\| \quad \text{for every } x \in A$$

by condition (ii) of 5.3.1. For $n = 1, 2, \dots$, let $Y_{n,x}$ be the set of $\zeta \in Z$ such that

$$\|\pi(\zeta)(x)\| \leq \|\pi(x)\| - \frac{1}{n}.$$

We have

$$\|E_{Y_{n,x}} \pi(x)\| \leq \|\pi(x)\| - \frac{1}{n}$$

(A 77), so $Y_{n,x}$ is null by what precedes. Since this is true for every n , it follows that

$$\|\pi(\zeta)(x)\| = \|\pi(x)\|$$

outside a null set N_x . Let D be a countable dense subset of A . Then

$$N = \bigcup_{x \in D} N_x$$

is null. If $\zeta \notin N$, then $\|\pi(\zeta)(x)\| = \|\pi(x)\|$ for every $x \in D$, and hence for every $x \in A$; consequently $\ker \pi(\zeta) = \ker \pi$. By hypothesis (v), almost all the $\pi(\zeta)$ are equivalent to one and the same irreducible representation π_0 . Thus π is a multiple of π_0 (8.1.7), and is therefore of type I (5.4.11).

It remains to prove (ii) $\Rightarrow$ (iii), which will require fairly extensive preparations.

Bibliography: [11, 28, 46, 76, 77].

9.2. Preliminaries on Systems of Matrix Units

We give these preliminaries in greater detail than is needed below, in order to illuminate somewhat a later, very technical proof.

9.2.1. Equip $\mathbb{C}^p$ with its canonical structure as a Hilbert space of dimension p :

$$((z_1, \dots, z_p) \mid (z'_1, \dots, z'_p)) = z_1 \overline{z'_1} + \dots + z_p \overline{z'_p}.$$

Then the algebra $M_p(\mathbb{C})$ of complex p -by- p matrices is identified with the C^* -algebra $\mathcal{L}(\mathbb{C}^p)$, which is simple and has dimension p^2 . Let (e_{ij}) be the canonical basis of $M_p(\mathbb{C})$. We have

$$e_{kl}e_{ij} = \delta_l^i e_{kj}, \quad e_{ij}^* = e_{ji},$$

where δ_l^i is the Kronecker symbol.

9.2.2. More generally, let A be a C^* -algebra, and let $(f_{ij})_{1 \leq i, j \leq p}$ be a family of elements of A . If

$$f_{kl}f_{ij} = \delta_l^i f_{kj}, \quad f_{ij}^* = f_{ji},$$

this family is called a system of matrix units of order p . The linear combinations of the f_{ij} then form a C^* -subalgebra B of A . The linear map from $M_p(\mathbb{C})$ onto B which carries e_{ij} to f_{ij} for $1 \leq i, j \leq p$ is a homomorphism; since $M_p(\mathbb{C})$ is simple, this homomorphism is an isomorphism if $B \neq 0$.

Suppose that A is realized as a C^* -algebra of operators on a Hilbert space H . Since $f_{ii} = f_{ii}^* = f_{ii}^2$, the f_{ii} are projections. One verifies that f_{ij} is a partial isometry with initial projection f_{jj} and final projection f_{ii} . Put $v_j = f_{j1}$, $j = 1, \dots, p$. Then

$$f_{ij} = v_i v_j^*.$$

The v_j are partial isometries all having the initial projection $v_1 = f_{11}$; their final projections are pairwise orthogonal, that is, $v_i^* v_j = 0$ for $i \neq j$. Conversely, let $v_1, \dots, v_p$ be partial isometries in A all having v_1 as initial projection and satisfying $v_i^* v_j = 0$ for $i \neq j$. It is readily verified that, on putting $f_{ij} = v_i v_j^*$, the f_{ij} form a system of matrix units of order p . Geometrically, there are pairwise orthogonal closed vector subspaces $H_1, \dots, H_p$ of H such that v_j is a partial isometry with initial subspace H_1 and final subspace H_j , while v_1 is the projection onto H_1 .

9.2.3. Let $a_1, \dots, a_n$ be variables, each of which can take the values 0 and 1. Let

$$(v(a_1, \dots, a_n))_{a_1, \dots, a_n \in \{0,1\}}$$

be a system of partial isometries in A , all having $v(o_n)$ as initial projection, where o_n denotes the string $(0, \dots, 0)$ of n zeros, and such that

$$v(b_1, \dots, b_n)^* v(a_1, \dots, a_n) = 0 \quad \text{if} \quad (b_1, \dots, b_n) \neq (a_1, \dots, a_n).$$

Then the elements

$$v(b_1, \dots, b_n) v(a_1, \dots, a_n)^*$$

form a system of matrix units of order 2^n . If $H(a_1, \dots, a_n)$ is the final subspace of $v(a_1, \dots, a_n)$, the spaces $H(a_1, \dots, a_n)$ are pairwise orthogonal, $v(a_1, \dots, a_n)$ has $H(o_n)$ as its initial subspace, and $v(o_n)$ is the projection onto $H(o_n)$.

9.2.4. Let

$$(v(a_1, \dots, a_n))_{a_1, \dots, a_n \in \{0,1\}} \quad \text{and} \quad (v(a_1, \dots, a_{n+1}))_{a_1, \dots, a_{n+1} \in \{0,1\}}$$

be two systems of elements of A having the preceding properties. Suppose that:

1° the projection $v(o_n)$ is the sum of the projection $v(o_{n+1})$ and the final projection of $v(o_n, 1)$;

2°

$$v(a_1, \dots, a_{n+1}) = v(a_1, \dots, a_n) v(o_n, a_{n+1}).$$

If $H(a_1, \dots, a_n)$ and $H(a_1, \dots, a_{n+1})$ are the final subspaces of $v(a_1, \dots, a_n)$ and $v(a_1, \dots, a_{n+1})$, this implies that $H(a_1, \dots, a_n)$ is the Hilbert-space direct sum

of $H(a_1, \dots, a_n, 0)$ and $H(a_1, \dots, a_n, 1)$. We shall say that the system of the $v(a_1, \dots, a_{n+1})$ is obtained from the system of the $v(a_1, \dots, a_n)$ by bisection.

9.2.5. For $n = 0, 1, 2, \dots$, let

$$(v(a_1, \dots, a_n))_{a_1, \dots, a_n \in \{0,1\}}$$

be a system of elements of A having the properties of 9.2.3, each system being obtained from the preceding one by bisection. [For $n = 0$, the system reduces to the single element $v(\emptyset)$.] In particular, the following relations hold:

(i) if $j \leq k$ and $(a_1, \dots, a_j) \neq (b_1, \dots, b_j)$, then

$$v(a_1, \dots, a_j)^* v(b_1, \dots, b_k) = 0;$$

(ii) if $k \geq 1$, then

$$v(a_1, \dots, a_k) = v(a_1, \dots, a_{k-1})v(o_{k-1}, a_k);$$

(iii) if $j < k$, then

$$v(a_1, \dots, a_j)^* v(a_1, \dots, a_j)v(o_{k-1}, a_k) = v(o_{k-1}, a_k);$$

(iv)

$$v(o_k) \geq 0.$$

If $H(a_1, \dots, a_n)$ is the final subspace of $v(a_1, \dots, a_n)$, (i) means that $H(a_1, \dots, a_j)$ and $H(b_1, \dots, b_k)$ are orthogonal whenever $j \leq k$ and $(a_1, \dots, a_j) \neq (b_1, \dots, b_j)$; (iii) means that $H(o_j)$ contains $H(o_k)$ and $H(o_{k-1}, 1)$ for $j < k$.

9.2.6. In an arbitrary C^* -algebra one does not know how to construct systems of nonzero elements having all the properties of 9.2.5. Under certain fairly broad hypotheses, however (cf. 9.3.7), we shall succeed in constructing in A' systems of nonzero elements

$$(v(a_1, \dots, a_n))_{a_1, \dots, a_n \in \{0,1\}}, \quad n = 0, 1, 2, \dots,$$

satisfying conditions (i), (ii), (iii), and (iv). We shall now indicate some consequences of the properties (i), (ii), and (iii). We shall suppose that A is realized as a C^* -algebra of operators on a Hilbert space H . Put

$$e(n) = \sum_{a_1, \dots, a_n \in \{0,1\}} v(a_1, \dots, a_n)v(a_1, \dots, a_n)^*,$$

which is an element of A^+ . Let

$$E(n) = \overline{e(n)(H)} = H \ominus \ker e(n).$$

By (i), for fixed n the subspaces

$$v(a_1, \dots, a_n)(H) \quad (a_1, \dots, a_n \in \{0, 1\})$$

are pairwise orthogonal. Consequently $P_{E(n)}$ is the sum of the support projections of the operators

$$v(a_1, \dots, a_n)v(a_1, \dots, a_n)^*.$$

9.2.7. The spaces $E(n)$ form a decreasing sequence. Indeed, if $\xi \in H$ and $e(n)\xi = 0$, then

$$(v(a_1, \dots, a_n)v(a_1, \dots, a_n)^*\xi \mid \xi) = 0$$

for all $a_1, \dots, a_n$; hence $v(a_1, \dots, a_n)^*\xi = 0$ for all $a_1, \dots, a_n$, and therefore, by (ii), $v(a_1, \dots, a_{n+1})^*\xi = 0$ for all $a_1, \dots, a_{n+1}$. Thus $e(n+1)\xi = 0$.

9.2.8. The operators $v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*$ leave $E(n+1)$ invariant. Indeed, let $\eta \in H$ and $\xi = v(c_1, \dots, c_{n+1})\eta$. Then

$$\begin{aligned} & v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*\xi \\ &= v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*v(c_1, \dots, c_n)v(o_n, c_{n+1})\eta \\ &= \delta_{b_1}^{c_1} \cdots \delta_{b_n}^{c_n} v(a_1, \dots, a_n)v(o_n, c_{n+1})\eta \\ &= \delta_{b_1}^{c_1} \cdots \delta_{b_n}^{c_n} v(a_1, \dots, a_n, c_{n+1})\eta, \end{aligned} \tag{1}$$

which belongs to $E(n+1)$ by 9.2.6.

9.2.9. The operators $v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*$ induce on $E(n+1)$ a system of matrix units of order 2^n [thus $e(n)|_{E(n+1)}$ is a projection, and hence $e(n)|_{E(n+1)} = 1$ by 9.2.7]. Indeed, by (1),

$$\begin{aligned} & v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*v(a'_1, \dots, a'_n)v(b'_1, \dots, b'_n)^*\xi \\ &= \delta_{b_1}^{a'_1} \cdots \delta_{b_n}^{a'_n} v(a_1, \dots, a_n)v(b'_1, \dots, b'_n)^*v(c_1, \dots, c_n)v(o_n, c_{n+1})\eta \\ &= \delta_{b_1}^{a'_1} \cdots \delta_{b_n}^{a'_n} \delta_{b'_1}^{c_1} \cdots \delta_{b'_n}^{c_n} v(a_1, \dots, a_n)v(o_n, c_{n+1})\eta \\ &= \delta_{b_1}^{a'_1} \cdots \delta_{b_n}^{a'_n} v(a_1, \dots, a_n)v(b'_1, \dots, b'_n)^*\xi. \end{aligned}$$

Consequently,

$$\left[v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*|_{E(n+1)} \right] \left[v(a'_1, \dots, a'_n)v(b'_1, \dots, b'_n)^*|_{E(n+1)} \right]$$

$$= \delta_{b_1}^{a'_1} \cdots \delta_{b_n}^{a'_n} v(a_1, \dots, a_n) v(b'_1, \dots, b'_n)^*|_{E(n+1)}.$$

On the other hand,

$$v(a_1, \dots, a_n) v(b_1, \dots, b_n)^* \quad \text{and} \quad v(b_1, \dots, b_n) v(a_1, \dots, a_n)^*$$

are adjoints of one another and both leave $E(n+1)$ invariant; their restrictions to $E(n+1)$ are therefore adjoints of one another.

9.2.10. One has

$$\begin{aligned} & v(a_1, \dots, a_{n-1}) v(b_1, \dots, b_{n-1})^*|_{E(n+1)} \\ &= [v(a_1, \dots, a_{n-1}, 0) v(b_1, \dots, b_{n-1}, 0)^* \\ &\quad + v(a_1, \dots, a_{n-1}, 1) v(b_1, \dots, b_{n-1}, 1)^*]|_{E(n+1)}. \end{aligned}$$

Indeed, with the notation of 9.2.8,

$$\begin{aligned} & [v(a_1, \dots, a_{n-1}, 0) v(b_1, \dots, b_{n-1}, 0)^* + v(a_1, \dots, a_{n-1}, 1) v(b_1, \dots, b_{n-1}, 1)^*] \xi \\ &= \delta_{b_1}^{c_1} \cdots \delta_{b_{n-1}}^{c_{n-1}} \delta_0^{c_n} v(a_1, \dots, a_{n-1}, 0, c_{n+1}) \eta \\ &\quad + \delta_{b_1}^{c_1} \cdots \delta_{b_{n-1}}^{c_{n-1}} \delta_1^{c_n} v(a_1, \dots, a_{n-1}, 1, c_{n+1}) \eta \\ &= \delta_{b_1}^{c_1} \cdots \delta_{b_{n-1}}^{c_{n-1}} v(a_1, \dots, a_{n-1}, c_n, c_{n+1}) \eta, \end{aligned}$$

whereas

$$\begin{aligned} & v(a_1, \dots, a_{n-1}) v(b_1, \dots, b_{n-1})^* \xi \\ &= v(a_1, \dots, a_{n-1}) v(b_1, \dots, b_{n-1})^* v(c_1, \dots, c_{n-1}) v(o_{n-1}, c_n) v(o_n, c_{n+1}) \eta \\ &= \delta_{b_1}^{c_1} \cdots \delta_{b_{n-1}}^{c_{n-1}} v(a_1, \dots, a_{n-1}) v(o_{n-1}, c_n) v(o_n, c_{n+1}) \eta \\ &= \delta_{b_1}^{c_1} \cdots \delta_{b_{n-1}}^{c_{n-1}} v(a_1, \dots, a_{n-1}, c_n, c_{n+1}) \eta. \end{aligned}$$

9.2.11. The operators $v(a_1, \dots, a_n) v(b_1, \dots, b_n)^*$ leave $E(r)$ invariant for $r > n$ (which generalizes 9.2.8). We know this when $r - n = 1$. Suppose it has been established when $r - n = s \geq 1$, and consider $r = n + s + 1 \geq n + 2$. By the induction hypothesis, the operators $v(a_1, \dots, a_{n+1}) v(b_1, \dots, b_{n+1})^*$ leave $E(r)$ invariant. By 9.2.10, the operators $v(a_1, \dots, a_n) v(b_1, \dots, b_n)^*$ act on $E(n+2)$ as linear combinations of operators $v(a_1, \dots, a_{n+1}) v(b_1, \dots, b_{n+1})^*$, and hence leave $E(r) \subset E(n+2)$ invariant.

9.2.12. Let $M(n)$ be the set of linear combinations of the operators

$$v(a_1, \dots, a_n) v(b_1, \dots, b_n)^*,$$

and let $N(n)$ be the set of linear combinations of elements of $M(0) \cup M(1) \cup \dots \cup M(n)$. Every element of $N(n)$ leaves $E(n+1)$ invariant (9.2.11). Let Φ_n be the map $x \mapsto x|_{E(n+1)}$, where x ranges over $N(n)$. By 9.2.10,

$$\Phi_n(N(n)) = \Phi_n(M(n)).$$

Suppose in addition that $v(a_1, \dots, a_n) \neq 0$ for all n and all $a_1, \dots, a_n$. Since

$$v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*v(b_1, \dots, b_n, b_{n+1}) = v(a_1, \dots, a_n, b_{n+1})$$

[by (1), applied with $c_1 = b_1, \dots, c_n = b_n$], it follows that

$$v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*|_{E(n+1)} \neq 0.$$

In view of 9.2.9, there is an isomorphism Ψ_n from the involutive algebra $\Phi_n(M(n))$ onto $M_{2^n}(\mathbb{C})$.

For $x \in N(n)$, put

$$f_n(x) = 2^{-n} \operatorname{Tr}(\Psi_n(\Phi_n(x))).$$

Then f_n is a linear form on $N(n)$, and

$$f_n(x^*) = \overline{f_n(x)}, \quad f_n(x) \geq 0 \quad (x \geq 0),$$

$$f_n(v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*) = 2^{-n} \delta_{a_1}^{b_1} \dots \delta_{a_n}^{b_n}.$$

Successive applications of 9.2.10 show that

$$f_n(v(a_1, \dots, a_p)v(b_1, \dots, b_p)^*) = 2^{-p} \delta_{a_1}^{b_1} \dots \delta_{a_p}^{b_p} \quad (p \leq n).$$

We have $N(0) \subset N(1) \subset \dots$, and the preceding formulas show that the forms f_n extend one another. They therefore define a linear form f on

$$N = \bigcup_{n \geq 0} N(n)$$

with the following properties:

- 1° $f(x^*) = \overline{f(x)}$;
- 2° $f(x) \geq 0$ for $x \geq 0$;
- 3° For every $n, a_1, \dots, a_n, b_1, \dots, b_n$,

$$f(v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*) = 2^{-n} \delta_{a_1}^{b_1} \dots \delta_{a_n}^{b_n}.$$

9.2.13. For $n = 0, 1, \dots, r$, let $(v(a_1, \dots, a_n))_{a_1, \dots, a_n \in \{0,1\}}$ be a system of elements of A satisfying conditions (i), (ii), (iii), and (iv) of 9.2.5. Choose arbitrary elements $v(o_{r+1})$ and $v(o_r, 1)$ of A . For $a_1, \dots, a_{r+1} \in \{0, 1\}$ with $(a_1, \dots, a_r) \neq o_r$, put

$$v(a_1, \dots, a_{r+1}) = v(a_1, \dots, a_r)v(o_r, a_{r+1}).$$

This defines a system $(v(a_1, \dots, a_{r+1}))_{a_1, \dots, a_{r+1} \in \{0,1\}}$. Let us determine whether conditions (i)–(iv) still hold. It is necessary that

$$v(a_1, \dots, a_r)^*v(a_1, \dots, a_r)v(o_r, a_{r+1}) = v(o_r, a_{r+1}), \quad (1)$$

$$v(o_r, 1)^*v(o_{r+1}) = 0, \quad (2)$$

$$v(o_{r+1}) \geq 0, \quad (3)$$

by (iii), (i), and (iv), respectively. We show that (1)–(3) are sufficient. Suppose they hold. Let $j < k \leq r + 1$ and prove (iii). If $k \leq r$, it already holds. If $k = r + 1$ and $j = r$, it follows from (1). Suppose that $k = r + 1$ and $j < r$. By (1) and (3),

$$v(o_r, a_{r+1}) = v(o_r)^*v(o_r)v(o_r, a_{r+1}) = v(o_r)^2v(o_r, a_{r+1}).$$

Using (iii) for systems with at most r indices, we obtain

$$\begin{aligned} & v(a_1, \dots, a_j)^*v(a_1, \dots, a_j)v(o_r, a_{r+1}) \\ &= v(a_1, \dots, a_j)^*v(a_1, \dots, a_j)v(o_r)^2v(o_r, a_{r+1}) \\ &= v(o_r)v(o_r)v(o_r, a_{r+1}) = v(o_r, a_{r+1}). \end{aligned}$$

Let us prove (ii). It suffices to do so when $k = r + 1$, that is, to prove

$$v(a_1, \dots, a_{r+1}) = v(a_1, \dots, a_r)v(o_r, a_{r+1}).$$

This is precisely the definition of $v(a_1, \dots, a_{r+1})$ when $(a_1, \dots, a_r) \neq o_r$. The relation already noted,

$$v(o_r, a_{r+1}) = v(o_r)^2v(o_r, a_{r+1}),$$

implies

$$v(o_r, a_{r+1}) = v(o_r)v(o_r, a_{r+1}),$$

because $v(o_r) \geq 0$. Thus (ii) holds in every case. Let us prove (i). It suffices to consider $k = r + 1$. If $j < r + 1$, then

$$v(a_1, \dots, a_j)^*v(b_1, \dots, b_{r+1})$$

$$= v(a_1, \dots, a_j)^* v(b_1, \dots, b_r) v(o_r, b_{r+1}),$$

since (ii) is now proved; one need only apply (i) to systems having at most r indices. Suppose $j = r + 1$. By (ii),

$$\begin{aligned} & v(a_1, \dots, a_{r+1})^* v(b_1, \dots, b_{r+1}) \\ &= v(o_r, a_{r+1})^* v(a_1, \dots, a_r)^* v(b_1, \dots, b_r) v(o_r, b_{r+1}). \end{aligned} \quad (4)$$

This vanishes if $(a_1, \dots, a_r) \neq (b_1, \dots, b_r)$. Finally, suppose

$$(a_1, \dots, a_r) = (b_1, \dots, b_r), \quad a_{r+1} \neq b_{r+1}.$$

By (4) and (iii), which has already been proved,

$$v(a_1, \dots, a_{r+1})^* v(b_1, \dots, b_{r+1}) = v(o_r, a_{r+1})^* v(o_r, b_{r+1}),$$

and this vanishes by (2).

Bibliography: [46].

9.3. Some Lemmas

9.3.1. Lemma. *Let A be a unital C^* -algebra, let π be a nonzero irreducible representation of A , let H' be a finite-dimensional vector subspace of H_π , and let L be the set of states $x \mapsto (\pi(x)\xi \mid \xi)$ of A , where ξ ranges over the unit sphere of H' . Let K be the weakly closed convex hull of L . Let K_1 be the set of states of A that take the value 1 at every $x \in A$ such that $0 \leq x \leq 1$ and $\pi(x)$ restricts to the identity on H' . Then $K = K_1$.*

Clearly K_1 is convex and weakly closed, and $L \subset K_1$; hence $K \subset K_1$. Let l be an extreme point of K_1 . We shall show that $l \in K$. Together with the Krein–Milman theorem, this will imply $K_1 \subset K$.

Suppose that $l = \frac{1}{2}(l_1 + l_2)$, where l_1 and l_2 are states of A . If $x \in A$, $0 \leq x \leq 1$, and $\pi(x)$ restricts to the identity on H' , then

$$l_1(x) \leq 1, \quad l_2(x) \leq 1, \quad 1 = l(x) = \frac{1}{2}(l_1(x) + l_2(x)).$$

Thus $l_1(x) = l_2(x) = 1$, so $l_1, l_2 \in K_1$ and hence $l_1 = l_2 = l$. Therefore l is a pure state. Suppose that π_l is not equivalent to π . There is then a Hermitian element $y \in A$ such that $\pi(y)$ restricts to the identity on H' and $\pi_l(y)\xi_l = 0$ (2.8.3). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function for which $f(t)$ is 0, t , or 1 according as $t < 0$, $0 \leq t \leq 1$, or $t > 1$. Replacing y by $f(y)$, we may suppose $0 \leq y \leq 1$. But then

$$l(y) = (\pi_l(y)\xi_l \mid \xi_l) = 0,$$

which proves that $l \notin K_1$, a contradiction. Hence $\pi_l \sim \pi$, and l is the state defined by π and a unit vector $\xi \in H_\pi$. Write $\xi = \eta + \zeta$, with $\eta \in H'$ and $\zeta \in H_\pi \ominus H'$. There is a Hermitian element $x \in A$ such that $\pi(x)$ restricts to the identity on H' and $\pi(x)\zeta = 0$ (2.8.3); arguing as before, we may suppose $0 \leq x \leq 1$. Since $l \in K_1$,

$$1 = l(x) = (\pi(x)\xi \mid \xi) = (\eta \mid \xi) = (\eta \mid \eta).$$

But

$$1 = \|\xi\|^2 = \|\eta\|^2 + \|\zeta\|^2,$$

so $\zeta = 0$, $\xi = \eta \in H'$, and $l \in K$.

9.3.2. Lemma. *Retain the notation of Lemma 9.3.1. Let E be the set of states of A . For every $\delta > 0$ and every $x \in A$ such that $0 \leq x \leq 1$ and $\pi(x)$ restricts to the identity on H' , denote by $V_{x,\delta}$ the set of $f \in E$ such that $f(x) \geq 1 - \delta$. Then the $V_{x,\delta}$ form a fundamental system of neighborhoods of K in E for the weak topology.*

It is clear that $V_{x,\delta}$ is a closed neighborhood of K in E . Moreover, the intersection of all the $V_{x,\delta}$ is K by 9.3.1. Since E is compact, it remains only to prove that the family of the $V_{x,\delta}$ is downward directed. Let $x_1, x_2 \in A$ satisfy $0 \leq x_1, x_2 \leq 1$, with $\pi(x_1)$ and $\pi(x_2)$ both the identity on H' , and let $\delta_1, \delta_2 > 0$. Put

$$x = \frac{1}{2}(x_1 + x_2) \in A, \quad \delta = \inf(\delta_1, \delta_2).$$

Then $0 \leq x \leq 1$, and $\pi(x)$ is the identity on H' . If $f \in E$ satisfies $f(x) \geq 1 - \delta/2$, then

$$f(x_1 + x_2) \geq 2 - \delta, \quad f(x_1) \leq 1, \quad f(x_2) \leq 1;$$

hence $f(x_1), f(x_2) \geq 1 - \delta$, and therefore

$$V_{x,\delta/2} \subset V_{x_1,\delta_1} \cap V_{x_2,\delta_2}.$$

9.3.3. Lemma. *Let H be a Hilbert space, $T \in \mathcal{L}(H)$, $\xi \in H$, and $\varepsilon > 0$, with*

$$\|T\| \leq 1, \quad \|\xi\| \leq 1, \quad \operatorname{Re}(T\xi \mid \xi) \geq 1 - \frac{\varepsilon}{2}.$$

Then $\|T\xi - \xi\|^2 \leq \varepsilon$.

Indeed,

$$\|T\xi - \xi\|^2 = (T\xi \mid T\xi) - 2\operatorname{Re}(T\xi \mid \xi) + (\xi \mid \xi) \leq 1 - 2 + \varepsilon + 1 = \varepsilon.$$

9.3.4. Lemma. *Let $\varepsilon > 0$ and let n be a positive integer. There is a number $\delta(\varepsilon, n) > 0$ with the following property: if H is a Hilbert space and $T_1, \dots, T_n \in \mathcal{L}(H)$ and $\xi \in H$ satisfy*

$$0 \leq T_1, \dots, T_n \leq 1, \quad \|\xi\| \leq 1, \quad \operatorname{Re}(T_n T_{n-1} \cdots T_1 \xi \mid \xi) \geq 1 - \delta(\varepsilon, n),$$

then

$$(T_i \xi \mid \xi) \geq 1 - \varepsilon, \quad \|T_i \xi - \xi\| \leq \varepsilon \quad (i = 1, \dots, n).$$

Put

$$\delta(\varepsilon, 1) = \inf \left(1, \frac{\varepsilon^2}{2} \right), \quad \delta(\varepsilon, n) = \frac{\delta(\varepsilon, n-1)^2}{16}.$$

Then

$$\delta(\varepsilon, n) \leq \frac{\delta(\varepsilon, n-1)}{2}, \quad \delta(\varepsilon, 1) \leq \frac{\varepsilon^2}{2}, \quad \delta(\varepsilon, 1) \leq \varepsilon.$$

The lemma for $n = 1$ follows from 9.3.3. Let $n \geq 2$ and suppose the lemma true for $n - 1$ (with the chosen values of δ). If

$$\operatorname{Re}(T_n T_{n-1} \cdots T_1 \xi \mid \xi) \geq 1 - \delta(\varepsilon, n),$$

then

$$\begin{aligned} 1 - \delta(\varepsilon, n) &\leq \operatorname{Re}(T_n \cdots T_1 \xi \mid \xi) \\ &\leq (T_n T_{n-1} \cdots T_1 \xi \mid T_{n-1} \cdots T_1 \xi)^{1/2} (T_n \xi \mid \xi)^{1/2} \\ &\leq (T_n \xi \mid \xi)^{1/2}. \end{aligned}$$

Hence

$$(T_n \xi \mid \xi) \geq 1 - 2\delta(\varepsilon, n) \geq 1 - \varepsilon,$$

and, by 9.3.3,

$$\|T_n \xi - \xi\|^2 \leq 4\delta(\varepsilon, n) \leq \varepsilon^2.$$

It follows that

$$\begin{aligned} \operatorname{Re}(T_{n-1} \cdots T_1 \xi \mid \xi) &= \operatorname{Re}(T_n \cdots T_1 \xi \mid \xi) + \operatorname{Re}(T_{n-1} \cdots T_1 \xi \mid (1 - T_n)\xi) \\ &\geq \operatorname{Re}(T_n \cdots T_1 \xi \mid \xi) - \|\xi - T_n \xi\| \\ &\geq 1 - \delta(\varepsilon, n) - 2\delta(\varepsilon, n)^{1/2} \\ &\geq 1 - \frac{1}{2}\delta(\varepsilon, n-1) - \frac{1}{2}\delta(\varepsilon, n-1) \\ &= 1 - \delta(\varepsilon, n-1). \end{aligned}$$

The induction hypothesis gives $(T_i \xi \mid \xi) \geq 1 - \varepsilon$ and $\|T_i \xi - \xi\| \leq \varepsilon$ for $i = 1, \dots, n-1$.

9.3.5. For every $\varepsilon \in (0, 1]$, let f_ε denote the function equal to 0 on $(-\infty, 1 - \varepsilon]$, equal to 1 on $[1 - \varepsilon/2, +\infty)$, and linear on $[1 - \varepsilon, 1 - \varepsilon/2]$. For $\varepsilon \in (0, 1/2]$,

$$f_\varepsilon f_{2\varepsilon} = f_\varepsilon.$$

Lemma. Let A be an antiliminary C^* -algebra, let $d \in A^+$ with $\|d\| = 1$, and let $\tau \in (0, 1]$. There exist $w, w', d' \in A$ such that:

- (i) $\|w\| = \|w'\| = \|d'\| = 1$, $w \geq 0$, $d' \geq 0$, and $w'^*w = 0$;
- (ii) $f_\tau(d)w = w$ and $f_\tau(d)w' = w'$;
- (iii) $w^2d' = d'$ and $w'^*w'd' = d'$.

Put $\sigma = \tau/8$. Choose $u \in A$ and $c \in A$ with $\|u\| \leq 1$ and $0 \leq c \leq 1$, and put

$$d_0 = f_{2\sigma}(d)c f_{2\sigma}(d), \quad d_1 = f_{4\sigma}(d) - d_0.$$

We have $0 \leq d_0 \leq 1$, and hence $-1 \leq d_1 \leq 1$. Moreover,

$$f_{4\sigma}(d)d_0 = d_0 = d_0f_{4\sigma}(d),$$

so d_0 and $f_{4\sigma}(d)$ generate a commutative C^* -subalgebra B of A containing d_1 . If ρ is any character of B such that $\rho(d_0) \neq 0$, then

$$\rho(f_{4\sigma}(d)) = 1, \quad \rho(d_1) = 1 - \rho(d_0);$$

also $0 \leq \rho(d_0) \leq 1$. Thus, if $g : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function vanishing on $[0, 1/2]$, then

$$g(\rho(d_0))g(\rho(d_1)) = 0, \quad \text{that is,} \quad \rho(g(d_0)g(d_1)) = 0.$$

It follows that $g(d_0)g(d_1) = 0$.

Put

$$v = f_\sigma(d_1)u f_\sigma(d_0).$$

Then

$$\|v\| \leq 1, \quad v^*v = f_\sigma(d_0)u^* f_\sigma(d_1)^2 u f_\sigma(d_0).$$

Consequently,

$$f_{2\sigma}(d_0)v^*v = v^*v, \quad v^*f_{2\sigma}(d_1) = v^*,$$

and hence

$$v^*(v^*v) = v^*f_{2\sigma}(d_1)f_{2\sigma}(d_0)v^*v = 0.$$

Furthermore,

$$f_{8\sigma}(d)d_0 = d_0, \quad f_{8\sigma}(d)d_1 = d_1;$$

therefore $f_{8\sigma}(d)p(d_0) = p(d_0)$ for every polynomial p without constant term, and hence for every continuous function p vanishing at the origin; likewise $f_{8\sigma}(d)p(d_1) = p(d_1)$. Thus

$$f_{8\sigma}(d)v = v, \quad f_{8\sigma}(d)v^* = v^*.$$

Finally put

$$d' = f_{1/4}(v^*v), \quad w = f_{1/2}(v^*v)^{1/2}, \quad w' = v k(v^*v),$$

where $k : \mathbb{R} \rightarrow \mathbb{R}$ is the function equal to $(f_{1/2}(t)t^{-1})^{1/2}$ for $t \neq 0$, and equal to 0 for $t = 0$. We have $0 \leq d' \leq 1$ and $0 \leq w \leq 1$. Since $v^*(v^*v) = 0$, we have $w'^*w = 0$. Since $f_{8\sigma}(d)v = v$ and $f_{8\sigma}(d)v^* = v^*$,

$$f_{8\sigma}(d)w = w, \quad f_{8\sigma}(d)w' = w'.$$

Moreover,

$$w^2 = f_{1/2}(v^*v), \quad w'^*w' = k^2(v^*v)v^*v = f_{1/2}(v^*v),$$

so

$$w^2d' = f_{1/2}(v^*v)f_{1/4}(v^*v) = f_{1/4}(v^*v) = d',$$

and similarly $w'^*w'd' = d'$. On the other hand,

$$\|w'\| = \|w'^*w'\|^{1/2} \leq 1.$$

We shall prove that c and u can be chosen so that $\|d'\| \geq 1$. Together with (iii), this will imply

$$\|w\|, \|w'\| \geq 1, \quad \text{and hence} \quad \|w\| = \|w'\| = \|d'\| = 1,$$

and the proof will be complete. We have $f_\sigma(d) \neq 0$. Since A is antiliminary, there is an irreducible representation π of A such that $\pi(f_\sigma(d))$ is not compact (4.2.6). We can therefore find two orthogonal unit vectors ξ and η in the range of $\pi(f_\sigma(d))$. There is a Hermitian element $c \in A$ such that $\pi(c)\xi = \xi$ and $\pi(c)\eta = 0$ (2.8.3); arguing as in 9.3.1, we may suppose $0 \leq c \leq 1$. There is a unitary element $u \in A$ such that $\pi(u)\xi = \eta$ (2.8.3). Since

$$f_{2\sigma}(d)f_\sigma(d) = f_\sigma(d)$$

and $\xi, \eta \in \pi(f_\sigma(d))(H_\pi)$, we have

$$\pi(f_{2\sigma}(d))\xi = \xi, \quad \pi(f_{2\sigma}(d))\eta = \eta,$$

and likewise $\pi(f_{4\sigma}(d))\eta = \eta$. It follows successively that

$$\pi(d_0)\xi = \xi, \quad \pi(d_0)\eta = 0, \quad \pi(d_1)\eta = \eta,$$

$$\begin{aligned}
\pi(v^*v)\xi &= f_\sigma(\pi(d_0))\pi(u)^{-1}f_\sigma(\pi(d_1))^2\pi(u)f_\sigma(\pi(d_0))\xi \\
&= f_\sigma(\pi(d_0))\pi(u)^{-1}f_\sigma(\pi(d_1))^2\pi(u)\xi \\
&= f_\sigma(\pi(d_0))\pi(u)^{-1}f_\sigma(\pi(d_1))^2\eta \\
&= f_\sigma(\pi(d_0))\pi(u)^{-1}\eta = f_\sigma(\pi(d_0))\xi = \xi, \\
\pi(d')\xi &= \xi, \quad \|\pi(d')\| \geq 1, \quad \|d'\| \geq 1.
\end{aligned}$$

9.3.6. Lemma. *Let A be a unital C^* -algebra, and let $v_1, \dots, v_p, c$ be norm-one elements of A such that $c \geq 0$, $v_i^*v_i c = c$, and $v_i^*v_j = 0$ for $i \neq j$. Let s be a Hermitian element of A , and let $\varepsilon > 0$. There exist a Hermitian element t of A , which is a linear combination of the $v_i v_j^*$, an element $b \in A$ with $0 \leq b \leq 1$, and a number $\gamma > 0$ such that:*

(i) *if f is a state of A such that $f(b) \geq 1 - \gamma$, then $|f(s - t)| \leq \varepsilon$;*

(ii)

$$\left\| c \prod_{i=1}^p v_i^* b v_i \right\| = 1.$$

The commutative C^* -subalgebra of A generated by 1 and c has a pure state taking the value 1 at c . This pure state can be extended to a pure state m of A (2.10.1). Put

$$\alpha_{ij} = m(v_i^* s v_j), \quad t = \sum_{i,j} \alpha_{ij} v_i v_j^* \in A.$$

We have $\overline{\alpha_{ij}} = \alpha_{ji}$, so t is Hermitian. Moreover,

$$(\pi_m(c)\xi_m \mid \xi_m) = m(c) = 1 \geq \|\pi_m(c)\xi_m\| \|\xi_m\|,$$

and consequently $\pi_m(c)\xi_m = \xi_m$. Hence

$$\pi_m(v_i^* v_i)\xi_m = \pi_m(v_i^* v_i c)\xi_m = \pi_m(c)\xi_m = \xi_m.$$

Let H' be the finite-dimensional vector subspace of H_{π_m} spanned by the vectors $\pi_m(v_i)\xi_m$. We have

$$\begin{aligned}
&(\pi_m(t)\pi_m(v_k)\xi_m \mid \pi_m(v_l)\xi_m) \\
&= \sum_{i,j} \alpha_{ij} (\pi_m(v_j^* v_k)\xi_m \mid \pi_m(v_i^* v_l)\xi_m) \\
&= \alpha_{lk} (\xi_m \mid \xi_m) = \alpha_{lk} \\
&= (\pi_m(v_l^* s v_k)\xi_m \mid \xi_m) = (\pi_m(s)\pi_m(v_k)\xi_m \mid \pi_m(v_l)\xi_m),
\end{aligned}$$

so

$$P_{H'}\pi_m(t)P_{H'} = P_{H'}\pi_m(s)P_{H'}.$$

Let E be the set of states of A , and let W be the set of $f \in E$ such that $|f(s-t)| \leq \varepsilon$. Let K be the weakly closed convex hull of the states of A of the form $x \mapsto (\pi_m(x)\eta \mid \eta)$, where η ranges over the unit sphere of H' . By what precedes, W is a weak neighborhood of K in E . By 9.3.2, there exist $b \in A$, $0 \leq b \leq 1$, and $\gamma > 0$ such that $\pi_m(b)$ restricts to the identity on H' and W contains every $f \in E$ for which $f(b) \geq 1 - \gamma$. Furthermore, it is clear that

$$\left\| c \prod_{i=1}^p v_i^* b v_i \right\| \leq 1.$$

Finally,

$$\begin{aligned} \pi_m(c) \prod_{i=1}^p \pi_m(v_i^*) \pi_m(b) \pi_m(v_i) \xi_m \\ = \pi_m(c) \prod_{i=1}^p \pi_m(v_i^* v_i) \xi_m = \pi_m(c) \xi_m = \xi_m, \end{aligned}$$

and hence

$$\left\| \pi_m \left(c \prod_{i=1}^p v_i^* b v_i \right) \right\| \geq 1, \quad \left\| c \prod_{i=1}^p v_i^* b v_i \right\| \geq 1.$$

9.3.7. Lemma. *Let A be a unital antiliminary C^* -algebra. Let $(s_0, s_1, \dots)$ be a sequence of Hermitian elements of A . There exist nonzero elements $v(a_1, \dots, a_n)$ in the unit ball of A ($a_1, \dots, a_n \in \{0, 1\}$; $n = 0, 1, 2, \dots$) with the following properties:*

- (i) *if $j \leq k$ and $(a_1, \dots, a_j) \neq (b_1, \dots, b_j)$, then $v(a_1, \dots, a_j)^* v(b_1, \dots, b_k) = 0$;*
- (ii) *if $k \geq 1$, then $v(a_1, \dots, a_k) = v(a_1, \dots, a_{k-1})v(o_{k-1}, a_k)$;*
- (iii) *if $j < k$, then*

$$v(a_1, \dots, a_j)^* v(a_1, \dots, a_j) v(o_{k-1}, a_k) = v(o_{k-1}, a_k);$$

(iv) $v(\emptyset) = 1$ and $v(o_k) \geq 0$;

(v) *if*

$$e(j) = \sum_{a_1, \dots, a_j \in \{0, 1\}} v(a_1, \dots, a_j) v(a_1, \dots, a_j)^*,$$

then, for every $j \geq 0$, there is a linear combination t_j of the operators

$$v(a_1, \dots, a_j) v(b_1, \dots, b_j)^*$$

such that

$$\|e(j+1)(s_j - t_j)e(j+1)\| \leq \frac{1}{j+1}.$$

We shall construct not only elements $v(a_1, \dots, a_n)$ having these properties, but also elements $b(n) \in A^+$ such that $\|b(n)\| = 1$ and

$$(vi) \quad v(a_1, \dots, a_n)^* v(a_1, \dots, a_n) b(n) = b(n).$$

For $n = 0$, put $v(\emptyset) = b(0) = 1$. Suppose that the nonzero elements $v(a_1, \dots, a_j)$ in the unit ball of A and the norm-one elements $b(j) \in A^+$ have been constructed for $j \leq n$, so that properties (i)–(vi) hold [property (v), of course, only for $j \leq n-1$].

By 9.3.6, there exist a Hermitian element t_n of A that is a linear combination of the $v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*$, an element $b \in A$ with $0 \leq b \leq 1$, and a number $\gamma > 0$ such that: first, if f is a state of A and $f(b) \geq 1 - \gamma$, then

$$|f(s_n - t_n)| \leq \frac{1}{n+1};$$

second, $\|a\| = 1$, where

$$a = b(n) \prod_{a_1, \dots, a_n} v(a_1, \dots, a_n)^* b v(a_1, \dots, a_n),$$

the elements $v(a_1, \dots, a_n)$ being arranged in any order.

Apply 9.3.5 with $d = aa^*$ and with τ , for the moment, arbitrary in $(0, 1]$. Denote by $v(o_{n+1})$, $v(o_n, 1)$, and $b(n+1)$ the elements w, w', d' constructed in 9.3.5. By 9.3.5(ii),

$$\begin{aligned} & v(a_1, \dots, a_n)^* v(a_1, \dots, a_n) v(o_n, a_{n+1}) \\ &= v(a_1, \dots, a_n)^* v(a_1, \dots, a_n) f_\tau(aa^*) v(o_n, a_{n+1}). \end{aligned}$$

But (vi) implies

$$v(a_1, \dots, a_n)^* v(a_1, \dots, a_n) aa^* = aa^*,$$

and hence

$$v(a_1, \dots, a_n)^* v(a_1, \dots, a_n) f_\tau(aa^*) = f_\tau(aa^*).$$

Thus, using 9.3.5(ii) once more,

$$\begin{aligned} & v(a_1, \dots, a_n)^* v(a_1, \dots, a_n) v(o_n, a_{n+1}) \\ &= f_\tau(aa^*) v(o_n, a_{n+1}) = v(o_n, a_{n+1}). \end{aligned}$$

Furthermore, $v(o_n, 1)^*v(o_{n+1}) = 0$ by 9.3.5(i), and $v(o_{n+1}) \geq 0$. By 9.2.13, if we put

$$v(a_1, \dots, a_{n+1}) = v(a_1, \dots, a_n)v(o_n, a_{n+1}) \quad \text{when } (a_1, \dots, a_n) \neq o_n,$$

conditions (i)–(iv) of 9.3.7 still hold. Moreover, $b(n+1) \in A^+$ and $\|b(n+1)\| = 1$ by 9.3.5(i). We have

$$v(o_n, a_{n+1})^*v(o_n, a_{n+1})b(n+1) = b(n+1)$$

by 9.3.5(iii); and, when $(a_1, \dots, a_n) \neq o_n$, condition (iii) of 9.3.7, which has already been established, gives

$$\begin{aligned} & v(a_1, \dots, a_{n+1})^*v(a_1, \dots, a_{n+1})b(n+1) \\ &= v(o_n, a_{n+1})^*v(a_1, \dots, a_n)^*v(a_1, \dots, a_n)v(o_n, a_{n+1})b(n+1) \\ &= v(o_n, a_{n+1})^*v(o_n, a_{n+1})b(n+1) = b(n+1). \end{aligned}$$

Thus condition (vi) of 9.3.7 still holds; in particular, the $v(a_1, \dots, a_{n+1})$ are nonzero.

It remains to prove (v) for $j = n$. Suppose that A is realized as a C^* -algebra of operators on a Hilbert space H . Let $\xi \in H$ and let

$$\eta = v(a_1, \dots, a_{n+1})\xi$$

be a unit vector. By (ii) and (iii),

$$\begin{aligned} v(a_1, \dots, a_n)^*\eta &= v(a_1, \dots, a_n)^*v(a_1, \dots, a_n)v(o_n, a_{n+1})\xi \\ &= v(o_n, a_{n+1})\xi. \end{aligned}$$

Since $t \geq f_\tau(t) - \tau/2$ for every $t \geq 0$,

$$\begin{aligned} & (aa^*v(a_1, \dots, a_n)^*\eta \mid v(a_1, \dots, a_n)^*\eta) \\ &= (aa^*v(o_n, a_{n+1})\xi \mid v(o_n, a_{n+1})\xi) \\ &\geq (f_\tau(aa^*)v(o_n, a_{n+1})\xi \mid v(o_n, a_{n+1})\xi) - \frac{\tau}{2}, \end{aligned}$$

which, by 9.3.5(ii), is

$$\begin{aligned} & (v(o_n, a_{n+1})\xi \mid v(o_n, a_{n+1})\xi) - \frac{\tau}{2} \\ &\geq \|v(a_1, \dots, a_n)v(o_n, a_{n+1})\xi\|^2 - \frac{\tau}{2} = \|\eta\|^2 - \frac{\tau}{2} = 1 - \frac{\tau}{2}. \end{aligned}$$

Apply Lemma 9.3.4, choosing τ so that

$$\frac{\tau}{2} \leq \delta \left(\frac{\gamma}{2^{2n+3}}, 2^{n+1} + 2 \right).$$

We obtain

$$\begin{aligned} & (v(a_1, \dots, a_n)^* b v(a_1, \dots, a_n) v(a_1, \dots, a_n)^* \eta \mid v(a_1, \dots, a_n)^* \eta) \\ & \geq 1 - \frac{\gamma}{2^{2n+3}}. \end{aligned}$$

Since

$$v(a_1, \dots, a_n) v(a_1, \dots, a_n)^* \eta = v(a_1, \dots, a_n) v(o_n, a_{n+1}) \xi = \eta,$$

this reads

$$(b\eta \mid \eta) \geq 1 - \frac{\gamma}{2^{2n+3}}.$$

Likewise, if η' is a unit vector in the range of $v(b_1, \dots, b_{n+1})$, then

$$(b\eta' \mid \eta') \geq 1 - \frac{\gamma}{2^{2n+3}}.$$

If $(a_1, \dots, a_{n+1}) \neq (b_1, \dots, b_{n+1})$, then η and η' are orthogonal by (i), and therefore

$$\begin{aligned} |(b\eta \mid \eta')|^2 &= |((1-b)\eta \mid \eta')|^2 \\ &\leq ((1-b)\eta \mid \eta)((1-b)\eta' \mid \eta') \leq \left(\frac{\gamma}{2^{2n+3}} \right)^2. \end{aligned}$$

Now let $\sigma \in e(n+1)(H)$ be a unit vector. By the definition of $e(n+1)$,

$$\sigma = \sum_{a_1, \dots, a_{n+1} \in \{0,1\}} \sigma(a_1, \dots, a_{n+1}),$$

where $\sigma(a_1, \dots, a_{n+1})$ lies in the range of $v(a_1, \dots, a_{n+1})$. It follows that

$$\begin{aligned} (b\sigma \mid \sigma) &\geq \sum_{a_1, \dots, a_{n+1}} \left(1 - \frac{\gamma}{2} \right) \|\sigma(a_1, \dots, a_{n+1})\|^2 \\ &\quad - \frac{\gamma}{2^{2n+3}} \sum_{(a_1, \dots, a_{n+1}) \neq (b_1, \dots, b_{n+1})} \|\sigma(a_1, \dots, a_{n+1})\| \|\sigma(b_1, \dots, b_{n+1})\| \\ &\geq 1 - \frac{\gamma}{2} - \frac{\gamma}{2} = 1 - \gamma. \end{aligned}$$

Consequently,

$$|((s_n - t_n)\sigma \mid \sigma)| \leq \frac{1}{n+1}.$$

Thus

$$-\frac{1}{n+1} \leq e(n+1)(s_n - t_n)e(n+1) \leq \frac{1}{n+1},$$

and

$$\|e(n+1)(s_n - t_n)e(n+1)\| \leq \frac{1}{n+1}.$$

Bibliography: [46].

9.4. Completion of the Proof of the Theorem

To prove that (ii) implies (iii) in Theorem 9.1, we shall show that if A is a separable nonpostliminal C^* -algebra, then A has factorial representations that are not of type I. Since A has a nonzero antiliminary quotient (4.3.6), this will follow from the more precise result below.

Proposition. *Let A be a nonzero separable antiliminary C^* -algebra. Then A has a factorial representation of type II.*

By 4.3.9, we may suppose that A has an identity. Let $(s_0, s_1, s_2, \dots)$ be a dense sequence in the Hermitian part of A , and let the $v(a_1, \dots, a_n)$ have the properties of 9.3.7. Let $M(n)$ be the set of linear combinations of the $v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*$, let $N(n)$ be the set of linear combinations of elements of $M(0) \cup M(1) \cup \dots \cup M(n)$, and put

$$N = \bigcup_{n \geq 0} N(n).$$

By 9.2.12, there is a linear form f on N such that $f(x^*) = \overline{f(x)}$, $f(x) \geq 0$ for $x \geq 0$, and

$$f(v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*) = 2^{-n} \delta_{a_1}^{b_1} \dots \delta_{a_n}^{b_n}, \quad f(1) = 1.$$

By 2.10.1(i), there is a state g of A extending f . We shall prove that π_g is factorial of type II; that is, the von Neumann algebra $\mathfrak{B}$ generated by $\pi_g(A)$ is a factor of type II.

The operators $\pi_g(v(a_1, \dots, a_n))$ have properties (i), (ii), and (iii) of 9.2.5. Put

$$F(n) = \overline{\pi_g(e(n))(H)}.$$

By 9.2.7, the spaces $F(n)$ form a decreasing sequence. It is clear that $F(n)$ is invariant under every operator in $\pi_g(A)'$. Hence $F = \bigcap_n F(n)$ is invariant under

every operator in $\pi_g(A)'$, and so $P_F \in \pi_g(A)'' = \mathfrak{B}$. Moreover, F is invariant under every operator in $\pi_g(N)$ by 9.2.11. For every n ,

$$\begin{aligned} \|\pi_g(e(n))\xi_g\| \|\xi_g\| &\geq (\pi_g(e(n))\xi_g \mid \xi_g) = g(e(n)) = f(e(n)) \\ &= 2^n 2^{-n} = 1 \geq \|\pi_g(e(n))\xi_g\| \|\xi_g\|, \end{aligned}$$

and therefore $\pi_g(e(n))\xi_g = \xi_g$, so $\xi_g \in F(n)$. Thus $\xi_g \in F$. Now ξ_g is cyclic for $\pi_g(A)$ and, a fortiori, for $\mathfrak{B}$. Hence $P_F \in \mathfrak{B}$ has central support 1, and $\mathfrak{B}'$ is isomorphic to $\mathfrak{B}'_F$ (A 20). It suffices to prove that $\mathfrak{B}'_F$ is a factor of type II, and hence that $\mathfrak{B}_F$ is a factor of type II (A 52).

By 9.2.9 and 9.2.11, the operators

$$\pi_g(v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*)$$

induce on F a system of matrix units of order 2^n . Since $\pi_g(e(n))\xi_g = \xi_g$, this system is nonzero. There is therefore an isomorphism Ψ_n from the involutive algebra

$$\pi_g(M(n))|_F = \pi_g(N(n))|_F$$

onto $M_{2^n}(\mathbb{C})$. First, $\mathfrak{B}_F$ is infinite-dimensional as a vector space. We shall prove: first,

$$(aa'\xi_g \mid \xi_g) = (a'a\xi_g \mid \xi_g) \quad (a, a' \in \mathfrak{B}_F);$$

second, $\mathfrak{B}_F$ is a factor. This factor will be finite (A 33), and it cannot be of type I_q for finite q (A 49); it will therefore be of type II_1 , and the proof will be complete.

Let s be a Hermitian element of A . For every positive integer n , there is a $j \geq 2n$ such that $\|s - s_j\| \leq 1/(2n)$, and then a $t_j \in M(j)$ such that

$$\|e(j+1)(s_j - t_j)e(j+1)\| \leq \frac{1}{2n}.$$

It follows that

$$\|P_F \pi_g(s - t_j) P_F\| \leq \|\pi_g(e(j+1)(s - t_j)e(j+1))\| \leq \frac{1}{n},$$

which shows that $\pi_g(N)P_F$ is norm dense in $P_F \pi_g(A)P_F$.

If $x \in \pi_g(N(n))$, then

$$(x\xi_g \mid \xi_g) = 2^{-n} \operatorname{Tr}(\Psi_n(x|_F)).$$

It is enough to verify this when $x = \pi_g(v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*)$, in which case

$$(x\xi_g \mid \xi_g) = g(v(a_1, \dots, a_n)v(b_1, \dots, b_n)^*) = 2^{-n} \delta_{a_1}^{b_1} \cdots \delta_{a_n}^{b_n}.$$

Thus, for $x, x' \in \pi_g(N(n))$,

$$(xx'\xi_g \mid \xi_g) = (x'x\xi_g \mid \xi_g), \quad (1)$$

because

$$\mathrm{Tr}(\Psi_n(x|_F)\Psi_n(x'|_F)) = \mathrm{Tr}(\Psi_n(x'|_F)\Psi_n(x|_F)).$$

Since (1) holds for every n , it holds for $x, x' \in \pi_g(N)$, hence for $x, x' \in P_F\pi_g(A)P_F$ by the norm density just proved, and finally for $x, x' \in \mathfrak{B}_F$ because $P_F\pi_g(A)P_F$ is strongly dense in $P_F\mathfrak{B}P_F$.

Finally, let r be a projection in the center of $\mathfrak{B}_F$. Put $\eta = r\xi_g$ and $\zeta = (1-r)\xi_g$. Since $a \mapsto (a\xi_g \mid \xi_g)$ is a trace on $\mathfrak{B}_F$, it follows at once that $a \mapsto (a\eta \mid \eta)$ and $a \mapsto (a\zeta \mid \zeta)$ are traces t_1, t_2 on $\mathfrak{B}_F$. Because $\pi_g(N(n))|_F$ is isomorphic to $M_{2^n}(\mathbb{C})$, the restrictions of t_1, t_2 to $\pi_g(N(n))_F$ [the set of operators induced on F by those in $\pi_g(N(n))$] are proportional. Since t_1, t_2 are weakly continuous, they are proportional on $\mathfrak{B}_F$. We have $\eta \neq 0$ or $\zeta \neq 0$. If, for example, $\eta \neq 0$, then $(r\eta \mid \eta) \neq 0$, whereas $(r\zeta \mid \zeta) = 0$; hence $t_2 = 0$. Therefore

$$0 = t_2(1-r) = ((1-r)\zeta \mid \zeta) = \|\zeta\|^2, \quad \xi_g = r\xi_g.$$

Since ξ_g is cyclic for $\mathfrak{B}_F$, it follows that $r = 1$. This proves that $\mathfrak{B}_F$ is a factor.

Bibliography: [46].

9.5. Supplements

9.5.1. Let A be a separable C^* -algebra. If A/I is postliminal for every primitive ideal I of A , then A is postliminal. (Apply 9.1.)

9.5.2. Let A be a separable C^* -algebra. Then A is postliminal if and only if $\widehat{A}$ is a T_0 -space. (Apply 9.1.)

9.5.3. Let A be a separable C^* -algebra. Then A is liminal if and only if every point of $\widehat{A}$ is closed. (Apply 9.5.2 and 4.7.15.)

*** 9.5.4.** Let A be a separable C^* -algebra. The following conditions are equivalent:

- (i) A is antiliminary;
- (ii) A has an injective representation of type II;
- (iii) A has an injective representation of type III;
- (iv) there is a family (π_i) of irreducible representations of A such that $\bigoplus_i \pi_i$ is injective and $\pi_i(A) \cap C(H_{\pi_i}) = 0$;

- (v) there are families $(\pi_i)_{i \in I}$ and $(\rho_i)_{i \in I}$ of irreducible representations of A such that $\bigoplus \pi_i$ and $\bigoplus \rho_i$ are injective, $\ker \pi_i = \ker \rho_i$ for every i , and ρ_i is not equivalent to π_i for any i .

[46].

*** 9.5.5.**

- a. Let G be the group with two elements, let X be the compact group $G \times G \times G \times \cdots$, and let X' be the dense subgroup of X consisting of the elements $(g_1, g_2, \dots)$ all but finitely many of whose components equal e . Let Y be the quotient Borel space X/X' . There is no sequence of Borel subsets of Y separating the points of Y .
- b. Let A be a separable nonpostliminal C^* -algebra. There is a Borel injection of Y into $\widehat{A}$, where $\widehat{A}$ is endowed with its Mackey structure. [46]

9.5.6. Let A be a separable C^* -algebra. The following conditions are equivalent:

- (i) A is postliminal;
- (ii) for the Borel structure underlying its topology, $\widehat{A}$ is standard;
- (iii) for the Mackey structure, there is a sequence of Borel subsets of $\widehat{A}$ separating its points;
- (iv) the Mackey structure of $\widehat{A}$ is identical with the Borel structure underlying its topology.

Indeed, (i) implies (ii) and (iv) by 4.6.1; (ii) implies (iii) by 3.8.3; (iv) implies (iii) by 3.3.4 and 3.8.4; and (iii) implies (i) by 9.5.5. [12, 46].

*** 9.5.7.** Let H, H' be two infinite-dimensional real vector spaces, and let B be a bilinear form on $H \times H'$ putting H and H' in duality. Let K be a Hilbert space. For every $x \in H$ (respectively, $x' \in H'$), let $p(x)$ [respectively, $q(x')$] be a self-adjoint, not necessarily bounded, operator on K . Suppose that the $p(x)$ commute pairwise, that $p(\lambda_1 x_1 + \lambda_2 x_2)$ is the smallest closed extension of $\lambda_1 p(x_1) + \lambda_2 p(x_2)$, and that the analogous properties hold for q . Suppose that

$$e^{ip(x)} e^{iq(x')} = e^{iB(x, x')} e^{iq(x')} e^{ip(x)} \quad (x \in H, x' \in H').$$

If M, M' are vector subspaces of H, H' , let $\mathcal{A}(M, M')$ be the von Neumann algebra generated by the $e^{ip(x)}$ and $e^{iq(x')}$ ($x \in M, x' \in M'$). Let A be the C^* -algebra generated by the $\mathcal{A}(M, M')$ as M (respectively, M') ranges over the finite-dimensional vector subspaces of H (respectively, H'). Then A depends, up to isomorphism, only on (H, H', B) . The C^* -algebra A is antiliminal, has no closed two-sided ideals other than A and 0 , and has no traceable representation. If H is

a separable real Hilbert space, $H' = H$, and $B(x, x') = (x \mid x')$, then A has factorial representations of type II_∞ and factorial representations of type III. Irreducible representations of A can be constructed explicitly [40, 46, 139].

This is the physicists' problem of representing the commutation relations. The representation of the anticommutation relations [39] was translated by Mackey [92] into a problem concerning group representations.

9.5.8. Let A be a separable C^* -algebra. If there is a Borel subset of $\text{Irr}(A)$ that meets each equivalence class in exactly one point, then A is postliminal. (Use 7.2.3 and 9.5.6.) [91].

9.5.9. Let A be a C^* -algebra. Conditions (i), (ii), (iii), and (iv) of 9.1 are equivalent, but it is not known whether they are equivalent to (v). [253, 254, 292].

§ 10. — Continuous Fields of C^* -Algebras

Let A be a C^* -algebra, T a topological space, and B the set of continuous maps $f : T \rightarrow A$ such that $\sup_{t \in T} \|f(t)\| < +\infty$. If $f, g \in B$ and $\lambda \in \mathbb{C}$, define $f + g, \lambda f, fg, f^* \in B$ and $\|f\|$ by

$$\begin{aligned} (f + g)(t) &= f(t) + g(t), & (\lambda f)(t) &= \lambda f(t), \\ (fg)(t) &= f(t)g(t), & f^*(t) &= f(t)^* \end{aligned}$$

for every $t \in T$, and

$$\|f\| = \sup_{t \in T} \|f(t)\|.$$

It is easy to see that B thereby becomes a C^* -algebra.

If T is discrete, B is simply the product C^* -algebra $\prod_{t \in T} A_t$, where $A_t = A$ for every t . We have already introduced the general product $\prod_{t \in T} A_t$ of C^* -algebras, where T is a set and A_t is a C^* -algebra varying with t . We now define a construction of C^* -algebras that includes the preceding constructions: T will be a topological space, not necessarily discrete, and the A_t will vary with t .

The principal interest of this construction is that fairly broad classes of C^* -algebras can be obtained in this way with very simple A_t . This will enable us to solve, very partially, the second problem stated at the beginning of [Section 3](#).

10.1. Continuous Fields of Banach Spaces

10.1.1. Let T be a topological space and $(E(t))_{t \in T}$ a family of complex Banach spaces. A *vector field* is any element of $\prod_{t \in T} E(t)$; that is, any function x defined on T such that $x(t) \in E(t)$ for every $t \in T$. More generally, if $Y \subset T$, a vector field on Y is an element of $\prod_{t \in Y} E(t)$.

10.1.2. Definition. Let T be a topological space. A continuous field $\mathcal{E}$ of Banach spaces over T is a family $(E(t))_{t \in T}$ of Banach spaces, equipped with a set $\Gamma \subset \prod_{t \in T} E(t)$ of vector fields such that:

- (i) Γ is a complex vector subspace of $\prod_{t \in T} E(t)$;
- (ii) for every $t \in T$, the set of $x(t)$, where $x \in \Gamma$, is dense in $E(t)$;
- (iii) for every $x \in \Gamma$, the function $t \mapsto \|x(t)\|$ is continuous;
- (iv) if $x \in \prod_{t \in T} E(t)$ is a vector field such that, for every $t \in T$ and every $\varepsilon > 0$, there is an $x' \in \Gamma$ for which $\|x(t') - x'(t')\| \leq \varepsilon$ on a neighborhood of t , then $x \in \Gamma$.

The elements of Γ are called the *continuous vector fields* of $\mathcal{E}$. Definition 10.1.2 is analogous to the definition of measurable fields.

10.1.3. Let

$$\mathcal{E} = ((E(t))_{t \in T}, \Gamma), \quad \mathcal{E}' = ((E'(t))_{t \in T}, \Gamma')$$

be two continuous fields of Banach spaces over T . An *isomorphism* of $\mathcal{E}$ onto $\mathcal{E}'$ is a family $\varphi = (\varphi_t)_{t \in T}$ such that each φ_t is an isometric isomorphism of $E(t)$ onto $E'(t)$ and φ carries Γ onto Γ' .

10.1.4. Example. Let E be a Banach space, let T be a topological space, and let Γ be the set of continuous maps from T into E . Put $E(t) = E$ for every $t \in T$. Then $\mathcal{E} = ((E(t))_{t \in T}, \Gamma)$ is a continuous field of Banach spaces over T , called the *constant field over T defined by E* . (The reader will do well to keep this example constantly in mind in what follows.) A field isomorphic to a constant field is called *trivial*.

10.1.5. Example. In Definition 10.1.2, suppose T is discrete. Axioms (ii) and (iv) then imply that necessarily

$$\Gamma = \prod_{t \in T} E(t).$$

10.1.6. Let T be a topological space and $\mathcal{E} = ((E(t))_{t \in T}, \Gamma)$ a continuous field of Banach spaces over T . Let $Y \subset T$ and $t_0 \in Y$. A vector field x on Y is said to be *continuous at t_0* if, for every $\varepsilon > 0$, there is an $x' \in \Gamma$ such that $\|x(t) - x'(t)\| \leq \varepsilon$ on a neighborhood of t_0 . It is *continuous on Y* if it is continuous at every point of Y . When $Y = T$, this agrees with the definition already adopted for continuous vector fields, by axiom (iv).

10.1.7. Let Γ_Y be the set of continuous vector fields on Y . It is immediate that $((E(t))_{t \in Y}, \Gamma_Y)$ is a continuous field of Banach spaces over Y , called the field induced by $\mathcal{E}$ on Y and denoted by $\mathcal{E}|_Y$.

10.1.8. If every point of T has a neighborhood V such that $\mathcal{E}|_V$ is trivial, then $\mathcal{E}$ is said to be *locally trivial*.

10.1.9. Proposition. Let $((E(t))_{t \in T}, \Gamma)$ be a continuous field of Banach spaces.

- (i) Let $t_0 \in T$, let x be a vector field on T continuous at t_0 , and let $f : T \rightarrow \mathbb{C}$ be a function continuous at t_0 . Then the vector field

$$t \longmapsto (fx)(t) = f(t)x(t)$$

is continuous at t_0 .

(ii) If $y \in \Gamma$ and $g : T \rightarrow \mathbb{C}$ is continuous on T , then $gy \in \Gamma$.

Let $\varepsilon > 0$. There is an $x' \in \Gamma$ such that $\|x(t) - x'(t)\| \leq \varepsilon$ near t_0 . Near t_0 we have

$$|f(t) - f(t_0)| \leq \varepsilon, \quad |f(t)| \leq |f(t_0)| + 1, \quad \|x'(t)\| \leq \|x'(t_0)\| + 1.$$

Therefore, near t_0 ,

$$\begin{aligned} \|f(t)x(t) - f(t_0)x'(t)\| &\leq |f(t)|\|x(t) - x'(t)\| + |f(t) - f(t_0)|\|x'(t)\| \\ &\leq \varepsilon(|f(t_0)| + \|x'(t_0)\| + 2). \end{aligned}$$

This proves (i), and (ii) follows from (i).

10.1.10. Proposition. For every $t_0 \in T$ and every $\xi \in E(t_0)$, there is an $x \in \Gamma$ such that $x(t_0) = \xi$.

In view of axiom (ii) of Definition 10.1.2, there is a sequence (ξ_n) of nonzero vectors of $E(t_0)$ with the following properties:

- a. $\sum \|\xi_n\| < +\infty$;
- b. $\sum \xi_n = \xi$;
- c. for every n there is an $x_n \in \Gamma$ such that $x_n(t_0) = \xi_n$.

Let f_n be the function on T equal to 1 when $\|x_n(t)\| \leq \|\xi_n\|$, and equal to $\|\xi_n\|/\|x_n(t)\|$ when $\|x_n(t)\| > \|\xi_n\|$. This function is continuous, so $y_n = f_n x_n \in \Gamma$ by 10.1.9(ii). We have $y_n(t_0) = \xi_n$ and $\|y_n(t)\| \leq \|\xi_n\|$ for every $t \in T$; hence $\sum y_n(t)$ converges for every t to an element $x(t) \in E(t)$. Also $x(t_0) = \sum \xi_n = \xi$. For every $\varepsilon > 0$, there is a positive integer p such that

$$\left\| x(t) - \sum_{n=1}^p y_n(t) \right\| \leq \varepsilon \quad \text{on all of } T.$$

Thus $x \in \Gamma$ by axioms (i) and (iv) of Definition 10.1.2.

10.1.11. Lemma. Let T be paracompact, let Y be a closed subset of T , let $((E(t)))_{t \in T}$ be a continuous field of Banach spaces over T , let x (respectively, y) be a continuous vector field on T (respectively, on Y), and let $a > 0$ satisfy $\|x(t) - y(t)\| \leq a$ on Y . There exists $x' \in \Gamma$ such that $\|x'(t) - y(t)\| \leq a/2$ on Y and $\|x'(t) - x(t)\| \leq 2a$ on T .

There is a cover (V_i) of Y by open subsets of T and, for every i , an $x_i \in \Gamma$ such that $\|x_i(t) - y(t)\| \leq a/2$ on V_i . Replacing (V_i) by a refinement, we may suppose it locally finite. Let (η_i) be a family of nonnegative continuous functions on T subordinate to (V_i) , such that $\sum \eta_i = 1$ on Y . Put $w = \sum \eta_i x_i$, which is continuous

at every point of T , and hence belongs to Γ . If $t \in Y$, only the indices i for which $\eta_i(t) > 0$ contribute to $\sum \eta_i(t)x_i(t)$, and for those indices

$$\|x_i(t) - y(t)\| \leq \frac{a}{2}; \quad \text{hence} \quad \left\| \sum \eta_i(t)x_i(t) - y(t) \right\| \leq \frac{a}{2}.$$

Thus $\|w(t) - x(t)\| \leq 2a$ on a neighborhood Z of Y . Let $f : T \rightarrow [0, 1]$ be continuous, equal to 1 on Y and to 0 on $T - Z$. Put $x' = fw + (1 - f)x$. On Y ,

$$\|x'(t) - y(t)\| = \|w(t) - y(t)\| \leq \frac{a}{2}.$$

On T ,

$$\|x'(t) - x(t)\| = |f(t)| \|w(t) - x(t)\| \leq 2a.$$

10.1.12. Proposition. *Let T be paracompact, let Y be a closed subset of T , let $((E(t))_{t \in T}, \Gamma)$ be a continuous field of Banach spaces, and let y be a continuous vector field on Y . There is an $x \in \Gamma$ extending y .*

Arguing as in Lemma 10.1.11, first construct $x_0 \in \Gamma$ such that $\|x_0(t) - y(t)\| \leq 1$ on Y . Then, applying Lemma 10.1.11 successively, obtain a sequence $(x_n) \subset \Gamma$ such that

$$\|x_n(t) - y(t)\| \leq 2^{-n} \quad \text{on } Y, \quad \|x_n(t) - x_{n-1}(t)\| \leq 2^{-n+2} \quad \text{on } T.$$

The x_n converge uniformly to an $x \in \Gamma$, and $x(t) = y(t)$ on Y .

10.1.13. Proposition. *Let T be completely regular, let $(Y_i)_{i \in I}$ be an open cover of T , and put $Y_{ij} = Y_i \cap Y_j$. For every $i \in I$, let $\mathcal{E}_i = ((E_i(t))_{t \in Y_i}, \Gamma_i)$ be a continuous field of Banach spaces over Y_i . For all $i, j \in I$, let g_{ij} be an isomorphism of $\mathcal{E}_j|_{Y_{ij}}$ onto $\mathcal{E}_i|_{Y_{ij}}$. Suppose that $g_{ij}g_{jk} = g_{ik}$ for all $i, j, k \in I$. Then there exists, uniquely up to isomorphism, a continuous field $\mathcal{E}$ of Banach spaces over T with the following property: for every $i \in I$ there is an isomorphism h_i of $\mathcal{E}_i$ onto $\mathcal{E}|_{Y_i}$ such that $g_{ij} = h_i^{-1}h_j$ for all $i, j \in I$.*

Uniqueness is easy; we prove existence. For every $t \in T$, let $I(t) \neq \emptyset$ be the set of $i \in I$ such that $t \in Y_i$. If $i, j \in I(t)$, then $g_{ij}(t)$ is an isomorphism of $E_j(t)$ onto $E_i(t)$, and

$$g_{ij}(t)g_{jk}(t) = g_{ik}(t) \quad (i, j, k \in I(t)).$$

There is therefore a Banach space $E(t)$ and, for every $i \in I(t)$, an isomorphism $h_i(t)$ of $E_i(t)$ onto $E(t)$ such that

$$g_{ij}(t) = h_i(t)^{-1}h_j(t) \quad (i, j \in I(t)).$$

[For example, take $E(t) = E_{i_0}(t)$, where $i_0 \in I(t)$, and $h_i(t) = g_{i_0 i}(t)$.] There is a unique set Δ_i of vector fields on Y_i , with values in the $E(t)$, such that $(h_i(t))_{t \in Y_i}$ is

an isomorphism of $\mathcal{E}_i$ onto $\mathcal{F}_i = ((E(t))_{t \in Y_i}, \Delta_i)$. Moreover, $\mathcal{F}_i|_{Y_{ij}} = \mathcal{F}_j|_{Y_{ij}}$ for all $i, j \in I$.

Now let Γ be the set of all $x \in \prod_{t \in T} E(t)$ such that $x|_{Y_i} \in \Delta_i$ for every $i \in I$. It is immediate that Γ satisfies axioms (i), (iii), and (iv) of Definition 10.1.2. We show that it also satisfies axiom (ii). Let $t_0 \in T$ and $\xi \in E(t_0)$. Choose $i \in I$ with $t_0 \in Y_i$. Let $V \subset Y_i$ be a neighborhood of t_0 that is closed in T , and let $\eta : T \rightarrow \mathbb{R}$ be a continuous function equal to 1 at t_0 and to 0 on $T - V$. Choose $y \in \Delta_i$ with $y(t_0) = \xi$. Let $x \in \prod_{t \in T} E(t)$ be 0 on $T - V$ and ηy on V . Then $x(t_0) = \xi$. We shall show that $x \in \Gamma$. Let $j \in I$; we must show that $x|_{Y_j} \in \Delta_j$. It is enough to show that, for every $t_1 \in Y_j$, x agrees near t_1 with an element of Δ_j . This is immediate if $t_1 \notin V$. Suppose $t_1 \in V$. Then $t_1 \in Y_{ij}$. Since $\mathcal{F}_i|_{Y_{ij}} = \mathcal{F}_j|_{Y_{ij}}$ and x agrees near t_1 with an element of Δ_i , the assertion follows.

We have thus proved that $\mathcal{E} = ((E(t))_{t \in T}, \Gamma)$ is a continuous field of Banach spaces over T . It is easy to see that $\mathcal{E}|_{Y_i} = \mathcal{F}_i$, which proves the proposition. We say that $\mathcal{E}$ is obtained by *gluing* the $\mathcal{E}_i$ by means of the g_{ij} .

Bibliography: [19, 29, 52, 77, 108, 161, 169].

10.2. Total Subsets

10.2.1. Definition. Let $\mathcal{E} = ((E(t))_{t \in T}, \Gamma)$ be a continuous field of Banach spaces over T . A subset $\Lambda \subset \Gamma$ is called *total* if, for every $t \in T$, the set of $x(t)$ with $x \in \Lambda$ is total in $E(t)$. The field $\mathcal{E}$ is called *separable* if Γ contains a countable total subset.

If Λ is a countable total subset of Γ , the set Λ' of linear combinations of elements of Λ with rational complex coefficients is again countable, and $\Lambda'(t)$ is dense in $E(t)$ for every t .

10.2.2. Proposition. Let $((E(t))_{t \in T}, \Gamma)$ be a continuous field of Banach spaces over T . Let $\Lambda \subset \Gamma$ be a total subset, and let $\overline{\Lambda}$ be the vector subspace of Γ spanned by Λ . For a vector field $x \in \prod_{t \in T} E(t)$, the following properties are equivalent:

- (i) $x \in \Gamma$;
- (ii) for every $t_0 \in T$ and every $\varepsilon > 0$, there is an $x' \in \Gamma$ such that $\|x(t) - x'(t)\| \leq \varepsilon$ near t_0 ;
- (ii') for every $t_0 \in T$ and every $\varepsilon > 0$, there is an $x' \in \overline{\Lambda}$ such that $\|x(t) - x'(t)\| \leq \varepsilon$ near t_0 ;
- (iii) for every $x' \in \Gamma$, the function $t \mapsto \|x(t) - x'(t)\|$ is continuous;
- (iii') for every $x' \in \overline{\Lambda}$, the function $t \mapsto \|x(t) - x'(t)\|$ is continuous.

The implications (ii') $\Rightarrow$ (ii) $\Rightarrow$ (i) $\Rightarrow$ (iii) $\Rightarrow$ (iii') are immediate.

For (iii') $\Rightarrow$ (ii'), suppose (iii') holds. Let $t_0 \in T$ and $\varepsilon > 0$. Since $\overline{\Lambda}(t_0)$ is dense in $E(t_0)$, there is an $x' \in \overline{\Lambda}$ such that $\|x(t_0) - x'(t_0)\| < \varepsilon$. By (iii'), $\|x(t) - x'(t)\| < \varepsilon$ near t_0 , proving (ii').

10.2.3. Proposition. *Let T be a topological space, let $(E(t))_{t \in T}$ be a family of Banach spaces, and let $\Lambda \subset \prod_{t \in T} E(t)$ satisfy axioms (i), (ii), and (iii) of Definition 10.1.2, with Γ replaced by Λ . There is a unique subset $\Gamma \subset \prod_{t \in T} E(t)$ containing Λ and satisfying axioms (i)–(iv) of Definition 10.1.2: namely, the set of vector fields x satisfying condition (ii') of Proposition 10.2.2.*

Uniqueness follows from 10.2.2: if Γ exists, its elements are characterized by (ii'). Conversely, the set of vector fields x satisfying (ii') plainly satisfies axioms (i)–(iv) of Definition 10.1.2 and contains Λ .

10.2.4. Proposition. *Let $\mathcal{E} = ((E(t))_{t \in T}, \Gamma)$ and $\mathcal{E}' = ((E'(t))_{t \in T}, \Gamma')$ be two continuous fields of Banach spaces over T , and let Λ be a total subset of Γ . For each $t \in T$, let φ_t be an isometric isomorphism of $E(t)$ onto $E'(t)$. Then $\varphi = (\varphi_t)_{t \in T}$ is an isomorphism of $\mathcal{E}$ onto $\mathcal{E}'$ if and only if $\varphi(\Lambda) \subset \Gamma'$.*

Necessity is clear. Suppose $\varphi(\Lambda) \subset \Gamma'$ and prove that φ is an isomorphism. We may suppose that Λ is a vector subspace of Γ . If $x \in \Gamma$, then x is a limit of elements of Λ for local uniform convergence [10.2.2(ii')]; hence $\varphi(x)$ is a local uniform limit of elements of Γ' , and so $\varphi(x) \in \Gamma'$. Conversely, let $x' \in \Gamma'$, put $x = \varphi^{-1}(x')$, and let $t_0 \in T$, $\varepsilon > 0$. There is a $y \in \Gamma$ such that $y(t_0) = x(t_0)$. We have

$$\varphi(y) \in \Gamma', \quad \varphi(y)(t_0) = \varphi_{t_0}(x(t_0)) = x'(t_0),$$

and hence $\|\varphi(y)(t) - x'(t)\| \leq \varepsilon$ near t_0 , so $\|y(t) - x(t)\| \leq \varepsilon$ near t_0 . Thus $x \in \Gamma$, and $\varphi^{-1}(\Gamma') \subset \Gamma$.

10.2.5. Proposition. *Let T be locally compact, let $((E(t))_{t \in T}, \Gamma)$ be a continuous field of Banach spaces over T , let $x \in \Gamma$, let K be a compact subset of T , let $\varepsilon > 0$, and let Λ be a total subset of Γ . There exist compactly supported continuous complex functions $f_1, \dots, f_n$ on T and elements $x_1, \dots, x_n \in \Lambda$ such that*

$$\|x(t) - f_1(t)x_1(t) - \dots - f_n(t)x_n(t)\| \leq \varepsilon \quad \text{on } K.$$

Let $\overline{\Lambda}$ be the vector subspace of Γ spanned by Λ . By 10.2.2(ii'), there is a cover (V_i) of K by relatively compact open subsets of T , and elements $x'_i \in \overline{\Lambda}$, such that $\|x(t) - x'_i(t)\| \leq \varepsilon$ in V_i . Since K is compact, we may suppose the cover is

finite, say $(V_1, \dots, V_p)$. Let $(\eta_1, \dots, \eta_{p+1})$ be a continuous partition of unity on T subordinate to the open cover $(V_1, \dots, V_p, T - K)$. Then

$$\|x(t) - \eta_1(t)x'_1(t) - \dots - \eta_p(t)x'_p(t)\| \leq \varepsilon$$

throughout K . Expressing the x'_i as linear combinations of elements of Λ proves the proposition.

10.2.6. Let T be locally compact, let $\mathcal{E} = ((E(t))_{t \in T}, \Gamma)$ be a continuous field of Banach spaces over T , and let T' be the compact space obtained by adjoining a point at infinity ω to T . Put $E'(t) = E(t)$ for $t \in T$ and $E'(\omega) = 0$. Let Γ' be the set of $x \in \prod_{t \in T'} E'(t)$ such that $x|_T \in \Gamma$ and $\|x(t)\| \rightarrow 0$ at infinity on T .

Proposition. $\mathcal{E}' = ((E'(t))_{t \in T'}, \Gamma')$ is a continuous field of Banach spaces over T' , and $\mathcal{E}'|_T = \mathcal{E}$.

It is immediate that Γ' satisfies axioms (i), (iii), and (iv) of Definition 10.1.2. Let $t_0 \in T$ and $\xi \in E(t_0)$. There is an $x \in \Gamma$ such that $x(t_0) = \xi$. Multiplying x by a compactly supported function equal to 1 at t_0 , we may suppose that $\|x(t)\| \rightarrow 0$ at infinity on T ; hence there is at once an $x' \in \Gamma'$ with $x'(t_0) = \xi$. Let Γ_0 be the set of restrictions to T of elements of Γ' ; that is, the set of elements of Γ tending to 0 at infinity. What precedes shows that Γ_0 is a total subset of Γ . For every $t \in T$, let φ_t be the identity map on $E(t)$, and put $\varphi = (\varphi_t)_{t \in T}$. For every $x \in \Gamma_0$, $\varphi(x)$ is a continuous vector field relative to $\mathcal{E}'|_T$. Thus φ is an isomorphism of $\mathcal{E}$ onto $\mathcal{E}'|_T$ by 10.2.4.

10.2.7. Proposition. Let T be a separable metrizable space and $\mathcal{E} = ((E(t))_{t \in T}, \Gamma)$ a locally trivial continuous field of Banach spaces over T . If every $E(t)$ is separable, then $\mathcal{E}$ is separable.

There is a countable open cover (U_n) of T such that every $\mathcal{E}|_{U_n}$ is trivial, and then an open cover (V_n) of T such that $\overline{V_n} \subset U_n$ for every n . For every n , there is a sequence $(x_{nm})_m$ of bounded continuous vector fields on U_n such that, for every $t \in U_n$, the $x_{nm}(t)$ are dense in $E(t)$ (indeed, $\mathcal{E}|_{U_n}$ is identified with the constant field defined by a separable space). Let $\eta_n : T \rightarrow [0, 1]$ be continuous, equal to 1 on V_n , and to 0 on $T - U_n$. Let y_{nm} be the vector field on T equal to $\eta_n x_{nm}$ on U_n and to 0 on $T - U_n$. Then $y_{nm} \in \Gamma$: if $t \in U_n$, y_{nm} is continuous at t ; if $t \in T - U_n$ and $\varepsilon > 0$, there is a neighborhood of t on which $\|y_{nm}(t')\| \leq \varepsilon$. Let $t \in T$ and $\xi \in E(t)$. There is an n such that $t \in V_n$. Then ξ is a limit of vectors $x_{nm}(t) = y_{nm}(t)$. Thus the y_{nm} form a total subset of Γ .

Bibliography: [19, 29, 52, 77, 108, 161, 169].

10.3. Continuous Fields of C^* -Algebras

10.3.1. Definition. Let T be a topological space. A continuous field of C^* -algebras over T is a continuous field $((A(t))_{t \in T}, \Theta)$ of Banach spaces over T , each $A(t)$ being equipped with a multiplication and an involution that make it a C^* -algebra, and Θ being stable under multiplication and involution.

When we speak of isomorphisms of continuous fields of C^* -algebras, we shall of course mean isomorphisms compatible with the involutive-algebra structure on the fibers. The same convention applies to triviality and local triviality.

A continuous field $((A(t))_{t \in T}, \Theta)$ is called a *continuous field of elementary C^* -algebras* if every $A(t)$ is an elementary C^* -algebra (4.1.1).

10.3.2. Proposition. Let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be a continuous field of Banach spaces over T , each $A(t)$ being a C^* -algebra. For $\mathcal{A}$ to be a continuous field of C^* -algebras, it is enough that there be a total subset Λ of Θ stable under multiplication and involution.

Indeed, let $\bar{\Lambda}$ be the vector subspace of Θ spanned by Λ ; it is stable under multiplication and involution. Every element of Θ is a local uniform limit of elements of $\bar{\Lambda}$. Hence, if $x, y \in \Theta$, the vector field xy is a local uniform limit of elements of $\bar{\Lambda}$, so $xy \in \Theta$. Likewise $x^* \in \Theta$.

10.3.3. Proposition. Let $((A(t))_{t \in T}, \Theta)$ be a continuous field of C^* -algebras over T , let $x \in \Theta$ satisfy $xx^* = x^*x$, and let $f : \mathbb{C} \rightarrow \mathbb{C}$ be continuous with $f(0) = 0$. Then $t \mapsto f(x(t))$ belongs to Θ .

Let $t_0 \in T$. There is a neighborhood V of t_0 and a constant $M > 0$ such that $\|x(t)\| \leq M$ on V . Let (p_n) be a sequence of polynomials without constant terms in z and $\bar{z}$, converging uniformly to f on the disk $|z| \leq M$. Then $p_n(x, x^*) \in \Theta$, and $p_n(x(t), x^*(t))$ converges uniformly on V to $f(x(t))$. This proves the proposition. The field $t \mapsto f(x(t))$ is denoted by $f(x)$.

10.3.4. Corollary. If $x \in \Theta$ and $x(t) \geq 0$ for every t , then the field $t \mapsto x(t)^{1/2}$ is an element y of Θ satisfying $y = y^*$ and $y^2 = x$.

10.3.5. If every $A(t)$ has an identity 1_t , and if $t \mapsto 1_t$ belongs to Θ , then the hypothesis $f(0) = 0$ can be omitted in 10.3.3.

10.3.6. Proposition. Let $((A(t))_{t \in T}, \Theta)$ be a continuous field of C^* -algebras over T , let $x \in \Theta$ satisfy $xx^* = x^*x$, let $t_0 \in T$, and let U be a neighborhood of $\text{Sp}' x(t_0)$ in $\mathbb{C}$. There is a neighborhood V of t_0 in T such that $\text{Sp}' x(t) \subset U$ for every $t \in V$.

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be continuous, equal to 0 on $\text{Sp}' x(t_0)$ and to 1 on $\mathbb{C} - U$. Then $f(x)(t_0) = 0$, so $\|f(x)(t)\| < 1$ in a neighborhood V of t_0 . If $t \in V$, $\text{Sp}' x(t)$ cannot meet $\mathbb{C} - U$.

Bibliography: [19, 29, 77, 108, 169].

10.4. The C^* -Algebra Defined by a Continuous Field of C^* -Algebras

10.4.1. Let T be locally compact, let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be a continuous field of C^* -algebras over T , and let A be the set of $x \in \Theta$ such that $\|x(t)\|$ tends to 0 at infinity on T . Then A is an involutive subalgebra of Θ . For $x \in A$, put

$$\|x\| = \sup_{t \in T} \|x(t)\| < +\infty.$$

It is immediate that A is a C^* -algebra for this norm. It is called the C^* -algebra defined by $\mathcal{A}$. For every $t \in T$, the map $x \mapsto x(t)$ is a morphism of A onto $A(t)$.

10.4.2. Lemma. *Let I be a closed left or right ideal of A . For every $t \in T$, let $I(t)$ be the set of $x(t)$ with $x \in I$. Then I is the set of all $x \in A$ such that $x(t) \in I(t)$ for every $t \in T$.*

We argue for a right ideal. If $x \in I$, then $x(t) \in I(t)$ for every t . Conversely, let $x \in A$ satisfy $x(t) \in I(t)$ for every t , and let $\varepsilon > 0$. For every $t \in T$, there are $y_t \in I$ and $z_t \in A$ such that $x(t) = y_t(t)$ and $\|y_t(t) - y_t(t)z_t(t)\| \leq \varepsilon$ (1.7.2); hence $\|x(t) - y_t(t)z_t(t)\| \leq \varepsilon$ near t . Let K be the compact set of $t \in T$ for which $\|x(t)\| \geq \varepsilon$. There is a cover $(U_1, \dots, U_n)$ of K by open subsets of T , and, for each i , elements $y_i \in I$, $z_i \in A$, such that $\|x(t) - y_i(t)z_i(t)\| \leq \varepsilon$ on U_i . Put $U_0 = T - K$ and $y_0 = z_0 = 0$. Let $(f_0, f_1, \dots, f_n)$ be a continuous partition of unity subordinate to $(U_0, U_1, \dots, U_n)$. Since $f_i z_i \in A$ by 10.1.9(ii), we have $\sum f_i y_i z_i \in I$. Let $t \in T$, and let $i_1, \dots, i_p$ be the indices i such that $t \in U_i$. Then

$$\sum_i f_i(t) y_i(t) z_i(t) = f_{i_1}(t) y_{i_1}(t) z_{i_1}(t) + \dots + f_{i_p}(t) y_{i_p}(t) z_{i_p}(t),$$

and

$$\|x(t) - y_{i_1}(t) z_{i_1}(t)\|, \dots, \|x(t) - y_{i_p}(t) z_{i_p}(t)\| \leq \varepsilon.$$

Since $f_{i_1}(t) + \dots + f_{i_p}(t) = 1$, it follows that

$$\left\| x(t) - \sum_i f_i(t) y_i(t) z_i(t) \right\| \leq \varepsilon.$$

Thus $\|x - \sum_i f_i y_i z_i\| \leq \varepsilon$. Since I is closed, $x \in I$.

10.4.3. Theorem. *Let T be locally compact, let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be a continuous field of C^* -algebras over T , and let A be the C^* -algebra defined by $\mathcal{A}$. For $t \in T$ and $\pi \in \widehat{A(t)}$, let ρ_π be the irreducible representation $x \mapsto \pi(x(t))$ of A . Then $\pi \mapsto \rho_\pi$ is a bijection from the disjoint union $\coprod_{t \in T} \widehat{A(t)}$ onto $\widehat{A}$.*

Let $\pi \in \widehat{A(t)}$ and $\pi' \in \widehat{A(t')}$. If $t \neq t'$, there is a continuous complex function f on T such that $f(t) = 1$ and $f(t') = 0$. Choose $x \in A$ with $\pi(x(t)) \neq 0$. Then

$$\rho_\pi(fx) = \pi(x(t)) \neq 0, \quad \rho_{\pi'}(fx) = \pi'(0) = 0,$$

so ρ_π and $\rho_{\pi'}$ are inequivalent. Thus $\pi \mapsto \rho_\pi$ is injective. Now let $\rho \in \widehat{A}$, and let $I = \ker \rho$. Let $I(t)$ be the set of $x(t)$ with $x \in I$, and let Y be the set of $t \in T$ for which $I(t) \neq A(t)$. By 10.4.2, $Y \neq \emptyset$. Suppose Y contains two distinct points t_1, t_2 , and choose disjoint neighborhoods U_1, U_2 of them. Let K_i be the closed two-sided ideal of A consisting of the $x \in A$ that vanish outside U_i . Then $I \supset 0 = K_1 K_2$, so, for example, $I \supset K_1$ by 2.11.4. It follows that $I(t_1) \supset K_1(t_1) = A(t_1)$, contrary to $t_1 \in Y$. Hence Y consists of one point $t_0 \in T$. By 10.4.2, I is the set of $x \in A$ such that $x(t_0) \in I(t_0)$. Thus ρ is the composite of the morphism $x \mapsto x(t_0)$ and an irreducible representation of $A(t_0)$.

10.4.4. Corollary. *Let T be locally compact, let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be a continuous field of nonzero elementary C^* -algebras over T , and let A be the C^* -algebra defined by $\mathcal{A}$. For each $t \in T$, let ρ_t be the irreducible representation of A obtained by composing $x \mapsto x(t)$ with the irreducible representation of $A(t)$, unique up to equivalence. Then $t \mapsto \rho_t$ is a homeomorphism from T onto $\widehat{A}$.*

This map is bijective by 10.4.3. Let $Y \subset T$ and $t_0 \in T$. The intersection of the kernels of the ρ_t , $t \in Y$, is the set I_Y of $x \in A$ that vanish on Y . The point ρ_{t_0} lies in the closure of the set of ρ_t , $t \in Y$, if and only if $I_{\{t_0\}} \supset I_Y$. If $t_0 \in \overline{Y}$, this inclusion is clear. If $t_0 \notin \overline{Y}$, there is a continuous complex function on T equal to 1 at t_0 and to 0 on Y , hence an $x \in A$ nonzero at t_0 and zero on Y ; thus $I_{\{t_0\}} \not\supset I_Y$. This proves the corollary. We identify T with $\widehat{A}$ by this homeomorphism.

10.4.5. Corollary. *With the notation of Corollary 10.4.4, A is a liminal C^* -algebra with spectrum T .*

Indeed, for every $t \in T = \widehat{A}$, $\rho_t(A)$ consists of compact operators, since $A(t)$ is elementary.

10.4.6. Proposition. *Let T be compact, let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be a continuous field of C^* -algebras over T , and let $A = \Theta$ be the C^* -algebra defined by $\mathcal{A}$. Suppose that every $A(t)$ has an identity 1_t and that the field $t \mapsto 1_t$, denoted by 1 , belongs to A . Let B be a C^* -subalgebra of A . Suppose that, for every pair of distinct points $t_1, t_2 \in T$, there is an $x \in B$ such that $x(t_1) = 1_{t_1}$ and $x(t_2) = 0$ (and, if T consists of one point, suppose $1 \in B$). Then, for every continuous complex function f on T , the field $t \mapsto f(t)1_t$ belongs to B .*

Let $t_1, t_2 \in T$, $t_1 \neq t_2$. There is an $x \in B$ with $x(t_1) = 1_{t_1}$ and $x(t_2) = 0$. Replacing x by xx^* , we may suppose $x \in B^+$. Let $p : \mathbb{R} \rightarrow \mathbb{R}$ be continuous, equal to 0 in a neighborhood of 0 and to 1 on $[1, +\infty)$. Put $y = p(x)$. Then $y(t_1) = 1_{t_1}$, and $y(t)$ vanishes in a neighborhood of t_2 .

Let K be a nonempty compact subset of T not containing t_1 . There are open sets $U_1, \dots, U_n$ covering K and elements $y_1, \dots, y_n \in B$ such that $y_i(t_1) = 1_{t_1}$ and $y_i(t) = 0$ on U_i . Forming the product $y_1 \cdots y_n y_n^* \cdots y_1^*$, we see that there is a $z \in B^+$ such that $z(t_1) = 1_{t_1}$ and $z(t) = 0$ on K . The same conclusion is still true, and evident, when K is empty. Near t_1 we have $\|z(t) - 1_t\| \leq 1/2$, and hence $z(t) \geq \frac{1}{2}1_t$. Now let L be a compact subset of T disjoint from K . There are open sets $V_1, \dots, V_p$ covering L and elements $z_1, \dots, z_p \in B^+$ such that $z_i(t) = 0$ on K and $z_i(t) \geq \frac{1}{2}1_t$ on V_i . Thus $u = 2(z_1 + \cdots + z_p) \in B^+$, with $u(t) = 0$ on K and $u(t) \geq 1_t$ on L .

Let f be a continuous real function on T , and let $\varepsilon > 0$. Choose an open cover $(W_1, \dots, W_q)$ of T such that the oscillation of f on each W_i is at most ε , and choose a value λ_i of f on W_i . Choose an open cover $(W'_1, \dots, W'_q)$ of T such that $\overline{W'_i} \subset W_i$ for every i . There is a $v_i \in B^+$ such that $v_i(t) \geq 1_t$ on W'_i and $v_i(t) = 0$ on $T - W_i$. Put $v = v_1 + \cdots + v_q$. Then $v(t) \geq 1_t$ for every t , so by functional calculus (10.3.3) there is a $v' \in B^+$ such that $(v'^2 v)(t) = 1_t$ for every t . Put $w_i = v' v_i v' \in B^+$, so that $\sum w_i = v' v v' = 1$. Let $t \in T$; for example, suppose

$$t \in W_1, \dots, W_m, \quad t \notin W_{m+1}, \dots, W_q.$$

Then

$$\begin{aligned} f(t)1_t - \sum_{i=1}^q \lambda_i w_i(t) &= f(t) \sum_{i=1}^m w_i(t) - \sum_{i=1}^m \lambda_i w_i(t) \\ &= \sum_{i=1}^m [f(t) - \lambda_i] w_i(t), \end{aligned}$$

and hence

$$-\varepsilon 1_t = -\varepsilon \sum_{i=1}^m w_i(t) \leq \sum_{i=1}^m [f(t) - \lambda_i] w_i(t) \leq \varepsilon \sum_{i=1}^m w_i(t) = \varepsilon 1_t.$$

Therefore

$$\left\| f(t)1_t - \sum_{i=1}^q \lambda_i w_i(t) \right\| \leq \varepsilon \quad \text{for every } t.$$

Since ε is arbitrary, the field $t \mapsto f(t)1_t$ belongs to $\overline{B} = B$, and hence to B .

10.4.7. Proposition. *Let T be a locally compact space with a countable base, let $\mathcal{A}$ be a separable continuous field of C^* -algebras over T , and let A be the C^* -algebra defined by $\mathcal{A}$. Then A is separable.*

By 10.2.6, it is enough to treat the case in which T is compact with a countable base. Let $(f_1, f_2, \dots)$ be a dense sequence in the Banach space of continuous complex functions on T . Put $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$, and let $(x_1, x_2, \dots)$ be a total sequence in Θ . The $f_i x_j$ are elements of A . We shall prove that their linear combinations are dense in A , which establishes the proposition. Let $x \in A$ and $\varepsilon > 0$. By 10.2.5, there are continuous complex functions $g_1, \dots, g_p$ on T such that $\|x - g_1 x_1 - \dots - g_p x_p\| \leq \varepsilon$. Choosing indices $i_1, \dots, i_p$ suitably gives $\|x - f_{i_1} x_1 - \dots - f_{i_p} x_p\| \leq 2\varepsilon$.

Bibliography: [29, 77, 108, 167, 169].

10.5. Continuous Fields of C^* -Algebras Defined by Certain C^* -Algebras

10.5.1. Let A be a C^* -algebra whose spectrum T is assumed Hausdorff. Then T is locally compact (3.3.8). Two irreducible representations with the same kernel are equivalent (3.1.6), so the elements of T are identified with the primitive ideals of A . For every $t \in T$, put $A(t) = A/t$, a nonzero C^* -algebra. Every $x \in A$ defines a vector field $t \mapsto x'(t) \in A(t)$ on T , where $x'(t)$ is the canonical image of x in $A(t)$. Let Λ be the set of the fields $t \mapsto x'(t)$, $x \in A$. Then Λ is an involutive subalgebra of the set of all vector fields. For every $x \in A$, the function $t \mapsto \|x'(t)\|$ is continuous (3.3.9); and for every $t \in T$, the set of $x'(t)$, $x \in A$, is all of $A(t)$.

It follows from 10.2.3 that there is a unique subset $\Gamma \subset \prod_{t \in T} A(t)$ containing Λ such that $\mathcal{A} = ((A(t))_{t \in T}, \Gamma)$ is a continuous field of Banach spaces; Γ is the

set of vector fields that are local uniform limits of elements of Λ . By 10.3.2, $\mathcal{A}$ is a continuous field of C^* -algebras. We call $\mathcal{A}$ the *continuous field of C^* -algebras defined by A* . If A is liminal, $\mathcal{A}$ is a continuous field of elementary C^* -algebras. If A is separable, $\mathcal{A}$ is separable.

10.5.2. Theorem. *Let T be locally compact, let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be a continuous field of nonzero elementary C^* -algebras over T , and let A be the liminal C^* -algebra defined by $\mathcal{A}$, whose spectrum is T . Let $\mathcal{A}' = ((A'(t))_{t \in T}, \Theta')$ be the continuous field of nonzero elementary C^* -algebras defined by A . For every $t \in T$, let $\varphi(t)$ be the canonical isomorphism of $A(t)$ onto $A'(t)$. Then $(\varphi(t))_{t \in T}$ is an isomorphism of $\mathcal{A}$ onto $\mathcal{A}'$.*

The elements of A form a total family of vector fields continuous relative to $\mathcal{A}$, and their images under $(\varphi(t))_{t \in T}$ belong to Θ' by the definition of $\mathcal{A}'$. It remains only to apply 10.2.4.

10.5.3. The following lemma will be greatly generalized later (11.5.3).

Lemma. *Let T be locally compact, let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be a continuous field of elementary C^* -algebras over T , let A be the C^* -algebra defined by $\mathcal{A}$, and let B be a C^* -subalgebra of A with the following property: whenever $t_1, t_2 \in T$, $\xi_1 \in A(t_1)$, and $\xi_2 \in A(t_2)$, with $\xi_1 = \xi_2$ if $t_1 = t_2$, there is an $x \in B$ such that $x(t_1) = \xi_1$ and $x(t_2) = \xi_2$. Then $B = A$.*

By adjoining a point at infinity to T , we reduce to the case in which T is compact (10.2.6). Then $A = \Theta$, and B is a total subset of Θ . In view of 10.2.5, it is enough to prove that $\varphi x \in B$ whenever $x \in B$ and φ is a continuous complex function on T . Let $\varepsilon > 0$. We shall construct an element of B approximating φx to within $2\varepsilon[\sup |\varphi|^2]^{1/2}$, which will complete the proof. Put $y = x^*x$.

Let $t_0 \in T$. There are numbers α, β such that $0 < \alpha < \beta < \varepsilon$ and the spectrum of $y(t_0)$ does not meet $[\alpha, \beta]$ [because $A(t_0)$ is elementary]. Choose α', β' with $\alpha < \alpha' < \beta' < \beta$. There is a neighborhood of t_0 such that, for t in this neighborhood, $\text{Sp}' y(t)$ does not meet $[\alpha', \beta']$ (10.3.6). Let $p : \mathbb{R} \rightarrow \mathbb{R}$ be a non-negative continuous function equal to 0 on $(-\infty, \alpha']$ and to 1 on $[\beta', +\infty)$. Put $e = p(y) \in B^+$. Then, for every $t \in T$, $e(t)$ is an element of $A(t)^+$ with finite spectrum; and, throughout a closed neighborhood U of t_0 , $e(t)$ is idempotent and $\|e(t)y(t) - y(t)\| \leq \varepsilon$.

Let $((A(t))_{t \in U}, \Delta)$ be the continuous field of C^* -algebras induced by $\mathcal{A}$ on U . For $z \in \Delta$, the following conditions are equivalent:

- 1° $z(t) \in A'(t) = e(t)A(t)e(t)$ for every $t \in U$;
- 2° there is a $z' \in \Delta$ such that $z(t) = e(t)z'(t)e(t)$ for every $t \in U$.

Let Δ' be the set of $z \in \Delta$ satisfying these conditions. It is immediate that $\mathcal{A}' = ((A'(t))_{t \in U}, \Delta')$ is a continuous field of C^* -algebras over U ; for every $t \in U$, $e(t)$ is the identity of $A'(t)$. Let A' be the C^* -algebra defined by $\mathcal{A}'$. Let B' be the set of elements of A' that are restrictions to U of elements of B . The map $x \mapsto x|_U$ is a morphism from the C^* -algebra $A = \Theta$ onto the C^* -algebra Δ (10.1.12); the image of B under this morphism is a C^* -subalgebra of Δ , and hence B' is a C^* -subalgebra of A' . By the hypothesis on B , whenever $t_1, t_2 \in U$, $\xi_1 \in A'(t_1)$, and $\xi_2 \in A'(t_2)$, with $\xi_1 = \xi_2$ if $t_1 = t_2$, there is an $x_1 \in B$ such that $x_1(t_1) = \xi_1$ and $x_1(t_2) = \xi_2$. Thus the restriction of ex_1e to U is an element of B' taking the values ξ_1, ξ_2 at t_1, t_2 . We may therefore apply 10.4.6: for every continuous complex function ψ' on U , the field $t \mapsto \psi'(t)e(t)$ on U belongs to B' . Consequently, for every continuous complex function ψ on T , the field ψe agrees on U with an element of B .

Let V be a closed subset of T contained in the interior $\mathring{U}$ of U . We show that there is an $f \in B$ such that $f(t) = e(t)$ on V and $f(t) = 0$ on $T - \mathring{U}$. Let $t_1 \in T - \mathring{U}$ and $t_2 \in V$. Since $e(t_1)$ has finite spectrum, the hypothesis on B shows that there is a Hermitian $d \in B$ such that $d(t_1)$ is idempotent and

$$d(t_1)e(t_1) = e(t_1), \quad d(t_2) = 0.$$

Replacing d by $q(d)$, where q vanishes near 0 and equals 1 at 1, we may suppose that $d(t)$ vanishes near t_2 . Covering V by finitely many such neighborhoods and multiplying the corresponding elements d , we obtain a $d' \in B$ such that

$$d'(t_1)e(t_1) = e(t_1), \quad d'(t) = 0 \quad \text{on } V.$$

Then $d'' = e - d'e \in B$ agrees with e on V and vanishes at t_1 . By functional calculus, we obtain $d''' \in B$ that agrees with e on V and vanishes in a neighborhood of t_1 . Covering $T - \mathring{U}$ by finitely many such neighborhoods and multiplying the corresponding elements d''' , we obtain f .

Let θ be a continuous complex function on T vanishing outside V . We have seen that there is an element of B taking the value $\theta(t)e(t)$ on U . Multiplying it by the preceding element f gives an element of B taking the value $\theta(t)e(t)$ everywhere on T .

In summary, changing the notation slightly, we can associate with every $t \in T$ an $e_t \in B^+$ and a closed neighborhood U_t of t with the following properties:

- 1° $\|e_t(t')y(t') - y(t')\| \leq \varepsilon$ on U_t ;
- 2° if θ is a continuous complex function on T whose support is contained in $\mathring{U}_t$, then $\theta e_t \in B$.

Let $(U_{t_1}, \dots, U_{t_r})$ be a finite cover of T . Choose an open cover $(V_1, \dots, V_r)$ of T such that $\bar{V}_1 \subset \overset{\circ}{U}_{t_1}, \dots, \bar{V}_r \subset \overset{\circ}{U}_{t_r}$, and let $(\theta_1, \dots, \theta_r)$ be a continuous partition of unity subordinate to $(V_1, \dots, V_r)$. Then $\varphi\theta_i e_{t_i} \in B$. Put

$$z = \varphi x(\theta_1 e_{t_1} + \dots + \theta_r e_{t_r}) \in B.$$

Then

$$(z - \varphi x)^*(z - \varphi x) = |\varphi|^2 \sum_{i,j} \theta_i \theta_j (e_{t_i} y e_{t_j} - e_{t_i} y - y e_{t_j} + y).$$

Since

$$\|e_{t_i}(t')y(t') - y(t')\| \leq \varepsilon \quad \text{on } U_{t_i},$$

we have

$$\|(e_{t_i} y e_{t_j} - e_{t_i} y - y e_{t_j} + y)(t')\| \leq 2\varepsilon \quad \text{on } U_{t_i},$$

and therefore

$$\|z - \varphi x\|^2 \leq 2\varepsilon [\sup |\varphi|^2].$$

10.5.4. Theorem. *Let A be a liminal C^* -algebra whose spectrum T is Hausdorff. Let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be the continuous field of nonzero elementary C^* -algebras over T defined by A , and let A' be the C^* -algebra defined by $\mathcal{A}$. For every $x \in A$, let x' be the element of Θ defined by x . Then $x' \in A'$, and $x \mapsto x'$ is an isomorphism of A onto A' .*

If $x \in A$, the function $t \mapsto \|x'(t)\|$ tends to zero at infinity on T (3.3.7), and hence $x' \in A'$. It is clear that $x \mapsto x'$ is a morphism π of A into A' . By 2.7.3, this morphism is isometric. If $t \in T$, the set of all $x'(t)$, $x \in A$, is the whole of $A(t)$ by the definition of $A(t)$. If t_1, t_2 are distinct points of T , and if $\xi_1 \in A(t_1)$ and $\xi_2 \in A(t_2)$, there is an $x \in A$ such that $x'(t_1) = \xi_1$ and $x'(t_2) = \xi_2$ (4.2.5). Thus $\pi(A) = A'$ by 10.5.3.

10.5.5. Theorems 10.5.2 and 10.5.4 show that there is a canonical one-to-one correspondence between liminal C^* -algebras with Hausdorff spectrum (up to isomorphism) and continuous fields of nonzero elementary C^* -algebras over locally compact spaces (up to isomorphism, the notion of isomorphism used here allowing a homeomorphism of the base space).

Let A be a liminal C^* -algebra with Hausdorff spectrum, and let $\mathcal{A}$ be the field defined by A . If A is separable, $\widehat{A}$ has a countable base (3.3.4) and $\mathcal{A}$ is separable. The converse follows from 10.4.7.

10.5.6. Corollary. *Let A be a liminal C^* -algebra such that $\widehat{A}$ is Hausdorff, let $x \in A$, and let f be a bounded continuous complex function on $\widehat{A}$. There is a $y \in A$ such that $\pi(y) = f(\pi)\pi(x)$ for every $\pi \in \widehat{A}$.*

With the notation of 10.5.4, we may replace A by A' ; the corollary then follows from 10.1.9(ii).

10.5.7. Definition. Let T be a topological space and let $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ be a continuous field of elementary C^* -algebras over T . We say that $\mathcal{A}$ satisfies Fell's condition if, for every $t_0 \in T$, there are a neighborhood V of t_0 and a vector field p of $\mathcal{A}$, defined and continuous on V , such that $p(t)$ is a rank-one projection for every $t \in V$ (cf. 4.1.9).

It is clear that a locally trivial continuous field of nonzero elementary C^* -algebras satisfies Fell's condition. The converse, however, is false.

10.5.8. Proposition. Let A be a liminal C^* -algebra with Hausdorff spectrum, and let $\mathcal{A}$ be the continuous field of nonzero elementary C^* -algebras defined by A . Then A has continuous trace if and only if $\mathcal{A}$ satisfies Fell's condition.

If A has continuous trace, $\mathcal{A}$ satisfies Fell's condition [4.5.3(iii)]. Conversely, suppose that $\mathcal{A}$ satisfies Fell's condition, and let $\pi_0 \in \widehat{A}$. There are a compact neighborhood V of π_0 and a vector field p of $\mathcal{A}$, defined and continuous on V , such that $p(t)$ is a rank-one projection for every $t \in V$. Let $f : T \rightarrow \mathbb{R}$ be a continuous nonnegative function equal to 1 on a neighborhood of π_0 and to 0 on a neighborhood of $T - V$. The vector field equal to 0 on $T - V$ and to fp on V is continuous and tends to zero at infinity. It comes from an element $y \in A^+$ such that $\pi(y)$ is a rank-one projection in a neighborhood of π_0 . Thus A has continuous trace (4.5.4).

Bibliography: [19, 29, 77, 108, 167, 169].

10.6. Remarks on Elementary C^* -Algebras

10.6.1. Let H be a Hilbert space. For $\xi, \eta \in H$, throughout the remainder of this subsection we denote by $\theta_{\xi, \eta}$ the operator $\zeta \mapsto (\zeta | \eta)\xi$ on H . It has rank at most 1, and every operator of rank at most 1 is of this form. If $\xi', \eta' \in H$, then

$$\begin{aligned}\theta_{\xi, \eta} \theta_{\xi', \eta'}(\zeta) &= \theta_{\xi, \eta}((\zeta | \eta')\xi') \\ &= (\zeta | \eta')(\xi' | \eta)\xi = (\xi' | \eta)\theta_{\xi, \eta'}(\zeta),\end{aligned}$$

and hence

$$\theta_{\xi, \eta} \theta_{\xi', \eta'} = (\xi' | \eta)\theta_{\xi, \eta'}.$$

On the other hand, if $\zeta' \in H$, then

$$(\theta_{\xi, \eta}^*(\zeta) | \zeta') = (\zeta | \theta_{\xi, \eta}(\zeta')) = (\zeta | (\zeta' | \eta)\xi)$$

$$= (\eta \mid \zeta')(\zeta \mid \xi) = ((\zeta \mid \xi)\eta \mid \zeta') = (\theta_{\eta, \xi}(\zeta) \mid \zeta'),$$

so

$$\theta_{\xi, \eta}^* = \theta_{\eta, \xi}.$$

We have $\theta_{\xi, \xi} = \|\xi\|^2 P_{\mathbb{C}\xi}$; consequently every finite-rank operator on H is a linear combination of operators $\theta_{\xi, \eta}$ (and even of operators $\theta_{\xi, \xi}$).

10.6.2. Let $\xi_1, \dots, \xi_{2n}$ be variable vectors in H . If $\xi_1 \rightarrow \eta_1, \dots, \xi_{2n} \rightarrow \eta_{2n}$, it is clear that

$$\theta_{\xi_1, \xi_2} + \dots + \theta_{\xi_{2n-1}, \xi_{2n}} \longrightarrow \theta_{\eta_1, \eta_2} + \dots + \theta_{\eta_{2n-1}, \eta_{2n}}$$

uniformly. This being so, Lemma 3.5.6 shows that, if $(\xi_i \mid \xi_j) \rightarrow (\eta_i \mid \eta_j)$ for all i, j , then

$$\|\theta_{\xi_1, \xi_2} + \dots + \theta_{\xi_{2n-1}, \xi_{2n}}\| \longrightarrow \|\theta_{\eta_1, \eta_2} + \dots + \theta_{\eta_{2n-1}, \eta_{2n}}\|.$$

In other words,

$$\|\theta_{\xi_1, \xi_2} + \dots + \theta_{\xi_{2n-1}, \xi_{2n}}\|$$

is a continuous function of the scalar products $(\xi_i \mid \xi_j)$.

10.6.3. If H is a Hilbert space with a distinguished unit vector ξ , put

$$\alpha(H, \xi) = (A, p),$$

where A is the elementary C^* -algebra $\mathcal{LC}(H)$ and $p \in A$ is the projection onto $\mathbb{C}\xi$. Similarly, let $(A', p') = \alpha(H', \xi')$. Every isomorphism $U : (H, \xi) \rightarrow (H', \xi')$ (that is, every isomorphism of H onto H' carrying ξ to ξ') defines an isomorphism

$$\alpha(U) : (A, p) \longrightarrow (A', p')$$

(that is, an isomorphism of A onto A' carrying p to p'). If U_1, U_2 are two such isomorphisms and $\alpha(U_1) = \alpha(U_2)$, then $U_1 = U_2$: indeed, by 4.1.8, U_1 and U_2 differ by a scalar factor, and that factor is 1 because $U_1\xi = \xi' = U_2\xi$.

10.6.4. Retain the preceding notation H, ξ, A, p . If $a \in Ap$, then $a^*a \in pAp$; hence a^*a leaves $\mathbb{C}\xi$ invariant and vanishes on $H \ominus \mathbb{C}\xi$, so

$$\|a^*a\| = (a^*a\xi \mid \xi) = \|a\xi\|^2, \quad \text{whence} \quad \|a\| = \|a\xi\|.$$

Thus $a \mapsto a\xi$ is an isometric isomorphism of the normed vector space Ap onto the normed vector space H .

It follows that Ap may be regarded as a Hilbert subspace of A ; it is in fact well determined, since scalar products in a Hilbert space can be computed from norms. If $a, b \in Ap$, then $b^*a \in pAp$, and therefore

$$\text{Tr}(b^*a) = (b^*a\xi \mid \xi) = (a\xi \mid b\xi).$$

Thus the scalar product in the Hilbert space Ap is given by $(a \mid b) = \text{Tr}(b^*a)$.

Notice also that ApA is the set of elements of A of rank at most 1. This follows from

$$\theta_{\zeta, \eta} = \theta_{\zeta, \xi} \theta_{\xi, \xi} \theta_{\xi, \eta} \quad (\|\xi\| = 1).$$

10.6.5. If A is an elementary C^* -algebra with a rank-one projection p , put

$$\beta(A, p) = (K, \eta),$$

where K is the Hilbert space Ap (10.6.4) and η is the unit vector p of K . Similarly, let $(K', \eta') = \beta(A', p')$. Every isomorphism $V : (A, p) \rightarrow (A', p')$ defines an isomorphism

$$\beta(V) : (K, \eta) \longrightarrow (K', \eta').$$

If V_1, V_2 are two such isomorphisms and $\beta(V_1) = \beta(V_2)$, then $V_1 = V_2$: indeed, V_1 and V_2 coincide on Ap , hence on $pA = (Ap)^*$, hence on $ApA = (Ap)(pA)$, and consequently on A by 10.6.4.

10.6.6. Lemma.

- (i) Let H be a Hilbert space and ξ a unit vector of H . Put $\alpha(H, \xi) = (A, p)$, so that $\beta(\alpha(H, \xi)) = (Ap, p)$. For every $a \in Ap$, put $\varphi(a) = a\xi$. Then φ is an isomorphism of the Hilbert space Ap onto the Hilbert space H , and $\varphi(p) = \xi$.
- (ii) Let A be an elementary C^* -algebra and p a rank-one projection of A . Put $\beta(A, p) = (H, \xi)$ and $\alpha(H, \xi) = (A', p')$. For every $a \in A$, let $\psi(a)$ be the linear map of $H = Ap$ into itself defined by $\psi(a)y = ay$. Then ψ is an isomorphism of the C^* -algebra A onto the C^* -algebra A' , and $\psi(p) = p'$.

Part (i) was proved in 10.6.4.

Adopt the notation of (ii). There are a Hilbert space H_0 and a unit vector $\xi_0 \in H_0$ such that (A, p) is identified with $\alpha(H_0, \xi_0)$. Let φ_0 be the isomorphism of (H, ξ) onto (H_0, ξ_0) defined in (i). For every $a \in A$ and every $b \in H$,

$$\varphi_0(\psi(a)b) = \varphi_0(ab) = (ab)\xi_0 = a(\varphi_0(b)).$$

Thus $\psi(a) = \varphi_0^{-1}a\varphi_0$, and ψ is indeed an isomorphism of A onto A' carrying $P_{\mathbb{C}\xi_0} = p$ to $P_{\mathbb{C}\xi} = p'$. This proves the lemma.

The isomorphisms φ and ψ will be called *canonical*.

10.6.7. Adopt the notation of 10.6.6(ii). If $a, a' \in A$, then $ap \in H$ and $a'^*p \in H$, so we may form θ_{ap, a'^*p} , which is an element of A' . We have

$$\psi(ap a') = \theta_{ap, a'^*p}.$$

Indeed, for every $b \in A$,

$$\psi(ap a')(bp) = ap a' bp = a[\text{Tr}(p a' bp)]p,$$

and by 10.6.4 this is equal to

$$(bp \mid a'^*p)ap = \theta_{ap, a'^*p}(bp).$$

10.6.8. Let H, H_1 be Hilbert spaces, and let ξ, ξ_1 be unit vectors of H, H_1 , respectively. Put

$$\begin{aligned} (A, p) &= \alpha(H, \xi), & (H', \xi') &= \beta(A, p), \\ (A_1, p_1) &= \alpha(H_1, \xi_1), & (H'_1, \xi'_1) &= \beta(A_1, p_1). \end{aligned}$$

Let

$$\varphi : (H', \xi') \longrightarrow (H, \xi), \quad \varphi_1 : (H'_1, \xi'_1) \longrightarrow (H_1, \xi_1)$$

be the canonical isomorphisms. Let U be an isomorphism of (H, ξ) onto (H_1, ξ_1) . Then $U' = \beta(\alpha(U))$ is an isomorphism of (H', ξ') onto (H'_1, ξ'_1) , and

$$U\varphi = \varphi_1 U'.$$

Indeed, if $a \in H' = Ap$, then

$$(U\varphi)(a) = U(a\xi) = UaU^{-1}\xi_1 = (\alpha(U)a)\xi_1 = \varphi_1(\alpha(U)(a)) = \varphi_1 U'(a).$$

There is an analogous result if one starts with $(A, p), (A_1, p_1)$ and an isomorphism of (A, p) onto (A_1, p_1) .

Bibliography: [19].

10.7. Continuous Field of Elementary C^* -Algebras Defined by a Continuous Field of Hilbert Spaces

10.7.1. Let T be a topological space, and let $\mathcal{H} = ((H(t))_{t \in T}, \Gamma)$ be a continuous field of Hilbert spaces over T . If $x, y \in \Gamma$, the function $t \mapsto (x(t) \mid y(t))$ is continuous because

$$4(x(t) \mid y(t)) = \|x(t) + y(t)\|^2 - \|x(t) - y(t)\|^2$$

$$+ i\|x(t) + iy(t)\|^2 - i\|x(t) - iy(t)\|^2.$$

10.7.2. For every $t \in T$, let $A(t) = \mathcal{LC}(H(t))$. For $x, y \in \Gamma$, define $\theta_{x,y} \in \prod_{t \in T} A(t)$ by $\theta_{x,y}(t) = \theta_{x(t),y(t)}$. By 10.6.1,

$$\theta_{x,y}^* = \theta_{y,x}, \quad \theta_{x,y} \theta_{x',y'} = \theta_{z,y'},$$

where $z(t) = (x'(t) \mid y(t))x(t)$. It follows that the set Λ of fields

$$\theta_{x_1,x_2} + \theta_{x_3,x_4} + \cdots + \theta_{x_{2n-1},x_{2n}}, \quad x_1, x_2, \dots, x_{2n} \in \Gamma,$$

is an involutive subalgebra of $\prod_{t \in T} A(t)$. By 10.6.1, $\Lambda(t)$ is the set of finite-rank operators on $H(t)$ and is therefore dense in $A(t)$. By 10.6.2,

$$\|\theta_{x_1,x_2}(t) + \cdots + \theta_{x_{2n-1},x_{2n}}(t)\|$$

is a continuous function of the $(x_i(t) \mid x_j(t))$, and hence of t . There is therefore (10.2.3) a unique set $\Theta \subset \prod_{t \in T} A(t)$ containing Λ such that $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ is a continuous field of Banach spaces. Since Λ is an involutive algebra, $\mathcal{A}$ is in fact a continuous field of elementary C^* -algebras (10.3.2), called the field *associated with* $\mathcal{H}$. It will be denoted by $\mathcal{A}(\mathcal{H})$. By 10.2.4, it is immediate that, if $Y \subset T$, then

$$\mathcal{A}(\mathcal{H}|Y) = \mathcal{A}(\mathcal{H})|Y.$$

10.7.3. If $a \in \Theta$ and $x \in \Gamma$, the field $ax : t \mapsto a(t)x(t)$ is an element of Γ . Indeed, since every element of Θ is a local uniform limit of elements of Λ (10.2.3), it is enough to verify this for $a \in \Lambda$, and hence for $a = \theta_{y,z}$, where $y, z \in \Gamma$; but $\theta_{y,z}x$ is the field $t \mapsto (x(t) \mid z(t))y(t)$.

10.7.4. Let (x_α) be a family of elements of Γ such that, for every $t \in T$, the $x_\alpha(t)$ are dense in $H(t)$. Then, for every $t \in T$, the $\theta_{x_\alpha, x_\beta}(t)$ are dense in the set of operators of rank at most 1 on $H(t)$. It follows that, if $\mathcal{H}$ is separable, $\mathcal{A}(\mathcal{H})$ is separable.

10.7.5. To study the correspondence between continuous fields of Hilbert spaces and continuous fields of elementary C^* -algebras, it will be convenient to make the following definitions.

Let $\mathcal{H}$ be a continuous field of Hilbert spaces over T , endowed with a continuous vector field x such that $\|x(t)\| = 1$ for every t . Put

$$\alpha(\mathcal{H}, x) = (\mathcal{A}, p),$$

where $\mathcal{A} = \mathcal{A}(\mathcal{H})$ and $p = \theta_{x,x}$ (so that p is a continuous field, relative to $\mathcal{A}$, of rank-one projections). Every isomorphism $\varphi : (\mathcal{H}, x) \rightarrow (\mathcal{H}', x')$ defines, in the evident way, an isomorphism

$$\alpha(\varphi) : \alpha(\mathcal{H}, x) \longrightarrow \alpha(\mathcal{H}', x').$$

It follows from 10.6.3 that $\alpha(\varphi_1) = \alpha(\varphi_2)$ implies $\varphi_1 = \varphi_2$.

Let $\mathcal{A}$ be a continuous field of elementary C^* -algebras over T , endowed with a continuous field p of rank-one projections. Put

$$\beta(\mathcal{A}, p) = (\mathcal{H}, x),$$

where $\mathcal{H}$ and x are defined as follows. Write $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$. Let $H(t)$ be the vector subspace $A(t)p(t)$ of $A(t)$; this is a Hilbert space (10.6.4). Let Γ be the set of $x \in \Theta$ such that $x(t) \in H(t)$ for every $t \in T$. It is clear that $\mathcal{H} = ((H(t))_{t \in T}, \Gamma)$ satisfies axioms (i), (iii), and (iv) of 10.1.2. Moreover, if $a \in \Theta$, then ap is an element of Γ , so that $\Gamma(t) = A(t)p(t) = H(t)$. Thus $\mathcal{H}$ is a continuous field of Hilbert spaces over T . We put $x = p$; then $x \in \Gamma$ and $\|x(t)\| = 1$ for every t .

If $\mathcal{A}$ is separable, then $\mathcal{H}$ is separable. Indeed, if $(a_1, a_2, \dots)$ is a sequence in Θ such that the $a_i(t)$ are dense in $A(t)$ for every t , then $(a_1p, a_2p, \dots)$ is a sequence in Γ such that the $(a_ip)(t)$ are dense in $H(t)$ for every t . Every isomorphism $\varphi : (\mathcal{A}, p) \rightarrow (\mathcal{A}', p')$ defines, in the evident way, an isomorphism

$$\beta(\varphi) : \beta(\mathcal{A}, p) \longrightarrow \beta(\mathcal{A}', p').$$

It follows from 10.6.5 that $\beta(\varphi_1) = \beta(\varphi_2)$ implies $\varphi_1 = \varphi_2$.

10.7.6. Lemma.

- (i) Let $\mathcal{H}$ be a continuous field of Hilbert spaces over T , endowed with a continuous vector field x such that $\|x(t)\| = 1$ for every t . Let $(\mathcal{H}', x') = \beta(\alpha(\mathcal{H}, x))$. Write $\mathcal{H} = ((H(t))_{t \in T}, \Gamma)$ and $\mathcal{H}' = ((H'(t))_{t \in T}, \Gamma')$. For every $t \in T$, let φ_t be the canonical isomorphism of $H'(t)$ onto $H(t)$ (10.6.6). Then $\varphi = (\varphi_t)_{t \in T}$ is an isomorphism of $\mathcal{H}'$ onto $\mathcal{H}$ such that $\varphi(x') = x$.
- (ii) Let $\mathcal{A}$ be a continuous field of elementary C^* -algebras over T , endowed with a continuous field p of rank-one projections. Let $(\mathcal{A}', p') = \alpha(\beta(\mathcal{A}, p))$. Write $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ and $\mathcal{A}' = ((A'(t))_{t \in T}, \Theta')$. For every $t \in T$, let ψ_t be the canonical isomorphism of $A(t)$ onto $A'(t)$ (10.6.6). Then $\psi = (\psi_t)_{t \in T}$ is an isomorphism of $\mathcal{A}$ onto $\mathcal{A}'$ such that $\psi(p) = p'$.

Adopt the notation of (i). It is clear that $\varphi(x') = x$. Let $y \in \Gamma$ and $y' = \theta_{y,x} \in \Gamma'$. Then

$$\varphi_t(y'(t)) = \theta_{y,x}(t)x(t) = y(t).$$

Thus $\varphi^{-1}(y) = y' \in \Gamma'$, which proves that φ^{-1} is an isomorphism of $\mathcal{H}$ onto $\mathcal{H}'$ (10.2.4).

Adopt the notation of (ii). It is clear that $\psi(p) = p'$. To show that ψ is an isomorphism of $\mathcal{A}$ onto $\mathcal{A}'$, it is enough to show that $\psi(a) \in \Theta'$ for every a of the form $a'pa''$, with $a', a'' \in \Theta$, since fields of this form constitute a total subset of Θ (10.6.4). But, if $a = a'pa''$, then $\psi(a) = \theta_{a'p, a''*p} \in \Theta'$ by 10.6.7.

10.7.7. Proposition. *Let T be a topological space and $\mathcal{A}$ a continuous field of elementary C^* -algebras over T . The following conditions are equivalent:*

- (i) *for every $t_0 \in T$, there are a neighborhood V of t_0 and a continuous field $\mathcal{H}$ of nonzero Hilbert spaces over V such that $\mathcal{A}|V$ is isomorphic to $\mathcal{A}(\mathcal{H})$;*
- (ii) *$\mathcal{A}$ satisfies Fell's condition.*

(i) $\Rightarrow$ (ii). Suppose (i) holds. Let $t_0 \in T$. Let V_0 be a neighborhood of t_0 and let $\mathcal{H} = ((H(t))_{t \in V_0}, \Gamma)$ be a continuous field of nonzero Hilbert spaces over V_0 such that $\mathcal{A}|V_0$ is isomorphic to $\mathcal{A}(\mathcal{H})$. Choose a nonzero $\xi \in H(t_0)$, and a continuous vector field x of $\mathcal{H}$ such that $x(t_0) = \xi$. The set V of $t \in V_0$ for which $x(t) \neq 0$ is a neighborhood of t_0 ; put $y(t) = \|x(t)\|^{-1}x(t)$ for $t \in V$. The vector field $\theta_{y,y}$ of $\mathcal{A}(\mathcal{H})$ is defined and continuous on V , and $\theta_{y,y}(t)$ is a rank-one projection for every $t \in V$. There is therefore a vector field p of $\mathcal{A}$, defined and continuous on V , whose values $p(t)$ are rank-one projections.

(ii) $\Rightarrow$ (i). Suppose that $\mathcal{A}$ satisfies Fell's condition, and let $t_0 \in T$. There are a neighborhood V of t_0 and a continuous field p of rank-one projections of $\mathcal{A}$, defined on V . Put $\beta(\mathcal{A}|V, p) = (\mathcal{H}, x)$. By 10.7.6, $\mathcal{A}|V$ is isomorphic to $\mathcal{A}(\mathcal{H})$.

10.7.8. Proposition 10.7.7 is a strictly local result. Even if $\mathcal{A}$ satisfies Fell's condition, in general there is no continuous field $\mathcal{H}$ of Hilbert spaces over T such that $\mathcal{A}$ is isomorphic to $\mathcal{A}(\mathcal{H})$. The existence of $\mathcal{H}$ is obstructed by a cohomology class, which we shall now specify.

10.7.9. Lemma. *Let $\mathcal{H}$ and $\mathcal{H}'$ be two continuous fields of nonzero Hilbert spaces over T , let $\mathcal{A}$ and $\mathcal{A}'$ be their associated fields of C^* -algebras, and let f be an isomorphism of $\mathcal{A}$ onto $\mathcal{A}'$. For every $t \in T$, there are a neighborhood V of t and an isomorphism of $\mathcal{H}|V$ onto $\mathcal{H}'|V$ that induces $f|V$.*

There are a neighborhood V of t and a continuous field p of rank-one projections of $\mathcal{A}|V$ (10.7.7). Then $p' = f(p)$ is a continuous field of rank-one projections of $\mathcal{A}'|V$. Choose a continuous vector field x of $\mathcal{H}$ such that $p(t)x(t) \neq 0$. In a neighborhood of t we have $p(t')x(t') \neq 0$. Replacing x by the field

$$t' \mapsto \|p(t')x(t')\|^{-1}p(t')x(t'),$$

we obtain, in a neighborhood of t , a continuous vector field y of $\mathcal{H}$ such that $\|y(t')\| = 1$ and p is the field of projections defined by y . Similarly, there is a continuous vector field y' of $\mathcal{H}'$ in a neighborhood of t such that $\|y'(t')\| = 1$ and p' is the field of projections defined by y' . Since the assertion to be proved is local, we may henceforth suppose that y, y', p, p' are all defined on the whole of T . Then $\beta(f)$ is an isomorphism of

$$\beta(\mathcal{A}, p) = \beta(\alpha(\mathcal{H}, y)) \quad \text{onto} \quad \beta(\mathcal{A}', p') = \beta(\alpha(\mathcal{H}', y')).$$

By 10.7.6 and 10.6.8, there is an isomorphism g of $(\mathcal{H}, y)$ onto $(\mathcal{H}', y')$ such that $\beta(f) = \beta(\alpha(g))$. Hence $f = \alpha(g)$ (10.7.5).

10.7.10. Lemma. *Let $\mathcal{H} = ((H(t))_{t \in T}, \Gamma)$ be a continuous field of Hilbert spaces over T , and let $\mathcal{A} = \mathcal{A}(\mathcal{H})$. An automorphism φ of $\mathcal{H}$ induces the identity automorphism of $\mathcal{A}$ if and only if it has the form $(u(t)1_t)_{t \in T}$, where 1_t denotes the identity map of $H(t)$ and u is a continuous map of T into $\mathbb{U}$, the group of complex numbers of modulus 1.*

The condition is plainly sufficient. Conversely, if φ induces the identity automorphism of $\mathcal{A}$, then every φ_t is a homothety with ratio $u(t) \in \mathbb{U}$ (4.1.8), and it remains to prove that u is continuous. For every $t_0 \in T$ there is a continuous vector field x of $\mathcal{H}$ such that $x(t) \neq 0$ on a neighborhood V of t_0 . On V ,

$$u(t) = (\varphi_t x(t) \mid x(t)) \|x(t)\|^{-2},$$

so u is continuous at t_0 .

10.7.11. Lemma. *Let T be paracompact and let $\mathcal{A}$ be a continuous field of elementary C^* -algebras over T satisfying Fell's condition. There exist:*

- (i) *an open cover $(T_i)_{i \in I}$ of T ;*
- (ii) *for each $i \in I$, a continuous field $\mathcal{H}_i$ of Hilbert spaces over T_i and an isomorphism h_i of $\mathcal{A}(\mathcal{H}_i)$ onto $\mathcal{A}|_{T_i}$;*
- (iii) *for all $i, j \in I$, an isomorphism g_{ij} of $\mathcal{H}_j|_{T_{ij}}$ onto $\mathcal{H}_i|_{T_{ij}}$ that induces the isomorphism $h_i^{-1} h_j$ of $\mathcal{A}(\mathcal{H}_j|_{T_{ij}})$ onto $\mathcal{A}(\mathcal{H}_i|_{T_{ij}})$.*

Here $T_i \cap T_j = T_{ij}$, and below similarly $T_i \cap T_j \cap T_k = T_{ijk}$, and so forth.

There are an open cover $(U_\alpha)_{\alpha \in \Lambda}$ of T and, for every $\alpha \in \Lambda$, a continuous field $\mathcal{K}_\alpha$ of Hilbert spaces over U_α and an isomorphism k_α of $\mathcal{A}(\mathcal{K}_\alpha)$ onto $\mathcal{A}|_{U_\alpha}$ (10.7.7). Since T is paracompact, we may suppose that (U_α) is locally finite. There is an open cover $(V_\alpha)_{\alpha \in \Lambda}$ of T such that $\overline{V}_\alpha \subset U_\alpha$ for every α . Fix provisionally $\alpha \in \Lambda$ and $t \in V_\alpha$. There are an open neighborhood $W \subset V_\alpha$ of t and a finite subset Λ' of Λ such that $W \cap \overline{V}_\beta = \emptyset$ for $\beta \notin \Lambda'$. Let $\beta \in \Lambda'$. There are an open

neighborhood $W_\beta \subset W$ of t and an isomorphism of $\mathcal{K}_\beta|_{W_\beta \cap V_\beta}$ onto $\mathcal{K}_\alpha|_{W_\beta \cap V_\beta}$ that induces $k_\alpha^{-1}k_\beta$ on $W_\beta \cap V_\beta$. This is clear if $t \notin \overline{V}_\beta$; if $t \in \overline{V}_\beta$, then $t \in U_\beta \cap U_\alpha$, and the assertion follows from 10.7.9. Since Λ' is finite, we obtain an open neighborhood $W_{\alpha,t} \subset V_\alpha$ of t and, for every $\beta \in \Lambda$, an isomorphism of $\mathcal{K}_\beta|_{W_{\alpha,t} \cap V_\beta}$ onto $\mathcal{K}_\alpha|_{W_{\alpha,t} \cap V_\beta}$ inducing $k_\alpha^{-1}k_\beta$ there. Let the index set I in the lemma be the set of pairs $i = (\alpha, t)$ with $\alpha \in \Lambda$ and $t \in V_\alpha$, and put

$$T_i = W_{\alpha,t}, \quad \mathcal{H}_i = \mathcal{K}_\alpha|_{W_{\alpha,t}}, \quad h_i = k_\alpha|_{W_{\alpha,t}}.$$

The existence of the g_{ij} follows at once from the preceding construction.

In the remainder of this subsection we shall use the language of sheaves. For this subject the reader may consult R. Godement, *Théorie des faisceaux* (Act. Sci. Ind., no. 1252, Hermann, Paris, 1958).

10.7.12. Lemma. *Retain the notation of 10.7.11.*

(i) *For all $i, j, k \in I$, there is a unique continuous function $u_{ijk} : T_{ijk} \rightarrow \mathbb{U}$ such that*

$$g_{ij}g_{jk} = u_{ijk}g_{ik}.$$

(ii) *The u_{ijk} define a 2-cocycle of (T_i) with values in the sheaf $\mathcal{U}$ of germs of continuous maps of T into $\mathbb{U}$.*

(iii) *The image γ of this cocycle in the cohomology group $H^2(T, \mathcal{U})$ depends only on $\mathcal{A}$.*

If $i, j, k \in I$, then $g_{ij}g_{jk}g_{ik}^{-1}$ induces on $\mathcal{A}(\mathcal{H}_i|_{T_{ijk}})$ the automorphism

$$h_i^{-1}h_jh_j^{-1}h_kh_k^{-1}h_i = 1.$$

Thus (i) follows from 10.7.10.

If $i, j, k, l \in I$, then on T_{ijkl} ,

$$\begin{aligned} u_{jkl}u_{ikl}^{-1}u_{ijl} &= (g_{jl}^{-1}g_{jk}g_{kl})(g_{kl}^{-1}g_{ik}^{-1}g_{il})(g_{il}^{-1}g_{ij}g_{jl}) \\ &= g_{jl}^{-1}(g_{jk}g_{ik}^{-1})g_{ij}g_{jl} \\ &= g_{jl}^{-1}(u_{ijk}g_{ij}^{-1})g_{ij}g_{jl} = u_{ijk}, \end{aligned}$$

which proves (ii).

We prove (iii). Let $(T_\lambda)_{\lambda \in \Lambda}$, $\mathcal{K}_\lambda$, h_λ , $g_{\lambda\mu}$ be data analogous to those of 10.7.11, and let $(u_{\lambda\mu\nu})$ be the corresponding cocycle. To prove that (u_{ijk}) and $(u_{\lambda\mu\nu})$ define the same element of $H^2(T, \mathcal{U})$, we may replace (T_i) by a refinement, replacing $\mathcal{H}_i$, h_i , g_{ij} by their restrictions to the members of the new cover, and do the same for (T_λ) . This allows us to suppose, as in the proof of 10.7.11, that for every $i \in I$

and every $\lambda \in \Lambda$ there is an isomorphism $g_{i\lambda}$ of $\mathcal{H}_i|_{T_{i\lambda}}$ onto $\mathcal{K}_\lambda|_{T_{i\lambda}}$ inducing the isomorphism $h_\lambda^{-1}h_i$ of the associated fields of C^* -algebras. Let Λ' be the disjoint union of I and Λ . The preceding data define a cover $(T_a)_{a \in \Lambda'}$, fields $\mathcal{H}_a$, isomorphisms h_a , and transition maps g_{ab} satisfying the properties of 10.7.11, and hence functions u_{abc} , $a, b, c \in \Lambda'$. The class in $H^2(T, \mathcal{U})$ defined by (u_{abc}) is at once the class defined by (u_{ijk}) and the class defined by $(u_{\lambda\mu\nu})$. This proves the lemma.

10.7.13. Let $\mathcal{R}$ be the sheaf of germs of continuous maps from T into $\mathbb{R}$. The map $t \mapsto \exp(2\pi it)$ of $\mathbb{R}$ onto $\mathbb{U}$ defines a morphism $\mathcal{R} \rightarrow \mathcal{U}$. Its kernel is the constant sheaf $\mathbb{Z}$. Since every continuous function $g : T \rightarrow \mathbb{U}$ is locally of the form $\exp(2\pi if)$, where f is a continuous map of T into $\mathbb{R}$, we have an exact sequence of sheaves

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{R} \longrightarrow \mathcal{U} \longrightarrow 1,$$

and hence an exact cohomology sequence

$$\cdots \longrightarrow H^n(T, \mathbb{Z}) \longrightarrow H^n(T, \mathcal{R}) \longrightarrow H^n(T, \mathcal{U}) \longrightarrow H^{n+1}(T, \mathbb{Z}) \longrightarrow \cdots.$$

The existence of continuous partitions of unity on T shows that $\mathcal{R}$ is a fine sheaf, so $H^n(T, \mathcal{R}) = 0$ for $n > 0$. It follows in particular that the canonical morphism

$$H^2(T, \mathcal{U}) \longrightarrow H^3(T, \mathbb{Z})$$

is an isomorphism.

10.7.14. Definition. Let T be paracompact and let $\mathcal{A}$ be a continuous field of elementary C^* -algebras over T satisfying Fell's condition. Let γ be the element of $H^2(T, \mathcal{U})$ defined in Lemma 10.7.12. Its canonical image in $H^3(T, \mathbb{Z})$ will be denoted by $\delta(\mathcal{A})$.

10.7.15. Theorem. Let T be paracompact and let $\mathcal{A}$ be a continuous field of elementary C^* -algebras over T satisfying Fell's condition. There exists a continuous field $\mathcal{H}$ of Hilbert spaces over T such that $\mathcal{A}$ is isomorphic to $\mathcal{A}(\mathcal{H})$ if and only if $\delta(\mathcal{A}) = 0$.

Suppose first that $\mathcal{A}$ is isomorphic to $\mathcal{A}(\mathcal{H})$. In constructing $\delta(\mathcal{A})$, we may take as the cover of T the cover consisting of the single open set T . Every 2-cocycle relative to this cover is cohomologous to zero, and so $\delta(\mathcal{A}) = 0$.

Conversely, suppose that $\delta(\mathcal{A}) = 0$. There are $T_i, \mathcal{H}_i, h_i, g_{ij}, u_{ijk}$, with $i, j, k \in I$, having the properties of 10.7.11 and 10.7.12, and continuous functions $v_{ij} : T_{ij} \rightarrow \mathbb{U}$ such that

$$u_{ijk} = v_{jk}v_{ik}^{-1}v_{ij} \quad (i, j, k \in I).$$

Put $g'_{ij} = v_{ij}^{-1}g_{ij}$. The isomorphisms $g'_{ij} : \mathcal{H}_j|_{T_{ij}} \rightarrow \mathcal{H}_i|_{T_{ij}}$ satisfy

$$g'_{ij}g'_{jk} = v_{ij}^{-1}g_{ij}g_{jk}v_{jk}^{-1} = u_{ijk}v_{ij}^{-1}v_{jk}^{-1}g_{ik} = g'_{ik}.$$

We may therefore define a continuous field $\mathcal{H}$ of Hilbert spaces over T by gluing the $\mathcal{H}_i$ by means of the g'_{ij} (10.1.13). In other words, we may henceforth suppose that $\mathcal{H}_i = \mathcal{H}|_{T_i}$ and that the g'_{ij} are the identity automorphisms. Then $h_i : \mathcal{A}(\mathcal{H})|_{T_i} \rightarrow \mathcal{A}|_{T_i}$ and $h_j : \mathcal{A}(\mathcal{H})|_{T_j} \rightarrow \mathcal{A}|_{T_j}$ coincide on T_{ij} . Thus the h_i define an isomorphism of $\mathcal{A}(\mathcal{H})$ onto $\mathcal{A}$.

Bibliography: [19, 29].

10.8. Locally Trivial Fields of Elementary C^* -Algebras

10.8.1. Lemma. *Let H be an infinite-dimensional Hilbert space. There exist:*

- a. *a unit vector $\xi \in H$;*
- b. *for every $t \in [0, 1]$, a closed vector subspace H_t of H ;*
- c. *for every $t \in (0, 1]$, a linear isometric map U_t of H_t onto H ;*

with the following properties:

- (i) *the projection $P_t = P_{H_t}$ is a strongly continuous function of $t \in [0, 1]$, and $H_1 = H$, $H_0 = \{0\}$;*
- (ii) *the operators $U_t P_t$ and U_t^{-1} are strongly continuous functions of $t \in (0, 1]$, and $U_1 = 1$;*
- (iii) $\|P_t \xi\|^2 = t$.

Suppose first that $\dim H = \aleph_0$, so that H may be identified with $L^2([0, 1])$. Let H_t be the set of $f \in L^2([0, 1])$ such that $f(x) = 0$ for $x \geq t$. If $t \in (0, 1]$ and $f \in H_t$, define $U_t f \in H$ by

$$(U_t f)(x) = \sqrt{t} f(tx).$$

Finally, take for ξ the constant function equal to 1. Properties (i) and (iii) are immediate. For $f \in H_t$,

$$\|U_t f\|^2 = \int_0^1 t |f(tx)|^2 dx = \int_0^t |f(x')|^2 dx' = \|f\|^2,$$

and it is clear that U_t is a linear isometric map of H_t onto H ; if $g \in H$, then, for $0 \leq x \leq t$,

$$(U_t^{-1} g)(x) = \frac{1}{\sqrt{t}} g\left(\frac{x}{t}\right).$$

For $f \in H$, the expression

$$\|U_t P_t f - U_{t'} P_{t'} f\|^2 = \int_0^1 \left| \sqrt{t} f(tx) - \sqrt{t'} f(t'x) \right|^2 dx$$

tends to zero as $t' \rightarrow t$. On the other hand,

$$\begin{aligned}\|U_t^{-1}f - U_{t'}^{-1}f\|^2 &= 2\|f\|^2 - 2\operatorname{Re}(U_t^{-1}f \mid U_{t'}^{-1}f) \\ &= 2\|f\|^2 - 2\operatorname{Re} \int_0^{\min(t,t')} \frac{1}{\sqrt{t}} f\left(\frac{x}{t}\right) \frac{1}{\sqrt{t'}} \overline{f\left(\frac{x}{t'}\right)} dx.\end{aligned}$$

If, for example, $t \leq t'$, this gives

$$\|U_t^{-1}f - U_{t'}^{-1}f\|^2 = 2\|f\|^2 - 2\operatorname{Re} \int_0^1 \sqrt{\frac{t}{t'}} f(x') \overline{f\left(\frac{t}{t'}x'\right)} dx',$$

and this tends to zero as $t' \rightarrow t$. This proves (ii).

If now $\dim H$ is an arbitrary infinite cardinal, we may identify H with a Hilbert space $K \otimes K'$, where K, K' are Hilbert spaces and $\dim K = \aleph_0$. There are in K operators P_t, U_t and a vector ξ having properties (i), (ii), and (iii); it is enough to consider on H the operators $P_t \otimes 1, U_t \otimes 1$, and the vector $\xi \otimes \xi'$, where ξ' is a unit vector of K' .

10.8.2. Lemma. *Let H be an infinite-dimensional Hilbert space, let G be the unitary group of H with the strong topology, and let S be the unit sphere of H with the norm topology. Then G and S are contractible.*

Let H_t, P_t, U_t, ξ have the properties of 10.8.1. For $U \in G$ and $t \in (0, 1]$, put

$$\Phi(U, t) = (1 - P_t) + U_t^{-1} U U_t P_t.$$

The operator $\Phi(U, t)$ induces the identity map on $H \ominus H_t$ and, on H_t , the unitary operator whose restriction is $U_t^{-1} U U_t$. Thus $\Phi(U, t) \in G$, and $\Phi(U, 1) = U$. Put $\Phi(U, 0) = 1$. We show that $\Phi : G \times [0, 1] \rightarrow G$ is continuous. On $G \times (0, 1]$ this follows at once from Lemma 10.8.1. Moreover, if $(U', t') \in G \times (0, 1]$ tends to $(U, 0) \in G \times [0, 1]$, then $P_{t'}$ converges strongly to zero and, for every $f \in H$,

$$\|U_{t'}^{-1} U' U_{t'} P_{t'} f\| = \|P_{t'} f\| \longrightarrow 0.$$

Hence $\Phi(U', t')$ converges strongly to $1 = \Phi(U, 0)$. Thus G is contractible.

For $f \in S$ and $t \in (0, 1]$, put

$$\Psi(f, t) = \sqrt{t} U_t^{-1} f + (1 - P_t) \xi.$$

We have

$$\|\Psi(f, t)\|^2 = t\|U_t^{-1} f\|^2 + \|(1 - P_t) \xi\|^2 = t + 1 - t = 1.$$

Thus $\Psi(f, t) \in S$, and $\Psi(f, 1) = f$. Put $\Psi(f, 0) = \xi$. The map $\Psi : S \times [0, 1] \rightarrow S$ is continuous on $S \times (0, 1]$. Moreover, if $(f', t') \in S \times (0, 1]$ tends to $(f, 0)$, then $\|\sqrt{t'}U_{t'}^{-1}f'\| \rightarrow 0$, and hence $\Psi(f', t') \rightarrow \xi = \Psi(f, 0)$. Thus S is contractible.

10.8.3. A continuous field $((A(t))_{t \in T}, \Theta)$ of elementary C^* -algebras is said to have *rank a* if every $A(t)$ has rank a (4.1.9).

10.8.4. Theorem. *Let T be a paracompact space and a an infinite cardinal. The map $\mathcal{A} \mapsto \delta(\mathcal{A})$ defines a one-to-one correspondence between $H^3(T, \mathbb{Z})$ and the locally trivial continuous fields of elementary C^* -algebras of rank a over T , up to isomorphism.*

Choose a Hilbert space H of dimension a . Let G be its unitary group, let $\mathcal{G}$ be the sheaf of germs of continuous maps of T into G , and let $\mathcal{U}$ be the sheaf of germs of continuous maps of T into $\mathbb{U}$.

1° Let $\mathcal{A}, \mathcal{A}'$ be two locally trivial continuous fields of elementary C^* -algebras of rank a over T . Suppose that $\delta(\mathcal{A}) = \delta(\mathcal{A}')$. We shall prove that $\mathcal{A}$ and $\mathcal{A}'$ are isomorphic.

There exist: (a) a locally finite open cover $(T_i)_{i \in I}$ of T ; (b) for every $i \in I$, continuous fields $\mathcal{H}_i, \mathcal{H}'_i$ of Hilbert spaces over T_i , and isomorphisms

$$h_i : \mathcal{A}(\mathcal{H}_i) \longrightarrow \mathcal{A}|_{T_i}, \quad h'_i : \mathcal{A}(\mathcal{H}'_i) \longrightarrow \mathcal{A}'|_{T_i};$$

(c) for all $i, j \in I$, isomorphisms

$$g_{ij} : \mathcal{H}_j|_{T_{ij}} \longrightarrow \mathcal{H}_i|_{T_{ij}}, \quad g'_{ij} : \mathcal{H}'_j|_{T_{ij}} \longrightarrow \mathcal{H}'_i|_{T_{ij}}$$

that induce the isomorphisms $h_i^{-1}h_j$ and $h_i'^{-1}h'_j$, respectively. If $u_{ijk}, u'_{ijk} : T_{ijk} \rightarrow \mathbb{U}$ are defined by

$$g_{ij}g_{jk} = u_{ijk}g_{ik}, \quad g'_{ij}g'_{jk} = u'_{ijk}g'_{ik},$$

then (u_{ijk}) and (u'_{ijk}) both represent the classes $\delta(\mathcal{A})$ and $\delta(\mathcal{A}')$, respectively. Replacing (T_i) by a refinement, we may therefore suppose that there is a 1-cochain (v_{ij}) with values in $\mathcal{U}$ such that

$$u'_{ijk} = u_{ijk}v_{ij}v_{jk}v_{ik}^{-1}.$$

Replacing the g'_{ij} by $v_{ij}^{-1}g'_{ij}$, we reduce to the case $u'_{ijk} = u_{ijk}$.

Since $\mathcal{A}$ and $\mathcal{A}'$ are locally trivial, we may suppose, after refining (T_i) once more, that $\mathcal{H}_i$ and $\mathcal{H}'_i$ are both the constant field over T_i defined by H . Then g_{ij} and g'_{ij} are continuous maps of T_{ij} into G with its strong topology, and hence define continuous sections $\gamma_{ij}, \gamma'_{ij}$ of $\mathcal{G}$ over T_{ij} . Choose an open cover $(S_i)_{i \in I}$ of

T such that $\bar{S}_i \subset T_i$ for every $i \in I$. Let L be the set of pairs (J, β) , where $J \subset I$, $\beta = (\beta_i)_{i \in J}$, and, for every $i \in J$, β_i is a continuous section of $\mathcal{G}$ over $\bar{S}_i$, such that

$$\gamma'_{ij} = \beta_i \gamma_{ij} \beta_j^{-1} \quad \text{on } \bar{S}_i \cap \bar{S}_j \quad (i, j \in J).$$

Order L inductively, and let (J, β) be a maximal element. We prove that $J = I$. Suppose that $i \in I - J$, and put

$$R = \bar{S}_i \cap \bigcup_{j \in J} \bar{S}_j.$$

If $t \in R$, and $j_1, j_2 \in J$ are such that $t \in \bar{S}_{j_1} \cap \bar{S}_{j_2}$, then

$$\begin{aligned} \gamma'_{ij_2}(t) \beta_{j_2}(t) \gamma_{ij_2}(t)^{-1} &= u_{ij_1 j_2}(t)^{-1} \gamma'_{ij_1}(t) \gamma'_{j_1 j_2}(t) \beta_{j_2}(t) \gamma_{j_1 j_2}(t)^{-1} \gamma_{ij_1}(t)^{-1} u_{ij_1 j_2}(t) \\ &= \gamma'_{ij_1}(t) (\gamma'_{j_1 j_2}(t) \beta_{j_2}(t) \gamma_{j_1 j_2}(t)^{-1}) \gamma_{ij_1}(t)^{-1} \\ &= \gamma'_{ij_1}(t) \beta_{j_1}(t) \gamma_{ij_1}(t)^{-1}. \end{aligned}$$

It follows that one can define a continuous section ζ of $\mathcal{G}$ over R such that $\gamma'_{ij} = \zeta \gamma_{ij} \beta_j^{-1}$ for every $j \in J$. Since G is contractible (10.8.2), $\mathcal{G}$ is soft; hence ζ extends to a continuous section β_i of $\mathcal{G}$ over $\bar{S}_i$. This contradicts the maximality of (J, β) . Thus $J = I$.

Each β_i now defines an automorphism a_i of $\mathcal{A}(\mathcal{H}_i | S_i)$, and

$$h_i'^{-1} h_j' = a_i h_i^{-1} h_j a_j^{-1} \quad \text{on } S_{ij},$$

or equivalently

$$h_i' a_i h_i^{-1} = h_j' a_j h_j^{-1} \quad \text{on } S_{ij}.$$

Thus the $h_i' a_i h_i^{-1}$ define an isomorphism of $\mathcal{A}$ onto $\mathcal{A}'$.

2° Let $(T_i)_{i \in I}$ be a locally finite open cover of T , and let (u_{ijk}) be an alternating 2-cocycle of this cover with values in $\mathcal{U}$. We shall construct a locally trivial continuous field $\mathcal{A}$ of elementary C^* -algebras of rank a over T such that $\delta(\mathcal{A})$ is the element of $H^3(T, \mathbb{Z})$ defined by (u_{ijk}) . This will complete the proof of the theorem.

Choose an open cover $(T'_i)_{i \in I}$ such that $\bar{T}'_i \subset T_i$. Let L be the set of pairs (J, γ) , where $J \subset I$, $\gamma = (\gamma_{ij})_{i, j \in J}$, and for all $i, j \in J$, γ_{ij} is a continuous section of $\mathcal{G}$ over $\bar{T}'_i \cap \bar{T}'_j$, satisfying

$$\gamma_{ij} \gamma_{jk} = \gamma_{ik} u_{ijk} \quad (i, j, k \in J).$$

Order L inductively, and let (J, γ) be a maximal element. We show that $J = I$.

Suppose that $i \in I - J$. Let M be the set of pairs (K, γ') , where $K \subset J$, $\gamma' = (\gamma'_{ij})_{j \in K}$, and, for every $j \in K$, γ'_{ij} is a continuous section of $\mathcal{G}$ over $\overline{T'_i} \cap \overline{T'_j}$, such that

$$\gamma'_{ij} \gamma_{jk} = \gamma'_{ik} u_{ijk} \quad (j, k \in K).$$

Let (K, γ') be a maximal element of M . We prove that $K = J$. If $j \in J - K$, one can define a section β of $\mathcal{G}$ over

$$\overline{T'_i} \cap \overline{T'_j} \cap \bigcup_{k \in K} \overline{T'_k}$$

such that

$$\beta \gamma_{jk} = \gamma'_{ik} u_{ijk} \quad (k \in K).$$

Indeed, if $k, l \in K$, define on the corresponding intersections β_k, β_l by

$$\beta_k \gamma_{jk} = \gamma'_{ik} u_{ijk}, \quad \beta_l \gamma_{jl} = \gamma'_{il} u_{ijl}.$$

On $\overline{T'_i} \cap \overline{T'_j} \cap \overline{T'_k} \cap \overline{T'_l}$ one obtains

$$\begin{aligned} \beta_l &= \gamma'_{il} u_{ijl} \gamma_{jl}^{-1} \\ &= \gamma'_{ik} \gamma_{kl} u_{ikl}^{-1} u_{ijl} \gamma_{kl}^{-1} \gamma_{jk}^{-1} u_{jkl} \\ &= \gamma'_{ik} \gamma_{jk}^{-1} u_{ijl} u_{ikl}^{-1} u_{jkl} = \gamma'_{ik} \gamma_{jk}^{-1} u_{ijk} = \beta_k. \end{aligned}$$

Since $\mathcal{G}$ is soft, β extends to a continuous section of $\mathcal{G}$ over $\overline{T'_i} \cap \overline{T'_j}$, contradicting the maximality of (K, γ') . Thus $K = J$. Consequently there is a family $(\gamma'_{ij})_{j \in J}$ such that each γ'_{ij} is a continuous section of $\mathcal{G}$ over $\overline{T'_i} \cap \overline{T'_j}$ and

$$\gamma'_{ij} \gamma_{jk} = \gamma'_{ik} u_{ijk} \quad (j, k \in J).$$

The sections γ_{ij} and γ'_{ij} then contradict the maximality of (J, γ) . Hence $J = I$.

Let $\mathcal{H}_i$ be the constant field over T'_i defined by H , and put $\mathcal{A}_i = \mathcal{A}(\mathcal{H}_i)$. For $i, j \in I$, the section γ_{ij} defines an isomorphism $f_{ij} : \mathcal{A}_j|_{T'_{ij}} \rightarrow \mathcal{A}_i|_{T'_{ij}}$, and $f_{ij} f_{jk} = f_{ik}$. There is therefore a continuous field $\mathcal{A}$ of elementary C^* -algebras over T and, for every $i \in I$, an isomorphism h_i of $\mathcal{A}_i$ onto $\mathcal{A}|_{T'_i}$ such that $f_{ij} = h_i^{-1} h_j$ (10.1.13). To compute $\delta(\mathcal{A})$, we may use the cover (T'_i) , the $\mathcal{H}_i$, the h_i , and the restrictions of the γ_{ij} to T'_{ij} . It follows that $\delta(\mathcal{A})$ is the element of $H^3(T, \mathbb{Z})$ defined by (u_{ijk}) .

10.8.5. We shall now see that the local-triviality hypothesis made in Theorem 10.8.4 is sometimes automatically satisfied.

Lemma. Let T be a topological space, let $\mathcal{H} = ((H(t))_{t \in T}, \Gamma)$ be a separable continuous field of Hilbert spaces over T , and let H_0 be a Hilbert space of dimension $\aleph_0$. For every $t \in T$ there exist a closed vector subspace $K(t)$ of H_0 and an isomorphism $U(t)$ of the Hilbert space $H(t)$ onto the Hilbert space $K(t)$, with the following properties:

- a. if Δ is the set of continuous maps $x : T \rightarrow H_0$ such that $x(t) \in K(t)$ for every $t \in T$, then $((K(t))_{t \in T}, \Delta)$ is a continuous field $\mathcal{K}$ of Hilbert spaces over T ;
- b. $(U(t))_{t \in T}$ is an isomorphism of $\mathcal{H}$ onto $\mathcal{K}$.

Let $(x_1, x_2, \dots)$ be a total subset of Γ . Multiplying the x_i by suitable continuous functions, we may suppose that

$$\|x_i(t)\| \leq \frac{1}{i} \quad \text{for every } i \text{ and every } t.$$

Let $(\varepsilon_1, \varepsilon_2, \dots)$ be an orthonormal basis of H_0 . For every $t \in T$, there is a continuous linear map $S(t) : H_0 \rightarrow H(t)$ such that $S(t)\varepsilon_i = x_i(t)$ for every i . We have

$$\|S(t)\|^2 \leq \text{Tr}(S(t)^* S(t)) \leq \sum_{i=1}^{\infty} i^{-2},$$

and $S(t)(H_0)$ is dense in $H(t)$. Let $t \mapsto x(t) = \sum_{i=1}^{\infty} \lambda_i(t) \varepsilon_i$ be a continuous map of T into H_0 . Then

$$t \mapsto S(t)x(t) = \sum_{i=1}^{\infty} \lambda_i(t) x_i(t)$$

belongs to Γ , since the λ_i are continuous functions and

$$\left\| \sum_{i=n+1}^{\infty} \lambda_i(t) x_i(t) \right\| \leq \sum_{i=n+1}^{\infty} |\lambda_i(t)| \|x_i(t)\| \leq \|x(t)\| \left(\sum_{i=n+1}^{\infty} i^{-2} \right)^{1/2}.$$

We show that $t \mapsto S(t)^* S(t)x(t)$ is a continuous map of T into H_0 . It is enough to prove this when $x(t) = \varepsilon_i$ for every t . For every $z \in H_0$,

$$\begin{aligned} \|S(t)^* S(t)\varepsilon_i - z\|^2 &= \|S(t)^* S(t)\varepsilon_i\|^2 + \|z\|^2 - 2 \text{Re}(S(t)^* S(t)\varepsilon_i | z) \\ &= \sum_{j=1}^{\infty} |(S(t)^* S(t)\varepsilon_i | \varepsilon_j)|^2 + \|z\|^2 - 2 \text{Re}(S(t)^* S(t)\varepsilon_i | z) \\ &= \sum_{j=1}^{\infty} |(x_i(t) | x_j(t))|^2 + \|z\|^2 - 2 \text{Re}(x_i(t) | S(t)z). \end{aligned}$$

This is a continuous function of t by what precedes, and it follows that $t \mapsto S(t)^* S(t)\varepsilon_i$ is continuous.

Let $z \in \Gamma$. For every $t_0 \in T$ and every $\varepsilon > 0$, there is a continuous map $x : T \rightarrow H_0$ such that

$$\|S(t_0)x(t_0) - z(t_0)\| < \varepsilon,$$

and hence, in a neighborhood of t_0 ,

$$\|S(t)^*S(t)x(t) - S(t)^*z(t)\| < \left(\sum_{i=1}^{\infty} i^{-2}\right)^{1/2} \varepsilon.$$

This proves that $t \mapsto S(t)^*z(t)$ is continuous. Let $K(t)$ be the closure of $S(t)^*(H(t))$ in H_0 . What precedes proves that the set Δ in the statement of the lemma satisfies axiom (ii) of 10.1.2; axioms (i), (iii), and (iv) are evident, and this establishes (a). The map $t \mapsto S(t)^*S(t)$ from T into $\mathcal{L}(H_0)$ is strongly continuous, and therefore so is

$$t \mapsto (S(t)^*S(t))^{1/2} = |S(t)|.$$

Let $S(t) = V(t)|S(t)|$ be the polar decomposition of $S(t)$. Since $|S(t)|(H_0)$ and $S(t)^*(H(t))$ are dense in $K(t)$, while $S(t)(H_0)$ is dense in $H(t)$, we see that $V(t)|K(t)$ is an isomorphism of $K(t)$ onto $H(t)$. Thus $U(t) = V(t)^{-1}$ carries a total family in Δ onto a total family in Γ , and (b) follows.

10.8.6. We recall a result of E. Michael (*Ann. of Math.*, vol. 64, 1956, pp. 562–580, Theorem 1.2). Let T be a paracompact space of finite dimension (one says that T has dimension at most n if every finite open cover of T has a finite open refinement of order at most n). Let A be a closed subset of T , let Y be a complete metric space, and let φ be a map of T into the set of nonempty closed subsets of Y . Suppose that φ is lower semicontinuous, that is, whenever $\varphi(t_0)$ meets an open subset Y' of Y , $\varphi(t)$ meets Y' for all t sufficiently near t_0 . Suppose that $\varphi(t)$ is contractible for every $t \in T$. Finally, suppose that there is an $\varepsilon_0 > 0$ such that, for every $t \in T$ and every open ball β in Y of radius at most ε_0 , $\varphi(t) \cap \beta$ is contractible. Then every continuous map $f : A \rightarrow Y$ such that $f(t) \in \varphi(t)$ for every $t \in A$ extends to a continuous map $f' : T \rightarrow Y$ such that $f'(t) \in \varphi(t)$ for every $t \in T$.

10.8.7. Lemma. *Let T be a paracompact space of finite dimension, and let $\mathcal{H} = ((H(t))_{t \in T}, \Gamma)$ be a separable continuous field of Hilbert spaces over T , each $H(t)$ having dimension $\aleph_0$. Then $\mathcal{H}$ is trivial.*

Let H_0 be a Hilbert space of dimension $\aleph_0$. By 10.8.5, we may suppose that every $H(t)$ is a closed vector subspace of H_0 and that Γ is the set of continuous

maps $x : T \rightarrow H_0$ such that $x(t) \in H(t)$ for every t . Let $(y_1, y_2, \dots)$ be a sequence in Γ such that, for every $t \in T$, the $y_i(t)$ are dense in $H(t)$. We shall construct a sequence $(x_1, x_2, \dots)$ in Γ such that, for every $t \in T$, $(x_i(t))$ is an orthonormal basis of $H(t)$. Then, if $U(t)$ is the isomorphism of $H(t)$ onto H_0 carrying $(x_i(t))$ onto a fixed orthonormal basis of H_0 , $(U(t))_{t \in T}$ will be an isomorphism of $\mathcal{H}$ onto the constant field defined by H_0 (10.2.4).

Suppose that elements $x_1, \dots, x_n$ of Γ have been constructed with the following properties: 1° for every $t \in T$, $(x_1(t), x_2(t), \dots, x_n(t))$ is an orthonormal system; 2° there are continuous complex functions $f_{i1}, f_{i2}, \dots, f_{in}$ on T such that

$$\|y_i(t) - f_{i1}(t)x_1(t) - \dots - f_{in}(t)x_n(t)\| \leq \frac{1}{i} \quad (i = 1, 2, \dots, n).$$

We shall construct $x_{n+1} \in \Gamma$ so that $(x_1, \dots, x_{n+1})$ has the same properties with n replaced by $n + 1$. This permits us to construct $(x_1, x_2, \dots)$ inductively. The system $(x_1(t), x_2(t), \dots)$ will indeed be total in $H(t)$, since its linear combinations approximate $y_i(t)$ to within $1/i$ and the $y_i(t)$ are dense in $H(t)$.

Let $K(t)$ be the closed vector subspace of $H(t)$ orthogonal to $x_1(t), \dots, x_n(t)$, and let $\varphi(t)$ be the unit sphere of $K(t)$. Since $\dim K(t) = \aleph_0$, $\varphi(t)$ is contractible (10.8.2). Moreover, the intersection of $\varphi(t)$ with every open ball of radius less than 1 in H_0 is contractible, since it is homeomorphic to an open ball of H_0 by stereographic projection. We show that the map $t \mapsto \varphi(t)$ from T into the set of nonempty closed subsets of H_0 is lower semicontinuous. Let U be an open subset of H_0 meeting $\varphi(t_0)$, and choose $\xi \in \varphi(t_0) \cap U$. There is an $x \in \Gamma$ such that $x(t_0) = \xi$. Define x' by

$$x'(t) = x(t) - \sum_{i=1}^n (x(t) | x_i(t)) x_i(t) \in K(t).$$

We have $x'(t_0) = \xi$, so $x'(t) \neq 0$ for t sufficiently near t_0 . Then

$$t \mapsto x''(t) = \frac{x'(t)}{\|x'(t)\|}$$

is a continuous map of a neighborhood V of t_0 into H_0 , with $x''(t) \in \varphi(t)$ for $t \in V$ and $x''(t_0) = \xi$. For t sufficiently near t_0 , we have $x''(t) \in U$, and hence $\varphi(t) \cap U \neq \emptyset$. We may therefore apply 10.8.6.

Define $z \in \Gamma$ by

$$y_{n+1}(t) = \sum_{i=1}^n (y_{n+1}(t) | x_i(t)) x_i(t) + z(t).$$

Then $z(t) \in K(t)$ for every t . Let T' be the closed set of $t \in T$ such that

$$\|z(t)\| \geq \frac{1}{2(n+1)}.$$

By 10.8.6, the map $t \mapsto z(t)/\|z(t)\|$ from T' into H_0 extends to a continuous map $x_{n+1} : T \rightarrow H_0$ such that $x_{n+1}(t) \in \varphi(t)$ for every $t \in T$. Let $f : T \rightarrow \mathbb{R}$ be the continuous function equal to $\|z(t)\|$ on T' and to $1/[2(n+1)]$ on $T - T'$. We have

$$\begin{aligned} \left\| y_{n+1}(t) - \sum_{i=1}^n (y_{n+1}(t) | x_i(t)) x_i(t) - f(t) x_{n+1}(t) \right\| \\ = \|z(t) - f(t) x_{n+1}(t)\|, \end{aligned}$$

and this is zero on T' and at most

$$\frac{2}{2(n+1)} = \frac{1}{n+1}$$

on $T - T'$.

10.8.8. Theorem. *Let T be a paracompact space of finite dimension, and let $\mathcal{A}$ be a separable continuous field of elementary C^* -algebras over T , of rank $\aleph_0$, satisfying Fell's condition. Then $\mathcal{A}$ is locally trivial.*

Let $t_0 \in T$. There are a closed neighborhood V of t_0 and a continuous vector field p over V , relative to $\mathcal{A}$, such that $p(t)$ is everywhere on V a rank-one projection. Put $\beta(\mathcal{A}|V, p) = (\mathcal{H}, x)$ and $\mathcal{H} = ((H(t))_{t \in V}, \Gamma)$. Then $\mathcal{H}$ is separable (10.7.5), every $H(t)$ has dimension $\aleph_0$, and V is paracompact and finite-dimensional. Thus $\mathcal{H}$ is trivial (10.8.7). But $\mathcal{A}|V$ is isomorphic to $\mathcal{A}(\mathcal{H})$ (10.7.6), and is therefore trivial.

Bibliography: [19].

10.9. Application to C^* -Algebras with Continuous Trace

10.9.1. Let A be a separable continuous-trace C^* -algebra, and let $\mathcal{A}$ be the continuous field of elementary C^* -algebras over $\widehat{A}$ defined by A . It satisfies Fell's condition (10.5.8). Moreover, $\widehat{A}$ is locally compact with a countable base (3.3.4, 3.3.8, and 4.5.3), and is therefore paracompact. We may thus form the element $\delta(\mathcal{A})$ of $H^3(\widehat{A}, \mathbb{Z})$ (10.7.14). It will be denoted by $\delta(A)$.

10.9.2. Let T be a locally compact space and let $\mathcal{H}$ be a continuous field of nonzero Hilbert spaces over T . Then $\mathcal{A}(\mathcal{H})$ satisfies Fell's condition (10.7.7). Let

A be the C^* -algebra defined by $\mathcal{A}(\mathcal{H})$ (10.4.1). Then A has spectrum T (10.4.4) and continuous trace (10.5.2 and 10.5.8). We shall say that A is *defined by* $\mathcal{H}$.

10.9.3. Theorem. *Let A be a separable continuous-trace C^* -algebra. Then A is isomorphic to the C^* -algebra defined by a continuous field of nonzero Hilbert spaces over $\widehat{A}$ if and only if $\delta(A) = 0$.*

Let $\mathcal{H}$ be a continuous field of nonzero Hilbert spaces over $\widehat{A}$ such that A is defined by $\mathcal{A}(\mathcal{H})$. Then $\delta(A) = \delta(\mathcal{A}(\mathcal{H}))$ (10.5.2), and hence $\delta(A) = 0$ (10.7.15).

Conversely, suppose that $\delta(A) = 0$. Let $\mathcal{A}$ be the continuous field of C^* -algebras defined by A . There is a continuous field $\mathcal{H}$ of Hilbert spaces over $\widehat{A}$ such that $\mathcal{A}$ is isomorphic to $\mathcal{A}(\mathcal{H})$ (10.7.15). Thus A is isomorphic to the C^* -algebra defined by $\mathcal{A}(\mathcal{H})$ (10.5.4).

10.9.4. Definition. *Let a be a cardinal. A C^* -algebra A is called homogeneous of degree a if every irreducible representation of A has dimension a .*

10.9.5. Theorem. *Let T be a locally compact space with a countable base and of finite dimension.*

- (i) *Let A be a separable continuous-trace C^* -algebra, homogeneous of degree $\aleph_0$, with spectrum T . The continuous field of elementary C^* -algebras over T defined by A is locally trivial.*
- (ii) *The map $A \mapsto \delta(A)$ defines a one-to-one correspondence between $H^3(T, \mathbb{Z})$ and the separable continuous-trace C^* -algebras homogeneous of degree $\aleph_0$ with spectrum T , up to isomorphism.*

Assertion (i) follows from 10.8.8.

Let A, A' be two separable continuous-trace C^* -algebras, homogeneous of degree $\aleph_0$, with spectrum T . Let $\mathcal{A}, \mathcal{A}'$ be the continuous fields of C^* -algebras over T defined by A, A' . Suppose that $\delta(A) = \delta(A')$, that is, $\delta(\mathcal{A}) = \delta(\mathcal{A}')$. The fields $\mathcal{A}$ and $\mathcal{A}'$ have rank $\aleph_0$ and are locally trivial by (i). Hence $\mathcal{A}$ and $\mathcal{A}'$ are isomorphic (10.8.4), and therefore A and A' are isomorphic.

Let $\delta \in H^3(T, \mathbb{Z})$. There is a locally trivial continuous field $\mathcal{A}$ of elementary C^* -algebras of rank $\aleph_0$ over T such that $\delta(\mathcal{A}) = \delta$ (10.8.4). This field satisfies Fell's condition, and hence defines a continuous-trace C^* -algebra A with spectrum T , homogeneous of degree $\aleph_0$ (10.5.8). Moreover, $\mathcal{A}$ is separable (10.2.7), and hence A is separable (10.4.7). The continuous field of C^* -algebras defined by A is isomorphic to $\mathcal{A}$ (10.5.2), and therefore $\delta(A) = \delta$.

10.9.6. Corollary. *Let T be a locally compact space with a countable base, of finite dimension, such that $H^3(T, \mathbb{Z}) = 0$. Let A_0 be an elementary C^* -algebra of rank $\aleph_0$, and let A be the C^* -algebra of continuous maps $t \mapsto x(t)$ from T into A_0 such that $\|x(t)\|$ tends to zero at infinity. Then every separable continuous-trace C^* -algebra, homogeneous of degree $\aleph_0$ and with spectrum T , is isomorphic to A .*

Indeed, A is separable (10.4.7), has continuous trace (10.5.8), has spectrum T (10.4.4), and is homogeneous of degree $\aleph_0$ (10.4.4). It is now enough to apply 10.9.5(ii).

Bibliography: [19, 29].

10.10. Complements

10.10.1. Let $(A_i)_{i \in I}$ be a family of C^* -algebras. The restricted product of the A_i is the C^* -algebra defined by a continuous field of C^* -algebras over the discrete space I .

10.10.2. Let T be a locally compact space, let B be a C^* -algebra, and let A be the C^* -algebra of continuous maps from T into B that tend to zero at infinity. Then $\widehat{A}$ is canonically identified with $T \times \widehat{B}$. More generally, in the situation of 10.4.3, the topology of $\widehat{A}$ can be described explicitly [29].

10.10.3. Let A be a C^* -algebra. Form the compact space $\mathcal{X}(A)$ and the canonical injection of $\text{Prim}(A)$ into $\mathcal{X}(A)$ (1.9.13). Identify $\text{Prim}(A)$ with a subset of $\mathcal{X}(A)$ [in general the topology induced on $\text{Prim}(A)$ by that of $\mathcal{X}(A)$ is not the Jacobson topology, because the latter is not always Hausdorff]. The closure T of $\text{Prim}(A)$ in $\mathcal{X}(A)$ is called the *regularized spectrum* of A . For every $t \in T$, let $I(t)$ be the closed two-sided ideal of A consisting of those $x \in A$ on which the seminorm t vanishes; put $A(t) = A/I(t)$, and let $x(t)$ be the canonical image of $x \in A$ in $A(t)$. The set Γ of vector fields $t \mapsto x(t)$, $x \in A$, has the following properties: 1° Γ is an involutive algebra of vector fields; 2° the set of $x(t)$, $x \in A$, is all of $A(t)$; 3° for every $x \in A$, the function $t \mapsto \|x(t)\|$ is continuous on T ; 4° the set of these fields is complete for uniform convergence. For $((A(t))_{t \in T}, \Gamma)$ to be a continuous field of C^* -algebras, it is necessary and sufficient that $\text{Prim}(A)$ be Hausdorff for the Jacobson topology [29].

10.10.4. Let $T = \{0, 1, 2, 3, \dots, \infty\}$ with its usual topology. For $t \in T$, define $A(t)$ as follows: if $t = \infty$, let $A(t) = \mathbb{C}$; if $t \neq \infty$, let $A(t)$ be the C^* -algebra of two-

rowed and two-columned complex matrices. Let Θ be the set of vector fields $t \mapsto x(t) \in A(t)$ having the following properties [write $x(t) = (x_{ij}(t))_{i,j=1,2}$ for $t \neq \infty$]:

$$\begin{aligned} \lim_{n \rightarrow \infty} x_{12}(n) &= \lim_{n \rightarrow \infty} x_{21}(n) = \lim_{n \rightarrow \infty} x_{22}(2n) = 0, \\ \lim_{n \rightarrow \infty} x_{11}(n) &= \lim_{n \rightarrow \infty} x_{22}(2n+1) = x(\infty). \end{aligned}$$

Then $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ is a continuous field of C^* -algebras. Let $A = \Theta$ be the C^* -algebra defined by $\mathcal{A}$. We have $\widehat{A} = T$. The ideal $\mathfrak{m}(A)$ (4.5.2) is the set of $x \in A$ such that $x(\infty) = 0$. The algebra A is liminal and has generalized continuous trace (4.7.12), but it does not have continuous trace; $\mathcal{A}$ does not satisfy Fell's condition at ∞ [27].

10.10.5. Let $H = L^2_{\mathbb{C}}([0, 1])$. For every finite sequence $s = (s_1, \dots, s_n)$, where $n = 1, 2, \dots$ and each s_i is 0 or 1, put

$$k_s = \sum_{i=1}^n s_i / 2^i, \quad I_s = [k_s, k_s + 2^{-n}].$$

If s and t are two such sequences of n terms, denote by $Q_{s,t}$ the operator on H that carries $f \in H$ into $g \in H$, where

$$g(x) = 0 \quad \text{for } x \notin I_s, \quad g(k_s + u) = f(k_t + u) \quad \text{if } 0 \leq u \leq 2^{-n}.$$

Let $A(n)$ be the C^* -algebra of operators generated by the $Q_{s,t}$, with s, t ranging over the sequences of n terms. Then $A(n)$ is isomorphic to the C^* -algebra of complex matrices with 2^n rows and 2^n columns. We have $A(n) \subset A(n+1)$. Let A be the C^* -subalgebra of $\mathcal{L}(H)$ generated by the $A(n)$. Let $A(\infty)$ be the commutative C^* -algebra of multiplication operators by continuous complex functions on $[0, 1]$. We have $A(\infty) \subset A$.

Let $T = \{1, 2, \dots, \infty\}$ with its usual topology. The $A(t)$, $t \in T$, have now been defined. Let Θ be the set of continuous maps $x : T \rightarrow A$ such that $x(t) \in A(t)$ for every $t \in T$. Then $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ is a continuous field of C^* -algebras. Let $A = \Theta$ be the C^* -algebra defined by $\mathcal{A}$. The elements of $\widehat{A}$ are the representations

$$\pi_n : x \mapsto x(n) \quad (n < \infty)$$

and the representations

$$\pi_\lambda : x \mapsto \lambda(x(\infty)), \quad \lambda \in \widehat{A(\infty)}.$$

The C^* -algebra A is liminal and has generalized continuous trace. The sequence $(\pi_n)_{n=1,2,\dots}$ converges simultaneously to all the π_λ (Fell, unpublished).

10.10.6. Let A be a C^* -algebra.

- a. The space $\text{Prim}(A)$ is discrete if and only if A is the restricted product of a family of C^* -algebras having no nontrivial closed two-sided ideals.
- b. The algebra A is dual if and only if it is liminal and has discrete spectrum [74].

10.10.7. Let A be a liminal (respectively, postliminal) C^* -algebra, let T be compact, and let B be the C^* -algebra of continuous maps from T into A . Then B is liminal (respectively, postliminal). [If A is liminal, apply 10.4.3. If A is postliminal, let (I_ρ) be a composition series of A such that the $I_{\rho+1}/I_\rho$ are liminal. Let J_ρ be the set of continuous maps from T into I_ρ . Then the J_ρ form a composition series of B , and $J_{\rho+1}/J_\rho$ is isomorphic to the C^* -algebra of continuous maps from T into $I_{\rho+1}/I_\rho$.] [77].

10.10.8. Let A be a liminal C^* -algebra, let U be an open dense subset of $\widehat{A}$, and let I be the closed two-sided ideal of A such that $\widehat{I} = U$. Let $\mathcal{A} = ((A(t))_{t \in U}, \Theta)$ be the continuous field of elementary C^* -algebras over U defined by I . For every $x \in A$ and $t \in U$, let $x(t)$ be the canonical image of x in $A(t)$ [the image of A under the representation t is the same as that of I , namely $\mathcal{LC}(H_t)$]. For every $x \in A$, let x' be the field $t \mapsto x(t)$ over U . Then $x \mapsto x'$ is injective. In general the field x' is not an element of Θ , but, for every $y \in I$, the fields $x'y$ and yx' belong to Θ . Suppose that A has generalized continuous trace and that $I = J(A)$ (4.7.12). Then $x' \in \Theta$ for every $x \in A$. [For every $y \in I$, the function $t \mapsto \|x(t) - y(t)\|$ is continuous on U , because the points of U are separated in $\widehat{A}$ (3.9.4).]

10.10.9. Let S be the two-dimensional sphere and let $T = S \times S \times S \times \dots$ be the compact product space. There is over T a continuous field $\mathcal{H} = ((H(t))_{t \in T}, \Gamma)$ of Hilbert spaces of dimension $\aleph_0$ which is not locally trivial at any point of T . Let A be the continuous-trace C^* -algebra defined by $\mathcal{A}(\mathcal{H})$. Then A has no nonzero closed two-sided ideal I isomorphic to the C^* -algebra of continuous maps $X \rightarrow A_0$ tending to zero at infinity, where X is a locally compact space and A_0 an elementary C^* -algebra [19].

10.10.10. Let T be a topological space, and let

$$\mathcal{A}_1 = ((A_1(t))_{t \in T}, \Theta_1), \quad \mathcal{A}_2 = ((A_2(t))_{t \in T}, \Theta_2)$$

be two continuous fields of elementary C^* -algebras over T satisfying Fell's condition. For every $t \in T$, let $A(t) = A_1(t) \otimes A_2(t)$ (2.12.15). Let Γ be the set of vector fields of the form

$$t \longmapsto a_1(t) \otimes a_2(t) + b_1(t) \otimes b_2(t) + \cdots + k_1(t) \otimes k_2(t),$$

where $a_1, b_1, \dots, k_1 \in \Theta_1$ and $a_2, b_2, \dots, k_2 \in \Theta_2$. There is a unique continuous field of elementary C^* -algebras $\mathcal{A} = ((A(t))_{t \in T}, \Theta)$ over T such that $\Theta \supset \Gamma$. We write $\mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$. This field satisfies Fell's condition. If T is paracompact, then

$$\delta(\mathcal{A}_1 \otimes \mathcal{A}_2) = \delta(\mathcal{A}_1) + \delta(\mathcal{A}_2).$$

If $\mathcal{A}_1$ is separable and $\mathcal{A}_2$ is locally trivial of infinite rank, then $\mathcal{A}_1 \otimes \mathcal{A}_2$ is locally trivial [18].

10.10.11. Problem. Let $((A(t))_{t \in T}, \Theta)$ be a continuous field of postliminal C^* -algebras over T . For every $t \in T$, let $B(t)$ be the largest liminal ideal of $A(t)$. Let Θ' be the set of $x \in \Theta$ such that $x(t) \in B(t)$ for every t . Is $((B(t))_{t \in T}, \Theta')$ a continuous field of C^* -algebras?

10.10.12. Let A be a C^* -algebra, let $x \in A$, and let f be a bounded continuous complex function on $\widehat{A}$. There is a $y \in A$ such that

$$\pi(y) = f(\pi)\pi(x) \quad \text{for every } \pi \in \widehat{A}$$

[346]. Cf. also [313].

§ 11. — Extension to C^* -Algebras of the Stone–Weierstrass Theorem

Let A be the C^* -algebra of continuous complex functions tending to zero at infinity on a locally compact space. The Stone–Weierstrass theorem gives conditions under which a C^* -subalgebra B of A is equal to A . But the C^* -algebras of the form A are, up to isomorphism, precisely the commutative C^* -algebras. In this section we shall extend the Stone–Weierstrass theorem to arbitrary C^* -algebras.

11.1. The Case of Postliminal C^* -Algebras

11.1.1. Definition. Let A be a C^* -algebra and B a C^* -subalgebra of A . We say that B is a rich C^* -subalgebra of A if the following conditions hold:

- (i) for every irreducible representation π of A , the restriction $\pi|_B$ is irreducible;
- (ii) if π and π' are inequivalent irreducible representations of A , then $\pi|_B$ and $\pi'|_B$ are inequivalent.

It follows that, if π is a nonzero irreducible representation of A , then $\pi|_B$ is nonzero.

11.1.2. Lemma. Let A be a C^* -algebra, let B be a rich C^* -subalgebra of A , and let π be an irreducible representation of B . There is an irreducible representation π' of A on H_π such that $\pi = \pi'|_B$.

There are a Hilbert space $H' \supset H_\pi$ and an irreducible representation π' of A on H' such that $\pi(x) = \pi'(x)|_{H_\pi}$ for every $x \in B$ (2.10.2). Since B is rich, $\pi'|_B$ is irreducible; and H_π is invariant under $\pi'(B)$, so $H_\pi = H'$.

11.1.3. Lemma. Let A be a C^* -algebra, let B be a rich C^* -subalgebra of A , and let I be a closed two-sided ideal of A .

- (i) $B/(B \cap I)$ is a rich C^* -subalgebra of A/I ;
- (ii) $B \cap I$ is a rich C^* -subalgebra of I .

Assertion (i) follows at once from the definitions. We prove (ii). Let π be an irreducible representation of I and let us prove that $\pi|_{B \cap I}$ is irreducible. We may suppose that $\pi \neq 0$. There is an irreducible representation ρ of A on H_π extending π (2.10.4). Since B is rich, $\rho|_B$ is irreducible. As $B \cap I$ is a closed two-sided ideal of B , $\pi|_{B \cap I}$ is zero or irreducible (2.11.2). If $\pi|_{B \cap I} = 0$, then $\rho|_B$ defines an irreducible representation of $B/(B \cap I)$ which, by (i) and 11.1.2, extends to an irreducible representation of A/I . In other words, $\rho|_B$ extends to

an irreducible representation ρ' of A that vanishes on I . Since B is rich, ρ' is equivalent to ρ , and hence $\pi = \rho|I = 0$, contrary to the hypothesis. Thus $\pi|B \cap I$ is irreducible.

Now let π, π' be two irreducible representations of I and suppose that $\pi|B \cap I \simeq \pi'|B \cap I$. Let ρ, ρ' be representations of A on $H_\pi, H_{\pi'}$ extending π, π' (2.10.4). If $\pi|B \cap I$ and $\pi'|B \cap I$ are zero, the preceding argument shows that π and π' are zero. Otherwise, $\rho|B \simeq \rho'|B$ (2.11.2), so $\rho \simeq \rho'$ because B is rich, and hence $\pi \simeq \pi'$.

11.1.4. Lemma. *Let A be a liminal C^* -algebra with Hausdorff spectrum, and let B be a rich C^* -subalgebra of A . Then $B = A$.*

Let $\mathcal{A}$ be the continuous field of elementary C^* -algebras over $\widehat{A}$ defined by A . Then A is identified with the C^* -algebra defined by $\mathcal{A}$ (10.5.4). If $\pi \in \widehat{A}$, then $\pi|B \in \widehat{B}$, and hence

$$\pi(A) = \pi(B) = \mathcal{LC}(H_\pi).$$

If π and π' are two inequivalent irreducible representations of A , then $\pi|B$ and $\pi'|B$ are inequivalent. Thus, if $T \in \mathcal{LC}(H_\pi)$ and $T' \in \mathcal{LC}(H_{\pi'})$, there is an $x \in B$ such that $\pi(x) = T$ and $\pi'(x) = T'$ (4.2.5). Hence $B = A$ by 10.5.3.

11.1.5. Lemma. *Let A be a C^* -algebra, let I be its largest postliminal ideal, and let B be a rich C^* -subalgebra of A . Then $B \supset I$.*

There is a composition series $(I_\rho)_{\rho \leq \alpha}$ of I such that the $I_{\rho+1}/I_\rho$ are continuous-trace C^* -algebras (4.5.5), and hence liminal C^* -algebras with Hausdorff spectrum (4.5.3). We have $I_0 = 0 \subset B$. Suppose that there is a least ordinal ρ for which $I_\rho \not\subset B$. Then $I_{\rho'} \subset B$ for $\rho' < \rho$, so ρ is not a limit ordinal. Let $\rho = \rho' + 1$. By 11.1.3(i), $B/I_{\rho'}$ is a rich C^* -subalgebra of $A/I_{\rho'}$. By 11.1.3(ii), $(B \cap I_\rho)/I_{\rho'}$ is a rich C^* -subalgebra of $I_\rho/I_{\rho'}$. By 11.1.4,

$$(B \cap I_\rho)/I_{\rho'} = I_\rho/I_{\rho'},$$

whence $B \supset I_\rho$, a contradiction. Therefore $I = I_\alpha \subset B$.

11.1.6. Proposition. *In a postliminal C^* -algebra A , every rich C^* -subalgebra is equal to A .*

This is a special case of 11.1.5.

11.1.7. Let E, F be two vector spaces in duality, with $A \subset E$ and $B \subset F$. We say that A separates B if, for every pair (x, y) of distinct points of B , there is an $a \in A$ such that $\langle a, x \rangle \neq \langle a, y \rangle$.

Lemma. *Let A be a C^* -algebra and let B be a C^* -subalgebra that separates $P(A) \cup \{0\}$. Then B is a rich C^* -subalgebra of A .*

Let $\pi \in \widehat{A}$. Suppose that H_π is the Hilbert direct sum of two nonzero subspaces H_1, H_2 invariant under $\pi|B$. Choose nonzero $\xi_1 \in H_1, \xi_2 \in H_2$ such that $\|\xi_1 + \xi_2\| = 1$. Then

$$x \longmapsto (\pi(x)(\xi_1 + \xi_2) \mid \xi_1 + \xi_2), \quad x \longmapsto (\pi(x)(\xi_1 - \xi_2) \mid \xi_1 - \xi_2)$$

are pure states f_1, f_2 of A . If $x \in B$, then

$$\begin{aligned} f_1(x) &= (\pi(x)\xi_1 \mid \xi_1) + (\pi(x)\xi_2 \mid \xi_2) \\ &= (\pi(x)\xi_1 \mid \xi_1) + (\pi(x)(-\xi_2) \mid -\xi_2) = f_2(x). \end{aligned}$$

Since B separates $P(A)$, we have $f_1 = f_2$. Thus $\xi_1 + \xi_2$ is proportional to $\xi_1 - \xi_2$ (2.5.7), so ξ_1 and ξ_2 are proportional, which is absurd. Therefore $\pi|B$ is irreducible.

Let π_1, π_2 be two irreducible representations of A , and suppose there is an isomorphism $U : H_{\pi_1} \rightarrow H_{\pi_2}$ carrying $\pi_1|B$ into $\pi_2|B$. Let ξ_1 be a unit vector of H_{π_1} , let $\xi_2 = U(\xi_1)$, and let $f_i \in P(A) \cup \{0\}$ be the functional defined by π_i and ξ_i . We have $f_1(x) = f_2(x)$ for $x \in B$, and hence $f_1 = f_2$. Thus either π_1 and π_2 are both zero, or they are both nonzero. In the latter case ξ_i is cyclic for π_i , and $f_1 = f_2$ implies $\pi_1 \simeq \pi_2$ (2.4.1).

11.1.8. Theorem. *Let A be a postliminal C^* -algebra and B a C^* -subalgebra of A . If B separates $P(A) \cup \{0\}$, then $B = A$.*

This follows from 11.1.6 and 11.1.7.

It is not known whether the hypothesis that A be postliminal is essential. For arbitrary C^* -algebras we shall obtain later (11.5.2) a less precise result than 11.1.8.

Bibliography: [45, 77].

This subsection follows an unpublished account by Fell.

11.2. Abundance of Pure States in Certain C^* -Algebras

11.2.1. Lemma. *Let H be a Hilbert space, let $C = \mathcal{LC}(H)$, let A be a C^* -subalgebra of $\mathcal{L}(H)$ containing 1, and let f be a state of A that vanishes on $A \cap C$.*

- (i) *f is a weak limit of vector states of A (2.4.2).*
- (ii) *If A acts irreducibly on H , then f is a weak limit of pure states of A .*

Part (ii) follows at once from (i); we prove (i).

Suppose first that $A \supset C$. Then f defines a state f' of A/C , itself a weak limit of convex combinations of pure states of A/C (2.5.6). Hence f is a weak limit of states of the form

$$\lambda_1 f_1 + \cdots + \lambda_n f_n,$$

where $\lambda_1, \dots, \lambda_n \geq 0$, $\lambda_1 + \cdots + \lambda_n = 1$, and $f_1, \dots, f_n$ are pure states of A vanishing on C . It is therefore enough to consider the case in which f itself has this form. Let $x_1, \dots, x_s$ be Hermitian elements of A with $x_1 = 1$. We shall construct unit vectors $\xi_1, \dots, \xi_n$ in H such that $(x_i \xi_j \mid \xi_k) = 0$ for $j < k$ and

$$|f_j(x_i) - \omega_{\xi_j}(x_i)| \leq 1 \quad \text{for all } i, j.$$

Suppose that the ξ_j have been constructed for $j < m$. Let K be the vector subspace of H spanned by the $x_i \xi_j$, with $1 \leq i \leq s$ and $j < m$, and let $K' = H \ominus K$. We have $P_K \in C \subset A$, and hence $P_{K'} A P_{K'}$ is a subalgebra of A . Moreover, $f_m|_{P_{K'} A P_{K'}}$ is a state of $P_{K'} A P_{K'}$, since

$$f_m(P_{K'}) = f_m(P_K) + f_m(P_{K'}) = f_m(1) = 1.$$

If

$$f_m|_{P_{K'} A P_{K'}} = \lambda g_1 + (1 - \lambda) g_2, \quad 0 < \lambda < 1,$$

where g_1, g_2 are states of $P_{K'} A P_{K'}$, let $\tilde{g}_i$ be the state $x \mapsto g_i(P_{K'} x P_{K'})$ of A . Since f_m vanishes on C , we have $f_m = \lambda \tilde{g}_1 + (1 - \lambda) \tilde{g}_2$; therefore $\tilde{g}_1 = \tilde{g}_2$ because f_m is pure, and hence $g_1 = g_2$. Thus $f_m|_{P_{K'} A P_{K'}}$ is a pure state of $P_{K'} A P_{K'}$. By 3.4.1, it is a weak limit of forms ω_{ξ_α} , where the ξ_α are unit vectors in K' . Thus, for $1 \leq i \leq s$,

$$f_m(x_i) = f_m(P_{K'} x_i P_{K'})$$

is a limit of

$$(P_{K'} x_i P_{K'} \xi_\alpha \mid \xi_\alpha) = (x_i \xi_\alpha \mid \xi_\alpha).$$

We can therefore find $\xi_m \in K'$ such that

$$|f_m(x_i) - \omega_{\xi_m}(x_i)| \leq 1 \quad (1 \leq i \leq s).$$

The construction of the ξ_j can thus be continued inductively. Put

$$\xi = \sqrt{\lambda_1} \xi_1 + \cdots + \sqrt{\lambda_n} \xi_n.$$

Since $x_1 = 1$, we have $(\xi_j \mid \xi_k) = 0$ for $j \neq k$, so ξ is a unit vector. Since x_i is Hermitian, we have $(x_i \xi_j \mid \xi_k) = 0$ for $j \neq k$, and hence

$$\begin{aligned} \left| \sum_{j=1}^n \lambda_j f_j(x_i) - \omega_\xi(x_i) \right| &= \left| \sum_{j=1}^n \lambda_j f_j(x_i) - \sum_{j,k=1}^n (x_i \sqrt{\lambda_j} \xi_j \mid \sqrt{\lambda_k} \xi_k) \right| \\ &= \left| \sum_{j=1}^n \lambda_j f_j(x_i) - \sum_{j=1}^n \lambda_j (x_i \xi_j \mid \xi_j) \right| \leq \sum_{j=1}^n \lambda_j = 1. \end{aligned}$$

This proves that $\sum_{j=1}^n \lambda_j f_j$ is a weak limit of vector states of A .

Now consider the general case. Let A_1 be the C^* -algebra $A + C$. Then A_1/C is identified with $A/(A \cap C)$, and hence f defines a state of A_1/C ; in other words, f extends to a state f' of A_1 that vanishes on C . By what precedes, f' is a weak limit of vector states of A_1 , and consequently f is a weak limit of vector states of A .

11.2.2. Let A be a C^* -algebra. For every representation θ of A , temporarily denote by C_θ the set of $x \in A$ such that $\theta(x)$ is compact. We have

$$C_\theta = \theta^{-1}(\mathcal{LC}(H_\theta)),$$

so C_θ is a closed two-sided ideal of A .

Lemma. *Let A be an antiliminal C^* -algebra with identity, and let $P'(A)$ be the set of states that vanish on at least one C_θ , $\theta \in \widehat{A}$. Then $P(A)$ and $P'(A)$ have the same weak closure.*

Let x be a Hermitian element of A such that $f(x) \geq 0$ for every $f \in P'(A)$. Let $\theta \in \widehat{A}$, and let π_θ be a representation of A with kernel C_θ . Every state associated with π_θ belongs to $P'(A)$ and is therefore nonnegative on x ; hence $\pi_\theta(x) \geq 0$. But

$$\bigcap_{\theta \in \widehat{A}} C_\theta = 0$$

because A is antiliminal (4.2.6). Thus $\bigoplus_{\theta \in \widehat{A}} \pi_\theta$ is isometric, and consequently $x \geq 0$. By 3.4.1, the weak closure of $P'(A)$ contains $P(A)$.

If $g \in P'(A)$, there is a $\theta \in \widehat{A}$ such that $g(C_\theta) = 0$. Thus g defines a state g' of $\theta(A)$ that vanishes on $\theta(A) \cap \mathcal{LC}(H_\theta)$. By 11.2.1, g' is a weak limit of pure states of $\theta(A)$, and hence g is a weak limit of pure states of A . Thus the weak closure of $P(A)$ contains $P'(A)$.

11.2.3. Let A be a C^* -algebra. Throughout the remainder of this subsection, $\overline{P(A)}$ will denote the weak closure of $P(A)$ in the dual of A , and $E(A)$ will denote the set of states of A .

Lemma. *Let A be an antiliminal C^* -algebra with identity, and let $f \in \overline{P(A)}$. Every state of A that vanishes on $\ker \pi_f$ belongs to $\overline{P(A)}$.*

By 3.4.3, it is enough to prove that the state

$$x \longmapsto \lambda_1 f(y_1^* x y_1) + \cdots + \lambda_n f(y_n^* x y_n),$$

where

$$\lambda_1, \dots, \lambda_n \geq 0, \quad y_1, \dots, y_n \in A, \quad \sum_i \lambda_i f(y_i^* y_i) = 1,$$

belongs to $\overline{P(A)}$. But f is a weak limit of states $f_\alpha \in P'(A)$ (11.2.2), and the state

$$x \longmapsto \lambda_1 f_\alpha(y_1^* x y_1) + \cdots + \lambda_n f_\alpha(y_n^* x y_n)$$

plainly belongs to $P'(A)$, and hence to $\overline{P(A)}$ (11.2.2).

11.2.4. Lemma. *Let A be an antiliminal C^* -algebra with identity. Suppose that any two nonzero closed two-sided ideals of A have nonzero intersection. Then $E(A) = \overline{P(A)}$.*

The condition on the ideals of A means that any two nonempty open subsets of $\widehat{A}$ have nonempty intersection (3.2.2). Let R be the equivalence relation on $P(A)$ defined by the canonical map $P(A) \rightarrow \widehat{A}$. If $f \in P(A)$, the elements of $P(A)$ congruent to f modulo R are the states $x \mapsto f(u^* x u)$, where u belongs to the unitary group G of A (2.8.6). Since $\widehat{A}$ is the quotient space of $P(A)$ by R , any two nonempty open subsets of $P(A)$ saturated for R have nonempty intersection. Let now $f_1, f_2 \in P(A)$ and $\lambda \in [0, 1]$. It follows from what precedes that $\lambda f_1 + (1 - \lambda) f_2$ is a weak limit of states

$$x \longmapsto \lambda f(x) + (1 - \lambda) f(u^* x u), \tag{1}$$

where $f \in P(A)$ and $u \in G$. But the state (1) vanishes on $\ker \pi_f$, and hence belongs to $\overline{P(A)}$ (11.2.3). Thus $\lambda f_1 + (1 - \lambda) f_2 \in \overline{P(A)}$. By induction, every convex combination of pure states of A belongs to $\overline{P(A)}$. Therefore $E(A) \subset \overline{P(A)}$ (2.5.6).

Bibliography: [45].

11.3. Statement of the Theorem

11.3.1. Theorem. *Let A be a C^* -algebra with identity and let B be a C^* -subalgebra of A containing 1. If B separates $\overline{P(A)}$, then $B = A$.*

The proof will be given in 11.5.1.

11.3.2. Lemma. *Let A and B be as in 11.3.1. If B separates $E(A)$, then $B = A$.*

Let A_h (respectively, B_h) be the set of Hermitian elements of A (respectively, B). Suppose that $A \neq B$, and hence $A_h \neq B_h$. By the Hahn–Banach theorem there is a nonzero continuous Hermitian linear functional f on A such that $f(B) = 0$. Let $f = g - g'$, where g, g' are positive functionals on A (2.6.4). We have

$$0 = f(1) = g(1) - g'(1), \quad \text{so} \quad g(1) = g'(1) \neq 0.$$

Multiplying f, g, g' by the same scalar, we may suppose that $g(1) = g'(1) = 1$, and hence $g, g' \in E(A)$. But then B does not separate g and g' , although $g \neq g'$.

11.3.3. By 11.2.4 and 11.3.2, Theorem 11.3.1 is proved when A is an antiliminal C^* -algebra in which the intersection of any two nonzero closed two-sided ideals is nonzero. We shall reduce the general case to this special case.

Bibliography: [45].

11.4. Some Lemmas

11.4.1. Lemma. *Let A be a C^* -algebra with identity, and let B be a C^* -subalgebra of A containing 1 and separating $\overline{P(A)}$. Let K_1, K_2 be two closed two-sided ideals of A , and put*

$$K = B \cap (K_1 + K_2), \quad K' = (B \cap K_1) + (B \cap K_2).$$

Then $K = K'$.

It is clear that $K \supset K'$. Let f be a pure state of B such that $f(K') = 0$. We shall show that $f(K) = 0$, which will complete the proof (2.7.3).

There is an $f' \in P(A)$ extending f [2.10.1(ii)]; since B separates $P(A)$, this extension is unique. By 2.10.1(iii), for every Hermitian element x of A ,

$$f'(x) = \sup_{\substack{y=y^* \in B \\ y \leq x}} f(y). \quad (1)$$

Let $x \in K_1^+$. If $y = y^* \in B$ and $y \leq x$, and if φ denotes the canonical morphism of A onto A/K_1 , then $\varphi(y) \leq \varphi(x) = 0$. Hence $\varphi(y^+) = \varphi(y)^+ = 0$, so $y^+ \in K_1$; therefore $f(y^+) = 0$ and $f(y) \leq 0$. Equation (1) now shows that $f'(x) = 0$. Thus f' vanishes on K_1 , and similarly on K_2 ; hence $f(K) = 0$.

11.4.2. Lemma. *Let A and B be as in 11.4.1, and let π_1, π_2 be representations of A such that $\pi_1(A) = \pi_1(B)$ and $\pi_2(A) = \pi_2(B)$. Then*

$$(\pi_1 \oplus \pi_2)(A) = (\pi_1 \oplus \pi_2)(B).$$

Put $K_i = \ker \pi_i$. We have $A = B + K_1 = B + K_2$, whence first

$$K_1 \subset (B \cap (K_1 + K_2)) + K_2.$$

In view of 11.4.1,

$$K_1 \subset (B \cap K_1) + (B \cap K_2) + K_2 = (B \cap K_1) + K_2.$$

It follows that

$$K_1 \subset (B \cap K_1) + (K_1 \cap K_2),$$

and hence

$$A = B + K_1 = B + (K_1 \cap K_2),$$

which proves the lemma.

11.4.3. Lemma. *Let A be an antiliminal C^* -algebra with identity, and let B be a C^* -subalgebra of A containing 1 and separating $\overline{P(A)}$. If $f_1, f_2 \in \overline{P(A)}$, then*

$$(\pi_{f_1} \oplus \pi_{f_2})(A) = (\pi_{f_1} \oplus \pi_{f_2})(B).$$

If $f \in \overline{P(A)}$, every state of A that vanishes on $\ker \pi_f$ belongs to $\overline{P(A)}$ (11.2.3). Thus $\pi_f(B)$ separates the states of $\pi_f(A)$, and hence $\pi_f(B) = \pi_f(A)$ (11.3.2). It is now enough to apply 11.4.2.

11.4.4. Lemma. *Let H be a Hilbert space, let $T \in \mathcal{L}(H)$ satisfy $0 \leq T \leq 1$, let $\varepsilon \in [0, 1]$, and let ξ be a unit vector of H such that $(T\xi | \xi) \geq 1 - \varepsilon$.*

(i) *If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a nonnegative Borel function such that $f \geq 1$ on $[1 - \varepsilon^{1/2}, 1]$, then*

$$(f(T)\xi | \xi) \geq 1 - \varepsilon^{1/2}.$$

(ii) *If $T' \in \mathcal{L}(H)$, then*

$$|(T'\xi | \xi) - (T'T\xi | \xi)| \leq 2\varepsilon^{1/2}\|T'\|.$$

(iii) We have

$$T + 3\varepsilon^{1/2} \geq P_{\mathbb{C}\xi}.$$

(iv) If S is a Hermitian operator on H such that $S \geq T$, and if $g : \mathbb{R} \rightarrow \mathbb{R}$ is a nonnegative Borel function such that $g \geq 1$ on $[\varepsilon, +\infty)$, then

$$(g(S)\xi \mid \xi) \geq 1 - 4\varepsilon^{1/2}.$$

Let

$$T = \int_0^1 \lambda dE_\lambda$$

be the spectral decomposition of T , and put $1 - \varepsilon^{1/2} = \alpha$. We have

$$\begin{aligned} 1 - \varepsilon^{1/2}(E_\alpha \xi \mid \xi) &= (1 - \varepsilon^{1/2})(E_\alpha \xi \mid \xi) + ((1 - E_\alpha)\xi \mid \xi) \\ &\geq \int_0^\alpha \lambda d(E_\lambda \xi \mid \xi) + \int_\alpha^1 \lambda d(E_\lambda \xi \mid \xi) \\ &= (T\xi \mid \xi) \geq 1 - \varepsilon, \end{aligned}$$

whence $(E_\alpha \xi \mid \xi) \leq \varepsilon^{1/2}$, and

$$(f(T)\xi \mid \xi) \geq ((1 - E_\alpha)\xi \mid \xi) \geq 1 - \varepsilon^{1/2}.$$

This proves (i).

If $T' \in \mathcal{L}(H)$, then, in view of 9.3.3,

$$\begin{aligned} |(T'\xi \mid \xi) - (T'T\xi \mid \xi)| &= |(T'(\xi - T\xi) \mid \xi)| \\ &\leq \|T'\| \|\xi - T\xi\| \leq (2\varepsilon)^{1/2} \|T'\|, \end{aligned}$$

which proves (ii).

Let $\zeta \in H$ be a unit vector. Write $\zeta = \lambda\xi_1 + \mu\xi$, where $\lambda, \mu \in \mathbb{C}$ and ξ_1 is a unit vector of H orthogonal to ξ . Let P be the orthogonal projection onto $\mathbb{C}\xi_1 + \mathbb{C}\xi$, and let

$$\begin{pmatrix} p & q \\ \bar{q} & r \end{pmatrix}$$

be the matrix of PTP relative to the basis (ξ_1, ξ) . We have

$$r = (PTP\xi \mid \xi) = (T\xi \mid \xi) \in [1 - \varepsilon, 1].$$

Since $0 \leq PTP \leq P$, we have $p \in [0, 1]$ and

$$|q|^2 \leq (1 - p)(1 - r) \leq \varepsilon.$$

Therefore

$$\begin{aligned}
 ((T - P_{\mathbb{C}\xi})\zeta \mid \zeta) &= (p\lambda + q\mu)\bar{\lambda} + (\bar{q}\lambda + (r-1)\mu)\bar{\mu} \\
 &\geq p|\lambda|^2 + (r-1)|\mu|^2 - 2|q\lambda\mu| \\
 &\geq 0 - \varepsilon - 2\varepsilon^{1/2} \geq -3\varepsilon^{1/2},
 \end{aligned}$$

which proves (iii).

Let

$$S = \int_0^{+\infty} \lambda dF_\lambda$$

be the spectral decomposition of S . Write $\xi = \rho\xi_1 + \sigma\xi_2$, where $\rho, \sigma \in \mathbb{C}$ and ξ_1, ξ_2 are unit vectors of $(1 - F_\varepsilon)(H)$ and $F_\varepsilon(H)$, respectively. Let Q be the orthogonal projection onto $\mathbb{C}\xi_1 + \mathbb{C}\xi_2$. We have

$$QSQ = a_1 P_{\mathbb{C}\xi_1} + a_2 P_{\mathbb{C}\xi_2}, \quad a_1 \geq \varepsilon \geq a_2 \geq 0.$$

Consequently, by (iii),

$$\begin{aligned}
 4\varepsilon^{1/2} &\geq a_2 + 3\varepsilon^{1/2} = ((QSQ + 3\varepsilon^{1/2})\xi_2 \mid \xi_2) \\
 &\geq ((QTQ + 3\varepsilon^{1/2})\xi_2 \mid \xi_2) \geq (P_{\mathbb{C}\xi}\xi_2 \mid \xi_2) = |(\xi \mid \xi_2)|^2 = |\sigma|^2.
 \end{aligned}$$

It follows that

$$(g(S)\xi \mid \xi) \geq ((1 - F_\varepsilon)\xi \mid \xi) = |\rho|^2 = 1 - |\sigma|^2 \geq 1 - 4\varepsilon^{1/2}.$$

11.4.5. Lemma. *Let A and B be as in 11.4.3, let K be a closed two-sided ideal of A , let $f \in \overline{P(A)}$, and let $\varepsilon \in [0, 1/4)$. Suppose that there is an $x \in K$ such that $0 \leq x \leq 1$ and $f(x) \geq 1 - \varepsilon$. Then there is a $y \in B \cap K$ such that $0 \leq y \leq 1$ and $f(y) \geq 1 - \varepsilon^{1/2}$.*

By 11.4.3, there exists $x' \in B$ such that $\pi_f(x) = \pi_f(x')$; replacing x' by $\frac{1}{2}(x' + x'^*)$, we may suppose that x' is Hermitian.

If $x' - x \leq \frac{1}{2}$, let $p: \mathbb{R} \rightarrow \mathbb{R}$ be the function equal to 0 on $(-\infty, \frac{1}{2}]$, equal to 1 on $[1 - \varepsilon^{1/2}, +\infty)$, and linear on $[\frac{1}{2}, 1 - \varepsilon^{1/2}]$; put $y = p(x') \in B$. We have $0 \leq y \leq 1$. On the other hand, if ω is the canonical homomorphism of A onto A/K , then

$$\omega(x') = \omega(x' - x) \leq \frac{1}{2}, \quad \omega(y) = p(\omega(x')) = 0,$$

so that $y \in K$. Finally,

$$\pi_f(y) = p(\pi_f(x')) = p(\pi_f(x)),$$

and hence

$$f(y) = (\pi_f(y)\xi_f \mid \xi_f) \geq 1 - \varepsilon^{1/2}$$

by 11.4.4(i).

If nothing is known about $x' - x$, we shall modify x' so as to reduce to the preceding case. If $g \in \overline{P(A)}$, there exists $t_g \in B$ such that

$$\pi_f(t_g) = \pi_f((x' - x)^+) = 0, \quad \pi_g(t_g) = \pi_g((x' - x)^+) \quad (11.4.3);$$

replacing t_g by $\frac{1}{2}(t_g + t_g^*)^+$, we may suppose that $t_g \geq 0$. Let W_g be the set of $h \in \overline{P(A)}$ such that

$$h(t_g) > h(x' - x) - \frac{1}{2};$$

this is a neighborhood of g in $\overline{P(A)}$, since

$$g(t_g) = g((x' - x)^+) \geq g(x' - x) > g(x' - x) - \frac{1}{2}.$$

Let $(W_{g_1}, \dots, W_{g_n})$ be a finite cover of $\overline{P(A)}$, which is compact, and put

$$x'' = x' - \sum_{i=1}^n t_{g_i} \in B;$$

we have $\pi_f(x'') = \pi_f(x') = \pi_f(x)$. Moreover, if $h \in \overline{P(A)}$, then $h \in W_{g_i}$ for some i , and therefore

$$\begin{aligned} h(x'' - x) &= h(x' - x) - h(t_{g_1}) - \dots - h(t_{g_n}) \\ &\leq h(x' - x) - h(t_{g_i}) < \frac{1}{2}, \end{aligned}$$

hence $x'' - x \leq \frac{1}{2}$, and the proof is complete.

11.4.6. Lemma. *Let A and B be as in 11.4.3, let K be a closed two-sided ideal of A , let $x_1, \dots, x_n$ be elements of K such that $0 \leq x_i \leq 1$, and let $\varepsilon \in (0, \frac{1}{4})$. Let Q be the set of $f \in \overline{P(A)}$ such that $f(x_i) \geq 1 - \varepsilon$ for at least one i . There exists $y \in B \cap K$ such that $0 \leq y \leq 1$ and*

$$f(y) \geq 1 - 6\varepsilon^{1/4} \quad (f \in Q).$$

Let $f \in Q$. There is an index i_f such that $f(x_{i_f}) \geq 1 - \varepsilon$, and then there is $y_f \in B \cap K$ such that

$$0 \leq y_f \leq 1, \quad f(y_f) \geq 1 - \varepsilon^{1/2} \quad (11.4.5).$$

Let W_f be the set of $g \in Q$ such that $g(y_f) \geq 1 - 2\varepsilon^{1/2}$; this is a neighborhood of f in Q . Let $(W_{f_1}, \dots, W_{f_s})$ be a finite cover of Q , and put $x = y_{f_1} + \dots + y_{f_s}$. Let $p: \mathbb{R} \rightarrow \mathbb{R}$ be the function equal to 0 on $(-\infty, 0]$, equal to 1 on $[2\varepsilon^{1/2}, +\infty)$, and linear on $[0, 2\varepsilon^{1/2}]$. Put $y = p(x)$. Then $y \in B \cap K$ and $0 \leq y \leq 1$. If $f \in Q$, then, for example, $f \in W_{f_1}$, and hence $f(y_{f_1}) \geq 1 - 2\varepsilon^{1/2}$. Consequently,

$$f(y) \geq 1 - 4(2\varepsilon^{1/2})^{1/2} \geq 1 - 6\varepsilon^{1/4}$$

by 11.4.4(iv).

11.4.7. Lemma. *Let A and B be as in 11.4.3, let K be a closed two-sided ideal of A , let $f_1, \dots, f_n$ be states of A such that each $f_i|_K$ is a state of K , and let $\varepsilon > 0$. There exists $y \in B \cap K$ such that $0 \leq y \leq 1$ and*

$$f_1(y) \geq 1 - \varepsilon, \dots, f_n(y) \geq 1 - \varepsilon.$$

We may suppose that $(\varepsilon/12)^4 \in (0, \frac{1}{4})$. For $1 \leq i \leq n$, there exists $x_i \in K$ such that $0 \leq x_i \leq 1$ and

$$f_i(x_i) > 1 - (\varepsilon/12)^5.$$

Let Q be the set of $f \in \overline{P(A)}$ such that $f(x_i) \geq 1 - (\varepsilon/12)^4$ for at least one i . By 11.4.6, there exists $y \in B \cap K$ such that $0 \leq y \leq 1$ and

$$f(y) \geq 1 - \frac{\varepsilon}{2} \quad (f \in Q).$$

There are convex combinations

$$\lambda_{i1}^\alpha f_{i1}^\alpha + \dots + \lambda_{ip}^\alpha f_{ip}^\alpha$$

of pure states of A which converge weakly to f_i . Let J_α be the set of indices $j \in \{1, 2, \dots, p\}$ such that

$$f_{ij}^\alpha(x_i) \geq 1 - (\varepsilon/12)^4, \quad \beta_\alpha = \sum_{j \in J_\alpha} \lambda_{ij}^\alpha.$$

Since

$$(\lambda_{i1}^\alpha f_{i1}^\alpha + \dots + \lambda_{ip}^\alpha f_{ip}^\alpha)(x_i) \leq \beta_\alpha + (1 - (\varepsilon/12)^4)(1 - \beta_\alpha),$$

we have, for sufficiently large α ,

$$\begin{aligned} 1 - (\varepsilon/12)^5 &\leq \beta_\alpha + (1 - (\varepsilon/12)^4)(1 - \beta_\alpha), \\ (\varepsilon/12)^4 \beta_\alpha &\geq (\varepsilon/12)^4 - (\varepsilon/12)^5, \end{aligned}$$

$$\beta_\alpha \geq 1 - \frac{\varepsilon}{12}.$$

For $j \in J_\alpha$, we have $f_{ij}^\alpha \in Q$, and therefore $f_{ij}^\alpha(y) \geq 1 - \varepsilon/2$. Consequently, for sufficiently large α ,

$$\left(\sum_j \lambda_{ij}^\alpha f_{ij}^\alpha \right)(y) \geq \beta_\alpha \left(1 - \frac{\varepsilon}{2} \right) \geq \left(1 - \frac{\varepsilon}{12} \right) \left(1 - \frac{\varepsilon}{2} \right) \geq 1 - \varepsilon,$$

and, on passing to the limit, $f_i(y) \geq 1 - \varepsilon$.

Bibliography: [45].

11.5. Proof of the Theorem

11.5.1. Let A be a unital C^* -algebra, and let B be a C^* -subalgebra containing 1 and separating $\overline{P(A)}$. Suppose that $B \neq A$. We shall derive a contradiction.

Let M be the set of continuous Hermitian linear functionals on A which vanish on B and have norm at most 1. By the Hahn–Banach theorem, $M \neq 0$. Moreover, M is convex and weakly compact; by the Krein–Milman theorem, M has at least one extreme point $m \neq 0$. Clearly $\|m\| = 1$. Let I be the largest two-sided ideal of A contained in $\ker m$; this ideal is closed. We have $B + I \subset \ker m$, and hence $(B + I)/I \neq A/I$. Every state of A/I which is a weak limit of pure states of A/I defines a state of A which is a weak limit of pure states of A . Thus $(B + I)/I$ separates $\overline{P(A/I)}$.

We shall show that A/I is antiliminal. Let J be a closed two-sided ideal of A containing I such that J/I is postliminal. By 11.1.5 and 11.1.7,

$$J/I \subset (B + I)/I,$$

whence $J \subset B + I \subset \ker m$, and therefore $J \subset I$ by the definition of I . Thus $J/I = 0$, proving that A/I is antiliminal. This first shows that it is enough to establish Theorem 11.3.1 for antiliminal C^* -algebras. We shall therefore add to our previous hypotheses the assumption that A is antiliminal.

We shall prove that, in A/I , the intersection of two nonzero closed two-sided ideals is nonzero. By 11.3.3, this will give the desired contradiction.

Let K, L be two closed two-sided ideals of A not contained in I , and let us prove that $K \cap L \not\subset I$. There are positive functionals f_1, f_2 on A such that

$$f_1 - f_2 = m, \quad \|f_1\| + \|f_2\| = \|m\| = 1 \quad (2.6.4).$$

Moreover, for $i = 1, 2$, there are positive functionals g_i, h_i on A such that

$$f_i = g_i + h_i, \quad h_i(K) = 0, \quad \|g_i|_K\| = \|g_i\| \quad (2.11.7).$$

Let $x \in B$. For every $\varepsilon > 0$, there exists $y \in B \cap K$ such that

$$|g_i(x) - g_i(xy)| \leq \varepsilon \quad (i = 1, 2)$$

by 11.4.7 and 11.4.4(ii). Since $xy \in K \cap B$, we have

$$\begin{aligned} |g_1(x) - g_2(x)| &\leq |g_1(xy) - g_2(xy)| + 2\varepsilon \\ &= |f_1(xy) - f_2(xy)| + 2\varepsilon = 2\varepsilon. \end{aligned}$$

Thus $(g_1 - g_2)(B) = 0$, and consequently $(h_1 - h_2)(B) = 0$. On the other hand,

$$\begin{aligned} 1 = \|m\| &= \|g_1 - g_2 + h_1 - h_2\| \\ &\leq \|g_1 - g_2\| + \|h_1 - h_2\| \\ &\leq \|g_1\| + \|h_1\| + \|g_2\| + \|h_2\| \\ &= \|f_1\| + \|f_2\| = 1. \end{aligned}$$

Hence, if

$$\lambda = \|g_1 - g_2\|, \quad \mu = \|h_1 - h_2\|,$$

then $\lambda + \mu = 1$. Since $K \not\subset I$, we have $K \not\subset \ker m$, and therefore $g_1 - g_2$ cannot be proportional to $h_1 - h_2$, which vanishes on K . This proves first that $\lambda \neq 0$. If $\mu \neq 0$, the equality

$$m = \lambda(\lambda^{-1}(g_1 - g_2)) + \mu(\mu^{-1}(h_1 - h_2))$$

contradicts the fact that m is extreme in M . Thus $\mu = 0$ and $m = g_1 - g_2$. Let $z \in L$ be such that $m(z) \neq 0$. There exists $z' \in K$ such that

$$|g_1(z) - g_1(zz')| \quad \text{and} \quad |g_2(z) - g_2(zz')|$$

are arbitrarily small, by 11.4.7 and 11.4.4(ii), and hence such that

$$m(zz') = g_1(zz') - g_2(zz') \neq 0.$$

But $zz' \in K \cap L$, so $K \cap L \not\subset I$.

11.5.2. Corollary. *Let A be a C^* -algebra and B a C^* -subalgebra separating $\overline{P(A)} \cup \{0\}$. Then $B = A$.*

Let $\tilde{A} = A + \mathbb{C}e$ be the C^* -algebra obtained from A by adjoining an identity element e , and let $\tilde{B} = B + \mathbb{C}e$. Let $f \in \overline{P(\tilde{A})}$. Then f is a weak limit of pure states f_α of $\tilde{A}$. If $f|_A$ is nonzero, we may suppose that the $f_\alpha|_A$ are nonzero; they are then pure states of A (2.11.8), and hence in this case $f|_A \in \overline{P(A)}$. Thus in every case $f|_A \in \overline{P(A)} \cup \{0\}$. If f and g are two distinct elements of $\overline{P(\tilde{A})}$, then $f|_A$ and $g|_A$ are distinct and belong to $\overline{P(A)} \cup \{0\}$, and are therefore separated by B . Thus $\tilde{B}$ separates $\overline{P(\tilde{A})}$. Theorem 11.3.1 shows that $\tilde{B} = \tilde{A}$, whence $B = A$.

11.5.3. Corollary. *Let T be a locally compact space, let $\mathcal{A} = ((A(t)), \Theta)$ be a continuous field of C^* -algebras over T , let A be the C^* -algebra defined by $\mathcal{A}$, and let B be a C^* -subalgebra of A such that, for all $t_1, t_2 \in T$ and $\xi_1 \in A(t_1)$, $\xi_2 \in A(t_2)$, with $\xi_1 = \xi_2$ if $t_1 = t_2$, there exists $x \in B$ satisfying*

$$x(t_1) = \xi_1, \quad x(t_2) = \xi_2.$$

Then $B = A$.

By adjoining a point at infinity to T , we may suppose that T is compact (10.2.6).

Let $f \in \overline{P(A)}$. Then f is a weak limit of pure states f_α of A . By 10.4.3, there are $t_\alpha \in T$ and a pure state g_α of $A(t_\alpha)$ such that

$$f_\alpha(x) = g_\alpha(x(t_\alpha)) \quad (x \in A).$$

We may suppose that the t_α converge to a point $t \in T$. Let $x \in A$ satisfy $x(t) = 0$. Then $\|x(t_\alpha)\| \rightarrow \|x(t)\| = 0$, and hence $g_\alpha(x(t_\alpha)) \rightarrow 0$, so $f(x) = 0$. This proves that there is a positive functional g on $A(t)$ such that

$$f(x) = g(x(t)) \quad (x \in A).$$

Now let f_1, f_2 be two distinct points of $\overline{P(A)} \cup \{0\}$, and let us prove that B separates f_1 and f_2 ; then 11.5.3 follows from 11.5.2. Let $t_1, t_2 \in T$, and let g_i be a positive functional on $A(t_i)$, such that

$$f_i(x) = g_i(x(t_i)) \quad (x \in A).$$

At least one of g_1, g_2 is nonzero, say g_1 . If $t_1 = t_2$, then $g_1 \neq g_2$, so there exists $\xi \in A(t_1)$ such that $g_1(\xi) \neq g_2(\xi)$; then there is $x \in B$ with $x(t_1) = \xi$, and $f_1(x) \neq f_2(x)$. If $t_1 \neq t_2$, there is $y \in B$ such that $y(t_2) = 0$ and $g_1(y(t_1)) \neq 0$, whence $f_1(y) \neq f_2(y)$.

Bibliography: [29, 45, 167].

11.6. Complements

***11.6.1.** Let H be a Hilbert space, let A be a C^* -subalgebra of $\mathcal{L}(H)$ such that $1 \in A$ (respectively, $1 \notin A$), and let V be the weak closure of the set of vector states of A . Then V is the set of functionals

$$a(\omega_\xi|_A) + bg,$$

where $a, b \in [0, 1]$, $a + b = 1$ (respectively, $a + b \leq 1$), ξ is a unit vector of $\overline{A(H)}$, and g is a state of A such that $g(A \cap \mathcal{LC}(H)) = 0$. [45, 46]

***11.6.2.** Let A be a unital C^* -algebra, let K be its largest postliminal two-sided ideal, let B be a C^* -subalgebra of A containing 1, let S be the set of pure states f of A such that $f|_K$ is a (pure) state of K , and let T be the set of states which are weak limits of pure states of A vanishing on K . If B separates $S \cup T$, then $B = A$. [45]

***11.6.3.** Let A be a von Neumann algebra, let B be an antiliminal C^* -subalgebra of A , weakly dense in A and containing the center $\mathfrak{Z}$ of A . Then $\overline{P(B)}$ is the set of states f of B such that $f|_{\mathfrak{Z}}$ is a character of $\mathfrak{Z}$. In particular, if F is a continuous factor, then $\overline{P(F)} = E(F)$. [45]

***11.6.4.** Let A be a von Neumann algebra and $\mathfrak{Z}$ its center. Then $\overline{P(A)}$ is the set of functionals

$$ag + (1 - a)h,$$

where $a \in [0, 1]$, $g, h \in E(A)$, the restriction $(ag + (1 - a)h)|_{\mathfrak{Z}}$ is a character of $\mathfrak{Z}$, there is an abelian projection e of A such that $ag(e) = a$, and $(1 - a)h$ vanishes on every abelian projection of A . [45]^a

***11.6.5.** Let A be a von Neumann algebra, and let B be a C^* -subalgebra of A containing 1 and weakly dense in A . Then $\overline{P(B)}$ is the set of restrictions to B of the elements of $\overline{P(A)}$. [45]

11.6.6. Let A be a unital C^* -algebra. The following conditions are equivalent:

- (i) $\overline{P(A)} = E(A)$;
- (ii) A is antiliminal and the zero ideal is prime.

^a*Translator's note.* The source prints f in the first occurrence of this convex combination, but subsequently uses h ; the latter is required by the stated variables.

[The implication (ii) $\Rightarrow$ (i) follows from 11.2.4. Let I, J be nonzero closed two-sided ideals of A such that $I \cap J = 0$. Let f (respectively, g) be a pure state of A whose restriction to I (respectively, J) is nonzero. Then the restriction of g (respectively, f) to I (respectively, J) is zero, and $\frac{1}{2}(f + g) \notin \overline{P(A)}$.] [169]

§ 12. — Enveloping von Neumann Algebra of a C^* -Algebra

12.1. Bidual of a C^* -Algebra

12.1.1. Proposition. *Let H be a Hilbert space, let A be a C^* -subalgebra of $\mathcal{L}(H)$, and let B be the weak closure of A . Suppose that $1 \in B$ and that every norm-continuous linear functional f on A is ultrastrongly continuous, and hence has a unique ultrastrongly continuous extension $\tilde{f}$ to B .*

- (i) *The map $f \mapsto \tilde{f}$ is an isometric isomorphism from the Banach dual A' of A onto the predual of B .*
- (ii) *For each $y \in B$, let $\hat{y}$ be the linear functional $f \mapsto \tilde{f}(y)$ on A' . Then $y \mapsto \hat{y}$ is an isometric isomorphism from B onto the Banach dual A'' of A' , whose restriction to A is the canonical injection of A into A'' .*

If $f \in A'$, then $\tilde{f}$ belongs to the predual of B by hypothesis. Let A_1, B_1 be the unit balls of A, B . Since A_1 is ultrastrongly dense in B_1 (A 13),

$$\sup_{y \in B_1} |\tilde{f}(y)| = \sup_{x \in A_1} |f(x)|,$$

that is, $\|\tilde{f}\| = \|f\|$. Conversely, every element of the predual of B is the ultrastrongly continuous extension of its restriction to A . This proves (i). In view of (i) and A 23, $y \mapsto \hat{y}$ is an isometric isomorphism of B onto A'' ; if $y \in A$, then $\hat{y}$ is the functional $f \mapsto f(y)$ on A' , that is, the canonical image of A in A'' .

12.1.2. Corollary. *Let H be a Hilbert space and let A be the C^* -algebra of compact operators on H .*

- (i) *Every linear functional $f \in A'$ is ultrastrongly continuous on A , and hence has a unique ultrastrongly continuous extension $\tilde{f}$ to $\mathcal{L}(H)$.*
- (ii) *The map $f \mapsto \tilde{f}$ is an isometric isomorphism from A' onto the predual of $\mathcal{L}(H)$.*

- (iii) For every $y \in \mathcal{L}(H)$, let $\widehat{y}$ be the linear functional $f \mapsto \widetilde{f}(y)$ on A' . Then $y \mapsto \widehat{y}$ is an isometric isomorphism from $\mathcal{L}(H)$ onto A'' , whose restriction to A is the canonical injection of A into A'' .

Assertion (i) follows from 2.6.4 and 4.1.3, and (ii), (iii) are then special cases of 12.1.1.

12.1.3. Corollary. Let A be a C^* -algebra, let π be the universal representation of A (2.7.6), and let B be the weak closure of $\pi(A)$, which is a von Neumann algebra on H_π .

- (i) Every normal positive functional on B is a functional ω_ξ , with $\xi \in H_\pi$. Every ultrastrongly continuous linear functional on B is weakly continuous.
(ii) If $f \in A'$, there is one and only one weakly continuous linear functional $\widetilde{f}$ on B such that

$$\widetilde{f}(\pi(x)) = f(x) \quad (x \in A).$$

- (iii) The map $f \mapsto \widetilde{f}$ is an isometric isomorphism from the dual A' of A onto the predual of B ; it transforms the positive functionals on A into the normal positive functionals on B . For every $f \in A'$ one has

$$\widetilde{f}^* = (\widetilde{f})^*.$$

- (iv) For every $y \in B$, let $\widehat{y}$ be the linear functional $f \mapsto \widetilde{f}(y)$ on A' . Then $y \mapsto \widehat{y}$ is an isometric isomorphism from B onto A'' , whose composite with π is the canonical injection of A into A'' .
(v) This isomorphism is bicontinuous for the weak operator topology on B and the topology $\sigma(A'', A')$ on A'' .

Let f be a normal positive functional on B . Then $f|_{\pi(A)}$ is a positive functional on $\pi(A)$. By the construction of π , there is $\xi \in H_\pi$ such that

$$f(\pi(x)) = (\pi(x)\xi \mid \xi) \quad (x \in A).$$

The equality $f(y) = \omega_\xi(y)$, valid for $y \in \pi(A)$, remains true by continuity for $y \in B$. Since every ultrastrongly continuous linear functional on B is a linear combination of normal positive functionals on B (A 25), this proves (i).

We have just observed that every positive functional on $\pi(A)$ is weakly continuous. The same therefore holds for every continuous linear functional on $\pi(A)$ (2.6.4), which proves (ii). By 12.1.1, $f \mapsto \widetilde{f}$ is an isometric isomorphism from A' onto the predual of B .^b We have seen that, if $f \geq 0$, then $\widetilde{f}$ is a functional ω_ξ ,

^bTranslator's note. The source prints A here; B is required by 12.1.1 and by the definition of the predual.

and is therefore normal and positive, and that all normal positive functionals on B arise in this way. If $f \in A'$, then, for $x \in A$,

$$\widetilde{f}^*(\pi(x)) = f^*(x) = \overline{f(x^*)} = \overline{\widetilde{f}(\pi(x)^*)}.$$

Thus, by weak continuity,

$$\widetilde{f}^*(y) = \overline{\widetilde{f}(y^*)} \quad (y \in B),$$

that is, $\widetilde{f}^* = (\widetilde{f})^*$. This proves (iii), and (iv) is a consequence of 12.1.1. By (i) and (iii), the weak operator topology on B is defined by the linear functionals $y \mapsto \widetilde{f}(y)$, where $f \in A'$; the isomorphism $y \mapsto \widehat{y}$ carries this topology into $\sigma(A'', A')$, since $\widetilde{f}(y) = \widehat{y}(f)$. This proves (v).

12.1.4. Definition. *The algebra B is called the enveloping von Neumann algebra of A , and π the canonical homomorphism of A into B .*

We sometimes identify A with $\pi(A)$ and B with A'' .

12.1.5. The pair (B, π) solves a universal problem. Indeed:

Proposition. *Let A be a C^* -algebra, let B be the enveloping von Neumann algebra of A , and let $\pi: A \rightarrow B$ be the canonical homomorphism. Let ρ be a representation of A . There exists one and only one normal representation $\widetilde{\rho}$ of B on H_ρ such that*

$$\widetilde{\rho}(\pi(x)) = \rho(x) \quad (x \in A).$$

The image $\widetilde{\rho}(B)$ is the weak closure of $\rho(A)$.

A normal representation is ultraweakly continuous (A 27); since $\pi(A)$ is ultraweakly dense in B , this proves the uniqueness of $\widetilde{\rho}$. To prove its existence, we may suppose that ρ admits a cyclic vector, since every nondegenerate representation of A is a Hilbert direct sum of representations admitting cyclic vectors (2.2.7). Then there is a positive functional f on A such that ρ may be identified with π_f (2.4.6). By the construction of π , there is a closed vector subspace K of H_π , invariant under $\pi(A)$, such that ρ may be identified with the representation $x \mapsto \pi(x)|_K$. We may therefore take $\widetilde{\rho}$ to be the representation $y \mapsto y|_K$ of B on K . Finally, the image of B under a normal representation is weakly closed (A 27), which proves the last assertion of the proposition.

12.1.6. Recall that, if E is a Banach space and $E', E'', \dots$ denote its successive duals, the canonical injection of E into E'' has as its transpose a norm-one projection, called canonical, from E''' onto E' . The kernel of this projection is the set of continuous linear functionals on E'' which vanish on E .

Now let A be a von Neumann algebra, let A' be its dual, and let A'' be its bidual, identified with the enveloping von Neumann algebra of A . Recall that A is identified with the dual P' of its predual P , a closed vector subspace of A' .

Proposition. *Let ρ be the normal representation of A'' on A extending the identity map of A (12.1.5). If A and A'' are identified with P' and P''' , respectively, then ρ is the canonical projection from P''' onto P' . The kernel of this projection is a weakly closed two-sided ideal of A'' .*

The canonical projection π from P''' onto P' is continuous for

$$\sigma(P''', P'') = \sigma(A'', A') \quad \text{and} \quad \sigma(P', P) = \sigma(A, P).$$

In view of 12.1.3(i),(v), ρ is continuous for $\sigma(A'', A')$ and $\sigma(A, P)$. Now π and ρ both reduce on A to the identity map, and A is dense in A'' for $\sigma(A'', A')$. Hence $\pi = \rho$. The kernel of ρ is a two-sided ideal of A'' , ultraweakly closed and therefore weakly closed.

12.1.7. Corollary 12.1.3 reduces questions concerning continuous linear functionals on a C^* -algebra to questions concerning ultraweakly continuous linear functionals on a von Neumann algebra. The latter questions are even, in certain respects, more general, since the predual of a von Neumann algebra A is not always the dual of a C^* -subalgebra of A .

Bibliography: [55, 141, 150].

12.2. Polar Decomposition of a Linear Functional

12.2.1. Let A be a C^* -algebra, let A' be its dual, let $f \in A'$, and let $x \in A$. We denote by $x \cdot f$ and $f \cdot x$ the linear functionals

$$y \mapsto f(xy), \quad y \mapsto f(yx)$$

on A , which are elements of A' . We have

$$\|x \cdot f\| \leq \|x\| \|f\|, \quad \|f \cdot x\| \leq \|x\| \|f\|.$$

The operators $f \mapsto f \cdot x$ (respectively, $f \mapsto x \cdot f$) define a representation of A (respectively, of the opposite algebra) in the non-Hilbertian vector space A' . For $x, y \in A$ and $f \in A'$, we have

$$\begin{aligned} (x \cdot f)^*(y) &= \overline{(x \cdot f)(y^*)} = \overline{f(xy^*)} \\ &= f^*(yx^*) = (f^* \cdot x^*)(y), \end{aligned}$$

and hence

$$(x \cdot f)^* = f^* \cdot x^*.$$

If A is a von Neumann algebra and f is ultraweakly continuous (respectively, weakly continuous), then so are $x \cdot f$ and $f \cdot x$.

12.2.2. Lemma. *Let A be a unital C^* -algebra, let A_1 be the ball $\|x\| \leq 1$ in A , and let t be an extreme point of A_1 . Then t^*t is idempotent.*

We have $0 \leq t^*t \leq 1$ and $\|t^*t\| = 1$. Let B be the commutative C^* -subalgebra of A generated by 1 and t^*t , let Ω be its compact spectrum, let B' be the algebra of continuous complex-valued functions on Ω , let φ be the Gelfand isomorphism of B onto B' , and put $f = \varphi(t^*t)$. Then $0 \leq f \leq 1$, and f takes the value 1. Suppose that there is $\omega \in \Omega$ such that $0 < f(\omega) < 1$. There is then a continuous function $g: \Omega \rightarrow \mathbb{R}$, with $g \geq 0$, such that

$$fg \neq 0, \quad \sup f(1+g)^2 = \sup f(1-g)^2 = 1$$

(it is enough to take the support of g in a sufficiently small neighborhood of ω , and $|g|$ sufficiently small). Thus there exists $u \in B^+$ such that

$$\|t^*t(1+u)^2\| = \|t^*t(1-u)^2\| = 1.$$

Hence $t(1+u), t(1-u) \in A_1$. The equality

$$t = \frac{1}{2}(t(1+u) + t(1-u))$$

implies, since t is extreme, that $t = t(1+u) = t(1-u)$, whence $tu = 0$ and $fg = 0$, a contradiction. Therefore f takes only the values 0 and 1, so t^*t is a projection.

12.2.3. Lemma. *Let A be a von Neumann algebra, let e be a projection of A , and let f belong to the predual of A .*

(i)

$$\|f\|^2 \geq \|e \cdot f\|^2 + \|(1-e) \cdot f\|^2.$$

(ii) *If $\|f\| = \|e \cdot f\|$, then $f = e \cdot f$.*

Assertion (ii) follows from (i). We prove (i). Let H be the Hilbert space on which A acts. Let P be the predual of A , let Q be the predual of $\mathcal{L}(H)$, and let L be the orthogonal of A in Q . Since A is closed in $\mathcal{L}(H)$ for $\sigma(\mathcal{L}(H), Q)$, A is the orthogonal of L in $\mathcal{L}(H)$. Thus A is canonically identified with the dual of the Banach space Q/L . Let $f \in P$. By the Hahn–Banach theorem, f has an extension

$f' \in Q$. The element $f' + L$ of Q/L defines the functional f on A . Hence $\|f\|$ is the norm of $f' + L$ in Q/L , that is,

$$\|f\| = \inf_{g \in f' + L} \|g\|.$$

There is therefore a functional $g \in Q$ extending f and satisfying $\|g\| \leq \|f\| + \varepsilon$ (this is, incidentally, 12.5.2). If we prove that

$$\|g\|^2 \geq \|e \cdot g\|^2 + \|(1 - e) \cdot g\|^2,$$

we shall deduce

$$(\|f\| + \varepsilon)^2 \geq \|e \cdot f\|^2 + \|(1 - e) \cdot f\|^2,$$

and (i) will follow from the arbitrariness of ε . We are thus reduced to the case $A = \mathcal{L}(H)$. Since the weakly continuous linear functionals on $\mathcal{L}(H)$ are norm-dense in Q , we may suppose that f is weakly continuous. There are therefore two orthonormal systems

$$(\xi_1, \dots, \xi_n), \quad (\xi'_1, \dots, \xi'_n)$$

in H , and numbers $\lambda_1 \geq 0, \dots, \lambda_n \geq 0$, such that

$$\|f\| = \lambda_1 + \dots + \lambda_n, \quad f(x) = \sum_i \lambda_i (x \xi_i \mid \xi'_i) \quad (x \in A)$$

(A 24). Then

$$(e \cdot f)(x) = \sum_i \lambda_i (x \xi_i \mid e \xi'_i), \quad ((1 - e) \cdot f)(x) = \sum_i \lambda_i (x \xi_i \mid (1 - e) \xi'_i).$$

Consequently,

$$\begin{aligned} & \|e \cdot f\|^2 + \|(1 - e) \cdot f\|^2 \\ & \leq \left(\sum_i \lambda_i \|e \xi'_i\| \right)^2 + \left(\sum_i \lambda_i \|(1 - e) \xi'_i\| \right)^2 \\ & = \sum_{i=1}^n \lambda_i^2 (\|e \xi'_i\|^2 + \|(1 - e) \xi'_i\|^2) \\ & \quad + 2 \sum_{1 \leq i < j \leq n} \lambda_i \lambda_j (\|e \xi'_i\| \|e \xi'_j\| + \|(1 - e) \xi'_i\| \|(1 - e) \xi'_j\|) \end{aligned}$$

$$\leq \sum_{i=1}^n \lambda_i^2 + 2 \sum_{1 \leq i < j \leq n} \lambda_i \lambda_j = \left(\sum_{i=1}^n \lambda_i \right)^2 = \|f\|^2.$$

12.2.4. Theorem. *Let A be a von Neumann algebra and let f belong to the predual of A .*

- (i) *There exists a pair (p, u) with the following properties:*
 - a. *p is a normal positive functional on A , and $\|p\| = \|f\|$;*
 - b. *u is a partial isometry in A whose final projection is the support of p (A 26);*
 - c. *$f = u \cdot p$ and $p = u^* \cdot f$.*
- (ii) *Let p' be a normal positive functional on A and u' a partial isometry in A whose final projection is majorized by the support of p' , with*

$$f = u' \cdot p', \quad p' = u'^* \cdot f.$$

Then $p' = p$ and $u' = u$.

We may suppose that $\|f\| = 1$. Let B be the ball $\|x\| \leq 1$ in A , and let B' be the set of $x \in B$ such that $f(x) = 1$. Since B is ultraweakly compact and f is ultraweakly continuous, there are $x \in B$ such that $|f(x)| = 1$. Multiplying x by a suitable scalar shows that $B' \neq \emptyset$; moreover, B' is convex and ultraweakly compact. There is therefore an extreme point v in B' . This point is also extreme in B , for if $v = \frac{1}{2}(s + t)$ with $s, t \in B$, then

$$|f(s)| \leq 1, \quad |f(t)| \leq 1,$$

and

$$1 = f(v) = \frac{1}{2}(f(s) + f(t)), \quad \text{whence} \quad f(s) = f(t) = 1.$$

Thus $s, t \in B'$, and $s = t = v$ because v is extreme in B' . By 12.2.2, v^*v is a projection, so v is a partial isometry.

Put $p = v \cdot f$. We have $\|p\| \leq \|v\| \|f\| = 1$. Since $p(1) = f(v) = 1$, p is positive (2.1.9), has norm 1, and is normal because it is ultraweakly continuous. Let e be the support of p . We have

$$p(v^*v) = f(vv^*v) = f(v) = 1,$$

and hence $v^*v \geq e$. Put $u = ev^*$. Then

$$uu^* = e(v^*v)e = e,$$

so u is a partial isometry with final projection e . For every $x \in A$,

$$(u^* \cdot f)(x) = f(u^*x) = f(vex) = p(ex) = p(x),$$

and therefore $p = u^* \cdot f$. Put $u^*u = e'$. We have $\|e' \cdot f\| \leq 1$ and

$$(e' \cdot f)(u^*) = f(u^*uu^*) = f(u^*) = f(ve) = p(e) = 1.$$

Thus $\|e' \cdot f\| = 1 = \|f\|$, and hence $e' \cdot f = f$ by 12.2.3(ii). Consequently, for every $x \in A$,

$$(u \cdot p)(x) = p(ux) = f(u^*ux) = f(e'x) = (e' \cdot f)(x) = f(x),$$

and therefore $f = u \cdot p$. This proves (i).

12.2.5. To prove (ii), we first establish the following proposition.

Proposition. *Let A be a unital C^* -algebra, let f be a continuous linear functional on A , let p be a positive functional on A , and let a be an element of A such that*

$$\|a\| \leq 1, \quad a \cdot p = f, \quad \|p\| = \|f\|.$$

If $b \in A$ satisfies

$$\|b\| \leq 1, \quad b \cdot f \geq 0, \quad \|b \cdot f\| = \|f\|,$$

then $b \cdot f = p$.

We may suppose that $\|p\| = \|f\| = 1$. We have

$$\begin{aligned} (\xi_p \mid \pi_p(b^*a^*)\xi_p) &= (\pi_p(ab)\xi_p \mid \xi_p) \\ &= p(ab) = f(b) = (b \cdot f)(1) = \|b \cdot f\| = 1. \end{aligned}$$

Since $\|b^*a^*\| \leq 1$, equality in the Cauchy–Schwarz inequality shows that

$$\pi_p(b^*a^*)\xi_p = \xi_p.$$

Thus, for every $x \in A$,

$$\begin{aligned} (b \cdot f)(x) &= f(bx) = p(abx) = (\pi_p(abx)\xi_p \mid \xi_p) \\ &= (\pi_p(x)\xi_p \mid \pi_p(b^*a^*)\xi_p) = (\pi_p(x)\xi_p \mid \xi_p) = p(x), \end{aligned}$$

which proves $b \cdot f = p$.

12.2.6. We return to the proof of 12.2.4. Let (p', u') be a pair having the properties in (ii). We have $\|u'^*\| \leq 1$, $u'^* \cdot f \geq 0$, and

$$\|p'\| = \|u'^* \cdot f\| \leq \|f\| = \|u' \cdot p'\| \leq \|p'\|,$$

so $\|u'^* \cdot f\| = \|f\|$. Proposition 12.2.5, applied with $a = u$ and $b = u'^*$, gives

$$p = u'^* \cdot f = p'.$$

We shall prove that $u = u'$. Put $x = uu'^*$. Then

$$p(x) = (u \cdot p)(u'^*) = f(u'^*) = (u'^* \cdot f)(1) = p(1) = 1.$$

Thus $p(x^*) = 1$ and

$$1 = p(x)^2 \leq p(xx^*) \leq 1,$$

so $p(xx^*) = 1$, and

$$p((e - x)(e - x)^*) = 1 - 1 - 1 + 1 = 0.$$

Now $exe = x$, since the final projections of u and u' are majorized by e . Since p is faithful on eAe , it follows that $x = e$. Let H be the Hilbert space on which A acts, and let e_1 be the initial projection of u . For $\xi \in e(H)$,

$$\|u(u'^*\xi)\| = \|e\xi\| = \|\xi\|, \quad \text{hence} \quad \|u(u'^*\xi)\| \geq \|u'^*\xi\|.$$

Thus $u'^*\xi \in e_1(H)$. Since u maps $e_1(H)$ isometrically onto $e(H)$, the equality

$$uu'^*\xi = \xi = uu^*\xi$$

implies $u'^*\xi = u^*\xi$. Moreover, u^* and u'^* vanish on $H \ominus e(H)$. Hence $u'^* = u^*$ and $u' = u$.

12.2.7. Definition. With the notation of 12.2.4, p is called the absolute value of f and is denoted by $|f|$. The equality $f = u \cdot |f|$ is called the polar decomposition of f .

12.2.8. Definition. Let A be a C^* -algebra, let E be the enveloping von Neumann algebra of A , and let f be a continuous linear functional on A . The polar decomposition $f = u \cdot p$ of f , regarded as an element of the predual of E , is called the enveloping polar decomposition of f . The functional p , regarded as a positive functional on A , is called the absolute value of f and is denoted by $|f|$.

If A is a von Neumann algebra and f belongs to its predual, then f has both a polar decomposition and an enveloping polar decomposition; in general these are distinct (12.5.7). We shall nevertheless see that the two possible definitions of $|f|$ coincide. Let $f = u \cdot p$ be the polar decomposition of f , and identify p with a normal positive functional on E (12.1.3(iii)). We have

$$u^* \in A \subset E, \quad \|u^*\| \leq 1, \quad \|p\| = \|f\|.$$

The equality $p(x) = f(u^*x)$, valid for $x \in A$, extends by continuity to every $x \in E$, where f is identified with a weakly continuous linear functional on E . Proposition 12.2.5, applied to E and $b = u^*$, then proves that p is the absolute value of f in the sense of 12.2.8.

12.2.9. Proposition. *Let A be a C^* -algebra, let f be a continuous linear functional on A , and let p be a positive functional on A . In order that $p = |f|$, it is necessary and sufficient that*

$$\|p\| = \|f\|, \quad |f(x)|^2 \leq \|f\|p(x^*x) \quad (x \in A).$$

Let $f = u \cdot |f|$ be the enveloping polar decomposition of f . If $p = |f|$, then $\|p\| = \|f\|$, and, for every $x \in A$,

$$|f(x)|^2 = |p(ux)|^2 \leq p(u^*u)p(x^*x) \leq \|p\|p(x^*x) = \|f\|p(x^*x).$$

Conversely, suppose that $\|p\| = \|f\|$ and that

$$|f(x)|^2 \leq \|f\|p(x^*x) \quad (x \in A).$$

Then

$$|f(x)|^2 \leq \|f\|(\pi_p(x^*x)\xi_p \mid \xi_p) = \|f\| \|\pi_p(x)\xi_p\|^2$$

for every $x \in A$, and hence for every x in the enveloping von Neumann algebra E of A ; here π_p is extended to a normal representation of E , the function $x \mapsto \|\pi_p(x)\xi_p\|$ is continuous for the ultrastrong topology, and A is ultrastrongly dense in E . Applying this inequality to u^*x gives

$$|f|(x)^2 \leq \|f\| \|\pi_p(u^*x)\xi_p\|^2 \leq \|f\| \|\pi_p(x)\xi_p\|^2 \quad (x \in E).$$

There is therefore a vector ξ in the representation space of π_p such that $\|\xi\| \leq \|f\|^{1/2}$ and

$$|f|(x) = (\pi_p(x)\xi_p \mid \xi).$$

Consequently,

$$(\xi_p \mid \xi) = |f|(1) = \|f\| = \|p\|^{1/2}\|f\|^{1/2} \geq \|\xi_p\| \|\xi\|,$$

so $\xi = \xi_p$ and

$$|f|(x) = (\pi_p(x)\xi_p \mid \xi_p) = p(x).$$

Bibliography: [21, 65, 128, 131, 163].

12.3. Decomposition of a Hermitian Functional into Positive and Negative Parts

12.3.1. Proposition. *Let A be a von Neumann algebra, let f, f' be two normal positive functionals on A , let e, e' be their supports, and put $g = f - f'$.*

(i) *The equality*

$$\|g\| = \|f\| + \|f'\|$$

holds if and only if $ee' = 0$.

(ii) *Suppose this condition holds. Then*

$$|g| = f + f',$$

the support of $|g|$ is $e + e'$, and $g = (e - e') \cdot |g|$ is the polar decomposition of g ; moreover,

$$f = e \cdot g, \quad f' = -e' \cdot g.$$

Suppose that $ee' = 0$. In order that a projection d of A satisfy $(f + f')(d) = 0$, it is necessary and sufficient that $f(d) = f'(d) = 0$, that is, that $de = de' = 0$. Thus $f + f'$ has support $e + e'$. On the other hand, $e - e'$ is a Hermitian partial isometry with initial and final projection $e + e'$. For every $x \in A$, we have

$$\begin{aligned} ((e - e') \cdot (f + f'))(x) &= (f + f')(ex - e'x) \\ &= f(ex) - f'(e'x) = f(x) - f'(x) = g(x), \\ ((e - e')^* \cdot (f - f'))(x) &= (f - f')(ex - e'x) \\ &= f(ex) + f'(e'x) = f(x) + f'(x). \end{aligned}$$

By 12.2.4(ii), this proves that $|g| = f + f'$ and that $g = (e - e') \cdot |g|$ is the polar decomposition of g . We have

$$e \cdot g = e \cdot f = f, \quad e' \cdot g = -e' \cdot f' = -f'.$$

Finally,

$$\|g\| = \|f + f'\| = \|f\| + \|f'\|.$$

Conversely, suppose that $\|g\| = \|f\| + \|f'\|$. The set B of $x \in A$ satisfying $x = x^*$ and $\|x\| \leq 1$ is ultraweakly compact, and g is ultraweakly continuous. There is therefore $x \in B$ such that $g(x) = \|g\|$. It follows that

$$f(x^+) + f'(x^-) - f(x^-) - f'(x^+) = g(x) = \|g\| = \|f\| + \|f'\|.$$

But

$$f(x^+) \leq \|f\|, \quad f'(x^-) \leq \|f'\|, \quad -f(x^-) - f'(x^+) \leq 0,$$

and hence

$$f(x^+) = \|f\|, \quad f'(x^-) = \|f'\|.$$

Let s, s' be the supports of x^+, x^- . Then

$$\|f\| = f(x^+) \leq f(s) \leq \|f\|,$$

so $f(1 - s) = 0$ and $s \geq e$; similarly, $s' \geq e'$. Since $ss' = 0$, we have $ee' = 0$.

12.3.2. Definition. *If the conditions of 12.3.1(i) are satisfied, f and f' are said to be mutually singular.*

12.3.3. Theorem. *Let A be a von Neumann algebra, and let g be an ultraweakly continuous Hermitian linear functional on A . There exists one and only one pair (f, f') of mutually singular normal positive functionals on A such that $g = f - f'$.*

Let P be the predual of A , let A' be its dual, and let A'' be its bidual, identified with its enveloping von Neumann algebra. By 2.6.4, there are positive functionals f, f' on A , identifiable with normal positive functionals on A'' , such that

$$g = f - f', \quad \|g\| = \|f\| + \|f'\|.$$

Applying 12.3.1 to A'' , we obtain $|g| = f + f'$. But $|g| \in P$ (12.2.8). Hence $f, f' \in P$, so that f and f' are normal.

Suppose that $g = f_1 - f'_1$, with f_1, f'_1 mutually singular normal positive functionals. Let e, e', e_1, e'_1 be the supports of f, f', f_1, f'_1 . By 12.3.1(ii),

$$f_1 + f'_1 = |g| = f + f', \quad \text{whence} \quad f = f_1, \quad f' = f'_1.$$

12.3.4. Corollary. *Let A be a C^* -algebra and let g be a continuous Hermitian linear functional on A . There exists one and only one pair (f, f') of positive functionals on A such that*

$$g = f - f', \quad \|g\| = \|f\| + \|f'\|.$$

Existence was already known. Uniqueness follows from 12.3.3 and 12.1.3(iii).

12.3.5. The functionals f, f' of 12.3.4 are called the positive and negative parts of g , and are denoted by g^+ and g^- . By 12.3.1,

$$|g| = g^+ + g^-.$$

12.3.6. Let A be a C^* -algebra, let E be the enveloping von Neumann algebra of A , and let f be a positive functional on A . Then f , regarded as a normal positive functional on E , has a support e , which is a projection of E . This is called the *enveloping support* of f . [If A is a von Neumann algebra and f is normal, then f

has a support and an enveloping support which are in general distinct (12.5.7).] If f, f' are two positive functionals on the C^* -algebra A , the equality

$$\|f - f'\| = \|f\| + \|f'\|$$

is equivalent to the enveloping supports of f and f' being orthogonal (12.3.1(i)). One then says that f and f' are mutually singular, which generalizes Definition 12.3.2.

Bibliography: [55, 70, 131, 150, 151, 152].

12.4. Positive Part of an Ideal in a C^* -Algebra

12.4.1. Proposition. *Let A be a C^* -algebra, let $\mathcal{I}$ be the set of closed left ideals of A , and let $\mathcal{J}$ be the set of closed subsets J of A^+ having the following properties:*

- (i) $x, y \in J \implies x + y \in J$;
- (ii) $x \in J, y \in A^+, \text{ and } y \leq x \implies y \in J$.

Then the map $I \mapsto I^+ = I \cap A^+$ is a bijection from $\mathcal{I}$ onto $\mathcal{J}$.

First observe that, if $I \in \mathcal{I}$ and $x \in I^+$, then $x^{1/2} \in I^+$, for $I \cap I^*$ is a C^* -subalgebra of A and $I^+ \subset I \cap I^*$.

Next observe that, if $J \in \mathcal{J}$ and $x \in J$, then $x^{1/2} \in J$. Indeed, there is a sequence of continuous functions $f_n: \text{Sp } x \rightarrow \mathbb{R}$, with $f_n \geq 0$, converging uniformly to $t \mapsto \sqrt{t}$ and vanishing in a neighborhood of 0. There is a sequence of constants $\lambda_n \geq 0$ such that $f_n(t) \leq \lambda_n t$ on $\text{Sp } x$. Thus

$$0 \leq f_n(x) \leq \lambda_n x, \quad f_n(x) \in J, \quad x^{1/2} \in J.$$

We interrupt the proof to establish the following lemma.

12.4.2. Lemma. *Regard A as extended inside its enveloping von Neumann algebra E . Let $I \in \mathcal{I}$, and let EI^+ be the set of products ax , where $a \in E$ and $x \in I^+$. Then*

$$I = (EI^+) \cap A.$$

Let $y \in I$. Then $y = u|y|$ for some $u \in E$. Moreover, $|y| = (y^*y)^{1/2} \in I^+$ by an earlier observation. Thus $y \in EI^+$, and hence $I \subset (EI^+) \cap A$. Conversely, let $a \in E$, $x \in I^+$, and suppose that $ax \in A$. We must prove that $ax \in I$. Now a is in the closure of A for $\sigma(E, A')$, and so ax is in the closure of $Ax \subset I$ for $\sigma(A, A')$. Since I is a norm-closed vector subspace of A , it is closed for $\sigma(A, A')$. Thus $ax \in I$.

12.4.3. We now return to the proof of 12.4.1. Let $I \in \mathcal{I}$. If $x, y \in I^+$, then $x+y \in I^+$. Let $z \in A^+$ satisfy $z \leq x$. There exists $s \in E$ such that $z^{1/2} = sx^{1/2}$ (A 8); since $x^{1/2} \in I^+$ by an earlier observation,

$$z = (sx^{1/2})^*(sx^{1/2}) = (x^{1/2}s^*s)x^{1/2} \in EI^+,$$

and therefore $z \in I^+$ by 12.4.2. Thus $I^+ \in \mathcal{J}$. The map $I \mapsto I^+$ from $\mathcal{I}$ to $\mathcal{J}$ is injective by 12.4.2.

We prove that it is surjective. Let $J \in \mathcal{J}$, and put $I = (EJ) \cap A$. Plainly $I^+ \supset J$. Let $a \in E$ and $x \in J$ be such that $ax \in A^+$, and let us prove that $ax \in J$. We have

$$(ax)^2 = xa^*ax \leq \|a\|^2x^2 \leq \|a\|^2\|x\|x.$$

Hence $(ax)^2 \in J$, and therefore $ax \in J$ by an earlier observation. Thus $I^+ = J$.

We shall prove that I is a left ideal of A . Clearly $AI \subset I$. Let $x, y \in I$. Then $x^*x, y^*y \in I^+ = J$, and hence $x^*x + y^*y \in J$. Since

$$(x+y)^*(x+y) \leq 2(x^*x + y^*y),$$

it follows that

$$|x+y|^2 = (x+y)^*(x+y) \in J, \quad \text{and hence} \quad |x+y| \in J.$$

But $x+y = u|x+y|$ for some $u \in E$, so $x+y \in (EJ) \cap A = I$. Finally, we show that I is closed. If (x_n) is a sequence of points of I converging to $x \in A$, then $x_n^*x_n \in I^+ = J$ converges to x^*x . Thus $|x|^2 = x^*x \in J$, $|x| \in J$, and $x \in (EJ) \cap A = I$.

12.4.4. Proposition 12.4.1 remains valid if left ideals are replaced by right ideals. Moreover, if I is a left ideal of A , then I^* is a right ideal and

$$I^+ = (I^*)^+.$$

Bibliography: [21].

12.5. Complements

12.5.1. Let A be a C^* -algebra. In the dual A' of A , let (f_α) be a family indexed by a directed set and converging weakly to f . If $\|f_\alpha\| \rightarrow \|f\|$, then $|f_\alpha|$ converges weakly to $|f|$. [21]

12.5.2. Let H be a Hilbert space, let A be a von Neumann algebra on H , and let f be an ultraweakly continuous linear functional on A . There is an ultraweakly continuous linear functional g on $\mathcal{L}(H)$ extending f and such that

$$\|f\| = \|g\|.$$

[If $f \geq 0$, there are vectors $\xi_i \in H$ such that $\sum_i \|\xi_i\|^2 < +\infty$ and

$$f(x) = \sum_i (x\xi_i \mid \xi_i) \quad (x \in A).$$

Put

$$g(x) = \sum_i (x\xi_i \mid \xi_i) \quad (x \in \mathcal{L}(H));$$

then $\|g\| = g(1) = f(1) = \|f\|$. In the general case, use the polar decomposition of f .] [128, 131]

12.5.3. Let A be a C^* -algebra, let E be the enveloping von Neumann algebra of A , and let f, g be two pure states of A . In order that $\pi_f \simeq \pi_g$, it is necessary and sufficient that the supports of f and g , regarded as normal positive functionals on E , be equivalent relative to E . [168]

***12.5.4.** Let A be a C^* -algebra. In order that the dual of A be separable in norm, it is necessary and sufficient that A admit a countable composition series (I_ρ) such that the quotients $I_{\rho+1}/I_\rho$ are separable dual C^* -algebras. [168] (See also *Mathematical Reviews*, 1963, p. 808.)

12.5.5. In order that a factor, regarded as a Banach space, be the bidual of a C^* -algebra, it is necessary and sufficient that it be of type I. [55, 152]

12.5.6. Let A be a von Neumann algebra, let A'' be its enveloping von Neumann algebra, let P be the predual of A , and let $f \in P$ be identified with a linear functional on A'' . If $x \in A''$, then

$$x \cdot f \in P, \quad f \cdot x \in P.$$

[Let ρ be the canonical projection from A'' onto A . If $y \in A$, then, since f vanishes on $\ker \rho$,

$$f(xy) = f(\rho(xy)) = f(\rho(x)y),$$

and therefore $x \cdot f = \rho(x) \cdot f \in P$.] (Grothendieck, unpublished.)

12.5.7. Let H be a Hilbert space, let $(\xi_1, \xi_2, \dots)$ be an orthonormal basis of H , and let A be the von Neumann algebra on H generated by the projections $P_i = P_{\mathbb{C}\xi_i}$. The positive functional

$$\sum_i \lambda_i P_i \mapsto \sum_i i^{-2} \lambda_i$$

is normal and has support 1. Let g be the positive functional

$$\sum_i \lambda_i P_i \mapsto \lim_i \lambda_i,$$

where a generalized Banach limit is used. Regard f and g as normal positive functionals on the enveloping algebra E of A . Put

$$Q_n = P_n + P_{n+1} + \dots \in A \subset E.$$

The Q_n have a strong limit Q in E , which is a projection of E . We have $g(Q_n) = 1$ for every n , so $g(Q) = 1$ and $Q \neq 0$. On the other hand, $f(Q_n) \rightarrow 0$, so $f(Q) = 0$. Thus the support of f relative to E is not 1.

The polar decomposition of f therefore differs from its enveloping polar decomposition.

12.5.8. Let A be a von Neumann algebra, let f, f' be two mutually singular normal positive functionals on A , and put $g = f - f'$. Then

$$\|f\| = \sup_{\substack{x \in A^+ \\ \|x\| \leq 1}} g(x), \quad \|f'\| = \sup_{\substack{x \in A^+ \\ \|x\| \leq 1}} -g(x).$$

[If $x \in A^+$ and $\|x\| \leq 1$, then $g(x) \leq f(x) \leq \|f\|$. On the other hand, if e is the support of f , then $g(e) = f(e) = \|f\|$.]

12.5.9. Let A be a von Neumann algebra. Let $\mathcal{E}$ be the set of weakly closed left ideals of A . Let $\mathcal{E}'$ be the set of weakly closed subsets P of A^+ such that:

- 1° $x, y \in P \implies x + y \in P$;
- 2° $x \in P$ and $\lambda \geq 0 \implies \lambda x \in P$;
- 3° $x \in P, y \in A^+$, and $y \leq x \implies y \in P$.

Then $L \mapsto L \cap A^+$ is a bijection from $\mathcal{E}$ onto $\mathcal{E}'$. [21]

12.5.10. Let A be a C^* -algebra. Let $\mathcal{E}$ be the set of closed two-sided ideals of A . Let $\mathcal{E}'$ be the set of closed subsets P of A^+ such that:

- 1° $x, y \in P \implies x + y \in P$;
- 2° $x \in P$ and $\lambda \geq 0 \implies \lambda x \in P$;

3° $x \in P$ and $y \in A \implies y^*xy \in P$.

Then $I \mapsto I \cap A^+$ is a bijection from $\mathcal{E}$ onto $\mathcal{E}'$. [21]

12.5.11. Let A be a C^* -algebra, let B be its enveloping von Neumann algebra, and let $\pi: A \rightarrow B$ be the canonical homomorphism. Define A_1, B_1, π_1 similarly. Let $\varphi: A \rightarrow A_1$ be a homomorphism. There exists one and only one homomorphism $\psi: B \rightarrow B_1$ such that

$$\pi_1 \circ \varphi = \psi \circ \pi.$$

If B, B_1 are identified with A'', A_1'' , then ψ is the bitranspose of φ .

Applications to Group Representations

§ 13. — Unitary Representations of Locally Compact Groups

13.1. Elementary Definitions Concerning Representations

13.1.1. Definition. *Let G be a topological group and H a Hilbert space. A continuous unitary representation of G on H is a homomorphism from G into the unitary group of H which is continuous for the strong topology.*

In other words, a continuous unitary representation of G on H is a map π from G into the unitary group of H such that

$$\pi(st) = \pi(s)\pi(t)$$

and such that, for every $\xi \in H$, the map $s \mapsto \pi(s)\xi$ from G into H is continuous for the strong topology of H . The identity $\pi(st) = \pi(s)\pi(t)$ implies

$$\pi(e) = 1, \quad \pi(s^{-1}) = \pi(s)^{-1} = \pi(s)^*,$$

where e will always denote the identity element of the group under consideration.

The functions

$$s \longmapsto \varphi_{\xi, \eta}(s) = (\pi(s)\xi \mid \eta) \quad (s \in G; \xi, \eta \in H \text{ fixed})$$

are called the *coefficients* of π . We have

$$\varphi_{\xi, \eta}(s) = \overline{\varphi_{\eta, \xi}(s^{-1})}.$$

13.1.2. On the unitary group of H , the strong and weak topologies coincide. Thus throughout the preceding discussion the strong topology may be replaced by the weak topology. If, on the other hand, continuity for the norm topology were imposed, a much more restrictive definition, of no interest for what follows, would result. Of course, if $\dim H < +\infty$, all these notions of continuity coincide.

13.1.3. The Hilbert dimension of H is called the dimension of π and is denoted by $\dim \pi$. The space H is called the space of π and is denoted by H_π .

By analogy with the case of involutive algebras, the following notions are also defined without difficulty: equivalent representations, class of representations, intertwining operator, intertwining number, Hilbert direct sum of representations, multiple of a representation, subrepresentation, representation contained in a representation, and cyclic vector. [The vector ξ is called cyclic for π if $\pi(G)\xi$ is total in H_π .] We use the same notation as in the case of algebras, and these notions have the same elementary properties; we leave their verification to the reader.

There is no need here to define the essential subspace of a representation. Everything proceeds as if continuous unitary representations were automatically nondegenerate.

13.1.4. Let π be a continuous unitary representation of G . The following conditions are equivalent:

- (i) the only closed vector subspaces of H_π invariant under $\pi(G)$ are 0 and H_π ;
- (ii) the commutant of $\pi(G)$ in $\mathcal{L}(H_\pi)$ consists only of the scalars;
- (iii) every nonzero vector of H_π is cyclic for π .

If these conditions hold and, moreover, $H_\pi \neq 0$, then π is said to be *topologically irreducible*, or simply *irreducible*. [We shall never consider the notion of algebraic irreducibility, except when $\dim \pi < +\infty$, in which case it is equivalent to topological irreducibility.] We denote by $\widehat{G}$ the set of classes of irreducible representations of G .

Exactly as in [Section 5](#), we define disjoint, factor, quasi-equivalent, type I, type II, and so on, multiplicity-free, and multiplicity- n representations. All the arguments and all the results of [Section 5](#) extend immediately to group representations. [One may also use the correspondence between representations of G and representations of $L^1(G)$ which will be established in [13.3](#), at least when G is locally compact.]

13.1.5. We shall now define for group representations operations which have no meaning for representations of involutive algebras.

Let π, π' be continuous unitary representations of G . For every $s \in G$, form, on the Hilbert tensor product $H_\pi \otimes H_{\pi'}$, the unitary operator $\pi(s) \otimes \pi'(s)$. It is immediate that

$$s \longmapsto \pi(s) \otimes \pi'(s)$$

is a continuous unitary representation of G on $H_\pi \otimes H_{\pi'}$, called the *tensor product* of π and π' , and denoted by $\pi \otimes \pi'$.

Let π be a continuous unitary representation of G on H , and let $\overline{H}$ be the conjugate Hilbert space of H . Each $\overline{\pi(s)}$ is a unitary operator on $\overline{H}$, and it is immediate that $s \mapsto \overline{\pi(s)}$ is also a continuous unitary representation of G on $\overline{H}$, called the *conjugate representation* of π and denoted by $\overline{\pi}$.

13.1.6. Let G be a locally compact group equipped with a left Haar measure ds . For every $s \in G$, let $\lambda(s)$ be the operator on $L^2(G)$ defined by

$$(\lambda(s)f)(x) = f(s^{-1}x) \quad (f \in L^2(G), x \in G).$$

One checks immediately that λ is a continuous unitary representation of G on $L^2(G)$, called the *left regular representation*. Let $\Delta: G \rightarrow \mathbb{R}$ be the modular function of G . For every $s \in G$, let $\rho(s)$ be the operator on $L^2(G)$ defined by

$$(\rho(s)f)(x) = \Delta(s)^{1/2} f(xs) \quad (f \in L^2(G), x \in G).$$

One checks immediately that ρ is a continuous unitary representation of G on $L^2(G)$, called the *right regular representation*.

The representations λ and ρ are injective.

For every $f \in L^2(G)$, define $f' \in L^2(G)$ by

$$f'(x) = \Delta(x)^{-1/2} f(x^{-1}).$$

Then $f \mapsto f'$ is an isomorphism of the Hilbert space $L^2(G)$ onto itself, and, for every $s \in G$,

$$\begin{aligned} (\lambda(s)f)'(x) &= \Delta(x)^{-1/2} f(s^{-1}x^{-1}) \\ &= \Delta(s)^{1/2} \Delta(xs)^{-1/2} f((xs)^{-1}) = (\rho(s)f')(x). \end{aligned}$$

Thus the isomorphism $f \mapsto f'$ transforms λ into ρ , so $\lambda \simeq \rho$. One therefore sometimes speaks simply of the *regular representation* of G .

For every $f \in L^2(G)$, let $\overline{f} \in \overline{L^2(G)}$ be the complex conjugate function. Then $f \mapsto \overline{f}$ is an isomorphism from the Hilbert space $L^2(G)$ onto its conjugate Hilbert space which transforms λ into $\overline{\lambda}$. Thus

$$\lambda \simeq \overline{\lambda}, \quad \rho \simeq \overline{\rho}.$$

13.1.7. Let a be a cardinal and let H be a Hilbert space of Hilbert dimension a . The representation $s \mapsto 1$ of G on H is called the *trivial representation* of G on H ,

or the trivial representation of G of dimension a . It is denoted by 1_H , or simply by 1 when there is no ambiguity concerning H .

13.1.8. Proposition. *Let G_1, G_2 be topological groups, let $G = G_1 \times G_2$, and let σ be an irreducible continuous unitary representation of G . Suppose that every continuous unitary factor representation of G_1 is of type I. Then, for $i = 1, 2$, there is an irreducible continuous unitary representation σ_i of G_i such that σ is equivalent to the representation*

$$(s_1, s_2) \mapsto \sigma_1(s_1) \otimes \sigma_2(s_2)$$

on the Hilbert space $H_{\sigma_1} \otimes H_{\sigma_2}$.

Let $\mathfrak{U}_i$ be the von Neumann algebra generated by $\sigma(G_i)$. Then $\mathfrak{U}_1$ and $\mathfrak{U}_2$ commute and together generate the von Neumann algebra $\mathcal{L}(H_\sigma)$. Every element of the center of $\mathfrak{U}_1$ commutes with $\mathfrak{U}_1$ and $\mathfrak{U}_2$, and is therefore a scalar. Hence $\mathfrak{U}_1$ is a factor, of type I by the hypothesis on G_1 . There are Hilbert spaces H_1, H_2 such that H_σ is identified with $H_1 \otimes H_2$ and

$$\mathfrak{U}_1 = \mathcal{L}(H_1) \otimes \mathbb{C} \quad (\text{A } 36).$$

For every $s_1 \in G_1$, the operator $\sigma(s_1)$ has the form $\sigma_1(s_1) \otimes 1$, where $\sigma_1(s_1)$ is a unitary operator on H_1 , and σ_1 is a continuous unitary representation of G_1 on H_1 . The operators $\sigma_1(s_1)$ generate the von Neumann algebra $\mathcal{L}(H_1)$, so σ_1 is irreducible. We have

$$\mathfrak{U}_2 \subset \mathbb{C} \otimes \mathcal{L}(H_2) \quad (\text{A } 18),$$

and therefore, for every $s_2 \in G_2$, the operator $\sigma(s_2)$ has the form $1 \otimes \sigma_2(s_2)$, where σ_2 is a continuous unitary representation of G_2 on H_2 . If $T \in \mathcal{L}(H_2)$ commutes with the $\sigma_2(s_2)$, then $1 \otimes T$ commutes with $\mathfrak{U}_1$ and $\mathfrak{U}_2$, and is therefore a scalar. Thus T is a scalar and σ_2 is irreducible.

Bibliography: [89, 108].

13.2. The Involutive Algebra $L^1(G)$

13.2.1. Let G be a locally compact group, and let $M^1(G)$ be the algebra, for the convolution product, of bounded complex measures on G . It is a Banach algebra whose identity is the Dirac measure ε_e at the point e . If, for every $\mu \in M^1(G)$, μ^* is defined by

$$d\mu^*(s) = d\overline{\mu}(s^{-1}),$$

then $M^1(G)$ becomes an involutive algebra and $\|\mu^*\| = \|\mu\|$. Thus $M^1(G)$ is an involutive Banach algebra.

13.2.2. Fix once and for all a left Haar measure ds on G . Associating with each $f \in L^1(G)$ the measure

$$d\mu(s) = f(s) ds \in M^1(G)$$

gives an isometric homomorphism Φ from the Banach algebra $L^1(G)$ into the Banach algebra $M^1(G)$. For $g \in \mathcal{K}(G)$, the set of continuous compactly supported complex-valued functions on G , and with Δ denoting the modular function of G , we have

$$\begin{aligned} \int g(s) d\mu^*(s) &= \int \overline{g(s^{-1})} d\mu(s) = \int g(s^{-1}) \overline{f(s)} ds \\ &= \int g(s) \overline{f(s^{-1})} \Delta(s^{-1}) ds. \end{aligned}$$

For every complex-valued function f on G , define f^* by

$$f^*(s) = \overline{f(s^{-1})} \Delta(s^{-1}).$$

The preceding calculation then shows that $d\mu^*(s) = f^*(s)ds$. Thus $f \mapsto f^*$ is an isometric involution on $L^1(G)$, and Φ is a homomorphism of the involutive algebra $L^1(G)$ into the involutive algebra $M^1(G)$. We identify $L^1(G)$ with $\Phi(L^1(G))$. In general, $L^1(G)$ is not a C^* -algebra.

13.2.3. For every complex-valued function f on G , define $\check{f}$ and $\tilde{f}$ by

$$\check{f}(s) = f(s^{-1}), \quad \tilde{f}(s) = \overline{f(s^{-1})}.$$

Thus $f^* = \tilde{f} \Delta^{-1}$. If $a \in G$, put

$${}_a f(s) = f(as), \quad f_a(s) = f(sa).$$

Writing ε_a for the Dirac measure at a , we have

$$\varepsilon_a * f = {}_{a^{-1}} f, \quad f * \varepsilon_a = \Delta(a^{-1}) f_{a^{-1}}.$$

The formula

$$(\varepsilon_a * f)^* = f^* * \varepsilon_a^* = f^* * \varepsilon_{a^{-1}}$$

therefore gives

$$({}_{a^{-1}} f)^* = \Delta(a)(f^*)_a.$$

13.2.4. If G is discrete, $L^1(G)$ has an identity element. If G is separable, $L^1(G)$ is separable.

13.2.5. Let $s \in G$. Let I be the set of neighborhoods of s , ordered by the relation opposite to inclusion. For each $i \in I$, let u_i be a positive function in $L^1(G)$, vanishing outside i and having integral 1. Then, for every $f \in L^1(G)$,

$$\|u_i * f - \varepsilon_s * f\|_1 \longrightarrow 0, \quad \|f * u_i - f * \varepsilon_s\|_1 \longrightarrow 0.$$

[It is enough to verify this for $f \in \mathcal{K}(G)$, when it follows immediately from uniform continuity.] Applying this with $s = e$ shows that $L^1(G)$ has an approximate identity.

Bibliography: [86, 108].

13.3. Representations of G and Representations of $L^1(G)$

13.3.1. Proposition. *Let π be a continuous unitary representation of G . For every $\mu \in M^1(G)$, put*

$$\pi(\mu) = \int \pi(s) d\mu(s) \in \mathcal{L}(H_\pi).$$

Then $\mu \mapsto \pi(\mu)$ is a representation of the involutive algebra $M^1(G)$ on H_π , whose restriction to $L^1(G)$ is nondegenerate.

The assertion that $\mu \mapsto \pi(\mu)$ is a representation of the involutive algebra $M^1(G)$ follows from easy calculations. For example, if $\mu \in M^1(G)$ and $\xi, \eta \in H_\pi$, then

$$\begin{aligned} (\eta | \pi(\mu^*)\xi) &= \overline{(\pi(\mu^*)\xi | \eta)} = \overline{\int (\pi(s^{-1})\xi | \eta) d\bar{\mu}(s)} \\ &= \int (\pi(s)\eta | \xi) d\mu(s) = (\pi(\mu)\eta | \xi) = (\eta | \pi(\mu)^*\xi), \end{aligned}$$

so $\pi(\mu^*) = \pi(\mu)^*$. On the other hand, let $s \in G$, $\xi \in H_\pi$, and $\varepsilon > 0$. With the notation of 13.2.5, there is $i \in I$ such that

$$\|\pi(s')\xi - \pi(s)\xi\| \leq \varepsilon \quad (s' \in i),$$

whence $\|\pi(u_i)\xi - \pi(s)\xi\| \leq \varepsilon$. Thus

$$\pi(u_i) \longrightarrow \pi(s) \tag{1}$$

for the strong operator topology. Applying this with $s = e$, we see that the representation of $L^1(G)$ is nondegenerate.

13.3.2. The representations obtained from $M^1(G)$ and $L^1(G)$ are said to be associated with the given representation of G .

13.3.3. For $s \in G$, we have $\pi(\varepsilon_s) = \pi(s)$. Thus, if $f \in L^1(G)$,

$$\pi({}_a f) = \pi(\varepsilon_{a^{-1}} * f) = \pi(a^{-1})\pi(f),$$

and

$$\pi(f_a) = \pi(\Delta(a^{-1})f * \varepsilon_{a^{-1}}) = \Delta(a^{-1})\pi(f)\pi(a^{-1}).$$

13.3.4. Proposition. *Let H be a Hilbert space and let π be a nondegenerate representation of the involutive algebra $L^1(G)$ on H . Then π is associated with one and only one continuous unitary representation of G .*

Uniqueness follows from formula (1). We prove existence. Let H' be the vector subspace of H spanned by the $\pi(f)(H)$. By hypothesis, H' is dense in H . Let $s \in G$. With the notation of 13.2.5,

$$\|u_i * f - \varepsilon_s * f\|_1 \longrightarrow 0, \quad \|\pi(u_i)\pi(f) - \pi(\varepsilon_s * f)\| \longrightarrow 0.$$

Thus $\pi(u_i)$ converges, for the topology of pointwise convergence on H' , to an operator $\pi(s)$ on H' satisfying

$$\pi(s)\pi(f) = \pi(\varepsilon_s * f). \quad (1)$$

Since $\|\pi(u_i)\| \leq \|u_i\| = 1$, $\pi(s)$ extends uniquely to a continuous linear operator on H , still denoted by $\pi(s)$, such that $\|\pi(s)\| \leq 1$. For $s, t \in G$ and $f \in L^1(G)$, formula (1) gives

$$\begin{aligned} \pi(st)\pi(f) &= \pi(\varepsilon_{st} * f) = \pi(\varepsilon_s * \varepsilon_t * f) \\ &= \pi(s)\pi(\varepsilon_t * f) = \pi(s)\pi(t)\pi(f). \end{aligned}$$

Hence $\pi(st) = \pi(s)\pi(t)$ on H' , and therefore on H . Formula (1) shows that $\pi(e) = 1$ and that $\pi(s)\xi$ depends continuously on s for $\xi \in H'$, and hence for $\xi \in H$. Finally, since $\pi(s)$ and $\pi(s^{-1}) = \pi(s)^{-1}$ decrease norms, $\pi(s)$ is unitary. This proves that π is a continuous unitary representation of G .

We now show that the associated representation of $L^1(G)$ is the original representation. Let $f, g \in L^1(G)$. Since

$$f * g = \int f(s)(\varepsilon_s * g) ds$$

in $L^1(G)$, we have

$$\begin{aligned}\pi(f)\pi(g) &= \pi(f * g) = \int f(s)\pi(\varepsilon_s * g) ds \\ &= \int f(s)\pi(s)\pi(g) ds = \left(\int f(s)\pi(s) ds \right) \pi(g).\end{aligned}$$

Thus

$$\pi(f) = \int f(s)\pi(s) ds.$$

13.3.5. Propositions 13.3.1 and 13.3.4 establish a bijective correspondence between continuous unitary representations of G and nondegenerate representations of $L^1(G)$. Let π be a continuous unitary representation of G , and let π' be the associated representation of $L^1(G)$. The closed vector subspaces invariant under $\pi(G)$ and $\pi'(L^1(G))$ are the same, and the cyclic vectors for $\pi(G)$ and $\pi'(L^1(G))$ are the same, in particular by formula (1). Thus $\pi(G)$ and $\pi'(L^1(G))$ have the same commutant and generate the same von Neumann algebra. In particular, topological irreducibility of π' is equivalent to that of π . We have therefore established a bijective correspondence between irreducible continuous unitary representations of G and topologically irreducible representations of $L^1(G)$. Similarly, saying that π is a factor representation, of type I, and so on, is equivalent to saying that π' is a factor representation, of type I, and so on. If π_1, π_2 are continuous unitary representations of G , and π'_1, π'_2 the associated representations of $L^1(G)$, then the intertwining operators of π_1 and π_2 are the same as those of π'_1 and π'_2 , and

$$\pi'_1 \oplus \pi'_2 = (\pi_1 \oplus \pi_2)',$$

and so forth. Henceforth the same letter will denote a continuous unitary representation of G and the associated representation of $L^1(G)$.

13.3.6. Let λ be the left regular representation of G on $L^2(G)$. If $f \in L^1(G)$, then $\lambda(f)$ is the operator

$$g \mapsto f * g$$

on $L^2(G)$. This representation of $L^1(G)$ is called the left regular representation of $L^1(G)$ on $L^2(G)$. If $\lambda(f) = 0$, then $f * g = 0$ for every $g \in \mathcal{K}(G)$, and hence $f = 0$ by 13.2.5. Thus the left regular representation of $L^1(G)$ on $L^2(G)$ is injective.

If G is commutative and $\widehat{G}$ denotes the dual group of G , the Plancherel isomorphism from $L^2(G)$ onto $L^2(\widehat{G})$ transforms $\lambda(f)$ into multiplication by $\mathcal{F}f$, the Fourier transform of f , on $L^2(\widehat{G})$. Hence

$$\|\lambda(f)\| = \sup_{t \in \widehat{G}} |(\mathcal{F}f)(t)|.$$

In general,

$$\int |f(s)| ds \neq \sup |\mathcal{F}f|,$$

so λ is not isometric; thus $L^1(G)$ is not a C^* -algebra.

Bibliography: [86, 108].

13.4. Positive Forms on $L^1(G)$ and Functions of Positive Type

13.4.1. Section 13.3 leads us to study the representations of $L^1(G)$, or, what amounts to the same thing by 2.4, the continuous positive forms on $L^1(G)$. Now a continuous linear form on $L^1(G)$ is defined by an element of $L^\infty(G)$. This leads us to introduce certain bounded functions on G .

Definition. A continuous complex-valued function φ on G is said to be of positive type if, for arbitrary elements $s_1, \dots, s_n$ of G , the matrix

$$(\varphi(s_i^{-1}s_j))_{1 \leq i, j \leq n}$$

is positive Hermitian. In other words, for arbitrary $s_1, \dots, s_n \in G$ and $\alpha_1, \dots, \alpha_n \in \mathbb{C}$, one has

$$\sum_{i,j=1}^n \alpha_i \overline{\alpha_j} \varphi(s_i^{-1}s_j) \geq 0. \quad (1)$$

13.4.2. The sum of two continuous functions of positive type is of positive type. If φ is continuous and of positive type and $\lambda \geq 0$, then $\lambda\varphi$ is of positive type. For examples of continuous functions of positive type, see 13.6.3.

13.4.3. In formula (1), take $n = 2$, $s_1 = e$, and $s_2 = s \in G$. The matrix

$$\begin{pmatrix} \varphi(e) & \varphi(s) \\ \varphi(s^{-1}) & \varphi(e) \end{pmatrix}$$

must be positive Hermitian. This implies

$$\varphi(s^{-1}) = \overline{\varphi(s)}, \quad |\varphi(s)| \leq \varphi(e)$$

for every $s \in G$. In particular, φ is bounded, and $\|\varphi\|_\infty = \varphi(e)$.

13.4.4. Proposition. *Let φ be a continuous complex-valued function on G . The following conditions are equivalent:*

- (i) φ is of positive type;
- (ii) φ is bounded and, for every bounded complex measure μ on G , one has $\langle \varphi, \mu^* * \mu \rangle \geq 0$, in other words,

$$\iint \varphi(y^{-1}z) d\bar{\mu}(y) d\mu(z) \geq 0;$$

- (iii) for every $f \in \mathcal{K}(G)$, one has $\langle \varphi, f^* * f \rangle \geq 0$, in other words,

$$\iint \varphi(y^{-1}z) \overline{f(y)} f(z) dy dz \geq 0.$$

(ii) $\Rightarrow$ (i). Suppose condition (ii) holds. Let $s_1, \dots, s_n \in G$ and $\alpha_1, \dots, \alpha_n \in \mathbb{C}$, and let μ be the measure on G having mass α_i at s_i . Then

$$0 \leq \iint \varphi(y^{-1}z) d\bar{\mu}(y) d\mu(z) = \sum_{i,j=1}^n \bar{\alpha}_i \alpha_j \varphi(s_i^{-1}s_j),$$

so φ is of positive type.

- (i) $\Rightarrow$ (iii). Suppose that φ is of positive type. Let $f \in \mathcal{K}(G)$. The function

$$(y, z) \mapsto \varphi(y^{-1}z) \overline{f(y)} f(z)$$

on $G \times G$ is continuous and has compact support S ; S is contained in a set $K \times K$, where K is a compact subset of G . The measure induced on K by Haar measure is the vague limit of positive finitely supported measures ν_i of bounded norms; hence the measure induced on $K \times K$ by Haar measure on $G \times G$ is the vague limit of the measures $\nu_i \otimes \nu_i$. Now, if ν_i is given by the masses $\beta_1, \dots, \beta_n$ at the points $s_1, \dots, s_n$, then

$$\iint \varphi(y^{-1}z) \overline{f(y)} f(z) d\nu_i(y) d\nu_i(z) = \sum_{i,j=1}^n \varphi(s_i^{-1}s_j) \overline{f(s_i)} f(s_j) \beta_i \beta_j \geq 0,$$

and, on passing to the limit,

$$\iint \varphi(y^{-1}z) \overline{f(y)} f(z) dy dz \geq 0.$$

(iii) $\Rightarrow$ (ii). Suppose condition (iii) holds. Let μ be a compactly supported complex measure on G . For every $f \in \mathcal{K}(G)$, $\mu * f$ belongs to $\mathcal{K}(G)$, and consequently

$$\begin{aligned} 0 &\leq \langle \varphi, f^* * \mu^* * \mu * f \rangle \\ &= \iiint \varphi(xyzt) f^*(x) f(t) dx dt d\mu^*(y) d\mu(z) \\ &= \iint d\mu^*(y) d\mu(z) \iint \varphi(xyzt) f^*(x) f(t) dx dt. \end{aligned}$$

Taking $f \geq 0$, with integral 1 and support in an increasingly small neighborhood of e , the integral

$$\iint \varphi(xyzt) f^*(x) f(t) dx dt$$

tends to $\varphi(yz)$ uniformly on every compact subset of $G \times G$. It follows on passing to the limit that $\langle \varphi, \mu^* * \mu \rangle \geq 0$. Arguing as in (ii) $\Rightarrow$ (i), we first see that φ is of positive type, and hence bounded. Now let μ be any bounded measure on G . Then μ is the norm limit of compactly supported measures ν . Since φ is bounded, $\langle \varphi, \mu^* * \mu \rangle$ is the limit of $\langle \varphi, \nu^* * \nu \rangle$, and is therefore nonnegative by what precedes.

13.4.5. Theorem.

- (i) Let $\varphi \in L^\infty(G)$, and let ω be the continuous linear form on $L^1(G)$ defined by φ . Then ω is positive if and only if φ is locally almost everywhere equal to a continuous function of positive type.
- (ii) A complex-valued function ψ on G is continuous and of positive type if and only if there exist a continuous unitary representation π of G and a vector $\xi \in H_\pi$ (which may be assumed cyclic for π) such that

$$\psi(s) = (\pi(s)\xi \mid \xi).$$

One then has $\|\psi\|_\infty = (\xi \mid \xi)$.

- (iii) Let π and π' be continuous unitary representations of G , and let ξ (respectively, ξ') be a cyclic vector for π (respectively, π'). If

$$(\pi(s)\xi \mid \xi) = (\pi'(s)\xi' \mid \xi') \quad (s \in G),$$

there exists an isomorphism from H_π onto $H_{\pi'}$ which transforms π into π' and ξ into ξ' .

Suppose that φ is locally almost everywhere equal to a continuous function of positive type. Then, for every $f \in L^1(G)$,

$$\int \varphi(s)(f^* * f)(s) ds \geq 0$$

by 13.4.4. Thus ω is a positive form on $L^1(G)$. Conversely, suppose that $\omega \geq 0$. Form the representation π_ω of $L^1(G)$ and the vector ξ_ω . By 13.3.4, π_ω is associated with a continuous unitary representation of G on the space of π_ω , which we again denote by π_ω . Then, for every $f \in L^1(G)$,

$$\int \varphi(s)f(s) ds = \omega(f) = (\pi_\omega(f)\xi_\omega | \xi_\omega) = \int (\pi_\omega(s)\xi_\omega | \xi_\omega)f(s) ds,$$

and therefore

$$\varphi(s) = (\pi_\omega(s)\xi_\omega | \xi_\omega) \quad \text{locally almost everywhere.} \quad (1)$$

If φ is continuous, equality (1) holds everywhere, since its right-hand side is a continuous function of s . To complete the proof of (i) and (ii), it is enough to prove that, if π is a continuous unitary representation of G and $\xi \in H_\pi$, then $s \mapsto (\pi(s)\xi | \xi)$ is of positive type. But if $\mu \in M^1(G)$, then

$$0 \leq \|\pi(\mu)\xi\|^2 = (\pi(\mu^* * \mu)\xi | \xi) = \int (\pi(s)\xi | \xi) d(\mu^* * \mu)(s),$$

which proves the assertion by 13.4.4.

Now adopt the hypotheses and notation of (iii). We have

$$(\pi(f)\xi | \xi) = (\pi'(f)\xi' | \xi') \quad \text{for every } f \in L^1(G).$$

Consequently there exists an isomorphism Φ from H_π onto $H_{\pi'}$ which transforms ξ into ξ' and $\pi(f)$ into $\pi'(f)$ for every $f \in L^1(G)$ [2.4.1(ii)]. By 13.3.4, Φ transforms $\pi(s)$ into $\pi'(s)$ for every $s \in G$.

13.4.6. Let π be a continuous unitary representation of G , and let $\xi \in H_\pi$. The function $s \mapsto (\pi(s)\xi | \xi)$ is called the function of positive type defined by π and ξ . With π fixed and ξ varying in H_π , one obtains the functions of positive type associated with π . If S is a set of representations of G , the functions of positive type associated with S are the functions of positive type associated with the various elements of S .

Let φ be a continuous function of positive type on G . It defines a positive form ω on $L^1(G)$, and hence a pair (π_ω, ξ_ω) , where π_ω may be regarded as a

continuous unitary representation of G . This pair is also denoted by $(\pi_\varphi, \xi_\varphi)$, and is said to be defined by φ . Up to isomorphism, it is characterized by the facts that

$$\varphi(s) = (\pi_\varphi(s)\xi_\varphi \mid \xi_\varphi) \quad (s \in G)$$

and that ξ_φ is cyclic for π_φ .

13.4.7. Proposition. *Let φ be a continuous function of positive type on G . If $s, t \in G$, then*

$$|\varphi(s) - \varphi(t)|^2 \leq 2\varphi(e)[\varphi(e) - \operatorname{Re} \varphi(s^{-1}t)].$$

Indeed, put $\pi = \pi_\varphi$ and $\xi = \xi_\varphi$. Then

$$\begin{aligned} |\varphi(s) - \varphi(t)|^2 &= |((\pi(s) - \pi(t))\xi \mid \xi)|^2 \\ &\leq \|\xi\|^2 \|\pi(s)\xi - \pi(t)\xi\|^2 \\ &= \varphi(e)[\|\pi(s)\xi\|^2 + \|\pi(t)\xi\|^2 - 2\operatorname{Re}(\pi(s)\xi \mid \pi(t)\xi)] \\ &= \varphi(e)[2\|\xi\|^2 - 2\operatorname{Re}(\pi(s^{-1}t)\xi \mid \xi)] \\ &= 2\varphi(e)[\varphi(e) - \operatorname{Re} \varphi(s^{-1}t)]. \end{aligned}$$

13.4.8. Proposition. *Every translate of a continuous function of positive type is a linear combination of four continuous functions of positive type.*

Let φ be a continuous function of positive type on G , put $\pi = \pi_\varphi$ and $\xi = \xi_\varphi$, and let $a, b \in G$. Set $\pi(a)\xi = \eta$ and $\pi(b)^{-1}\xi = \zeta$. Then

$$\begin{aligned} 4\varphi(bsa) &= 4(\pi(bsa)\xi \mid \xi) = 4(\pi(s)\eta \mid \zeta) \\ &= (\pi(s)(\eta + \zeta) \mid \eta + \zeta) - (\pi(s)(\eta - \zeta) \mid \eta - \zeta) \\ &\quad + i(\pi(s)(\eta + i\zeta) \mid \eta + i\zeta) - i(\pi(s)(\eta - i\zeta) \mid \eta - i\zeta). \end{aligned}$$

13.4.9. Proposition. *Let φ, φ' be two continuous functions of positive type on G . Then $\varphi\varphi'$ is of positive type. More precisely, put $\pi = \pi_\varphi$, $\xi = \xi_\varphi$, $\pi' = \pi_{\varphi'}$, and $\xi' = \xi_{\varphi'}$. Then*

$$\varphi(s)\varphi'(s) = ((\pi \otimes \pi')(s)(\xi \otimes \xi') \mid \xi \otimes \xi').$$

Indeed,

$$\begin{aligned} ((\pi \otimes \pi')(s)(\xi \otimes \xi') \mid \xi \otimes \xi') &= (\pi(s)\xi \otimes \pi'(s)\xi' \mid \xi \otimes \xi') \\ &= (\pi(s)\xi \mid \xi)(\pi'(s)\xi' \mid \xi') = \varphi(s)\varphi'(s). \end{aligned}$$

13.4.10. Proposition. Let φ be a continuous function of positive type on G , and put $\pi = \pi_\varphi$. Every continuous function of positive type associated with π is the uniform limit on G of functions of the form

$$s \mapsto \sum_{i,j=1}^n \lambda_i \bar{\lambda}_j \varphi(s_j^{-1} s s_i), \quad s_1, \dots, s_n \in G, \quad \lambda_1, \dots, \lambda_n \in \mathbb{C}.$$

Let $\xi = \xi_\varphi$, $\eta \in H_\pi$, and $\varepsilon > 0$. There exist $s_1, \dots, s_n \in G$ and $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ such that

$$\left\| \eta - \sum_{i=1}^n \lambda_i \pi(s_i) \xi \right\| \leq \varepsilon.$$

Then, for every $s \in G$,

$$\left| (\pi(s)\eta \mid \eta) - \left(\pi(s) \left(\sum_{i=1}^n \lambda_i \pi(s_i) \xi \right) \mid \sum_{i=1}^n \lambda_i \pi(s_i) \xi \right) \right| \leq \varepsilon \|\eta\| + \varepsilon (\|\eta\| + \varepsilon).$$

But the left-hand side is

$$\left| (\pi(s)\eta \mid \eta) - \sum_{i,j=1}^n \lambda_i \bar{\lambda}_j \varphi(s_j^{-1} s s_i) \right|.$$

13.4.11. Let $f \in L^2(G)$. If λ denotes the left regular representation of G , then

$$\begin{aligned} (\overline{\lambda(s)f} \mid \bar{f}) &= \int \overline{f(s^{-1}t)} f(t) dt \\ &= \int f(t) \tilde{f}(t^{-1}s) dt = (f * \tilde{f})(s). \end{aligned}$$

Thus $f * \tilde{f}$ is a continuous function of positive type associated with $\bar{\lambda}$. If $f \in \mathcal{K}(G)$, then $f * \tilde{f} \in \mathcal{K}(G)$; by passage to the limit, if $f \in L^2(G)$, then $f * \tilde{f}$ is the uniform limit of functions in $\mathcal{K}(G)$, that is, it tends to zero at infinity.

Bibliography: [43, 50, 63, 108, 133, 134].

13.5. Weak Convergence and Compact Convergence of Continuous Functions of Positive Type

13.5.1. Lemma. *Let A be a bounded subset of $L^\infty(G)$, and let $f \in L^1(G)$. If $\varphi \in A$ converges weakly to $\varphi_0 \in A$, then $f * \varphi$ converges to $f * \varphi_0$ for the topology of compact convergence.*

Indeed,

$$(f * \varphi)(s) = \int f(t)\varphi(t^{-1}s) dt = \int f(st)\varphi(t^{-1}) dt = \langle \check{\varphi}, {}_s f \rangle.$$

As φ tends weakly to φ_0 while remaining in A , $\langle \check{\varphi}, g \rangle$ tends to $\langle \check{\varphi}_0, g \rangle$ uniformly on every compact subset of $L^1(G)$. But, as s ranges over a compact subset of G , the set of the ${}_s f$ is strongly compact in $L^1(G)$, since the map $s \mapsto {}_s f$ from G into $L^1(G)$ is strongly continuous.

13.5.2. Theorem. *Let P_1 be the set of continuous functions φ of positive type on G such that $\varphi(e) = 1$. On P_1 , the weak topology $\sigma(L^\infty(G), L^1(G))$ coincides with the topology of compact convergence.*

Since $\|\varphi\|_\infty = 1$ for every $\varphi \in P_1$, it is clear that, on P_1 , the topology of compact convergence is finer than the weak topology.

Now let $\varphi_0 \in P_1$, let K be a compact subset of G , and let $\varepsilon > 0$. We shall prove that if $\varphi \in P_1$ belongs to a suitable weak neighborhood of φ_0 , then

$$|\varphi(s) - \varphi_0(s)| \leq \varepsilon + 4\sqrt{\varepsilon} \quad (s \in K).$$

This will complete the proof.

There exists a compact neighborhood V of e in G such that

$$|\varphi_0(s) - 1| = |\varphi_0(s) - \varphi_0(e)| \leq \varepsilon \quad (s \in V).$$

Let χ be the characteristic function of V , and let $a > 0$ be the measure of V . Let $\mathcal{V}$ be the weak neighborhood of φ_0 in P_1 defined by the condition

$$|\langle \varphi - \varphi_0, \chi \rangle| \leq \varepsilon a,$$

that is, by the condition

$$\left| \int_V (\varphi(s) - \varphi_0(s)) ds \right| \leq \varepsilon a.$$

For $\varphi \in \mathcal{V}$, we have

$$\begin{aligned} \left| \int_V (1 - \varphi(s)) ds \right| &\leq \left| \int_V (1 - \varphi_0(s)) ds \right| + \left| \int_V (\varphi(s) - \varphi_0(s)) ds \right| \\ &\leq 2\varepsilon a. \end{aligned}$$

On the other hand, for $\varphi \in \mathcal{V}$ and $s \in G$,

$$\begin{aligned} |(a^{-1}\chi * \varphi)(s) - \varphi(s)| &= \left| a^{-1} \int \chi(t)\varphi(t^{-1}s) dt - \varphi(s) \right| \\ &= \left| a^{-1} \int_V \varphi(t^{-1}s) dt - a^{-1} \int_V \varphi(s) dt \right| \\ &\leq a^{-1} \int_V |\varphi(t^{-1}s) - \varphi(s)| dt. \end{aligned}$$

By 13.4.7, this is at most

$$\begin{aligned} a^{-1} \int_V \sqrt{2} (1 - \operatorname{Re} \varphi(t))^{1/2} dt &\leq \sqrt{2} a^{-1} \left(\int_V (1 - \operatorname{Re} \varphi(t)) dt \right)^{1/2} \left(\int_V 1 dt \right)^{1/2} \\ &\leq \sqrt{2} a^{-1} \sqrt{2\varepsilon a} \sqrt{a} = 2\sqrt{\varepsilon}. \end{aligned}$$

There now exists (13.5.1) a weak neighborhood $\mathcal{V}'$ of φ_0 in P_1 such that $\varphi \in \mathcal{V}'$ implies

$$|(a^{-1}\chi * \varphi_0)(s) - (a^{-1}\chi * \varphi)(s)| \leq \varepsilon \quad \text{for every } s \in K.$$

Thus, for $\varphi \in \mathcal{V} \cap \mathcal{V}'$,

$$|\varphi(s) - \varphi_0(s)| \leq \varepsilon + 4\sqrt{\varepsilon} \quad \text{for every } s \in K.$$

Bibliography: [121, 187].

13.6. Pure Functions of Positive Type

13.6.1. Definition. A continuous function φ of positive type on G is called pure if π_φ is irreducible.

13.6.2. This is equivalent to saying that the positive form on $L^1(G)$ defined by φ is pure (2.5.4). In view of 13.4.5, it is also equivalent to saying that, in every decomposition $\varphi = \varphi_1 + \varphi_2$ of φ as the sum of two continuous functions of positive type, φ_1 and φ_2 are proportional to φ .

13.6.3. If G is commutative, the continuous irreducible unitary representations of G are the characters of G . The pure continuous functions of positive type on G are therefore the functions of the form $\lambda\chi$, where $\lambda \geq 0$ and χ is a character of G .

13.6.4. Theorem. *Let φ be a continuous function of positive type on G such that $\varphi(e) = 1$. Then, for the topology of compact convergence, φ is a limit of functions of the form*

$$\lambda_1\varphi_1 + \cdots + \lambda_n\varphi_n,$$

where $\varphi_1, \dots, \varphi_n$ are pure continuous functions of positive type equal to 1 at e , and $\lambda_1, \dots, \lambda_n$ are numbers ≥ 0 such that $\lambda_1 + \cdots + \lambda_n = 1$.

The positive form ψ on $L^1(G)$ defined by φ is the weak limit of convex combinations ψ_i of pure states of A and the zero form (2.5.5). Since

$$\liminf_i \|\psi_i\| \geq \|\psi\| = 1 \quad \text{and} \quad \|\psi_i\| \leq 1,$$

we may even suppose, after multiplying the ψ_i by suitable scalars, that $\|\psi_i\| = 1$. Then ψ_i is the positive form on $L^1(G)$ defined by a function $\lambda_1\varphi_1 + \cdots + \lambda_n\varphi_n$, where the φ_j are pure continuous functions of positive type, $\varphi_1(e) = \cdots = \varphi_n(e) = 1$, the λ_j are nonnegative, and $\lambda_1 + \cdots + \lambda_n = 1$. It remains only to apply 13.5.2.

13.6.5. Corollary. *Every continuous complex-valued function on G is, for the topology of compact convergence, a limit of linear combinations of pure functions of positive type.*

It plainly suffices to prove the corollary for a function $f \in \mathcal{K}(G)$. Such a function is a uniform limit on G of functions $f * g$, with $g \in \mathcal{K}(G)$. On the other hand,

$$\begin{aligned} 4f * g &= (f + g) * (f + g)^\sim - (f - g) * (f - g)^\sim \\ &\quad + i(f + ig) * (f + ig)^\sim - i(f - ig) * (f - ig)^\sim. \end{aligned}$$

Finally, every function of the form $h * h^\sim$, with $h \in \mathcal{K}(G)$, is a continuous function of positive type (13.4.11), to which 13.6.4 can be applied.

13.6.6. Corollary. *For every $s \in G$ distinct from e , there exists a continuous irreducible unitary representation π of G such that $\pi(s) \neq 1$.*

There is a continuous complex-valued function on G taking distinct values at s and e , and hence, by 13.6.5, a pure continuous function φ of positive type on G taking distinct values at s and e . We have

$$(\pi_\varphi(s)\xi_\varphi \mid \xi_\varphi) = \varphi(s) \neq \varphi(e) = (\xi_\varphi \mid \xi_\varphi),$$

so $\pi_\varphi(s) \neq 1$, and π_φ is irreducible.

13.6.7. Corollary 13.6.6, due to Gelfand and Raikov, says that a locally compact group has “enough” continuous irreducible unitary representations. This corollary would have failed had we restricted ourselves to finite-dimensional continuous unitary representations. This justifies considering infinite-dimensional continuous unitary representations.

13.6.8. We saw in 13.6.4 how to reconstruct all continuous functions of positive type from the pure continuous functions of positive type. We shall now give a result of the same kind, but in integral form.

Suppose that G is separable. Let B be the convex set of continuous functions of positive type on G that are at most 1 at e . For the weak topology, B is compact and is separable because $L^1(G)$ is separable (B 7). For every $s \in G$, the function $\varphi \mapsto \varphi(s)$ on B , which is bounded in modulus by 1, is Borel, because it is the limit of a sequence of continuous functions

$$\varphi \mapsto \int_G \varphi(t) f_n(t) dt;$$

here the f_n are chosen in $\mathcal{K}(G)$, nonnegative, of integral 1, and with supports in successively smaller neighborhoods of s . On the other hand, the set P of pure continuous functions of positive type equal to 1 at e is the set of extreme points of B with 0 removed (2.5.5); it is therefore a G_δ in B (B 13). With this understood, we have the following result.

Proposition. *Let φ be a continuous function of positive type on G such that $\varphi(e) = 1$. There exists a positive measure of norm 1 on B , concentrated on P , such that*

$$\varphi(s) = \int_P \zeta(s) d\mu(\zeta) \quad \text{for every } s \in G.$$

There exists a positive measure μ of norm 1 on B , concentrated on $P \cup \{0\}$, such that

$$\varphi = \int_{P \cup \{0\}} \zeta d\mu(\zeta),$$

where the integral is taken in the weak sense (B 13). We may plainly suppose that $\mu(\{0\}) = 0$, and hence that μ is concentrated on P . Let $f \in L^1(G)$. We have

$$\int_G \varphi(s) f(s) ds = \int_B d\mu(\zeta) \int_G \zeta(s) f(s) ds. \quad (1)$$

The function $(\zeta, s) \mapsto \zeta(s)$ is continuous on $P \times G$ by 13.5.2, and is therefore measurable on $B \times G$ for the product measure $d\mu(\zeta) ds$; moreover it is bounded in modulus by 1. Thus $(\zeta, s) \mapsto \zeta(s)f(s)$ is integrable for $d\mu(\zeta) ds$, and (1) may be written

$$\int_G \varphi(s)f(s) ds = \int_G f(s) ds \int_B \zeta(s) d\mu(\zeta).$$

Consequently,

$$\varphi(s) = \int_B \zeta(s) d\mu(\zeta) \quad (2)$$

almost everywhere on G . On the other hand, if $(s_1, s_2, \dots)$ is a sequence of elements of G tending to s , then $\zeta(s_n) \rightarrow \zeta(s)$ for every $\zeta \in B$, and hence

$$\int_B \zeta(s_n) d\mu(\zeta) \longrightarrow \int_B \zeta(s) d\mu(\zeta)$$

by the Lebesgue theorem. Thus $\int_B \zeta(s) d\mu(\zeta)$ depends continuously on s , so (2) holds everywhere on G .

Bibliography: [43, 50, 52, 108, 134, 185].

13.7. Measures of Positive Type

13.7.1. Definition. A complex measure μ on G is said to be of positive type if

$$\langle \mu, f * f^\sim \rangle \geq 0 \quad (1)$$

for every $f \in \mathcal{K}(G)$. We then write $\mu \gg 0$.

13.7.2. If $\mu \gg 0$, then $\mu^* = \mu$. Indeed,

$$\langle \mu^*, f * f^\sim \rangle = \overline{\langle \mu, (f * f^\sim)^\sim \rangle} = \overline{\langle \mu, f * f^\sim \rangle} = \langle \mu, f * f^\sim \rangle$$

for every $f \in \mathcal{K}(G)$. Hence, by polarization,

$$\langle \mu^*, f * g^\sim \rangle = \langle \mu, f * g^\sim \rangle$$

for all $f, g \in \mathcal{K}(G)$, and therefore, by passage to the limit, $\langle \mu^*, h \rangle = \langle \mu, h \rangle$ for every $h \in \mathcal{K}(G)$.

13.7.3. Relation (1) may be written

$$\iint f(s) \overline{f(t^{-1}s)} ds d\mu(t) \geq 0$$

for every $f \in \mathcal{K}(G)$, or, on replacing f by $\bar{f}$,

$$\int_G (\mu * f)(s) \overline{f(s)} ds \geq 0$$

for every $f \in \mathcal{K}(G)$.

13.7.4. Suppose that μ is bounded. If λ is the left regular representation of G , the operator $\lambda(\mu)$ is defined. By 13.7.3, saying that $\mu \gg 0$ means that $(\lambda(\mu)f | f) \geq 0$ for every $f \in \mathcal{K}(G)$. Since $\mathcal{K}(G)$ is dense in $L^2(G)$,

$$\mu \gg 0 \iff \lambda(\mu) \geq 0.$$

13.7.5. Let Δ be the modular function of G , and let $f \in \mathcal{K}(G)$. We have

$$\begin{aligned} (f * f^*)(s) \Delta(s)^{1/2} &= \int f(t) \overline{f(s^{-1}t)} \Delta(s^{-1}t) \Delta(s)^{1/2} dt \\ &= \int f(t) \Delta(t)^{1/2} \overline{f(s^{-1}t)} \Delta(s^{-1}t)^{1/2} dt \\ &= ((f \Delta^{1/2}) * (f \Delta^{1/2})^\sim)(s). \end{aligned}$$

Thus, on replacing $f \Delta^{1/2}$ by f , condition (1) is also equivalent to

$$\langle \Delta^{1/2} \mu, f * f^* \rangle \geq 0 \quad \text{for every } f \in \mathcal{K}(G).$$

13.7.6. Definition. A locally integrable function φ on G is said to be of positive type if the measure $\Delta^{-1/2}(s) \varphi(s) ds$ on G is of positive type. We then write $\varphi \gg 0$.

By 13.7.5, this is equivalent to

$$\langle \varphi, f * f^* \rangle \geq 0 \quad \text{for every } f \in \mathcal{K}(G).$$

For continuous φ , this gives precisely the earlier notion (13.4.4). If φ and ψ are two locally integrable functions on G such that $\psi - \varphi \gg 0$, we write $\psi \gg \varphi$.

13.7.7. If $\varphi \gg 0$, then the measures satisfy

$$\Delta^{-1/2}(s) \varphi(s) ds = (\Delta^{-1/2}(s) \varphi(s) ds)^* = \Delta^{1/2}(s) \overline{\varphi(s^{-1})} d(s^{-1}) = \Delta^{-1/2}(s) \overline{\varphi(s^{-1})} ds,$$

and hence

$$\varphi = \varphi^\sim \quad \text{locally almost everywhere.}$$

13.7.8. Let φ be a complex-valued function on G such that $\Delta^{-1/2} \varphi \in L^1(G)$. By 13.7.4, $\varphi \gg 0$ if and only if

$$\lambda(\Delta^{-1/2} \varphi) \geq 0,$$

where λ denotes the left regular representation of G .

13.7.9. Let μ be a measure of positive type on G . For $f, g \in \mathcal{K}(G)$, put

$$(f \mid g)_\mu = \langle \mu, g^\sim * f \rangle = \iint f(t^{-1}s) \overline{g(t^{-1})} dt d\mu(s).$$

Then $\mathcal{K}(G)$ becomes a pre-Hilbert space. Let H_μ be the separated Hilbert-space completion of this pre-Hilbert space. For $s \in G$ and $f \in \mathcal{K}(G)$, put

$$\sigma(s)f = (\varepsilon_s * f)\Delta(s)^{1/2},$$

that is,

$$(\sigma(s)f)(x) = f(s^{-1}x)\Delta(s)^{1/2}.$$

We have

$$\begin{aligned} \|\sigma(s)f\|_\mu^2 &= \iint \overline{(\sigma(s)f)(t^{-1})} (\sigma(s)f)(t^{-1}u) dt d\mu(u) \\ &= \iint \overline{f(s^{-1}t^{-1})} \Delta(s)^{1/2} f(s^{-1}t^{-1}u) \Delta(s)^{1/2} dt d\mu(u) \\ &= \iint \overline{f(t^{-1})} f(t^{-1}u) \Delta(s) \Delta(s)^{-1} dt d\mu(u) \\ &= \iint \overline{f(t^{-1})} f(t^{-1}u) dt d\mu(u) = \|f\|_\mu^2. \end{aligned}$$

Thus $\sigma(s)$ defines a unitary operator on H_μ , which we again denote by $\sigma(s)$. It is clear that σ is a unitary representation of G on H_μ . Moreover, as $s \rightarrow s_0$, $\sigma(s)f$ tends uniformly to $\sigma(s_0)f$, while its support remains contained in a fixed compact set. It follows readily that the unitary representation σ is continuous.

13.7.10. Proposition. Let V be a neighborhood of e , and let μ be a measure of positive type on G such that

$$\langle \mu, f^\sim * f \rangle \leq K \left(\int f(x) dx \right)^2 \quad (1)$$

for all nonnegative $f \in \mathcal{K}(G)$ vanishing outside V , where K is independent of f . Then

$$d\mu(s) = \varphi(s)\Delta^{-1/2}(s) ds,$$

where φ is a continuous function of positive type.

There is a family (f_i) of nonnegative functions in $\mathcal{K}(G)$, indexed by an increasing directed set, vanishing outside V , such that $\int f_i(x) dx = 1$ and

$$\int f_i(x)g(x) dx \longrightarrow g(e) \quad \text{for every } g \in \mathcal{K}(G).$$

By (1), $\|f_i\|_\mu^2 \leq K$ for every i . On the other hand, for every $g \in \mathcal{K}(G)$,

$$\begin{aligned} (f_i | g)_\mu &= \iint \overline{g(t^{-1})} f_i(t^{-1}s) dt d\mu(s) \\ &= \int d\mu(s) \int \overline{g(t^{-1}s^{-1})} f_i(t^{-1}) dt \longrightarrow \int \overline{g(s^{-1})} d\mu(s). \end{aligned}$$

Hence, if Λ denotes the canonical map from $\mathcal{K}(G)$ into H_μ , Λf_i converges weakly to an element $\xi \in H_\mu$ such that

$$(\xi | \Lambda g) = \int \overline{g(s^{-1})} d\mu(s) \quad (g \in \mathcal{K}(G)). \quad (2)$$

Put

$$\varphi(s) = (\sigma(s)\xi | \xi) \quad (s \in G).$$

Then φ is a continuous function of positive type on G (13.4.5). Moreover, (2) implies, for every $t \in G$,

$$\begin{aligned} (\sigma(t)\xi | \Lambda g) &= (\xi | \sigma(t^{-1})\Lambda g) \\ &= \int \overline{(\sigma(t^{-1})g)(s^{-1})} d\mu(s) \\ &= \int \overline{g(ts^{-1})} \Delta^{-1/2}(t) d\mu(s). \end{aligned}$$

Thus, for every $f \in \mathcal{K}(G)$,

$$\begin{aligned} \int f(t)(\sigma(t)\xi | \Lambda g) dt &= \iint f(t) \overline{g(ts^{-1})} \Delta^{-1/2}(t) dt d\mu(s) \\ &= \iint f(t^{-1}) \overline{g(t^{-1}s^{-1})} \Delta^{-1/2}(t) dt d\mu(s) \\ &= \iint \overline{g(t^{-1})} f(t^{-1}s) \Delta^{1/2}(s^{-1}t) dt d\mu(s) \\ &= \langle \mu, g^\sim * (\Delta^{1/2} f) \rangle = (\Delta^{1/2} f | g)_\mu. \end{aligned}$$

It follows that

$$\int f(t)(\sigma(t)\xi | \xi) dt = (\Delta^{1/2} f | \xi).$$

In view of (2) and $\mu^* = \mu$, this says

$$\int f(t)\varphi(t) dt = \int \Delta^{1/2}(t^{-1}) f(t^{-1}) d\bar{\mu}(t) = \int \Delta^{-1/2}(t) f(t) d\mu(t).$$

Since this holds for every $f \in \mathcal{K}(G)$, we conclude that

$$d\mu(t) = \Delta^{-1/2}(t)\varphi(t) dt.$$

13.7.11. Corollary. *Let φ be a continuous function of positive type, and let ψ be a locally integrable function of positive type such that $\varphi \gg \psi$. Then ψ is locally almost everywhere equal to a continuous function of positive type.*

Indeed, for every $f \in \mathcal{K}(G)$,

$$\int \psi(s)\Delta^{-1/2}(s)(f^\sim * f)(s) ds \leq \int \varphi(s)\Delta^{-1/2}(s)(f^\sim * f)(s) ds.$$

If $f \in \mathcal{K}(G)$ is nonnegative and vanishes outside a compact symmetric neighborhood V of e , the right-hand side is at most

$$\begin{aligned} & \sup_{s \in V^2} |\varphi(s)\Delta^{-1/2}(s)| \iint f(t^{-1})f(t^{-1}s) dt ds \\ &= \sup_{s \in V^2} |\varphi(s)\Delta^{-1/2}(s)| \sup_{t \in V} \Delta(t) \left(\int f(t) dt \right)^2, \end{aligned}$$

and it remains only to apply 13.7.10.

Bibliography: [43, 50].

13.8. Square-Integrable Functions of Positive Type

13.8.1. Throughout §13.8, λ denotes the left regular representation of G . Let $f \in L^2(G)$. For every $g \in \mathcal{K}(G)$, $\lambda(g)f \in L^2(G)$. If there is a finite constant M such that

$$\|\lambda(g)f\|_2 \leq M\|g\|_2 \quad \text{for every } g \in \mathcal{K}(G),$$

we shall say that f is a *bounded element* of $L^2(G)$. The map $g \mapsto \lambda(g)f$ then extends uniquely to a continuous linear operator on $L^2(G)$, denoted by $\rho(f)$. For every $g \in \mathcal{K}(G)$,

$$\begin{aligned} (\rho(f)\bar{g} \mid \bar{g}) &= (\lambda(\bar{g})f \mid \bar{g}) = \int (\bar{g} * f)(s)g(s) ds \\ &= \iint \overline{g(t)}f(t^{-1}s)g(s) ds dt = \langle f, g^\sim * g \rangle. \end{aligned}$$

Thus $f \gg 0$ if and only if $\rho(f) \geq 0$ (13.7.6).

13.8.2. Now let f be an element of $L^2(G)$ of positive type, but not necessarily bounded. The operator $g \mapsto \lambda(g)f$, defined on $\mathcal{K}(G)$ and taking values in $L^2(G)$, is nonnegative by the same calculation as in 13.8.1. We denote by $\rho(f)$ its Friedrichs extension (B 23), which is self-adjoint and nonnegative. When f is bounded, this is the operator $\rho(f)$ defined in 13.8.1.

13.8.3. Lemma. *Let f be an element of $L^2(G)$ of positive type.*

- (i) *For every $s \in G$, $\rho(f)$ commutes with $\lambda(s)$.*
- (ii) *If h belongs to the domain D of $\rho(f)$, then $\rho(f)h = h * f$.*

For $g \in \mathcal{K}(G)$ and $s \in G$, we have

$$\varepsilon_s * g \in \mathcal{K}(G), \quad \varepsilon_s * (g * f) = (\varepsilon_s * g) * f,$$

that is, $\lambda(s)\rho(f)g = \rho(f)\lambda(s)g$. Hence

$$\rho(f)|_{\mathcal{K}(G)} = \lambda(s)\rho(f)\lambda(s)^{-1}|_{\mathcal{K}(G)},$$

and consequently

$$\rho(f) = \lambda(s)\rho(f)\lambda(s)^{-1}$$

by B 23.

On the other hand, on D the operator $\rho(f)$ coincides with the adjoint of $\rho(f)|_{\mathcal{K}(G)}$ (B 23). Thus, if $h \in D$ and $g \in \mathcal{K}(G)$,

$$(\rho(f)h | g) = (h | \rho(f)g) = \int h(s) ds \int \overline{g(t)} f(t^{-1}s) dt. \quad (1)$$

Now $h(s)\overline{g(t)}f(t^{-1}s)$ is measurable on $G \times G$ for $ds dt$, and vanishes outside a countable union of sets integrable for $ds dt$. Moreover,

$$\int^* ds \int^* |h(s)g(t)f(t^{-1}s)| dt \leq \int |h(s)|(|g| * |f|)(s) ds < +\infty,$$

because $|h| \in L^2(G)$ and $|g| * |f| \in L^2(G)$. Thus the function is integrable for $ds dt$, and (1) becomes

$$(\rho(f)h | g) = \int \left[\int h(s)f(s^{-1}t) ds \right] \overline{g(t)} dt = \int (h * f)(t) \overline{g(t)} dt,$$

whence $\rho(f)h = h * f$ almost everywhere.

13.8.4. Lemma. *Let $a, b \in L^2(G)$ be two bounded elements of positive type such that $\rho(a)$ and $\rho(b)$ commute. Then $a * b$ is a continuous function of positive type and a bounded element of $L^2(G)$. We have*

$$\rho(a * b) = \rho(a)\rho(b), \quad (a | b) \geq 0.$$

If, in addition, $a \ll b$, then

$$\|b - a\|_2^2 \leq \|b\|_2^2 - \|a\|_2^2.$$

The convolution product of two functions in $L^2(G)$ is continuous on G . Let (a_n) be a sequence of functions in $\mathcal{K}(G)$ tending to a in $L^2(G)$. Then $a_n * b = \rho(b)a_n$ tends to $\rho(b)a$ in $L^2(G)$. On the other hand,

$$\|a_n * b - a * b\|_\infty \longrightarrow 0,$$

and hence $a * b = \rho(b)a \in L^2(G)$.

Let $f \in \mathcal{K}(G)$. We have $f * a_n \in \mathcal{K}(G)$, and $f * a_n$ tends to $f * a$ in $L^2(G)$. Therefore

$$(f * a_n) * b = \rho(b)(f * a_n)$$

tends to $\rho(b)(f * a) = \rho(b)\rho(a)f$ in $L^2(G)$. On the other hand, $f * (a_n * b)$ tends to $f * (a * b)$ in $L^2(G)$. Thus

$$f * (a * b) = \rho(b)\rho(a)f = \rho(a)\rho(b)f \quad \text{for every } f \in \mathcal{K}(G),$$

which proves that $a * b$ is bounded and that $\rho(a * b) = \rho(a)\rho(b)$. Since $\rho(a)$ and $\rho(b)$ are nonnegative and commute, $\rho(a)\rho(b)$ is nonnegative; hence $a * b$ is of positive type. We have

$$(a \mid b) = \int a(s)b(s^{-1}) ds = (a * b)(e) \geq 0.$$

Finally, if $a \ll b$, put $c = b - a \gg 0$. Then c is bounded, and by what precedes,

$$(a \mid a) \leq (a \mid a) + (a \mid c) = (a \mid b),$$

so

$$\|b - a\|_2^2 = \|b\|_2^2 + \|a\|_2^2 - 2(a \mid b) \leq \|b\|_2^2 - \|a\|_2^2.$$

13.8.5. Lemma. Let $a_1, a_2, \dots$ be bounded elements of positive type in $L^2(G)$ such that $a_1 \ll a_2 \ll \dots$ and the $\rho(a_i)$ commute pairwise. If

$$\sup_i \|a_i\|_2 < +\infty,$$

then the a_i have a strong limit in $L^2(G)$.

By 13.8.4, the numbers $\|a_i\|_2$ form an increasing, and therefore convergent, sequence. Another application of 13.8.4 shows that the a_i form a Cauchy sequence.

13.8.6. Theorem. *Let φ be a continuous square-integrable function of positive type on G . There exists a square-integrable function ψ of positive type on G such that*

$$\varphi = \psi * \psi = \psi * \psi^\sim.$$

First suppose that φ is bounded. Multiplying φ by a positive constant if necessary, we may suppose that $0 \leq \rho(\varphi) \leq 1$. Let $p_1, p_2, \dots$ be a sequence of non-negative polynomials on $[0, 1]$, vanishing at 0, and converging uniformly and increasingly to $\sqrt{t}$ on $[0, 1]$. By 13.8.4, we may form

$$\psi_1 = p_1(\varphi), \quad \psi_2 = p_2(\varphi), \quad \dots,$$

where the multiplication used is convolution. These elements belong to $L^2(G)$ and are bounded, and

$$\rho(\psi_1) = p_1(\rho(\varphi)), \quad \rho(\psi_2) = p_2(\rho(\varphi)), \quad \dots$$

We see that

$$0 \leq \rho(\psi_1) \leq \rho(\psi_2) \leq \dots,$$

and hence $0 \ll \psi_1 \ll \psi_2 \ll \dots$; moreover, the $\rho(\psi_n)$ commute pairwise. Since $p_n(t)^2 \leq t$ on $[0, 1]$, we have $\rho(\psi_n)^2 \leq \rho(\varphi)$, and hence $\psi_n * \psi_n \ll \varphi$. Consequently,

$$(\psi_n * \psi_n)(e) = \|\psi_n\|_2^2 \leq \varphi(e).$$

Thus the ψ_n converge strongly to an element $\psi \in L^2(G)$ (13.8.5). At the same time, $\rho(\psi_n) = p_n(\rho(\varphi))$ converges in norm to $\rho(\varphi)^{1/2}$. For $f \in \mathcal{K}(G)$, $f * \psi_n = \rho(\psi_n)f$ tends in $L^2(G)$ both to $f * \psi$ and to $\rho(\varphi)^{1/2}f$. Hence ψ is bounded and of positive type, and

$$\rho(\psi) = \rho(\varphi)^{1/2}.$$

This implies

$$\rho(\varphi) = \rho(\psi)^2 = \rho(\psi * \psi),$$

and therefore $\varphi = \psi * \psi$.

We now pass to the general case. Let

$$\rho(\varphi) = \int_0^{+\infty} \zeta dE_\zeta$$

be the spectral decomposition of $\rho(\varphi)$. By 13.8.3(i), the projections E_ζ commute with the $\lambda(s)$, $s \in G$. Put $\varphi_\zeta = E_\zeta \varphi$. For every $g \in \mathcal{K}(G)$,

$$(g * \varphi_\zeta^\sim)(t) = \int g(s) \varphi_\zeta^\sim(s^{-1}t) ds = \int g(s) \overline{\varphi_\zeta(t^{-1}s)} ds$$

$$\begin{aligned}
&= (g \mid \lambda(t)E_\zeta\varphi) = (E_\zeta g \mid \lambda(t)\varphi) \\
&= \int (E_\zeta g)(s) \overline{\varphi(t^{-1}s)} ds = \int (E_\zeta g)(s) \varphi(s^{-1}t) ds \\
&= (E_\zeta g * \varphi)(t).
\end{aligned}$$

By 13.8.3(ii), this is equal to $(\rho(\varphi)E_\zeta g)(t)$. Hence $\varphi_\zeta^\sim$ is bounded and

$$\rho(\varphi_\zeta^\sim) = \rho(\varphi)E_\zeta \geq 0,$$

so $\varphi_\zeta^\sim = \varphi_\zeta \gg 0$. Moreover, $\rho(\varphi_\zeta) \leq \rho(\varphi)$; hence $\varphi_\zeta \ll \varphi$, and therefore φ_ζ is continuous by 13.7.11. By the first part of the proof there is a bounded element ω_ζ of positive type such that

$$\varphi_\zeta = \omega_\zeta * \omega_\zeta.$$

Take in particular $\zeta = 1, 2, \dots, n, \dots$. The $\rho(\varphi_n)$ commute pairwise and increase; consequently so do

$$\rho(\omega_n) = \rho(\varphi_n)^{1/2}.$$

Thus $\omega_1 \ll \omega_2 \ll \dots$. On the other hand,

$$\|\omega_n\|_2^2 = \varphi_n(e) \leq \varphi(e),$$

so the ω_n have a strong limit $\omega \gg 0$ (13.8.5). Then $\varphi_n = \omega_n * \omega_n$ converges uniformly on G to $\omega * \omega$, while $\varphi_n = E_n\varphi$ converges strongly to φ in $L^2(G)$. Hence $\varphi = \omega * \omega$.

Bibliography: [50].

13.9. C^* -Algebra of a Locally Compact Group

13.9.1. Since $L^1(G)$ is an involutive Banach algebra with an approximate identity, we may form its enveloping C^* -algebra (2.7.2). This C^* -algebra is called the C^* -algebra of G and is denoted by $C^*(G)$.

For $f \in L^1(G)$, put

$$\|f\|' = \sup_{\pi} \|\pi(f)\| \leq \|f\|_1,$$

where π ranges over the nondegenerate representations of $L^1(G)$ or, equivalently, over the continuous unitary representations of G . Then $f \mapsto \|f\|'$ is a seminorm on $L^1(G)$ (2.7.1), and indeed a norm, since $L^1(G)$ has an injective representation (13.3.6). The C^* -algebra of G is precisely the completion of $L^1(G)$ for this norm.

13.9.2. If G is discrete, $C^*(G)$ has an identity element. If G is separable, $C^*(G)$ is separable (13.2.4).

13.9.3. By 2.7.4 and 13.3.5, there is a bijective correspondence between continuous unitary representations of G and nondegenerate representations of $C^*(G)$. Everything said in 13.3.5 remains valid with $L^1(G)$ replaced by $C^*(G)$.

The left regular representation of G corresponds to a representation of $C^*(G)$ called the left regular representation of $C^*(G)$ on $L^2(G)$.

13.9.4. The group G is called liminal, postliminal, antiliminal, or of type I, according as $C^*(G)$ is liminal, postliminal, antiliminal, or of type I.

For G to be liminal, it is necessary and sufficient that, for every irreducible continuous unitary representation π of G and every $f \in L^1(G)$, the operator $\pi(f)$ be compact.

Suppose that G is postliminal. Then, for every irreducible continuous unitary representation π of G , the norm closure of $\pi(L^1(G))$ contains $\mathcal{LC}(H_\pi)$ (4.3.7). The converse is true if G is separable (9.1).

For G to be of type I, it is necessary and sufficient that, for every continuous unitary representation π of G , the von Neumann algebra generated by $\pi(G)$ be of type I. If G is postliminal, then G is of type I (5.5.2). If G is separable, the following conditions are equivalent:

- 1° G is of type I;
- 2° for every factor continuous unitary representation π of G , the factor generated by $\pi(G)$ is of type I;
- 3° G is postliminal.

This follows from 9.1.

Bibliography: [27, 77].

13.10. Hilbert Algebra of a Locally Compact Group

Throughout this section, G denotes a unimodular locally compact group.

13.10.1. Under convolution and the involution $f \mapsto f^*$, $\mathcal{K}(G)$ is an involutive algebra. Endow $\mathcal{K}(G)$ with the scalar product

$$(f \mid g) = \int f(s) \overline{g(s)} ds.$$

It is easy to see that $\mathcal{K}(G)$ then becomes a Hilbert algebra; its completed Hilbert space is precisely $L^2(G)$. The completed Hilbert algebra A of bounded elements (A 57) is called the Hilbert algebra of G ; one has

$$\mathcal{K}(G) \subset A \subset L^2(G).$$

It is clear that A is stable under $f \mapsto \bar{f}$, and hence also under $f \mapsto \tilde{f} = (\bar{f})^*$.

13.10.2. A continuous linear operator on $L^2(G)$ commutes with the left translation operators if and only if it commutes with the operators of left convolution by the elements of $\mathcal{K}(G)$ (this follows, for example, from 13.3.5 applied to the left regular representation of G). Thus $\mathfrak{U}(A)$ is the von Neumann algebra generated by the left translation operators on $L^2(G)$. Similarly, $\mathfrak{B}(A)$ is the von Neumann algebra generated by the right translation operators on $L^2(G)$.

13.10.3. If $f \in A$, recall that U_f, V_f denote the continuous linear operators on $L^2(G)$ extending left and right multiplication by f in A . If $f \in A$ and $g \in L^2(G)$, then

$$U_f g = f * g, \quad V_f g = g * f.$$

Indeed, first suppose that $g \in \mathcal{K}(G)$, and let (f_n) be a sequence of elements of $\mathcal{K}(G)$ converging to f in mean square. Then $V_g f_n = f_n * g$ converges in mean square both to $V_g f = U_f g$ and to $f * g$; hence $U_f g = f * g$. In the general case, let (g_n) be a sequence of elements of $\mathcal{K}(G)$ converging to g in mean square. Then $U_f g_n$ converges to $U_f g$ in mean square, and $f * g_n$ converges uniformly on G to $f * g$. Since $U_f g_n = f * g_n$ by the first part of the argument, we obtain $U_f g = f * g$.

13.10.4. Recall that the map $f \mapsto f^*$ on $L^2(G)$ commutes with every Hermitian element of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$, the common center of $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$ (A 54). Recall also that $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$ are each other's commutants and are semifinite von Neumann algebras (A 60).

13.10.5. Proposition. *Let G be a unimodular locally compact group and A its Hilbert algebra. The following conditions are equivalent:*

- (i) *The von Neumann algebra $\mathfrak{U}(A)$ is finite;*
- (i') *the von Neumann algebra $\mathfrak{B}(A)$ is finite;*
- (ii) *there exists in G a fundamental system of compact neighborhoods of e invariant under the inner automorphisms of G .*

Conditions (i) and (i') are equivalent for every Hilbert algebra (A 63).

Suppose that there exists a fundamental system $(V_i)_{i \in I}$ of compact neighborhoods of e invariant under the inner automorphisms of G . Let f_i be the charac-

teristic function of V_i . The f_i are central elements of A [they belong to the center of $L^1(G)$]. On the other hand, every $f \in L^2(G)$ is a strong limit of elements $f_i * f$. Thus the characteristic projection of A is 1 (A 62). Hence $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$ are finite von Neumann algebras (A 63).

Conversely, suppose that $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$ are finite von Neumann algebras. The characteristic projection of A is 1 (A 63). Let $s \in G$, $s \neq e$. The operator $f \mapsto {}_s f$ on $L^2(G)$ is not the identity; it commutes with right translations, while the set of right translates of the elements of $L^2(G)$ central relative to A is total in $L^2(G)$ (A 62). Hence there exists an $f \in L^2(G)$ central relative to A such that ${}_s f \neq f$. Since f is central, ${}_t f = f_t$ for every $t \in G$ (A 62); hence f is invariant under the inner automorphisms of G . Consider the function

$$t \mapsto g(t) = ({}_t f \mid f)$$

on G . It is continuous, invariant under inner automorphisms, tends to zero at infinity, and $g(s) \neq g(e)$. If W is a compact neighborhood of $g(e)$ in $\mathbb{C}$ containing neither 0 nor $g(s)$, then the condition $g(t) \in W$ defines a compact neighborhood of e , invariant under inner automorphisms and not containing s . Thus the intersection of the compact neighborhoods of e invariant under inner automorphisms is $\{e\}$. Consequently every neighborhood of e contains a compact neighborhood invariant under inner automorphisms.

Bibliography: [53, 54, 95, 96, 136, 137].

13.11. Complements

13.11.1. Let G be a locally compact group and $f \in L^1(G)$. If $f(s) \geq 0$ for every $s \in G$, the norm of f in $L^1(G)$ is the same as its norm in $C^*(G)$. (Consider the trivial one-dimensional representation of G .) [27].

13.11.2. Let G be a locally compact group, π a continuous unitary representation of G , S a set of continuous unitary representations of G , and K the set of $\xi \in H_\pi$ such that the function

$$s \mapsto (\pi(s)\xi \mid \xi)$$

is, uniformly on each compact set, a limit of sums of functions of positive type associated with S . Then K is a closed vector subspace of H_π invariant under $\pi(G)$. [33].

13.11.3. Let G be a locally compact group, λ the left regular representation of G , and π a continuous unitary representation of G . Then

$$\lambda \otimes \pi \simeq (\dim \pi)\lambda.$$

[Consider the isomorphism from $L^2(G) \otimes H_\pi$ onto $L^2(G) \otimes H_\pi = L^2_{H_\pi}(G)$ which sends $f \otimes \xi$ to the function $s \mapsto f(s)\pi(s^{-1})\xi$.] [31].

13.11.4. Let G be the locally compact group of affine transformations of $\mathbb{R}$, and let G' be the closed normal subgroup of translations. If π is a continuous unitary representation of G and χ is a nontrivial character of G' , then $\pi|_{G'}$ does not contain χ . (If $\pi|_{G'}$ contained χ , then, by the action of the inner automorphisms of G on G' , $\pi|_{G'}$ would contain all the nontrivial characters of G' ; the corresponding subspaces of H_π would be pairwise orthogonal.) It follows that a nontrivial character of G' cannot be extended to a continuous function of positive type on G . (Douady, unpublished.)

13.11.5. Let G be a locally compact group and Δ its modular function. If a continuous function of positive type on G is integrable with respect to

$$\Delta(s)^{-1/2} ds,$$

then it is square-integrable with respect to ds . [50].

***13.11.6.** Let G be a locally compact group, and let E be the set of linear combinations of continuous functions of positive type. For $\varphi \in E$, let K_φ^g (respectively K_φ^d) be the closed convex hull of the set of left (respectively right) translates of φ in the space of bounded continuous complex-valued functions on G , endowed with the uniform norm. Then K_φ^g and K_φ^d contain one and the same constant $M(\varphi)$, and no other constant. For a continuous function φ of positive type to satisfy $M(\varphi) = 0$, it is necessary and sufficient that π_φ have no nonzero finite-dimensional subrepresentation. [50].

13.11.7. Let G_1, G_2 be two topological groups.

- a. Let π_1 be an irreducible continuous unitary representation of G_1 , and let π_2, π'_2 be continuous unitary representations of G_2 . If the representations

$$(s_1, s_2) \mapsto \pi_1(s_1) \otimes \pi_2(s_2) \quad \text{and} \quad (s_1, s_2) \mapsto \pi_1(s_1) \otimes \pi'_2(s_2)$$

are equivalent, then π_2 and π'_2 are equivalent.

- b. Let π be a continuous unitary representation of $G_1 \times G_2$. If π is a factor representation, then $\pi|_{G_1}$ and $\pi|_{G_2}$ are factor representations.

- c. Let π be a continuous unitary representation of $G_1 \times G_2$, and put $\pi_1 = \pi|_{G_1}$ and $\pi_2 = \pi|_{G_2}$. Suppose that π_1 is a type I factor representation. Then there exist $\pi'_1 \simeq \pi_1$ and $\pi'_2 \simeq \pi_2$ such that π is equivalent to the representation

$$(s_1, s_2) \longmapsto \pi'_1(s_1) \otimes \pi'_2(s_2).$$

- d. If G_1 and G_2 are of type I, then $G_1 \times G_2$ is of type I. [89]

13.11.8. Let G be a locally compact group and π a unitary representation of G . Suppose that, for all $\xi, \eta \in H_\pi$, the function $s \mapsto (\pi(s)\xi \mid \eta)$ is measurable for Haar measure on G . We may define

$$\pi(f) = \int f(s)\pi(s) ds$$

for every $f \in L^1(G)$. Let $K \subset H_\pi$ be the essential subspace for the representation π of $L^1(G)$. Then K and $H_\pi \ominus K$ are invariant under $\pi(G)$. The representation $s \mapsto \pi(s)|_K$ of G is continuous. For $\xi, \eta \in H_\pi \ominus K$, one has

$$(\pi(s)\xi \mid \eta) = 0$$

locally almost everywhere on G . The space $H_\pi \ominus K$ is zero or nonseparable. It may be nonzero. [140].

13.11.9. Let G be a topological group and π a continuous unitary representation of G . In

$$H_\pi \otimes H_\pi \otimes \cdots \otimes H_\pi \quad (n \text{ factors}),$$

the symmetric tensors span a closed vector subspace K invariant under $\rho = \pi \otimes \pi \otimes \cdots \otimes \pi$. The subrepresentation $\rho|_K$ of ρ is called the n th symmetric tensor power of π . The antisymmetric tensor powers are defined similarly.

***13.11.10.** Let G be a connected real Lie group and π an irreducible unitary representation of G continuous for the norm topology on operators. Then π is finite-dimensional. [143].

13.11.11. Let G be a topological group. A unitary representation π of G is called *real* if there exists a closed real vector subspace K of H_π such that H_π is the direct sum of K and iK , the scalar product is real on K , and $\pi(G)$ leaves K invariant (in other words, H_π is the complexification of a real Hilbert space K , and π is the complexification of a representation of G on K). Equivalently, there exists a

conjugation J on H_π which commutes with $\pi(G)$. If π is real, then $\pi \simeq \overline{\pi}$, but the converse is false.

***13.11.12.** A connected real semisimple Lie group and a connected real nilpotent Lie group are liminal (cf. 15.5.6). A real algebraic linear group is postliminal. Fairly broad classes of postliminal solvable Lie groups are known, but Mautner has given an example of a solvable Lie group which is not postliminal. The non-commutative connected solvable real Lie group of dimension 2 is postliminal but not liminal. A countable discrete group is of type I if and only if it is an extension of a finite group by an abelian group. Problems: Is a p -adic algebraic group postliminal? Is the C^* -algebra of a liminal real Lie group of generalized continuous trace? [5, 7, 10, 30, 59, 79, 81, 146, 155, 209].

***13.11.13.** Let G be a locally compact group and π an irreducible continuous unitary representation of G . Then $\pi(L^1(G))$ is not, in general, algebraically irreducible.

13.11.14.

- a. Let G be a discrete group, A its Hilbert algebra, $\mathfrak{U} = \mathfrak{U}(A)$, which is a finite von Neumann algebra, and t the natural trace on $\mathfrak{U}^+$ defined by A . Since the operator $1 \in \mathfrak{U}$ corresponds to the characteristic function of e , one has $\mathfrak{m}_t = \mathfrak{n}_t = \mathfrak{U}$, and every element of $\mathfrak{U}$ is of the form U_x , where $x \in L^2(G)$. We shall again denote by t the linear extension of t to $\mathfrak{U}$. If $U_x \in \mathfrak{U}$, with $x \in L^2(G)$, then

$$t(U_x) = x(e).$$

- b. Let E_{I_i} (respectively E_{II}) be the largest projection in the center of $\mathfrak{U}$ such that the corresponding induced algebra is of type I_i (respectively of type II), where $i = 1, 2, \dots$. One has

$$E_{II} + \sum_i E_{I_i} = 1.$$

Put $t(E_{II}) = r$ and $t(E_{I_i}) = r_i$; then

$$r + \sum_i r_i = 1.$$

- c. Let C be the commutator subgroup of G , and let φ_C be its characteristic function. If C is infinite, then $r_1 = 0$. If C is finite, then

$$r_1 = (\text{Card } C)^{-1},$$

and E_{I_1} is the element of $\mathfrak{U}$ defined by the function $(\text{Card } C)^{-1} \varphi_C$ on G .

- d. If G is infinite and C coincides with the center of G and has prime order p , then $\mathfrak{U}$ is the product of an abelian von Neumann algebra and $p - 1$ factors of type II_1 . One has

$$r_1 = \frac{1}{p}, \quad r = \frac{p-1}{p}.$$

- e. Let (G_i) be an infinite family of noncommutative finite groups. Let G be the set of elements of $\prod_i G_i$ all but finitely many of whose components are equal to e , and regard G as a discrete group. Then $\mathfrak{U}$ is of type II_1 .
- f. Let G be a discrete group, and let G_0 be the subgroup of G which is the union of the finite conjugacy classes of G . If G/G_0 is infinite, then $\mathfrak{U}$ is of type II_1 . If $G_0 = \{e\}$ and G is infinite, then $\mathfrak{U}$ is a factor of type II_1 .
[76, 90, 98, 99]

§ 14. — Irreducible Square-Integrable Representations

Throughout this section, G denotes a unimodular locally compact group.

14.1. Definition of Square-Integrable Representations

14.1.1. Lemma. *Let φ be a continuous square-integrable function of positive type on G . The representation π_φ is contained in the left regular representation.*

There exists $\psi \in L^2(G)$ such that $\varphi = \overline{\psi} * \check{\psi}$ (13.8.6). In other words,

$$\varphi(s) = \int \overline{\psi(t)} \psi(s^{-1}t) dt = (\lambda(s)\psi \mid \psi),$$

where λ denotes the left regular representation of G . Hence π_φ is equivalent to the subrepresentation of λ which has ψ as a cyclic vector [13.4.5(iii)].

14.1.2. Lemma. *Let λ be the left regular representation of G , and let π be a representation contained in λ , with $H_\pi \neq 0$. There exists a nonzero square-integrable function of positive type associated with π .*

Let A be the Hilbert algebra of G , and let K be a closed vector subspace of $L^2(G)$ invariant under $\lambda(G)$. Then

$$P_K \in \mathfrak{U}(A)' = \mathfrak{B}(A).$$

Consequently $P_K(A) \subset A$ (A 59). If $K \neq 0$, there is therefore a nonzero $\psi \in K \cap A$. The subrepresentation of λ corresponding to K has as an associated function of positive type

$$s \longmapsto (\pi(s)\psi \mid \psi) = \overline{\psi} * \check{\psi}.$$

Now $\overline{\psi} \in A$, and hence $\overline{\psi} * \check{\psi} \in L^2(G)$.

14.1.3. Definition. *An irreducible continuous unitary representation π of G is called square-integrable if there exists $\xi \in H_\pi$, $\xi \neq 0$, such that the coefficient*

$$s \longmapsto (\pi(s)\xi \mid \xi)$$

of π is square-integrable on G .

[We shall see that in this case all the coefficients of π are square-integrable (14.3.2).]

By 14.1.1 and 14.1.2, these representations are precisely the irreducible representations contained in the regular representation.

14.1.4. Suppose that G is commutative. An element χ of $\widehat{G}$ is a character of G , and $|\chi|^2 = 1$. Thus, if G is compact, every element of $\widehat{G}$ is square-integrable (cf. §15); and if G is not compact, no element of $\widehat{G}$ is square-integrable.

Bibliography: [51, 60].

14.2. Square-Integrable Representations and Minimal Bi-Invariant Subspaces of $L^2(G)$

14.2.1. Let A be the Hilbert algebra of G , and let K be a closed vector subspace of $L^2(G)$. To say that K is invariant under the left (respectively right) regular representation of G is equivalent to saying that

$$P_K \in \mathfrak{B}(A) \quad (\text{respectively } P_K \in \mathfrak{U}(A)).$$

To say that K is invariant under both the left and right regular representations, or, as we shall say, *bi-invariant*, is equivalent to saying that P_K belongs to $\mathfrak{U}(A) \cap \mathfrak{B}(A)$, the common center of $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$. We shall establish a bijective correspondence between the classes of square-integrable irreducible representations of G and certain minimal bi-invariant subspaces of $L^2(G)$, in other words, certain minimal projections of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$.

14.2.2. Proposition. *Let λ be the left regular representation of G , and let H be a closed vector subspace of $L^2(G)$ minimally invariant under λ , so that the subrepresentation σ of λ on H is irreducible. Let $E = P_H \in \mathfrak{B}(A)$, and let F be the central support of E in $\mathfrak{U}(A) \cap \mathfrak{B}(A)$.*

- (i) F is a minimal projection of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$.
- (ii) $\mathfrak{U}(A)_F$ and $\mathfrak{B}(A)_F$ are type I factors commuting with one another.
- (iii) Every element of $F(L^2(G))$ belongs to A .

Assertion (i) follows from 5.2.7. By condition (v) of 5.3.1, the subrepresentation of λ defined by F is quasi-equivalent to σ , and is therefore a type I factor representation (5.4.11), which proves (ii). Finally, (iii) follows from A 65 and A 67.

14.2.3. Proposition. *Retain the notation of 14.2.2. Let σ' be another irreducible subrepresentation of λ , and let F' be the corresponding minimal projection of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$. The representations σ and σ' are equivalent (respectively inequivalent) if and only if F and F' are equal (respectively orthogonal).*

If $F = F'$, then σ and σ' are quasi-equivalent [condition (v) of 5.3.1], and hence equivalent [5.3.3(ii)]. If F and F' are orthogonal, then σ and σ' are disjoint [condition (iii) of 5.2.1].

14.2.4. We call F [respectively $F(L^2(G))$] the central projection (respectively the bi-invariant subspace) associated with σ , or with any equivalent representation.

14.2.5. Proposition. *Let F be a minimal projection of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$ such that $\mathfrak{U}(A)_F$ and $\mathfrak{B}(A)_F$, which are factors, are of type I. There exists at least one minimal projection E of $\mathfrak{B}(A)$ dominated by F . The subrepresentation σ of λ defined by E is irreducible and square-integrable; its class does not depend on the choice of E , and the associated central projection is F .*

Every type I factor has minimal projections (A 36), which proves the existence of E . Since σ is an irreducible subrepresentation of λ , it is square-integrable (14.1.3); its class does not depend on the choice of E (14.2.3); and its associated central projection is evidently F .

Bibliography: [51, 60].

14.3. Coefficients of Square-Integrable Representations

14.3.1. Proposition. *Let σ be an irreducible subrepresentation of the left regular representation of G . For $\xi, \eta \in H_\sigma$, let*

$$s \longmapsto \varphi_{\xi, \eta}^\sigma(s) = (\sigma(s)\xi \mid \eta)$$

be the corresponding coefficient of σ . Then

$$\check{\varphi}_{\xi, \eta}^\sigma = \xi * \eta^* \in L^2(G).$$

The closed vector subspace of $L^2(G)$ spanned by the $\check{\varphi}_{\xi, \eta}^\sigma$, $\xi, \eta \in H_\sigma$, is the bi-invariant subspace associated with σ .

Indeed,

$$\begin{aligned} \check{\varphi}_{\xi, \eta}^\sigma(s) &= (\xi \mid \sigma(s)\eta) = \int \xi(t) \overline{\eta(s^{-1}t)} dt \\ &= \int \xi(t) \eta^*(t^{-1}s) dt = (\xi * \eta^*)(s). \end{aligned}$$

Let K be the bi-invariant subspace associated with σ . Then ξ and η , and hence η^* , belong to K , and therefore to the Hilbert algebra A of G [14.2.2(iii)]. Thus

$$\xi * \eta^* = V_{\eta^*} \xi \in K.$$

On the other hand, since H_σ is invariant under $\mathfrak{U}(A)$, the set of the $\xi * \eta^*$, $\xi, \eta \in H_\sigma$, is stable under left or right multiplication by elements of A . If K' denotes the closed vector subspace spanned by this set, then K' is a closed bi-invariant vector subspace of $L^2(G)$ contained in K . Since K is a minimal closed bi-invariant vector subspace [14.2.2(i)], one has $K' = K$.

14.3.2. Corollary. *Let σ be a square-integrable irreducible representation of G . All the coefficients of σ are square-integrable.*

14.3.3. Theorem. *Let σ be a square-integrable irreducible representation of G . For $\xi, \eta \in H_\sigma$, put $\varphi_{\xi, \eta}(s) = (\sigma(s)\xi \mid \eta)$, $s \in G$. There exists a unique constant d_σ , $0 < d_\sigma < +\infty$, such that*

$$\int \varphi_{\xi, \eta}(s) \overline{\varphi_{\xi', \eta'}(s)} ds = d_\sigma^{-1} (\xi \mid \xi') (\eta \mid \eta'), \quad (1)$$

$$\varphi_{\xi, \eta} * \varphi_{\xi', \eta'} = d_\sigma^{-1} (\xi \mid \eta') \varphi_{\xi', \eta}, \quad (2)$$

for all $\xi, \eta, \xi', \eta' \in H_\sigma$. In particular, if φ is a function of positive type associated with σ and equal to 1 at e , then

$$\|\varphi\|_2^2 = d_\sigma^{-1}, \quad (3)$$

$$\varphi * \varphi = d_\sigma^{-1} \varphi. \quad (4)$$

Let A be the Hilbert algebra of G . We may suppose that σ is the subrepresentation of the left regular representation defined by a minimal projection E of $\mathfrak{B}(A)$. Using the equality $\check{\varphi}_{\xi, \eta} = \xi * \eta^* = V_{\eta^*} \xi$ of 14.3.1, we have

$$\begin{aligned} \int \varphi_{\xi, \eta}(s) \overline{\varphi_{\xi', \eta'}(s)} ds &= \int \check{\varphi}_{\xi, \eta}(s^{-1}) \overline{\check{\varphi}_{\xi', \eta'}(s^{-1})} ds \\ &= (V_{\eta^*} \xi \mid V_{\eta'^*} \xi') = (V_{\eta'} V_{\eta^*} \xi \mid \xi') \\ &= (E V_{\eta'} V_{\eta^*} E \xi \mid \xi'). \end{aligned}$$

Since E is a minimal projection of $\mathfrak{B}(A)$, there exists $\lambda_{\eta, \eta'} \in \mathbb{C}$ such that

$$E V_{\eta'} V_{\eta^*} E = \lambda_{\eta, \eta'} E.$$

Consequently

$$\int \varphi_{\xi, \eta}(s) \overline{\varphi_{\xi', \eta'}(s)} ds = \lambda_{\eta, \eta'} (\xi \mid \xi').$$

By 13.1.1, the integral is also equal to

$$\int \varphi_{\eta', \xi'}(s^{-1}) \overline{\varphi_{\eta, \xi}(s^{-1})} ds = \lambda_{\xi', \xi} (\eta' \mid \eta).$$

Thus $\lambda_{\eta, \eta'} = \lambda_\sigma(\eta' \mid \eta)$, where λ_σ is a finite constant. Taking $\xi = \xi' = \eta = \eta' \neq 0$ shows that $\lambda_\sigma > 0$. It is then enough to put $\lambda_\sigma^{-1} = d_\sigma$ to obtain (1).

In particular,

$$\begin{aligned} (\varphi_{\xi, \eta} * \varphi_{\xi', \eta'})(s) &= \int (\sigma(t)\xi \mid \eta)(\sigma(t^{-1}s)\xi' \mid \eta') dt \\ &= \int (\sigma(t)\xi \mid \eta) \overline{(\sigma(t)\eta' \mid \sigma(s)\xi')} dt \\ &= d_\sigma^{-1}(\xi \mid \eta')(\sigma(s)\xi' \mid \eta) = d_\sigma^{-1}(\xi \mid \eta')\varphi_{\xi', \eta}(s). \end{aligned}$$

14.3.4. Definition. The constant d_σ is called the formal dimension of σ .

Its value depends on the choice of Haar measure on G . We shall see in 15.2.3 that, if G is compact and the chosen Haar measure has total mass 1, then d_σ is the dimension of σ in the usual sense. This explains the terminology adopted.

14.3.5. Proposition. Let λ, ρ be the left and right regular representations of G , let σ be an irreducible subrepresentation of λ , let $K \supset H_\sigma$ be the bi-invariant subspace of $L^2(G)$ associated with σ , let λ_1, ρ_1 be the subrepresentations of λ, ρ defined by K , and let $\overline{H}_\sigma$ be the conjugate Hilbert space of H_σ .

- (i) There exists one and only one isomorphism Φ from the Hilbert space $H_\sigma \otimes \overline{H}_\sigma$ onto K which sends $\xi \otimes \overline{\eta}$ to $d_\sigma^{1/2} \check{\varphi}_{\xi, \eta}$ for every $\xi, \eta \in H_\sigma$.
- (ii) Let 1_{H_σ} and $1_{\overline{H}_\sigma}$ be the trivial representations of G on H_σ and $\overline{H}_\sigma$. Then Φ transforms $\sigma \otimes 1_{\overline{H}_\sigma}$ into λ_1 , and $1_{H_\sigma} \otimes \overline{\sigma}$ into ρ_1 .
- (iii) $\lambda_1 \simeq (\dim \sigma)\sigma$ and $\rho_1 \simeq (\dim \sigma)\overline{\sigma}$.

The map

$$(\xi, \eta) \longmapsto d_\sigma^{1/2} \check{\varphi}_{\xi, \eta}$$

is linear in ξ and conjugate-linear in η . Hence there is a linear map from the algebraic tensor product of H_σ and $\overline{H}_\sigma$ onto a dense vector subspace of K which sends $\xi \otimes \overline{\eta}$ to $d_\sigma^{1/2} \check{\varphi}_{\xi, \eta}$ (14.3.1). If $\xi, \xi', \eta, \eta' \in H_\sigma$, then, by 14.3.3,

$$(\xi \otimes \overline{\eta} \mid \xi' \otimes \overline{\eta'}) = (\xi \mid \xi')(\overline{\eta} \mid \overline{\eta'}) = (d_\sigma^{1/2} \check{\varphi}_{\xi, \eta} \mid d_\sigma^{1/2} \check{\varphi}_{\xi', \eta'}).$$

Thus the preceding linear map extends to an isomorphism Φ from $H_\sigma \otimes \overline{H}_\sigma$ onto K . The uniqueness in (i) is immediate, since the $\xi \otimes \overline{\eta}$ form a total set in $H_\sigma \otimes \overline{H}_\sigma$. For $s \in G$ and $\xi, \eta \in H_\sigma$, one has

$$\Phi(\sigma(s)\xi \otimes \overline{\eta}) = \check{\varphi}_{\sigma(s)\xi, \eta}$$

and

$$\check{\varphi}_{\sigma(s)\xi, \eta}(t) = (\sigma(t^{-1})\sigma(s)\xi \mid \eta) = (\sigma(t^{-1}s)\xi \mid \eta)$$

$$= \check{\varphi}_{\xi, \eta}(s^{-1}t) = (\lambda(s)\check{\varphi}_{\xi, \eta})(t).$$

Thus Φ transforms $\sigma \otimes 1_{\overline{H_\sigma}}$ into λ_1 ; similarly, it transforms $1_{H_\sigma} \otimes \overline{\sigma}$ into ρ_1 . Finally, (iii) follows from (ii).

14.3.6. Proposition. *Retain the notation of 14.3.5. If $H_\sigma \otimes \overline{H_\sigma}$ is canonically identified with the set of Hilbert–Schmidt operators on H_σ , then*

$$\Phi(u^*) = \Phi(u)^*, \quad \Phi(uv) = d_\sigma^{1/2} \Phi(u) * \Phi(v)$$

for all $u, v \in H_\sigma \otimes \overline{H_\sigma}$.

It is enough to verify this for $u = \xi \otimes \overline{\eta}$ and $v = \xi' \otimes \overline{\eta'}$, where $\xi, \eta, \xi', \eta' \in H_\sigma$. Now

$$\Phi((\xi \otimes \overline{\eta})^*) = \Phi(\eta \otimes \overline{\xi}) = d_\sigma^{1/2} \check{\varphi}_{\eta, \xi} = (d_\sigma^{1/2} \check{\varphi}_{\xi, \eta})^*,$$

and

$$\begin{aligned} \Phi((\xi \otimes \overline{\eta})(\xi' \otimes \overline{\eta'})) &= \Phi((\xi' \mid \eta)\xi \otimes \overline{\eta'}) \\ &= d_\sigma^{1/2} (\xi' \mid \eta) \check{\varphi}_{\xi, \eta'} \\ &= d_\sigma^{1/2} d_\sigma \check{\varphi}_{\xi, \eta} * \check{\varphi}_{\xi', \eta'} \\ &= d_\sigma^{1/2} \Phi(\xi \otimes \overline{\eta}) * \Phi(\xi' \otimes \overline{\eta'}). \end{aligned}$$

14.3.7. Theorem. *Let σ, σ' be inequivalent square-integrable irreducible representations of G . For $\xi, \eta \in H_\sigma$ and $\xi', \eta' \in H_{\sigma'}$, one has*

$$\int \varphi_{\xi, \eta}^\sigma(s) \overline{\varphi_{\xi', \eta'}^{\sigma'}(s)} ds = 0, \quad \check{\varphi}_{\xi, \eta}^\sigma * \check{\varphi}_{\xi', \eta'}^{\sigma'} = 0.$$

The bi-invariant subspaces K and K' of $L^2(G)$ associated with σ and σ' are orthogonal (14.2.3). Moreover, $\check{\varphi}_{\xi, \eta}^\sigma \in K$ and $\check{\varphi}_{\xi', \eta'}^{\sigma'} \in K'$ (14.3.1), whence

$$(\varphi_{\xi, \eta}^\sigma \mid \varphi_{\xi', \eta'}^{\sigma'}) = (\check{\varphi}_{\xi, \eta}^\sigma \mid \check{\varphi}_{\xi', \eta'}^{\sigma'}) = 0.$$

Finally, K and K' are invariant under $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$, where A denotes the Hilbert algebra of G ; hence

$$\check{\varphi}_{\xi', \eta'}^{\sigma'} * \check{\varphi}_{\xi, \eta}^\sigma \in K \cap K' = 0, \quad \check{\varphi}_{\xi, \eta}^\sigma * \check{\varphi}_{\xi', \eta'}^{\sigma'} = 0.$$

Bibliography: [51, 60].

14.4. Formal Dimension and Trace

14.4.1. Lemma. *Let H be a Hilbert space, B the Hilbert algebra of Hilbert–Schmidt operators on H , $b \mapsto U_b$ the canonical map from B into $\mathfrak{U}(B)$, E a minimal projection of $\mathfrak{B}(B)$, and $\sigma(b)$ the operator induced by U_b on $E(B)$. For every $b \in B$,*

$$\|b\|^2 = \text{Tr}(\sigma(b)\sigma(b)^*).$$

The algebra $\mathfrak{U}(B)$ is a type I factor, and $b \mapsto U_b$ is an isomorphism from the Hilbert algebra B onto the Hilbert algebra of Hilbert–Schmidt operators relative to $\mathfrak{U}(B)$ (A 61). On the other hand, $T \mapsto T_E$ is an isomorphism from $\mathfrak{U}(B)$ onto $\mathcal{L}(E(B))$. Hence $b \mapsto \sigma(b)$ is an isomorphism from the Hilbert algebra B onto the Hilbert algebra of Hilbert–Schmidt operators on $E(B)$. Therefore

$$\text{Tr}(\sigma(b)\sigma(b)^*) = \text{Tr}(bb^*) = \|b\|^2.$$

14.4.2. Proposition. *Let σ be a square-integrable irreducible representation of G , and let F be the central projection associated with σ . For every $f \in L^1(G) \cap L^2(G)$, one has*

$$\int |(Ff)(s)|^2 ds = d_\sigma \text{Tr}(\sigma(f)\sigma(f^*)).$$

Let A be the Hilbert algebra of G , and put $K = F(L^2(G))$. Then $P_K(A) = A \cap K = K$ (14.2.2), and K is a Hilbert algebra whose associated von Neumann algebras are $\mathfrak{U}(A)_F$ and $\mathfrak{B}(A)_F$. Let E be a minimal projection of $\mathfrak{B}(A)$ dominated by F , so that σ is identified with the subrepresentation of the left regular representation defined by E . Every $f \in A$ defines an operator $U_f \in \mathfrak{U}(A)$, and U_f induces on $H_\sigma = E(L^2(G))$ an operator which we shall denote by $\sigma(f)$; this agrees with the usual notation if $f \in L^1(G) \cap L^2(G)$. In fact, we shall prove the equality in the proposition for every $f \in A$. One has $U_{Ff} = FU_f$, and hence $\sigma(Ff) = \sigma(f)$. Neither side of the equality to be proved is therefore changed if f is replaced by Ff . We shall henceforth suppose that $f \in K$.

Resume the notation of 14.3.5. There exists $u \in H_\sigma \otimes \overline{H}_\sigma$ such that $f = \Phi(u)$, and $\|f\|_2 = \|u\|$. Moreover, if $v \in H_\sigma \otimes \overline{H}_\sigma$, then

$$\Phi(u) * \Phi(v) = d_\sigma^{-1/2} \Phi(uv) \quad (14.3.6).$$

Thus Φ^{-1} transforms left multiplication by f on K into left multiplication by $d_\sigma^{-1/2}u$ on $H_\sigma \otimes \overline{H}_\sigma$. In view of 14.4.1, we have

$$\text{Tr}(\sigma(f)\sigma(f)^*) = \|d_\sigma^{-1/2}u\|^2 = d_\sigma^{-1}\|u\|^2 = d_\sigma^{-1}\|f\|_2^2.$$

14.4.3. Proposition. Let $f, f' \in L^1(G) \cap L^2(G)$ and put $g = f * f'$. For $t \in G$, let g^t denote the function $s \mapsto g(tst^{-1})$ on G . Let σ be a square-integrable irreducible representation of G .

- (i) The operator $\sigma(g)$ is of trace class.
- (ii) For every unit vector ξ of H_σ , the function $t \mapsto (\sigma(g^t)\xi \mid \xi)$ on G is integrable, and

$$\int (\sigma(g^t)\xi \mid \xi) dt = d_\sigma^{-1} \operatorname{Tr} \sigma(g).$$

By linearity, it is enough to consider the case $f' = f^*$. Since $\sigma(f)$ is a Hilbert–Schmidt operator (14.4.2), $\sigma(g) = \sigma(f)\sigma(f)^*$ is a positive trace-class operator. Let (ξ_i) be an orthonormal basis of H_σ consisting of eigenvectors of $\sigma(g)$. Write $\sigma(g)\xi_i = \lambda_i\xi_i$, so that $\sum_i \lambda_i < +\infty$. For every $t \in G$,

$$\begin{aligned} (\sigma(g^t)\xi \mid \xi) &= (\sigma(t^{-1})\sigma(g)\sigma(t)\xi \mid \xi) = (\sigma(g)\sigma(t)\xi \mid \sigma(t)\xi) \\ &= \sum_i (\sigma(g)\sigma(t)\xi \mid \xi_i) \overline{(\sigma(t)\xi \mid \xi_i)} = \sum_i \lambda_i |(\sigma(t)\xi \mid \xi_i)|^2. \end{aligned}$$

Consequently, by 14.3.3,

$$\int (\sigma(g^t)\xi \mid \xi) dt = \sum_i \lambda_i \int |(\sigma(t)\xi \mid \xi_i)|^2 dt = \sum_i \lambda_i d_\sigma^{-1} = d_\sigma^{-1} \operatorname{Tr} \sigma(g).$$

Bibliography: [60].

14.5. Integrable Representations

14.5.1. Proposition. Let σ be an irreducible continuous unitary representation of G . The following conditions are equivalent:

- (i) There exists $\xi \in H_\sigma$, $\xi \neq 0$, such that the function $s \mapsto (\sigma(s)\xi \mid \xi)$ is integrable on G .
- (ii) There exists a dense vector subspace H' of H_σ such that, for $\eta, \zeta \in H'$, the function $s \mapsto (\sigma(s)\eta \mid \zeta)$ is integrable on G .

Suppose that (i) holds. Let $g, h \in \mathcal{K}(G)$. For every $s \in G$,

$$\begin{aligned} (\sigma(g)\xi \mid \sigma(s)\sigma(h)\xi) &= \iint g(t) \overline{h(u)} (\sigma(t)\xi \mid \sigma(s)\sigma(u)\xi) dt du \\ &= \iint g(t) \overline{h(u)} (\xi \mid \sigma(t^{-1}su)\xi) dt du. \end{aligned}$$

Hence

$$\begin{aligned}
 & \int |(\sigma(g)\xi \mid \sigma(s)\sigma(h)\xi)| ds \\
 & \leq \iiint |g(t)h(u)| |(\xi \mid \sigma(t^{-1}su)\xi)| ds dt du \\
 & = \iiint |g(t)h(u)| |(\xi \mid \sigma(s)\xi)| ds dt du \\
 & = \int |g(t)| dt \int |h(u)| du \int |(\xi \mid \sigma(s)\xi)| ds < +\infty.
 \end{aligned}$$

Thus (ii) holds with $H' = \sigma(\mathcal{K}(G))\xi$. [This subspace is nonzero and invariant under $\sigma(G)$, and is therefore dense in H_σ , since σ is irreducible.]

14.5.2. Definition. *An irreducible continuous unitary representation of G is called integrable if it satisfies the equivalent conditions of 14.5.1.*

With the notation of 14.5.1, the function $s \mapsto (\sigma(s)\xi \mid \xi)$ on G , being bounded and integrable, is square-integrable. Thus an integrable irreducible representation is square-integrable. The converse is false.

Bibliography: [60].

14.6. Complements

14.6.1. Let G be a locally compact group, π a continuous unitary representation of G , and $p, q \in [1, +\infty]$ such that $p^{-1} + q^{-1} = 1$. The following conditions are equivalent:

- a. There exists a finite constant M such that

$$\|\pi(f)\| \leq M\|f\|_p \quad \text{for } f \in L^1(G) \cap L^p(G),$$

where $\|f\|_p$ denotes the norm in $L^p(G)$.

- b. For all $\xi, \eta \in H_\pi$, the function $s \mapsto (\pi(s)\xi \mid \eta)$ belongs to $L^q(G)$.

[84].

14.6.2. Let G be a unimodular locally compact group, let σ be a square-integrable irreducible continuous unitary representation of G , and let K_σ be the associated bi-invariant subspace. Then $\bar{\sigma}$ is square-integrable and irreducible, and its associated bi-invariant subspace is the image of K_σ under the map $f \mapsto \bar{f}$, and also under the map $f \mapsto \check{f}$ from $L^2(G)$ onto $L^2(G)$.

***14.6.3.** The square-integrable irreducible continuous unitary representations of connected semisimple real Lie groups are beginning to be well understood. Connected complex semisimple Lie groups and simply connected nilpotent real Lie groups have no square-integrable irreducible continuous unitary representations. [9, 60, 61].

§ 15. — Representations of Compact Groups

Throughout this section, G denotes a compact group. The Haar measure chosen on G will always be the one of total mass 1. One has

$$L^1(G) \supset L^2(G), \quad \|f\|_2 \geq \|f\|_1 \quad \text{for every } f \in L^2(G).$$

15.1. Complete Reducibility

15.1.1. Lemma. *There exists in G a fundamental system of neighborhoods of e invariant under inner automorphisms.*

It is enough to show that, if $s \in G$ and $s \neq e$, there exists a compact neighborhood V of e , invariant under inner automorphisms, such that $s \notin V$. Let W be a compact neighborhood of s such that $e \notin W$. The image of $G \times W$ under the map

$$(t, w) \mapsto twt^{-1}$$

is a compact neighborhood W' of s , invariant under inner automorphisms, such that $e \notin W'$. The complement of W' is therefore an open neighborhood of e , invariant under inner automorphisms, whose closure does not contain s .

15.1.2. Lemma. *The Hilbert algebra A of G is equal to $L^2(G)$. Let (F_i) be the family of pairwise orthogonal minimal projections of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$. Then A is the Hilbert sum of the $A_i = F_i(A)$; the A_i are pairwise annihilating self-adjoint two-sided ideals of A ; and each A_i is isomorphic to the Hilbert algebra of Hilbert–Schmidt operators on a finite-dimensional Hilbert space.*

One has

$$A \supset L^1(G) \cap L^2(G) = L^2(G), \quad \text{and hence } A = L^2(G).$$

By 15.1.1 and 13.10.5, $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$ are finite von Neumann algebras. The lemma now follows from A 65 and A 68.

15.1.3. Theorem. *Let π be a continuous unitary representation of G . Then π is a Hilbert sum of finite-dimensional irreducible representations.*

Retain the notation of 15.1.2. Let H_i be the closed vector subspace of H_π spanned by the $\pi(f)\xi$, $f \in A_i$, $\xi \in H_\pi$. Since $A = L^2(G)$ is dense in $L^1(G)$, the sum of the H_i is dense in H_π . If $H_\pi \neq 0$, there is an i such that $H_i \neq 0$. For $j \neq i$, one has $A_j A_i = 0$, and therefore

$$\pi(A_j)\pi(A_i)(H_\pi) = 0, \quad \pi(A_j)(H_i) = 0.$$

Let ξ be a nonzero element of H_i . The finite-dimensional subspace $\pi(A_i)\xi$ is equal to $\pi(A)\xi$, and hence to $\pi(L^1(G))\xi$; it is nonzero and invariant under $\pi(G)$. This subspace contains a nonzero subspace invariant under $\pi(G)$ on which the corresponding subrepresentation of G is irreducible (2.3.5).

Let (K_α) be a maximal family of pairwise orthogonal, finite-dimensional subspaces of H_π , invariant under $\pi(G)$, such that the corresponding subrepresentations of G are irreducible. Then $H_\pi \ominus (\bigoplus_\alpha K_\alpha)$ is invariant under $\pi(G)$ and is therefore zero, by what precedes and the maximality of the family (K_α) . Hence

$$H_\pi = \bigoplus_\alpha K_\alpha.$$

15.1.4. Corollary. *Every irreducible continuous unitary representation of G is finite-dimensional.*

15.1.5. Corollary. *The C^* -algebra of G is liminal.*

This follows from 15.1.4, since every linear operator on a finite-dimensional Hilbert space is compact.

15.1.6. Corollary. *Let (σ_α) be the family of finite-dimensional irreducible continuous unitary representations of G . Then*

$$\bigcap_\alpha \ker \sigma_\alpha = \{e\}.$$

This follows from 13.6.6 and 15.1.4.

15.1.7. Corollary. *G is a projective limit of compact Lie groups.*

Let (ρ_α) be the family of finite-dimensional continuous unitary representations of G . The family $(\ker \rho_\alpha)$ is decreasingly directed, and

$$\bigcap_\alpha \ker \rho_\alpha = \{e\} \quad (15.1.6).$$

Thus G is the projective limit of the groups $G/\ker \rho_\alpha$, themselves isomorphic to $\rho_\alpha(G)$. Since $\dim \rho_\alpha < +\infty$, the unitary group of H_{ρ_α} is a Lie group; since $\rho_\alpha(G)$ is a closed subgroup of this group, $\rho_\alpha(G)$ is a Lie group.

15.1.8. Corollary. *Let π be a continuous unitary representation of G .*

(i) *There exist cardinals n_σ , $\sigma \in \widehat{G}$, such that*

$$\pi = \bigoplus_{\sigma \in \widehat{G}} n_\sigma \sigma.$$

- (ii) The subspace $H_{n_\sigma\sigma}$ of H_π is independent of the chosen decomposition of π . It is the closed vector subspace of H_π spanned by the spaces of the subrepresentations equivalent to σ .
- (iii) The cardinals n_σ are independent of the chosen decomposition of π .

Assertion (i) follows at once from 15.1.3. Let σ' be a subrepresentation of π equivalent to $\sigma \in \widehat{G}$. Then σ' is disjoint from τ for $\tau \in \widehat{G}$, $\tau \neq \sigma$ (5.2.2); consequently $H_{\sigma'}$ is orthogonal to $H_{n_\tau\tau}$ for $\tau \neq \sigma$ [condition (iii) of 5.2.1]. Hence $H_{\sigma'} \subset H_{n_\sigma\sigma}$, which proves (ii). One has

$$\dim H_{n_\sigma\sigma} = n_\sigma \dim \sigma,$$

so (iii) follows from (ii). Another intrinsic description of the $H_{n_\sigma\sigma}$ will be given in 15.3.12.

15.1.9. With the notation of 15.1.8, we say that σ occurs n_σ times in π .

Bibliography: [58, 82, 86, 104].

15.2. Irreducible Representations of a Compact Group

15.2.1. By what precedes, the study of the continuous unitary representations of G reduces to that of the irreducible representations. The latter are obviously square-integrable, so the results of §14 apply to them. In particular, they are all subrepresentations of the regular representation.

15.2.2. Theorem. Let $A = L^2(G)$ be the Hilbert algebra of G , let (F_i) be the family of minimal projections of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$, let λ, ρ be the left and right regular representations of G , let λ_i, ρ_i be the subrepresentations of λ, ρ defined by F_i , let σ_i be the class of irreducible representations whose associated central projection is F_i , and put $\delta_i = \dim \sigma_i$.

- (i) λ is the Hilbert sum of the λ_i , and ρ is the Hilbert sum of the ρ_i .
- (ii) λ_i is the Hilbert sum of δ_i representations of class σ_i , and ρ_i is the Hilbert sum of δ_i representations of class $\overline{\sigma_i}$.

We know that $L^2(G)$ is the Hilbert sum of the $F_i(L^2(G))$ (15.1.2), which proves (i); and (ii) is a special case of 14.3.5(iii).

15.2.3. Proposition. Let σ be an irreducible continuous unitary representation of G . The formal dimension d_σ of σ is equal to $\dim \sigma$.

The algebra $\sigma(\mathcal{K}(G))$ is irreducible on H_σ , and hence equals $\mathcal{L}(H_\sigma)$, since $\dim H_\sigma < +\infty$. Let $f \in \mathcal{K}(G)$ be such that $\sigma(f) = 1$, and put $g = f * f^*$. For every $t \in G$, let g^t be the function $s \mapsto g(tst^{-1})$ on G . Then

$$\sigma(g^t) = \sigma(t)^{-1} \sigma(f * f^*) \sigma(t) = 1.$$

Let ξ be a unit vector of H_σ . By 14.4.3,

$$d_\sigma^{-1}(\dim \sigma) = d_\sigma^{-1} \operatorname{Tr} \sigma(g) = \int (\sigma(g^t) \xi \mid \xi) dt = 1,$$

whence $d_\sigma = \dim \sigma$.

15.2.4. Theorem. *For every $f \in L^2(G)$, one has*

$$\int_G |f(s)|^2 ds = \sum_{\sigma \in \widehat{G}} (\dim \sigma) \operatorname{Tr}(\sigma(f) \sigma(f^*)).$$

For every $\sigma \in \widehat{G}$, let F_σ be the central projection on $L^2(G)$ associated with σ . By 15.1.2, one has

$$L^2(G) = \bigoplus_{\sigma \in \widehat{G}} F_\sigma(L^2(G)), \quad \|f\|_2^2 = \sum_{\sigma \in \widehat{G}} \|F_\sigma f\|_2^2.$$

By 14.4.2 and 15.2.3,

$$\|F_\sigma f\|_2^2 = (\dim \sigma) \operatorname{Tr}(\sigma(f) \sigma(f)^*).$$

15.2.5. For every $\sigma \in \widehat{G}$, choose an orthonormal basis

$$(\xi_1, \xi_2, \dots, \xi_{\dim \sigma})$$

of H_σ , and put

$$\varphi_{i,j,\sigma}(s) = (\sigma(s) \xi_i \mid \xi_j).$$

Then, by 15.1.2 and 14.3.5(i), the functions

$$(\dim \sigma)^{1/2} \varphi_{i,j,\sigma} \quad (\sigma \in \widehat{G}, \quad i, j = 1, 2, \dots, \dim \sigma)$$

form an orthonormal basis of $L^2(G)$. By 14.3.3 and 14.3.7, one has

$$\varphi_{i,j,\sigma} * \varphi_{i',j',\sigma} = (\dim \sigma)^{-1} \delta_i^{j'} \varphi_{i',j,\sigma}, \quad \varphi_{i,j,\sigma} * \varphi_{i',j',\sigma'} = 0 \quad \text{if } \sigma \neq \sigma'.$$

Bibliography: [86, 108, 181].

15.3. Characters of Compact Groups

15.3.1. Definition. Let σ be an irreducible continuous unitary representation of G . The character of σ , denoted by χ_σ , is the continuous function

$$s \longmapsto \text{Tr } \sigma(s)$$

on G . The function $(\dim \sigma)^{-1} \chi_\sigma$ is called the normalized character.

When G is compact and commutative, this recovers the usual notion of a character.

15.3.2. The character of σ depends only on the class of σ . If $(\xi_1, \dots, \xi_n)$ is an orthonormal basis of H_σ , then

$$\chi_\sigma(s) = (\sigma(s)\xi_1 \mid \xi_1) + \dots + (\sigma(s)\xi_n \mid \xi_n),$$

which proves that χ_σ is a function of positive type [13.4.5(ii)], equal to $\dim \sigma$ at e .

We shall say that a function on a group is *central* if it is invariant under the inner automorphisms of that group. Definition 15.3.1 immediately shows that χ_σ is central.

15.3.3. By 15.2.5, if $\sigma, \sigma' \in \widehat{G}$ are inequivalent, then

$$(\chi_\sigma \mid \chi_{\sigma'}) = 0, \tag{1}$$

$$\chi_\sigma * \chi_{\sigma'} = 0. \tag{2}$$

On the other hand, retaining the notation of 15.3.2,

$$(\chi_\sigma * \chi_\sigma)(s) = n^{-1} [(\sigma(s)\xi_1 \mid \xi_1) + \dots + (\sigma(s)\xi_n \mid \xi_n)],$$

whence

$$\chi_\sigma * \chi_\sigma = (\dim \sigma)^{-1} \chi_\sigma, \tag{3}$$

$$\|\chi_\sigma\|_2 = 1. \tag{4}$$

15.3.4. Let K_σ be the bi-invariant subspace of $L^2(G)$ associated with σ . As a Hilbert algebra, K_σ is isomorphic to the algebra of Hilbert–Schmidt operators on a finite-dimensional Hilbert space (15.1.2). This algebra has a one-dimensional center. Now

$$\check{\chi}_\sigma = \overline{\chi_\sigma} = \chi_{\bar{\sigma}}$$

is a central function on G belonging to K_σ (14.3.1). Thus the center of K_σ is $\mathbb{C}\chi_{\bar{\sigma}}$. By 15.1.2, the center of the Hilbert algebra $L^2(G)$ is the Hilbert sum of the $\mathbb{C}\chi_\sigma$. In view of formula (4), the χ_σ form an orthonormal basis of the center of $L^2(G)$.

15.3.5. By formula (3), the operator of convolution by $(\dim \sigma)\chi_{\bar{\sigma}}$ on $L^2(G)$ is a nonzero projection. Since $\chi_{\bar{\sigma}} = \chi_\sigma^*$, this projection is Hermitian. It belongs to $\mathfrak{U}(A) \cap \mathfrak{B}(A)$, where A is the Hilbert algebra of G , and its range is contained in K_σ . Since P_{K_σ} is a minimal projection of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$, the operator of convolution by $(\dim \sigma)\chi_{\bar{\sigma}}$ is P_{K_σ} . Thus, for every $f \in L^2(G)$,

$$f = \sum_{\sigma \in \widehat{G}} (\dim \sigma)(f * \chi_{\bar{\sigma}}),$$

the series converging in $L^2(G)$. If f is central, this expansion reduces to the expansion of f in the orthonormal basis (χ_σ) of the center of $L^2(G)$:

$$f = \sum_{\sigma \in \widehat{G}} (f | \chi_\sigma) \chi_\sigma.$$

Let f be a continuous central complex-valued function on G . In view of 15.1.1, f is a uniform limit on G of functions of the form $f * g$, where g is central and belongs to $L^2(G)$. Since f and g are limits in $L^2(G)$ of finite linear combinations of characters of G , every continuous central complex-valued function on G is a uniform limit on G of finite linear combinations of characters of G .

For G equal to the group of real numbers modulo 1, these facts reduce to familiar facts about Fourier series.

15.3.6. For every finite-dimensional continuous unitary representation π of G , one can define the character χ_π of π as in 15.3.1.

Let

$$\pi = n_1 \sigma_1 \oplus \cdots \oplus n_p \sigma_p$$

be the decomposition of π into irreducible representations, where the σ_i are pairwise inequivalent and the n_i are integers. Then

$$(\chi_\pi | \chi_{\sigma_i}) = n_i (\chi_{\sigma_i} | \chi_{\sigma_i}) = n_i.$$

Thus knowledge of χ_π determines the n_i . In other words, two finite-dimensional continuous unitary representations of G are equivalent if and only if they have the same character.

If π and π' are two finite-dimensional continuous unitary representations of G , then

$$\chi_{\pi \otimes \pi'} = \chi_\pi \chi_{\pi'}.$$

Henceforth, when we speak of characters of a compact group, we shall mean only characters of irreducible representations.

15.3.7. Proposition. *Let μ be a central measure on G , that is, one which commutes with every measure under convolution. For $\mu \gg 0$, it is necessary and sufficient that $\mu(\chi) \geq 0$ for every character χ of G .*

If $\mu \gg 0$, then

$$\mu(\chi) = (\dim \sigma) \mu(\check{\chi} * \chi),$$

where σ is a representation with character χ . Now $\mu(\check{\chi} * \chi) = \mu(\chi * \chi^*) \geq 0$.

Conversely, suppose that $\mu(\chi) \geq 0$ for every character χ . Let λ be the left regular representation of G . To prove that $\mu \gg 0$, it is enough to prove that $\lambda(\mu) \geq 0$ (13.7.4). Let A be the Hilbert algebra of G , let (F_i) be the family of minimal projections of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$, and put $K_i = F_i(L^2(G))$. Since μ is central,

$$\lambda(\mu) \in \mathfrak{U}(A) \cap \mathfrak{B}(A),$$

and the restriction of $\lambda(\mu)$ to each K_i is a scalar λ_i . Let χ_i be the character belonging to K_i . If σ_i is a representation with character χ_i , then

$$\begin{aligned} \lambda_i(\chi_i | \chi_i) &= (\lambda(\mu)\chi_i | \chi_i) = (\mu * \chi_i | \chi_i) \\ &= (\mu * \chi_i * \check{\chi}_i)(e) = (\dim \sigma_i)^{-1}(\mu * \chi_i)(e) \\ &= (\dim \sigma_i)^{-1}\mu(\check{\chi}_i) \geq 0. \end{aligned}$$

15.3.8. Proposition. *Let $f \in L^2(G)$ be a central function on G . Write*

$$f = \sum_{\sigma \in \widehat{G}} \lambda_\sigma \chi_\sigma, \quad \sum_{\sigma \in \widehat{G}} |\lambda_\sigma|^2 < +\infty \quad (15.3.4).$$

For f to be continuous and of positive type, it is necessary and sufficient that $\lambda_\sigma \geq 0$ for every $\sigma \in \widehat{G}$ and

$$\sum_{\sigma \in \widehat{G}} \lambda_\sigma \dim \sigma < +\infty.$$

The condition is sufficient because χ_σ is of positive type and is bounded in absolute value by $\dim \sigma$. Suppose that f is continuous and of positive type. Then

$f = g * g^*$ for some $g \in L^2(G)$ (13.8.6). For every $\sigma \in \widehat{G}$, let K_σ be the bi-invariant subspace of $L^2(G)$ associated with σ . One has

$$g = \sum_{\sigma \in \widehat{G}} \mu_\sigma g_\sigma,$$

where

$$g_\sigma \in K_\sigma, \quad \|g_\sigma\|_2 = 1, \quad \sum_{\sigma \in \widehat{G}} |\mu_\sigma|^2 < +\infty.$$

Since $g_\sigma * g_{\sigma'}^* = 0$ for $\sigma \neq \sigma'$, and

$$\|u * v\|_\infty \leq \|u\|_2 \|v\|_2 \quad (u, v \in L^2(G)),$$

f is the sum of the uniformly convergent series

$$\sum_{\sigma \in \widehat{G}} |\mu_\sigma|^2 (g_\sigma * g_\sigma^*).$$

Moreover, $g_\sigma * g_\sigma^* \in K_\sigma$, so that $|\mu_\sigma|^2 (g_\sigma * g_\sigma^*)$ is the orthogonal projection of f onto K_σ and is therefore central. We may suppose that g_σ is central if $\mu_\sigma = 0$. Thus $g_\sigma * g_\sigma^*$ is central, and

$$(g_\sigma * g_\sigma^*)(e) = \|g_\sigma\|_2^2 = 1.$$

By 15.3.4,

$$g_\sigma * g_\sigma^* = (\dim \sigma)^{-1} \chi_{\bar{\sigma}},$$

and hence

$$\lambda_{\bar{\sigma}} = |\mu_\sigma|^2 (\dim \sigma)^{-1},$$

which completes the proof.

15.3.9. Corollary. *In the convex set of continuous central functions f of positive type on G such that $f(e) \leq 1$, the extreme points are 0 and the normalized characters.*

15.3.10. Proposition. *Let ψ be a continuous complex-valued function on G . For ψ to be a normalized character, it is necessary and sufficient that $\psi \neq 0$ and*

$$\int \psi(xsx^{-1}t) dx = \psi(s)\psi(t) \quad (s, t \in G). \quad (1)$$

Suppose that there exists $\sigma \in \widehat{G}$ such that $\psi = (\dim \sigma)^{-1} \chi_\sigma$. Put

$$S = \int \sigma(xsx^{-1}) dx \in \mathcal{L}(H_\sigma).$$

For every $t \in G$, one has

$$\sigma(t)S = \int \sigma(txsx^{-1}) dx = \int \sigma(xs(t^{-1}x)^{-1}) dx = S\sigma(t).$$

Since σ is irreducible, S is scalar. Moreover,

$$\text{Tr } S = \int \chi_\sigma(xsx^{-1}) dx = \int \chi_\sigma(s) dx = \chi_\sigma(s).$$

Consequently

$$\int \sigma(xsx^{-1}) dx = (\dim \sigma)^{-1} \chi_\sigma(s) 1. \quad (2)$$

It follows that

$$\int \sigma(xsx^{-1}t) dx = \left(\int \sigma(xsx^{-1}) dx \right) \sigma(t) = (\dim \sigma)^{-1} \chi_\sigma(s) \sigma(t).$$

Taking traces of both sides gives (1).

Suppose now that $\psi \neq 0$ and that ψ satisfies (1). Since $\psi \neq 0$, there is a continuous irreducible unitary representation σ of G such that $\bar{\sigma}(\psi) \neq 0$. We have

$$\begin{aligned} \psi(s)\bar{\sigma}(\psi) &= \int \psi(s)\psi(t)\bar{\sigma}(t) dt = \iint \psi(xsx^{-1}t)\bar{\sigma}(t) dx dt \\ &= \iint \psi(t)\bar{\sigma}(xs^{-1}x^{-1}t) dx dt \\ &= \int \psi(t) \left(\int \bar{\sigma}(xs^{-1}x^{-1}) dx \right) \bar{\sigma}(t) dt. \end{aligned}$$

By (2),

$$\begin{aligned} \psi(s)\bar{\sigma}(\psi) &= \int \psi(t)(\dim \sigma)^{-1} \chi_{\bar{\sigma}}(s^{-1}) \bar{\sigma}(t) dt \\ &= (\dim \sigma)^{-1} \chi_\sigma(s) \bar{\sigma}(\psi), \end{aligned}$$

and hence $\psi = (\dim \sigma)^{-1} \chi_\sigma$.

15.3.11. Lemma. *Let σ and σ' be continuous irreducible unitary representations of G . If σ and σ' are inequivalent, then $\sigma'(\chi_{\bar{\sigma}}) = 0$. If σ and σ' are equivalent, then*

$$\sigma'(\chi_{\bar{\sigma}}) = (\dim \sigma)^{-1} 1.$$

Let λ be the left regular representation of G . Then $\lambda((\dim \sigma)\chi_{\bar{\sigma}})$ is the projection onto the bi-invariant subspace K_{σ} of $L^2(G)$ associated with σ (15.3.5). On the other hand, we may suppose that σ' is a subrepresentation of λ (15.2.1), and $H_{\sigma'}$ is orthogonal to K_{σ} , or is contained in K_{σ} , according as σ' is inequivalent or equivalent to σ (14.2.3).

15.3.12. Theorem. *Let π be a continuous unitary representation of G , and let $\pi = \bigoplus_{\sigma \in \widehat{G}} n_{\sigma} \sigma$ be its decomposition into irreducible representations. For every $\sigma \in \widehat{G}$, the orthogonal projection of H_{π} onto $H_{n_{\sigma} \sigma}$ is $\pi((\dim \sigma)\chi_{\bar{\sigma}})$.*

This follows at once from 15.3.11.

Bibliography: [50, 86, 108, 181].

15.4. Representations of Finite Groups

15.4.1. Proposition. *Let G be a finite group of order n . Let r be the number of conjugacy classes of elements of G . Then $\widehat{G}$ has r elements, and*

$$\sum_{\sigma \in \widehat{G}} (\dim \sigma)^2 = n.$$

For each $\sigma \in \widehat{G}$, let K_{σ} be the bi-invariant subspace of $L^2(G)$ associated with σ . We have

$$\dim K_{\sigma} = (\dim \sigma)^2, \quad n = \dim L^2(G) = \sum_{\sigma \in \widehat{G}} \dim K_{\sigma},$$

by 15.2.1 and 15.2.2. Thus $\sum_{\sigma \in \widehat{G}} (\dim \sigma)^2 = n$. On the other hand, the number of elements of $\widehat{G}$ equals the number of characters of G , that is, the dimension of the center Z of $L^2(G)$ (15.3.4); the characteristic functions of the conjugacy classes of elements of G form a basis of the vector space Z . Hence $\dim Z = r$.

15.4.2. Let $C_1, C_2, \dots, C_r$ be the distinct conjugacy classes in G , and let $\sigma_1, \sigma_2, \dots, \sigma_r$ be the distinct elements of $\widehat{G}$. The *character table* of G is the matrix $(\chi_i^j)_{1 \leq i, j \leq r}$, where χ_i^j is the value on C_i of the character of σ_j . Let h_i be the number of elements of C_i . By formulas (1) and (4) of 15.3.3,

$$\frac{1}{n} \sum_i h_i \chi_i^j \overline{\chi_i^{j'}} = \begin{cases} 0, & j \neq j', \\ 1, & j = j'. \end{cases} \quad (1)$$

In other words, the matrix $(\sqrt{h_i/n} \chi_i^j)$ is unitary. Consequently

$$\sum_j \sqrt{\frac{h_i}{n}} \chi_i^j \sqrt{\frac{h_{i'}}{n}} \overline{\chi_{i'}^j} = \delta_{i'}^{i'},$$

or

$$\frac{1}{n} \sum_j \chi_i^j \overline{\chi_{i'}^j} = \begin{cases} 0, & i \neq i', \\ 1/h_i, & i = i'. \end{cases} \quad (2)$$

15.4.3. The following proposition requires very different methods.

Proposition. *The dimension of an irreducible representation of G divides the order of G .*

Retain the notation of 15.4.2. Put $d_j = \dim \sigma_j$.

a. Let $s \in C_i$. Then $\chi_i^j = \text{Tr } \sigma_j(s)$ is the sum of the eigenvalues of $\sigma_j(s)$. Since $\sigma_j(s)$ has finite order, these eigenvalues are roots of unity. Thus χ_i^j is an algebraic integer.

b. Let f_i be the characteristic function of C_i . We have

$$f_i * f_{i'} = \sum_{i''=1}^r c_{ii'}^{i''} f_{i''},$$

where $c_{ii'}^{i''}$ is the number of ways in which an element of $C_{i''}$ can be written as the product of an element of C_i and an element of $C_{i'}$. Thus the $c_{ii'}^{i''}$ are rational integers. On the other hand, for every $\sigma \in \widehat{G}$,

$$\sigma(f_i) \sigma(f_{i'}) = \sum_{i''=1}^r c_{ii'}^{i''} \sigma(f_{i''}).$$

Identifying $\sigma(f_i)$, $\sigma(f_{i'})$, and $\sigma(f_{i''})$ with scalars, we see that $\sigma(f_i)$ is an eigenvalue of the matrix $(c_{ii'}^{i''})_{1 \leq i', i'' \leq r}$, and hence is an algebraic integer. Now

$$\sigma_j(f_i) = \frac{1}{d_j} \text{Tr } \sigma_j(f_i) = \frac{1}{d_j} \sum_{s \in C_i} \text{Tr } \sigma_j(s) = \frac{h_i}{d_j} \chi_i^j.$$

Thus $(h_i/d_j) \chi_i^j$ is an algebraic integer.

c. By formula (1),

$$\frac{n}{d_j} = \sum_i \left(\frac{h_i}{d_j} \chi_i^j \right) \overline{\chi_i^j}$$

is an algebraic integer; on the other hand it is rational, and is therefore a rational integer.

The bibliography concerning representations of finite groups is enormous. See, for example, M. Hall, *The Theory of Groups*, The Macmillan Company, New York, 1959.

15.5. Use of Compact Subgroups of Arbitrary Groups

15.5.1. Let G' be a locally compact group and G a compact subgroup of G' . If $f \in L^1(G)$, we may identify f with a measure on G , hence with a compactly supported measure on G' . We shall do so in this subsection.

Lemma. *Let π be a continuous irreducible unitary representation of G' , put $\rho = \pi|_G$, and let $\rho = \bigoplus_{\sigma \in \widehat{G}} n_\sigma \sigma$ be the decomposition of ρ into irreducible representations. Suppose that n_σ is finite for every $\sigma \in \widehat{G}$.*

- (i) *If χ is a character of G and $f \in L^1(G')$, then $\pi(f * \chi)$ is an operator of finite rank.*
- (ii) *The linear combinations of the $f * \chi$, where $f \in L^1(G')$ and χ is a character of G , are dense in $L^1(G')$.*
- (iii) *$\pi(C^*(G')) = \mathcal{LC}(H_\pi)$.*

Let $\sigma \in \widehat{G}$. By 15.3.12, $\rho(\chi_{\bar{\sigma}})$ has rank $n_\sigma \dim \sigma$. Thus, for every $f \in L^1(G')$, $\pi(f * \chi_{\bar{\sigma}}) = \pi(f)\rho(\chi_{\bar{\sigma}})$ has finite rank. This proves (i).

There are arbitrarily small neighborhoods V_i of e in G invariant under the inner automorphisms of G . Let φ_i be the characteristic function of V_i in G . By 15.3.5, φ_i is a limit in $L^2(G)$, hence in $M^1(G)$, of finite linear combinations of characters of G . If $f \in L^1(G')$, then $f * \varphi_i$ can be made arbitrarily close to f in $L^1(G')$. This proves (ii).

Finally, (iii) follows from (i), (ii), and 4.1.11.

15.5.2. Theorem. *Let G' be a locally compact group and G a compact subgroup of G' . Suppose that, for every $\pi \in \widehat{G'}$ and every $\sigma \in \widehat{G}$, σ occurs only finitely many times in $\pi|_G$. Then G' is liminal.*

This follows at once from 15.5.1 (iii).

15.5.3. Lemma. *Let Γ be a topological group, let σ be a representation of Γ of finite dimension d , and let n be an integer greater than d . Then $n\sigma$ has no cyclic vector.*

Indeed, $n\sigma = \sigma \otimes \tau$, where τ denotes the trivial representation of Γ of dimension n . Every vector of $H_\sigma \otimes H_\tau$ can be written $\xi = \xi_1 \otimes \eta_1 + \cdots + \xi_d \otimes \eta_d$,

where $(\xi_1, \dots, \xi_d)$ is a basis of H_σ . Let H be the vector subspace of H_τ spanned by $\eta_1, \dots, \eta_d$. Then $H \neq H_\tau$ and $[(\sigma \otimes \tau)(\Gamma)](\xi) \subset H_\sigma \otimes H$; hence ξ is not cyclic.

15.5.4. We shall call a topological group *linear* if it admits a continuous injective linear representation, not necessarily unitary, in a finite-dimensional complex vector space.

15.5.5. Let n be a positive integer, and let M_n be the algebra of complex n -rowed, n -columned matrices. Denote by $r(n)$ the least integer r for which the identity

$$\sum_{\alpha \in \mathfrak{S}_r} \varepsilon_\alpha X_{\alpha(1)} \cdots X_{\alpha(r)} = 0 \quad (1)$$

holds in M_n (cf. 3.6.1). Recall that $r(n+1) > r(n)$.

Lemma. *Let G' be a connected linear semisimple real Lie group, let G be a maximal compact subgroup of G' , let $\sigma \in \widehat{G}$, and put $d = \dim \sigma$. Let χ be the character of σ and $\psi = d\chi$, so that ψ is an idempotent of the algebra $\mathcal{K}(G)$ and $\psi * \mathcal{K}(G') * \psi$ is a subalgebra A of $\mathcal{K}(G')$. Then identity (1) holds in A , with $r = r(d^2)$.*

Let τ be a linear representation, not necessarily unitary, of G' in a finite-dimensional complex vector space V . If $\xi \in V$ and $\xi' \in V'$ (the dual of V), the function $s \mapsto \langle \tau(s)\xi, \xi' \rangle$ is called a coefficient of τ . Let C be the set of linear combinations of coefficients of continuous finite-dimensional linear representations of G' . By considering the conjugate of such a representation and the tensor product of two such representations, we see that C is an involutive algebra of continuous functions on G' [for the product $(f, g) \mapsto fg$ and the involution $f \mapsto \bar{f}$]. Since G' is linear, this algebra separates the points of G' . By the Stone–Weierstrass theorem, every continuous complex function on G' is a uniform limit on every compact set of elements of C . Thus, if $f \in \mathcal{K}(G')$ and $f \neq 0$, there is a $g \in C$ such that $\int f(s)g(s)ds \neq 0$. Consequently, if $f \in \mathcal{K}(G')$ and $f \neq 0$, there is a continuous finite-dimensional linear representation τ of G' , not necessarily unitary, such that $\tau(f) \neq 0$; and, since G' is connected and semisimple, we may suppose τ irreducible. To prove the lemma it is therefore enough to establish the following: if τ is an irreducible finite-dimensional representation of G' , not necessarily unitary, and $r = r(d^2)$, then identity (1) holds in $\tau(A)$.

The space V of τ may be given a Hilbert-space structure invariant under $\tau|G$ (B 34). Let $\tau|G = \bigoplus_{\rho \in \widehat{G}} n_\rho \rho$ be the decomposition of $\tau|G$ into irreducible representations. Then $\tau(\psi)$ is the projection onto the space of $n_\sigma \sigma$ (15.3.12), and so has rank $n_\sigma d$. Consequently $\tau(A)$ may be identified with a subalgebra of the algebra of $n_\sigma d$ by $n_\sigma d$ matrices, and it remains only to prove that $n_\sigma \leq d$.

There is a closed connected solvable subgroup N of G' such that every element of G' is the product of an element of G and an element of N . By Lie's theorem, there is a nonzero $\xi \in V$ such that $\tau|_N$ leaves $\mathbb{C}\xi$ invariant. Since ξ is cyclic for τ , we see that ξ is cyclic for $\tau|_G$. Hence $n_\sigma\sigma$, which is a subrepresentation of $\tau|_G$, has a cyclic vector. Thus $n_\sigma \leq \dim \sigma = d$ by 15.5.3.

15.5.6. Theorem. *Let G' be a connected linear semisimple real Lie group and G a maximal compact subgroup of G' .*

(i) *If $\pi \in \widehat{G'}$ and $\sigma \in \widehat{G}$, then σ occurs at most $\dim \sigma$ times in $\pi|_G$.*

(ii) *G' is liminal.*

Let $\pi \in \widehat{G'}$, and write $\pi|_G = \bigoplus_{\sigma \in \widehat{G}} n_\sigma \sigma$. Let $\sigma \in \widehat{G}$, put $d = \dim \sigma$, and let $E \subset H_\pi$ be the space of $n_\sigma \sigma$. By 15.5.5, identity (1) holds in the algebra $P_E \pi(\mathcal{K}(G')) P_E$, with $r = r(d^2)$. Since π is irreducible, $\pi(\mathcal{K}(G'))$ is strongly dense in $\mathcal{L}(H_\pi)$. Hence $P_E \pi(\mathcal{K}(G')) P_E$ is strongly dense in $P_E \mathcal{L}(H_\pi) P_E$, and identity (1) holds in $\mathcal{L}(E)$, with $r = r(d^2)$. Since $r(d^2 + 1) > r(d^2)$, E cannot contain a vector subspace of dimension $d^2 + 1$. Thus $\dim E \leq d^2$, and consequently $n_\sigma \leq d$. This proves (i), and (ii) follows from (i) and 15.5.2.

Bibliography: [54, 59, 146].

15.6. Complements

In this subsection G denotes a compact group.

15.6.1. Every continuous unitary representation of G that has a cyclic vector is a subrepresentation of the regular representation [90].

15.6.2. A continuous complex function f on G is called *almost invariant* if the linear combinations of its translates ${}_s f$ form a finite-dimensional vector space. The coefficients of finite-dimensional representations are such functions. Every continuous complex function on G is a uniform limit of linear combinations of these coefficients [86].

15.6.3. Let H be a Hilbert space, and let π be a representation in H of the involutive subalgebra $\mathcal{K}(G)$ of $L^1(G)$. There is a unique continuous unitary representation $\tilde{\pi}$ of G in H such that

$$\pi(f) = \int \tilde{\pi}(s) f(s) ds$$

for every $f \in \mathcal{K}(G)$.

15.6.4. Let G be a compact group and φ a continuous function of positive type on G . There are continuous pure functions of positive type $\varphi_1, \varphi_2, \dots$ on G , and constants $\lambda_1, \lambda_2, \dots \geq 0$, such that

$$\sum_n \lambda_n < +\infty, \quad \varphi_1(e) = \varphi_2(e) = \dots = 1, \quad \varphi = \sum_n \lambda_n \varphi_n,$$

where the series converges uniformly on G [50].

15.6.5. All connected compact real Lie groups are known. For such a group, the list of continuous irreducible unitary representations is also known: these representations are classified by means of a finite number of integers. The simplest of the noncommutative connected compact real Lie groups is the group G of rotations of ordinary three-dimensional space. If $\tilde{G}$ is its universal two-sheeted covering, then $\tilde{G}$ has a continuous irreducible unitary representation π of dimension 2, and the other continuous irreducible unitary representations of $\tilde{G}$ are the symmetric tensor powers of π .

The list of finite groups is very far from being known, so the results are less complete in that case. But a great many particular results are known.

§ 16. — Almost-Periodic Functions

In this section, G denotes a topological group. We shall see that the search for finite-dimensional continuous unitary representations of G reduces to that for finite-dimensional continuous unitary representations of a compact group Σ associated with G . In truth, even for very simple groups G , the group Σ is often complicated, so this will not give us a practical method of determining the finite-dimensional continuous unitary representations of G . But the properties of [Section 15](#), applied to Σ , will imply remarkable properties of a certain class of functions on G , called almost-periodic functions.

16.1. Compact Group Associated with a Topological Group

16.1.1. Theorem. *Let G be a topological group. There exist a compact group Σ and a continuous homomorphism $\alpha : G \rightarrow \Sigma$ having the following property: for every compact group Σ' and every continuous homomorphism $\alpha' : G \rightarrow \Sigma'$, there is a unique continuous homomorphism $\beta : \Sigma \rightarrow \Sigma'$ such that $\alpha' = \beta \circ \alpha$.*

Moreover, this property determines the pair (Σ, α) up to isomorphism.

Let (π_i) be the family of finite-dimensional continuous unitary representations of G . Let U_i be the unitary group of H_{π_i} , let U be the compact product group of the U_i , let α be the continuous homomorphism $s \mapsto (\pi_i(s))$ of G into U , and let Σ be the closure of $\alpha(G)$ in U , which is a compact subgroup of U . We show that (Σ, α) has the property in the theorem. Let Σ' be a compact group and $\alpha' : G \rightarrow \Sigma'$ a continuous homomorphism. The uniqueness of a continuous homomorphism $\beta : \Sigma \rightarrow \Sigma'$ such that $\alpha' = \beta \circ \alpha$ is immediate, since $\alpha(G)$ is dense in Σ . We prove the existence of β . The intersection of the kernels of the finite-dimensional unitary representations of Σ' is reduced to e ([15.1.6](#)); hence Σ' can be identified with a subgroup of a group $\prod_j V_j$, where each V_j is the unitary group of a finite-dimensional Hilbert space. Then α' is identified with a homomorphism $s \mapsto (\rho_j(s))$, where each ρ_j is a continuous homomorphism of G into V_j . Since ρ_j is equivalent to one of the representations π_i , there is a continuous homomorphism β_j of Σ into V_j such that $\rho_j = \beta_j \circ \alpha$. The β_j define a continuous homomorphism β of Σ into $\prod_j V_j$ such that $\alpha' = \beta \circ \alpha$, whence $\beta(\Sigma) \subset \Sigma'$.

The uniqueness of the pair (Σ, α) up to isomorphism is an easy general property of universal problems. Let (Σ_1, α_1) be another solution. There are contin-

uous homomorphisms $\beta : \Sigma \rightarrow \Sigma_1$ and $\beta_1 : \Sigma_1 \rightarrow \Sigma$ such that $\alpha_1 = \beta \circ \alpha$ and $\alpha = \beta_1 \circ \alpha_1$; hence $\alpha = (\beta_1 \circ \beta) \circ \alpha$. Thus $\beta_1 \circ \beta$ is the identity map of Σ (by uniqueness of β when α and α' are given). Similarly, $\beta \circ \beta_1$ is the identity map of Σ_1 . Hence β is an isomorphism of Σ onto Σ_1 that carries α to α_1 .

16.1.2. Definition. *The group Σ is called the compact group associated with G , and α is called the canonical homomorphism of G into Σ .*

Thus $\overline{\alpha(G)} = \Sigma$. With the notation of 16.1.1, $\beta(\Sigma)$ is the closure of $\alpha'(G)$. If G is compact, we may take $\Sigma = G$ and α to be the identity map of G .

16.1.3. Proposition. *For every finite-dimensional continuous unitary representation ρ of Σ , put $\rho' = \rho \circ \alpha$, which is a finite-dimensional continuous unitary representation of G . This defines a bijection from the set of classes of finite-dimensional continuous unitary representations of Σ onto the set of classes of finite-dimensional continuous unitary representations of G .*

If ρ' is equivalent to ρ'_1 , then ρ is equivalent to ρ_1 (because $\overline{\alpha(G)} = \Sigma$), so the map in question is injective. On the other hand, if π is a finite-dimensional continuous unitary representation of G , there is, by 16.1.1, a continuous unitary representation ρ of Σ in H_π such that $\pi = \rho \circ \alpha$; hence the map in question is surjective.

16.1.4. When G is a locally compact commutative group, the pair (Σ, α) is given by the following proposition.

Proposition. *Let G be a locally compact commutative group and $\widehat{G}$ its locally compact dual group. Let G' be the compact commutative group whose dual $\widehat{G'}$ is the group $\widehat{G}$ endowed with the discrete topology. Let φ be the continuous homomorphism of G into G' whose dual $\widehat{\varphi}$ is the identity homomorphism of $\widehat{G'}$ into $\widehat{G}$. Then G' may be identified with the compact group associated with G , and φ with the canonical homomorphism of G into G' .*

If $\chi' \in \widehat{G'}$ is trivial on $\overline{\varphi(G)}$, then $\widehat{\varphi}(\chi')$ is trivial on G ; hence $\widehat{\varphi}(\chi') = 1$ and $\chi' = 1$. This proves that $\overline{\varphi(G)} = G'$. Now let H be a compact group and ψ a continuous homomorphism of G into H . We shall prove that there is a unique continuous homomorphism β of G' into H such that $\psi = \beta \circ \varphi$; this will establish the proposition. Replacing H by $\psi(H)$, we reduce to the case where H is compact and commutative. Then the dual $\widehat{\psi}$ of ψ is a homomorphism from the discrete group $\widehat{H}$ into $\widehat{G}$. We are reduced to proving that there is a unique homomor-

phism $\widehat{\beta}$ from $\widehat{H}$ into $\widehat{G'}$ such that $\widehat{\psi} = \widehat{\varphi} \circ \widehat{\beta}$; this is evident because $\widehat{\varphi}$ is the identity map of $\widehat{G'}$ onto $\widehat{G}$.

Bibliography: [86, 181].

16.2. Almost-Periodic Functions

16.2.1. Theorem. *Let G be a topological group, Σ the associated compact group, α the canonical homomorphism of G into Σ , $E(G)$ the vector space of bounded continuous complex functions on G , furnished with the norm of uniform convergence, and $f \in E(G)$. The following conditions are equivalent:*

- (i) *The set of ${}_s f$, $s \in G$, is relatively compact in $E(G)$.*
- (ii) *The set of f_t , $t \in G$, is relatively compact in $E(G)$.*
- (iii) *The set of ${}_s f_t$, $s, t \in G$, is relatively compact in $E(G)$.*
- (iv) *There is a continuous complex function g on Σ such that $f = g \circ \alpha$.*
- (v) *f is a uniform limit on G of linear combinations of coefficients of finite-dimensional continuous irreducible unitary representations of G .*

[Since $E(G)$ is complete, “relatively compact” may be replaced by “precompact” in (i), (ii), and (iii).]

(iv) $\Rightarrow$ (iii): Let $g \in \mathcal{K}(\Sigma)$ and $f = g \circ \alpha$. The map $(\sigma, \tau) \mapsto {}_\sigma g_\tau$ from $\Sigma \times \Sigma$ into the Banach space $\mathcal{K}(\Sigma)$ is continuous; hence the set of the ${}_\sigma g_\tau$ is compact. Its image in $E(G)$ under the isometric map $h \mapsto h \circ \alpha$ is compact and contains the set of the ${}_s f_t$, $s, t \in G$.

(iii) $\Rightarrow$ (i) and (ii): This is evident.

(i) $\Rightarrow$ (iv) [the proof of (ii) $\Rightarrow$ (iv) is analogous]: For every $g \in E(G)$ and every $s \in G$, put $\psi(s)g = {}_{s^{-1}}g$. Then $\psi(s)$ is an isometric linear map of $E(G)$ onto $E(G)$, and $\psi(ss') = \psi(s)\psi(s')$. Let A be the set of the ${}_s f$, $s \in G$. It is clear that every $\psi(s)$ induces a bijection of A onto A , hence also a bijection $\varphi(s)$ of $\overline{A}$ onto $\overline{A}$. The $\varphi(s)$ are isometries, and $\overline{A}$ is compact if condition (i) holds. Thus $\varphi(G)$ is relatively compact in the set $C(\overline{A}, \overline{A})$ of continuous maps of $\overline{A}$ into itself, endowed with the topology of uniform convergence (Ascoli’s theorem). Consequently, the closure of $\varphi(G)$ in $C(\overline{A}, \overline{A})$ is a compact group Γ of homeomorphisms of $\overline{A}$ (B 16). The map φ is a homomorphism of G into Γ . We show that it is continuous. It is enough to prove that, as $s \rightarrow e$, $\varphi(s)$ tends to $\varphi(e)$ in the topology of uniform convergence on $\overline{A}$, or even only in the topology of pointwise convergence on $\overline{A}$ [which is equivalent here because $\varphi(G)$ is equicontinuous]. Let $g \in \overline{A}$ and let V be an open neighborhood of g in $\overline{A}$. Then $\overline{A} \setminus V$ is compact. If $h \in \overline{A} \setminus V$, then

$|g(u) - h(u)| > 0$ at some point u of G ; hence there are an open neighborhood V_h of h in $\overline{A}$ and a neighborhood W_h of e in G such that $\varphi(s)g \notin V_h$ for $s \in W_h$. We can cover $\overline{A} \setminus V$ by finitely many neighborhoods $V_{h_1}, \dots, V_{h_n}$; then, for $s \in W_{h_1} \cap \dots \cap W_{h_n}$, we have $\varphi(s)g \notin V_{h_1} \cup \dots \cup V_{h_n}$, and hence $\varphi(s)g \in V$. Thus φ is a continuous homomorphism of G into Γ . By the definition of Σ and α , there is a continuous homomorphism β of Σ into Γ such that $\varphi = \beta \circ \alpha$. The function $\sigma \mapsto (\beta(\sigma)f)(e)$ on Σ is continuous, and

$$f(s^{-1}) = (\varphi(s)f)(e) = (\beta(\alpha(s))f)(e),$$

so condition (iv) holds.

(iv) $\Rightarrow$ (v): Suppose that $f = g \circ \alpha$, where $g \in \mathcal{K}(\Sigma)$. Then g is a uniform limit on Σ of linear combinations of coefficients of continuous irreducible unitary representations, necessarily finite-dimensional, of Σ (13.6.5 and 15.1.4). Hence f is a uniform limit on G of linear combinations of coefficients of finite-dimensional continuous irreducible unitary representations of G .

(v) $\Rightarrow$ (iv): Suppose that, for every n , there is a linear combination f_n of coefficients of finite-dimensional continuous irreducible unitary representations of G such that $\|f - f_n\|_\infty \leq 1/n$. There is a function g_n on Σ , a linear combination of coefficients of continuous irreducible unitary representations of Σ , such that $f_n = g_n \circ \alpha$ (16.1.3). We have $\|g_m - g_n\|_\infty = \|f_m - f_n\|_\infty \rightarrow 0$ as $m, n \rightarrow +\infty$; hence the g_n converge uniformly to a continuous function g on Σ , and $f = g \circ \alpha$.

16.2.2. Definition. *An almost-periodic function on G is a bounded continuous complex function on G satisfying the equivalent conditions of 16.2.1.*

We denote the set of these functions by $\text{PP}(G)$.

16.2.3. By condition (iv) of 16.2.1, a function in $\text{PP}(G)$ is uniformly continuous for the right and left uniform structures of G . The sum, product, and so forth of two almost-periodic functions is almost-periodic; $\text{PP}(G)$ is closed in $E(G)$.

Bibliography: [86, 181].

16.3. Mean of an Almost-Periodic Function

16.3.1. Theorem. *Let $f \in \text{PP}(G)$. Let K be the closed convex hull in $E(G)$ (notation of 16.2.1) of the set of the ${}_s f$ as s ranges over G . Then K contains exactly one constant $M(f)$. If f' is the function on Σ corresponding to f , then*

$$M(f) = \int_{\Sigma} f'(s) ds,$$

the Haar measure on Σ being chosen to have total mass 1.

Through the canonical isomorphism of $\text{PP}(G)$ onto $\mathcal{K}(\Sigma)$, we reduce to the case $G = \Sigma$. Let therefore $f \in \mathcal{K}(\Sigma)$ and let $\varepsilon > 0$. There is a neighborhood V of e in Σ such that $st^{-1} \in V$ implies $|f(s) - f(t)| \leq \varepsilon$. Let $(V_i s_i)_{1 \leq i \leq n}$ be a finite covering of Σ , and let $(h_i)_{1 \leq i \leq n}$ be a partition of unity subordinate to this covering. Put $c_i = \int_{\Sigma} h_i(s) ds$. Then

$$\left| \int_{\Sigma} f(s) ds - \sum_i c_i f(s_i t) \right| = \left| \sum_i \int_{\Sigma} h_i(s) [f(st) - f(s_i t)] ds \right|,$$

and this is bounded by ε , since $|f(st) - f(s_i t)| \leq \varepsilon$ whenever $h_i(s) \neq 0$. Moreover, $\sum_i c_i = 1$. Thus the constant $\int_{\Sigma} f(s) ds$ belongs to K . Finally, all translates of f have the same integral on Σ , and hence all functions of K have the same integral on Σ ; if a constant belongs to K , its value is necessarily $\int_{\Sigma} f(s) ds$.

16.3.2. Definition. *The constant $M(f)$ of 16.3.1 is called the mean of f .*

16.3.3. The map $f \mapsto M(f)$ is a linear form on $\text{PP}(G)$. We have $M(f) \geq 0$ for $f \geq 0$, $M(1) = 1$, and $M({}_s f) = M(f_s) = M(f)$ for every $s \in G$. All this follows from the equality $M(f) = \int_{\Sigma} f'(s) ds$ in 16.3.1.

16.3.4. For $f, g \in \text{PP}(G)$, put $(f \mid g) = M(f \bar{g})$. Then $\text{PP}(G)$ becomes a separated pre-Hilbert space, canonically isomorphic to $\mathcal{K}(\Sigma)$ regarded as a vector subspace of $L^2(\Sigma)$. Let C be the set of classes of finite-dimensional continuous irreducible unitary representations of G . For each $\sigma \in C$, choose an orthonormal basis $(\xi_1, \dots, \xi_{\dim \sigma})$ of H_{σ} , and put

$$\varphi_{i,j,\sigma}(s) = (\sigma(s)\xi_i \mid \xi_j).$$

Then the functions $(\dim \sigma)^{1/2} \varphi_{i,j,\sigma}$, where $\sigma \in C$ and $1 \leq i, j \leq \dim \sigma$, form an orthonormal basis of the pre-Hilbert space $\text{PP}(G)$. This follows from 15.2.5 and 16.1.3.

Bibliography: [86, 181].

16.4. Groups Embeddable in a Compact Group

16.4.1. Lemma. *The following three subgroups of G are equal:*

- 1° *the kernel of the canonical homomorphism $G \rightarrow \Sigma$;*
- 2° *the intersection of the kernels of all continuous homomorphisms of G into compact groups;*
- 3° *the intersection of the kernels of the finite-dimensional continuous irreducible unitary representations of G .*

Denote these three sets by N_1, N_2, N_3 . We have $N_2 \supset N_1$ by the definition of Σ and of the homomorphism $G \rightarrow \Sigma$. Clearly $N_3 \supset N_2$. Finally, if $s \in N_3$, its canonical image in Σ belongs to the kernel of every continuous irreducible unitary representation of Σ (16.1.3), and hence equals e (13.6.6); thus $N_3 \subset N_1$.

16.4.2. Definition. *A topological group is said to be embeddable in a compact group if the subgroup in 16.4.1 is reduced to e .*

16.4.3. A subgroup of a group embeddable in a compact group is embeddable in a compact group. A product of groups embeddable in compact groups is embeddable in a compact group. Hence an inverse limit of groups embeddable in compact groups is embeddable in a compact group. The quotient of an arbitrary topological group by the normal subgroup of 16.4.1 is embeddable in a compact group. A locally compact commutative group is embeddable in a compact group, since the subgroup defined by condition 3° of 16.4.1 is reduced to e .

16.4.4. Proposition. *Let G be a topological group generated by a compact neighborhood of e , and suppose that G is embeddable in a compact group. Then G is an inverse limit of Lie groups locally isomorphic to compact groups.*

Let $\mathcal{F}$ be the filter base on G consisting of the kernels of the finite-dimensional continuous unitary representations of G . Since G is embeddable in a compact group, the intersection of the elements of $\mathcal{F}$ is $\{e\}$. By hypothesis, there is a compact symmetric neighborhood U of e that generates G . The trace of $\mathcal{F}$ on the compact set U^3 converges to e . Hence there is an $N_0 \in \mathcal{F}$ such that $U^3 \cap N_0 \subset U$. For $N \in \mathcal{F}$, $N \subset N_0$, put

$$K_N = U^2 \cap N \subset U.$$

The K_N have the following properties.

- a. K_N is a compact subgroup of G , since $K_N K_N^{-1} \subset U^2 \cap N = K_N$.

b. K_N is normal in $U^\infty = G$, for, if $s \in U$, then

$$sK_Ns^{-1} \subset U^3 \cap N \subset U \cap N \subset K_N.$$

Since $\bigcap_N K_N = \{e\}$, G is the inverse limit of the G/K_N .

c. K_N is open in N , so N/K_N is a discrete subgroup of G/K_N ; consequently G/K_N is locally isomorphic to G/N . Now G/N has a finite-dimensional continuous injective unitary representation, and is therefore a Lie group. Moreover, the Lie algebra $\mathfrak{g}$ of G/N is isomorphic to a subalgebra of a compact Lie algebra. Hence $\mathfrak{g}$ is a product of a commutative Lie algebra and a compact Lie algebra, and is therefore the Lie algebra of a compact group C ; thus G/K_N is locally isomorphic to C .

16.4.5. Lemma. *Let G be a connected locally compact group having a fundamental system of neighborhoods of e invariant under inner automorphisms. Let A and B be closed subgroups of G such that: 1° A is contained in the center of G ; 2° $A \subset B$, and B/A is contained in the center of G/A . Then B is contained in the center of G .*

Let $x \in B$, $y \in G$, let χ be a character of the commutative group A , and let N be the kernel of χ . We have $xyx^{-1}y^{-1} \in A$. We shall show that $\chi(xyx^{-1}y^{-1}) = 1$. Since χ is arbitrary, this will prove that $xyx^{-1}y^{-1} = e$ and establish the lemma. There is a neighborhood V of e in G such that $|\chi(u) - 1| \leq 1$ for $u \in V \cap A$, and a neighborhood V' of e in G , invariant under inner automorphisms, such that $V'V'^{-1} \subset V$. For every $z \in V'$, the element $xzx^{-1}z^{-1}$ is an element a of A , and the relation $xzx^{-1} = az$ gives by induction $x^n zx^{-n} = a^n z$. Hence

$$a^n = (x^n zx^{-n})z^{-1} \in V'V'^{-1} \subset V,$$

and consequently $a^n \in V \cap A$ and $|\chi(a^n) - 1| \leq 1$. Since this holds for every n , it follows that $\chi(a) = 1$; that is, x and z commute modulo N . Since G is connected, y is a product of elements of V' , so x and y commute modulo N .

16.4.6. Theorem. *Let G be a connected locally compact group, and let Z be its center. The following properties are equivalent:*

- (i) G is embeddable in a compact group;
- (ii) G is the product of a compact group and a group $\mathbb{R}^n$;
- (iii) G is an inverse limit of Lie groups locally isomorphic to compact groups;
- (iv) G/Z is compact;
- (v) G has a fundamental system of neighborhoods of e invariant under inner automorphisms.

Moreover, if G has these properties, every connected group locally isomorphic to G also has them.

(i) $\Rightarrow$ (iii): This follows from 16.4.4.

(iii) $\Rightarrow$ (ii): Suppose condition (iii) holds. There is in G a filter base converging to e , formed of closed normal subgroups N such that the groups $G/N = G_N$ are Lie groups locally isomorphic to compact groups. Moreover, G_N is connected. Let G_N^1, G_N^2 be the connected closed normal subgroups of G_N corresponding to the center and the derived algebra of the reductive Lie algebra of G_N . Then $G_N = G_N^1 G_N^2$, the groups G_N^1 and G_N^2 commute, and G_N^2 is compact. If $N' \subset N$, the canonical map of $G_{N'}$ onto G_N maps $G_{N'}^1$ onto G_N^1 and $G_{N'}^2$ onto G_N^2 . Let $A \subset G$ be the inverse limit of the G_N^1 and $K \subset G$ the inverse limit of the G_N^2 . Then A is commutative and connected, K is compact, A and K commute, and $G = AK$. We have $A = \mathbb{R}^n \times K_1$, with K_1 compact (B 25). Thus $G = \mathbb{R}^n K_2$, with a compact subgroup K_2 commuting with $\mathbb{R}^n$. Necessarily $K_2 \cap \mathbb{R}^n = \{e\}$, and hence G is isomorphic to $\mathbb{R}^n \times K_2$.

(ii) $\Rightarrow$ (v): This follows from 15.1.1.

(v) $\Rightarrow$ (iv): Suppose condition (v) holds. For each $s \in G$, let $j(s)$ be the inner automorphism of G defined by s . The hypothesis implies that the right and left uniform structures of G coincide and that $j(G)$ is an equicontinuous group of homeomorphisms of G . There is a compact neighborhood U of e invariant under inner automorphisms. For each $s \in G$, there is a positive integer n such that $s \in U^n$, since G is connected; the conjugacy class of s is then contained in U^n , and is therefore relatively compact. Let $\mathcal{T}$ be the topology of compact convergence on the set $C(G, G)$ of continuous maps of G into itself. By Ascoli's theorem, $j(G)$ is relatively compact in $C(G, G)$ for $\mathcal{T}$. Thus the closure of $j(G)$ is a group Γ of homeomorphisms of G , compact for $\mathcal{T}$ (B 16). Finally, j is continuous as a map of G into Γ , and therefore the injective homomorphism of G/Z into Γ induced by j is a topological isomorphism onto its image. Thus G/Z is embeddable in a compact group. By the implication (i) $\Rightarrow$ (ii), already established, we conclude that $G/Z = A \times K$, with A commutative and K compact. By 16.4.5, A is reduced to $\{e\}$; hence G/Z is compact.

(iv) $\Rightarrow$ (i): If G/Z is compact, then Z is generated by a compact neighborhood of e (B 24), and hence is of the form $K \times L$, with K compact and $L = \mathbb{R}^n \times \mathbb{Z}^p$ (B 25). Let s be an element of G distinct from e . We shall prove, thereby establishing (iv) $\Rightarrow$ (i), that there is a continuous homomorphism φ of G into a compact group such that $\varphi(s) \neq e$. There is a closed subgroup M of L such that $s \notin M$ and L/M

is compact. Then G/M , which is an extension of G/Z by $K \times (L/M)$, is compact, and it is enough to take for φ the canonical homomorphism of G onto G/M .

Finally, if G has property (v), every connected group locally isomorphic to G also has it.

In place of “embeddable in a compact group,” one also says “representable in a compact group” or “maximally almost-periodic.”

Bibliography: [2, 181].

16.5. Complements

16.5.1. A group embeddable in a compact group may have infinite-dimensional continuous irreducible unitary representations [50].

* **16.5.2.** Let G be a locally compact group. We use again the notation of 13.11.6.

a. For $\varphi, \psi \in E$ and $x \in G$, the function $t \mapsto \psi(t)\varphi(t^{-1}x)$ is an element ω_x of E . Put $M(\omega_x) = (\varphi \times \psi)(x)$. Then $\varphi \times \psi \in \text{PP}(G)$.

b. Every continuous function of positive type can be written $\varphi_1 + \varphi_2$, where φ_1 is of positive type and almost-periodic, and φ_2 is of positive type and satisfies $M(\varphi_2 \overline{\varphi_2}) = 0$.

c. Every almost-periodic function of positive type can be written $\sum \lambda_i \varphi_i$, where $\lambda_i > 0$, $\sum \lambda_i < +\infty$, and φ_i is a pure function of positive type corresponding to a finite-dimensional representation [50].

16.5.3. *a.* Let G be a locally compact group and G_0 the connected component of e . Suppose that G/G_0 is compact. The following conditions are equivalent: (i) G is embeddable in a compact group; (ii) G_0 is embeddable in a compact group; (iii) G is a semidirect product of a compact subgroup K and a normal subgroup V isomorphic to $\mathbb{R}^n$, every element of V commuting with every element of the identity component of K ; (iv) G has a fundamental system of compact neighborhoods of e invariant under inner automorphisms.

b. There are totally disconnected locally compact groups embeddable in compact groups that do not satisfy condition (iv) of part a [85, 103].

16.5.4. Problem: prove the equivalence of conditions (i), (ii), (iv), and (v) of 16.4.6 without passing through Lie groups.

§ 17. — Characters of a Locally Compact Group

17.1. Definitions

17.1.1. Let G be a locally compact group. A *trace* (respectively a *bitrace*, a *character*) of G means a semifinite lower semicontinuous trace (respectively a maximal bitrace, a character) of $C^*(G)$.

17.1.2. A *traced representation* of G means a pair (π, t) with the following properties: (i) π is a continuous unitary representation of G ; (ii) t is a faithful normal trace on $\mathfrak{U}^+$, where $\mathfrak{U}$ denotes the von Neumann algebra generated by $\pi(G)$; (iii) $\pi(C^*(G)) \cap \mathfrak{M}_t$ generates the von Neumann algebra $\mathfrak{U}$.

Traced representations of G are identified with traced representations of $C^*(G)$. Quasi-equivalence of two traced representations is defined in the evident manner (cf. 6.6.2). With every traced representation of G there is canonically associated a trace of G .

17.1.3. A trace f of G defines the objects $\mathfrak{M}_f, \mathfrak{N}_f, N_f, \Lambda_f, H_f, J_f, \lambda_f, \rho_f, \mathfrak{U}_f, \mathfrak{B}_f, t_f$ (Section 6.2 and 6.4.5). The representation λ_f of $C^*(G)$ is identified with a representation of G , also denoted by λ_f . The representation $\bar{\rho}_f^0$ of $C^*(G)$ in $\bar{H}_f$ is the transform of λ_f by the isomorphism J_f of H_f onto $\bar{H}_f$ (6.2.2); it is identified with a representation of G , also denoted by $\bar{\rho}_f^0$, which is the transform by J_f of the representation λ_f of G . Thus $\bar{\rho}_f^0 \simeq \lambda_f$. A trace of G is said to be of type I, type II, and so on, if λ_f is of type I, type II, and so on.

The preceding discussion and 17.1.2 define a bijection from the set of traces of G onto the set of quasi-equivalence classes of traced representations of G .

17.1.4. Let π be a continuous unitary representation of G , and let $\mathfrak{U}$ be the von Neumann algebra generated by $\pi(G)$. The representation π is called *traceable* if there is a trace t on $\mathfrak{U}^+$ such that (π, t) is a traced representation. Traceable representations of G are identified with traceable representations of $C^*(G)$.

17.1.5. Let π be a continuous factor unitary representation of G , and let $\mathfrak{F}$ be the factor generated by $\pi(G)$. The following conditions are equivalent: (i) π is traceable; (ii) $\mathfrak{F}$ is semifinite and $\pi(C^*(G))$ contains a nonzero Hilbert–Schmidt operator relative to $\mathfrak{F}$ (6.7.2).

Let f be a trace of G . The corresponding representation of G is a factor representation if and only if f is a character (6.7.3). Hence there is a canonical bijection between: *a.* the set of characters of G considered up to a positive scalar factor; *b.* the set of quasi-equivalence classes of traceable factor representations of G (6.7.4).

17.1.6. Suppose G is postliminal. Every continuous irreducible unitary representation π of G is traceable (6.7.5) and has a normalized character f_π , defined by $f_\pi(x) = \text{Tr } \pi(x)$ for $x \in C^*(G)^+$. If π' is another continuous irreducible unitary representation of G , then π and π' are equivalent if and only if $f_\pi = f_{\pi'}$ (6.7.6). Every character of G has the form λf_π , where $\lambda > 0$ and π is a continuous irreducible unitary representation of G (6.7.6). Let $f = f_\pi$. Then $H_f, \lambda_f, \bar{\rho}_f^0, \mathfrak{U}_f, \mathfrak{B}_f$ are canonically identified with $H_\pi \otimes \overline{H_\pi}, \pi \otimes 1, 1 \otimes \bar{\pi}, \mathcal{L}(H_\pi) \otimes \mathbb{C}$, and $\mathbb{C} \otimes \mathcal{L}(\overline{H_\pi})$, respectively; J_f is identified with the canonical involution of $H_\pi \otimes \overline{H_\pi}$; and t_f is identified with the trace

$$T \otimes 1 \longmapsto \text{Tr } T \quad \text{on } (\mathcal{L}(H_\pi) \otimes \mathbb{C})^+$$

(6.7.7).

Bibliography: [54].

17.2. Character Defined by a Measure or a Distribution

17.2.1. Lemma. *Let A be a C^* -algebra, let A' be a dense involutive subalgebra of A , and let s be a complex-valued function on $A' \times A'$ satisfying the following conditions:*

- (i) *s is a positive Hermitian sesquilinear form;*
- (ii) *$s(y, x) = s(x^*, y^*)$ for $x, y \in A'$;*
- (iii) *$s(zx, y) = s(x, z^*y)$ for $x, y, z \in A'$;*
- (iv) *$s(zx, zx) \leq \|z\|^2 s(x, x)$ for $x, z \in A'$;*
- (v) *the elements xy , where $x, y \in A'$, are dense in A' for the pre-Hilbert structure defined by s .*

Then there is a unique maximal bitrace of A extending s .

Uniqueness follows from 6.5.3. We prove existence. Let N be the set of $x \in A'$ such that $s(x, x) = 0$. As in 6.2.2, one verifies that N is a self-adjoint two-sided ideal of A' and that A'/N is a Hilbert algebra for the inner product $(\cdot | \cdot)$ induced by s . Let H be the Hilbert-space completion of A'/N , and let Λ be the canonical map of A' into H . Hypothesis (iv) implies that, for every $z \in A'$, there is

a unique bounded linear operator $\lambda(z)$ on H such that $\lambda(z)\Lambda x = \Lambda(zx)$ for every $x \in A'$; moreover $\|\lambda(z)\| \leq \|z\|$. It is immediate that λ is a representation of A' in H ; it extends uniquely to a representation, still denoted by λ , of A in H . Then $\lambda(A)$ and $\lambda(A')$ generate the same von Neumann algebra, namely $\mathfrak{U}(A'/N)$. Let θ be the natural trace on $\mathfrak{U}(A'/N)^+$ defined by the Hilbert algebra A'/N . Then $\mathfrak{n}_\theta \supset \lambda(A')$, so (λ, θ) is a traced representation of A . Let s_1 be the corresponding maximal bitrace of A . For $x \in A'$, we have $\lambda(x) \in \mathfrak{n}_\theta$, so $x \in \mathfrak{n}_{s_1}$, and

$$s_1(x, x) = \theta(\lambda(x)\lambda(x)^*) = s(x, x).$$

Thus s_1 extends s .

17.2.2. If G is a locally compact group, $\mathcal{K}(G)$ is an involutive subalgebra of $L^1(G)$, dense in $L^1(G)$ and hence in $C^*(G)$.

Proposition. *Let G be a unimodular locally compact group and μ a measure on G . For $f, g \in \mathcal{K}(G)$, put $s(f, g) = \mu(g^* * f)$. Then s satisfies conditions (i)–(v) of 17.2.1 if and only if μ is a central measure of positive type.*

It is clear that s is sesquilinear, so condition (i) of 17.2.1 holds if and only if $\mu(f^* * f) \geq 0$ for every $f \in \mathcal{K}(G)$, that is, if and only if $\mu \gg 0$. Suppose henceforth that this is so. Condition (ii) holds if and only if $\mu(g^* * f) = \mu(f * g^*)$ for all $f, g \in \mathcal{K}(G)$, that is, if and only if μ is central. Condition (iii) reads

$$\mu(g^* * (h * f)) = \mu((h^* * g)^* * f)$$

for all $f, g, h \in \mathcal{K}(G)$; it holds automatically. Let $f \in \mathcal{K}(G)$ and let K be a compact neighborhood of the support of f . There is a $g \in \mathcal{K}(G)$ such that $f * g$ has support in K and $\|f - f * g\|_\infty$ is arbitrarily small; it follows at once that condition (v) holds. We saw in 13.7.9 that μ defines a unitary representation σ of G . The space H_σ is obtained precisely by furnishing $\mathcal{K}(G)$ with the inner product $s(f, g)$ and passing to the separated completed Hilbert space. If Λ denotes the canonical map of $\mathcal{K}(G)$ into H_σ , then $\sigma(x)\Lambda f = \Lambda(\varepsilon_x * f)$ for every $x \in G$ and $f \in \mathcal{K}(G)$. It follows that $\sigma(g)\Lambda f = \Lambda(g * f)$ for $f, g \in \mathcal{K}(G)$, and hence

$$\begin{aligned} s(g * f, g * f) &= (\sigma(g)\Lambda f \mid \sigma(g)\Lambda f) \\ &\leq \|\sigma(g)\|^2 (\Lambda f \mid \Lambda f) \leq \|g\|_1^2 s(f, f), \end{aligned}$$

so condition (iv) holds.

17.2.3. Let μ be a central measure of positive type on the unimodular group G . By 17.2.1 and 17.2.2, the form $(f, g) \mapsto \mu(g^* * f)$ on $\mathcal{K}(G) \times \mathcal{K}(G)$ extends to a

unique maximal bitrace μ' of $C^*(G)$. Let σ' be the representation of $C^*(G)$, or of G , defined by μ' , and retain the notation σ of 17.2.2. By 6.3.6, $H_{\sigma'} = H_\sigma$; moreover, for $f \in \mathcal{K}(G)$, $\sigma'(f)$ is induced by the operator of left convolution by f in $\mathcal{K}(G)$, so $\sigma'(f) = \sigma(f)$. Thus $\sigma' = \sigma$.

In fact, μ' defines a traced representation (σ, t) of G . For $f \in \mathcal{K}(G)$, we have $\sigma(f) \in \mathfrak{n}_t$ and

$$t(\sigma(f)\sigma(f)^*) = \mu'(f, f) = \mu(f^* * f).$$

The map $\mu \mapsto \mu'$ is injective, since knowledge of μ on the functions $g * f, f, g \in \mathcal{K}(G)$, determines μ completely.

One sometimes identifies μ, μ' , and the associated trace, so it is meaningful to say that a central measure μ of positive type on G is a character of G . By the preceding discussion, this is so if and only if the continuous unitary representation of G obtained from μ by the procedure of 13.7.9 is a factor representation. More particularly, it is meaningful to say that a continuous central function φ of positive type on G is a character [we identify φ with the measure $\varphi(x) dx$].

17.2.4. We spell out a special case important for applications.

Proposition. *Let G be a unimodular postliminal group, let σ be a continuous irreducible unitary representation of G , and let μ be a central measure of positive type on G . The representation σ has normalized character μ if and only if $\sigma(f)$ is a Hilbert–Schmidt operator for every $f \in \mathcal{K}(G)$ and, for $f, g \in \mathcal{K}(G)$,*

$$\mathrm{Tr}(\sigma(g)^*\sigma(f)) = \mu(g^* * f).$$

If this is so, σ is quasi-equivalent to the representation of G obtained from μ by the procedure of 13.7.9.

17.2.5. Suppose G is unimodular. The Dirac measure ε_e at e is central and of positive type. It defines a maximal bitrace s of $C^*(G)$. Then $\mathcal{K}(G)$ is dense in $\mathfrak{n}_s$ for the pre-Hilbert structure of $\mathfrak{n}_s$; for $f, g \in \mathcal{K}(G)$,

$$s(f, g) = \int f(x) \overline{g(x)} dx.$$

The space H_s is $L^2(G)$, λ_s is the left regular representation, $\bar{\rho}_s^\circ$ is the conjugate of the right regular representation, and J_s is the map $f \mapsto f^*$ of $L^2(G)$ onto $\overline{L^2(G)}$. The completed Hilbert algebra of $\mathfrak{n}_s/N_s$ is the Hilbert algebra A of G ; the von Neumann algebra $\mathfrak{U}_s$ is $\mathfrak{U}(A)$, that is, the von Neumann algebra generated by

the left-translation operators on $L^2(G)$; $\mathfrak{U}_s$ has a natural trace t_s , and (λ_s, t_s) is a traced representation of G . If $f \in L^1(G) \cap L^2(G)$, then

$$t_s(\lambda_s(f)\lambda_s(f)^*) < +\infty;$$

and, for $f, g \in L^1(G) \cap L^2(G)$,

$$t_s(\lambda_s(f)\lambda_s(g)^*) = \int f(x)\overline{g(x)} dx.$$

17.2.6. Let G be a real Lie group. Denote by $\mathcal{D}(G)$ the set of infinitely differentiable compactly supported complex functions on G . This set is dense in $L^1(G)$, and hence in $C^*(G)$.

A distribution μ on G is called *central* if it is invariant under the inner automorphisms of G , or, equivalently, if $\mu(f * g) = \mu(g * f)$ for all $f, g \in \mathcal{D}(G)$. A distribution μ on G is said to be of *positive type* if $\mu(f^* * f) \geq 0$ for every $f \in \mathcal{D}(G)$.

Proposition. *Let G be a unimodular real Lie group and μ a distribution on G . For $f, g \in \mathcal{D}(G)$, put $s(f, g) = \mu(g^* * f)$. Then s satisfies conditions (i)–(v) of 17.2.1 if and only if μ is a central distribution of positive type.*

The proof, entirely analogous to that of 17.2.2, is left to the reader.

17.2.7. Retain the hypotheses of the proposition in 17.2.6. By 17.2.1, the form s extends to a unique maximal bitrace μ' of $C^*(G)$, which defines a traced representation (σ, t) of G . The representation σ is obtained by furnishing $\mathcal{D}(G)$ with the inner product $s(f, g)$, passing to the separated completed Hilbert space, and making G act by left translation on $\mathcal{D}(G)$. For $f \in \mathcal{D}(G)$, we have $\sigma(f) \in \mathfrak{n}_t$ and

$$t(\sigma(f)\sigma(f)^*) = \mu'(f, f) = \mu(f^* * f).$$

The map $\mu \mapsto \mu'$ is injective.

One sometimes identifies μ, μ' , and the associated trace, so it is meaningful to say that a central distribution of positive type on G is a character of G . In particular:

17.2.8. Proposition. *Let G be a unimodular postliminal Lie group, let σ be a continuous irreducible unitary representation of G , and let μ be a central distribution of positive type on G . The representation σ has normalized character μ if and only if $\sigma(f)$ is a Hilbert–Schmidt operator for every $f \in \mathcal{D}(G)$ and, for $f, g \in \mathcal{D}(G)$,*

$$\mathrm{Tr}(\sigma(g)^*\sigma(f)) = \mu(g^* * f).$$

If this is so, σ is quasi-equivalent to the representation of G obtained from μ by the procedure of 17.2.7.

Bibliography: [54, 115].

17.3. Characters of Finite Type

17.3.1. Proposition. *Let G be a unimodular locally compact group, let s be a character of G , and let (π, t) be a traced factor representation of G associated with s .*

- (i) *The character s is of finite type if and only if it is defined by a continuous central function χ of positive type on G .*
- (ii) *If this is so, then $\chi(x) = t'(\pi(x))$, where t' denotes the linear extension of t to the von Neumann algebra generated by $\pi(G)$.*

The character s is of finite type if and only if it extends to a central positive linear form on $C^*(G)$ (6.8.1 and 6.8.5). The central positive linear forms on $C^*(G)$ correspond bijectively to the continuous central positive linear forms on $L^1(G)$ (2.7.5), and hence also to the continuous central functions of positive type on G . Thus, if s is of finite type, there is a continuous central function χ of positive type on G such that

$$s(f * f^*) = \langle \chi, f * f^* \rangle \quad (f \in \mathcal{K}(G));$$

in this sense s is defined by χ . The converse is evident. Suppose s is defined by χ . The function $x \mapsto t'(\pi(x))$ on G is continuous. [Indeed, t' is normal, hence ultra-strongly continuous (A 25), and therefore strongly continuous on the unit ball (A 1).] For all $f, g \in \mathcal{K}(G)$, we have

$$\begin{aligned} \int t'(\pi(x))(g^* * f)(x) dx &= t' \left(\int \pi(x)(g^* * f)(x) dx \right) \\ &= t'(\pi(g^* * f)) = \int \chi(x)(g^* * f)(x) dx. \end{aligned}$$

Thus the measures $t'(\pi(x)) dx$ and $\chi(x) dx$ are equal, and hence $t'(\pi(x)) = \chi(x)$ for every $x \in G$.

We shall identify the finite-type characters of G with continuous central functions of positive type on G .

17.3.2. Suppose G is compact. Every factor representation of G is a multiple of an irreducible representation π (15.1.8), and $\dim \pi < +\infty$. By 17.3.1, every character of G in the sense of 17.1.1 is identified with a function $s \mapsto \lambda \operatorname{Tr} \pi(s)$ on G , where

$\lambda > 0$ and π is an irreducible representation of G . Up to the scalar factor λ , this is precisely the definition in 15.3.1.

17.3.3. Suppose G is commutative. Every factor representation of G is a multiple of a one-dimensional representation. Thus the characters of G in the sense of 17.1.1 are identified with the functions $\lambda\chi$, where $\lambda > 0$ and χ is a character of G in the usual sense.

17.3.4. Proposition. *Let G be a unimodular locally compact group. There is a canonical bijection between:*

- a. *the set of quasi-equivalence classes of continuous finite-type factor unitary representations of G ;*
- b. *the set of finite-type characters of G equal to 1 at e .*

This follows from 6.8.6.

17.3.5. Proposition. *Let G be a unimodular locally compact group, and let C be the set of continuous central functions φ of positive type on G such that $\varphi(e) \leq 1$.*

- (i) *C is convex and compact for the weak topology $\sigma(L^\infty(G), L^1(G))$.*
- (ii) *The extreme points of C are 0 and the finite-type characters of G equal to 1 at e .*
- (iii) *C is the weakly closed convex hull of 0 and of the set of finite-type characters equal to 1 at e .*

This follows from 6.8.7.

17.3.6. Proposition. *Suppose that G has a fundamental system of compact neighborhoods of e invariant under inner automorphisms.*

- (i) *For every $s \in G, s \neq e$, there is a finite-type character χ of G such that $\chi(s) \neq \chi(e)$.*
- (ii) *For every $s \in G, s \neq e$, there is a continuous finite-type factor unitary representation π of G such that $\pi(s) \neq 1$.*

Let $s \in G, s \neq e$. There is a compact symmetric neighborhood V of e , invariant under inner automorphisms, such that $s \notin V^2$. Let f be the characteristic function of V . Then $g = f * f^*$ is a continuous central function of positive type on G , equal to 0 at s and to $\lambda = \|f\|_2^2 > 0$ at e . By 17.3.5, $\lambda^{-1}g$ is a weak limit of positive linear combinations h_i of finite-type characters such that $\|h_i\|_\infty \leq 1$. Since

$$\|\lambda^{-1}g\|_\infty \leq \liminf_i \|h_i\|_\infty,$$

we may suppose that $\|h_i\|_\infty = 1$ for every i ; then h_i tends to $\lambda^{-1}g$ uniformly on every compact set (13.5.2). Thus $h_i(e) \neq h_i(s)$ for some i , proving (i).

Let χ be a finite-type character of G such that $\chi(s) \neq \chi(e)$, and let (π, t) be the corresponding finite-type traced factor representation. Denote by t' the linear extension of t to the von Neumann algebra generated by $\pi(G)$. We have

$$t'(\pi(s)) = \chi(s) \neq \chi(e) = t'(\pi(e));$$

hence $\pi(s) \neq \pi(e) = 1$.

17.3.7. It is not known whether the converse of 17.3.6 is true. However:

Proposition.

- (i) *Let G be a locally compact group. Suppose that, for every $s \in G$, $s \neq e$, there is a continuous finite-type factor unitary representation π of G such that $\pi(s) \neq 1$. Then:*
 - (★) *if K is a compact subset of G with $e \notin K$, there is a neighborhood of e disjoint from K and invariant under inner automorphisms.*
- (ii) *If a locally compact group G satisfies (★) and is generated by a compact set, then G has a fundamental system of compact neighborhoods of e invariant under inner automorphisms.*

Let s be a point of G distinct from e . There is a continuous finite-type factor unitary representation π of G such that $\pi(s) \neq 1$. Let t be the canonical trace on the factor generated by $\pi(G)$. The function

$$x \longmapsto \varphi(x) = t[(\pi(x)^* - 1)(\pi(x) - 1)] = t(2 - \pi(x) - \pi(x^{-1}))$$

on G is continuous (cf. the proof of 17.3.1) and central; $\varphi(e) = 0$ and $\varphi(s) \neq 0$. Hence there is a closed neighborhood V_s of s , invariant under inner automorphisms, such that $e \notin V_s$. If K is a compact subset of G with $e \notin K$, then K can be covered by finitely many sets $V_{s_1}, \dots, V_{s_n}$, and the complement of $V_{s_1} \cup \dots \cup V_{s_n}$ is a neighborhood of e disjoint from K and invariant under inner automorphisms. This proves (i). Assertion (ii) follows at once from the next lemma.

17.3.8. Lemma. *Let G be a locally compact group satisfying condition (★) of 17.3.7. Let V be a neighborhood of e and K a compact subset of G . There is a neighborhood of e contained in V and invariant under the inner automorphisms defined by the elements of K .*

Shrinking V , we may suppose it open and relatively compact. Then, enlarging K , we may suppose $V \subset K$ and $K = K^{-1}$. Put $K' = K^3 \cap (G \setminus V)$, which is compact. We have $e \notin K'$. By (★) there is a neighborhood W of e , disjoint from K' and invariant under inner automorphisms. We shall show, and this will complete the proof, that if $s \in K$, then $V \cap W$ is invariant under the automorphism

$g \mapsto sgs^{-1}$ of G . Let $x \in V \cap W$. We have $sxs^{-1} \in W$ and $sxs^{-1} \in KVK^{-1} \subset K^3$. If $sxs^{-1} \notin V$, then $sxs^{-1} \in K^3 \cap (G \setminus V) = K'$, a contradiction. Hence $sxs^{-1} \in V \cap W$.

17.3.9. A connected group is generated by every neighborhood of the identity. Thus, among locally compact groups having sufficiently many continuous finite-type factor unitary representations, the only connected groups are, by 17.3.7 and 16.4.6, products of groups $\mathbb{R}^n$ with compact groups. These are not very interesting examples, since in fact the continuous irreducible unitary representations of these groups are finite-dimensional (13.1.8). But note that, by 17.3.6, every discrete group has sufficiently many continuous finite-type factor unitary representations.

Later, in 18.7.9, we shall obtain results concerning the existence of characters that are not necessarily of finite type.

Bibliography: [53, 54].

17.4. Complements

17.4.1. *a.* Let G be a locally compact group having a fundamental system (V_i) of compact neighborhoods of e invariant under inner automorphisms. Let f be a character of G such that $\mathfrak{m}_f$ is dense in $C^*(G)$. Then f is of finite type. [Let φ_i be the characteristic function of V_i . The operator $\lambda_f(\varphi_i)$ is scalar and nonzero for some i . Since φ_i is a limit in $C^*(G)$ of elements of $\mathfrak{m}_f$, the operator 1 is a norm limit of trace-class operators relative to $\mathfrak{U}_f$. Thus $\mathfrak{U}_f$ is a finite factor.] [54].

** b.* A countable discrete group may have characters that are not of finite type [57].

** 17.4.2.* There are countable discrete antiliminal groups for which the space of finite-type characters of norm 1 is not locally compact [57].

17.4.3. Let G be a connected locally compact group, let π be a continuous unitary representation of G , and let A be the von Neumann algebra generated by $\pi(G)$. Then the largest projection E in the center of A such that A_E is of type II_1 is 0 [72, 140].

17.4.4. Let G be a connected locally compact group having a family of unitary representations (π_i) with the following properties: 1° for every $s \in G$, there is an i such that $\pi_i(s) \neq 1$; 2° the maps $s \mapsto \pi_i(s)$ are norm-continuous. Then G is a product of a compact group and a group $\mathbb{R}^n$. (Use 16.4.6 and 17.3.8.) [72, 143].

17.4.5. Let G be a connected noncompact simple real Lie group. The only continuous unitary representations π of G for which $\pi(G)$ is contained in a finite factor are the trivial representations. This applies in particular to finite-dimensional representations [140].

* **17.4.6.** Let G be a connected semisimple real Lie group and π a continuous irreducible unitary representation of G . If f is a compactly supported element of $L^2(G)$, then $\pi(f)$ is a Hilbert–Schmidt operator. If f is infinitely differentiable and compactly supported on G , then $\pi(f)$ is a trace-class operator, and $f \mapsto \text{Tr } \pi(f)$ is a distribution χ_π on G . Thus π has a character, namely the distribution χ_π . Harish-Chandra has recently announced that χ_π is defined by a locally integrable function on G , analytic on the set of regular elements of G . For the classical complex simple groups, χ_π has been calculated by Gelfand and Naimark [59, 60, 61, 115].

* **17.4.7.** Let G be a simply connected nilpotent real Lie group and π a continuous irreducible unitary representation of G . If f is infinitely differentiable on G and rapidly decreasing at infinity, then $\pi(f)$ is a trace-class operator, and $f \mapsto \text{Tr } \pi(f)$ is a tempered distribution χ_π on G . Thus π has a character, namely the distribution χ_π . In general this distribution is not defined by a function, or even by a measure [6, 7, 8, 53, 81].

17.4.8. Let G be a locally compact group in which e has a fundamental system of compact neighborhoods invariant under inner automorphisms. If μ is a central measure of positive type on G , the corresponding unitary representation of G is of finite type [53].

§ 18. — Dual of a Locally Compact Group

18.1. Definition of the Dual

18.1.1. Let G be a locally compact group. There is a canonical bijection from $C^*(G)^\wedge$ onto $\widehat{G}$ (13.9.3). By transporting the topology of $C^*(G)^\wedge$ under this bijection, one obtains a topology on $\widehat{G}$.

Definition. The topological space so obtained is called the dual of G .

Henceforth $\widehat{G}$ will denote this topological space. The spaces $\widehat{G}$ and $C^*(G)^\wedge$ are often identified.

18.1.2. The space $\widehat{G}$ is a locally quasi-compact Baire space (3.4.13 and 3.3.8). If G is discrete, $C^*(G)$ has an identity, so $\widehat{G}$ is quasi-compact (3.1.8). If G is separable, $\widehat{G}$ is separable (3.3.4). If G is postliminal, $\widehat{G}$ has a dense open locally compact subset (4.4.5).

18.1.3. Let π be a continuous unitary representation of G , and let S be a set of continuous unitary representations of G . We say that π is *weakly contained* in S if π , regarded as a representation of $C^*(G)$, is weakly contained in S , regarded as a set of representations of $C^*(G)$.

We shall now translate 3.4.4, 3.4.9, and 3.4.10 into the language of groups. For this translation we use: 1° Proposition 2.7.5, which passes from positive forms on $C^*(G)$ to continuous positive forms on $L^1(G)$; 2° Theorems 13.4.5 and 13.5.2, which pass from positive forms on $L^1(G)$ to continuous functions of positive type on G . We obtain:

18.1.4. Proposition. Let π be a continuous unitary representation of G , and let S be a set of continuous unitary representations of G . The following conditions are equivalent:

- (i) π is weakly contained in S ;
- (ii) every function of positive type associated with π is a uniform limit on every compact set of sums of functions of positive type associated with S .

If π has a cyclic vector ξ , these conditions are also equivalent to:

- (ii') the function $s \mapsto (\pi(s)\xi \mid \xi)$ is a uniform limit on every compact set of sums of functions of positive type associated with S .

18.1.5. Proposition. Let $\pi \in \widehat{G}$ and $S \subset \widehat{G}$. The following conditions are equivalent:

- (i) $\pi \in \overline{S}$;
- (ii) π is weakly contained in S ;
- (iii) one of the nonzero functions of positive type associated with π is a uniform limit on every compact set of functions of positive type associated with S ;
- (iv) every function of positive type associated with π is a uniform limit on every compact set of functions of positive type associated with S .

18.1.6. The equivalence (i) $\Leftrightarrow$ (iii) of 18.1.5 proves in particular that, if G is commutative, the topology defined in 18.1.1 is the usual topology of the dual group of G .

18.1.7. Definition. If π is a continuous unitary representation of G , the support of π is the set of $\sigma \in \widehat{G}$ that are weakly contained in π . If φ is a continuous function of positive type on G , the analyzer of φ is the support of π_φ .

If S is the support of π , then S is closed in $\widehat{G}$, and π is weakly contained in S (3.4.6).

18.1.8. Proposition. Let φ be a continuous function of positive type on G , and let $A \subset \widehat{G}$ be its analyzer.

- (i) φ is a uniform limit on every compact set of sums of pure functions of positive type associated with the elements of A .
- (ii) Every function of positive type associated with an element of A is a uniform limit on every compact set of functions

$$s \mapsto \sum_{i,j=1}^n \lambda_i \overline{\lambda_j} \varphi(s_j^{-1} s s_i),$$

where $s_1, \dots, s_n \in G$ and $\lambda_1, \dots, \lambda_n \in \mathbb{C}$.

Since π_φ is weakly contained in A , (i) follows from 18.1.4. Let $\pi \in A$. If π and π_φ are regarded as representations of $C^*(G)$, then $\ker \pi \supset \ker \pi_\varphi$; hence every state associated with π is a weak limit of states associated with π_φ [3.4.2 (ii)]. Assertion (ii) now follows from 13.4.10.

18.1.9. Let n be a cardinal, let H_n be the standard Hilbert space of dimension n , and let $\text{Rep}_n(G)$ be the set of continuous unitary representations of G in H_n . There is a canonical bijection from $\text{Rep}_n(G)$ onto $\text{Rep}'_n(C^*(G))$, the set of nondegenerate representations of $C^*(G)$ in H_n . The inverse image under this bijection of the topology of $\text{Rep}'_n(C^*(G))$ (3.5.2) is a certain topology on $\text{Rep}_n(G)$, which we

now describe directly. By 3.5.8, this will give a new definition of the topology of $\widehat{G}_n$, the subspace of $\widehat{G}$ formed by those $\sigma \in \widehat{G}$ for which $\dim \sigma = n$.

Proposition. *Let $\pi_i, \pi \in \text{Rep}_n(G)$, and let π'_i, π' be the corresponding elements of $\text{Rep}'_n(C^*(G))$. The following conditions are equivalent:*

- (i) $\pi'_i \rightarrow \pi'$;
- (ii) for every $\xi \in H_n$ and every compact subset K of G ,

$$\|\pi_i(s)\xi - \pi(s)\xi\| \longrightarrow 0 \quad \text{uniformly on } K;$$

- (iii) for every $\xi, \eta \in H_n$ and every compact subset K of G ,

$$|(\pi_i(s)\xi \mid \eta) - (\pi(s)\xi \mid \eta)| \longrightarrow 0 \quad \text{uniformly on } K.$$

(ii) $\Rightarrow$ (iii): This is evident.

(iii) $\Rightarrow$ (i): Suppose (iii) holds. For every $f \in \mathcal{K}(G)$,

$$|(\pi'_i(f)\xi \mid \eta) - (\pi'(f)\xi \mid \eta)| \leq \int |f(s)| |(\pi_i(s)\xi \mid \eta) - (\pi(s)\xi \mid \eta)| ds,$$

and the right-hand side tends to zero. Since $\mathcal{K}(G)$ is dense in $C^*(G)$, $\pi'_i \rightarrow \pi'$ by 3.5.4.

(i) $\Rightarrow$ (ii): Suppose (i) holds, and prove (ii). It is enough to do so when ξ has the form $\pi(f)\eta$, with $f \in L^1(G)$ and $\eta \in H_n$, since vectors of this form are dense in H_n . Now

$$\begin{aligned} \|\pi_i(s)\pi(f)\eta - \pi(s)\pi(f)\eta\| &\leq \|\pi_i(s)\pi_i(f)\eta - \pi(s)\pi(f)\eta\| \\ &\quad + \|\pi_i(s)\pi(f)\eta - \pi_i(s)\pi_i(f)\eta\| \\ &= \|\pi_i(s^{-1}f)\eta - \pi(s^{-1}f)\eta\| + \|\pi_i(f)\eta - \pi(f)\eta\|. \end{aligned}$$

The set of maps $g \mapsto \pi_i(g)\eta$ from $L^1(G)$ into H_n is equicontinuous for fixed η . On the compact set of translates ${}_s f$, $s \in K$, pointwise convergence of these maps is therefore equivalent to uniform convergence. Thus $\|\pi_i(s^{-1}f)\eta - \pi(s^{-1}f)\eta\| \rightarrow 0$ uniformly on K . On the other hand, $\|\pi_i(f)\eta - \pi(f)\eta\| \rightarrow 0$.

18.1.10. We denote by $\text{Irr}_n(G)$ the set of continuous irreducible unitary representations of G in H_n . It is furnished with the topology induced by that of $\text{Rep}_n(G)$. If G is separable, $\text{Rep}_n(G)$ and $\text{Irr}_n(G)$ are Polish spaces (3.7.4 and 3.7.5).

Bibliography: [27, 50, 91].

18.2. Fourier Transform

18.2.1. Definition. Let $\mu \in M^1(G)$. For every $\pi \in \widehat{G}$, put $(\mathcal{F}\mu)(\pi) = \pi(\mu)$, which is a bounded linear operator on H_π . The function $\pi \mapsto (\mathcal{F}\mu)(\pi)$ on $\widehat{G}$ is called the Fourier transform of μ .

Recall that the space of π is defined only up to isomorphism, which is rather inconvenient.

If G is commutative, $(\mathcal{F}\mu)(\pi)$ is scalar multiplication by $\int \pi(s) d\mu(s)$ on the one-dimensional space H_π , and the usual definition of the Fourier transform is recovered by identifying this scalar multiplication with its ratio.

18.2.2. We have immediately

$$\begin{aligned} \mathcal{F}(\alpha\mu) &= \alpha\mathcal{F}\mu & (\alpha \in \mathbb{C}, \mu \in M^1(G)), \\ \mathcal{F}(\mu + \mu') &= \mathcal{F}\mu + \mathcal{F}\mu' & (\mu, \mu' \in M^1(G)), \\ \mathcal{F}(\mu * \mu') &= (\mathcal{F}\mu)(\mathcal{F}\mu') & (\mu, \mu' \in M^1(G)), \\ \mathcal{F}(\mu^*) &= (\mathcal{F}\mu)^* & (\mu \in M^1(G)), \\ \sup_{\pi \in \widehat{G}} \|(\mathcal{F}\mu)(\pi)\| &\leq \|\mu\| & (\mu \in M^1(G)). \end{aligned}$$

18.2.3. The Fourier transform is injective. Indeed, let μ be a nonzero element of $M^1(G)$. There is an $f \in \mathcal{K}(G)$ such that $g = f * \mu$ is a nonzero element of $L^1(G)$. Hence there is a $\pi \in \widehat{G}$ such that $\pi(f)\pi(\mu) = \pi(f * \mu) \neq 0$ (2.7.3), and therefore $(\mathcal{F}\mu)(\pi) = \pi(\mu) \neq 0$.

18.2.4. If $f \in L^1(G)$, we denote by $\mathcal{F}f$ the Fourier transform of the measure $f(s) ds$. In other words, $(\mathcal{F}f)(\pi) = \pi(f)$. If $\varepsilon > 0$, the function $\pi \mapsto \|(\mathcal{F}f)(\pi)\|$ is less than ε outside a quasi-compact subset of $\widehat{G}$ (3.3.7).

Bibliography: [22, 23, 24, 53, 83, 101, 111, 137, 145, 147, 148, 162, 171].

18.3. Reduced Dual

18.3.1. Definition. The reduced dual of G is the support of the regular representation of G .

This reduced dual is a closed subset $\widehat{G}_r$ of $\widehat{G}$.

18.3.2. Let λ be the left regular representation of G . Regarded as a representation of $C^*(G)$, it has a certain kernel N . Then $\widehat{G}_r$ is the set of $\sigma \in \widehat{G} = C^*(G)^\wedge$

whose kernel contains N . In other words, $\widehat{G}_r$ is the spectrum of the C^* -algebra $C^*(G)/N$, itself isomorphic to the C^* -algebra $\lambda(C^*(G))$. This C^* -algebra $\lambda(C^*(G))$ is the norm closure in $\mathcal{L}(L^2(G))$ of the set of operators of left convolution by elements of $L^1(G)$.

Recall that $N \cap L^1(G) = 0$ (13.3.6).

18.3.3. Suppose G is unimodular. Every square-integrable irreducible representation π of G is contained in the regular representation, and therefore belongs to $\widehat{G}_r$. In particular, if G is compact, $\widehat{G} = \widehat{G}_r$.

18.3.4. Suppose G is commutative. Let $f \in L^1(G)$, let χ be a character of G , and let λ be the regular representation of G . We have $|\chi(f)| \leq \|\lambda(f)\|$ (13.3.6), and this inequality extends by continuity to $f \in C^*(G)$. Thus χ is weakly contained in λ , so $\widehat{G}_r = \widehat{G}$.

For other examples, see 18.3.9, 18.9.8, and 18.9.11.

18.3.5. Proposition. a. Let φ be a continuous function of positive type on G . The following conditions are equivalent:

- (i) φ is a uniform limit on every compact set of functions of the form $f * \widetilde{f}$, where $f \in \mathcal{K}(G)$;
- (ii) φ is a uniform limit on every compact set of continuous compactly supported functions of positive type;
- (iii) φ is a uniform limit on every compact set of continuous square-integrable functions of positive type;
- (iv) φ is associated with a representation weakly contained in the regular representation.

b. The functions satisfying (i)–(iv) form a weakly closed convex cone Q in $L^\infty(G)$.

c. If $\varphi \in Q$ and ψ is a continuous function of positive type, then $\varphi\psi \in Q$.

(i) $\Rightarrow$ (ii) $\Rightarrow$ (iii): This is evident.

(iii) $\Rightarrow$ (i): It is enough to show that if φ is a continuous square-integrable function of positive type, then φ is a uniform limit on G of functions $f * \widetilde{f}$ with $f \in \mathcal{K}(G)$. There is a $\psi \in L^2(G)$ such that $\varphi = \psi * \widetilde{\psi}$ (13.8.6). Let (f_n) be a sequence in $\mathcal{K}(G)$ tending to ψ in $L^2(G)$. Then $f_n * \widetilde{f_n}$ tends uniformly to $\psi * \widetilde{\psi} = \varphi$.

(i) $\Rightarrow$ (iv): Let λ be the left regular representation of G . If $f \in \mathcal{K}(G)$, then $f * \widetilde{f}$ is a function of positive type associated with λ (13.4.11). If φ satisfies (i), then π_φ is weakly contained in λ (18.1.4).

(iv) $\Rightarrow$ (ii): If φ satisfies (iv), then φ is a uniform limit on every compact set of sums of functions of positive type associated with λ (18.1.4). A function of

positive type associated with λ has the form $\psi * \widetilde{\psi}$, with $\psi \in L^2(G)$, and hence is a uniform limit on G of functions of the form $f * \widetilde{f}$, with $f \in \mathcal{K}(G)$. A sum of such functions is a uniform limit on G of continuous compactly supported functions of positive type.

This proves a.

It is clear that the functions satisfying (i)–(iv) form a convex cone Q in $L^\infty(G)$. Let Q_1 be the intersection of Q with the unit ball of $L^\infty(G)$. To prove that Q is weakly closed, it is enough to prove that Q_1 is weakly closed (B 8). Let φ be a function in $L^\infty(G)$ weakly adherent to Q_1 , and let ω be the positive form on $C^*(G)$ defined by φ . Then ω is a weak limit of positive forms on $C^*(G)$ defined by elements of Q_1 [2.7.5 (iii)]. Thus π_ω is weakly contained in λ (3.4.9), and hence $\varphi \in Q_1$.

Finally, c follows from characterization (ii) of the elements of Q and from the fact that the product of two continuous functions of positive type is of positive type (13.4.9).

18.3.6. Proposition. *Let G be a locally compact group. The following conditions are equivalent:*

- (i) *the dual of G equals the reduced dual;*
- (i') *the one-dimensional trivial representation of G belongs to the reduced dual;*
- (ii) *every continuous function of positive type on G is a uniform limit on every compact set of functions of the form $f * \widetilde{f}$, where $f \in \mathcal{K}(G)$;*
- (ii') *the function 1 is a uniform limit on every compact set of functions of the form $f * \widetilde{f}$, where $f \in \mathcal{K}(G)$;*
- (iii) *for every bounded measure μ of positive type on G and every continuous function φ of positive type on G ,*

$$\int \varphi(x) d\mu(x) \geq 0;$$

- (iii') *for every bounded measure μ of positive type on G ,*

$$\int d\mu(x) \geq 0.$$

The equivalence (i) $\Leftrightarrow$ (ii) follows from 18.3.5. The definition of measures of positive type shows that (ii) $\Rightarrow$ (iii). Suppose (iii) holds. Let φ be a continuous function of positive type on G , and use the notation Q of 18.3.5. To show that $\varphi \in Q$, it is enough to prove that φ belongs to the bipolar of Q . Let $f \in L^1(G)$ belong to the polar of Q . Since Q is a cone, $\operatorname{Re}\langle f, q \rangle \leq 0$ for every $q \in Q$, and in particular $\operatorname{Re}\langle f, g * \widetilde{g} \rangle \leq 0$ for every $g \in \mathcal{K}(G)$. Thus the measure $-(f + f^*)(s) ds$

is of positive type. By (iii), $\operatorname{Re} \int f(s)\varphi(s) ds \leq 0$, which proves that φ belongs to the bipolar of Q . Hence (iii) $\Rightarrow$ (ii).

The equivalences (i') $\Leftrightarrow$ (ii') $\Leftrightarrow$ (iii') are established similarly. Clearly (iii) $\Rightarrow$ (iii'). Finally, if $1 \in Q$, every continuous function of positive type belongs to Q by 18.3.5 (c).

18.3.7. Lemma. *If G is a semidirect product of two groups H and K satisfying the conditions of 18.3.6, then G also satisfies these conditions.*

Let C be a compact subset of G . There are compact subsets C' and C'' of H and K , respectively, such that $C \subset C'C''$, and we may suppose that $e \in C'$. Let $\varepsilon > 0$. There is an $x \in \mathcal{K}(H)$ such that $|x * \tilde{x} - 1| \leq \varepsilon$ on C' . Let C'_0 be the support of x . The set C''_0 of elements $h^{-1}kh$, where $h \in C'_0$ and $k \in C''$, is compact in the normal subgroup K . There is a $y \in \mathcal{K}(K)$ such that $|y * \tilde{y} - 1| \leq \varepsilon$ on C''_0 . Define $z \in \mathcal{K}(G)$ by

$$z(hk) = x(h)y(k) \quad (h \in H, k \in K).$$

For $h' \in H$ and $k' \in K$, we have

$$\begin{aligned} (z * \tilde{z})(h'k') &= \int z(h'k'g) \overline{z(g)} dg \\ &= \iint_{H \times K} x(h'h)y((h^{-1}k'h)k) \overline{x(h)} \overline{y(k)} dh dk \\ &= \int_H (y * \tilde{y})(h^{-1}k'h)x(h'h) \overline{x(h)} dh. \end{aligned}$$

Consequently,

$$\begin{aligned} |(z * \tilde{z})(h'k') - 1| &\leq \left| \int_H (y * \tilde{y})(h^{-1}k'h)x(h'h) \overline{x(h)} dh - \int_H x(h'h) \overline{x(h)} dh \right| \\ &\quad + \left| \int_H x(h'h) \overline{x(h)} dh - 1 \right| \\ &\leq \int_{C'_0} |(y * \tilde{y})(h^{-1}k'h) - 1| |x(h'h)| |x(h)| dh + |(x * \tilde{x})(h') - 1|. \end{aligned}$$

If $h'k' \in C$, then $h' \in C'$ and $k' \in C''$, so $h^{-1}k'h \in C''_0$ for $h \in C'_0$. Therefore

$$\begin{aligned} |(z * \tilde{z})(h'k') - 1| &\leq \varepsilon \int |x(h'h)| |x(h)| dh + \varepsilon \\ &\leq \varepsilon \int |x(h)|^2 dh + \varepsilon \end{aligned}$$

$$= \varepsilon [(x * \tilde{x})(e) + 1] \leq \varepsilon(2 + \varepsilon).$$

18.3.8. Lemma. *Let G be a locally compact group satisfying the conditions of 18.3.6, and let N be a closed normal subgroup of G . Then G/N satisfies the conditions of 18.3.6.*

Let $G^* = G/N$, and let φ be the canonical homomorphism of G onto G^* . Let C^* be a compact subset of G/N and let $\varepsilon > 0$. There is a compact subset C of G such that $\varphi(C) \supset C^*$, and we may suppose that $e \in C$. Then there is an $f \in \mathcal{K}(G)$ such that $|f * \tilde{f} - 1| \leq \varepsilon$ on C and $(f * \tilde{f})(e) = 1$, that is, $\|f\|_2 = 1$. Let F be the function on G^* defined by

$$F(s^*) = \left(\int_N |f(sn)|^2 dn \right)^{1/2}. \quad (1)$$

where dn denotes a left Haar measure on N , and where $s^* = \varphi(s)$ [the right-hand side of (1) plainly depends only on s^*]. Then $F \in \mathcal{K}(G^*)$ and

$$\|F\|_2^2 = \int_{G^*} ds^* \int_N |f(sn)|^2 dn = \int_G |f(s)|^2 ds = 1. \quad (2)$$

In view of (2), for every $s' \in G$ we have

$$\begin{aligned} |(F * \tilde{F})(s'^*) - 1|^2 &= \left| \int_{G^*} F(s'^* s^*) \overline{F(s^*)} ds^* - 1 \right|^2 \\ &= \left| \int_{G^*} (F(s'^* s^*) - F(s^*)) \overline{F(s^*)} ds^* \right|^2 \\ &\leq \int_{G^*} |F(s'^* s^*) - F(s^*)|^2 ds^* \\ &= \int_{G^*} \left| \left(\int_N |f(s' sn)|^2 dn \right)^{1/2} - \left(\int_N |f(sn)|^2 dn \right)^{1/2} \right|^2 ds^*. \end{aligned}$$

Applying in $L^2(N)$ the inequality $(\|a\| - \|b\|)^2 \leq \|a - b\|^2$, the last expression is at most

$$\begin{aligned} \int_{G^*} ds^* \int_N |f(s' sn) - f(sn)|^2 dn &= \int_G |f(s's) - f(s)|^2 ds \\ &= 2 - 2 \operatorname{Re} \int_G f(s's) \overline{f(s)} ds \\ &= 2[1 - \operatorname{Re}(f * \tilde{f})(s')]. \end{aligned}$$

Thus, if $s' \in C$, then $|(F * \tilde{F})(s'^*) - 1|^2 \leq 2\varepsilon$. Consequently $|F * \tilde{F} - 1| \leq \sqrt{2\varepsilon}$ on C^* .

18.3.9. Proposition. *Let G be a connected real Lie group and R its radical. If G/R is compact, then $\widehat{G} = \widehat{G}_r$.*

Let G' be the universal covering group of G ; it is the semidirect product of a compact group C and a simply connected solvable group S . Since G is a quotient of G' , it is enough to prove that $\widehat{G'} = \widehat{G'}_r$ (18.3.8). Now $\widehat{C} = \widehat{C}_r$ (18.3.3), and by 18.3.7 we are reduced to showing that a simply connected solvable group S satisfies $\widehat{S} = \widehat{S}_r$. We argue by induction on the dimension n of S . The assertion is evident for $n = 0$. Let $n > 0$, and suppose the assertion established when $\dim S < n$. Then S is the semidirect product of a group isomorphic to $\mathbb{R}$ and a simply connected solvable group of dimension $n - 1$; it now suffices to apply 18.3.4, 18.3.7, and the induction hypothesis.

Bibliography: [4, 27, 154].

18.4. Reduced Dual and Integrable Representations

18.4.1. Proposition. *Let G be a locally compact unimodular group and σ a square-integrable irreducible representation of G . Then $\{\sigma\}$ is closed in $\widehat{G}_r$.*

For $f \in \mathcal{K}(G)$, $\sigma(f)$ is a Hilbert–Schmidt operator (14.4.2), and therefore $\sigma(C^*(G)) \subset \mathcal{LC}(H_\sigma)$. Consequently the kernel of σ in $C^*(G)$ is a maximal closed two-sided ideal I of $C^*(G)$, and every irreducible representation of $C^*(G)$ having kernel I is equivalent to σ (4.1.10 and 4.1.11); hence $\{I\}$ is closed in $\text{Prim } C^*(G)$ (3.1.4), and $\{\sigma\}$ is closed in $(C^*(G))^\wedge = \widehat{G}$. Moreover, $\sigma \in \widehat{G}_r$ (18.3.3).

18.4.2. Proposition. *Let G be a locally compact unimodular group and σ an integrable irreducible representation of G . Then $\{\sigma\}$ is open and closed in $\widehat{G}_r$.*

We already know that $\{\sigma\}$ is closed in $\widehat{G}_r$ (18.4.1). Let λ be the left regular representation of G ; then $\lambda(C^*(G)) = B$ is a C^* -algebra of operators on $L^2(G)$ whose spectrum is $\widehat{G}_r$ (18.3.2). We may identify σ with a subrepresentation of λ , since σ is square-integrable. Let $K \supset H_\sigma$ be the bi-invariant subspace of $L^2(G)$ associated with σ , and let λ_1 be the subrepresentation of λ defined by K . For every $f \in L^1(G)$, $\lambda_1(f)$ is the restriction of $\lambda(f)$ to K ; thus, if λ_1 is identified with a representation of B , it is simply the map $T \mapsto T|K$ from B into $\mathcal{L}(K)$. The kernel of σ in B equals that of λ_1 , since $\lambda_1 \simeq (\dim \sigma) \cdot \sigma$ [14.3.5 (iii)]; this kernel is therefore the set I of $T \in B$ such that $T|K = 0$. Let J be the set of $T \in B$ such that $T|(L^2(G) \ominus K) = 0$. Let ξ be a nonzero vector of H_σ such that the function $s \mapsto \varphi(s) = (\lambda(s)\xi \mid \xi)$ on G is integrable. We have $\check{\varphi} \in K$ (14.3.1). Since $\check{\varphi} \in$

$L^1(G)$, we can form $\lambda(\check{\varphi})$, which is a nonzero element of B . For every $g \in L^2(G)$, $\lambda(\check{\varphi})g = \check{\varphi} * g \in K$, because K is bi-invariant; hence $\lambda(\check{\varphi})(L^2(G) \ominus K) = 0$. It follows that $J \neq 0$.

Now let $\tau \in \widehat{G}_r$, regarded as an irreducible representation of B . Its kernel in B is a primitive ideal containing $0 = I \cap J$, and hence contains I or J (2.11.4). If $\ker \tau \supset I$, then τ is adherent to $\{\sigma\}$ and thus is equal to σ . This proves that

$$\bigcap_{\tau \in \widehat{G}_r, \tau \neq \sigma} \ker \tau \supset J \neq 0.$$

Thus $\widehat{G}_r \setminus \{\sigma\}$ is not dense in $\widehat{G}_r$. Consequently $\{\sigma\}$ contains a nonempty open subset of $\widehat{G}_r$, and $\{\sigma\}$ is open in $\widehat{G}_r$.

18.4.3. Corollary. *If G is compact, $\widehat{G}$ is discrete.*

Since every irreducible representation of G is integrable, this follows from 18.4.2.

Bibliography: [13, 50].

18.5. Mackey Borel Structure

18.5.1. Let G be a separable locally compact group. For every cardinal $n \leq \aleph_0$, we equip $\text{Rep}_n(G)$ and $\text{Irr}_n(G)$ with the Borel structures underlying their topologies. With the notation of 18.1.9, the bijections

$$\text{Rep}_n(G) \longrightarrow \text{Rep}'_n(C^*(G)), \quad \text{Irr}_n(G) \longrightarrow \text{Irr}'_n(C^*(G))$$

are Borel isomorphisms. We then define $\text{Rep}(G)$ and $\text{Irr}(G)$ as the Borel sum spaces of the $\text{Rep}_n(G)$ and the $\text{Irr}_n(G)$; all these spaces are standard.

18.5.2. The functions $\pi \mapsto (\pi(s)\xi \mid \eta)$ on $\text{Rep}_n(G)$, where $s \in G$ and $\xi, \eta \in H_n$, define on $\text{Rep}_n(G)$ a Borel structure identical with that of 18.5.1. Indeed, these functions are continuous (18.1.9), and hence this structure is coarser than that of 18.5.1. It is then enough to observe that the functions $\pi \mapsto (\pi(s_k)\xi_p \mid \eta_q)$, where (s_k) is dense in G and $(\xi_p), (\eta_q)$ are dense in H_n , separate the points of $\text{Rep}_n(G)$, and to apply B 21.

18.5.3. The *Mackey Borel structure* on $\widehat{G}$ is the quotient structure of that on $\text{Irr}(G)$ under the canonical map $\text{Irr}(G) \rightarrow \widehat{G}$. This structure is identified with the

Mackey Borel structure of $(C^*(G))^\wedge$. It is finer than the Borel structure underlying the topology of $\widehat{G}$ (3.8.3), and is identical with it if G is postliminal (4.6.1). The points of $\widehat{G}$ are Borel sets for the Mackey structure (3.8.4).

Bibliography: [91].

18.6. Quasi-Dual

18.6.1. Let G be a separable locally compact group, let n be a cardinal, and let H_n be the model Hilbert space of dimension n . We denote by $\text{Fac}_n(G)$ the set of continuous unitary factor representations of G on H_n , by $\text{Fac}(G)$ the union of the $\text{Fac}_n(G)$ for $n = 1, 2, \dots, \aleph_0$, by $\text{FacI}_n(G)$ the set of $\pi \in \text{Fac}(G)$ for which $\pi(G)''$ is a factor of type I_n , and by $\text{FacI}(G)$ the set of $\pi \in \text{Fac}(G)$ which are of type I. All these sets are Borel subsets of $\text{Rep}(G)$ and hence are standard Borel spaces (7.1.4 and 7.3.4). They are identified with $\text{Fac}_n(C^*(G))$, and so on.

18.6.2. We denote by $\check{G}$ the set of quasi-equivalence classes of continuous unitary factor representations of G . The Mackey Borel structure on $\check{G}$ is the quotient structure of that on $\text{Fac}(G)$ under the canonical map $\text{Fac}(G) \rightarrow \check{G}$. The Borel space $\check{G}$ is called the *quasi-dual* of G . It is identified with $(C^*(G))^\vee$. Every point of $\check{G}$ is a Borel set in $\check{G}$ (7.2.4).

18.6.3. Let $\check{G}_I$ be the set of quasi-equivalence classes of the $\pi \in \text{Fac}(G)$ which are of type I. Then $\check{G}_I$ is identified with $(C^*(G))_I^\vee$. It is a Borel subset of $\check{G}$, and the canonical bijection from $\widehat{G}$ onto $\check{G}_I$ is a Borel isomorphism (7.3.6).

18.6.4. Let $\text{Facf}_n(G)$ be the set of $\pi \in \text{Fac}_n(G)$ which are of finite type, let $\text{Facf}(G)$ be the union of the $\text{Facf}_n(G)$, and let $\check{G}_f$ be the canonical image of $\text{Facf}(G)$ in $\check{G}$. The sets $\text{Facf}(G)$ and $\text{Facf}_n(G)$ are Borel in $\text{Fac}(G)$, $\check{G}_f$ is Borel in $\check{G}$, all these spaces are standard (7.4.3 and 7.4.4), and they are identified with $\text{Facf}(C^*(G))$, $\text{Facf}_n(C^*(G))$, and $(C^*(G))_f^\vee$.

Let C be the set of continuous central functions φ of positive type on G such that $\varphi(e) \leq 1$. Then C is compact metrizable for the weak topology. The set D of finite-type characters of G equal to 1 at e is a G_δ in C (17.3.5 and B 13), and hence is a Polish space. Equip $\check{G}_f$ with the topology transported from D by the canonical bijection of D onto $\check{G}_f$ (17.3.4). Then $\check{G}_f$ becomes a Polish space, and the Borel structure underlying its topology is the structure induced by that of $\check{G}$ (7.4.3).

Bibliography: [25, 57].

18.7. Integration and Disintegration of Representations

18.7.1. Let Z be a Borel space, let μ be a positive measure on Z , and let $\zeta \mapsto H(\zeta)$ be a μ -measurable field of Hilbert spaces on Z . For every $\zeta \in Z$, let $\pi(\zeta)$ be a continuous unitary representation of G on $H(\zeta)$; we say that $\zeta \mapsto \pi(\zeta)$ is a field of continuous unitary representations of G . This field is called measurable if, for every $s \in G$, the field of operators $\zeta \mapsto \pi(\zeta)(s)$ is measurable.

18.7.2. Suppose that G is separable. Suppose that Z is the union of pairwise disjoint Borel sets $Z_1, Z_2, \dots, Z_{\aleph_0}$, and that on Z_n the field $\zeta \mapsto H(\zeta)$ reduces to the constant field $\zeta \mapsto H_n$, where H_n is the model Hilbert space of dimension n . Let $\zeta \mapsto \pi(\zeta)$ be a field of continuous unitary representations of G on the $H(\zeta)$. This field is measurable if and only if $\zeta \mapsto \pi(\zeta)$ is almost everywhere equal to a Borel map of Z into $\text{Rep}(G)$. The proof is exactly the same as that of 8.1.8.

18.7.3. Proposition. *Let $\zeta \mapsto \pi(\zeta)$ be a field of continuous unitary representations of G . For every $\zeta \in Z$, let $\pi'(\zeta)$ be the corresponding representation of $C^*(G)$. Suppose G is separable. The field $\zeta \mapsto \pi(\zeta)$ is measurable if and only if the field $\zeta \mapsto \pi'(\zeta)$ is measurable.*

First reduce to the case in which the field $\zeta \mapsto H(\zeta)$ is the constant field defined by a fixed Hilbert space H (A 72). One may then suppose that H is a model Hilbert space H_n . To say that the field $\zeta \mapsto \pi(\zeta)$ [respectively, $\zeta \mapsto \pi'(\zeta)$] is measurable means that it is almost everywhere equal to a Borel map of Z into $\text{Rep}_n(G)$ [respectively, into $\text{Rep}_n(C^*(G))$], by 18.7.2 [respectively, 8.1.8]. The proposition now follows from 18.5.1.

18.7.4. Proposition. *Retain the hypotheses of 18.7.3, and suppose the fields $\zeta \mapsto \pi(\zeta)$ and $\zeta \mapsto \pi'(\zeta)$ are measurable. Let*

$$\pi' = \int_Z^{\oplus} \pi'(\zeta) d\mu(\zeta).$$

Let π be the continuous unitary representation of G corresponding to π' . Then, for every $s \in G$,

$$\pi(s) = \int_Z^{\oplus} \pi(\zeta)(s) d\mu(\zeta).$$

Let $(V_1, V_2, \dots)$ be a sequence of compact neighborhoods of s whose intersection is $\{s\}$. Let f_n be a nonnegative continuous function, supported in V_n , with integral 1. Then $\pi'(f_n)$ converges strongly to $\pi(s)$, $\pi'(\zeta)(f_n)$ converges strongly to $\pi(\zeta)(s)$ for every ζ , and

$$\pi'(f_n) = \int_Z^{\oplus} \pi'(\zeta)(f_n) d\mu(\zeta).$$

Hence $\pi(s) = \int_Z^{\oplus} \pi(\zeta)(s) d\mu(\zeta)$ (A 79).

18.7.5. With the preceding notation, π is called the Hilbert integral of the $\pi(\zeta)$, and we write

$$\pi = \int_Z^{\oplus} \pi(\zeta) d\mu(\zeta).$$

18.7.6. This having been established, all of § 8, from Proposition 8.1.6 through 8.7, may be copied verbatim, with A replaced everywhere by G (a separable locally compact group) and “representation” by “continuous unitary representation.”

18.7.7. We shall not translate 8.8.1. Applying 8.8.2 gives, in particular, the following:

Proposition. *Let G be a separable locally compact unimodular group; let λ and ρ be the left and right regular representations of G ; let $\mathfrak{U}$ and $\mathfrak{B}$ be the von Neumann algebras on $L^2(G)$ generated by $\lambda(G)$ and $\rho(G)$; let J be the map $f \mapsto f^*$ of $L^2(G)$ onto itself; and let t be the natural trace on $\mathfrak{U}^+$ defined by ε_e (17.2.5).*

(i) *There exist: (1) a standard positive measure μ on $\check{G}$; (2) for every $\zeta \in \check{G}$, a character $f(\zeta)$; and (3) a measurable-field structure on the field $\zeta \mapsto H_{f(\zeta)}$, with the following properties:*

a. *$L^2(G)$ is identified with $\int_{\check{G}}^{\oplus} H_{f(\zeta)} d\mu(\zeta)$; the fields*

$$\begin{aligned} \zeta &\mapsto J_{f(\zeta)}, & \zeta &\mapsto \lambda_{f(\zeta)}, & \zeta &\mapsto \bar{\rho}_{f(\zeta)}^{\circ}, \\ \zeta &\mapsto \mathfrak{U}_{f(\zeta)}, & \zeta &\mapsto \mathfrak{B}_{f(\zeta)}, & \zeta &\mapsto t_{f(\zeta)}. \end{aligned}$$

are measurable, and

$$\begin{aligned} J &= \int_{\check{G}}^{\oplus} J_{f(\zeta)} d\mu(\zeta), & \lambda &= \int_{\check{G}}^{\oplus} \lambda_{f(\zeta)} d\mu(\zeta), & \rho &= \int_{\check{G}}^{\oplus} \bar{\rho}_{f(\zeta)}^{\circ} d\mu(\zeta), \\ \mathfrak{U} &= \int_{\check{G}}^{\oplus} \mathfrak{U}_{f(\zeta)} d\mu(\zeta), & \mathfrak{B} &= \int_{\check{G}}^{\oplus} \mathfrak{B}_{f(\zeta)} d\mu(\zeta), & t &= \int_{\check{G}}^{\oplus} t_{f(\zeta)} d\mu(\zeta). \end{aligned}$$

The algebra $\mathfrak{U} \cap \mathfrak{B}$ is the algebra of diagonalizable operators.

b. $\lambda_{f(\zeta)} \in \zeta$ and $\bar{\rho}_{f(\zeta)}^0 \in \zeta$ almost everywhere.

c. For every $x \in C^*(G)^+$, the function $\zeta \mapsto f(\zeta)(x)$ is measurable and

$$f(x) = \int_{\check{G}} f(\zeta)(x) d\mu(\zeta).$$

(ii) Suppose a measure μ' , characters $f'(\zeta)$, and a measurable field structure on $\zeta \mapsto H_{f'(\zeta)}$ are given, with the same properties as in (i). Then μ and μ' are equivalent; write $\mu' = d \cdot \mu$, where $d(\zeta) > 0$ almost everywhere. We have $f(\zeta) = d(\zeta)f'(\zeta)$ almost everywhere. After modifying the $f'(\zeta)$ on a null set, there is an isomorphism from the measurable field $\zeta \mapsto H_{f(\zeta)}$ onto the measurable field $\zeta \mapsto H_{f'(\zeta)}$ which transforms $J_{f(\zeta)}$ into $J_{f'(\zeta)}$, $\lambda_{f(\zeta)}$ into $\lambda_{f'(\zeta)}$, $\bar{\rho}_{f(\zeta)}^0$ into $\bar{\rho}_{f'(\zeta)}^0$, $\mathfrak{U}_{f(\zeta)}$ into $\mathfrak{U}_{f'(\zeta)}$, $\mathfrak{B}_{f(\zeta)}$ into $\mathfrak{B}_{f'(\zeta)}$, and $t_{f(\zeta)}$ into $d(\zeta)t_{f'(\zeta)}$.

18.7.8. Corollary. Retain the preceding notation. Let $u \in L^1(G) \cap L^2(G)$. For almost every $\zeta \in \check{G}$, $u \in \mathfrak{n}_{f(\zeta)}$; the function

$$\zeta \mapsto t_{f(\zeta)}(\lambda_{f(\zeta)}(u)\lambda_{f(\zeta)}(u)^*)$$

is μ -integrable, and

$$\int_G |u(s)|^2 ds = \int_{\check{G}} t_{f(\zeta)}(\lambda_{f(\zeta)}(u)\lambda_{f(\zeta)}(u)^*) d\mu(\zeta).$$

We have $u \in C^*(G)$ and $u * u^* \in C^*(G)^+$. Hence the function

$$\zeta \mapsto f(\zeta)(u * u^*) = t_{f(\zeta)}(\lambda_{f(\zeta)}(u)\lambda_{f(\zeta)}(u)^*)$$

is measurable, and its integral with respect to μ is

$$f(u * u^*) = \varepsilon_e(u * u^*) = \int_G |u(s)|^2 ds < +\infty.$$

Thus, for almost every ζ ,

$$t_{f(\zeta)}(\lambda_{f(\zeta)}(u)\lambda_{f(\zeta)}(u)^*) < +\infty,$$

and consequently $u \in \mathfrak{n}_{f(\zeta)}$.

18.7.9. Corollary. Let G be a separable locally compact unimodular group, and let $s \in G$, $s \neq e$. There exists a traceable continuous unitary factor representation τ of G such that $\tau(s) \neq 1$.

With the notation of 18.7.7, we have $\lambda(s) \neq 1$, and hence $\lambda_{f(\zeta)}(s) \neq 1$ for a non-null set of values of ζ . But the $\lambda_{f(\zeta)}$ are traceable factor representations.

18.7.10. Let (u_i) be a sequence of elements of $\mathcal{K}(G)$. By 8.8.3, the representation τ of 18.7.9 may be chosen so that, for every i , $\tau(u_i)$ is a Hilbert–Schmidt operator relative to the factor generated by $\tau(G)$.

Bibliography: [25, 52, 53, 57, 87, 88, 89, 90, 91, 95, 96, 107, 108, 109, 110, 137, 162].

18.8. Plancherel Measure

18.8.1. Let G be a separable postliminal locally compact group.

Let $\zeta \mapsto K(\zeta)$ be the canonical field of Hilbert spaces over $\widehat{G}$ (8.6.1). Choose a field $\zeta \mapsto \pi(\zeta)$ of continuous unitary representations of G on the $K(\zeta)$ such that $\pi(\zeta)$ belongs to the class ζ for every ζ , and which is measurable for every positive measure on $\widehat{G}$ (8.6.2). For simplicity we shall write ζ in place of $\pi(\zeta)$. Let $\overline{K(\zeta)}$ be the Hilbert space conjugate to $K(\zeta)$, let J_ζ be the canonical involution of $K(\zeta) \otimes \overline{K(\zeta)}$, and let t_ζ be the trace $T \otimes 1 \mapsto \text{Tr } T$ on $(\mathcal{L}(K(\zeta)) \otimes \mathbb{C})^+$.

Theorem. Suppose in addition that G is unimodular. Let λ and ρ be the left and right regular representations of G , let $\mathfrak{U}$ and $\mathfrak{B}$ be the von Neumann algebras on $L^2(G)$ generated by $\lambda(G)$ and $\rho(G)$, let J be the map $f \mapsto f^*$ on $L^2(G)$, and let t be the natural trace on $\mathfrak{U}^+$ defined by ε_e (17.2.5). There exist a positive measure μ on $\widehat{G}$ and an isomorphism W from $L^2(G)$ onto

$$\int_{\widehat{G}}^{\oplus} (K(\zeta) \otimes \overline{K(\zeta)}) d\mu(\zeta)$$

with the following properties:

(i) W transforms

$$\begin{array}{ll} J & \text{into} \int_{\widehat{G}}^{\oplus} J_\zeta d\mu(\zeta), & \lambda & \text{into} \int_{\widehat{G}}^{\oplus} (\zeta \otimes 1) d\mu(\zeta), \\ \rho & \text{into} \int_{\widehat{G}}^{\oplus} (1 \otimes \bar{\zeta}) d\mu(\zeta), & \mathfrak{U} & \text{into} \int_{\widehat{G}}^{\oplus} (\mathcal{L}(K(\zeta)) \otimes \mathbb{C}) d\mu(\zeta), \\ \mathfrak{B} & \text{into} \int_{\widehat{G}}^{\oplus} (\mathbb{C} \otimes \mathcal{L}(\overline{K(\zeta)})) d\mu(\zeta), & t & \text{into} \int_{\widehat{G}}^{\oplus} t_\zeta d\mu(\zeta), \end{array}$$

and $\mathfrak{U} \cap \mathfrak{B}$ into the algebra of diagonalizable operators.

(ii) If $x \in C^*(G)^+$, the function $\zeta \mapsto \text{Tr } \zeta(x)$ on $\widehat{G}$ is lower semicontinuous, and

$$t(x) = \int_{\widehat{G}} \text{Tr } \zeta(x) d\mu(\zeta).$$

In particular, if $u \in L^1(G) \cap L^2(G)$, then

$$\int_G |u(s)|^2 ds = \int_{\widehat{G}} \text{Tr}(\zeta(u)\zeta(u)^*) d\mu(\zeta). \quad (1)$$

This follows from 8.8.5, applied to the trace t . Formula (1) is called the *Plancherel formula*.

18.8.2. Theorem. Let G be a separable unimodular postliminal locally compact group. There exists one and only one positive measure μ on $\widehat{G}$ such that, for every $u \in L^1(G) \cap L^2(G)$,

$$\int_G |u(s)|^2 ds = \int_{\widehat{G}} \text{Tr}(\zeta(u)\zeta(u)^*) d\mu(\zeta).$$

The measure μ of 18.8.1 has this property. Let μ' be another positive measure on $\widehat{G}$ having this property. For $x \in C^*(G)^+$, put

$$f(x) = \int_{\widehat{G}} \text{Tr } \zeta(x) d\mu'(\zeta).$$

It is clear that f is a trace on $C^*(G)^+$. If $(x_1, x_2, \dots)$ is a sequence of elements of $C^*(G)^+$ tending to x , then $\|\zeta(x_n) - \zeta(x)\| \rightarrow 0$ for every $\zeta \in \widehat{G}$; hence, by Fatou's lemma,

$$\begin{aligned} \liminf f(x_n) &\geq \int_{\widehat{G}} \liminf \text{Tr } \zeta(x_n) d\mu'(\zeta) \\ &\geq \int_{\widehat{G}} \text{Tr } \zeta(x) d\mu'(\zeta) = f(x). \end{aligned}$$

Thus f is lower semicontinuous. We have $f(u * u^*) < +\infty$ for every $u \in L^1(G) \cap L^2(G)$, so m_f is dense in $C^*(G)$. The bitraces corresponding to f and t agree on $L^1(G) \cap L^2(G)$, and hence agree everywhere (6.5.3). Consequently

$$t(x) = \int_{\widehat{G}} \text{Tr } \zeta(x) d\mu'(\zeta)$$

for every $x \in C^*(G)^+$, whence $\mu = \mu'$ (8.8.6).

18.8.3. Definition. The measure μ of 18.8.2 is called the *Plancherel measure* on $\widehat{G}$ associated with the Haar measure of G .

Suppose the Haar measure is multiplied by a scalar $k > 0$. Let $u \in L^1(G) \cap L^2(G)$. For every $\zeta \in \widehat{G}$, $\zeta(u)$ is multiplied by k , and hence $\text{Tr}(\zeta(u)\zeta(u)^*)$ is multiplied by k^2 . We see that the Plancherel measure is therefore multiplied by k^{-1} .

18.8.4. The support of the Plancherel measure μ equals the support of the regular representation of G (8.6.9 and 18.8.1), that is, the reduced dual of G (18.3.1).

If K is a quasi-compact subset of $\widehat{G}$, then $\mu(K) < +\infty$. Indeed, for every $\pi_0 \in K$ there is a $u \in L^1(G) \cap L^2(G)$ such that $\pi_0(u) \neq 0$; hence there is an $\alpha > 0$ such that $\text{Tr}(\pi(u)\pi(u)^*) \geq \alpha$ on an open neighborhood V_{π_0} of π_0 . By the Plancherel formula, $\mu(V_{\pi_0}) < +\infty$. But K can be covered by finitely many of the sets V_{π_0} .

18.8.5. Proposition. *Let G be a separable unimodular postliminal locally compact group, equipped with a Haar measure, and let μ be the corresponding Plancherel measure on $\widehat{G}$. Let $\zeta_0 \in \widehat{G}$. The representation ζ_0 is square-integrable if and only if $\mu(\{\zeta_0\}) > 0$, and $\mu(\{\zeta_0\})$ then equals the formal dimension of ζ_0 .*

If $\mu(\{\zeta_0\}) > 0$, then $\zeta_0 \otimes 1$ is contained in the regular representation λ [8.6.8 (ii) and 18.8.1]; thus ζ_0 is contained in λ , and so ζ_0 is square-integrable (14.1.3).

Suppose that ζ_0 is square-integrable. Use the notation of 18.8.1. Let F be the central projection associated with ζ_0 (14.2.4). It is diagonalizable; let Y be the μ -measurable subset of $\widehat{G}$ such that F corresponds to the characteristic function φ_Y of Y . Then the subrepresentation of λ defined by F is, on the one hand, quasi-equivalent to ζ_0 (14.3.5) and, on the other hand, equivalent to

$$\int_{\widehat{G}}^{\oplus} (\zeta \otimes 1) \varphi_Y(\zeta) d\mu(\zeta)$$

(18.8.1). Thus $\varphi_Y \cdot \mu$ is equivalent to the Dirac measure at ζ_0 (8.4.4), and hence $\mu(\{\zeta_0\}) > 0$. Furthermore, for $f \in L^1(G) \cap L^2(G)$, by 8.8.7,

$$\begin{aligned} \int_G |Ff(s)|^2 ds &= \int_{\widehat{G}} \varphi_Y(\zeta) \text{Tr}(\zeta(f)^* \zeta(f)) d\mu(\zeta) \\ &= \mu(\{\zeta_0\}) \text{Tr}(\zeta_0(f)^* \zeta_0(f)). \end{aligned}$$

On the other hand, if d denotes the formal dimension of ζ_0 , then

$$\int_G |Ff(s)|^2 ds = d \text{Tr}(\zeta_0(f)^* \zeta_0(f)) \quad (14.4.2).$$

We may choose $f \in L^1(G) \cap L^2(G)$ such that $\zeta_0(f) \neq 0$; therefore $\mu(\{\zeta_0\}) = d$.

18.8.6. If G is separable and commutative, the classical Plancherel theorem shows that the Plancherel measure is precisely the suitably normalized Haar measure of the dual group $\widehat{G}$.

Bibliography: [16, 53, 60, 62, 81, 94, 95, 96, 101, 107, 137, 148, 162, 171, 179].

18.9. Supplements

18.9.1. Let G be a separable unimodular postliminal locally compact group, let $\widehat{G}_r$ be its reduced dual, and let $\pi \in \widehat{G}$.

a. If $\{\pi\}$ is open in $\widehat{G}_r$, then π is square-integrable. [For π belongs to the support of the regular representation; hence, if μ denotes Plancherel measure, $\mu(\{\pi\}) > 0$.]

b. It may happen that $\pi \in \widehat{G}_r$ is square-integrable, that $\{\pi\}$ is open in $\widehat{G}$, and that π is not integrable. It may happen that π is square-integrable and that $\{\pi\}$ is not open in $\widehat{G}$. It may happen that $\{\pi\}$ is open in $\widehat{G}$ and that $\pi \notin \widehat{G}_r$.

c. Problems: If π is integrable, is $\{\pi\}$ open in $\widehat{G}$? If π is square-integrable, is $\{\pi\}$ open in $\widehat{G}_r$? [13, 283].

***18.9.2.** The Plancherel formula can be generalized to nonseparable unimodular postliminal locally compact groups. [16].

18.9.3. Let G be a locally compact group. Every continuous complex-valued function on G is a uniform limit on every compact set of linear combinations of functions $f * \widetilde{f}$, $f \in \mathcal{K}(G)$; these functions are associated with the regular representation. Compare this with 18.1.4 and with the fact that, for some G , not every element of $\widehat{G}$ is weakly contained in the regular representation. [27].

***18.9.4.** Let G be a connected semisimple real Lie group. The set of $f \in L^1(G)$ for which the rank of $\pi(f)$ remains finite and bounded as π ranges over $\widehat{G}$ is a self-adjoint two-sided ideal of $L^1(G)$ dense in $L^1(G)$. Thus 4.7.11 may be applied. [27].

18.9.5. Let G be a locally compact group. If the regular representation of G weakly contains a finite-dimensional continuous unitary representation, then $\widehat{G} = \widehat{G}_r$. [33].

18.9.6. Let G be a separable unimodular postliminal locally compact group, let μ be the Plancherel measure on $\widehat{G}$, and let E be the set of $\pi \in \widehat{G}$ such that $\pi(C^*(G)) = \mathcal{LC}(H_\pi)$, that is, such that $\{\pi\}$ is closed in $\widehat{G}$. Then $\mu(\widehat{G} \setminus E) = 0$. What can be said about E topologically?

18.9.7. Let G be a separable locally compact group in which e has a neighborhood basis consisting of compact neighborhoods invariant under inner automorphisms. In 18.7.7, almost all the characters $f(\zeta)$ are of finite type. (Use 8.8.3 and 17.4.1.)

***18.9.8.** Let G be the free group on two generators, equipped with the discrete topology. Then $\widehat{G}_r \neq \widehat{G}$; the set of pure functions of positive type is dense in the set of functions of positive type for compact convergence, and the disintegration of the regular representation λ into irreducible representations can be carried out in two completely distinct ways. The factor representation λ admits a disintegration (over a standard Borel space) into representations which are not factor representations almost everywhere. (One may observe that the von Neumann algebras associated with the Hilbert algebra of G are factors of type II_1 , and apply 5.7.5, 5.7.6, and 11.2.4.) [4, 185, 199].

18.9.9. Let G be a locally compact group, let H be a closed normal subgroup of G , let π be a continuous unitary factor representation of G , and put $\rho = \pi|_H$. Then ρ is either of type I, of type II_1 , of type II_∞ , or of type III. If ρ is of type I, it has multiplicity n for a uniquely determined n . The measure class on $\widehat{H}$ associated with ρ is invariant and ergodic under the action of G on $\widehat{H}$ (the group G acts on H by inner automorphisms and hence on $\widehat{H}$ by transport of structure). [92].

18.9.10. Let G be a locally compact group and H an open subgroup of G . If $\widehat{G} = \widehat{G}_r$, then $\widehat{H} = \widehat{H}_r$. [154].

***18.9.11.** Let G be a connected noncompact simple real Lie group. Then $\widehat{G} \neq \widehat{G}_r$. [154].

18.9.12. Let G be a locally compact group. The map $\pi \mapsto \overline{\pi}$ is a homeomorphism of $\widehat{G}$ onto $\widehat{G}$.

18.9.13. The topological space $\widehat{G}$ has been computed explicitly in only a small number of cases. Here are some examples. If G is the noncommutative simply connected nilpotent real Lie group of dimension 3, then $\widehat{G}$ may be identified with $(\mathbb{R} \setminus \{0\}) \cup \mathbb{R}^2$; the topology is that of the sum space of $\mathbb{R} \setminus \{0\}$ and $\mathbb{R}^2$, except that a point of $\mathbb{R} \setminus \{0\}$ which tends to zero in the ordinary sense tends in $\widehat{G}$ to every point of $\mathbb{R}^2$. If $G = \mathrm{SL}(2, \mathbb{C})$, then $\widehat{G}$ may be identified with the subset of $\mathbb{R}^2$ formed by the pairs (n, ν) , where $n = 1, 2, \dots$ and $\nu \in \mathbb{R}$, the pairs $(0, \nu)$, where $\nu \geq 0$, and the pairs $(x, 0)$, where $-1 \leq x \leq 0$; the topology is that induced by the topology

of $\mathbb{R}^2$, except that a point of $\widehat{G}$ which tends to $(-1, 0)$ in the ordinary sense tends in $\widehat{G}$ to both $(-1, 0)$ and $(2, 0)$. [8, 27].

For $G = \mathrm{SL}(2, \mathbb{C})$, the structure of $C^*(G)$ has been completely determined. [29].

For most groups, one is still at the stage of determining the set $\widehat{G}$. This work is nearly complete for the complex semisimple Lie groups, and in that case the Plancherel measure is also known; less progress has been made for real semisimple Lie groups. The fundamental work here is that of Gelfand, Graev, Harish-Chandra, and Naimark.

18.9.14. If G is commutative, $\widehat{G}$ carries a group structure. The extension of this fact to the noncommutative case is the existence, for every pair (π, π') of elements of $\widehat{G}$, of a measure class and multiplicities on $\widehat{G}$, namely the measure class and multiplicities associated with $\pi \otimes \pi'$ (when G is separable and postliminal). This measure class and these multiplicities have been computed in only a small number of cases.

18.9.15. Let G be a locally compact group, let π_1, π_2 be continuous unitary representations of G , and let S_1, S_2 be sets of continuous unitary representations of G . If π_k is weakly contained in S_k for $k = 1, 2$, then $\pi_1 \otimes \pi_2$ is weakly contained in the set of $\rho_1 \otimes \rho_2$ with $\rho_1 \in S_1$ and $\rho_2 \in S_2$. [33].

***18.9.16.** Let G be a locally compact group, let π be a finite-dimensional irreducible continuous unitary representation of G , let S be a set of continuous unitary representations of G , and let T be the set of all $\rho \otimes \bar{\pi}$ with $\rho \in S$. The set S weakly contains π if and only if T weakly contains the one-dimensional trivial representation. Both parts of this assertion may fail if $\dim \pi = +\infty$. [33].

Appendix A

Von Neumann Algebras

This Appendix is a list of results concerning von Neumann algebras which are used in the book. No proofs are given, but only references to *Les algèbres d'opérateurs dans l'espace hilbertien*, second edition, Gauthier–Villars, Paris, 1969 (designated by [A v N]).

Throughout, H denotes a Hilbert space.

A 1. The *norm topology* on $\mathcal{L}(H)$ is the topology induced by the norm $T \mapsto \|T\|$. The *strong*, *weak*, *ultrastrong*, and *ultraweak* topologies are the four topologies defined, respectively, by the following seminorms:

$$T \mapsto \|T\xi\| \quad (\xi \in H),$$

for the strong topology;

$$T \mapsto |(T\xi \mid \eta)| \quad (\xi, \eta \in H),$$

for the weak topology;

$$T \mapsto \left(\sum_{i=1}^{\infty} \|T\xi_i\|^2 \right)^{1/2} \quad \left(\xi_1, \xi_2, \dots \in H, \quad \sum_i \|\xi_i\|^2 < +\infty \right),$$

for the ultrastrong topology; and

$$T \mapsto \left| \sum_{i=1}^{\infty} (T\xi_i \mid \eta_i) \right| \quad \left(\begin{array}{l} \xi_1, \xi_2, \dots, \eta_1, \eta_2, \dots \in H, \\ \sum_i \|\xi_i\|^2 < +\infty, \quad \sum_i \|\eta_i\|^2 < +\infty \end{array} \right),$$

for the ultraweak topology.

On a bounded subset of $\mathcal{L}(H)$, the strong and ultrastrong topologies coincide, and the weak and ultraweak topologies coincide. ([A v N], pp. 30–34.)

A 2. Let M be a subset of $\mathcal{L}(H)$. The *commutant* of M , denoted M' , is the set of elements of $\mathcal{L}(H)$ which commute with the elements of M . Put $(M')' = M''$ (the *bicommutant* of M). The set M' is a subalgebra of $\mathcal{L}(H)$ containing 1. The inclusion $M \subset N$ implies $M' \supset N'$. We have $M' = M'''$ and $M \subset M''$.

A 3. Let A be an involutive subalgebra of $\mathcal{L}(H)$ and let A_1 be the unit ball of A . The following eight conditions are equivalent: A (respectively, A_1) is weakly

closed, A (respectively, A_1) is strongly closed, A (respectively, A_1) is ultrastrongly closed, and A (respectively, A_1) is ultraweakly closed. When these conditions hold, A contains a projection E greater than every other projection in A . For every $T \in A$, one has $ET = TE = T$. The elements of A'' are precisely the operators $T + \lambda 1$, where $T \in A$ and $\lambda \in \mathbb{C}$. ([A v N], p. 41, Theorem 2.)

A 4. Let A be an involutive subalgebra of $\mathcal{L}(H)$. The following nine conditions are equivalent: (1) each of the eight conditions in A 3, with the additional hypothesis $1 \in A$; (2) $A = A''$. When they hold, A is called a *von Neumann algebra* on H ([A v N], p. 42). The set $\mathcal{L}(H)$ is a von Neumann algebra on H . The set of scalar operators on H is a von Neumann algebra on H , denoted $\mathbb{C}_H$, or simply $\mathbb{C}$.

A 5. If M is a self-adjoint subset of $\mathcal{L}(H)$, then M' is a von Neumann algebra ([A v N], p. 2).

A 6. If $\mathcal{A}$ is a von Neumann algebra, then $\mathcal{A}$ and $\mathcal{A}'$ are each the commutant of the other. Their common center is $\mathcal{A} \cap \mathcal{A}'$. If this center consists only of the scalar operators, $\mathcal{A}$ is called a *factor*. The algebra $\mathcal{A}$ is a factor if and only if $\mathcal{A}'$ is a factor ([A v N], p. 3).

A 7. Let $\mathcal{A}$ be a von Neumann algebra on H , and let T be a Hermitian element of $\mathcal{L}(H)$. The operator T belongs to $\mathcal{A}$ if and only if its spectral projections belong to $\mathcal{A}$ ([A v N], p. 3).

A 8. Let $\mathcal{A}$ be a von Neumann algebra, and let $S, T \in \mathcal{A}^+$ with $S \leq T$. There exists an $R \in \mathcal{A}$ such that $S^{1/2} = RT^{1/2}$ ([A v N], p. 11, Lemma 2).

A 9. Let $\mathcal{A}$ be a von Neumann algebra and $\mathfrak{m}$ a two-sided ideal of $\mathcal{A}$. Then $\mathfrak{m}$ is self-adjoint and is the set of linear combinations of elements of $\mathfrak{m}^+ = \mathfrak{m} \cap \mathcal{A}^+$. The set of $T \in \mathcal{A}$ such that $TT^* \in \mathfrak{m}$ is a two-sided ideal $\mathfrak{n}$ of $\mathcal{A}$ for which the ideal $\mathfrak{n}^2$ equals $\mathfrak{m}$; it is denoted $\mathfrak{m}^{1/2}$ ([A v N], pp. 10–12).

Every weakly closed two-sided ideal of $\mathcal{A}$ has the form $E\mathcal{A}$, where E is a projection of $\mathcal{A} \cap \mathcal{A}'$ ([A v N], p. 42, Corollary 3).

A 10. Let $\mathcal{A}$ be a von Neumann algebra and E a projection of $\mathcal{A}$. Among the projections of $\mathcal{A} \cap \mathcal{A}'$ which majorize E , there is a least one. It is called the *central support* of E ([A v N], p. 6).

A 11. Let M be a subset of $\mathcal{L}(H)$. There is a least von Neumann algebra $\mathcal{A}$ on H which contains M ; it is called the von Neumann algebra generated by M . If $M = M^*$, then $\mathcal{A} = M''$ ([A v N], p. 2).

A 12. Let A be an involutive subalgebra of $\mathcal{L}(H)$. The weak, strong, ultraweak, and ultrastrong closures of A coincide; denote this closure by $\mathcal{B}$. Let K be the closed vector subspace of H spanned by the vectors $T\xi$, where $T \in A$ and $\xi \in H$. Then P_K is the greatest projection in $\mathcal{B}$. If $K = H$, then $\mathcal{B} = A''$ is the von Neumann algebra generated by A ([A v N], p. 42).

A 13. Let A and B be involutive subalgebras of $\mathcal{L}(H)$, with $A \subset B$. If A is strongly dense in B , then the unit ball A_1 of A is strongly dense in the unit ball B_1 of B ([A v N], p. 43, Theorem 3). If, in addition, H is separable, every element of B_1 is the strong limit of a sequence of elements of A_1 ([A v N], p. 32).

A 14. Let A be an involutive subalgebra of $\mathcal{L}(H)$ and let $\xi \in H$. The vector ξ is called *separating* for A if $T \in A$ and $T\xi = 0$ imply $T = 0$. It is called *cyclic* for A if $A\xi$ is dense in H . If ξ is cyclic for A , then ξ is separating for A' . The converse holds if A is a von Neumann algebra ([A v N], pp. 5–6).

A 15. Let $\mathcal{A}$ be a von Neumann algebra on H , and let $E = P_K$ be a projection of $\mathcal{A}$. Let $\mathcal{B}$ be the set of $T \in \mathcal{A}$ such that $ET = TE = T$. The operators $T|_K$, with $T \in \mathcal{B}$, form a von Neumann algebra on K , called the *reduction* of $\mathcal{A}$, and denoted $\mathcal{A}_E$ or $\mathcal{A}_K$. Let $E' = P_{K'}$ be a projection of $\mathcal{A}'$. The operators $T|_{K'}$, with $T \in \mathcal{A}$, form a von Neumann algebra on K' , said to be *induced* by $\mathcal{A}$ on K' , and denoted $\mathcal{A}_{E'}$ or $\mathcal{A}_{K'}$. The notations $\mathcal{A}_E, \mathcal{A}_{E'}$ are consistent when the projection belongs to $\mathcal{A} \cap \mathcal{A}'$. The map $T \mapsto T|_{K'}$, $T \in \mathcal{A}$, is called the *induction* of $\mathcal{A}$ on $\mathcal{A}_{E'}$. The von Neumann algebras $(\mathcal{A}_{E'})'$ and $(\mathcal{A}')_{E'}$ are equal and are denoted simply by $\mathcal{A}'_{E'}$ ([A v N], pp. 16–18). If $\mathfrak{Z}$ is the center of $\mathcal{A}$, the center of $\mathcal{A}_E$ is $\mathfrak{Z}_E$ ([A v N], p. 18, Corollary).

A 16. Let (H_i) be a family of Hilbert spaces, let H be their Hilbert sum, and let $\mathcal{A}_i$ be a von Neumann algebra on H_i . For every family (T_i) such that $T_i \in \mathcal{A}_i$ for all i and $\sup_i \|T_i\| < +\infty$, form the operator $(x_i) \mapsto (T_i x_i)$ on H . The set of all operators so obtained is a von Neumann algebra on H , called the *product* of the $\mathcal{A}_i$ and denoted $\prod_i \mathcal{A}_i$ ([A v N], p. 19).

A 17. Let H_1, H_2 be two Hilbert spaces. On the algebraic tensor product H_0 of H_1 and H_2 there is a unique pre-Hilbert space structure such that

$$(\xi_1 \otimes \xi_2 \mid \eta_1 \otimes \eta_2) = (\xi_1 \mid \eta_1)(\xi_2 \mid \eta_2)$$

for $\xi_1, \eta_1 \in H_1$ and $\xi_2, \eta_2 \in H_2$. This structure is separated. The Hilbert space H obtained by completing H_0 is called the *Hilbert tensor product* of H_1 and H_2 and is denoted $H_1 \otimes H_2$. If $T_1 \in \mathcal{L}(H_1)$ and $T_2 \in \mathcal{L}(H_2)$, their algebraic tensor product acts continuously on H_0 . Its continuous extension to H is denoted $T_1 \otimes T_2$ ([A v N], p. 21).

There is a family $(K_i)_{i \in I}$ of closed vector subspaces of $H_1 \otimes H_2$, of which $H_1 \otimes H_2$ is the Hilbert sum, and isomorphisms $U_i : H_1 \rightarrow K_i$, with the following properties: (1) for every $T \in \mathcal{L}(H_1)$, $T \otimes 1$ leaves every K_i invariant and $(T \otimes 1)|_{K_i} = U_i T U_i^{-1}$; (2) $\text{Card } I = \dim H_2$ ([A v N], pp. 22–24).

A 18. Let $\mathcal{A}_1$ be a von Neumann algebra on H_1 and $\mathcal{A}_2$ a von Neumann algebra on H_2 . Operators of the form

$$R_1 \otimes R_2 + S_1 \otimes S_2 + \cdots + T_1 \otimes T_2,$$

where $R_1, \dots, T_1 \in \mathcal{A}_1$ and $R_2, \dots, T_2 \in \mathcal{A}_2$, form an involutive subalgebra A_0 of $\mathcal{L}(H_1 \otimes H_2)$. The von Neumann algebra on $H_1 \otimes H_2$ generated by A_0 is called the *von Neumann tensor-product algebra* of $\mathcal{A}_1$ and $\mathcal{A}_2$, and is denoted $\mathcal{A}_1 \otimes \mathcal{A}_2$ ([A v N], p. 24). If A is an involutive subalgebra of $\mathcal{L}(H_1)$, the von Neumann algebra generated by $A \otimes \mathbb{C}_{H_2}$ has commutant $A' \otimes \mathcal{L}(H_2)$ ([A v N], p. 24).

A 19. Let $\mathcal{A}_1, \mathcal{A}_2$ be von Neumann algebras on H_1, H_2 . An isomorphism from $\mathcal{A}_1$ onto $\mathcal{A}_2$ is simply an isomorphism of the underlying involutive algebras; such an isomorphism is automatically isometric. Of course, an isomorphism $H_1 \rightarrow H_2$ of Hilbert spaces defines an isomorphism (called *spatial*) from every von Neumann algebra on H_1 onto a von Neumann algebra on H_2 . In general, however, an isomorphism need not be spatial ([A v N], p. 9).

A 20. Let $\mathcal{A}$ be a von Neumann algebra and E' a projection of $\mathcal{A}'$. The induction $T \mapsto T_{E'}$ from $\mathcal{A}$ onto $\mathcal{A}_{E'}$ is an isomorphism if and only if the central support of E' is 1 ([A v N], p. 18, Proposition 2).

A 21. Let $\mathcal{A}$ be a von Neumann algebra on H , and let H' be a Hilbert space. The map $T \mapsto T \otimes 1$ is an isomorphism of $\mathcal{A}$ onto the von Neumann algebra $\mathcal{A} \otimes \mathbb{C}_{H'} \subset \mathcal{L}(H \otimes H')$. Such an isomorphism is called an *amplification* ([A v N], p. 24).

A 22. Let $\mathcal{A}_1, \mathcal{A}_2$ be von Neumann algebras on H_1, H_2 , and let Φ be an isomorphism of $\mathcal{A}_1$ onto $\mathcal{A}_2$. There exist a Hilbert space K and a projection $E' \in (\mathcal{A}_1 \otimes \mathbb{C}_K)'$ of central support 1 such that Φ is the composite of the amplification $T \mapsto T \otimes 1_K$, the induction $T \otimes 1_K \mapsto (T \otimes 1_K)_{E'}$, and a spatial isomorphism. If H_1 and H_2 are separable, K may be taken separable ([A v N], pp. 55–56).

A 23. Let $\mathcal{A}$ be a von Neumann algebra on H , and let $\mathcal{A}'$ denote the Banach dual of the Banach space $\mathcal{A}$. For $f \in \mathcal{A}'$, ultraweak continuity and ultrastrong continuity of f are equivalent; the set of such forms is a vector subspace P of $\mathcal{A}'$, closed in the norm of $\mathcal{A}'$. Likewise, for $f \in \mathcal{A}'$, weak continuity and strong continuity are equivalent; the set of such forms is a vector subspace of P dense in P in the norm of $\mathcal{A}'$. The canonical bilinear form on $\mathcal{A} \times \mathcal{A}'$ induces a bilinear form on $\mathcal{A} \times P$ for which $\mathcal{A}$, with its norm, is the Banach dual of P . The space P is called the *predual* of $\mathcal{A}$ ([A v N], pp. 34–40).

A 24. Let f be a weakly continuous linear form on $\mathcal{L}(H)$. There are two orthonormal systems $(\xi_1, \dots, \xi_n)$ and $(\xi'_1, \dots, \xi'_n)$ in H , and numbers $\lambda_1 \geq 0, \dots, \lambda_n \geq 0$, such that $\|f\| = \lambda_1 + \dots + \lambda_n$ and

$$f(T) = \sum_{i=1}^n \lambda_i (T \xi_i \mid \xi'_i) \quad (T \in \mathcal{L}(H)).$$

([A v N], pp. 36–37.)

A 25. Let $\mathcal{A}$ be a von Neumann algebra. Every bounded increasing directed subset of $\mathcal{A}^+$ has a supremum in $\mathcal{A}^+$ ([A v N], p. 4). For a positive form f on $\mathcal{A}$, the following conditions are equivalent: (1) for every increasing directed set $\mathcal{F} \subset \mathcal{A}^+$ with supremum $T \in \mathcal{A}^+$, $f(T)$ is the supremum of $f(\mathcal{F})$; (2) f belongs to the predual of $\mathcal{A}$. Such a form f is called *normal*. Every element of the predual of $\mathcal{A}$ is a linear combination of positive normal forms ([A v N], pp. 50–51).

A 26. Let $\mathcal{A}$ be a von Neumann algebra, and let f be a positive normal form on $\mathcal{A}$. There is a greatest projection F of $\mathcal{A}$ such that $f(F) = 0$. One has $f(TF) = f(FT) = 0$ for every $T \in \mathcal{A}$. The projection $E = 1 - F$ is called the *support* of f . For every $T \in \mathcal{A}$, $f(T) = f(ET) = f(TE) = f(ETE)$ ([A v N], p. 57).

A 27. Let $\mathcal{A}$ and $\mathcal{B}$ be two von Neumann algebras, and let φ be a morphism from $\mathcal{A}$ into $\mathcal{B}$. The morphism φ is called *normal* if, for every increasing directed set $\mathcal{F} \subset \mathcal{A}^+$ with supremum $T \in \mathcal{A}^+$, $\varphi(T)$ is the supremum of $\varphi(\mathcal{F})$. Then φ is

ultrastrongly and ultraweakly continuous, and $\varphi(\mathcal{A})$ is weakly closed ([A v N], p. 53, Theorem 2, and p. 54, Corollary 2).

A 28. Let $\mathcal{A}$ be a von Neumann algebra. A *trace* on $\mathcal{A}^+$ is a function φ on $\mathcal{A}^+$, with values ≥ 0 finite or infinite, which has the following properties:

- (i) if $S, T \in \mathcal{A}^+$, then $\varphi(S + T) = \varphi(S) + \varphi(T)$;
- (ii) if $S \in \mathcal{A}^+$ and $\lambda \geq 0$, then $\varphi(\lambda S) = \lambda \varphi(S)$, with the convention $0 \cdot (+\infty) = 0$;
- (iii) if $S \in \mathcal{A}^+$ and U is a unitary operator in $\mathcal{A}$, then $\varphi(USU^{-1}) = \varphi(S)$.

The trace φ is called *finite* if $\varphi(S) < +\infty$ for every $S \in \mathcal{A}^+$. It is called *semifinite* if, for every $S \in \mathcal{A}^+$, $\varphi(S)$ is the supremum of the numbers $\varphi(T)$ for $T \in \mathcal{A}^+$ such that $T \leq S$ and $\varphi(T) < +\infty$.

(Everything preceding is a special case of 6.1.1.)

The trace φ is called *faithful* if $S \in \mathcal{A}^+$ and $\varphi(S) = 0$ imply $S = 0$. It is called *normal* if, for every increasing directed set $\mathcal{F} \subset \mathcal{A}^+$ with supremum $S \in \mathcal{A}^+$, $\varphi(S)$ is the supremum of $\varphi(\mathcal{F})$.

A 29. Let $\mathcal{A}$ be a von Neumann algebra, let ω, ω' be normal traces defined on $\mathcal{A}^+$, and let B be an involutive subalgebra of $\mathcal{A}$ weakly dense in the von Neumann algebra $\mathcal{A}$. Suppose that

$$\omega(TT^*) = \omega'(TT^*) < +\infty \quad (T \in B).$$

Then $\omega = \omega'$ ([A v N], p. 203, Lemma 1).

A 30. Let (e_i) be an orthonormal basis of H . For $T \in \mathcal{L}(H)^+$, put

$$\text{Tr}(T) = \sum_i (Te_i | e_i).$$

Then Tr is a faithful normal semifinite trace on $\mathcal{L}(H)^+$, independent of the orthonormal basis chosen ([A v N], p. 94, Theorem 5).

A 31. A von Neumann algebra $\mathcal{A}$ is called *finite* (respectively, *semifinite*) if, for every nonzero $T \in \mathcal{A}^+$, there exists a finite (respectively, semifinite) normal trace f on $\mathcal{A}^+$ such that $f(T) \neq 0$. It is called *properly infinite* (respectively, *purely infinite*) if there is no finite (respectively, semifinite) normal trace on $\mathcal{A}^+$ other than 0 ([A v N], p. 97, Definition 5).

On a semifinite von Neumann algebra there exists a faithful normal semifinite trace ([A v N], p. 99, Proposition 9).

A 32. Let $\mathcal{F}$ be a semifinite factor, and let f be a faithful normal semifinite trace on $\mathcal{F}^+$. Every normal trace on $\mathcal{F}^+$ has the form λf , where $\lambda \in [0, +\infty]$, with the

convention $0 \cdot (+\infty) = 0$ ([A v N], p. 81, Corollary 2, and p. 90, Corollary). The ideals $\mathfrak{m}_f, \mathfrak{n}_f$ are independent of the choice of f . The elements of $\mathfrak{m}_f$ (respectively, $\mathfrak{n}_f$) are called the *trace elements* (respectively, the *Hilbert–Schmidt elements*) relative to the semifinite factor $\mathcal{F}$ ([A v N], p. 101). When $\mathcal{F} = \mathcal{L}(H)$, one simply speaks of trace-class operators and Hilbert–Schmidt operators. If $T, T' \in \mathcal{L}(H)$ and T is trace-class, then

$$|\operatorname{Tr}(TT')| \leq (\operatorname{Tr} |T|) \|T'\|$$

([A v N], p. 106).

A 33. Let $\mathcal{F}$ be a factor. If $\mathcal{F}^+$ carries a nonzero finite normal trace f , then $\mathcal{F}$ is finite (for f is faithful by [A v N], p. 83, Corollary 2).

A 34. Let $\mathcal{A}$ be a von Neumann algebra. A projection E of $\mathcal{A}$ is called *minimal* if $E \neq 0$ and the only projections of $\mathcal{A}$ majorized by E are 0 and E . Equivalently, $E \neq 0$ and the reduced algebra $\mathcal{A}_E$ contains only scalar operators ([A v N], p. 123).

A projection E of $\mathcal{A}$ is called *abelian* if the reduced algebra $\mathcal{A}_E$ is commutative ([A v N], p. 123, Definition 3).

A 35. Let $\mathcal{A}$ be a von Neumann algebra. The following conditions on $\mathcal{A}$ are equivalent: (i) $\mathcal{A}$ is isomorphic to a von Neumann algebra $\mathcal{B}$ whose commutant $\mathcal{B}'$ is commutative; (ii) every nonzero projection of $\mathcal{A}$ majorizes a nonzero abelian projection; (iii) $\mathcal{A}$ has an abelian projection whose central support is 1. If these conditions hold, $\mathcal{A}$ is said to be of *type I*. Every type I algebra is semifinite ([A v N], p. 122, Proposition 2; p. 123, Theorem 1; p. 126, no. 4).

A 36. Let $\mathcal{A}$ be a von Neumann algebra on H . The following conditions are equivalent: (i) $\mathcal{A}$ is a type I factor; (ii) $\mathcal{A}$ is a factor and has minimal projections; (iii) there are Hilbert spaces H_1, H_2 and an isomorphism of H onto $H_1 \otimes H_2$ which transforms $\mathcal{A}$ into $\mathcal{L}(H_1) \otimes \mathbb{C}_{H_2}$; (iv) there are a family (K_i) of closed vector subspaces whose Hilbert sum is H and isomorphisms U_i from one Hilbert space K onto the K_i , with the following property: $\mathcal{A}$ is the set of $T \in \mathcal{L}(H)$ reduced by the K_i for which all the $T|_{K_i}$ are transported, by the U_i , from one and the same element of $\mathcal{L}(K)$ ([A v N], p. 124, Corollary 3, and pp. 21–23).

A 37. Every automorphism of a type I factor $\mathcal{A}$ is inner and is defined by a unitary element of $\mathcal{A}$ ([A v N], p. 241, Corollary 2; this may also be regarded as a consequence of 4.1.5).

A 38. A von Neumann algebra $\mathcal{A}$ is called *continuous* if there is no nonzero projection $E \in \mathcal{A} \cap \mathcal{A}'$ such that $\mathcal{A}_E$ is of type I. It is called of *type II* if it is semifinite and continuous. A purely infinite algebra is continuous; such an algebra is also called of *type III*. If the space on which $\mathcal{A}$ acts is nonzero, $\mathcal{A}$ cannot be of two different types ([A v N], pp. 121, 122, 126).

A 39. Let $\mathcal{A}$ be a von Neumann algebra. There are projections E_I, E_{II}, E_{III} in $\mathcal{A} \cap \mathcal{A}'$, characterized by the following properties: E_I, E_{II}, E_{III} are mutually orthogonal and have sum 1, while $\mathcal{A}_{E_I}$ is of type I, $\mathcal{A}_{E_{II}}$ of type II, and $\mathcal{A}_{E_{III}}$ of type III.

The projection E_I (respectively, E_{II}, E_{III}) is the greatest projection $E \in \mathcal{A} \cap \mathcal{A}'$ for which $\mathcal{A}_E$ is of type I (respectively, II, III); $E_I + E_{II}$ is the greatest projection $E \in \mathcal{A} \cap \mathcal{A}'$ for which $\mathcal{A}_E$ is semifinite; and $E_{II} + E_{III}$ is the greatest projection $E \in \mathcal{A} \cap \mathcal{A}'$ for which $\mathcal{A}_E$ is continuous ([A v N], pp. 98 and 122).

A 40. A finite type II algebra is called a *type II_1 von Neumann algebra*; a properly infinite type II algebra is called a *type II_∞ von Neumann algebra*. If $\mathcal{A}$ is a von Neumann algebra, there are projections E_{II_1}, E_{II_∞} characterized as follows: they are orthogonal and have sum E_{II} , and the corresponding algebras induced by $\mathcal{A}$ are of types II_1 and II_∞ ([A v N], pp. 98 and 122).

A 41. Let $\mathcal{A}$ be a von Neumann algebra and let E, F be two projections of $\mathcal{A}$. They are called *equivalent relative to $\mathcal{A}$* , written $E \sim F$, if there is a $U \in \mathcal{A}$ such that $U^*U = E$ and $UU^* = F$. We write $E < F$, or $F > E$, if there is a projection of $\mathcal{A}$, equivalent to E , which is majorized by F ([A v N], p. 215, Definition 1).

A 42. The relation $E \sim F$ is an equivalence relation. The relation $E < F$ is a preorder. If $E < F$ and $F < E$, then $E \sim F$ ([A v N], pp. 215–216).

A 43. Equivalent projections have the same central support ([A v N], p. 215).

A 44. Let E, F be two projections of $\mathcal{A}$, and let E_1, F_1 be their central supports. If E_1 and F_1 are not orthogonal, there are equivalent projections E', F' of $\mathcal{A}$ majorized respectively by E, F ([A v N], p. 217, Lemma 1).

A 45. Let H be the space on which $\mathcal{A}$ acts, and let $T \in \mathcal{A}$. The projections onto $H \ominus \ker T$ and onto $\overline{T(H)}$ belong to $\mathcal{A}$ and are equivalent ([A v N], p. 216, Proposition 2).

A 46. If $\mathcal{A}$ is a factor and E, F are projections of $\mathcal{A}$, then either $E < F$ or $F < E$ ([A v N], p. 218, Corollary 1).

A 47. A von Neumann algebra $\mathcal{A}$ is called *homogeneous* if it contains a family (E_i) of abelian, equivalent, pairwise orthogonal projections whose sum is 1. Such an algebra is of type I. The cardinal n of the family (E_i) depends only on $\mathcal{A}$, and $\mathcal{A}$ is said to be of *type* I_n . Equivalently, $\mathcal{A} = \mathcal{B} \otimes \mathcal{L}(K)$, where $\mathcal{B}$ is a commutative von Neumann algebra and K is a Hilbert space of dimension n ([A v N], pp. 239–240).

A 48. A von Neumann algebra $\mathcal{A}$ is continuous if and only if every projection of $\mathcal{A}$ is the sum of two orthogonal equivalent projections of $\mathcal{A}$ ([A v N], p. 219, Corollary 3).

A 49. A factor is of exactly one of the types I , II_1 , II_∞ , and III (except on the space $H = 0$). If it is of type I, it is of type I_n for a uniquely determined n . The only factors which are finite-dimensional as vector spaces are those of type I_n with $n < +\infty$. (This follows, for example, from A 48.)

A 50. Let $\mathcal{A}$ be a type I von Neumann algebra. There is a family $(E_i)_{i \in I}$ of nonzero, pairwise orthogonal projections of $\mathcal{A} \cap \mathcal{A}'$, with sum 1, such that the $\mathcal{A}_{E_i}$ are of types I_{n_i} with the n_i pairwise distinct. The E_i are uniquely determined up to a bijection of the index set ([A v N], p. 240).

A 51. Every isomorphism between type III von Neumann algebras acting on separable spaces is spatial ([A v N], p. 301, Corollary 7).

A 52. Let $\mathcal{A}$ be a von Neumann algebra. The projections E_I (respectively, E_{II}, E_{III}) associated with $\mathcal{A}$ and with $\mathcal{A}'$ are equal. The algebra $\mathcal{A}$ is of type I (respectively, of type II, of type III, semifinite, continuous) if and only if $\mathcal{A}'$ is of type I (respectively, of type II, of type III, semifinite, continuous) ([A v N], p. 104, Corollaries 1, 2, 3; p. 123, Theorem 1; p. 124, Corollary 1).

A 53. Let $\mathcal{A}$ be a von Neumann algebra and E a projection of $\mathcal{A}$ or of $\mathcal{A}'$. If $\mathcal{A}$ is of type I (respectively, of type II, of type III, semifinite, continuous), then $\mathcal{A}_E$ is of type I (respectively, of type II, of type III, semifinite, continuous) ([A v N], p. 103, Proposition 11; p. 104, Corollary 4; p. 125, Proposition 4).

A 54. A *Hilbert algebra* is an involutive algebra A equipped with an inner product which makes A a separated pre-Hilbert space and for which the following axioms hold:

- (i) $(x \mid y) = (y^* \mid x^*)$ for $x, y \in A$;
- (ii) $(xy \mid z) = (y \mid x^*z)$ for $x, y, z \in A$;
- (iii) for every $x \in A$, the map $y \mapsto xy$ from A into A is continuous;
- (iv) the elements $xy, x, y \in A$, are dense in A .

One has $(yx \mid z) = (y \mid zx^*)$ for $x, y, z \in A$, and the map $y \mapsto yx$ from A into A is continuous.

Let H be the Hilbert completion of A , let J be the involution of H which extends the map $x \mapsto x^*$ on A , and let U_x, V_x be the elements of $\mathcal{L}(H)$ which extend left and right multiplication by x on A . Then $x \mapsto U_x$ (respectively, $x \mapsto V_x$) is a nondegenerate representation of A (respectively, of the opposite algebra) on H . The U_x commute with the V_y . For every $x \in A$, $JU_xJ = V_{x^*}$. The weak closure of the set of U_x (respectively, V_x) is a von Neumann algebra $\mathfrak{U}(A)$ (respectively, $\mathfrak{B}(A)$), called the left-associated (respectively, right-associated) algebra of A . One has

$$\mathfrak{B}(A) = \mathfrak{U}(A)' = JU(A)J, \quad \mathfrak{U}(A) = \mathfrak{B}(A)' = J\mathfrak{B}(A)J.$$

The maps $x \mapsto U_x$ and $x \mapsto V_x$ are called the canonical maps of A into $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$. If $C \in \mathfrak{U}(A) \cap \mathfrak{B}(A)$, then $JCJ = C^*$ ([A v N], pp. 69–73).

A 55. Let A be a Hilbert algebra and A' an involutive subalgebra of A . Then A' is a Hilbert algebra.

If the $U_x, x \in A'$, are strongly dense in $\mathfrak{U}(A)$, then A' is dense in A .

If A' is dense in A , then $\mathfrak{U}(A') = \mathfrak{U}(A)$ and $\mathfrak{B}(A') = \mathfrak{B}(A)$ ([A v N], p. 72, Proposition 1).

A 56. Retain the preceding notation. For $a \in H$, the following conditions are equivalent: (i) the map $x \mapsto V_x a$ from A into H is continuous; (ii) the map $x \mapsto U_x a$ from A into H is continuous. Then a is called *bounded* relative to A . Let U_a (respectively, V_a) be the continuous extension to H of the map $x \mapsto V_x a$ (respectively, $x \mapsto U_x a$) from A into H . If $a \in A$, then a is bounded and the notations U_a, V_a agree with the preceding ones. An element $a \in H$ is bounded if and only if Ja is bounded. The Hilbert algebra A is called *completed* if every bounded element belongs to A ([A v N], pp. 72–74).

A 57. Let A be a Hilbert algebra dense in the Hilbert space H . If $a, b \in H$ are bounded, set

$$U_a b = V_b a = ab, \quad Ja = a^*.$$

Then the set of bounded elements becomes a Hilbert algebra A' containing A as a dense Hilbert subalgebra. The Hilbert algebra A' is completed, and $\mathfrak{U}(A') = \mathfrak{U}(A)$, $\mathfrak{B}(A') = \mathfrak{B}(A)$ ([A v N], p. 74).

A 58. The maps $a \mapsto U_a$ and $a \mapsto V_a$, $a \in A'$, are injective ([A v N], p. 70).

A 59. The operators U_a (respectively, V_a), where $a \in A'$, form a self-adjoint two-sided ideal of $\mathfrak{U}(A)$ (respectively, $\mathfrak{B}(A)$). For $a \in A'$, $T \in \mathfrak{U}(A)$, and $T' \in \mathfrak{B}(A)$, one has

$$Ta \in A', \quad T'a \in A', \quad U_{Ta} = TU_a, \quad V_{T'a} = T'V_a$$

([A v N], p. 72, Proposition 2).

A 60. For $S \in \mathfrak{U}(A)^+$, put $f(S) = (a \mid a)$ if $S^{1/2} = U_a$ with $a \in A'$, and put $f(S) = +\infty$ otherwise. Then f is a faithful normal semifinite trace on $\mathfrak{U}(A)^+$. The two-sided ideal of $\mathfrak{U}(A)$ formed by the U_a , $a \in A'$, is the set of $T \in \mathfrak{U}(A)$ such that $f(T^*T) < +\infty$, that is, $\mathfrak{n}_f$. If $a, b \in A'$, then

$$(a \mid b) = f'(U_b^* U_a),$$

where f' denotes the linear extension of the trace to $\mathfrak{m}_f$. There are analogous results for $\mathfrak{B}(A)$. The von Neumann algebras $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$ are semifinite. The trace f on $\mathfrak{U}(A)^+$ and the analogous trace on $\mathfrak{B}(A)^+$ are called the *natural traces* defined by A . These traces are unchanged when A is replaced by A' ([A v N], pp. 85–88).

A 61. Let $\mathcal{A}$ be a von Neumann algebra, let f be a faithful normal semifinite trace on $\mathcal{A}^+$, and let f' be the linear extension of f to $\mathfrak{m}_f$. With the inner product $(S \mid T) = f'(T^*S)$, $\mathfrak{n}_f$ is a completed Hilbert algebra. Let H_f be the Hilbert completion of $\mathfrak{n}_f$, and let J_f be the continuous involution of H_f extending that of $\mathfrak{n}_f$. For $R \in \mathcal{A}$, left and right multiplication by R on $\mathfrak{n}_f$ extend to continuous linear operators $\lambda_f(R)$, $\rho_f(R)$, and $\rho_f(R) = J_f \lambda_f(R) J_f$. The map λ_f is an isomorphism of $\mathcal{A}$ onto $\mathfrak{U}(\mathfrak{n}_f)$ which extends the canonical map of $\mathfrak{n}_f$ into $\mathfrak{U}(\mathfrak{n}_f)$ and transforms f into the natural trace defined on $\mathfrak{U}(\mathfrak{n}_f)^+$ by $\mathfrak{n}_f$ ([A v N], p. 88, Theorem 2).

A 62. Let A be a Hilbert algebra dense in a Hilbert space H , and let $a \in H$. The following conditions are equivalent: (i) $(a \mid xy) = (a \mid yx)$ for all $x, y \in A$; (ii) $Ta = JT^*Ja$ for every $T \in \mathfrak{U}(A)$. If a satisfies these conditions, it is called *central relative to A* . Elements of A which commute with every element of A are central relative to A . Let Z be the set of elements central relative to A . The transforms of the elements of Z by $\mathfrak{U}(A)$ (respectively, $\mathfrak{B}(A)$) span a closed vector subspace

Y (respectively, Y') of H . One has $Y = Y'$, and $P_Y \in \mathfrak{U}(A) \cap \mathfrak{B}(A)$ is called the *characteristic projection* of A ([A v N], pp. 75–76).

A 63. With the notation of A 62, the following conditions are equivalent: (i) $\mathfrak{U}(A)$ is a finite von Neumann algebra; (ii) $\mathfrak{B}(A)$ is a finite von Neumann algebra; (iii) $P_Y = 1$ ([A v N], p. 100, Theorem 6).

A 64. Let (A_i) be a family of Hilbert algebras, and let H_i be the Hilbert completion of A_i . Let A be the direct sum of the A_i , which is in the evident way an involutive algebra and a pre-Hilbert space. Then A is a Hilbert algebra, called the *direct-sum Hilbert algebra* of the A_i . The Hilbert completion H of A is the Hilbert sum of the H_i . One has

$$\mathfrak{U}(A) = \prod_i \mathfrak{U}(A_i), \quad \mathfrak{B}(A) = \prod_i \mathfrak{B}(A_i).$$

If E_i is the projection of H onto H_i , then

$$E_i \in \mathfrak{U}(A) \cap \mathfrak{B}(A), \quad \mathfrak{U}(A_i) = \mathfrak{U}(A)_{E_i}, \quad \mathfrak{B}(A_i) = \mathfrak{B}(A)_{E_i}$$

([A v N], p. 76).

A 65. Let A be a completed Hilbert algebra with Hilbert completion H , let (E_i) be a family of pairwise orthogonal projections in $\mathfrak{U}(A) \cap \mathfrak{B}(A)$ with sum 1, and put $H_i = E_i(H)$ and $A_i = E_i(A) = H_i \cap A$. Then A_i is an involutive subalgebra of A and a completed Hilbert algebra dense in H_i . The A_i annihilate one another and hence are two-sided ideals of A . One has

$$\mathfrak{U}(A_i) = \mathfrak{U}(A)_{E_i}, \quad \mathfrak{B}(A_i) = \mathfrak{B}(A)_{E_i}.$$

The direct-sum Hilbert algebra of the A_i is a dense involutive subalgebra of A . The natural trace of $\mathfrak{U}(A)^+$ induces on $\mathfrak{U}(A)_{E_i}^+ = \mathfrak{U}(A_i)^+$ the natural trace of $\mathfrak{U}(A_i)^+$ ([A v N], p. 77).

A 66. Let A be the involutive algebra of Hilbert–Schmidt operators on the Hilbert space H . With the inner product $(T \mid T') = \text{Tr}(T'^*T)$, it is a complete Hilbert algebra. Let $\overline{H}$ be the Hilbert space conjugate to H . There is a unique isomorphism Φ from the Hilbert space A onto $H \otimes \overline{H}$ which transforms the operator $\xi \mapsto (\xi \mid \eta)\zeta$, $\zeta, \eta \in H$, into $\zeta \otimes \overline{\eta}$. Identifying A with $H \otimes \overline{H}$ by Φ , one has, for $\xi, \eta \in H$ and $S \in \mathcal{L}(H)$,

$$(\xi \otimes \overline{\eta})^* = \eta \otimes \overline{\xi}, \quad S \cdot (\xi \otimes \overline{\eta}) = S\xi \otimes \overline{\eta}, \quad (\xi \otimes \overline{\eta}) \cdot S = \xi \otimes \overline{S^*\eta}.$$

([A v N], pp. 95–97.)

A 67. Let A be a completed Hilbert algebra such that $\mathfrak{U}(A)$ is a factor. The following conditions are equivalent:

- (i) A is complete;
- (ii) $\mathfrak{U}(A)$ is of type I;
- (iii) after multiplication of the norm of A by a constant, A is isomorphic to the Hilbert algebra of Hilbert–Schmidt operators on a suitable Hilbert space. The preceding Hilbert space is determined by A up to isomorphism ([A v N], p. 127).

A 68. Let A be a complete Hilbert algebra, and let (E_i) be the family of minimal projections of $\mathfrak{U}(A) \cap \mathfrak{B}(A)$.

- (i) The Hilbert space A is the Hilbert sum of the $A_i = E_i(A)$.
- (ii) The A_i are pairwise annihilating self-adjoint two-sided ideals of A .
- (iii) After multiplication of the norm of A_i by a constant, A_i is isomorphic to the Hilbert algebra of Hilbert–Schmidt operators on a suitable Hilbert space.
- (iv) If $\mathfrak{U}(A)$ and $\mathfrak{B}(A)$ are finite von Neumann algebras, each A_i is finite-dimensional.

([A v N], p. 127, Proposition 7.)

A 69. Let Z be a Borel space (B 19) and let μ be a positive measure on Z (B 30). A μ -measurable field of Hilbert spaces on Z is a pair

$$\mathcal{E} = ((H(\zeta))_{\zeta \in Z}, \Gamma),$$

where $(H(\zeta))_{\zeta \in Z}$ is a family of Hilbert spaces indexed by Z and Γ is a set of vector fields satisfying the following conditions:

- (i) Γ is a vector subspace of $\prod_{\zeta \in Z} H(\zeta)$;
- (ii) there is a sequence $(x_1, x_2, \dots)$ of elements of Γ such that, for every $\zeta \in Z$, the $x_n(\zeta)$ form a total sequence in $H(\zeta)$;
- (iii) for every $x \in \Gamma$, the function $\zeta \mapsto \|x(\zeta)\|$ is μ -measurable;
- (iv) let x be a vector field; if, for every $y \in \Gamma$, the function $\zeta \mapsto (x(\zeta) \mid y(\zeta))$ is μ -measurable, then $x \in \Gamma$.

The elements of Γ are then called the *measurable vector fields* of $\mathcal{E}$. If $x, y \in \Gamma$, the function $\zeta \mapsto (x(\zeta) \mid y(\zeta))$ is measurable ([A v N], p. 142).

A 70. Let Z, Z' be Borel spaces; let μ, μ' be positive measures on Z, Z' ; and let

$$\mathcal{E} = ((H(\zeta)), \Gamma), \quad \mathcal{E}' = ((H'(\zeta')), \Gamma')$$

be μ - and μ' -measurable fields of Hilbert spaces. Let $\eta : Z \rightarrow Z'$ be a Borel isomorphism which transforms μ into μ' . An η -isomorphism from $\mathcal{E}$ onto $\mathcal{E}'$ is a family $(V(\zeta))_{\zeta \in Z}$ having the following properties: (i) for every $\zeta \in Z$, $V(\zeta)$ is an isomorphism of $H(\zeta)$ onto $H'(\eta(\zeta))$; (ii) a vector field $\zeta \mapsto x(\zeta) \in H(\zeta)$ on Z is μ -measurable if and only if the field

$$\eta(\zeta) \mapsto V(\zeta)x(\zeta) \in H'(\eta(\zeta))$$

on Z' is μ' -measurable ([A v N], p. 143).

If $Z = Z'$ and $\mu = \mu'$, an η -isomorphism for which η is the identity map of Z is simply called an isomorphism from $\mathcal{E}$ onto $\mathcal{E}'$ ([A v N], p. 143).

A 71. Let Z be a Borel space, let μ be a positive measure on Z , and let H_0 be a separable Hilbert space. Put $H(\zeta) = H_0$ for every $\zeta \in Z$. Let Γ be the set of measurable maps of Z into H_0 , that is, the maps $x : Z \rightarrow H_0$ such that $\zeta \mapsto (x(\zeta) | a)$ is measurable for every $a \in H_0$. Then $\mathcal{E} = ((H(\zeta)), \Gamma)$ is a measurable field of Hilbert spaces on Z , called the *constant field* defined by H_0 ([A v N], p. 143). A field isomorphic to a constant field (cf. A 70) is called *trivial* ([A v N], p. 143).

A 72. Retain the notation of A 69. Let $p = 1, 2, \dots, \aleph_0$. The set

$$Z_p = \{\zeta \in Z : \dim H(\zeta) = p\}$$

is measurable. Let $\mathcal{E}_p$ be the restriction, defined in the evident way, of $\mathcal{E}$ to Z_p . Then every $\mathcal{E}_p$ is trivial ([A v N], pp. 144–145, Propositions 1 and 3).

A 73. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a μ -measurable field of Hilbert spaces on Z . A vector field x is called *square-integrable* if $x \in \Gamma$ and

$$\int_Z \|x(\zeta)\|^2 d\mu(\zeta) < +\infty.$$

If x, y are square-integrable, so are $x + y$ and λx , and the function $\zeta \mapsto (x(\zeta) | y(\zeta))$ is integrable; put

$$(x | y) = \int_Z (x(\zeta) | y(\zeta)) d\mu(\zeta).$$

The square-integrable vector fields then form a Hilbert space H , called the *Hilbert integral* of the $H(\zeta)$ and denoted

$$\int_Z^\oplus H(\zeta) d\mu(\zeta).$$

(Fields equal almost everywhere are identified.) An element x of this space is also written $\int_Z^\oplus x(\zeta) d\mu(\zeta)$. If μ is standard, H is separable ([A v N], pp. 145–146 and p. 149, Corollary).

A 74. Use the notation of A 70. The map which transforms the field

$$x \in H = \int_Z^\oplus H(\zeta) d\mu(\zeta)$$

into the field

$$\eta(\zeta) \mapsto V(\zeta)x(\zeta) \in H' = \int_{Z'}^\oplus H'(\zeta') d\mu'(\zeta')$$

is an isomorphism of H onto H' , denoted $\int_Z^\oplus V(\zeta) d\mu(\zeta)$ ([A v N], p. 159, Definition 2).

A 75. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a μ -measurable field of Hilbert spaces on Z , let $\tilde{\mu}$ be a positive measure on Z equivalent to μ , and put

$$\rho(\zeta) = \frac{d\tilde{\mu}(\zeta)}{d\mu(\zeta)}.$$

The map which assigns to every $x \in H = \int_Z^\oplus H(\zeta) d\mu(\zeta)$ the field

$$\zeta \mapsto \rho(\zeta)^{-1/2}x(\zeta) \in \tilde{H} = \int_Z^\oplus H(\zeta) d\tilde{\mu}(\zeta)$$

is an isomorphism from H onto $\tilde{H}$, called the *canonical isomorphism* ([A v N], p. 148).

A 76. Let Z be a Borel space, let μ be a positive measure on Z , let H_0 be a separable Hilbert space, and let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be the constant field on Z defined by H_0 . There is a unique isomorphism (called canonical) from $L^2(Z, \mu) \otimes H_0$ onto $\int_Z^\oplus H(\zeta) d\mu(\zeta)$ which, for every $f \in L^2(Z, \mu)$ and every $\xi \in H_0$, transforms $f \otimes \xi$ into the field $\zeta \mapsto f(\zeta)\xi$ ([A v N], p. 153, Corollary).

A 77. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a measurable field of Hilbert spaces on Z . For every $\zeta \in Z$, let $T(\zeta) \in \mathcal{L}(H(\zeta))$. The field $\zeta \mapsto T(\zeta)$ is called a *measurable field of operators* if, for every $x \in \Gamma$, the field $\zeta \mapsto T(\zeta)x(\zeta)$ is measurable. In that case the function $\zeta \mapsto \|T(\zeta)\|$ is measurable. Suppose, in addition, that this function is essentially bounded; the field $\zeta \mapsto T(\zeta)$ is then called *essentially bounded*. Let

$H = \int_Z^\oplus H(\zeta) d\mu(\zeta)$. For every $x \in H$, the field $Tx : \zeta \mapsto T(\zeta)x(\zeta)$ belongs to H . One has $T \in \mathcal{L}(H)$, and $\|T\|$ is the essential supremum of $\zeta \mapsto \|T(\zeta)\|$. Put

$$T = \int_Z^\oplus T(\zeta) d\mu(\zeta).$$

The $T(\zeta)$ are determined by T up to null sets. Operators on H of the form $\int_Z^\oplus T(\zeta) d\mu(\zeta)$ are called *decomposable* ([A v N], pp. 156–159).

A 78. If

$$S = \int_Z^\oplus S(\zeta) d\mu(\zeta), \quad T = \int_Z^\oplus T(\zeta) d\mu(\zeta)$$

are decomposable, then

$$\begin{aligned} S + T &= \int_Z^\oplus (S(\zeta) + T(\zeta)) d\mu(\zeta), & ST &= \int_Z^\oplus S(\zeta)T(\zeta) d\mu(\zeta), \\ \lambda S &= \int_Z^\oplus \lambda S(\zeta) d\mu(\zeta), & S^* &= \int_Z^\oplus S(\zeta)^* d\mu(\zeta). \end{aligned}$$

([A v N], p. 159, Proposition 3.)

A 79. Let

$$T_i = \int_Z^\oplus T_i(\zeta) d\mu(\zeta) \quad (i = 1, 2, \dots), \quad T = \int_Z^\oplus T(\zeta) d\mu(\zeta)$$

be decomposable operators. If T_i converges strongly to T , there is a subsequence (T_{n_i}) such that $T_{n_i}(\zeta)$ converges strongly to $T(\zeta)$ almost everywhere. If $T_i(\zeta)$ converges strongly to $T(\zeta)$ almost everywhere and $\sup_i \|T_i\| < +\infty$, then T_i converges strongly to T ([A v N], p. 160, Proposition 4).

A 80. Operators of the form $\int_Z^\oplus T(\zeta) d\mu(\zeta)$, where $T(\zeta)$ is scalar for every ζ , are called *diagonalizable operators*. Let $\mathfrak{Z}$ be the set of diagonalizable operators. Then $\mathfrak{Z}$ is a commutative von Neumann algebra, and $\mathfrak{Z}'$ is the set of decomposable operators ([A v N], p. 162, Definition 3; p. 163, Proposition 7; p. 164, Corollary).

A 81. Suppose that $\mathcal{E} = ((H(\zeta)), \Gamma)$ is the constant field defined by H_0 . Let $S \in \mathcal{L}(H_0)$ and put $S(\zeta) = S$ for every ζ . The canonical isomorphism from $L^2(Z, \mu) \otimes H_0$ onto $\int_Z^\oplus H(\zeta) d\mu(\zeta)$ transforms $1 \otimes S$ into $\int_Z^\oplus S(\zeta) d\mu(\zeta)$ ([A v N], pp. 169–170).

A 82. Let Z, Z' be Borel spaces and let μ, μ' be positive measures on Z, Z' . Let

$$\mathcal{E} = ((H(\zeta))_{\zeta \in Z}, \Gamma), \quad \mathcal{E}' = ((H'(\zeta'))_{\zeta' \in Z'}, \Gamma')$$

be μ - and μ' -measurable fields of Hilbert spaces. Let I be a countable set, and let η be a Borel isomorphism of Z onto Z' which transforms μ into μ' . For every $i \in I$, let $\zeta \mapsto T_i(\zeta)$ be a measurable field of operators relative to $\mathcal{E}$, and let $\zeta' \mapsto T'_i(\zeta')$ be a measurable field of operators relative to $\mathcal{E}'$. Suppose that, for every $\zeta \in Z$, there is an isomorphism from $H(\zeta)$ onto $H'(\eta(\zeta))$ which, for every $i \in I$, transforms $T_i(\zeta)$ into $T'_i(\eta(\zeta))$. Suppose μ and μ' are standard. Under these hypotheses, there is an η -isomorphism (A 70) from $\mathcal{E}$ onto $\mathcal{E}'$ which, for every $i \in I$, transforms the field $\zeta \mapsto T_i(\zeta)$ into the field $\zeta \mapsto T'_i(\eta(\zeta))$.

(This reduces at once to the case $Z = Z'$, $\mu = \mu'$, and η the identity map of Z . Then see [A v N], p. 166, Lemma 2.)

A 83. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a μ -measurable field of Hilbert spaces on Z , let

$$T_i = \int_Z^{\oplus} T_i(\zeta) d\mu(\zeta) \quad (i \in I)$$

be a family of decomposable operators on $\int_Z^{\oplus} H(\zeta) d\mu(\zeta)$, let H_0 be a separable Hilbert space, and let $(S_i)_{i \in I}$ be a family of elements of $\mathcal{L}(H_0)$. Suppose that, for every $\zeta \in Z$, there is an isomorphism of $H(\zeta)$ onto H_0 which transforms every $T_i(\zeta)$ into S_i . Suppose μ is standard. Then there is an isomorphism from $\int_Z^{\oplus} H(\zeta) d\mu(\zeta)$ onto $L^2(Z, \mu) \otimes H_0$ which, for every i , transforms T_i into $1 \otimes S_i$ ([A v N], p. 167, Theorem 2).

A 84. Let H be a separable Hilbert space and let $\mathfrak{B} \subset \mathcal{L}(H)$ be a commutative von Neumann algebra. There exist a standard Borel space Z , a bounded positive measure μ on Z , a measurable field $\zeta \mapsto H(\zeta)$ of nonzero Hilbert spaces on Z , and an isomorphism from H onto $\int_Z^{\oplus} H(\zeta) d\mu(\zeta)$ which transforms $\mathfrak{B}$ into the algebra of diagonalizable operators ([A v N], p. 210, Theorem 2).

A 85. Let Z be a Borel space, let μ be a standard positive measure on Z , let $\mathcal{E} = (H(\zeta))$ be a μ -measurable field of nonzero Hilbert spaces on Z , put

$$H = \int_Z^{\oplus} H(\zeta) d\mu(\zeta),$$

and let $\mathfrak{B}$ be the algebra of diagonalizable operators on H . Define similarly $Z_1, \mu_1, \mathcal{E}_1 = (H_1(\zeta_1)), H_1, \mathfrak{B}_1$. Let U be an isomorphism from H onto H_1 which transforms $\mathfrak{B}$ into $\mathfrak{B}_1$. There exist: (i) a Borel μ -null set N in Z and a Borel μ_1 -null set N_1 in Z_1 ; (ii) a Borel isomorphism η from $Z \setminus N$ onto $Z_1 \setminus N_1$ which transforms μ into a measure $\tilde{\mu}_1$ equivalent to μ_1 ; and (iii) an η -isomorphism $(V(\zeta))$

from $\mathcal{E}|_{Z \setminus N}$ onto $\mathcal{E}_1|_{Z_1 \setminus N_1}$, such that U is the composite of $\int^\oplus V(\zeta) d\mu(\zeta)$ and the canonical isomorphism from

$$\int_{Z_1}^\oplus H_1(\zeta_1) d\tilde{\mu}_1(\zeta_1) \quad \text{onto} \quad H_1$$

([A v N], p. 212, Theorem 4).

A 86. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a measurable field of Hilbert spaces, put

$$H = \int_Z^\oplus H(\zeta) d\mu(\zeta), \quad T_i = \int_Z^\oplus T_i(\zeta) d\mu(\zeta),$$

where (T_i) is a sequence of decomposable operators, let $\mathcal{A}$ be the von Neumann algebra generated by the T_i , let $\mathcal{A}(\zeta)$ be the von Neumann algebra generated by the $T_i(\zeta)$, and let $\mathfrak{Z}$ be the algebra of diagonalizable operators. The algebra $\mathfrak{Z}$ is a maximal commutative von Neumann subalgebra of $\mathcal{A}'$ if and only if $\mathcal{A}(\zeta) = \mathcal{L}(H(\zeta))$ almost everywhere ([A v N], p. 172, Corollary 1).

A 87. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a measurable field of Hilbert spaces on Z . For every $\zeta \in Z$, let $\mathcal{A}(\zeta)$ be a von Neumann algebra on $H(\zeta)$. The field $\zeta \mapsto \mathcal{A}(\zeta)$ is called a *measurable field of von Neumann algebras* on Z if there is a sequence

$$\zeta \mapsto T_1(\zeta), \quad \zeta \mapsto T_2(\zeta), \quad \dots$$

of measurable fields of operators such that, for every $\zeta \in Z$, $\mathcal{A}(\zeta)$ is the von Neumann algebra generated by the $T_i(\zeta)$. The set of decomposable operators

$$T = \int_Z^\oplus T(\zeta) d\mu(\zeta) \quad \text{such that} \quad T(\zeta) \in \mathcal{A}(\zeta) \text{ for every } \zeta$$

is a von Neumann algebra $\mathcal{A}$ on $H = \int_Z^\oplus H(\zeta) d\mu(\zeta)$, denoted

$$\mathcal{A} = \int_Z^\oplus \mathcal{A}(\zeta) d\mu(\zeta).$$

The $\mathcal{A}(\zeta)$ are determined by $\mathcal{A}$ up to null sets. Von Neumann algebras of this form are called *decomposable*. Let $\mathfrak{Z}$ be the algebra of diagonalizable operators. Suppose that μ is standard. A von Neumann algebra $\mathcal{A}$ is decomposable if and only if

$$\mathfrak{Z} \subset \mathcal{A} \subset \mathfrak{Z}'$$

([A v N], pp. 173–174).

A 88. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a measurable field of Hilbert spaces, let $T_i = \int_Z^\oplus T_i(\zeta) d\mu(\zeta)$ be a sequence of decomposable operators, let $\mathfrak{B}$ be the algebra of diagonalizable operators, let $\mathcal{A}(\zeta)$ be the von Neumann algebra generated by the $T_i(\zeta)$, and let $\mathcal{A}$ be the von Neumann algebra generated by $\mathfrak{B}$ and the T_i . Then

$$\mathcal{A} = \int_Z^\oplus \mathcal{A}(\zeta) d\mu(\zeta)$$

([A v N], p. 171, Theorem 1).

A 89. Let $\mathcal{A} = \int_Z^\oplus \mathcal{A}(\zeta) d\mu(\zeta)$ be a decomposable von Neumann algebra, and let $\mathfrak{B}$ be the algebra of diagonalizable operators. If $\mathfrak{B}$ is the center of $\mathcal{A}$, then $\mathcal{A}(\zeta)$ is a factor almost everywhere ([A v N], p. 174, Theorem 3).

A 90. Let $\mathcal{A} = \int_Z^\oplus \mathcal{A}(\zeta) d\mu(\zeta)$ be a decomposable von Neumann algebra, and suppose μ is standard. The algebra $\mathcal{A}$ is of type I if and only if $\mathcal{A}(\zeta)$ is of type I almost everywhere ([A v N], p. 183, Corollary 2 to Proposition 7).

A 91. Let

$$\mathcal{A} = \int_Z^\oplus \mathcal{A}(\zeta) d\mu(\zeta), \quad \mathcal{A}_1 = \int_Z^\oplus \mathcal{A}_1(\zeta) d\mu(\zeta)$$

be two decomposable von Neumann algebras. For every $\zeta \in Z$, let $\Phi(\zeta)$ be an isomorphism from $\mathcal{A}(\zeta)$ onto $\mathcal{A}_1(\zeta)$. The field of isomorphisms $\zeta \mapsto \Phi(\zeta)$ is called *measurable* if, for every $T = \int_Z^\oplus T(\zeta) d\mu(\zeta) \in \mathcal{A}$, the field $\zeta \mapsto \Phi(\zeta)(T(\zeta))$ is measurable. If

$$\Phi(T) = \int_Z^\oplus \Phi(\zeta)(T(\zeta)) d\mu(\zeta),$$

then Φ is an isomorphism from $\mathcal{A}$ onto $\mathcal{A}_1$, denoted $\int_Z^\oplus \Phi(\zeta) d\mu(\zeta)$ ([A v N], pp. 183 and 185).

A 92. Let $\mathcal{A} = \int_Z^\oplus \mathcal{A}(\zeta) d\mu(\zeta)$ be a decomposable von Neumann algebra. For every $\zeta \in Z$, let $f(\zeta)$ be a trace on $\mathcal{A}(\zeta)^+$. The field of traces $\zeta \mapsto f(\zeta)$ is called *measurable* if, for every measurable field of operators $\zeta \mapsto T(\zeta) \in \mathcal{A}(\zeta)^+$, the function $\zeta \mapsto f(\zeta)(T(\zeta))$ is measurable. If $T = \int_Z^\oplus T(\zeta) d\mu(\zeta) \in \mathcal{A}^+$, put

$$f(T) = \int_Z f(\zeta)(T(\zeta)) d\mu(\zeta).$$

Then f is a trace on $\mathcal{A}^+$, denoted $\int_Z^\oplus f(\zeta) d\mu(\zeta)$ ([A v N], p. 198).

A 93. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a measurable field of Hilbert spaces on Z . For every $\zeta \in Z$, let $A(\zeta)$ be a Hilbert algebra whose Hilbert completion is $H(\zeta)$. The field $\zeta \mapsto A(\zeta)$ of Hilbert algebras is called *μ -measurable* when the following conditions hold:

- (i) if $x \in \Gamma$ and $x(\zeta) \in A(\zeta)$ for every ζ , then the field $x^* : \zeta \mapsto x(\zeta)^*$ is measurable;
- (ii) if $x, y \in \Gamma$ and $x(\zeta), y(\zeta) \in A(\zeta)$ for every ζ , then the field $xy : \zeta \mapsto x(\zeta)y(\zeta)$ is measurable;
- (iii) there is a sequence $(x_1, x_2, \dots)$ of elements of Γ such that $x_i(\zeta) \in A(\zeta)$ for every i and every ζ , and such that $(x_1(\zeta), x_2(\zeta), \dots)$ is total in $H(\zeta)$ for every ζ .

([A v N], p. 187, Definition 1.)

A 94. Retain the notation of A 93, and put $H = \int_Z^\oplus H(\zeta) d\mu(\zeta)$. Let A be the set of $x \in H$ such that $x(\zeta) \in A(\zeta)$ for every ζ and the field of operators $\zeta \mapsto U_{x(\zeta)}$, which is automatically measurable, is essentially bounded. Then, with the operations $x \mapsto x^*$, $(x, y) \mapsto xy$, and the inner product induced by that of H , A is a Hilbert algebra dense in H , denoted

$$A = \int_Z^\oplus A(\zeta) d\mu(\zeta).$$

One then has

$$\mathfrak{U}(A) = \int_Z^\oplus \mathfrak{U}(A(\zeta)) d\mu(\zeta), \quad \mathfrak{B}(A) = \int_Z^\oplus \mathfrak{B}(A(\zeta)) d\mu(\zeta).$$

If J (respectively, $J(\zeta)$) is the involution of H (respectively, $H(\zeta)$) defined by A (respectively, $A(\zeta)$), then

$$J = \int_Z^\oplus J(\zeta) d\mu(\zeta).$$

The natural trace on $\mathfrak{U}(A)^+$ defined by A is $\int_Z^\oplus t(\zeta) d\mu(\zeta)$, where $t(\zeta)$ denotes the natural trace on $\mathfrak{U}(A(\zeta))^+$ defined by $A(\zeta)$ [1, pp. 188–190 and p. 199, Theorem 1].

A 95. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a measurable field of Hilbert spaces over Z , let

$$H = \int_Z^\oplus H(\zeta) d\mu(\zeta),$$

let $\mathfrak{D}$ be the algebra of diagonalizable operators, and let A be a completed Hilbert algebra having H as its completed Hilbert space. If $\mathfrak{U}(A) \subset \mathfrak{D}'$, there exists a

measurable field $\zeta \mapsto A(\zeta)$ of completed Hilbert algebras having the $H(\zeta)$ as their completed Hilbert spaces, such that

$$A = \int_Z^\oplus A(\zeta) d\mu(\zeta)$$

[1, p. 194, Theorem 1].

A 96. Let Z be a Borel space. A Borel field $\mathcal{E}$ of Hilbert spaces over Z is a family $(H(\zeta))_{\zeta \in Z}$ of Hilbert spaces together with a set Γ of vector fields satisfying conditions (i)–(iv) of A 69, with the word “ μ -measurable” replaced everywhere by “Borel” [1, pp. 143–144]. Such fields have properties analogous to those of measurable fields, proved in the same way. In particular, A 71 and A 72 may be repeated with “measurable” replaced everywhere by “Borel.”

A 97. Let $\mathcal{E} = ((H(\zeta)), \Gamma)$ be a Borel field of Hilbert spaces over Z , and let μ be a positive measure on Z . There exists one and only one set $\Gamma' \supset \Gamma$ of vector fields such that $((H(\zeta)), \Gamma')$ is a μ -measurable field of Hilbert spaces; Γ' is the set of vector fields x such that, for every $y \in \Gamma$, the function

$$\zeta \mapsto (x(\zeta) \mid y(\zeta))$$

is μ -measurable. The μ -measurable field $((H(\zeta)), \Gamma')$ is said to be derived from the Borel field $((H(\zeta)), \Gamma)$ [1, p. 144].

A 98. Let Z be a Borel space, let $\zeta \mapsto H(\zeta)$ be a field of Hilbert spaces over Z , and let $\zeta \mapsto x_i(\zeta)$, $i = 1, 2, \dots$, be a sequence of vector fields with the following properties:

- (i) for every $\zeta \in Z$, the sequence of the $x_i(\zeta)$ is total in $H(\zeta)$;
- (ii) the functions $\zeta \mapsto (x_i(\zeta) \mid x_j(\zeta))$ are Borel.

Then there exists one and only one Borel-field structure on $\zeta \mapsto H(\zeta)$ for which the $\zeta \mapsto x_i(\zeta)$ are Borel vector fields [1, p. 145, Proposition 4, and p. 146, Remark].

Appendix B

Miscellaneous Results

This Appendix is an unordered list of very varied results used in the course of the text. The references are to the special bibliography at the end of the Appendix.

B 1. Let A be a complex Banach algebra. For every $x \in A$, $\|x^n\|^{1/n}$ has a limit $\rho(x)$ as $n \rightarrow +\infty$, called the spectral radius of x . One has $\rho(x) \leq \|x\|$, and $\rho(x)$ is the supremum of $|z|$ as z ranges over $\text{Sp}_A x$ [11, Theorems 1.4.1 and 1.6.4]. Every maximal regular left ideal of A is closed [11, Corollary 2.1.4].

B 2. Let A be a unital Banach algebra, let B be a closed subalgebra containing 1, and let $x \in B$. Then $\text{Sp}_B x \supset \text{Sp}_A x$, and every boundary point of $\text{Sp}_B x$ relative to $\mathbb{C}$ belongs to $\text{Sp}_A x$ [11, Theorem 1.6.12].

B 3. Let A be a commutative Banach algebra. Every character of A has norm at most 1. The set of characters of A , endowed with the weak topology, is a locally compact space S called the spectrum of A . For every $x \in A$, the function $\chi \mapsto \chi(x)$ on S is called the Gelfand transform of x and is denoted by $\mathcal{F}x$. The functions $\mathcal{F}x$ ($x \in A$) are continuous, separate the points of S , and, for every $\chi \in S$, there exists $x \in A$ such that $(\mathcal{F}x)(\chi) \neq 0$. If A has an identity element, the set of values of $\mathcal{F}x$ is $\text{Sp}_A x$ [11, p. 110, Theorem 3.1.6, Corollary 3.1.7, and Theorem 3.1.20].

B 4. Let A be a unital Banach algebra, let $x \in A$, and let f be a function holomorphic in a neighborhood of $\text{Sp}_A x$. There exists an element $f(x) \in A$ with the following properties:

- (i) $\text{Sp}_A f(x) = f(\text{Sp}_A x)$;
- (ii) if $f(z) = \sum_{n=0}^{\infty} c_n z^n$ is an entire function, then $f(x) = \sum_{n=0}^{\infty} c_n x^n$, the series converging normally.

See [11, pp. 157–158 and the references cited there]. The hypothesis made in [11] that A is commutative is not really used. Moreover, at the only place in the present book (1.3.9) where these results are applied, it would be easy to reduce matters to the case of commutative algebras.

B 5. Let E be a real locally convex space, let C be a closed convex cone with vertex 0, and let x be a point of E not belonging to C . There exists a continuous linear

form on E which is nonnegative on C and negative at x . [Indeed, there exist a continuous linear form f on E and a real number α such that $f(y) \geq \alpha$ on C and $f(x) < \alpha$. We have $0 = f(0) \geq \alpha$, and hence $f(x) < 0$. If $f(y) < 0$ at a point of C , then $f(\lambda y) < \alpha$ for sufficiently large $\lambda > 0$, which is absurd.]

B 6. Let E be a separated locally convex space, let P be a convex cone in E such that $P \cap (-P) = 0$, let M be a vector subspace of E , and let f be a linear form on M which is nonnegative on $M \cap P$. Suppose that M contains an interior point of P . Then f extends to a linear form on E which is nonnegative on P [4, p. 64, Corollary].

B 7. Let E be a separable Banach space. The unit ball of the dual of E is compact metrizable for the weak topology [5, p. 112, Proposition 2].

B 8. Let E be a Banach space, let E' be its dual, and let K be a convex subset of E' . Then K is weakly closed if and only if its intersection with every closed ball of E' is weakly closed [5, p. 74, Theorem 5].

B 9. Let H be a separable Hilbert space, and let B be the unit ball of $\mathcal{L}(H)$ endowed with the weak topology. Then B is compact metrizable [5, p. 25, Proposition 6, and p. 65, Corollary 3].

B 10. Let H be a separable Hilbert space. The set of unitary operators on H is a Polish space for the weak topology [7, p. 280, Lemma 4].

B 11. Let E, F be two locally convex spaces. Suppose that E is barrelled and that F is separated and quasi-complete. Then $\mathcal{L}(E, F)$, endowed with the topology of pointwise convergence, is separated and quasi-complete [5, p. 31, Corollary 2].

B 12. Let E, F be two topological vector spaces. Suppose that F is separated and quasi-complete. Then, in $\mathcal{L}(E, F)$ endowed with the topology of pointwise convergence, every closed equicontinuous subset is a complete uniform subspace [5, p. 30, Theorem 4].

B 13. Let E be a separated locally convex space, and let C be a metrizable compact convex subset of E . The set A of extreme points of C is a G_δ subset of C [8, pp. 118–119]. Let $x \in C$. There exists a positive measure μ of total mass 1 on C , concentrated on A (that is, such that $C \setminus A$ is μ -negligible), and such that

$$x = \int_C y \, d\mu(y),$$

in other words,

$$f(x) = \int_C f(y) d\mu(y)$$

for every continuous linear form f on E (G. Choquet's theorem; [9, Theorem]).

B 14. Let E be a separated locally convex space, let C be a compact convex subset of E , and let A be the set of extreme points of C . Then C is the closed convex hull of A (the Krein–Milman theorem). If a subset A' of C has closed convex hull equal to C , then $\overline{A'}$ contains A .

We shall prove that A is a Baire space. (This unpublished result, as well as the proof which follows, is due to G. Choquet.) We may suppose that E is real. For every continuous linear form f on E and every real number α , let $U_{f,\alpha}$ (respectively, $F_{f,\alpha}$) denote the set of $x \in C$ such that $f(x) < \alpha$ (respectively, $f(x) \leq \alpha$).

Let $x \in A$. We first show that the set of the $F_{f,\alpha}$ such that $x \in U_{f,\alpha}$ is a fundamental system of neighborhoods of x . By the Hahn–Banach theorem, the intersection of these $F_{f,\alpha}$ is $\{x\}$; since the $F_{f,\alpha}$ are compact, it is enough to show that the family of the $F_{f,\alpha}$ such that $x \in U_{f,\alpha}$ is downward directed. Let $f_1, \alpha_1, f_2, \alpha_2$ be such that

$$x \in U_{f_1,\alpha_1} \cap U_{f_2,\alpha_2}.$$

Let C_1, C_2 be the complements in C of U_{f_1,α_1} and U_{f_2,α_2} . Since C_1 and C_2 are compact, their convex hull C_3 is compact. It does not contain x , because x is extreme. Hence there exist a continuous linear form f on E and a real number α such that $f(x) < \alpha$ and $f(y) > \alpha$ for $y \in C_3$. Then $F_{f,\alpha}$ meets neither C_1 nor C_2 ; that is,

$$F_{f,\alpha} \subset U_{f_1,\alpha_1} \cap U_{f_2,\alpha_2}, \quad x \in U_{f,\alpha}.$$

Let (V_n) be a sequence of open dense subsets of A . We must prove that $\bigcap_n V_n$ is dense in A , that is, that it meets every nonempty open subset V of A . Let U_n, U be open subsets of C , with U_n dense in C , such that $U_n \cap A = V_n$ and $U \cap A = V$. We may suppose that the U_n and the V_n are decreasing. We may also suppose that U is of the form U_{f_1,α_1} .

With this understood, we shall prove the existence of $(f_1, \alpha_1), (f_2, \alpha_2), \dots$ with the following properties:

$$F_{f_{n+1},\alpha_{n+1}} \subset U_{f_n,\alpha_n} \cap U_{n+1}, \quad U_{f_n,\alpha_n} \cap A \neq \emptyset \quad (n = 1, 2, \dots).$$

The initial pair (f_1, α_1) has already been chosen. Suppose that the pairs (f_i, α_i) have been constructed for $1 \leq i \leq n$. There is $x \in U_{f_n,\alpha_n} \cap A \cap U_{n+1}$. Then

$U_{f_n, \alpha_n} \cap U_{n+1}$ is a neighborhood of x in C , and so there exists (f_{n+1}, α_{n+1}) such that

$$x \in U_{f_{n+1}, \alpha_{n+1}} \subset F_{f_{n+1}, \alpha_{n+1}} \subset U_{f_n, \alpha_n} \cap U_{n+1}.$$

Since $x \in A$, one has $U_{f_{n+1}, \alpha_{n+1}} \cap A \neq \emptyset$, and the induction may continue.

The sets F_{f_n, α_n} form a decreasing sequence of nonempty compact sets, and hence their intersection F is nonempty. We have

$$F \subset U_{f_1, \alpha_1}, \quad F \subset \bigcap_n U_n.$$

Finally, F is compact and convex, and its complement in C is convex. An easy lemma then proves that F contains at least one extreme point y of C . (The set F has an extreme point x . If x is extreme in C , we are done. Otherwise, let δ be a line through x such that x is an interior point of the segment $C \cap \delta$; one then shows that one of the endpoints of $F \cap \delta$ is an extreme point of C .) The point y belongs to $A \cap U_n = V_n$ for every n , and to $A \cap U_{f_1, \alpha_1} = V$.

B 15. Every G_δ subset of a Polish space is a Polish space [2, p. 123, Theorem 1].

B 16. Let E be a locally compact space, let G be a group of homeomorphisms of E onto E , and let $C(E, E)$ be the set of continuous maps of E into E , endowed with the topology of compact convergence. Suppose that G is relatively compact in $C(E, E)$. Then the closure of G in $C(E, E)$ is a compact group of homeomorphisms of E onto E [3, p. 52, Corollary].

B 17. Let E be a Polish space, and let R be an equivalence relation on E . Suppose that the saturation in E of every open subset is Borel and that the equivalence classes are closed. Then there exists a Borel subset of E which meets each equivalence class in one and only one point [7, p. 279, Lemma 2].

B 18. Let E be a Baire space, and let $f : E \rightarrow \mathbb{R}$ be a lower semicontinuous function. Then f has a point of continuity. Indeed, on replacing f by $f(1 + |f|)^{-1}$, we may suppose that f is bounded. For every $x \in E$, let $\omega(x)$ be the oscillation of f at x . Then ω is upper semicontinuous, and hence the set E_n of points at which $\omega(x) \geq 1/n$ is closed. Suppose that E_n contains a nonempty open subset U . Put

$$\alpha = \sup_{x \in U} f(x).$$

There exists $x_0 \in U$ such that $f(x_0) > \alpha - 1/(2n)$. Throughout some neighborhood of x_0 one has

$$\alpha - \frac{1}{2n} < f(x) \leq \alpha,$$

and hence $\omega(x_0) \leq 1/(2n)$, a contradiction. Thus $E \setminus E_n$ is open and dense, so $\bigcap_n (E \setminus E_n)$ is nonempty. If $x \in \bigcap_n (E \setminus E_n)$, then $\omega(x) = 0$, and therefore f is continuous at x .

B 19. A Borel space is a set E together with a collection $\mathcal{B}$ of subsets of E having the following properties: $E \in \mathcal{B}$, $\emptyset \in \mathcal{B}$, and $\mathcal{B}$ is stable under countable unions, countable intersections, and passage to the complement. The elements of $\mathcal{B}$ are called the Borel subsets of E .

Let E be a set and let $\mathcal{A}$ be a collection of subsets of E . Among the collections $\mathcal{B}$ of subsets of E such that $\mathcal{B} \supset \mathcal{A}$ and $(E, \mathcal{B})$ is a Borel space, there is one, $\mathcal{B}_0$, smaller than all the others. The Borel structure of $(E, \mathcal{B}_0)$ is said to be generated by $\mathcal{A}$.

Let $(E, \mathcal{B}_1)$ and $(E, \mathcal{B}_2)$ be two Borel spaces. If $\mathcal{B}_2 \supset \mathcal{B}_1$, the Borel structure of $(E, \mathcal{B}_2)$ is said to be finer than that of $(E, \mathcal{B}_1)$.

Let $(E_i)_{i \in I}$ be a family of Borel spaces, let E be a set, and let $f_i: E \rightarrow E_i$ be maps. The Borel structure on E defined by the f_i is the structure generated by the sets $f_i^{-1}(A)$, where i ranges over I and A ranges over the Borel subsets of E_i .

Let E, F be two Borel spaces. A map $f: E \rightarrow F$ is said to be Borel if the inverse image under f of every Borel subset of F is a Borel subset of E .

Let E be a Borel space and let E' be a subset of E . The intersections with E' of the Borel subsets of E define a Borel structure on E' . The Borel space E' is called a Borel subspace of E .

Let E be a Borel space and let R be an equivalence relation on E . The subsets of E/R whose inverse images in E are Borel define a Borel structure on E/R . The Borel space E/R is called the quotient Borel space of E by R .

Let (E_α) be a family of Borel spaces, and let E be the disjoint union of the E_α . The subsets of E whose intersections with the E_α are Borel define a Borel structure on E . The Borel space E is called the sum Borel space of the E_α .

Let E be a topological space. The subsets of E which are Borel for the topology define a Borel structure on E , called the structure underlying the topology. If $E' \subset E$, the Borel structure induced on E' by that of E is the structure underlying the topology induced on E' by the topology of E .

For all the preceding material, see [10, pp. 136–137].

B 20. A Borel space E is said to be standard if its Borel structure underlies a topology of a Polish space. If E is countable and standard, every subset of E is Borel. If E is uncountable and standard, E is isomorphic to the Borel space $[0, 1]$ endowed with the Borel structure underlying its usual topology. Every

Borel subset of a standard space is standard. The sum of a sequence of standard spaces is standard [10, p. 138].

B 21. Let E_1, E_2 be two Borel spaces, and let $f: E_1 \rightarrow E_2$ be an injective Borel map. Suppose that:

(i) E_1 is standard;

(ii) there exists a sequence of Borel subsets of E_2 separating the points of E_2 and generating the Borel structure of E_2 (as is the case if E_2 is standard).

Then $f(E_1)$ is Borel in E_2 , and f is an isomorphism from E_1 onto $f(E_1)$ [10, p. 139, Theorem 3.2].

B 22. Let E_1, E_2 be two Borel spaces, and let $f: E_1 \rightarrow E_2$ be a bijective Borel map. Suppose that:

(i) E_2 is standard;

(ii) E_1 is a quotient of a standard Borel space.

Then f is an isomorphism from E_1 onto E_2 [10, p. 140, Theorem 4.2, and p. 141, Corollary].

B 23. Let H be a Hilbert space, and let S be a linear operator defined on a vector subspace D_S of H dense in H , with values in H , such that $(Sf | f) \geq 0$ for every $f \in D_S$. For $f, g \in D_S$, put

$$(f | g)_S = (Sf | g) + (f | g).$$

Every Cauchy sequence for $(\cdot | \cdot)_S$ is a Cauchy sequence in H , and two Cauchy sequences equivalent for $(\cdot | \cdot)_S$ are equivalent in H . Thus the completion K of D_S for $(\cdot | \cdot)_S$ is identified with a vector subspace of H . Let D be the intersection of K with the domain of definition of S^* , and let S' be the restriction of S^* to D . Then S' is self-adjoint and nonnegative. It is called the Friedrichs extension of S . If U is a unitary operator on H such that $USU^{-1} = S$, then $US'U^{-1} = S'$ [12, pp. 35–36].

B 24. Let G be a locally compact group and let H be a closed subgroup such that the homogeneous space G/H is compact. If G is generated by a compact neighborhood of e , the same is true of H [6, p. 78, Lemma 3].

B 25. Let G be a commutative locally compact group generated by a compact neighborhood of e . Then G is isomorphic to a group $K \times \mathbb{R}^n \times \mathbb{Z}^p$, where K is a compact commutative group [13, p. 110].

B 26. Let A be an algebra and let $a, b \in A$. Then

$$\text{Sp}'(ab) = \text{Sp}'(ba)$$

(this is due to N. Jacobson). Indeed, let $\tilde{A}$ be the algebra obtained from A by adjoining an identity element e . It is enough to prove that, if $\lambda \neq 0$ is such that $ab - \lambda e$ has an inverse u in $\tilde{A}$, then $ba - \lambda e$ is invertible in $\tilde{A}$. But

$$\begin{aligned} (ba - \lambda e)(bua - e) &= b(abu)a - ba - \lambda bua + \lambda e \\ &= b(\lambda u + e)a - ba - \lambda bua + \lambda e = \lambda e, \end{aligned}$$

and similarly $(bua - e)(ba - \lambda e) = \lambda e$. Since $\lambda \neq 0$, this indeed shows that $ba - \lambda e$ is invertible.

B 27. Let A be an algebra over a field k of characteristic zero. Let π, π' be irreducible representations of A on finite-dimensional vector spaces over k . If

$$\text{Tr } \pi(x) = \text{Tr } \pi'(x) \quad (x \in A),$$

then π and π' are equivalent [1, p. 136, Proposition 3].

B 28. Let T be a locally compact space, and let μ be a bounded complex measure on T . If $\|\mu\| = \mu(1)$, then $\mu \geq 0$. Indeed, put $\nu = |\mu|$. We have $\|\mu\| = \|\nu\| = \nu(1)$ and $\mu = f \cdot \nu$, where f is a complex ν -measurable function such that $|f(t)| = 1$ almost everywhere. Let $\alpha < 1$, and let T_α be the set of $t \in T$ such that $\alpha \leq \text{Re } f(t) \leq 1$. Then

$$\begin{aligned} \|\nu\| &= \mu(1) = \text{Re} \int_{T \setminus T_\alpha} f(t) d\nu(t) + \text{Re} \int_{T_\alpha} f(t) d\nu(t) \\ &\leq \alpha \int_{T \setminus T_\alpha} d\nu(t) + \int_{T_\alpha} d\nu(t) \\ &= \int_T d\nu(t) - (1 - \alpha) \int_{T \setminus T_\alpha} d\nu(t) \\ &= \|\nu\| - (1 - \alpha) \int_{T \setminus T_\alpha} d\nu(t). \end{aligned}$$

Thus $T \setminus T_\alpha$ is ν -negligible. Since this is true for every $\alpha < 1$, one has $\text{Re } f(t) \geq 1$ almost everywhere, hence $f(t) = 1$ almost everywhere and $\mu = \nu \geq 0$.

B 29. Let A be a normed algebra. An approximate identity of A is a family $(u_i)_{i \in I}$ of elements of A , indexed by an upward directed set, with the following properties:

- a. $\|u_i\| \leq 1$ for every i ;
- b. $\|u_i x - x\| \rightarrow 0$ and $\|x u_i - x\| \rightarrow 0$ for every $x \in A$.

B 30. Let X be a Borel space, and let $\mathcal{B}$ be the set of Borel subsets of X . A positive measure on X is a map $\mu: \mathcal{B} \rightarrow [0, +\infty]$ such that:

1° if $X_1, X_2, \dots$ are pairwise disjoint elements of $\mathcal{B}$, then

$$\mu(X_1 \cup X_2 \cup \dots) = \mu(X_1) + \mu(X_2) + \dots ;$$

2° X is the union of a sequence of Borel subsets $Y_1, Y_2, \dots$ such that $\mu(Y_i) < +\infty$ for every i .

A positive measure μ on X is said to be standard if there exists a Borel subset N of X such that $\mu(N) = 0$ and $X \setminus N$ is a standard Borel space.

B 31. A topological space is said to be separable if its topology has a countable base. When a space is metrizable, saying that it is separable amounts to saying that it has a countable dense subset. A compact metrizable space is separable.

B 32. In a topological space, a set is called a G_δ if it is the intersection of a countable family of open sets.

B 33. A character of a commutative algebra is a nonzero homomorphism from that algebra into the base field.

B 34. Let V be a finite-dimensional complex vector space, let G be a compact group, and let π be a continuous homomorphism from G into the linear group of V . There exists on V a scalar product invariant under $\pi(G)$ which makes V a Hilbert space [6, p. 70, Proposition 1].

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The first part of this bibliography extends approximately through 1963. The articles are arranged alphabetically by author. The second part extends approximately from 1964 to 1968. The articles are arranged by year. For some time it has become increasingly artificial to separate work devoted to von Neumann algebras from work devoted to C^* -algebras. The division made between the present book and the second edition of my book on von Neumann algebras (Cahiers Scientifiques, fasc. XXV) is therefore rather arbitrary.

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Index of Notation

$\mathbb{C}$: 1.1.1.

$\bar{z}$ (z a complex number): 1.1.2.

$\mathcal{L}(H)$ (H a Hilbert space): 1.1.2.

$\mathrm{Sp}_A x, \mathrm{Sp} x$ (x an element of a unital algebra A): 1.1.5.

$\mathrm{Sp}'_A x, \mathrm{Sp}' x$ (x an element of an algebra A): 1.1.6.

M' (M a subset of an algebra): 1.1.9.

f^* (f a linear functional on an involutive algebra): 1.1.10.

$\mathbb{R}$: 1.3.9.

$f(x)$ (x a normal element of a C^* -algebra; f a complex-valued function): 1.5.2, 1.5.7.

$x^+, x^-, |x|$ (x a Hermitian element of a C^* -algebra): 1.5.7.

A^+ (A a C^* -algebra): 1.6.5.

$x \geq y, y \leq x$ (x, y elements of a C^* -algebra): 1.6.5.

$f \geq g, g \leq f$ (f, g linear functionals): 2.1.1.

H_π (π a representation): 2.2.1, 13.1.3.

$\pi \simeq \pi'$ (π, π' representations): 2.2.1, 13.1.3.

$\bigoplus_i \pi_i, \pi_1 \oplus \cdots \oplus \pi_n$ (π_i representations): 2.2.3, 13.1.3.

$c\pi$ (c a cardinal; π a representation): 2.2.3, 13.1.3.

π_K, π_E (π a representation; K a vector subspace of H_π invariant under π ; E the projection onto K): 2.2.4, 13.1.3.

$\pi' \leq \pi, \pi \geq \pi'$ (π, π' representations): 2.2.4, 13.1.3.

$\bar{\pi}^0$ (π a representation): 2.2.8.

$\widehat{A}$ (A an involutive algebra): 2.3.2.

ω_ξ (ξ a vector in a Hilbert space): 2.4.2.

π_f, ξ_f (f a positive functional): 2.4.4.

$P(A)$ (A an involutive normed algebra): 2.5.2.

$\mathrm{Prim}(A)$ (A an algebra): 2.9.7.

$\widehat{A}_I, \widehat{A}^I$ (A a C^* -algebra; I a closed two-sided ideal): 2.11.2.

$\mathrm{Prim}_I(A), \mathrm{Prim}^I(A)$ (A a C^* -algebra; I a closed two-sided ideal): 2.11.5.

$P_I(A), P^I(A)$ (A a C^* -algebra; I a closed two-sided ideal): 2.11.8.

$\mathrm{Prim}(\varphi), \widehat{\varphi}$ (φ an automorphism of an algebra): 3.1.9.

$\mathrm{Rep}_n(A), \mathrm{Irr}_n(A), \widehat{A}_n$ (A a C^* -algebra): 3.5.1.

$\mathrm{Rep}(A), \mathrm{Irr}(A)$ (A a separable C^* -algebra): 3.8.1.

$\mathcal{LC}(H)$ (H a Hilbert space): 4.1.1.

$\mathfrak{m}(A)$ (A a C^* -algebra): 4.5.2.

$J(A)$ (A a C^* -algebra): 4.7.12.

$\pi \perp \pi'$ (π, π' representations): 5.2.2.

$\pi \approx \pi'$ (π, π' representations): 5.3.2.

$\mathfrak{m}_f$ (f a trace): 6.1.2, 17.1.3.

$\mathfrak{n}_s$ (s a bitrace): 6.2.1.

$N_s, A_s, H_s, J_s, \lambda_s, \rho_s$ (s a bitrace): 6.2.2.

$\mathfrak{U}_s, \mathfrak{V}_s$ (s a bitrace): 6.2.3.

t_s (s a bitrace): 6.2.4.

$N_f, A_f, H_f, J_f, \lambda_f, \rho_f, \mathfrak{U}_f, \mathfrak{V}_f, t_f$ (f a trace): 6.4.5, 17.1.3.

$\mathfrak{m}_s$ (s a bitrace): 6.4.5.

$\text{Fac}_n(A), \text{Fac}(A)$ (A a separable C^* -algebra): 7.1.3.

$\check{A}$ (A a separable C^* -algebra): 7.2.1.

$\text{FacI}_n(A), \text{FacI}(A), \check{A}_I$ (A a separable C^* -algebra): 7.3.1.

$\text{Facf}_n(A), \text{Facf}(A), \check{A}_f$ (A a separable C^* -algebra): 7.4.1.

$\int^\oplus \pi(\zeta) d\mu(\zeta)$ [$\zeta \mapsto \pi(\zeta)$ a measurable field of representations]: 8.1.3.

$\int_{\hat{A}}^\oplus \zeta d\mu(\zeta)$ (A a separable postliminal C^* -algebra; μ a positive measure on $\hat{A}$): 8.6.3.

$\theta_{\xi, \eta}$ (ξ, η vectors in a Hilbert space): 10.6.1.

$\alpha(H, \xi)$ (H a Hilbert space; $\xi \in H$): 10.6.3.

$\beta(A, p)$ (A an elementary C^* -algebra; $p \in A$): 10.6.5.

$\mathcal{A}(\mathcal{H})$ ($\mathcal{H}$ a continuous field of Hilbert spaces): 10.7.2.

$\alpha(\mathcal{H}, x)$ ($\mathcal{H}$ a continuous field of Hilbert spaces; x a continuous vector field): 10.7.5.

$\beta(\mathcal{A}, p)$ ($\mathcal{A}$ a continuous field of elementary C^* -algebras; p a continuous field of projections): 10.7.5.

$\mathcal{U}$: 10.7.10.

$\delta(\mathcal{A})$ ($\mathcal{A}$ a continuous field of elementary C^* -algebras satisfying Fell's condition): 10.7.14.

$\delta(A)$ (A a separable continuous-trace C^* -algebra): 10.9.1.

$E(A)$ (A a C^* -algebra): 11.2.3.

g^+, g^- (g a Hermitian linear functional): 12.3.5.

$\pi \otimes \pi'$ (π, π' representations): 13.1.5.

$\overline{\pi}$ (π a representation): 13.1.5.

1_H (H a Hilbert space): 13.1.7.

$M^1(G)$ (G a locally compact group): 13.2.1.

$\mathcal{K}(G)$ (G a locally compact group): 13.2.2.

$L^1(G)$ (G a locally compact group): 13.2.2.

$\check{f}, \widetilde{f}, f^*, f_s, {}_s f$ (f a function on a group): 13.2.3.

π_φ, ξ_φ (φ a continuous function of positive type): 13.4.6.

$\mu \gg 0$: 13.7.1, 13.7.6.

$C^*(G)$ (G a locally compact group): 13.9.1.

$PP(G)$ (G a topological group): 16.2.2.

$\widehat{G}$ (G a topological group): 13.1.4, 18.1.1.

$\text{Rep}_n(G)$ (G a topological group): 18.1.9.

$\text{Irr}_n(G)$ (G a topological group): 18.1.10.

$\widehat{G}_r$ (G a locally compact group): 18.3.1.

$\text{Rep}(G), \text{Irr}(G)$ (G a separable locally compact group): 18.5.1.

$\text{Fac}_n(G), \text{Fac}(G), \text{FacI}_n(G), \text{FacI}(G)$ (G a separable locally compact group): 18.6.1.

$\check{G}$ (G a separable locally compact group): 18.6.2.

$\check{G}_1$ (G a separable locally compact group): 18.6.3.

$\text{Facf}_n(G), \text{Facf}(G), \check{G}_f$ (G a separable locally compact group): 18.6.4.

$\mathfrak{m}^{1/2}$ ($\mathfrak{m}$ an ideal): A 9.

$\mathfrak{U}_E$ ($\mathfrak{U}$ a von Neumann algebra; E a projection): A 15.

$\prod_i \mathfrak{U}_i$ ($\mathfrak{U}_i$ von Neumann algebras): A 16.

$\mathfrak{U}_1 \otimes \mathfrak{U}_2$ ($\mathfrak{U}_1, \mathfrak{U}_2$ von Neumann algebras): A 18.

Tr : A 30.

$E \sim F$ (E, F projections): A 41.

$E < F, E > F$ (E, F projections): A 41.

$\mathfrak{U}(A), \mathfrak{B}(A)$ (A a Hilbert algebra): A 54.

$\int^\oplus H(\zeta) d\mu(\zeta)$ [$\zeta \mapsto H(\zeta)$ a measurable field of Hilbert spaces]: A 73.

$\int^\oplus V(\zeta) d\mu(\zeta)$ [$\zeta \mapsto V(\zeta)$ an isomorphism]: A 74.

$\int^\oplus T(\zeta) d\mu(\zeta)$ [$\zeta \mapsto T(\zeta)$ a measurable field of operators]: A 77.

$\int^{\oplus} \mathfrak{A}(\zeta) d\mu(\zeta)$ [$\zeta \mapsto \mathfrak{A}(\zeta)$ a measurable field of von Neumann algebras]: [A 87](#).

$\int^{\oplus} \Phi(\zeta) d\mu(\zeta)$ [$\zeta \mapsto \Phi(\zeta)$ a measurable field of isomorphisms]: [A 91](#).

$\int^{\oplus} f(\zeta) d\mu(\zeta)$ [$\zeta \mapsto f(\zeta)$ a measurable field of traces]: [A 92](#).

$\int^{\oplus} A(\zeta) d\mu(\zeta)$ [$\zeta \mapsto A(\zeta)$ a measurable field of Hilbert algebras]: [A 94](#).

$\rho(x)$ (x an element of a Banach algebra): [B 1](#).

$\mathcal{F}x$ (x an element of a commutative Banach algebra): [B 3](#).

G_{δ} : [B 32](#).

Terminological Index

- Abelian (projection): [A 35](#).
- Completed (Hilbert algebra): [A 56](#).
- Adjoint: [1.1.1](#).
- Involutive Banach algebra: [1.2.1](#).
- Von Neumann algebra: [A 4](#).
- Decomposable von Neumann algebra: [A 87](#).
- Von Neumann algebra generated by a set: [A 11](#).
- Enveloping von Neumann algebra: [12.1.4](#).
- Induced, reduced von Neumann algebra: [A 15](#).
- Hilbert algebra: [A 54](#).
- Hilbert algebra of a group: [14.1.1^c](#).
- Direct-sum Hilbert algebra: [A 64](#).
- Involutive algebra: [1.1.1](#).
- Involutive normed algebra: [1.2.1](#).
- Involutive normed algebra obtained by adjoining an identity element: [1.2.3](#).
- Algebraically equivalent (representations): [2.9.6](#).
- Ampliation: [A 21](#).
- Antiliminal (C^* -algebra, group): [4.3.1](#), [13.9.4](#).
- Borel map: [B 19](#).
- Canonical map $\widehat{A} \rightarrow \text{Prim}(A)$: [2.9.7](#).
- Canonical map $P(A) \rightarrow \widehat{A}$: [2.5.4](#).
- Self-adjoint (set): [1.1.1](#).
- Bi-invariant (subspace): [14.3.1](#).
- Bitrace: [6.2.1](#), [17.1.1](#).
- Bitrace associated with a trace: [6.4.1](#).
- Bitrace of type I, II, ...: [6.6.4](#).
- Maximal bitrace: [6.3.3](#).
- Bounded relative to a Hilbert algebra (element): [A 56](#).
- C^* -algebra: [1.3.1](#).
- Continuous-trace C^* -algebra: [4.5.2](#).
- Generalized continuous-trace C^* -algebra: [4.7.12](#).

^c*Translator's note.* The original index gives 14.1.1; the link is directed to 13.10.1, the relevant entry in the text.

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^d*Translator's note.* The original index gives 2.7.7; the link is directed to 2.7.2, where the enveloping C^* -algebra is defined.

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^f*Translator's note.* The original index gives [14.3.4](#); the link is directed to [14.2.4](#), where the term is introduced.

^g*Translator's note.* The original index gives [6.3.9](#); the link is directed to [Definition 6.3.5](#).

^h*Translator's note.* The original index gives [13.10.11](#); the link is directed to [13.11.9](#), where the term is defined.

ⁱ*Translator's note.* The original index gives [13.10.13](#); the link is directed to [13.11.11](#), where real representations are defined.

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^j*Translator's note.* The original index gives [14.2.3](#); the link is directed to Definition [14.1.3](#).

^k*Translator's note.* The original index gives [14.6.2](#); the link is directed to Definition [14.5.2](#).

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